



Systems and Interfaces Configuration Guide, Cisco SD-WAN Release 20.x

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CHAPTER 1

Read Me First



Note

To achieve simplification and consistency, the Cisco SD-WAN solution has been rebranded as Cisco Catalyst SD-WAN. In addition, from Cisco IOS XE SD-WAN Release 17.12.1a and Cisco Catalyst SD-WAN Release 20.12.1, the following component changes are applicable: **Cisco vManage** to **Cisco Catalyst SD-WAN Manager**, **Cisco vAnalytics** to **Cisco Catalyst SD-WAN Analytics**, **Cisco vBond** to **Cisco Catalyst SD-WAN Validator**, **Cisco vSmart** to **Cisco Catalyst SD-WAN Controller**, and **Cisco Controllers** to **Cisco Catalyst SD-WAN Control Components**. See the latest Release Notes for a comprehensive list of all the component brand name changes. While we transition to the new names, some inconsistencies might be present in the documentation set because of a phased approach to the user interface updates of the software product.

Related References

- [Cisco Catalyst SD-WAN Control Components Compatibility Matrix and Server Recommendations](#)
- [Cisco Catalyst SD-WAN Device Compatibility](#)

User Documentation

- [User Documentation for Cisco SD-WAN Release 20](#)

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CHAPTER 2

What's New in Cisco IOS XE (Catalyst SD-WAN)

[What's New in Cisco IOS XE Catalyst SD-WAN Release 17.x](#)

[What's New in Cisco SD-WAN \(vEdge\) Release 20.x](#)



CHAPTER 3

Cisco Catalyst SD-WAN Basic System Overview

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Cisco Catalyst SD-WAN Basic System Overview

Table 1: Feature History

Feature Name	Release Information	Description
CMAC-AES-128 Authentication for NTP Servers	Cisco Catalyst SD-WAN Control Components Release 20.14.1	Support for cipher-based message authentication code (CMAC) advanced encryption standard (AES) 128-bit (cmac-aes-128) authentication for network time protocol (NTP) server configuration for Cisco SD-WAN Control Components.
Statistics Database Upgrade	Cisco Catalyst SD-WAN Manager Release 20.16.1	Cisco SD-WAN Manager utilizes an upgraded database to improve performance

System and Interfaces Overview

Setting up the basic system-wide functionality of network devices is a simple and straightforward process. Basic parameters include defining host properties, such as name and IP address; setting time properties, including NTP; setting up user access to the devices; defining system log (syslog) parameters; and creating network interfaces.

In addition, the Cisco Catalyst SD-WAN software provides a number of management interfaces for accessing the Cisco Catalyst SD-WAN devices in the overlay network.

Host Properties

All devices have basic system-wide properties that specify information that the Cisco Catalyst SD-WAN software uses to construct a view of the network topology. Each device has a system IP address that provides a fixed location of the device in the overlay network. This address, which functions the same way as a router ID on a router, is independent of any of the interfaces and interface IP addresses on the device. The system IP address is one of the four components of the Transport Location (TLOC) property of each device.

A second host property that must be set on all devices is the IP address of the Cisco SD-WAN Validator for the network domain, or a Domain Name System (DNS) name that resolves to one or more IP addresses for Cisco SD-WAN Validators. A Cisco SD-WAN Validator automatically orchestrates the process of bringing up the overlay network, admitting a new device into the overlay, and providing the introductions that allow the device and Cisco SD-WAN Controllers to locate each other.

Two other system-wide host properties are required on all devices, except for the Cisco SD-WAN Validators, to allow the Cisco Catalyst SD-WAN software to construct a view of the topology—the domain identifier and the site identifier.

To configure the host properties, see [Cisco Catalyst SD-WAN Overlay Network Bring-Up Process](#).

Time and NTP

The Cisco Catalyst SD-WAN software implements the Network Time Protocol (NTP) to synchronize and coordinate time distribution across the Cisco Catalyst SD-WAN overlay network. NTP uses a intersection algorithm to select the applicable time servers and avoid issues caused due to network latency. The servers can also redistribute reference time using local routing algorithms and time daemons. NTP is defined in [RFC 5905, Network Time Protocol Version 4: Protocol and Algorithms Specification](#).

User Authentication and Access with AAA, RADIUS, and TACACS+

The Cisco Catalyst SD-WAN software uses Authentication, Authorization, and Accounting (AAA) to provide security for the devices on a network. AAA, in combination with RADIUS and Terminal Access Controller Access-Control System (TACACS+) user authentication, controls which users are allowed access to devices, and what operations they are authorized to perform after they are logged in or connected to the devices.

Authentication refers to the process by which users trying to access the devices are authenticated. To access devices, users log in with a username and a password. The local device can authenticate users. Alternatively, authentication can be performed by a remote device, either a RADIUS server or a TACACS+ server, or both in a sequence.

Authorization determines whether a user is authorized to perform a given activity on a device. In the Cisco Catalyst SD-WAN software, authorization is implemented using role-based access. Access is based on groups that are configured on the devices. A user can be a member of one or more groups. User-defined groups are

considered when performing authorization, that is, the Cisco Catalyst SD-WAN software uses group names received from RADIUS or TACACS+ servers to check the authorization level of a user. Each group is assigned privileges that authorize the group members to perform specific functions on the corresponding device. These privileges correspond to specific hierarchies of the configuration commands and the corresponding hierarchies of operational commands that members of the group are allowed to view or modify.

Beginning in Cisco SD-WAN Release 20.5.1, accounting generates a record of commands that a user executes on a device. Accounting is performed by a TACACS+ server.

For more information, see [Role-Based Access with AAA](#).

Authentication for WANs and WLANs

For wired networks (WANs), Cisco Catalyst SD-WAN devices can run IEEE 802.1X software to prevent unauthorized network devices from gaining access to the WAN. IEEE 802.1X is a port-based network access control (PNAC) protocol that uses a client-server mechanism to provide authentication for devices wishing to connect to the network. You can enable 802.1X on vEdge router interfaces to have the router act as an 802.1X authenticator, responsible for authorizing or denying access to network devices.

IEEE 802.1X authentication requires three components:

- **Requester:** Client device, such as a laptop, that requests access to the Wide-Area Network (WAN). In the Cisco Catalyst SD-WAN overlay network, a supplicant is any service-side device that is running 802.1X-compliant software. These devices send network access requests to the router.
- **Authenticator:** A network device that provides a barrier to the WAN. In the overlay network, you can configure an interface device to act as an 802.1X authenticator. The device supports both controlled and uncontrolled ports. For controlled ports, the Cisco Catalyst SD-WAN device acts as an 802.1X port access entity (PAE), allowing authorized network traffic and preventing unauthorized network traffic ingressing to and egressing from the controlled port. For uncontrolled ports, Cisco Catalyst SD-WAN, acting as an 802.1X PAE, transmits and receives Extensible Authentication Protocol over IEEE 802 (EAP over LAN, or EAPOL) frames.
- **Authentication server:** Host that is running authentication software that validates and authenticates requesters that want to connect to the WAN. In the overlay network, this host is an external RADIUS server. This RADIUS server authenticates each client connected to the 802.1X port interface Cisco Catalyst SD-WAN device and assigns the interface to a virtual LAN (VLAN) before the client is allowed to access any of the services offered by the router or by the LAN.

For wireless LANs (WLANs), routers can run IEEE 802.11i to prevent unauthorized network devices from gaining access to the WLANs. IEEE 802.11i implements Wi-Fi Protected Access (WPA) and Wi-Fi Protected Access II (WPA2) to provide authentication and encryption for devices that want to connect to a WLAN. WPA authenticates individual users on the WLAN using a username and a password. WPA uses the Temporal Key Integrity Protocol (TKIP), which is based on the RC4 cipher. WPA2 implements the NIST FIPS 140-2-compliant AES encryption algorithm along with IEEE 802.1X-based authentication, to enhance user access security over WPA. WPA2 uses the Counter Mode Cipher Block Chaining Message Authentication Code Protocol (CCMP), which is based on the AES cipher. Authentication is done by either using preshared keys or through RADIUS authentication.

Network Segmentation

The Layer 3 network segmentation in Cisco Catalyst SD-WAN is achieved through VPNs on Cisco vEdge devices.

Network Interfaces

In the Cisco Catalyst SD-WAN overlay network design, interfaces are associated with VPNs that translate to VRFs. The interfaces that participate in a VPN are configured and enabled in that VPN. Each interface can be present only in a single VPN.



Note Cisco IOS XE Catalyst SD-WAN devices use VRFs in place of VPNs. When you complete the configuration on Cisco SD-WAN Manager, the system automatically maps the VPN configurations to VRF configurations.

The overlay network has the following types of VPNs/VRFs:

- **VPN 0: Transport VPN**, that carries control traffic using the configured WAN transport interfaces. Initially, VPN 0 contains all the interfaces on a device except for the management interface, and all the interfaces are disabled.
- **VPN 512: Management VPN**, that carries out-of-band network management traffic among the Cisco Catalyst SD-WAN devices in the overlay network. The interface used for management traffic resides in VPN 512. By default, VPN 512 is configured and enabled on all Cisco Catalyst SD-WAN devices except for Cisco vEdge 100. For controller devices, by default, VPN 512 is not configured.
- **Service VPNs**: VPNs 1 through VPN 65535 except for VPN 0 and VPN 512. All service-side interfaces activated in these VPNs connect to a local or branch network that is generally located at the same site as the Cisco Catalyst SD-WAN router. These interfaces carry data traffic throughout the overlay network.

For each network interface, you can configure a number of interface-specific properties, such as DHCP clients and servers, VRRP, interface MTU and speed, and Point-to-Point Protocol over Ethernet (PPPoE). At a high level, for an interface to be operational, you must configure an IP address for the interface and mark it as operational (no shutdown). In practice, you always configure additional parameters for each interface.

Management and Monitoring Options

There are various ways in which you can manage and monitor a router. Management interfaces provide access to devices in the Cisco Catalyst SD-WAN overlay network, allowing you to collect information from the devices in an out-of-band fashion and to perform operations on the devices, such as configuring and rebooting them.

The following management interfaces are available:

- CLI
- IP Flow Information Export (IPFIX)
- RESTful API
- SNMP
- System logging (syslog) messages
- Cisco SD-WAN Manager

CLI

You can access a CLI on each device, and from the CLI, you configure overlay network features on the local device and gather operational status and information regarding that device. Using an available CLI, we strongly

recommend that you configure and monitor all the Cisco Catalyst SD-WAN network devices from Cisco SD-WAN Manager, which provides views of network-wide operations and device status, including detailed operational and status data. In addition, Cisco SD-WAN Manager provides straightforward tools for bringing up and configuring overlay network devices, including bulk operations for setting up multiple devices simultaneously.

You can access the CLI by establishing an SSH session to a Cisco Catalyst SD-WAN device. For a hardware vEdge router, you can also connect to the device's console port.

For a Cisco Catalyst SD-WAN device that is being managed by Cisco SD-WAN Manager, if you create or modify the configuration from the CLI, the changes are overwritten by the configuration that is stored in the Cisco SD-WAN Manager configuration database.

IPFIX

The IP Flow Information Export (IPFIX) protocol, also called cflowd, is a tool for monitoring the traffic flowing through Cisco Catalyst SD-WAN devices in the overlay network and exporting information about the traffic to a flow collector. The exported information is sent in template reports, that contain both information about the flow and the data extracted from the IP headers of the packets in the flow.

Cisco Catalyst SD-WAN cflowd performs 1:1 traffic sampling. Information about all the flows is aggregated in the cflowd records; flows are not sampled.



Note Cisco Catalyst SD-WAN devices do not cache any of the records that are exported to a collector.

The Cisco Catalyst SD-WAN cflowd software implements cflowd Version 10, as specified in RFC 7011 and RFC 7012.

For a list of elements exported by IPFIX, see [Traffic Flow Monitoring with Cflowd](#).

To enable the collection of traffic flow information, you must create data policies that identify the traffic of interest, and then direct that traffic to a cflowd collector. For more information, see [Traffic Flow Monitoring with Cflowd](#).

You can also enable cflowd visibility directly on Cisco Catalyst SD-WAN devices without configuring a data policy, so that you can perform traffic flow monitoring on the traffic coming to the device from all the VPNs in the LAN. You can then monitor the traffic from Cisco SD-WAN Manager or from the device's CLI.

RESTful API

The Cisco Catalyst SD-WAN software provides a RESTful API, which is a programmatic interface for controlling, configuring, and monitoring the Cisco Catalyst SD-WAN devices in an overlay network. You can access the RESTful API through Cisco SD-WAN Manager.

The Cisco Catalyst SD-WAN RESTful API calls expose the functionality of the Cisco Catalyst SD-WAN software and hardware to an application program. Such functionality includes the normal operations you perform to maintain the devices and the overlay network itself.

SNMP

The Simple Network Management Protocol (SNMP) allows you to manage all the Cisco Catalyst SD-WAN devices in the overlay network. The Cisco Catalyst SD-WAN software supports SNMP v2c.

You can configure basic SNMP properties—device name, location, contact, and community—that allow the device to be monitored by an SNMP Network Management System (NMS).

You can configure trap groups and SNMP servers to receive traps.

The object identifier (OID) for the internet port of the SNMP MIB is 1.3.6.1.

SNMP traps are asynchronous notifications that a Cisco Catalyst SD-WAN device sends to an SNMP management server. Traps notify the management server of events, whether normal or significant, that occur on the Cisco Catalyst SD-WAN device. By default, SNMP traps are not sent to an SNMP server. Note that for SNMPv3, the PDU type for notifications, is either SNMPv2c inform (InformRequest-PDU) or trap (Trapv2-PDU).

Syslog Messages

System logging operations use a mechanism that is similar to the UNIX **syslog** command to record system-wide, high-level operations that occur on the Cisco Catalyst SD-WAN devices in the overlay network. The log levels (priorities) of the messages are the same as those in standard UNIX commands, and you can configure the priority of the syslog messages that should be logged. Messages can be logged to files on the Cisco Catalyst SD-WAN device or to a remote host.

Cisco SD-WAN Manager

Cisco SD-WAN Manager is a centralized network management system that allows configuration and management of all the Cisco Catalyst SD-WAN devices in the overlay network, and provides a dashboard displaying the operations of the entire network and of individual devices in the network. Three or more Cisco SD-WAN Manager servers are consolidated into a Cisco SD-WAN Manager cluster to provide scalability and management support for up to 6,000 Cisco Catalyst SD-WAN devices, to distribute Cisco SD-WAN Manager functions across multiple devices, and to provide redundancy of network management operations.

Optional External Services

If you wish to configure additional external services, such as CSA/Cisco SSE, Threat Grid API/Advanced Malware Protection, or an Identity Provider (IdP), be sure to only enable services that are aligned with your compliance and regulatory requirements. For example, your IdP must be authorized at the impact baseline category you want to use.

Basic Settings for Cisco SD-WAN Manager

The System template is used to configure system-level Cisco SD-WAN Manager workflows.

Use the Settings screen to view the current settings and configure the setting for Cisco SD-WAN Manager parameters, including the organization name, Cisco SD-WAN Validators DNS name or IP address, certificate settings, and statistics collection.

The current setting for each item is displayed in the bar for each item, immediately following the name.

Configure Organization Name

Before you can generate a Certificate Signing Request (CSR), you must configure the name of your organization. The organization name is included in the CSR.

In public key infrastructure (PKI) systems, a CSR is sent to a certificate authority to apply for a digital identity certificate.

To configure the organization name:

1. From the Cisco SD-WAN Manager menu, choose **Administration > Settings**.
2. Click **Organization Name**. (If you are using Cisco Catalyst SD-WAN Manager Release 20.12.x or earlier, click **Edit**.)
3. In **Organization Name**, enter the name of your organization. The organization name must be identical to the name that is configured on the Cisco SD-WAN Validator.
4. In **Confirm Organization Name**, re-enter and confirm your organization name.
5. Click **Save**.



Note technology-guides

After the control connections are up and running, the organization name bar is no longer editable.

Configure Cisco SD-WAN Validator DNS Name or IP Address

1. From **Validator**, click **Edit**.
2. In **Validator DNS/IP Address: Port**, enter the DNS name that points to the Cisco SD-WAN Validator or the IP address of the Cisco SD-WAN Validator and the port number to use to connect to it.
3. Click **Save**.



Note The DNS cache timeout should be proportional to the number of Cisco Catalyst SD-WAN Validator IP addresses that DNS has to resolve, otherwise the control connection for Cisco SD-WAN Manager might not come up during a link failure. This is because, when there are more than six IP addresses (this is the recommended number since the default DNS cache timeout is currently two minutes) to check, the DNS cache timer expires even as the highest preferred interface tries all Cisco SD-WAN Validator IP addresses, before failing over to a different color. For instance, it takes about 20 seconds to attempt to connect to one IP address. So, if there are eight IP addresses to be resolved, the DNS cache timeout should be $20 \times 8 = 160$ seconds or three minutes.

Configure Controller Certificate Authorization Settings

Signed certificates are used to authenticate devices in the overlay network. Once authenticated, devices can establish secure sessions between each other. It is from the Cisco SD-WAN Manager that you generate these certificates and install them on the controller devices—Cisco SD-WAN Validator, Cisco SD-WAN Manager, and Cisco SD-WAN Controllers. You can use certificates signed by Symantec, or you can use enterprise root certificates.

The controller certification authorization settings establish how the certification generation for all controller devices will be done. They do not generate the certificates.

You need to select the certificate-generation method only once. The method you select is automatically used each time you add a device to the overlay network.

To have the Symantec signing server automatically generate, sign, and install certificates on each controller device:

1. From **Controller Certificate Authorization**, click **Edit**.
2. Click **Symantec Automated** (Recommended). This is the recommended method for handling controller signed certificates.
3. In the **Confirm Certificate Authorization Change** dialog box, click **Proceed** to confirm that you wish to have the Symantec signing server automatically generate, sign, and install certificates on each controller device.
4. Enter the first and last name of the requester of the certificate.
5. Enter the email address of the requester of the certificate. This address is required because the signed certificate and a confirmation email are sent to the requester via email; they are also made available through the customer portal.
6. Specify the validity period for the certificate. It can be 1, 2, or 3 years.
7. Enter a challenge phrase. The challenge phrase is your certificate password and is required when you renew or revoke a certificate.
8. Confirm your challenge phrase.
9. In **Certificate Retrieve Interval**, specify how often the Cisco SD-WAN Manager server checks if the Symantec signing server has sent the certificate.
10. Click **Save**.

To manually install certificates that the Symantec signing server has generated and signed:

1. From **Controller Certificate Authorization**, click **Edit**.
2. Click **Symantec Manual**.
3. In the **Confirm Certificate Authorization Change** dialog box, click **Proceed** to manually install certificates that the Symantec signing server has generated and signed.
4. Click **Save**.

To use enterprise root certificates:

1. From **Controller Certificate Authorization**, click **Edit**.
2. Click **Enterprise Root Certificate**.
3. In the **Confirm Certificate Authorization Change** dialog box, click **Proceed** to confirm that you wish to use enterprise root certificates.
4. In the **Certificate** box, either paste the certificate, or click **Select a file** and upload a file that contains the enterprise root certificate.
5. By default, the enterprise root certificate has the following properties: To view this information, issue the **show certificate signing-request decoded** command on a controller device, and check the output in the Subject line. For example:

- Country: United States
- State: California
- City: San Jose
- Organizational unit: ENB
- Organization: CISCO
- Domain Name: cisco.com
- Email: cisco-cloudops-sdwan@cisco.com

```
vSmart# show certificate signing-request decoded
...
Subject: C=US, ST=California, L=San Jose, OU=ENB, O=CISCO, CN=vsmart-uuid
.cisco.com/emailAddress=cisco-cloudops-sdwan@cisco.com
...
```

To change one or more of the default CSR properties:

- a. Click **Set CSR Properties**.
- b. Enter the domain name to include in the CSR. This domain name is appended to the certificate number (CN).
- c. Enter the organizational unit (OU) to include in the CSR.
- d. Enter the organization (O) to include in the CSR.
- e. Enter the city (L), state (ST), and two-letter country code (C) to include in the CSR.
- f. Enter the email address (emailAddress) of the certificate requester.
- g. Specify the validity period for the certificate. It can be 1, 2, or 3 years.

6. Click **Import & Save**.

Enforce Software Version on Devices

If you are using the Cisco Catalyst SD-WAN hosted service, you can enforce a version of the Cisco Catalyst SD-WAN software to run on a router when it first joins the overlay network.

To ensure that templates are in sync after an upgrade that enforces a software version, make sure of the following before you perform the upgrade:

- The bootflash and flash on the router must have enough free space to support the upgrade
- The version of the SD-WAN image that is on the device before the upgrade must be a lower version than the enforced SD-WAN version you specify in the following procedure
- For ZTP enforcement feature to start, initial onboarding has to be done through PNP/ZTP workflow.

To enforce a version of the Cisco Catalyst SD-WAN software to run on a router when it first joins the overlay network, follow these steps:

1. Ensure that the software image for the desired device software version is present in the Cisco SD-WAN Manager software image repository:
 - a. From the Cisco SD-WAN Manager menu, choose **Maintenance > Software Repository**.
The Software Repository screen opens and displays a table of software images. If the desired software image is present in the repository, continue with Step 2.
 - b. If you need to add a software image, click **Add New Software**.
 - c. Select the location from which to download the software images, either Cisco SD-WAN Manager, Remote Server, or Remote Server - Cisco SD-WAN Manager.
 - d. Select an x86-based or a MIPS-based software image.
 - e. To place the image in the repository, click **Add**.
2. From the Cisco SD-WAN Manager menu, choose **Administration > Settings**.
3. Click **Enforce Software Version (ZTP)**.
(In Cisco Catalyst SD-WAN Manager Release 20.12.x and earlier, locate **Enforce Software Version (ZTP)** and click **Edit**.)
4. For a specific platform, enable enforcing the software version.
5. Do one of the following:
 - Use an image on a local server:
 - a. In the **Image Location** field, choose **Local Server**.
 - b. In the **Version/Image Name** field, choose an image.
 - Use an image on a remote server:
 - a. In the **Image Location** field, choose **Remote Server**.
 - b. In the **Remote Server Name** field, choose a server.
 - c. In the **Image Filename** field, choose an image.
6. Click **Save**.

Configure a login page banner using a configuration group

You can configure the banner text for login pages.

Before you begin

On the **Configuration > Configuration Groups** page, choose **SD-WAN** as the solution type.

Procedure

Step 1 From the Cisco SD-WAN Manager menu, choose **Configuration > Configuration Groups**.

Step 2 Create and configure a [Banner](#) feature in a System profile.

Table 2: Banner Basic Settings

Field	Description
Type	Choose a feature from the drop-down list.
Feature Name*	Enter a name for the feature.
Description	Enter a description of the feature. The description can contain any characters and spaces.
Login	Enter the text to display before the login prompt. The string can be up to 2048 characters long. To insert a line break, type \n.
Message of the Day	On a Cisco IOS XE Catalyst SD-WAN device, enter the message-of-the-day text to display before the login banner. The string can be up to 2048 characters long. To insert a line break, type \n.

What to do next

Also see [Deploy a configuration group](#).

Configure a login page banner, using templates

Use the Banner template for Cisco Catalyst SD-WAN Validators, Cisco SD-WAN Managers, Cisco Catalyst SD-WAN Controllers, Cisco vEdge devices, and s.

You can configure two different banner text strings, one to be displayed before the CLI login prompt on a Cisco Catalyst SD-WAN device and the other to be displayed after a successful login to the device.

- To configure the banner text for login screens using Cisco SD-WAN Manager templates, create a Banner feature template to configure PIM parameters, as described in this topic.
- To configure a login banner for the Cisco SD-WAN Manager system, from the Cisco SD-WAN Manager menu, choose **Administration > Settings**.

Configure a Banner

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Device Templates**, and click **Create Template**.



Note In Cisco vManage Release 20.7.x and earlier releases, **Device Templates** is called **Device**.

3. From the **Create Template** drop-down list, select **From Feature Template**.
4. From the **Device Model** drop-down list, select the type of device for which you are creating the template.
5. Click **Additional Templates** or scroll to the **Additional Templates** section.
6. From the **Banner** drop-down list, click **Create Template**. The **Banner** template form is displayed. This form contains fields for naming the template, and the fields for defining Banner parameters.
7. In **Template Name**, enter a name for the template. The name can be up to 128 characters and can contain only alphanumeric characters.
8. In **Template Description**, enter a description of the template. The description can be up to 2048 characters and can contain only alphanumeric characters.

When you first open a feature template, for each parameter that has a default value, the scope is set to Default (indicated by a check mark), and the default setting or value is shown. To change the default or to enter a value, click the **Scope** drop-down list.

9. To set a banner, configure the following parameters:

Table 3: Parameters to be configured while setting a banner:

Parameter Name	Description
MOTD Banner	On a Cisco vEdge device enter message-of-the-day text to display after a successful login. The string can be up to 2048 characters long. To insert a line break, type \n.
Login Banner	Enter text to display before the login prompt. The string can be up to 2048 characters long. To insert a line break, type \n.

10. To save the feature template, click **Save**.

CLI equivalent:

```
banner{login text | motd text}
```

Release Information

Introduced in Cisco SD-WAN Manager NMS in Release 15.2.

Create a Custom Banner

To create a custom banner that is displayed after you log in to the Cisco SD-WAN Manager:

1. From **Banner**, click **Edit**.
2. In **Enable Banner**, click **Enabled**.
3. In **Banner Info**, enter the text string for the login banner or click **Select a File** to download a file that contains the text string.

4. Click **Save**.

Collect Device Statistics

Enable or disable the collection of statistics for devices in the overlay network. By default, the collection of statistics is enabled for all the devices in the overlay network.

1. From the Cisco SD-WAN Manager menu, choose **Administration > Settings**.
2. To modify the settings for collecting device statistics, click **Statistics Database Configuration**. (If you are using Cisco Catalyst SD-WAN Manager Release 20.12.x or earlier, click **Statistics Setting** and **Edit**).



Note From Cisco Catalyst SD-WAN Manager Release 20.16.1, upgrading the statistics database improves the performance of statistics processing. This upgrade enhances the Cisco SD-WAN Manager response time and scalability by reducing the software image size.

To delete the previous data from the statistics database, use the REST API provided in the [Cisco Developer Documentation](#).

By default, for every group of statistics (such as **Aggregated SAIE** and **AppHosting**), collection of statistics is enabled for all devices.

3. To enable the collection of a group of statistics for all devices, click **Enable All** for the particular group.
4. To disable the collection of a group of statistics for all devices, click **Disable All** for the particular group.
5. To collect statistics for all devices exclusively for Cisco SD-WAN Analytics, click **vAnalytics** for the specific group.
6. To enable or disable the collection of a group of statistics for specific devices in the overlay network, click **Custom** for the particular group.

In the **Select Devices** dialog box, depending on whether statistics collection is enabled or disabled for a device, the device is listed between **Enabled Devices** or **Disabled Devices** respectively.

- a. To enable statistics collection for one or more devices, choose the devices from **Disabled Devices** and move them to **Enabled Devices**.



Tip To choose all **Disabled Devices**, click **Select All**.

- b. To disable statistics collection for one or more devices, choose the devices from **Enabled Devices** and move them to **Disabled Devices**.



Tip To choose all **Enabled Devices**, click **Select All**.

- c. To save your selections, click **Done**.
To discard your selections, click **Cancel**.

7. To apply the modified settings, click **Save**.
To discard your changes, click **Cancel**.
To revert to the default settings, click **Restore Factory Default**.

Configure the Time Interval To Collect Device Statistics

1. From the Cisco SD-WAN Manager menu, choose **Administration > Settings**.
2. To modify the time interval at which device statistics are collected, click **Statistics Configuration**.
3. Enter the required **Collection Interval** in minutes.
 - Default value: 30 minutes
 - Minimum value: 5 minutes
 - Maximum value: 180 minutes
4. To apply the modified settings, click **Save**.
To discard your changes, click **Cancel**.
To revert to the default settings, click **Restore Factory Default**.



Note Starting from Cisco Catalyst SD-WAN Manager Release 20.9.6, Cisco SD-WAN Manager collects device statistics files at increased frequency, independent of any configured collection intervals.

Configure or Cancel an Cisco SD-WAN Manager Server Maintenance Window

You can set or cancel the start and end times and the duration of the maintenance window for the Cisco SD-WAN Manager server.

1. From the Cisco SD-WAN Manager menu, choose **Administration > Settings**.
2. Click **Maintenance Window**. (If you are using Cisco IOS XE Catalyst SD-WAN Release 17.12.x or earlier, click **Maintenance Window** and then click **Edit**.)
To cancel the maintenance window, click **Cancel**.
3. Click the **Start Date** and **Start Time** drop-down list. Select the date and time when the **Maintenance Window** will start.
4. Click the **End Date** and **EndTime** drop-down list. Select the date and time when the **Maintenance Window** will end.
5. Click **Save**. The start and end times and the duration of the maintenance window are displayed in the **Maintenance Window** bar.

Two days before the start of the window, the Cisco SD-WAN Manager Dashboard displays a maintenance window alert notification.

Configure Basic System Parameters Using a Configuration Group

Before you begin

On the **Configuration** > **Configuration Groups** page, choose **SD-WAN** as the solution type.

Procedure

- Step 1** From the Cisco SD-WAN Manager menu, choose **Configuration** > **Configuration Groups**.
- Step 2** Create and configure a Basic feature in a System profile.

- a. Configure basic settings.

Table 4: Basic Settings

Field	Description
Time Zone	Choose the time zone to use on the device.
Device Groups	Enter the names of one or more groups to which the device belongs, separated by commas.
Location	Enter a description of the location of the device. It can be up to 128 characters.
Description	Enter any additional descriptive information about the device.
Transport Gateway	(Minimum supported release: Cisco Catalyst SD-WAN Manager Release 20.13.1) Enable transport gateway functionality for the device. A transport gateway connects routers that may or may not have direct connectivity. One common use case for transport gateways is to provide connectivity between routers in disjoint networks, such as between public and private WANs. Another use case for transport gateway functionality is to use a transport gateway as the hub in a hub-and-spoke topology.

- b. Configure controller settings.
- c. Configure GPS.

Table 5: GPS

Field	Description
GPS Latitude	Enter the latitude of the device, in the format decimal-degrees.
GPS Longitude	Enter the longitude of the device, in the format decimal-degrees.

- d. Configure track settings.

Table 6: Track Settings

Field	Description
Track Transport	Enable this option to regularly check whether the DTLS connection between the device and a Cisco SD-WAN Validator is up. Default: Enabled
Track Default Gateway	Enable or disable tracking of default gateway. Gateway tracking determines, for static routes, whether the next hop is reachable before adding that route to the route table of the device. Default: Enabled
Track Interface Tag	Set the tag string to include in routes associated with a network that is connected to a non-operational interface. Range: 1 through 4294967295
Tracker DIA Stabilize Status	Enable this option to stabilize interface flaps by using the multiplier to update HTTP or ICMP tracker status from DOWN to UP.

- e. Configure advanced settings.

Table 7: Advanced

Field	Description
Port Hopping	Enable or disable port hopping. When a Cisco Catalyst SD-WAN device is behind a NAT, port hopping rotates through a pool of preselected OMP port numbers (called base ports) to establish DTLS connections with other Cisco Catalyst SD-WAN devices when a connection attempt is unsuccessful. The default base ports are 12346, 12366, 12386, 12406, and 12426. To modify the base ports, set a port offset value. Default: Enabled
Port Offset	Enter a number by which to offset the base port number. Configure this option when multiple Cisco Catalyst SD-WAN devices are behind a single NAT device, to ensure that each device uses a unique base port for DTLS connections. Values: 0 through 19
On Demand Tunnel	Enable dynamic on-demand tunnels between any two Cisco Catalyst SD-WAN spoke devices.
On Demand Tunnel Idle Timeout (In Minute)	Enter the on-demand tunnel idle timeout time. After the configured time, the tunnel between the spoke devices is removed. Range: 1 to 65535 minutes Default: 10 minutes

Field	Description
Control Session PPS	Enter a maximum rate of DTLS control session traffic to police the flow of control traffic. Range: 1 through 65535 pps Default: 300 pps
Multi Tenant	Enable this option to specify the device as multitenant.
Admin Tech On Failure	Enable this option to collect admin-tech information when the device reboots. Default: Enabled

What to do next

Also see [Deploy a configuration group](#).

Configure Basic System Parameters

Use the System template for all Cisco Catalyst SD-WAN devices.

To configure system-wide parameters using Cisco SD-WAN Manager templates:

1. Create a **System** feature template to configure system parameters.
2. Create an **NTP** feature template to configure NTP servers and authentication.
3. Configure the organization name and Cisco Catalyst SD-WAN Validator IP address on the Cisco SD-WAN Manager. These settings are appended to the device templates when the templates are pushed to devices.

Create System Template

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Device Templates**, and click **Create Template**.



Note In Cisco vManage Release 20.7.x and earlier releases, **Device Templates** is called **Device**.

3. From the **Create Template** drop-down list, select **From Feature Template**.
4. From the **Device Model** drop-down list, select the type of device for which you are creating the template.
5. To create a custom template for System, select the **Factory_Default_System_Template** and click **Create Template**.

The System template form is displayed. This form contains fields for naming the template, and fields for defining the System parameters.

6. In **Template Name**, enter a name for the template. The name can be up to 128 characters and can contain only alphanumeric characters.
7. In **Template Description**, enter a description of the template. The description can be up to 2048 characters and can contain only alphanumeric characters.

When you first open a feature template, for each parameter that has a default value, the scope is set to Default (indicated by a check mark), and the default setting or value is shown. To change the default or to enter a value, click the scope drop-down to the left of the parameter field and select one of the following:

Table 8:

Parameter Scope	Scope Description
Device Specific (indicated by a host icon)	Use a device-specific value for the parameter. For device-specific parameters, you cannot enter a value in the feature template. You enter the value when you attach a Cisco Catalyst SD-WAN device to a device template. When you click Device Specific, the Enter Key box opens. This box displays a key, which is a unique string that identifies the parameter in a CSV file that you create. This file is an Excel spreadsheet that contains one column for each key. The header row contains the key names (one key per column), and each row after that corresponds to a device and defines the values of the keys for that device. You upload the CSV file when you attach a Cisco Catalyst SD-WAN device to a device template. To change the default key, type a new string and move the cursor out of the Enter Key box. Examples of device-specific parameters are system IP address, host name, GPS location, and site ID.
Global (indicated by a globe icon)	Enter a value for the parameter, and apply that value to all devices. Examples of parameters that you might apply globally to a group of devices are DNS server, syslog server, and interface MTUs.

Basic System-Wide Configuration

To set up system-wide functionality on a Cisco Catalyst SD-WAN device, select the **Basic Configuration** tab and then configure the following parameters. Parameters marked with an asterisk are required.

Table 9:

Parameter Field	Description
Site ID* (on routers, Cisco SD-WAN Manager instances, and Cisco SD-WAN Controller)	Enter the identifier of the site in the Cisco Catalyst SD-WAN overlay network domain in which the device resides, such as a branch, campus, or data center. The site ID must be the same for all Cisco Catalyst SD-WAN devices that reside in the same site. <i>Range:</i> 1 through 4294967295 ($2^{32} - 1$)

Parameter Field	Description
System IP*	Enter the system IP address for the Cisco Catalyst SD-WAN device, in decimal four-part dotted notation. The system IP address provides a fixed location of the device in the overlay network and is a component of the device's TLOC address. It is used as the device's loopback address in the transport VPN (VPN 0). You cannot use this same address for another interface in VPN 0.
Timezone*	Select the timezone to use on the device.
Hostname	Enter a name for the Cisco Catalyst SD-WAN device. It can be up to 32 characters.
Location	Enter a description of the location of the device. It can be up to 128 characters.
Device Groups	Enter the names of one or more groups to which the device belongs, separated by commas.
Controller Groups	List the Cisco Catalyst SD-WAN Controller groups to which the router belongs.
Description	Enter any additional descriptive information about the device.
Console Baud Rate	Select the baud rate of the console connection on the router. <i>Values:</i> 1200, 2400, 4800, 9600, 19200, 38400, 57600, 115200 baud or bits per second (bps). <i>Default:</i> 115200 bps
Maximum OMP Sessions	Set the maximum number of OMP sessions that a router can establish to a Cisco Catalyst SD-WAN Controller. <i>Range:</i> 0 through 100. <i>Default:</i> 2
Dedicated Core for TCP Optimization (optional, on vEdge 1000 and 2000 routers only)	Click On to carve out a separate CPU core to use for performing TCP optimization.

To save the feature template, click **Save**.

CLI equivalent:

```

system
clock timezone timezone
console-baud-rate rate
controller-group-list numbers
description text
device-groups group-name
host-name string
location string
max-omp-sessions number
site-id site-id
system-ip ip-address
tcp-optimization-enabled

```

To configure the DNS name or IP address of the Cisco Catalyst SD-WAN Validator in your overlay network, go to **Administration > Settings** screen and click **Validator**. (If you are using Cisco Catalyst SD-WAN Manager Release 20.12.x or earlier, click **vBond**.)

Configure the GPS Location

To configure a device location, select the **GPS** tab and configure the following parameters. This location is used to place the device on the Cisco SD-WAN Manager network map. Setting the location also allows Cisco SD-WAN Manager to send a notification if the device is moved to another location.

Table 10:

Parameter Field	Description
Latitude	Enter the latitude of the device, in the format <i>decimal-degrees</i> .
Longitude	Enter the longitude of the device, in the format <i>decimal-degrees</i> .

To save the feature template, click **Save**.

CLI equivalent:

```
system gps-location (latitude decimal-degrees | longitude decimal-degrees)
```

Configure Interface Trackers for NAT Direct Internet Access

Table 11: Feature History

Feature Name	Release Information	Description
Support for Interface Status Tracking on Cisco vEdge Devices	Cisco vManage Release 17.2.2	This feature supports interface tracking on Cisco vEdge devices.
Dual Endpoint Support for Interface Status Tracking on Cisco vEdge Devices	Cisco SD-WAN Release 20.6.1 Cisco vManage Release 20.6.1	This feature allows you to configure tracker groups with dual endpoints using the Cisco System template and associate each template group to an interface. The dual endpoints provide redundancy for tracking the status of transport interfaces to avoid false negatives.

The DIA tracker helps determine if the internet or external network becomes unavailable. This feature is useful when NAT is enabled on a transport interface in VPN 0 to allow data traffic from the router to exit directly to the internet.

If the internet or external network becomes unavailable, the router continues to forward traffic based on the NAT route in the service VPN. Traffic that is forwarded to the internet gets dropped. To prevent the internet-bound traffic from being dropped, configure the DIA tracker on the edge router to track the status of the transport interface. The tracker periodically probes the interface IP address of the end point of the tunnel interface to determine the status of the transport interface. The tracker determines the status of the internet and returns the data to the attach points that are associated with the tracker.

When the tracker is configured on the transport interface, the interface IP address is used as a source IP address for probe packets.

IP SLA monitors the status of probes and measures the round trip time of these probe packets and compares the values with the configured latency in the probe. When the latency exceeds the configured threshold value, the tracker considers the network as unavailable.

If the tracker determines that the local internet is unavailable, the router withdraws the NAT route and reroutes the traffic based on the local routing configuration to overlay.

The local router continues to periodically check the status of the path to the interface. When it detects that the path is functioning again, the router reinstalls the NAT route to the internet.

Starting Cisco SD-WAN Release 20.6.1 you can configure a tracker group with dual endpoints on Cisco vEdge devices (using two trackers) and associate this tracker group to an interface. Dual endpoints help in avoiding false negatives that might be introduced regarding unavailability of the internal or external network.

Restrictions for Configuring Tracker Groups for Dual Endpoints

A tracker group with dual endpoints can only be configured on the following types of interfaces:

- Ethernet Interfaces
- Subinterfaces
- PPPoE Interfaces

Configure NAT DIA Tracker

To track the status of transport interfaces that connect to the internet (Network Address Translation Direct Internet Access (NAT DIA)), click **Tracker > Add New Tracker** and configure the following parameters:

Table 12:

Parameter Field	Description
Name	Name of the tracker. The name can be up to 128 alphanumeric characters. You can configure up to eight trackers.
Tracker Type	Choose an interface, static route, or a tracker group. Starting Cisco SD-WAN Release 20.6.1 you can configure a tracker group with dual endpoints on Cisco vEdge devices and associate this tracker group to an interface. Choose Tracker type as Interface for NAT DIA and dual endpoint tracker configuration.
Tracker Type: Tracker Elements	This field is displayed only if you chose Tracker Type as a tracker-group. Add the existing interface tracker names (separated by a space). When you add this tracker to the template, the tracker group is associated with these individual trackers and you can then associate the tracker group to an interface.

Parameter Field	Description
Tracker Type: Tracker Boolean	This field is displayed only if you chose Tracker Type as a tracker-group. Select AND or OR explicitly. An OR operation ensures that the transport interface status is reported as active if either one of the associated trackers of the tracker group report that the interface is active. If you select the AND operation, the transport interface status is reported as active if both the associated trackers of the tracker group report that the interface is active.
Threshold	How long to wait for the probe to return a response before declaring that the transport interface is down. <i>Range:</i> 100 through 1000 milliseconds. <i>Default:</i> 300 milliseconds.
Interval	How often probes are sent to determine the status of the transport interface. <i>Range:</i> 10 through 600 seconds. <i>Default:</i> 60 seconds (1 minute)
Multiplier	Number of times to resend probes before declaring that the transport interface is down. <i>Range:</i> 1 through 10. <i>Default:</i> 3
End Point Type: IP Address	IP address of the end point of the tunnel interface. This is the destination in the internet to which the router sends probes to determine the status of the transport interface. Note In Cisco SD-WAN Release 20.5.1 and later releases, if the tracker receives an HTTP response status code, which is less than 400, the endpoint is reachable. Prior to Cisco SD-WAN Release 20.5.1, the endpoint is reachable if the tracker receives an HTTP response status code of 200.
End Point Type: DNS Name	DNS name of the end point of the tunnel interface. This is the destination in the internet to which the router sends probes to determine the status of the transport interface.

To save a tracker, click **Add**.

To save the feature template, click **Save**.

Configure NAT DIA Tracker Using the CLI

Configure NAT DIA tracker

```

system
  tracker tracker-name
  endpoint-dns-name dns-name
  endpoint-ip ip-address
  interval seconds
  multiplier number
  threshold milliseconds

```

Configure tracker group and assign it to an interface



Note You can configure only one endpoint per tracker.

```

system
  tracker nat-tracker1
    endpoint-ip 10.1.1.1
  !
  tracker nat-tracker2
    endpoint-ip 10.2.2.2
  !
  tracker nat-tracker3
    tracker-type tracker-group
    boolean or
    tracker-elements nat-tracker1 nat-tracker2
  !
!
vpn 0
  interface ge0/1
    nat
    tracker nat-tracker3
  !
!

```

Verify dual endpoints configuration

vEdge1# **show running-config system | begin tracker**

```

tracker nat-tracker1
  endpoint-ip 10.1.1.1
!
tracker nat-tracker2
  endpoint-ip 10.2.2.2
!
tracker nat-tracker3
  boolean or
  tracker-type tracker-group
  tracker-elements nat-tracker1 nat-tracker2
!

```

vEdge1# **show tracker tracker-group**

VPN	INTERFACE	TRACKER NAME	BOOLEAN	STATUS	TRACKER ELEMENT NAME	TRACKER ELEMENT STATUS	TRACKER ELEMENT RTT
0	ge0_1	nat-tracker3	or	DOWN	nat-tracker1	DOWN	Timeout
					nat-tracker2	DOWN	Timeout

Apply Tracker to an Interface

To apply a tracker to an interface, configure it in the **VPN Interface Cellular**, **VPN Interface Ethernet**, **VPN Interface NAT Pool**, or **VPN Interface PPP** configuration templates. You can apply only one tracker to an interface.

A tracker group with dual endpoints can only be configured on the following types of interfaces:

- Ethernet Interfaces
- Subinterfaces

- PPPoE Interfaces

Monitor NAT DIA Endpoint Tracker Configuration

1. From the Cisco SD-WAN Manager menu, choose **Monitor** > **Devices**.
Cisco vManage Release 20.6.x and earlier: From the Cisco SD-WAN Manager menu, choose **Monitor** > **Network**.
2. Choose a device from the list of devices.
3. Click **Real Time**.
4. From the **Device Options** drop-down list, choose **Dual Endpoint Tracker Info**.

Configure Advanced Options

To configure additional system parameters, click **Advanced**:

Table 13:

Parameter Name	Description
Control Session Policer Rate	Specify a maximum rate of DTLS control session traffic, to police the flow of control traffic. <i>Range:</i> 1 through 65535 pps. <i>Default:</i> 300 pps
MTU of DTLS Tunnel	Specify the MTU size to use on the DTLS tunnels that send control traffic between Cisco Catalyst SD-WAN devices. <i>Range:</i> 500 through 2000 bytes. <i>Default:</i> 1024 bytes
Port Hopping	Click On to enable port hopping, or click Off to disable it. When a Cisco Catalyst SD-WAN device is behind a NAT, port hopping rotates through a pool of preselected OMP port numbers (called base ports) to establish DTLS connections with other Cisco Catalyst SD-WAN devices when a connection attempt is unsuccessful. The default base ports are 12346, 12366, 12386, 12406, and 12426. To modify the base ports, set a port offset value. To disable port hopping on an individual TLOC (tunnel interface), use the VPN Interface Ethernet configuration template. <i>Default:</i> Enabled (on routers); disabled (on Cisco SD-WAN Manager devices and Cisco Catalyst SD-WAN Controllers).
Port Offset	Enter a number by which to offset the base port number. Configure this option when multiple Cisco Catalyst SD-WAN devices are behind a single NAT device, to ensure that each device uses a unique base port for DTLS connections. <i>Values:</i> 0 through 19
DNS Cache Timeout	Specify when to time out the Cisco Catalyst SD-WAN Validator addresses that have been cached by the device. <i>Range:</i> 1 through 30 minutes. <i>Default:</i> 30 minutes
Track Transport	Click On to regularly check whether the DTLS connection between the device and a Cisco Catalyst SD-WAN Validator is up. Click Off to disable checking. By default, transport checking is enabled.

Parameter Name	Description
Local vBond (only on routers acting as vBond orchestrators)	Click On to configure the router to act as a Cisco Catalyst SD-WAN Validator. Then specify the DNS name for the Cisco Catalyst SD-WAN Validator or its IP address, in decimal four-part dotted notation.
Track Interface	Set the tag string to include in routes associated with a network that is connected to a non-operational interface. <i>Range:</i> 1 through 4294967295
Multicast Buffer	Specify the percentage of interface bandwidth that multicast traffic can use. <i>Range:</i> 5% through 100% <i>Default:</i> 20%
USB Controller (on vEdge 1000 and 2000 series routers only)	Click On to enable or click Off to disable the USB controller, which drives the external USB ports. If you enable the USB controller, the vEdge router reboots when you attach the device template to the device. <i>Default:</i> Disabled
Gateway Tracking	Click On to enable or click Off to Disable tracking of default gateway. Gateway tracking determines, for static routes, whether the next hop is reachable before adding that route to the device's route table. <i>Default:</i> Enabled
Host Policer Rate (on vEdge routers only)	Specify the maximum rate at which a policer delivers packets to the control plane. <i>Range:</i> 1000 through 20000 pps. <i>Default:</i> 5000 pps
ICMP Error Rate (on vEdge routers only)	Specify how many ICMP error messages a policer can generate or receive. <i>Range:</i> 1 through 200 pps <i>Default:</i> 100 pps
Allow Same-Site Tunnel (on vEdge routers only)	Click On to allow tunnels to be formed between vEdge routers in the same site. Note that no BFD sessions are established between the two collocated vEdge routers. <i>Default:</i> Off
Route Consistency Check (on vEdge routers only)	Click On to check whether the IPv4 routes in the device's route and forwarding table are consistent.
Collect Admin Tech on Reboot	Click On to collect admin-tech information when the device reboots.
Idle CLI Timeout in minutes	Set how long the CLI is inactive on a device before the user is logged out. If a user is connected to the device via an SSH connection, the SSH connection is closed after this time expires. <i>Default:</i> CLI session does not time out. The Idle CLI Timeout in minutes is represented as "exec-timeout" in the CLI equivalent shown next.
Eco-Friendly Mode (on vEdge Cloud routers only)	Click On to configure a Cloud router not to use its CPU minimally or not at all when the router is not processing any packets.

To save the feature template, click **Save**.

CLI equivalent:

```
system
  admin-tech-on-failure allow-same-site-tunnels
  control-session-pps rate eco-friendly-mode
  host-policer-pps rate

  icmp-error-pps rate
```

```

exec-timeout seconds multicast-buffer-percent percentage

port-hop port-offset number route-consistency-check
system-tunnel-mtu bytes timer
dns-cache-timeout minutes track-default-gateway
track-interface-tag number

track-transport upgrade-confirm minutes [no] usb-controller (on Cisco vEdge 1000 and
Cisco vEdge2000 routers only)
vbond (dns-name | ip-address) local (on Cisco vEdge routers acting as Cisco vBond
controllers)

```



Note Exec-Timeout governs the CLI logouts for devices in controller mode.

Release Information

Introduced in Cisco SD-WAN Manager in Release 15.2. In Releases 15.3.8 and 15.4.3, add Track Interface field. In Release 17.1.0, add Route Consistency Check and Collect Admin Tech on Reboot fields. In Release 17.2.0, add support for CLI idle timeout and eco-friendly mode. In Release 17.2.2, add support for interface status tracking.

Configure Global Using a Configuration Group

Before you begin

On the **Configuration > Configuration Groups** page, choose **SD-WAN** as the solution type.

Procedure

Step 1 From the Cisco SD-WAN Manager menu, choose **Configuration > Configuration Groups**.

Step 2 Create and configure a Global feature in a System profile.

a) Configure services.

Table 14: Services

Field	Description
HTTP Server	Enable or disable HTTP server.
HTTPS Server	Enable or disable secure HTTPS server.
FTP Passive	Enable or disable passive FTP.
Domain Lookup	Enable or disable Domain Name System (DNS) lookup.
ARP Proxy	Enable or disable proxy ARP.

Field	Description
RSH/RCP	Enable or disable remote shell (RSH) and remote copy (rcp) on the device.
Line Virtual Teletype (Configure Outbound Telnet)	Enable or disable outbound telnet.
Cisco Discovery Protocol (CDP)	Enable or disable Cisco Discovery Protocol (CDP).
Link Layer Discovery Protocol (LLDP)	Enable or disable Link Layer Discovery Protocol (LLDP).
Specify interface for source address	Enter the address of the source interface in all HTTPS client connections.

- b) Configure NAT 64 timeout.

Table 15: NAT 64

Field	Description
UDP Timeout	Specify the NAT64 translation timeout for UDP. Range: 1 to 536870 (seconds) Default: 300 seconds (5 minutes)
TCP Timeout	Specify the NAT64 translation timeout for TCP. Range: 1 to 536870 (seconds) Default: 3600 seconds (1 hour)

- c) Configure Authentication type.

Table 16: Authentication

Field	Description
HTTP Authentication	Choose the HTTP authentication mode. Accepted values: Local, AAA Default: Local

- d) Configure SSH Version.

Table 17: SSH Version

Field	Description
SSH Version	Choose the SSH version. Default: Disabled

- e) Configure Other Settings.

Table 18: Other Settings

Field	Description
TCP Keepalives (In)	Enable or disable generation of keepalive timers when incoming network connections are idle.
TCP Keepalives (Out)	Enable or disable generation of keepalive timers when outgoing network connections are idle.
TCP Small Servers	Enable or disable small TCP servers (for example, ECHO).
UDP Small Servers	Enable or disable small UDP servers (for example, ECHO).
Console Logging	Enable or disable console logging. By default, the router sends all log messages to its console port.
IP Source Routing	Enable or disable IP source routing. IP source routing is a feature that enables the originator of a packet to specify the path for the packet to use to get to the destination.
VTY Line Logging	Enable or disable the device to display log messages to a vty session in real time.
SNMP IFINDEX Persist	Enable or disable SNMP IFINDEX persistence, which provides an interface index (ifIndex) value that is retained and used when the device reboots.
Ignore BOOTP	Enable or disable BOOTP server. When enabled, the device listens for the BOOTP packet that comes in sourced from 0.0.0.0. When disabled, the device ignores these packets.
(optional) Interface statistics per minute	Minimum supported release: Cisco IOS XE Catalyst SD-WAN Release 26.1.1 Choose the time interval for interface statistics data collection: • 1 minute • 5 minutes (default)

Configure Global Parameters

Use the Global Settings template to configure a variety of global parameters for all Cisco IOS XE Catalyst SD-WAN devices, including:

- Various services, such as HTTP and Telnet
- NAT64 timeouts
- HTTP authentication mode
- TCP keepalive

- TCP and UDP small servers
- Console logging
- IP source routing
- VTY line logging
- SNMP IFINDEX persistence
- BOOTP server

Before applying the global parameters to a device, you can view the current configuration of the device and view the differences between the parameter values that you have set in the Global Settings template and the current values on a device.

To configure global settings using Cisco SD-WAN Manager:

1. Create a feature template to configure global settings.
2. Create a device template and include the Global Settings feature template.
3. (Recommended) Before applying the device template to a device, use the [Preview Device Configuration and View Configuration Differences, on page 188](#) feature to review the differences between the configuration currently on the device and the configuration to be sent to the device. This step is recommended because applying the device template overwrites the existing configuration on a device.

Limitations

Cisco Catalyst SD-WAN can apply the global settings feature template only to devices running Cisco IOS XE Catalyst SD-WAN Release Amsterdam 17.2.x or later.

Create Global Settings Feature Template

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Feature Templates**.



Note In Cisco vManage Release 20.7.x and earlier releases, **Feature Templates** is titled **Feature**.

3. Click **Add Template**.
4. In the left pane, select a device type.
5. Select the **Global Settings** template.
6. Provide a name and description for the template.
7. For each of the parameters, use the default or set custom values as desired.

Parameter	Description
Services	
HTTP Server	Enable or disable HTTP server.

Parameter	Description
HTTPS Server	Enable or disable secure HTTPS server.
Passive FTP	Enable or disable passive FTP.
IP Domain-Lookup	Enable or disable domain name server (DNS) lookup.
Arp Proxy	Enable or disable proxy ARP.
RSH/RCP	Enable or disable remote shell (RSH) and remote copy (RCP) on the device.
Telnet (Outbound)	Enable or disable outbound telnet.
CDP	
Other Settings	
TCP Keepalives (In)	Enable or disable generation of keepalive timers when incoming network connections are idle.
TCP Keepalives (Out)	Enable or disable generation of keepalive timers when outgoing network connections are idle.
TCP Small Servers	Enable or disable small TCP servers (for example, ECHO).
UDP Small Servers	Enable or disable small UDP servers (for example, ECHO).
Console Logging	Enable or disable console logging. By default, the router sends all log messages to its console port.
IP Source Routing	Enable or disable IP source routing. IP source routing is a feature that enables the originator of a packet to specify the path for the packet to use to get to the destination.
VTY Line Logging	Enable or disable the device to display log messages to a VTY session in real time.
SNMP IFINDEX Persist	Enable or disable SNMP IFINDEX persistence, which provides an interface index (ifIndex) value that is retained and used when the device reboots.
Ignore BOOTP	Enable or disable BOOTP server. When enabled, the device listens for the bootp packet that comes in sourced from 0.0.0.0. When disabled, the device ignores these packets.
NAT64	
UDP Timeout	NAT64 translation timeout for UDP Range: 1 to 65536 (seconds) Default: 300 seconds (5 minutes) Note Starting from Cisco IOS XE Catalyst SD-WAN Release 17.6.1a and Cisco vManage Release 20.6.1, the default UDP Timeout value for NAT64 has been changed to 300 seconds (5 minutes).

Parameter	Description
TCP Timeout	NAT64 translation timeout for TCP Range: 1 to 65536 (seconds) Default: 3600 seconds (1 hour) Note Starting from Cisco IOS XE Catalyst SD-WAN Release 17.6.1a and Cisco vManage Release 20.6.1, the default TCP Timeout value for NAT64 has been changed to 3600 seconds (1 hour).
HTTP Authentication	
HTTP Authentication	HTTP authentication mode Accepted values: Local, AAA Default: Local
SSH Version	
SSH version	Specify an SSH version. Default value: Version 2

8. Enter a name for the template and click **Save**.

CLI Equivalent

Services (enable):

```
system
 ip http server
 ip http secure-server
 ip ftp passive
 ip domain lookup
 ip arp proxy disable
 ip rcmd rsh-enable
 ip rcmd rcp-enable
 cdp run enable
```

Telnet outbound enable:

```
system
 line vty 0 4
   transport input telnet ssh
```

Services (disable):

```
system
 no ip http server
 no ip http secure-server
 no ip ftp passive
 no ip domain lookup
 no ip arp proxy disable
 no ip rcmd rsh-enable
 no ip rcmd rcp-enable
 no cdp run enable
```

Telnet outbound disable:

```
system
  line vty 0 4
    transport input ssh
```

Other settings (enable):

```
system
  service tcp-keepalives-in
  service tcp-keepalives-out
  service tcp-small-servers
  service udp-small-server
  logging console
  ip source-route
  logging monitor
  snmp-server ifindex persist
  ip bootp server
```

Other settings (disable):

```
system
  no service tcp-keepalives-in
  no service tcp-keepalives-out
  no service tcp-small-servers
  no service udp-small-server
  no logging console
  no ip source-route
  no logging monitor
  no snmp-server ifindex persist
  no ip bootp server
```

NAT 64:

```
system
  nat64 translation timeout udp timeout
  nat64 translation timeout tcp timeout
```

HTTP Authentication:

```
system
  ip http authentication {local | aaa}
```

Configure NTP Servers Using a Configuration Group

Before you begin

On the **Configuration > Configuration Groups** page, choose **SD-WAN** as the solution type.

Procedure

-
- Step 1** From the Cisco SD-WAN Manager menu, choose **Configuration > Configuration Groups**.
- Step 2** Create and configure an NTP feature in a System profile.
- a. Configure a server.

Table 19: Server

Field	Description
Add Server	
Hostname/IP address*	Enter the IP address of an NTP server, or a DNS server that knows how to reach the NTP server.
VPN to reach NTP Server*	Enter the number of the VPN that should be used to reach the NTP server, or the VPN in which the NTP server is located. If you have configured multiple NTP servers, they must all be located or be reachable in the same VPN. Range: 1 to 65525, excluding 512. For details see the VRF range behavior change described here .
Set authentication key for the server	Specify the MD5 key associated with the NTP server, to enable MD5 authentication. For the key to work, you must mark it as trusted in the Trusted Key field under Authentication .
Set NTP version*	Enter the version number of the NTP protocol software. Range: 1 to 4 Default: 4
Set interface to use to reach NTP server	Enter the name of a specific interface to use for outgoing NTP packets. The interface must be located in the same VPN as the NTP server. If it is not, the configuration is ignored.
Prefer this NTP server*	Enable this option if multiple NTP servers are at the same stratum level and you want one to be preferred. For servers at different stratum levels, Cisco Catalyst SD-WAN chooses the one at the highest stratum level.

- b. Configure Authentication.

Table 20: Authentication

Field	Description
Add Authentication Keys	
Key Id*	Enter an MD5 authentication key ID. Range: 1 to 65535
MD5 Value*	Enter an MD5 authentication key. Enter either a cleartext key or an AES-encrypted key.
Trusted Key	Enter the MD5 authentication key to designate the key as trustworthy. To associate this key with a server, enter the same value that you entered for the Set authentication key for the server field under Server .

- c. Configure advanced parameters.

Table 21: Advanced

Field	Description
Authoritative NTP Server	Choose Global from the drop-down list, and enable this option if you want to configure one or more supported routers as a primary NTP router.
Stratum	Enter the stratum value for the primary NTP router. The stratum value defines the hierarchical distance of the router from its reference clock. Valid values: Integers 1 to 15. If you do not enter a value, the system uses the router internal clock default stratum value, which is 8.
Source Interface	Enter the name of the exit interface for NTP communication. If configured, the system sends NTP traffic to this interface. For example, enter GigabitEthernet1 or Loopback0 .

What to do next

Also see [Deploy a configuration group](#).

Configure NTP Servers Using Cisco SD-WAN Manager

Configure NTP servers on your devices in order to synchronize time across all the devices in the Cisco overlay network. You can configure up to four NTP servers, and they must all be located or reachable in the same VPN.

Other devices are allowed to ask a Cisco Catalyst SD-WAN device for the time, but no devices are allowed to use a Cisco Catalyst SD-WAN device as an NTP server.



Note For the NTP to properly function when using VPN0 on the Cisco vEdge devices, you must configure **allow-service ntp** for the tunnel interface on the Cisco VPN Interface Ethernet template.

To configure an NTP server using Cisco SD-WAN Manager templates:

1. Create an NTP feature template to configure NTP parameters, as described in this section.
2. Configure the timezone in the System template.

Name the Template

1. From the Cisco SD-WAN Manager menu, choose **Configuration** > **Templates**.
2. Click **Device Templates**.



Note In Cisco vManage Release 20.7.x and earlier releases, **Device Templates** is titled **Device**.

3. From the **Create Template** drop-down list, choose **From Feature Template**.
4. From the **Device Model** drop-down list, choose the type of device for which you wish to create the template.
5. Click **Basic Information**.
6. From **Additional Cisco System Templates**, click **NTP**.
7. From the **NTP** drop-down list, choose **Create Template**.

The **Cisco NTP** template form is displayed. This form contains fields for naming the template, and fields for defining NTP parameters.

8. In **Template Name**, enter a name for the template.

The name can be up to 128 characters and can contain only alphanumeric characters.

9. In **Template Description**, enter a description of the template. The description can be up to 2048 characters and can contain only alphanumeric characters.

When you first open a feature template, for each parameter that has a default value, the scope is set to Default (indicated by a check mark), and the default setting or value is shown. To change the default value or to enter a value, click the scope drop-down list to the left of the parameter field and select one of the following:

Table 22: Setting Parameter Scope

Parameter Scope	Scope Description
Device Specific (indicated by a host icon)	Use a device-specific value for the parameter. For device-specific parameters, you cannot enter a value in the feature template. You enter the value when you attach a device to a device template. When you click Device Specific, the Enter Key box opens. This box displays a key, which is a unique string that identifies the parameter in a CSV file that you create. This file is an Excel spreadsheet that contains one column for each key. The header row contains the key names (one key per column), and each row after that corresponds to a device and defines the values of the keys for that device. You upload the CSV file when you attach a device to a device template. For more information, see Create a Template Variables Spreadsheet. To change the default key, type a new string and move the cursor out of the Enter Key box. Examples of device-specific parameters are system IP address, hostname, GPS location, and site ID.
Global (indicated by a globe icon)	Enter a value for the parameter, and apply that value to all devices. Examples of parameters that you might apply globally to a group of devices are DNS server, syslog server, and interface MTUs.

Configure an NTP Server

To configure an NTP server, click **Server**, and click **Add New Server**, and configure the following parameters. Parameters marked with an asterisk are required to configure an NTP server.

Table 23: Parameters for Configuring an NTP Server

Parameter Name	Description
Hostname/IP Address*	Enter the IP address of an NTP server, or a DNS server that knows how to reach the NTP server.
Authentication Key ID*	Specify the MD5 authentication key associated with the NTP server, to enable authentication. For the key to work, you must mark it as trusted in the Trusted Keys field, under Authentication . Note From Cisco Catalyst SD-WAN Control Components Release 20.14.1, you can use CMAC-AES authentication when configuring NTP servers for Cisco SD-WAN Control Components. This requires configuration using a CLI template.
VPN ID*	Enter the number of the VPN that should be used to reach the NTP server, or the VPN in which the NTP server is located. If you have configured multiple NTP servers, they must all be located or be reachable in the same VPN. Range: Range: 1 to 65525, excluding 512. For details see the VRF range behavior change described here .
Version*	Enter the version number of the NTP protocol software. The range is from 1 through 4. The default is 4.
Source Interface	Enter the name of a specific interface to use for outgoing NTP packets. The interface must be located in the same VPN as the NTP server. If it is not, the configuration is ignored.
Prefer	Click On if multiple NTP servers are at the same stratum level and you want one to be preferred. For servers at different stratum levels, the software chooses the one at the highest stratum level.

To add an NTP server, click **Add**.

To add another NTP server, click **Add New Server**. You can configure up to four NTP servers. The Cisco Catalyst SD-WAN software uses the server at the highest stratum level.

To edit an NTP server, click the pencil icon to the right of the entry.

To delete an NTP server, click the trash icon to the right of the entry.

To save the feature template, click **Save**.

CLI equivalent:

```
system ntp
  server (dns-server-address | ip-address)
  key key-id
  prefer
  source-interface interface-name
  version number
```



```
vpn vpn-id
```

Configure NTP Authentication Keys

To configure the authentication keys used to authenticate NTP servers, click **Authentication**, and then the **Authentication Key**. Then click **New Authentication Key**, and configure the following parameters. Parameters marked with an asterisk are required to configure the authentication keys.

Table 24: Parameters for Configuring NTP Authentication Keys

Parameter Name	Description
Authentication Key ID*	Enter the following values: • Authentication Key: Enter an authentication key ID. Valid range is from 1 to 65535. • Authentication Value: Enter either a cleartext key or an AES-encrypted key.
Authentication Value*	Enter an authentication key. For this key to be used, you must designate it as trusted. To associate a key with a server, enter the same value that you entered in the Authentication Key ID field under Server .

To configure the trusted keys used to authenticate NTP servers, under **Authentication**, click **Trusted Key**, and configure the following parameters.

Table 25: Parameters for Configuring Trusted Keys

Parameter Name	Description
Trusted Keys*	Enter the authentication key to designate the key as trustworthy. To associate this key with a server, enter the same value that you entered for the Authentication Key ID field under Server .

CLI equivalent:

```
system
 ntp
  keys
    authentication key-id md5 md5-key
    trusted key-id
```

Configure NTP using CLI

Configure Network-Wide Time with NTP

To coordinate and synchronize time across all devices in the Cisco Catalyst SD-WAN overlay network, configure the IP address or DNS server address of an NTP server on each device. If necessary, specify the VPN through which the server is reachable.

```
vEdge(config)# system ntp server (dns-server-address | ipv4-address)
vEdge(config-system)# ntp server (dns-server-address | ipv4-address) vpnvpn-id
```

You can configure up to four NTP servers, and they must all be located or reachable in the same VPN. The software uses the server at the highest stratum level. If more than one server is at the same stratum level, you can configure the preference to use a specific server:

```
vEdge(config-ntp)# ntp
server (dns-server-address | ipv4-address) prefer
```

You can configure an MD5 authentication key to use as a password to access an NTP server:

```
vEdge(config-system)# ntp keys
vEdge(config-keys)# authentication key-id md5 md5-key
```

key-id is a number that identifies the MD5 authentication key. It can be a number from 1 through 65535.

md5-key is the MD5 authentication key. You can enter it as cleartext or as an AES-encrypted key.

To use an MD5 authentication key for an NTP server, the key must be configured to be trusted:

```
vEdge(config-system)# ntp keys trusted key-id
```

Finally, associate the MD5 authentication key with the NTP time server:

```
vEdge(config-system)# ntp server (dns-server-address | ipv4-address) key key-id
```

You can configure NTP packets to exit from a specific interface on the router. The interface must be located in the same VPN as the NTP server. If it is not, the configuration is ignored.

```
vEdge(config-system)# ntp server (dns-server-address | ipv4-address) source-interface
interface-name
```

The following example displays configuration three NTP servers. One of the NTP servers is at the NTP pool project at the Network Time Foundation and uses no authentication. The other two are internal servers and are configured with MD5 authentication:

```
vEdge# show running-config system ntp
system
ntp
keys
  authentication 1001 md5 $4$KXLzYT9k6M8zj4BgLEFXKw==
  authentication 1002 md5 $4$KXLzYTxk6M8zj4BgLEFXKw==
  authentication 1003 md5 $4$KXLzYT1k6M8zj4BgLEFXKw==
  trusted 1001 1002
!
server 192.168.15.243
  key 1001
  vpn 512
  version 4
exit
server 192.168.15.242
  key 1002
  vpn 512
  version 4
exit
server us.pool.ntp.org
  vpn 512
  version 4
exit
!
!
```

Configuring NTP on a Cisco Catalyst SD-WAN device allows that device to contact NTP servers to synchronize time. Other devices are allowed to ask for the time, but no devices are allowed to use the Cisco Catalyst SD-WAN as an NTP server.

Configuring Time Using CLI on Cisco vEdge Device

Configure the Timezone

The default timezone on all Cisco vEdge devices is UTC. If your devices are located in multiple timezones (and even if they are not), we recommend that you use the default timezone, which is UTC, on all device so that the times in all logging and archive files are consistent.

To change the timezone on a device:

```
vEdge(config-system)# clock timezone timezone
```

Set the Time Locally

For Cisco vEdge devices that are part of a test or local network, you can set the time locally without using NTP because you do not need to ensure that time is synchronized across an entire network of devices. You can also set the time locally on any device as it is joining the network, in addition to configuring an NTP server, and this time will be overwritten by the official NTP time once the device contacts the NTP server.

To set the local time and date, issue the following operational commands:

```
vEdge# clock set time hh:mm:ss[.sss]  
vEdge# clock set date ccyy-mm-dd
```

You can also issue these commands as a single command:

```
vEdge# clock set date ccyy-mm-dd time hh:mm:ss[.sss]
```

or

```
vEdge# clock set time hh:mm:ss[.sss] date ccyy-mm-dd
```

To set the timezone, specify it in the configuration:

```
vEdge(config)# system clock timezone timezone
```

Configure GPS Using a Configuration Group

Before you begin

On the **Configuration > Configuration Groups** page, choose **SD-WAN** as the solution type.

Procedure

-
- Step 1** From the Cisco SD-WAN Manager menu, choose **Configuration > Configuration Groups**.
- Step 2** Create and configure a GPS feature in a Transport and Management profile.
- Configure GPS.

Table 26: GPS

Field	Description
Type	Choose a feature from the drop-down list.
Feature Name*	Enter a name for the feature. The name can be up to 128 characters and can contain only alphanumeric characters.
Description	Enter a description of the feature. The description can be up to 2,048 characters and can contain only alphanumeric characters.
GPS	Click On to enable the GPS feature on the router.
GPS Mode	Select the GPS mode: • MS-based: Use mobile station–based assistance, also called assisted GPS mode, when determining position. In this mode, cell tower data is used to enhance the quality and precision in determining location, which is useful when satellite signals are poor. • Standalone: Use satellite information when determining position.
NMEA	Click On to enable the use of NMEA streams to help with determining position. NMEA streams data from the router's cellular module to any marine device, such as a Windows-based PC, that is running a commercially available GPS-based application.
Source Address*	Enter the IP address of the router's interface that connects to the external device reading the NMEA.
Destination Address*	Enter the IP address of the external device's interface that's connected to router.
Destination Port*	Enter the number of the port to use to send NMEA data to the external device's interface.

What to do next

Also see [Deploy a configuration group](#).

Configure GPS Using CLI on Cisco vEdge Device

Configuring geographic location for a device by setting its latitude and longitude allows the device to be placed properly on the Cisco vManage network map.

To set a device's latitude and longitude:

```
vEdge(config-system) # gps-location latitude degrees.minutes-and-seconds longitude degrees.minutes-and-seconds
```

You can also set these values using two separate commands:

```
vEdge(config-system)# gps-location latitude degrees . minutes-and-seconds
vEdge(config-system)# gps-location longitude degrees . minutes-and-seconds
```

For example:

```
vEdge(config-system)# gps-location latitude 37.0000 longitude 122.0600
or
vEdge(config-system)# gps-location latitude 37.000
vEdge(config-system)# gps-location longitude 122.0600
vEdge(config-system)# show full-configuration
system
  host-name          vEdge
  gps-location latitude 36.972
  gps-location longitude 122.0263
  ...
```

You can also configure a text description of the device's location:

```
vEdge(config-system)# location "description of location"
```

For example:

```
vEdge(config-system)# location "UCSC in Santa Cruz, California"
vEdge(config-system)# show full-configuration
system
  host-name          vEdge
  location            "UCSC in Santa Cruz, California"
  gps-location latitude 37.0000
  gps-location longitude 122.0600
  ...
```

Configure System Logging

System logging operations use a mechanism similar to the UNIX syslog command to record system-wide, high-level operations that occur on Cisco Catalyst SD-WAN devices in the overlay network. The log levels (priorities) of the messages are the same as standard UNIX commands, and you can configure the priority of syslog messages. Cisco Catalyst SD-WAN devices can send log messages to a UNIX-style syslog service.

Cisco vEdge devices send syslog messages to syslog servers using UDP. TCP is not supported.

The syslog service accepts messages and stores them in files on the Cisco Catalyst SD-WAN device or to a remote host.



Note Disabling system logging to disk (`no system logging disk enable`) does not disable `vsyslog`.

Syslog Message Format, Syslog Message Levels, and System Log Files

Syslog Message Format

Syslog messages begin with a percent sign (%) and following are the syslog message formats:

- Syslog message format
`seq no:timestamp: %facility-severity-MENEMONIC:description (hostname-n)`

The field descriptions of syslog messages are:

Table 27: Field Descriptions of Syslog Message Format

Field	Description
facility	Sets the logging facility to a value other than 20, which UNIX systems expect.
severity	The importance or severity of the message is categorized by the numerical code from 0 through 7. A lower number in this range indicates greater severity of the system condition.
description	A text string that describes the condition of syslog server. This portion of the syslog message sometimes includes IP addresses, interface names, port numbers, or usernames.

Usually, the syslog messages are preceded by extra text.

- The following is an example of a system logging message preceded by a priority value, sequence number, and time stamp:

```
<45>10: polaris-user1: *Jun 21 10:76:84.100: %LINK-5-CHANGED: Interface GigabitEthernet0/0,
changed state to administratively down
```

Syslog Message Levels

All syslog messages are associated with priority levels that indicate the severity of syslog messages to save. The default priority value is "informational", so by default, all syslog messages are recorded. The priority level can be one of the following in order of decreasing severity:

- Emergency—System is unusable (corresponds to syslog severity 0).
- Alert—Ensure that you act immediately (corresponds to syslog severity 1).
- Critical—A serious condition (corresponds to syslog severity 2).
- Error—An error condition that does not fully impair system usability (corresponds to syslog severity 3).
- Warning—A minor error condition (corresponds to syslog severity 4).
- Notice—A normal, but significant condition (corresponds to syslog severity 5).
- Informational—Routine condition (the default) (corresponds to syslog severity 6).

System Log Files

All syslog messages that are at or above the default or configured priority value are recorded in a number of files in the `/var/log` directory on the local device of the syslog server. The following are the contents of the log files:

- `auth.log`—Login, logout, and superuser access events, and usage of authorization systems
- `kern.log`—Kernel messages

- `messages.log`—Consolidated log file that contains syslog messages from all sources.
- `vconfd.log`—All configuration-related syslog messages
- `vdebug.log`—All debug messages for modules whose debugging is turned on and all syslog messages that are above the default priority value. The debug log messages support various levels of logging based on the module. The different modules implement the logging levels differently. For example, the system manager (`sysmgr`) has two logging levels (on and off), while the chassis manager (`chmgr`) has four different logging levels (off, low, normal, and high). You cannot send debug messages to a remote host. Therefore, to enable debugging, use the **debug** operational command.
- `vsyslog.log`—All syslog messages from Cisco Catalyst SD-WAN processes (daemons) that are above the configured priority value. The default priority value is "informational", so by default, all "notice", "warning", "error", "critical", "alert", and "emergency" syslog messages are saved.
- `vmanage-syslog.log`—Cisco SD-WAN Manager NMS Audit log messages

The following are the standard LINUX files that Cisco Catalyst SD-WAN does not use and are available in the `/var/log` directory.

- `cron.log`
- `debug.log`
- `lpr.log`
- `mail.log`
- `syslog`

The messages sent to syslog files are not rate-limited and consequently:

- A storage limit of 10 log files with a capacity of up to 16 MB size is set for each syslog file.
 - When the storage capacity exceeds the 16 MB size limit, the log file is saved as a .GZ file along with the date appended to it.
 - When the storage limit exceeds 10 log files, the oldest log file is dropped.
- If many syslog messages are generated in a short span of time, the overflowing messages are buffered and queued to be stored in the syslog file.

For repeating syslog messages or identical messages that occur multiple times in succession, only one copy of the message is placed in the syslog file. The message is annotated to indicate the number of times the message occurred.

The maximum length of a log message is 1024 bytes. The longer messages are truncated.

The maximum length of a log message for Cisco SD-WAN Manager NMS audit logs is 1024 bytes. The longer messages are truncated into smaller fragments and each of these fragments are indicated by an identifier. The identifiers are, fragment 1/2, fragment 2/2, and so on. For example, a long audit log message when truncated into smaller fragments appears as:

```
local6.info: 18-Oct-2020 17:42:07 vm10 maintenance-fragment-1/2: {"logid":
"d9ed576a-43ae-49ce-921b-a51c1ed40698", "entry_time":
1576605512190, "statcycletime" 34542398334245, "logmodule": "maintenance", "logfeature":
"upgrade", "loguser": "admin", "logusersrcip":
"10.0.1.1", "logmessage": "Device validation Upgrade to version - Validation success",
"logdeviceid": "Validation", "auditdetails" :
```

```
[ "[18-Oct-2020 17:42:08 UTC] Published messages to vmanage(s)", "auditdetails": "[18-Oct-2020
17:42:07 UTC] Software image: vmanage-99.99.999-
x86_64.tar.gz", "Software image download may take up to 60}

local6.info: 18-Oct-2020 17:42:07 vm10 maintenance-fragment-2/2: { minutes", "logprocessid":
"software_install-7de0ec44-d290-4429-b24532435324", "tenant":, "default"}
```

The syslog messages related to AAA authentication and Netconf CLI access and usage are placed in the `auth.log` and `messages.log` files. Each time a Cisco SD-WAN Manager NMS logs into a router to retrieve statistics and status information and to push files to the router, the router generates AAA and Netconf log messages. So, over time, these messages can fill the log files. To prevent these messages from filling the log files, you can disable the logging of AAA and Netconf syslog messages by using the following commands from Cisco SD-WAN Manager NMS:



Note For information about available syslog messages on Cisco SD-WAN Manager, see [Syslog Messages](#).

How System Log Files Are Deleted

System log file cleanup in `/opt/data` and `/rootfs.rw` follows two strategies:

1. Daily Cron Job:

- Automatically deletes logs older than 15 days.
- Targets specific log patterns and directories.

2. Monitor and Truncate:

- Triggered when disk usage exceeds 90%.
- If the above step fails to recover disk space, the system deletes larger files over 1 MB that are less than 2 days old.



Note Critical system files and essential logs are excluded from cleanup, although active log files may be truncated if disk space becomes critically low.

Disable logging of AAA and Netconf Syslog Messages

1. `vManage# config`

Enters the configuration mode terminal

2. `vManage(config)# system aaa logs`

Configures the logging of AAA and Netconf system logging (syslog) messages

3. `vManage(config-logs)# audit-disable`

Disable logging of AAA events

4. `vManage(config-logs)# netconf-disable`

Disable logging of Netconf events

5. `vManage(config-logs) # commit`

Commit complete.

Configure Logging Using Cisco vManage

Use the Logging template for all Cisco Catalyst SD-WANs to configure logging to either the local hard drive or a remote host.

Navigate to the Template Screen and Name the Template

1. In vManage NMS, select the **Configuration ► Templates** screen.
2. In the **Device** tab, click **Create Template**.
3. From the **Create Template** drop-down, select **From Feature Template**.
4. From the **Device Model** drop-down, select the type of device for which you are creating the template.
5. To create a custom template for Logging, select the **Factory_Default_Logging_Template** and click **Create Template**. The Logging template form is displayed. The top of the form contains fields for naming the template, and the bottom contains fields for defining Logging parameters. You may need to click a tab or the plus sign (+) to display additional fields.

6. In the **Template Name** field, enter a name for the template. The name can be up to 128 characters and can contain only alphanumeric characters.
7. In the **Template Description** field, enter a description of the template. The description can be up to 2048 characters and can contain only alphanumeric characters.

When you first open a feature template, for each parameter that has a default value, the scope is set to Default (indicated by a check mark), and the default setting or value is shown. To change the default or to enter a value, click the scope drop-down to the left of the parameter field.

Minimum Logging Configuration

The following logging parameters are configured by default:

- Log event notification system log (syslog) messages are logged to a file on the local device's hard disk, at a priority level of "information."
- Log files are placed in the directory /var/log on the local device.
- Log files are readable by the "admin" user.

Configure Logging to the Local Disk

To configure logging of event notification system log messages to the local device's hard disk, select the **Disk** tab and configure the following parameters:

Table 28:

Parameter Name	Description
Enable Disk	Click On to allow syslog messages to be saved in a file on the local hard disk, or click Off to disallow it. By default, logging to a local disk file is enabled on all Viptela devices.
Maximum File Size	Enter the maximum size of syslog files. Syslog files are rotated on an hourly basis based on the file's size. When the file size exceeds configured value, the file is rotated and the syslogd process is notified. <i>Range:</i> 1 through 20 MB <i>Default:</i> 10 MB
Rotations	Enter the number of syslog files to create before discarding the oldest files. <i>Range:</i> 1 through 10 <i>Default:</i> 10
Priority	Select the priority level of the syslog message to save to the log files. The severity indicates the seriousness of the event that generated the message. The default priority value is "informational", so, by default, all syslog messages are recorded. The priority level can be one of the following (in order of decreasing severity): • Emergency—System is unusable (corresponds to syslog severity 0). • Alert—Action must be taken immediately (corresponds to syslog severity 1). • Critical—Critical: A serious condition (corresponds to syslog severity 2). • Error—An error condition that does not fully impair system usability (corresponds to syslog severity 3). • Warning—A minor error condition (corresponds to syslog severity 4). • Notice—A normal, but significant condition (corresponds to syslog severity 5). • Informational—Routine condition (the default) (corresponds to syslog severity 6).

To save the feature template, click **Save**.

CLI equivalent:

```
system
 logging
  disk
   enable
   file
    rotate numbersize megabytes priority priority
```

Configure Logging to Remote Servers

To configure logging of event notification system log messages to a remote server, click the **Server** tab. Then click **Add New Server** and configure the following parameters:

Table 29:

Parameter Name	Description
Hostname/IP Address	Enter the DNS name, hostname, or IP address of the system on which to store syslog messages. To add another syslog server, click the plus sign (+). To delete a syslog server, click the trash icon to the right of the entry.
VPN ID	Enter the identifier of the VPN in which the syslog server is located or through which the syslog server can be reached. <i>Range:</i> 0 through 65530
Source Interface	Enter the specific interface to use for outgoing system log messages. The interface must be located in the same VPN as the syslog server. Otherwise, the configuration is ignored. If you configure multiple syslog servers, the source interface must be the same for all of them.
Priority	Select the severity of the syslog message to save. The severity indicates the seriousness of the event that generated the message. <i>priority</i> can be one of the following: • Emergency—System is unusable (corresponds to syslog severity 0). • Alert— Action must be taken immediately (corresponds to syslog severity 1). • Critical—Critical: A serious condition (corresponds to syslog severity 2). • Error—An error condition that does not fully impair system usability (corresponds to syslog severity 3). • Warning—A minor error condition (corresponds to syslog severity 4). • Notice—A normal, but significant condition (corresponds to syslog severity 5). • Informational—Routine condition (the default) (corresponds to syslog severity 6). Click Add to save the logging server.

To edit a logging server, click the pencil icon to the right of the entry.

To remove a logging server, click the trash icon to the right of the entry.

To save the feature template, click **Save**.

CLI equivalent:

```
system
 logging
  server (dns-name | hostname | ip-address)
  priority priority
  source-interface interface-name
  vpn vpn-id
```

Release Information

Introduced in Cisco vManage NMS in Release 15.2.

Export Cisco SD-WAN Manager NMS Audit Log to Syslog Server

Table 30: Feature History

Feature Name	Release Information	Description
Export Cisco SD-WAN Manager Audit Log as Syslog	Cisco SD-WAN Release 20.3.1 Cisco vManage Release 20.3.1	The Cisco SD-WAN Manager exports audit logs in syslog message format to a configured external syslog server. This feature allows you to consolidate and store network activity logs in a central location.

On Cisco IOS XE Catalyst SD-WAN devices and Cisco vEdge devices, you can log event notification system log (syslog) messages to files on a local device, or to files on a remote host using CLI. These event notification logs are converted to system log files and exported to the syslog server. You can then retrieve system log information from the syslog server.

Configure System Logging Using CLI

Log Syslog Messages to a Local Device

By default, a priority level of “information” is enabled when you log syslog messages to a file on a local device. Use the following commands:

1. logging disk

Logs syslog messages on a hard disk

Example:

```
vm01(config-system)# logging disk
```

2. enable

Enables logging to a disk

Example:

```
vm01(config-logging-disk)# enable
```

3. file size size

Specifies the size of syslog files in megabytes (MB) By default, the syslog files are 10 MB. You can configure the size of syslog files to be 1–20 MB.

Example:

```
vm01(config-logging-disk)# file size 3
```

4. file rotate number

Rotates syslog files on an hourly basis based on the size of the file By default, 10 syslog files are created. You can configure the rotate command to be a number from 1 through 10.

Example:

```
vm01(config-logging-disk)# file rotate 3
```

For more information about logging disk commands, see the [logging disk](#) command.

Log Syslog Messages to a Remote Device

To log event notification system log (syslog) messages to a remote host, use the following commands:

1. **logging server**

Logs syslog messages to a remote host or syslog server. You can configure the name of the server by DNS name, hostname, or IP address. You can configure up to four syslog servers.

Example:

```
vm01(config-system)# logging server 192.168.0.1
```

2. (Optional) **vpn vpn-id**

Specifies the VPN ID of the syslog server.

3. (Optional) **source interface interface-name**

Specifies the source interface to reach the syslog server. The interface name can be a physical interface or a sub-interface (a VLAN-tagged interface). Ensure that the interface is located in the same VPN as the syslog server. Otherwise, the configuration is ignored. If you configure multiple syslog servers, the source interface must be the same for all of them.

Example:

```
vm01(config-server-192.168.0.1)# source interface eth0
```

4. **priority priority**

Specifies the severity of the syslog message to be saved. The default priority value is "informational" and by default, all syslog messages are recorded.

Example:

In the following example, set the syslog priority to log alert conditions.

```
vm01(config-server-192.168.0.1)# priority alert
```

If the syslog server is unreachable, the system suspends sending syslog messages for 180 seconds. When the server becomes reachable, logging resumes. For more information about logging server commands, see the [logging server](#) command.

View System Logging Information

To view system log settings after logging syslog messages to a remote host, use the **show logging** command. For example:

```
vm01(config-server-192.168.0.1)# show logging
```

```
System logging
  server 192.168.0.1
  source interface eth0
  exit
!
!
```

To view the contents of the syslog file, use the **show log** command. For example:

```
vm01(config-server-192.168.0.1)# show log nms/vmanage-syslog.log tail 10
```

To view the configured system logging settings from Cisco SD-WAN Manager, see [Audit Log](#).

To view device-specific syslog files from Cisco SD-WAN Manager, perform the following steps:

1. From the Cisco SD-WAN Manager menu, choose **Administration > Settings**, and ensure that you enable **Data Stream**.
2. From the Cisco SD-WAN Manager menu, choose **Monitor > Devices**, and choose a Cisco vEdge device
Cisco vManage Release 20.6.x and earlier: From the Cisco SD-WAN Manager menu, choose **Monitor > Network**, and choose a Cisco vEdge device.
3. Click **Troubleshooting**.
4. From **Logs**, click **Debug Log**.
5. From **Log Files**, select a name of the log file to view the log information.

SSH Terminal

Use the SSH Terminal screen to establish an SSH session to a router. From an SSH session, you can issue CLI commands on a router.

Establish an SSH Session to a Device

To establish an SSH session to a device:

1. From the Cisco SD-WAN Manager menu, choose **Tools > SSH Terminal**.
2. Select the device on which you wish to collect statistics:
 - a. Select the device group to which the device belongs.
 - b. If needed, sort the device list by its status, hostname, system IP, site ID, or device type.
 - c. Click the device to select it.
3. Enter the username and password to log in to the device.

You can now issue CLI commands to monitor or configure the device.

HTTP/HTTPS Proxy Server for Cisco SD-WAN Manager Communication with External Servers

Table 31: Feature History

Feature Name	Release Information	Description
HTTP/HTTPS Proxy Server for Cisco SD-WAN Manager Communication with External Servers	Cisco SD-WAN Release 20.5.1 Cisco vManage Release 20.5.1	Cisco SD-WAN Manager uses HTTP/HTTPS to access some web services and for some REST API calls. With this feature, you can channel the HTTP/HTTPS communication through an HTTP/HTTPS proxy server.
Cisco SD-WAN Manager HTTP/HTTPS Proxy Server Support Over IPv6	Cisco Catalyst SD-WAN Manager Release 20.13.1	You can now configure an IPv6 address when configuring an HTTP/HTTPS proxy server.

The following are some instances in which Cisco SD-WAN Manager uses an HTTP/HTTPS connection to an external server:

- Certificate request or renewal
- Cisco Plug and Play integration
- Smart Licensing Using Policy
- Cloud OnRamp
- Software image download
- Data upload to Cisco SD-WAN Analytics

In Cisco vManage Release 20.4.1 and earlier releases, you must permit this HTTP/HTTPS communication in the firewall configured on your on-premises Cisco SD-WAN Manager instance. Beginning Cisco vManage Release 20.5.1, you can channel the HTTP/HTTPS communication via an HTTP/HTTPS proxy server. With the HTTP/HTTPS proxy server configured, you can restrict HTTP/HTTPS communication with external servers while configuring the firewall and secure the system further.

Traffic is directed through the HTTP/HTTPS proxy server in the following cases:

- HTTPS connection for Symantec or Cisco automated certificate request or renewal
- REST API calls to URLs of the following domains:
 - cisco.com
 - amazonaws.com
 - microsoft.com
 - office.com

- microsoftonline.com

Once every 24 hours, Cisco SD-WAN Manager checks whether the configured HTTP/HTTPS proxy server is reachable. If the proxy server is unreachable, Cisco SD-WAN Manager raises the alarm `HTTPS proxy server {IP} not reachable`.

Restrictions

- When configured to communicate with external servers via an HTTP/HTTPS proxy server, Cisco SD-WAN Manager resolves FQDNs locally or through configured DNS servers, bypassing the proxy server. Cisco SD-WAN Manager then sends the HTTP/HTTPS connections resulting from the resolution to the proxy server. DNS queries for the resolution of external server FQDNs must be successful before Cisco SD-WAN Manager can send resulting HTTP/HTTPS connections to the HTTP/HTTPS proxy server.
- Use of the HTTP/HTTPS proxy server is not supported for communication between the SD-AVC container in Cisco SD-WAN Manager and external services.

Configure HTTP/HTTPS Proxy Server

Prerequisites

Enable out of band interface on single node using **Administration > Cluster Management** before configuring proxy server.

1. From the Cisco SD-WAN Manager menu, choose **Administration > Settings**.
2. Open **HTTP/HTTPS Proxy**.
3. For the **Enable HTTP/HTTPS Proxy** setting, click **Enabled**.
4. Enter the **HTTP/HTTPS Proxy IP Address** and **Port** number.

For releases before Cisco Catalyst SD-WAN Manager Release 20.13.1, enter an IPv4 address. For releases from Cisco Catalyst SD-WAN Manager Release 20.13.1, enter an IPv4 or IPv6 address.

5. Enter a **Non Proxy Host/IP List**.

This list is a pipe (|) separated list of IP addresses or hostnames that are not to be proxied.

6. Click **Save**.



Note Cisco SD-WAN Manager uses TCP port 7 echo request to validate reachability of the proxy server. Ensure that you configure your firewall and proxy server to allow the echo requests to make the destination host ports accessible.

Cisco SD-WAN Manager verifies that the HTTP/HTTPS proxy server is reachable and saves the server details in the configuration database. HTTP/HTTPS connections and REST API calls to external servers are directed through the proxy server.

If the HTTP/HTTPS proxy server is not reachable, Cisco SD-WAN Manager displays an error message on the GUI indicating the reason for failure.

Bulk API Rate Limit for a Cisco SD-WAN Manager Cluster

Table 32: Feature History

Feature Name	Release Information	Description
Bulk API Rate Limit for a Cisco SD-WAN Manager Cluster	Cisco vManage Release 20.10.1	For a Cisco SD-WAN Manager cluster, the rate limit for bulk APIs equals (rate-limit per node) * (number of nodes in the cluster). Cisco SD-WAN Manager distributes bulk API requests among the nodes in the cluster. With these changes, you can retrieve data faster from a Cisco SD-WAN Manager cluster through bulk APIs.

In Cisco vManage Release 20.9.x and earlier releases, you send bulk API requests to a node in the Cisco SD-WAN Manager cluster. The bulk API throughput is constrained by the rate-limit per node. To increase the throughput, you must send separate bulk API requests to each node in the cluster and the collate the API responses.

From Cisco vManage Release 20.10.1, send bulk API requests to the Cisco SD-WAN Manager cluster. Cisco SD-WAN Manager distributes the API requests among the clusters in the node. This distribution increases the rate limit to (rate-limit per node) * (number of nodes in the cluster), allowing you to retrieve more data in a shorter duration compared to a bulk API request addressed to a single node. With the distribution, you need not send separate bulk API requests to two or more nodes in the cluster or collate the API responses.

Configure Bulk API Rate Limit

1. Log in to one of the Cisco SD-WAN Manager nodes in the Cisco SD-WAN Manager cluster and configure the following command:

```
vManage# request nms server-proxy set ratelimit
```

2. The command-line displays the following prompt about the rate limit for non-bulk APIs:

```
Do you want to reconfigure rate limit for URL non bulk api [y/n] :
```

Enter **n**.

3. The command-line displays the following prompt about the rate limit for bulk APIs:

```
Do you want to reconfigure rate limit for URL bulk api /dataservice/data/device/statistics [y/n] :
```

Enter **y**.

4. Enter the per-node rate limit in response to a prompt similar to the following:

```
Enter the PER NODE rate limit for URL bulk api /dataservice/data/device/statistics [144 load balanced across all nodes at present] :
```

This prompt is from a three-node Cisco SD-WAN Manager cluster, with the bulk API rate limit configured to the default value of 48 requests per node. Across all the three nodes, the bulk API rate limit is (rate-limit/node) * 3, which is 144 requests.

Before you enter the rate limit, consider its effect on Cisco SD-WAN Manager resources.

5. Enter the unit time for which the rate limit applies in response to a prompt similar to the following.

You can apply a rate limit per second, minute, hour, or day. The default unit is minute.

```
Enter the rate limit unit (second, minute, hour, day) for URL bulk api
/dataservice/data/device/statistics [minute] :
```

Cisco vManage applies the rate limit on all the Cisco SD-WAN Manager instances in the cluster. The command line displays the following message:

```
Propagating rate limit update across all nodes. Please wait.
```

After the rate limit is applied, Cisco SD-WAN Manager prompts you to restart the server-proxy on all nodes and the command line returns to the privileged EXEC mode:

```
Done. Please restart server-proxy on all nodes using "request nms server-proxy restart"
command.
vManage#
```

6. Restart the server-proxy using the following command:
vManage# **request nms server-proxy restart**
7. Log in to the other Cisco SD-WAN Manager nodes in the cluster and restart the server-proxy using the **request nms server-proxy restart** command.

In the following example, the bulk API rate limit per node is set to 50 requests per minute.

```
vManage# request nms server-proxy set ratelimit
Do you want to reconfigure rate limit for URL non bulk api [y/n] : n
Do you want to reconfigure rate limit for URL bulk api /dataservice/data/device/statistics
[y/n] : y
Enter the PER NODE rate limit for URL bulk api /dataservice/data/device/statistics [144
load balanced across all nodes at present] : 50
Enter the rate limit unit (second, minute, hour, day) for URL bulk api
/dataservice/data/device/statistics [minute] : minute
Propagating rate limit update across all nodes. Please wait.
Done. Please restart server-proxy on all nodes using "request nms server-proxy restart"
command.
vManage# request nms server-proxy restart
```

View Bulk API Rate Limit

To view the bulk API rate limit, log in to any node in the Cisco SD-WAN Manager cluster and use the **show nms server-proxy ratelimit** command.

The following is a sample command output:

```
vManage# show nms server-proxy ratelimit
Non Bulk API: 100/second (per node)
Bulk API: 150/minute (across cluster)
```

This sample output is from three-node Cisco SD-WAN Manager cluster with the bulk API rate limit per node configured to 50 requests per minute. Therefore, the bulk API rate limit for the cluster is $50 \times 3 = 150$ requests per minute.



CHAPTER 4

Configure User Access and Authentication

Use the **Manage Users** screen to add, edit, or delete users and user groups from Cisco SD-WAN Manager.

Only a user logged in as the **admin** user or a user who has Manage Users write permission can add, edit, or delete users and user groups from Cisco SD-WAN Manager.

- [Configure Hardened Passwords , on page 62](#)
- [Configure User Login Options, on page 65](#)
- [Manage Users, on page 70](#)
- [Configure Users Using CLI, on page 70](#)
- [Manage a User Group, on page 72](#)
- [Creating Groups Using CLI, on page 73](#)
- [CiscoTAC User Access, on page 74](#)
- [Configure Sessions in Cisco SD-WAN Manager, on page 75](#)
- [Configuring RADIUS Authentication Using CLI, on page 77](#)
- [Configure SSH Authentication, on page 78](#)
- [Configure the Authentication Order, on page 79](#)
- [Configure NAS Attributes using CLI, on page 81](#)
- [Role-Based Access with AAA, on page 83](#)
- [Configure AAA Using a Configuration Group, on page 92](#)
- [Configuring AAA using Cisco SD-WAN Manager Template, on page 96](#)
- [Configuring Password Policy for AAA on Devices, on page 106](#)
- [Configuring IEEE 802.1X and IEEE 802.11i Authentication, on page 108](#)

Configure Hardened Passwords

Table 33: Feature History

Feature Name	Release Information	Description
Hardened Passwords	Cisco vManage Release 20.3.1	This feature enables password policy rules in Cisco SD-WAN Manager. After password policy rules are enabled, Cisco SD-WAN Manager enforces the use of strong passwords.
	Cisco SD-WAN Release 20.9.1 Cisco vManage Release 20.9.1	This feature lets you configure Cisco SD-WAN Manager to enforce predefined-medium security or high-security password criteria.

Enforce Strong Passwords

We recommend the use of strong passwords. You must enable password policy rules in Cisco SD-WAN Manager to enforce use of strong passwords.

After you enable a password policy rule, the passwords that are created for new users must meet the requirements that the rule defines. In addition, for releases from Cisco vManage Release 20.9.1, you are prompted to change your password the next time you log in if your existing password does not meet the requirements that the rule defines.

1. From the Cisco SD-WAN Manager menu, choose **Administration > Settings**.
2. Click **Password Policy**.
3. Perform one of these actions, based on your Cisco SD-WAN Manager release:
 - For releases before Cisco vManage Release 20.9.1, click **Enabled**.
 - For releases from Cisco vManage Release 20.9.1 click **Medium Security** or **High Security** to choose the password criteria.

By default, **Password Policy** is set to **Disabled**.

4. Click **Save**.

Password Requirements

Cisco SD-WAN Manager enforces the following password requirements after you have enabled the password policy rules:

- The following password requirements apply to releases before Cisco vManage Release 20.9.1:
 - Must contain a minimum of eight characters, and a maximum of 32 characters.

- Must contain at least one uppercase character.
 - Must contain at least one lowercase character.
 - Must contain at least one numeric character.
 - Must contain at least one of the following special characters: # ? ! @ \$ % ^ & * -.
 - Must not contain the full name or username of the user.
 - Must not reuse a previously used password.
 - Must contain different characters in at least four positions in the password.
- Minimum releases: Cisco SD-WAN Release 20.9.1, Cisco vManage Release 20.9.1:

Password Criteria	Requirements
Medium Security	• Must contain a minimum of 8 characters • Must contain no more than 32 characters • Must contain at least 1 lowercase character • Must contain at least 1 uppercase character • Must contain at least 1 numeric character • Must contain at least 1 of the following special characters: # ? ! @ \$ % ^ & * - • Must not be identical to any of the last 5 passwords used • Must not contain the full name or username of the user
High Security	• Must contain a minimum of 15 characters • Must contain no more than 32 characters • Must contain at least 1 lowercase character • Must contain at least 1 uppercase character • Must contain at least 1 numeric character • Must contain at least 1 of the following special characters: # ? ! @ \$ % ^ & * - • Must not be identical to any of the last 5 passwords used • Must not contain the full name or username of the user • Must have at least eight characters that are not in the same position they were in the old password

Password Attempts Allowed

You are allowed five consecutive password attempts before your account is locked. After six failed password attempts, you are locked out for 15 minutes. If you enter an incorrect password on the seventh attempt, you are not allowed to log in, and the 15-minute lock timer starts again.

If your account is locked, wait for 15 minutes for the account to automatically be unlocked. Alternatively, reach out to an administrator to reset the password, or have an administrator unlock your account.



Note Your account gets locked even if no password is entered multiple times. When you do not enter anything in the password field, it is considered as invalid or wrong password.

Password Change Policy



Note You must have enabled password policy rules first for strong passwords to take effect. For more information, see [Enforce Strong Passwords, on page 62](#).

When resetting your password, you must set a new password. You cannot reset a password using an old password.



Note In Cisco vManage Release 20.6.4, Cisco vManage Release 20.9.1 and later releases, a user that is logged out, or a user whose password has been changed locally or on the remote TACACS server cannot log in using their old password. The user can log in only using their new password.

Reset a Locked User

If a user is locked out after multiple password attempts, an administrator with the required rights can update passwords for this user.

There are two ways to unlock a user account, by changing the password or by getting the user account unlocked.



Note Only a **netadmin** user or a user with the User Management Write role can perform this operation.

To reset the password of a user who has been locked out:

1. In **Users (Administration > Manage Users)**, choose the user in the list whose account you want to unlock.
2. Click **...** and choose **Reset Locked User**.
3. Click **OK** to confirm that you want to reset the password of the locked user. Note that this operation cannot be undone.

Alternatively, you can click **Cancel** to cancel the operation.

Reset a Locked User Using the CLI

You can reset a locked user using the CLI as follows:

1. Log in to the device as an `admin` user.
2. Run the following command:

```
Device# request aaa unlock-user username
```
3. When prompted, enter a new password for the user.

Configure User Login Options

Table 34: Feature History

Feature Name	Release Information	Description
Inactivity Lockout		This feature lets you configure Cisco SD-WAN Manager to lock out users who have not logged in for a designated number of consecutive days.
Unsuccessful Login Attempts Lockout		This feature lets you configure Cisco SD-WAN Manager to lock out users who have made a designated number of consecutive unsuccessful login attempts within a designated period.
Duo Multifactor Authentication Support		This feature lets you configure Cisco SD-WAN Manager to require Duo multifactor authentication (MFA) to verify the identity of users before they can log in to Cisco SD-WAN Manager.

Beginning with Cisco IOS XE Catalyst SD-WAN Release 17.12.1a, you can access Cisco SD-WAN Manager with basic privileges even if TACACS user is not mapped to a group. Prior to Cisco IOS XE Catalyst SD-WAN Release 17.12.1a, the access to Cisco SD-WAN Manager was denied.

- Inactivity lockout: You can configure Cisco SD-WAN Manager to lock out users who have not logged in for a designated number of consecutive days. Locked out users cannot log in to Cisco SD-WAN Manager until an administrator unlocks their accounts.

See [Configure Account Lockout](#) , on page 66.

- Unsuccessful login lockout: You can configure Cisco SD-WAN Manager to prevent users who make a designated number of consecutive unsuccessful login attempts within a designated time period from logging in to Cisco SD-WAN Manager until a configured amount of time passes or an administrator unlocks their user accounts.

By default, Cisco SD-WAN Manager locks out users for 15 minutes after five consecutive unsuccessful login attempts within 15 minutes. After a lockout period expires, a user can log in with the correct user name and password.

See [Configure Unsuccessful Login Attempts Lockout](#), on page 67.

- Duo multifactor authentication: You can configure Cisco SD-WAN Manager to require the use of Duo multifactor authentication to verify identity before users can log in. Users must confirm a login attempt by using Duo multifactor authentication on their mobile devices.

See [Configure Duo Multifactor Authentication](#), on page 69.

Configure Account Lockout

Before You Begin

Cisco SD-WAN Manager marks locked out users as inactive, and they cannot log in again until an administrator unlocks their accounts in Cisco SD-WAN Manager.



Note To unlock a user account, see [Reset a Locked User](#).

Configure Account Lockout

1. From the Cisco SD-WAN Manager menu, choose **Administration > Settings**.
2. Click **Account Lockout** and enable the **Inactive days before locked out** option.

(In Cisco Catalyst SD-WAN Manager Release 20.12.x, locate the **Account Lockout**, click **Edit**, and enable **Inactive days before locked out**.)

3. Configure the following options:

Field	Description
Inactive days before account locked out	Enable this option and enter the number of consecutive inactive days after which Cisco SD-WAN Manager locks out a user. An inactive day is defined as a day on which a user does not log in to Cisco SD-WAN Manager. Valid values are 2 through 90.
Number of failed login attempts before lockout	Enter the number of failed login attempts after which Cisco SD-WAN Manager locks out a user. Possible values: 1 through 3600 Default: 3600

Field	Description
Duration within which the failed attempts are counted (minutes)	Enter the period, in minutes, during which the system counts consecutive unsuccessful login attempts. For example, if you set this period to 10 minutes, and set the number of failed login attempts before lockout to 5, Cisco SD-WAN Manager locks out a user if the user makes 5 consecutive unsuccessful login attempts within 10 minutes. Possible values: 1 through 60 Default: 60
Cooldown or Lockout period	This option controls whether Cisco SD-WAN Manager automatically resets a user who is locked because of unsuccessful login attempts. This option is enabled by default. If you disable it, an administrator must manually unlocks the account of a locked-out user. a. Click Enabled adjacent to Cooldown or Lockout period. b. In the Lockout Interval (minutes) field, enter the number of minutes after which Cisco SD-WAN Manager automatically resets a locked out user. Possible values: 1 through 60 Default: 15

4. Click **Save**.

Configure Unsuccessful Login Attempts Lockout

Before You Begin



Note From Cisco Catalyst SD-WAN Manager Release 20.13.1 or later, use the procedure described in [Configure Account Lockout](#), on page 66.

Cisco SD-WAN Manager prevents locked out users from logging in again until a configured amount of time has passed or an administrator unlocks their accounts in Cisco SD-WAN Manager.



Note To unlock a user account, see [Reset a Locked User](#).

Configure Unsuccessful Login Attempts Lockout

1. From the Cisco SD-WAN Manager menu, choose **Administration > Settings**.
2. Click **Account Lockout**
3. In the **Lockout on failed login attempts** row, click **Edit**.
4. Configure the following options:

Field	Description
Number of failed login attempts before lockout	Enter the number of failed login attempts after which Cisco SD-WAN Manager locks out a user. Possible values: 1 through 3600 Default: 3600
Duration within which the failed attempts are counted (minutes)	Enter the period, in minutes, during which the system counts consecutive unsuccessful login attempts. For example, if you set this period to 10 minutes, and set the number of failed login attempts before lockout to 5, Cisco SD-WAN Manager locks out a user if the user makes 5 consecutive unsuccessful login attempts within 10 minutes. Possible values: 1 through 60 Default: 60
Cooldown or Lockout period	This option controls whether Cisco SD-WAN Manager automatically resets a user who is locked because of unsuccessful login attempts. This option is enabled by default. If you disable it, an administrator must manually unlock the account of a locked-out user. a. Click Enabled adjacent to Cooldown or Lockout period. b. In the Lockout Interval (minutes) field, enter the number of minutes after which Cisco SD-WAN Manager automatically resets a locked out user. Possible values: 1 through 60 Default: 15

5. Click **Save**.

Configure Duo Multifactor Authentication

This feature requires that you have a Duo account with local users created on that account.

**Note**

- Duo MFA does not apply to the admin user by default. To enable Duo MFA for the admin user, enable the **DUO MFA Configuration** option, and then enter the [admin-auth-order](#) command from the CLI.
- Users do not see a message in Cisco SD-WAN Manager that an MFA request has been sent to a mobile device.

1. Log into the Duo Admin Panel.

2. Create an Auth API application.

This provides the Duo integration key, secret key, and API hostname information required to complete Duo MFA configuration. Read more about [Duo Auth API](#).

3. From the Cisco SD-WAN Manager menu, choose **Administration > Settings**.

4. Click **DUO MFA Configuration**. (If you are using Cisco Catalyst SD-WAN Manager Release 20.12.x or earlier, click **Edit**.)

5. Click **Enabled**.

6. Configure the following options:

Field	Description
Integration Key	Enter the integration key (Ikey) for your Duo account.
Secret Key	Enter the secret key (Skey) for your Duo account.
API Hostname	Enter the API hostname (api-hostname) for your Duo account.
Server proxy	(Read only) Shows the server proxy that is used to access the Duo server if Cisco SD-WAN Manager is behind a firewall. Set this server proxy with the system http proxy or the system https proxy command. Note If Cisco SD-WAN Manager is deployed on a cloud that can be reached by an external network, a server proxy should not be set.

7. Click **Save**.

8. If a Cisco SD-WAN Validator or a Cisco SD-WAN Controller does not have internet access, use the following commands in the CLI or the device template of the device to provide access to the Duo MFA feature.

These commands configure the device with proxy information about the device on which Duo MFA is enabled.

```
vm# config
vm(config)# system aaa
vm(config-aaa)# multi-factor-auth
vm(config-multi-factor-auth)# duo
vm(config-duo)# api-hostname name
vm(config-duo)# secret-key key
vm(config-duo)# integration-key key
vm(config-duo)# proxy proxy_url
vm(config-duo)# commit
```

Manage Users

From the Cisco SD-WAN Manager menu, choose **Administration > Manage Users** to add, edit, view, or delete users and user groups.

Please note the following:

- Only a user logged in as the **admin** user or a user who has Manage Users write permission can add, edit, or delete users and user groups from Cisco SD-WAN Manager.
- Each user group can have read or write permission for the features listed in this section. Write permission includes Read permission.
- All user groups, regardless of the read or write permissions selected, can view the information displayed in the Cisco SD-WAN Manager Dashboard.

Table 35: User Group Permissions for Different Device Types

Permissions	See This Section
User group permissions related to Cisco IOS XE Catalyst SD-WAN device configuration.	User Group Permissions: Cisco IOS XE Catalyst SD-WAN Devices
User group permissions related to Cisco Catalyst Wireless Gateway device configuration.	User Group Permissions: Cisco Catalyst Wireless Gateway Devices

Configure Users Using CLI

You can use the CLI to configure user credentials on each device. This way, you can create additional users and give them access to specific devices. The credentials that you create for a user by using the CLI can be different from the Cisco SD-WAN Manager credentials for the user. In addition, you can create different credentials for a user on each device. All users with the **netadmin** privilege can create a new user.

To create a user account, configure the username and password, and place the user in a group:

```
vEdge(config)# system aaa
vEdge(config)# user username password password
vEdge(config-aaa)# group group-name
```

The Username can be 1 to 128 characters long, and it must start with a letter. The name can contain only lowercase letters, the digits 0 through 9, hyphens (-), underscores (_), and periods (.). The name cannot contain any uppercase letters. Because some usernames are reserved, you cannot configure them. For a list of reserved usernames, see the **aaa** configuration command in the Cisco Catalyst SD-WAN Command Reference Guide.

The Password is the password for a user. Each username must have a password, and users are allowed to change their own password. The CLI immediately encrypts the string and does not display a readable version of the password. When a user logs in to a Cisco vEdge device, they have five chances to enter the correct password. After the fifth incorrect attempt, the user is locked out of the device, and must wait for 15 minutes before attempting to log in again.



Note Enclose any user passwords that contain the special character ! in double quotation marks (“”). If a double quotation is not included for the entire password, the config database (?) treats the special character as a space and ignores the rest of the password.

For example, if the password is C!sc0, use “C!sc0”.

Group name is the name of a standard Cisco Catalyst SD-WAN group (**basic**, **netadmin**, or **operator**) or of a group configured with the **usergroup** command (discussed below). If an admin user changes the permission of a user by changing their group, and if that user is currently logged in to the device, the user is logged out and must log back in again.

The factory-default password for the admin username is admin. We strongly recommend that you modify this password the first time you configure a Cisco vEdge device :

```
vEdge(config)# system aaa admin password password
```

Configure the password as an ASCII string. The CLI immediately encrypts the string and does not display a readable version of the password, for example:

```
vEdge(config-user-admin)# show config
system
aaa
  user admin
    password $1$xULc8yYH$k71cTjvKESmeIGgImNDaC.
  !
  user eve
    password $1$8z3q4qoU$F6DMBr9vPBF0s/sl45ax5.
    group    basic
  !
!
```

If you are using RADIUS to perform AAA authentication, you can configure a specific RADIUS server to verify the password:

```
vEdge(config)# system aaa radius-servers tag
```

The tag is a string that you defined with the **radius server tag** command, as described in the Cisco Catalyst SD-WAN Command Reference Guide.

Manage a User Group

Users are placed in groups, which define the specific configuration and operational commands that the users are authorized to view and modify. A single user can be in one or more groups. Cisco Catalyst SD-WAN software provides standard user groups, and you can create custom user groups, as needed:

- **basic**: Includes users who have permission to view interface and system information.
- **netadmin**: Includes the admin user, by default, who can perform all operations on the Cisco SD-WAN Manager. You can add other users to this group.
- **operator**: Includes users who have permission only to view information.
- Minimum supported release: Cisco vManage Release 20.9.1
- **network_operations**: Includes users who can perform non-security operations on Cisco SD-WAN Manager, such as viewing and modifying non-security policies, attaching and detaching device templates, and monitoring non-security data.
- Minimum supported release: Cisco vManage Release 20.9.1
- **security_operations**: Includes users who can perform security operations on Cisco SD-WAN Manager, such as viewing and modifying security policies, and monitoring security data.

Note: All user groups, regardless of the read or write permissions selected, can view the information displayed on the Cisco SD-WAN Manager Dashboard screen.

Delete a User Group

You can delete a user group when it is no longer needed. For example, you might delete a user group that you created for a specific project when that project ends.

1. From the Cisco SD-WAN Manager menu, choose **Administration > Manage Users**.
2. Click **User Groups**.
3. Click the name of the user group you wish to delete.



Note You cannot delete any of the default user groups—basic, netadmin, operator, network_operations, and security_operations.

4. Click **Trash** icon.
5. To confirm the deletion of the user group, click **OK**.

Edit User Group Privileges

You can edit group privileges for an existing user group. This procedure lets you change configured feature read and write permissions for the user group needed.

1. From the Cisco SD-WAN Manager menu, choose **Administration > Manage Users**.
2. Click **User Groups**.

3. Select the name of the user group whose privileges you wish to edit.



Note You cannot edit privileges for any of the default user groups—**basic**, **netadmin**, **operator**, **network_operations**, and **security_operations**.

4. Click **Edit**, and edit privileges as needed.
5. Click **Save**.

If an **admin** user changes the privileges of a user by changing their group, and if that user is currently logged in to the device, the user is logged out and must log back in again.

Creating Groups Using CLI

The Cisco Catalyst SD-WAN software provides default user groups: **basic**, **netadmin**, **operator**, **network_operations**, and **security_operations**. The username **admin** is automatically placed in the **netadmin** usergroup.

If needed, you can create additional custom groups and configure privilege roles that the group members have. To create a custom group with specific authorization, configure the group name and privileges:

```
vEdge(config)# system aaa usergroup group-name task privilege
```

group-name can be 1 to 128 characters long, and it must start with a letter. The name can contain only lowercase letters, the digits 0 through 9, hyphens (-), underscores (_), and periods (.). The name cannot contain any uppercase letters. Some group names are reserved, so you cannot configure them. For a list of them, see the **aaa** configuration command.

If a remote RADIUS or TACACS+ server validates authentication but does not specify a user group, the user is placed into the user group **basic**. If a remote server validates authentication and specifies a user group (say, X) using VSA Cisco SD-WAN-Group-Name, the user is placed into that user group only. However, if that user is also configured locally and belongs to a user group (say, Y), the user is placed into both the groups (X and Y).

In the **task** option, list the privilege roles that the group members have. The role can be one or more of the following: **interface**, **policy**, **routing**, **security**, and **system**.

In the following example, the **basic** user group has full access to the **system** and **interface** portions of the configuration and operational commands, and the **operator** user group can use all operational commands but can make no modifications to the configuration:

```
vEdge# show running-config system aaa
system
aaa
  usergroup basic
    task system read write
    task interface read write
  !
  usergroup operator
    task system read
    task interface read
    task policy read
    task routing read
    task security read
  !
```

```
user admin
password $1$tokPB7tf$vhR2JI9Sw1/dqgkq9S.
!
!
```

Ciscotac User Access

The Cisco Edge software provides two users—**ciscotacro** and **ciscotacrw**—that are for use only by the Cisco Support team. These users are available for both cloud and on-premises installations. They operate on a consent-token challenge and token response authentication in which a new token is required for every new login session. The ciscotacro and ciscotacrw users can use this token to log in to Cisco SD-WAN Manager web server as well as the SSH Terminal on Cisco SD-WAN Manager. These users can also access Cisco SD-WAN Validator, Cisco SD-WAN Controllers, and Cisco vEdge devices using the SSH Terminal on Cisco SD-WAN Manager.

The default CLI templates include the ciscotacro and ciscotacrw user configuration. These users are enabled by default. However, a customer can disable these users, if needed.

- **ciscotacro User:** This user is part of the operator user group with only read-only privileges. This user can only monitor a configuration but cannot perform any operation that will modify the configuration of the network.
- **ciscotacrw User:** This user is part of the netadmin user group with read-write privileges. This user can modify a network configuration. In addition, only this user can access the root shell using a consent token.

Use the **tools consent-token** command to authenticate the network administrator of an organization to access system shell. Starting Cisco Catalyst SD-WAN Control Components Release 20.12.x, the **request support ciscotac** command is deprecated.

Limitations

- Only 16 concurrent sessions are supported for the ciscotacro and ciscotacrw users.
- The session duration is restricted to four hours. It is not configurable.
- The inactivity timer functionality closes user sessions that have been idle for a specified period of time. This feature is enabled by default and the timeout value is 30 minutes. However, the user configuration includes the option of extending the inactivity timer.
- A customer can remove these two users. If removed, the customer can open a case and share temporary login credentials or share the screen with the Cisco Support team for troubleshooting an issue.

Configure Sessions in Cisco SD-WAN Manager

Table 36: Feature History

Feature History	Release Information	Description
Configure Sessions in Cisco SD-WAN Manager	Cisco SD-WAN Release 20.3.1 Cisco vManage Release 20.3.1	This feature lets you see all the HTTP sessions that are open within Cisco SD-WAN Manager. It gives you details about the username, source IP address, domain of the user, and other information. A user with User Management Write access, or a netadmin user can trigger a log out of any suspicious user's session.

Set a Client Session Timeout in Cisco SD-WAN Manager

You can set a client session timeout in Cisco SD-WAN Manager. When a timeout is set, such as no keyboard or keystroke activity, the client is automatically logged out of the system.



Note You can edit Client Session Timeout in a multitenant environment only if you have a Provider access.

1. From the Cisco SD-WAN Manager menu, choose **Administration** > **Settings**.
2. Click **User Sessions**.
3. Under **Client Session Timeout**, click **Session Timeout**.
4. Specify the timeout value, in minutes.
5. Click **Save**.

Set a Session Lifetime in Cisco SD-WAN Manager

You can specify how long to keep your session active by setting the session lifetime, in minutes. A session lifetime indicates the amount of time for which a session can be active. If you keep a session active without letting the session expire, you will be logged out of the session in 24 hours, which is the default session timeout value.

The default session lifetime is 1440 minutes or 24 hours.



Note You can edit Session Lifetime in a multitenant environment only if you have a Provider access.

1. From the Cisco SD-WAN Manager menu, choose **Administration** > **Settings**.

2. Click **User Sessions**.
3. In the **SessionLifeTime Timeout (minutes)** field, specify the session timeout value, in minutes, from the drop-down list.
4. Click **Save**.

Set the Server Session Timeout in Cisco SD-WAN Manager

You can configure the server session timeout in Cisco SD-WAN Manager. The server session timeout indicates how long the server should keep a session running before it expires due to inactivity. The default server session timeout is 30 minutes.



Note Server Session Timeout is not available in a multitenant environment even if you have a Provider access or a Tenant access.

1. From the Cisco SD-WAN Manager menu, choose **Administration > Settings**.
2. Click **User Sessions**.
3. In **Server Session Timeout Timeout(minutes)** field, specify the timeout value, in minutes.
4. Click **Save**.

Set the maximum sessions per user role

You can configure the maximum number of concurrent login sessions for each configured user role. This maximum value applies to all users assigned to that role.

For example, if the “Max user sessions” value for the netadmin role is set to 3, then each user assigned to the netadmin role can have up to 3 concurrent login sessions across the platform. If a fourth session is initiated by one of those users, an error message appears.

Range for “Max user sessions”: 1 to 255. Default: If undefined, there is no limit on the number of concurrent sessions for that role. For government installations, recommended values are 3 sessions for administrative roles and 2 sessions for conventional roles.

The value also applies to CLI sessions.

Change maximum sessions value for a user role

1. From the Cisco SD-WAN Manager menu, choose **Administration > Users and Access > Roles**.
The “Max user sessions value” is displayed for each defined role, including both default and custom roles.
2. Click the role name, edit the **Max user sessions** value, and click **Update**.



Note As of Release 20.18.1, the previous method for changing this value in **Settings > Trust and Privacy** is no longer supported. If a “Max user sessions” value has been defined on your system by that method, that value serves as the maximum value for all user roles until changed.

Configuring RADIUS Authentication Using CLI

The Remote Authentication Dial-In User Service (RADIUS) is a distributed client/server system that secures networks against unauthorized access. RADIUS clients run on supported Cisco devices and send authentication requests to a central RADIUS server, which contains all user authentication and network service access information.

To have a Cisco vEdge device use RADIUS servers for user authentication, configure one or up to 8 servers:

```
vEdge(config)# system radius
vEdge(config-radius)# server ip-address
vEdge(config-server)# secret-key password
vEdge(config-server)# priority number
vEdge(config-server)# auth-port port-number
vEdge(config-server)# acct-port port-number
vEdge(config-server)# source-interface interface-name
vEdge(config-server)# tag tag
vEdge(config-server)# vpn vpn-id
```

For each RADIUS server, you must configure, at a minimum, its IP address and a password, or key. You can specify the key as a clear text string up to 31 characters long or as an AES 128-bit encrypted key. The local device passes the key to the RADIUS server. The password must match the one used on the server. To configure more than one RADIUS server, include the **server** and **secret-key** commands for each server.

The remaining RADIUS configuration parameters are optional.

To set the priority of a RADIUS server, as a means of choosing or load balancing among multiple RADIUS servers, set a priority value for the server. The priority can be a value from 0 through 7. A server with a lower priority number is given priority over one with a higher number.

By default, the Cisco vEdge device uses port 1812 for authentication connections to the RADIUS server and port 1813 for accounting connections. To change these port numbers, use the **auth-port** and **acct-port** commands.

If the RADIUS server is reachable via a specific interface, configure that interface with the **source-interface** command.

You can tag RADIUS servers so that a specific server or servers can be used for AAA, IEEE 802.1X, and IEEE 802.11i authentication and accounting. Define the tag here, with a string from 4 to 16 characters long. Then associate the tag with the **radius-servers** command when you configure AAA, and when you configure interfaces for 802.1X and 802.11i.

If the RADIUS server is located in a different VPN from the Cisco vEdge device, configure the server's VPN number so that the Cisco vEdge device can locate it. If you configure multiple RADIUS servers, they must all be in the same VPN.

When a Cisco vEdge device is trying to locate a RADIUS server, it goes through the list of servers three times. To change this behavior, use the **retransmit** command, setting the number to a value from 1 to 1000:

```
vEdge(config-radius)# retransmit number
```

When waiting for a reply from the RADIUS server, a Cisco vEdge device waits 3 seconds before retransmitting its request. To change this time interval, use the **timeout** command, setting a value from 1 to 1000 seconds:

```
vEdge(config-radius)# timeout seconds
```

Configure SSH Authentication

Table 37: Feature History

Feature Name	Release Information	Description
Secure Shell Authentication Using RSA Keys	Cisco Catalyst SD-WAN Release 19.2.1	This feature helps configure RSA keys by securing communication between a client and a Cisco Catalyst SD-WAN server.

The Secure Shell (SSH) protocol provides secure remote access connection to network devices.

SSH supports user authentication using public and private keys. To enable SSH authentication, public keys of the users are stored in the home directory of authenticating user in the following location:

```
~<user>/.ssh/authorized_keys
```

A new key is generated on the client machine which owns the private-key. Any message encrypted using the public key of the SSH server is decrypted using the private key of the client.



Note By default, the SSH service on Cisco vEdge devices is always listening on both ports 22 and 830 on LAN. Cisco vManage uses these ports and the SSH service to perform device management. Due to this, any client machine that uses the Cisco vEdge device for internet access can attempt to SSH to the device. For each of the listening ports, we recommend that you create an ACL to block and/or allow access to Cisco vEdge devices and SSH connections for the listening ports.

Restrictions for SSH Authentication on Cisco Catalyst SD-WAN

- The range of SSH RSA key size supported by Cisco vEdge devices is from 2048 to 4096. SSH RSA key size of 1024 and 8192 are not supported.
- A maximum of 10 keys are required on Cisco vEdge devices.

SSH Authentication using vManage on Cisco vEdge Devices

1. In vManage NMS, select the **Configuration ► Templates** screen.
2. In the **Feature Templates** tab, click **Create Template**.



Note In Cisco vManage Release 20.7.x and earlier releases, **Feature Templates** is titled **Feature**.

3. From the **Device Model** check box, select the type of device for which you are creating the template.

4. From the **Basic Information** tab, choose **AAA** template.
5. From the **Local** section, **New User** section, enter the **SSH RSA Key**. You must enter the complete public key from the id_rsa.pub file in the SSH RSA Key text box.

Configure SSH Authentication using CLI on Cisco vEdge Devices

When a user is created in the `/home/<user>` directory, SSH authentication configures the following parameters:

- Create the `.ssh` directory with permissions 700
- Create the `authorized_keys` files in the directory with permission 600

When the public-key is copied and pasted in the key-string, the public key is validated using the **ssh-keygen** utility. The **key-string** and **key-type** fields can be added, updated, or deleted based on your requirement. Similarly, the key-type can be changed.

When a user associated with an SSH directory gets deleted, the **.ssh** directory gets deleted.

Types of Public Keys Supported on Cisco vEdge devices:

- SSH-RSA
- SSH-DSS
- ecdsa-sha2-nistp256
- ecdsa-sha2-nistp384
- ecdsa-sha2-nistp521
- ssh-ed25519

SSH Authentication using CLI

```
vm5(config)# system aaa user ssh-user password vip group tenantadmin
vm5(config-user-ssh-user)# pubkey-chain ssh-usertag key-string
AAAAB3NzaC1yc2EAAAADAQABAAQDAve2mZGpLkveIgZHWn6cjqsfTYTJogfPikgsBJDujfEMhUlhriWZ1h03sLwki290gZNNsJYM30Cy0TA7pEWpDDXQw/gd4/
Bh2TH09CBNEChdV0rA6K2fMwOZfmw2PvNFB10zV1ijjQaitd5Dge7Ar5HGTafIwVnku9HLQDZSfedt8cl/ftgn8sKQQuTccTpWFnYZkth978Bgm029v8/O5R
BdQOVtT3VBr9NNeC4egutS0yBNZeWBPfwrwcd4/aot38plF6jOoLDvUjn60CUJou9TIQIaSFg/dFFUB0twbtEOTUFMBeimRexT+cI3z8vM1D9tqfRDAI8EUegjU7BP
vm5(config-pubkey-chain-ssh-usertag)# commit
Commit complete.
```

Configure the Authentication Order

The authentication order dictates the order in which authentication methods are tried when verifying user access to a Cisco vEdge device through an SSH session or a console port. The default authentication order is **local**, then **radius**, and then **tacacs**. With the default authentication order, the authentication process occurs in the following sequence:

- The authentication process first checks whether a username and matching password are present in the running configuration on the local device.
- If local authentication fails, and if you have not configured authentication fallback (with the **auth-fallback** command), the authentication process stops. However, if you have configured authentication fallback,

the authentication process next checks the RADIUS server. For this method to work, you must configure one or more RADIUS servers with the **system radius server** command. If a RADIUS server is reachable, the user is authenticated or denied access based on that server's RADIUS database. If a RADIUS server is unreachable and if you have configured multiple RADIUS servers, the authentication process checks each server sequentially, stopping when it is able to reach one of them. The user is then authenticated or denied access based on that server's RADIUS database.

- If the RADIUS server is unreachable (or all the servers are unreachable), the authentication process checks the TACACS+ server. For this method to work, you must configure one or more TACACS+ servers with the **system tacacs server** command. If a TACACS+ server is reachable, the user is authenticated or denied access based on that server's TACACS+ database. If a TACACS+ server is unreachable and if you have configured multiple TACACS+ servers, the authentication process checks each server sequentially, stopping when it is able to reach one of them. The user is then authenticated or denied access based on that server's TACACS+ database.
- If the TACACS+ server is unreachable (or all TACACS+ servers are unreachable), user access to the local Cisco vEdge device is denied.

To modify the default order, use the **auth-order** command:

```
Device(config-system-aaa) # auth-order (local | radius | tacacs)
```

Specify one, two, or three authentication methods in the preferred order, starting with the one to be tried first. If you configure only one authentication method, it must be **local**.

To have the "admin" user use the authentication order configured in the **auth-order** command, use the following command:

```
Device(config-system-aaa) # admin-auth-order
```

If you do not include this command, the "admin" user is always authenticated locally.

You can configure authentication to fall back to a secondary or tertiary authentication mechanism when the higher-priority authentication method fails to authenticate a user, either because the user has entered invalid credentials or because the authentication server is unreachable (or all the servers are unreachable):

```
vEdge(config-system-aaa) # auth-fallback
```

Fallback to a secondary or tertiary authentication mechanism happens when the higher-priority authentication server fails to authenticate a user, either because the credentials provided by the user are invalid or because the server is unreachable.

The following examples illustrate the default authentication behavior and the behavior when authentication fallback is enabled:

- If the authentication order is configured as **radius local**:
 - With the default authentication, local authentication is used only when all RADIUS servers are unreachable. If an authentication attempt via a RADIUS server fails, the user is not allowed to log in even if they have provided the correct credentials for local authentication.
 - With authentication fallback enabled, local authentication is used when all RADIUS servers are unreachable or when a RADIUS server denies access to a user.
- If the authentication order is configured as **local radius**:
 - With the default authentication, RADIUS authentication is tried when a username and matching password are not present in the running configuration on the local device.

- With authentication fallback enabled, RADIUS authentication is tried when a username and matching password are not present in the running configuration on the local device. In this case, the behavior of two authentication methods is identical.
- If the authentication order is configured as **radius tacacs local**:
 - With the default authentication, TACACS+ is tried only when all RADIUS servers are unreachable, and local authentication is tried only when all TACACS+ servers are unreachable. If an authentication attempt via a RADIUS server fails, the user is not allowed to log in even if they have provided the correct credentials for the TACACS+ server. Similarly, if a TACACS+ server denies access, the user cannot log via local authentication.
 - With authentication fallback enabled, TACACS+ authentication is used when all RADIUS servers are unreachable or when a RADIUS server denies access a user. Local authentication is used next, when all TACACS+ servers are unreachable or when a TACACS+ server denies access to a user.

If a remote server validates authentication but does not specify a user group, the user is placed into the user group **basic**.

If a remote server validates authentication and specifies a user group (say, X), the user is placed into that user group only. However, if that user is also configured locally and belongs to a user group (say, Y), the user is placed into both the groups (X and Y).

If a remote server validates authentication and that user is not configured locally, the user is logged in to the vshell as the user **basic**, with a home directory of /home/basic.

If a remote server validates authentication and that user is configured locally, the user is logged in to the vshell under their local username (say, eve) with a home direction of /home/username (so, /home/eve).



Note Tags are used for grouping, describing, or finding devices. You can tag RADIUS and TACAC servers for authentication and accounting. You can add more than one tag to a device. Starting from Cisco vManage Release 20.9.1, following new tags are used in authentication:

- Viptela-User-Group: for user group definitions instead of Viptela-Group-Name.
- Viptela-Resource-Group: for resource group definitions.

The authentication fails if there is any space between keys and the values. For example, **key=value**.

Configure NAS Attributes using CLI

For RADIUS and TACACS+, you can configure Network Access Server (NAS) attributes for user authentication and authorization. To do this, you create a vendor-specific attributes (VSA) file, also called a RADIUS dictionary or a TACACS+ dictionary, on the RADIUS or TACACS+ server that contains the desired permit and deny commands for each user. The Cisco vEdge device retrieves this information from the RADIUS or TACACS+ server.

The VSA file must be named dictionary.viptela, and it must contain text in the following format:

```
localhost$ more dictionary.viptela
# -*- text -*-
#
```

```
# dictionary.viptela
#
#
# Version:      $Id$
#
VENDOR          Viptela                      41916

BEGIN-VENDOR     Viptela

ATTRIBUTE        Viptela-Group-Name          1      string
```

The Cisco SD-WAN software has three predefined user groups, as described above: **basic**, **netadmin**, and **operator**. These groups have the following permissions:

```
vEdge# show aaa usergroup
GROUP      USERS  TASK      PERMISSION
-----
basic      -      system    read
           -      interface read
netadmin   admin  system    read write
           -      interface read write
           -      policy    read write
           -      routing   read write
           -      security  read write
operator    -      system    read
           -      interface read
           -      policy    read
           -      routing   read
           -      security  read
```

To create new user groups, use this command:

```
vEdge(config)# system aaa usergroup
group-name task privilege
```

Here is a sample user configuration on a RADIUS server, which for FreeRADIUS would be in the file "users":

```
user1 Cleartext-password := "user123"
      Service-Type = NAS-Prompt-User,
      Viptela-Group-Name = operator,

user1 Cleartext-password := "user123"      Service-Type = NAS-Prompt-User,
Viptela-Group-Name = operator,
```

Then in the dictionary on the RADIUS server, add a pointer to the VSA file:

```
$INCLUDE /usr/share/freeradius/dictionary.viptela
```

For TACACS+, here is a sample configuration, which would be in the file tac_plus.conf:

```
group = test_group {
    default service = permit
    service = ppp protocol = ip {
        Viptela-Group-Name = operator
    }
}

user = user1 {
    pap = cleartext "user123"
    member = test_group
}
```



Note Starting from Cisco vManage Release 20.8.1, the unknown mandatory attributes from TACACS are not allowed. The authorization fails, when a client receives the configurations with the arguments that are not supported. For information about configuring ISE for Cisco SDWAN devices, see [RADIUS and TACACS-Based User Authentication and Authorization](#).

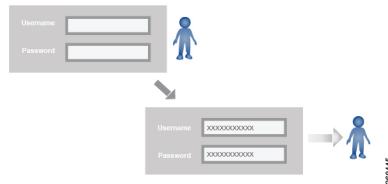
Role-Based Access with AAA

The Cisco Catalyst SD-WAN AAA software implements role-based access to control the authorization permissions for users on Cisco vEdge devices. Role-based access consists of three components:

- Users are those who are allowed to log in to a Cisco vEdge device.
- User groups are collections of users.
- Privileges are associated with each group. They define the commands that the group's users are authorized to issue.

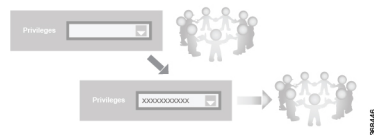
Users and User Groups

All users who are permitted to perform operations on a Cisco vEdge device must have a login account. For the login account, you configure a username and a password on the device itself. These allow the user to log in to that device. A username and password must be configured on each device that a user is allowed to access.

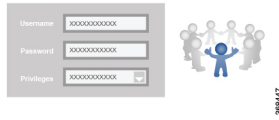


The Cisco Catalyst SD-WAN software provides one standard username, **admin**, which is a user who has full administrative privileges, similar to a UNIX superuser. By default, the **admin** username password is **admin**. You cannot delete or modify this username, but you can and should change the default password.

User groups pool together users who have common roles, or privileges, on the Cisco vEdge device. As part of configuring the login account information, you specify which user group or groups that user is a member of. You do not need to specify a group for the **admin** user, because this user is automatically in the user group **netadmin** and is permitted to perform all operations on the Cisco vEdge device.



The user group itself is where you configure the privileges associated with that group. These privileges correspond to the specific commands that the user is permitted to execute, effectively defining the role-based access to the Cisco Catalyst SD-WAN software elements.



The Cisco Catalyst SD-WAN software provides the following standard user groups:

- **basic**: The basic group is a configurable group and can be used for any users and privilege levels. This group is designed to include users who have permission to both view and modify information on the device.
- **operator**: The operator group is also a configurable group and can be used for any users and privilege levels. This group is designed to include users who have permission only to view information.
- **netadmin**: The netadmin group is a non-configurable group. By default, this group includes the **admin** user. You can add other users to this group. Users in this group are permitted to perform all operations on the device.

- Minimum supported release: Cisco vManage Release 20.9.1

network_operations: The **network_operations** group is a non-configurable group. Users in this group can perform all non-security-policy operations on the device and only view security policy information. For example, users can create or modify template configurations, manage disaster recovery, and create non-security policies such as application aware routing policy or CFlowD policy.

- Minimum supported release: Cisco vManage Release 20.9.1

security_operations: The **security_operations** group is a non-configurable group. Users in this group can perform all security operations on the device and only view non-security-policy information. For example, users can manage umbrella keys, licensing, IPS signatures auto update, TLS/SSL proxy settings, and so on.

Users of the **network_operations** group are authorized to apply policies to a device, revoke applied policies, and edit device templates. Users of the **security_operations** group require **network_operations** users to intervene on day-0 to deploy security policy on a device and on day-N to remove a deployed security policy. However, after a security policy is deployed on a device, **security_operations** users can modify the security policy without needing the **network_operations** users to intervene.



Note Only admin users can view running and local configuration. Users associated with predefined operator user group do not have access to the running and local configurations. The predefined user group operator has only read access for the template configuration. If you need only a subset of admin user privileges, then you need to create a new user group with the selected features from the features list with both read and write access and associate the group with the custom user.

Privileges for Role-Based Access

Role-based access privileges are arranged into five categories, which are called *tasks*:

- **Interface**—Privileges for controlling the interfaces on the Cisco vEdge device.
- **Policy**—Privileges for controlling control plane policy, OMP, and data plane policy.
- **Routing**—Privileges for controlling the routing protocols, including BFD, BGP, OMP, and OSPF.

- Security—Privileges for controlling the security of the device, including installing software and certificates. Only users belonging to the **netadmin** group can install software on the system.
- System—General systemwide privileges.

The tables in the following sections detail the AAA authorization rules for users and user groups. These authorization rules apply to commands issued from the CLI and to those issued from Netconf.

User Authorization Rules for Operational Commands

The user authorization rules for operational commands are based simply on the username. Any user who is allowed to log in to the Cisco vEdge device can execute most operational commands. However, only the **admin** user can issue commands that affect the fundamental operation of the device, such as installing and upgrading the software and shutting down the device.

Note that any user can issue the **config** command to enter configuration mode, and once in configuration mode, they are allowed to issue any general configuration command. Also, any user is allowed to configure their password by issuing the **system aaa user self password password** command and then committing that configuration change. For the actual commands that configure device operation, authorization is defined according to user group membership. See User Group Authorization Rules for Configuration Commands.

The following tables lists the AAA authorization rules for general CLI commands. All the commands are operational commands except as noted. Also, some commands available to the "admin" user are available only if that user is in the "netadmin" user group.

CLI Command	Any User	Admin User
clear history	X	X
commit confirm	X	X
complete-on-space	X	X
config	X	X
exit	X	X
file	X	X
help	X	X
[no] history	X	X
idle-timeout	X	X
job	X	X
logout	—	X (users in netadmin group only)
monitor	X	X
nslookup	X	X
paginate	X	X

CLI Command	Any User	Admin User
ping	X (users in netadmin group only)	X (users in netadmin group only)
poweroff	—	X(users in netadmin group only)
prompt1	X	X
prompt2	X	X
quit	X	X
reboot	—	X (users in netadmin group only)
request aaa request admin-tech request firmware request interface-reset request nms request reset request software	—	X (users in netadmin group only)
request execute request download request upload	X	X
request (everything else)	—	X
rollback (configuration mode command)	—	X (users in netadmin group only)
screen-length	X	X
screen-width	X	X
show cli	X	X
show configuration commit list	X	X
show history	X	X
show jobs	X	X
show parser dump	X	X
show running-config	X	X
show users	X	X
system aaa user <i>self</i> password <i>password</i> (configuration mode command) (Note: A user cannot delete themselves)		
tcpdump	X	X
timestamp	X	X
tools ip-route	X	X

CLI Command	Any User	Admin User
tools netstat	X	X
tools nping	X	X
tracert	X	X
vshell	X (The availability of vshell command is unavailable to all users that are not in netadmin group in Cisco vManage Release 20.9.5.)	X (The vshell AAA authorized access is limited only to users that are in netadmin group.)

User Group Authorization Rules for Operational Commands

The following table lists the user group authorization roles for operational commands.

Operational Command	Interface	Policy	Routing	Security	System
clear app		X			
clear app-route		X			
clear arp	X				
clear bfd			X		X
clear bgp			X		X
clear bridge	X				
clear cellular	X				
clear control				X	
clear crash					X
clear dhcp					X
clear dns					X
clear igmp			X		
clear installed-certificates				X	
clear interface	X				
clear ip			X		
clear notification					X
clear omp			X		

Operational Command	Interface	Policy	Routing	Security	System
clear orchestrator				X	
clear ospf			X		
clear pim			X		
clear policy		X			
clear pppoe	X				
clear system					X
clear tunnel				X	
clear wlan	X				
clear ztp				X	X
clock					X
debug bgp			X		
debug cellular	X				
debug cflowd		X			
debug chmgr					X
debug config-mgr					X
debug dhcp-client					X
debug dhcp-helper					X
debug dhcp-server					X
debug fpm		X			
debug ftm					X
debug igmp			X		
debug netconf					X
debug omp			X		
debug ospf			X		
debug pim			X		
debug resolver			X		
debug snmp					X
debug sysmgr					X

Operational Command	Interface	Policy	Routing	Security	System
debug transport					X
debug ttm					X
debug vdaemon				X	X
debug vrrp				X	
debug wlan	X				
request certificate				X	
request control-tunnel				X	
request controller				X	
request controller-upload				X	
request csr				X	
request device				X	
request device-upload				X	
request on-vbond-controller				X	
request port-hop				X	
request root-cert-chain				X	
request security				X	
request vedge				X	
request vedge-upload				X	
request vsmart-upload				X	
show aaa					X
show app		X			
show app-route		X			
show arp	X				
show bfd			X		X
show bgp			X		
show boot-partition					X
show bridge	X				

Operational Command	Interface	Policy	Routing	Security	System
show cellular	X				
show certificate				X	
show clock					X
show control				X	X
show crash					X
show debugs—same as debug commands					
show dhcp					X
show external-nat				X	X
show hardware					X
show igmp			X		
show interface	X				
show ip			X		X
show ipsec				X	
show licenses					X
show logging					X
show multicast			X		
show nms-server					X
show notification					X
show ntp					X
show omp		X	X		X
show orchestrator				X	
show ospf			X		
show pim			X		
show policer		X			
show policy		X			
show ppp	X				
show pppoe	X				

Operational Command	Interface	Policy	Routing	Security	System
show reboot					X
show security-info				X	
show software					X
show system					X
show transport					X
show tunnel				X	
show uptime					X
show users					X
show version					X
show vrrp	X				
show wlan	X				
show ztp				X	

User Group Authorization Rules for Configuration Commands

The following table lists the user group authorization rules for configuration commands.

Configuration Command	Interface	Policy	Routing	Security	System
apply-policy		X			
banner					X
bfd			X		X
bridge	X				
omp		X	X		X
policy		X			
security				X	X
snmp					X
system					X
vpn interface	X				
vpn ip			X		
vpn router			X		

Configuration Command	Interface	Policy	Routing	Security	System
vpn service			X		
vpn (everything else, including creating, deleting, and naming)					X
wlan	X				

Configure AAA Using a Configuration Group

Before you begin

On the **Configuration > Configuration Groups** page, choose **SD-WAN** as the solution type.

Procedure

Step 1 From the Cisco SD-WAN Manager menu, choose **Configuration > Configuration Groups**.

Step 2 Create and configure a AAA feature in a System profile.

- a. Configure users.

Table 38: Local

Field	Description
Enable AAA Authentication	Enable authentication parameters.
Accounting Group	Enable accounting parameters.
Add AAA User	
Name	Enter a name for the user. It can be 1 to 128 characters long, and it must start with a letter. The name can contain only lowercase letters, the digits 0 through 9, hyphens (-), underscores (_), and periods (.). The name cannot contain any uppercase letters. The following usernames are reserved, so you cannot configure them: backup, basic, bin, daemon, games, gnats, irc, list, lp, mail, man, news, nobody, proxy, quagga, root, sshd, sync, sys, uucp, and www-data. Also, names that start with viptela-reserved are reserved.
Password	Enter a password for the user. The password is an MD5 digest string, and it can contain any characters, including tabs, carriage returns, and linefeeds. For more information, see Section 9.4 in RFC 7950, The YANG 1.1 Data Modeling Language. Each username must have a password. Users are allowed to change their own passwords. The default password for the admin user is admin. We strongly recommended that you change this password.

Field	Description
Confirm Password	Re-enter the password for the user.
Privilege	Select between privilege level 1 or 15. • Level 1: User EXEC mode. Read-only, and access to limited commands, such as the ping command. • Level 15: Privileged EXEC mode. Full access to all commands, such as the reload command, and the ability to make configuration changes. By default, the EXEC commands at privilege level 15 are a superset of those available at privilege level 1.
Add Public Key Chain	
Key String*	Enter the authentication string for a key.
Key Type	Choose ssh-rsa .

b. Configure RADIUS servers.

Table 39: RADIUS

Field	Description
Address*	Enter the IP address of the RADIUS server host.
Acct Port	Enter the UDP port to use to send 802.1X and 802.11i accounting information to the RADIUS server. Range: 1 - 65534. Default: 1813
Auth Port	Enter the UDP destination port to use for authentication requests to the RADIUS server. Default: 1812 Range: 1 - 65534
Retransmit	Enter the number of times the device transmits each RADIUS request to the server before giving up. Default: 3 Range: 0 - 100
Timeout	Enter the number of seconds a device waits for a reply to a RADIUS request before retransmitting the request. Default: 5 seconds Range: 1 through 1000
Key*	Enter the key the Cisco IOS XE Catalyst SD-WAN device passes to the RADIUS server for authentication and encryption.
Key Type	Choose Protected Access Credential (PAC) key.

c. Configure TACACS servers.

Table 40: TACACS Server

Field	Description
Address*	Enter the IP address of the TACACS+ server host.
Port	Enter the UDP destination port to use for authentication requests to the TACACS+ server. If the server is not used for authentication, configure the port number to be 0. Default: 49
Timeout	Enter the number of seconds a device waits for a reply to a TACACS+ request before retransmitting the request. Default: 5 seconds Range: 1 through 1000
Key*	Enter the key the Cisco IOS XE Catalyst SD-WAN device passes to the TACACS+ server for authentication and encryption. You can type the key as a text string from 1 to 31 characters long, and it is immediately encrypted, or you can type an AES 128-bit encrypted key. The key must match the AES encryption key used on the TACACS+ server.

d. Configure accounting rules.

Table 41: Accounting

Field	Description
Rule Id*	Enter the accounting rule ID.
Method*	Specifies the accounting method list. Choose one of the following: • commands: Provides accounting information about specific, individual EXEC commands associated with a specific privilege level. • exec: Provides accounting records about user EXEC terminal sessions on the network access server, including username, date, and start and stop times. • network: Runs accounting for all network-related service requests. • system: Performs accounting for all system-level events not associated with users, such as reloads. Note When system accounting is used and the accounting server is unreachable at system startup time, the system will not be accessible for approximately two minutes.
Level	Choose the privilege level (1 or 15). Accounting records are generated only for commands entered by users with this privilege level.
Start Stop	Enable this option to if you want the system to send a start accounting notice at the beginning of an event and a stop record notice at the end of the event.

Field	Description
Use Server-group*	Choose a previously configured TACACS group. The parameters that this accounting rule defines are used by the TACACS servers that are associated with this group.

- e. Configure authorization parameters.

Table 42: Authorization

Field	Description
Server Auth Order*	Choose the authentication order. It dictates the order in which authentication methods are tried when verifying user access to a Cisco IOS XE Catalyst SD-WAN device through an SSH session or a console port.
Authorization Console	Enable this option to perform authorization for console access commands.
Authorization Config Commands	Enable this option to perform authorization for configuration commands.
Add Authorization Rule	
Rule Id*	Enter the authorization rule ID.
Method*	Choose Commands , which causes commands that a user enters to be authorized.
Level	Choose the privilege level (1 or 15) for commands to be authorized. Authorization is provided for commands entered by users with this privilege level.
If Authenticated	Enable this option to apply the authorization rule parameters only to the authenticated users. If you do not enable this option, the rule is applied to all users.
Use Server-group*	Choose a previously configured TACACS group. The parameters that this authorization rule defines are used by the TACACS servers that are associated with this group.

- f. Configure 802.1x parameters.

What to do next

Also see [Deploy a configuration group](#).

Configuring AAA using Cisco SD-WAN Manager Template

Table 43: Feature History

Feature Name	Release Information	Description
Authorization and Accounting	Cisco SD-WAN Release 20.5.1 Cisco vManage Release 20.5.1	This feature provides for the configuration of authorization, which authorizes commands that a user enters on a device before the commands can be executed, and accounting, which generates a record of commands that a user executes on a device.

Configuring AAA by using the Cisco SD-WAN Manager template lets you make configuration setting in Cisco SD-WAN Manager and then push the configuration to selected devices of the same type. This procedure is a convenient way to configure several of the same type of devices at one time.

Use the AAA template for Cisco Catalyst SD-WAN Validators, Cisco SD-WAN Manager instances, Cisco Catalyst SD-WAN Controllers, and Cisco vEdge device s.

Cisco vEdge device s support configuration of authentication, authorization, and accounting (AAA) in combination with RADIUS and TACACS+.



Note You must configure a local user with a secret key via the template if you are using PPP or using MLPPP with CHAP.

Navigating to the Template Screen and Naming the Template

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Device Templates**, and click **Create Template**.



Note In Cisco vManage Release 20.7.x and earlier releases, **Device Templates** is titled **Device**.

3. From the **Create Template** drop-down list, select **From Feature Template**.
4. From the **Device Model** drop-down list, select the type of device for which you are creating the template.
5. Select **Basic Information**.
6. To create a custom template for AAA, select **Factory_Default_AAA_Template** and click **Create Template**. The AAA template form is displayed. The top of the form contains fields for naming the template, and the bottom contains fields for defining AAA parameters.
7. In the **Template Name** field, enter a name for the template. The name can be up to 128 characters and can contain only alphanumeric characters.

8. In the **Template Description** field, enter a description of the template. The description can be up to 2048 characters and can contain only alphanumeric characters.

When you first open a feature template, for each parameter that has a default value, the scope is set to Default (indicated by a check mark), and the default setting or value is shown. To change the default or to enter a value, click the **Scope** drop-down list to the left of the parameter field and select one of the following:

Table 44:

Parameter Scope	Scope Description
Device Specific (indicated by a host icon)	Use a device-specific value for the parameter. For device-specific parameters, you cannot enter a value in the feature template. You enter the value when you attach a Cisco vEdge device to a device template . When you click Device Specific, the Enter Key box opens. This box displays a key, which is a unique string that identifies the parameter in a CSV file that you create. This file is an Excel spreadsheet that contains one column for each key. The header row contains the key names (one key per column), and each row after that corresponds to a device and defines the values of the keys for that device. You upload the CSV file when you attach a Cisco vEdge device to a device template. For more information, see Create a Template Variables Spreadsheet . To change the default key, type a new string and move the cursor out of the Enter Key box. Examples of device-specific parameters are system IP address, hostname, GPS location, and site ID.
Global (indicated by a globe icon)	Enter a value for the parameter, and apply that value to all devices. Examples of parameters that you might apply globally to a group of devices are DNS server, syslog server, and interface MTUs.

Configuring Authentication Order and Fallback

You can configure the authentication order and authentication fallback for devices. The authentication order specifies the order in which the system attempts to authenticate user, and provides a way to proceed with authentication if the current authentication method is unavailable. Fallback provides a mechanism for authentication if the user cannot be authenticated or if a RADIUS or TACACS+ server is unreachable.

To configure AAA authentication order and authentication fallback on a Cisco vEdge device, select the **Authentication** tab and configure the following parameters:

Table 45:

Parameter Name	Description
Authentication Order	The default order is local, then radius, and then tacacs. To change the default order of authentication methods that the software tries when verifying user access to a Cisco vEdge device: 1. Click the drop-down arrow to display the list of authentication methods. 2. In the list, click the up arrows to change the order of the authentication methods and click the boxes to select or deselect a method. If you select only one authentication method, it must be local.
Authentication Fallback	Click On to configure authentication to fall back from RADIUS or TACACS+ to the next priority authentication method if the user cannot be authenticated or if the RADIUS or TACACS+ servers are unreachable. With the default configuration (Off), authentication falls back only if the RADIUS or TACACS+ servers are unreachable.
Admin Authentication Order	Have the "admin" user use the authentication order configured in the Authentication Order parameter. If you do not configure the admin authentication order, the "admin" user is always authenticated locally.
Disable Netconf Logs	Click On to disable the logging of Netconf events. By default, these events are logged to the auth.info and messages log files.
Disable Audit Logs	Click On to disable the logging of AAA events. By default, these events are logged to the auth.info and messages log files.
RADIUS Server List	List the tags for one or two RADIUS servers. Separate the tags with commas. You set the tag under the RADIUS tab.

CLI equivalent:

```

system
aaa
admin-auth-order auth-fallback auth-order (local | radius | tacacs)
logs
[no] audit-disable
[no] netconf-disable
radius-servers tag

```

Configuring Local Access for Users and User Groups

You can configure local access to a device for users and user groups. Local access provides access to a device if RADIUS or TACACS+ authentication fails.

To configure local access for individual users, select **Local**.

To add a new user, from **Local** click + **New User**, and configure the following parameters:

Table 46:

Parameter Name	Description
Name	Enter a name for the user. It can be 1 to 128 characters long, and it must start with a letter. The name can contain only lowercase letters, the digits 0 through 9, hyphens (-), underscores (_), and periods (.). The name cannot contain any uppercase letters. The following usernames are reserved, so you cannot configure them: backup, basic, bin, daemon, games, gnats, irc, list, lp, mail, man, news, nobody, proxy, quagga, root, sshd, sync, sys, uucp, and www-data. Also, names that start with viptela-reserved are reserved. Note From Cisco Catalyst SD-WAN Manager Release 20.18.1, the character limit for local user accounts remains restricted to 32 characters. However, for TACACS users, usernames extend up to 128 characters.
Password	Enter a password for the user. Each username must have a password. Users are allowed to change their own passwords. The default password for the admin user is admin. We strongly recommended that you change this password.
Description	Enter a description for the user.
User Groups	Select from the list of configured groups. You must assign the user to at least one group. The admin user is automatically placed in the netadmin group and is the only member of this group.
SSH RSA Key(s)	Add SSH RSA Keys by clicking the + Add button. A new field is displayed in which you can paste your SSH RSA key. To remove a key, click the - button. Devices support a maximum of 10 SSH RSA keys.

Click **Add** to add the new user. Click + **New User** again to add additional users.

To configure local access for user groups, you first place the user into either the basic or operator group. The admin is automatically placed in the netadmin group. Then you configure user groups. To make this configuration, from **Local** select **User Group**.

Click + **New User Group**, and configure the following parameters:

Table 47:

Parameter Name	Description
Name	Name of an authentication group. It can be 1 to 128 characters long, and it must start with a letter. The name can contain only lowercase letters, the digits 0 through 9, hyphens (-), underscores (_), and periods (.). The name cannot contain any uppercase letters. The Cisco Catalyst SD-WAN software provides three standard user groups, basic, netadmin, and operator. The user admin is automatically placed in the group netadmin and is the only user in this group. All users learned from a RADIUS or TACACS+ server are placed in the group basic. All users in the basic group have the same permissions to perform tasks, as do all users in the operator group. The following groups names are reserved, so you cannot configure them: adm, audio, backup, bin, cdrom, dialout, dip, disk, fax, floppy, games, gnats, input, irc, kmem, list, lp, mail, man, news, nogroup, plugdev, proxy, quagga, quaggavty, root, sasl, shadow, src, sshd, staff, sudo, sync, sys, tape, tty, uucp, users, utmp, video, voice, and www-data. Also, group names that start with the string viptela-reserved are reserved.
Feature Type	Click Preset to display a list of preset roles for the user group. Click Custom to display a list of authorization tasks that have been configured.
Feature	The Preset list in the feature table lists the roles for the user group. These roles are Interface, Policy, Routing, Security, and System. Each role allows the user group to read or write specific portions of the device's configuration and to execute specific types of operational commands. Click the appropriate boxes for Read, Write, and None to assign privileges to the group for each role. The Custom list in the feature table lists the authorization tasks that you have created (see "Configure Authorization"). To associate a task with this user group, choose Read, Write, or both options. The Read option grants to users in this user group read authorization to XPath's as defined in the task. The Write option allows users in this user group write access to XPath's as defined in the task.

Click **Add** to add the new user group.

To add another user group, click + **New User Group** again.

To delete a user group, click the trash icon at the right side of the entry. You cannot delete the three standard user groups, basic, netadmin, and operator.

CLI equivalent:

```

system
aaa
  user username
  group group-name
  password password usergroup group-name
  task (interface | policy | routing | security | system) (read | write)

```

Configuring RADIUS Authentication

Configure RADIUS authentication if you are using RADIUS in your deployment.

To configure RADIUS authentication, select **RADIUS** and configure the following parameters:

Table 48:

Parameter Name	Description
Retransmit Count	Specify how many times to search through the list of RADIUS servers while attempting to locate a server. <i>Range: 1 through 1000 Default: 3</i>
Timeout	Specify how long to wait to receive a reply from the RADIUS server before retransmitting a request. <i>Range: 1 through 1000 Default: 5 seconds</i>

To configure a connection to a RADIUS server, from **RADIUS**, click + **New Radius Server**, and configure the following parameters:

Table 49:

Parameter Name	Description
Address	Enter the IP address of the RADIUS server host.
Tag	Enter a text string to identify the RADIUS server. The tag can be 4 to 16 characters long. The tag allows you to configure authentication for AAA, IEEE 802.1X, and IEEE 802.11i to use a specific RADIUS server or servers. For Cisco vEdge devices running Cisco Catalyst SD-WAN software, this field is ignored.
Authentication Port	Enter the UDP destination port to use for authentication requests to the RADIUS server. If the server is not used for authentication, configure the port number to be 0. <i>Default: Port 1812</i>
Accounting Port	Enter the UDP port to use to send 802.1X and 802.11i accounting information to the RADIUS server. <i>Range: 0 through 65535. Default: 1813.</i>
Key	This field is deprecated. Use the Secret Key field instead.
Secret Key	Enter the key the Cisco vEdge device passes to the RADIUS server for authentication and encryption. You can type the key as a text string from 1 to 31 characters long, and it is immediately encrypted, or you can type an AES 128-bit encrypted key. The key must match the AES encryption key used on the RADIUS server.
Source Interface	Enter the name of the interface on the local device to use to reach the RADIUS server.
VPN ID	Enter the number of the VPN in which the RADIUS server is located or through which the server can be reached. If you configure multiple RADIUS servers, they must all be in the same VPN.
Priority	Enter the priority of a RADIUS server. A server with a lower number is given priority. <i>Range: 0 through 7.</i> <i>Default: 0</i>

Click **Add** to add the new RADIUS server.

To add another RADIUS server, click + **New RADIUS Server** again.

To remove a server, click the trash icon.

CLI equivalent:

```
system radius
  retransmit number
  server ip-address
    acct-port port-number
    auth-port port-number
    priority number
    secret-key key
    source-interface interface-name
    tag tag
    vpn vpn-id
    timeout seconds
```

Configuring TACACS+ Authentication

Configure TACACS+ authentication if you are using TACACS+ in your deployment.

To configure the device to use TACACS+ authentication, select **TACACS** and configure the following parameters:

Table 50:

Parameter Name	Description
Timeout	Enter how long to wait to receive a reply from the TACACS+ server before retransmitting a request. <i>Range: 1 through 1000Default: 5 seconds</i>
Authentication	Set the type of authentication to use for the server password. The default authentication type is PAP. You can change it to ASCII.

To configure a connection to a TACACS+ server, from **TACACS**, click + **New TACACS Server**, and configure the following parameters:

Table 51:

Parameter Name	Description
Address	Enter the IP address of the TACACS+ server host.
Authentication Port	Enter the UDP destination port to use for authentication requests to the TACACS+ server. If the server is not used for authentication, configure the port number to be 0. <i>Default: Port 49</i>
Key (Deprecated)	This field is deprecated. Use the Secret Key field instead.

Parameter Name	Description
Secret Key	Enter the key the Cisco vEdge device passes to the TACACS+ server for authentication and encryption. You can type the key as a text string from 1 to 31 characters long, and it is immediately encrypted, or you can type an AES 128-bit encrypted key. The key must match the AES encryption key used on the TACACS+ server.
Source Interface	Enter the name of the interface on the local device to use to reach the TACACS+ server.
VPN ID	VPN in which the TACACS+ server is located or through which the server can be reached. If you configure multiple TACACS+ servers, they must all be in the same VPN.
Priority	Set the priority of a TACACS+ server. A server with lower priority number is given priority over one with a higher number. <i>Range: 0 through 7</i> <i>Default: 0</i>

Click **Add** to add the new TACACS server.

To add another TACACS server, click + **New TACACS Server** again.

To remove a server, click the trash icon.

CLI equivalent:

```

system tacacs
 authentication password-authentication
 server ip-address
   auth-port port-number
   priority number
   key key
   source-interface interface-name
   vpn vpn-id
 timeout seconds

```

Configure Authorization and Accounting

Table 52: Feature History

Feature Name	Release Information	Description
Authorization and Accounting	Cisco SD-WAN Release 20.5.1 Cisco vManage Release 20.5.1	This feature provides for the configuration of authorization, which authorizes commands that a user enters on a device before the commands can be executed, and accounting, which generates a record of commands that a user executes on a device.

Restrictions to Configure Authorization and Accounting

Minimum supported release: Cisco IOS XE Catalyst SD-WAN Release 17.18.1a

- If you enter a configuration and press enter before selecting a value from an enumeration, the CLI presents a choice sub-menu. In this scenario, the final chosen value is not sent for authorization.
- The load merge and load override commands are not available when authorization is configured.

- The individual commands configured via load or rollback are not authorized or accounted.

Configuring Authorization

You can configure authorization, which causes the device to authorize commands that users enter on a device before the commands can be executed.

Configuring authorization involves creating one or more tasks. A task consists of a set of operational commands and a set of configuration commands. Operational commands are show commands and exec commands. Configuration commands are the XPath of configuration commands.

You define the default user authorization action for each command type. The default action can be accept or deny. You also can define user authorization accept or deny actions for individual commands or for XPath strings within a command type. In this way, you can override the default action for specific commands as needed.

A task is mapped to a user group, so all users in the user group are granted the authorizations that the command sets in the task define.

To configure authorization, choose the **Authorization** tab, click **+ New Task**, and configure the following parameters:

Table 53:

Parameter Name	Description
Name	Enter a unique name for the task
+ Add Oper	Click to add a set of operational commands. In the Add Oper window that pops up: 1. From the Default action drop-down list, choose the default authorization action for operational commands. Choose accept to grant user authorization by default, or choose deny to prevent user authorization by default. 2. To designate specific operational commands for which user authorization is granted or denied authorization, click + Add Oper to expand the Add Oper area. In the Oper field that displays, click accept to grant user authorization for a command, or click deny to prevent user authorization for a command, and enter the command in the CLI field. Then click Add in the Add Oper area. Do not include quotes or a command prompt when entering a command. For example, config terminal is a valid entry, but "config terminal" is not valid. Repeat this Step 2 as needed to designate other commands. The actions that you specify here override the default action. In this way, you can designate specific commands that are not authorized when the default action is accept, and designate specific commands that are authorized when the default action is deny. To remove a specific command, click the trash icon on the right side of its line in the table at the bottom of the Add Oper window. 3. Click Add at the bottom right of the Add Oper window.

Parameter Name	Description
+ Add Config	Click to add a set of XPath strings for configuration commands. In the Add Config window that pops up: - From the Default action drop-down list, choose the default authorization action for configuration commands. Choose accept to grant user authorization by default, or choose deny to prevent user authorization by default. - To designate specific configuration command XPath strings for which user is granted or denied authorization Click + Add Config to expand the Add Config area. In the Config field that displays, click accept to grant user authorization for an XPath, or click deny to prevent user authorization for an XPath, and enter the XPath string in the CLI field. Then click Add in the Add Config area. To display the XPath for a device, enter the show running-config display xpath command on the device. Do not include quotes or a command prompt when entering an XPath string. Repeat this Step 2 as needed to designate other XPath strings. The actions that you specify here override the default action. In this way, you can designate specific XPath strings that are not authorized when the default action is accept, and designate specific XPath strings that are authorized when the default action is deny. To remove a specific command, click the trash icon on the right side of its line in the table at the bottom of the Add Config window. - Click Add at the bottom right of the Add Config window.

To remove a task, click the trash icon on the right side of the task line.

After you create a tasks, perform these actions:

- Create or update a user group. Use the Custom feature type to associate one or more tasks with the user group by assigning read, write, or both privileges to each task. See [Configure Local Access for Users and User Groups](#).



Note A user group can be associated with either a predefined task or with user-defined tasks. Associating a user group with a combination of both predefined and user-defined tasks is not supported.

- Add users to the user group. These users then receive the authorization for operational and configuration commands that the tasks that are associated with the user group define. See [Configure Local Access for Users and User Groups](#).

If a user is attached to multiple user groups, the user receives the authorization access that is configured for the last user group that was created.

CLI equivalent:

```
system aaa
  accounting
```

```

task name
  config
    default-action {accept | deny}
    accept "xpath"
    deny "xpath"
  oper-exec
    default-action {accept | deny}
    accept "command"
    deny "command-id"
  usergroup group-name
    task authorization-task {read | write}

```

Configuring Accounting

You can configure accounting, which causes a TACACS+ server to generate a record of commands that a user executes on a device.



Note Accounting does not generate a record of CLI commands for Cisco SD-WAN Manager template configuration.

Prerequisites

- The TACACS+ server must be configured with a secret key on the **TACACS** tab
- The TACACS+ server must be configured as first in the authentication order on the **Authentication** tab

To configure accounting, choose the **Accounting** tab and configure the following parameter:

Table 54:

Parameter Name	Description
Enable/disable user accounting	Click On to enable the accounting feature. Click Off to disable this feature.

CLI equivalent:

```

system aaa
  accounting

```

Configuring Password Policy for AAA on Devices

In Cisco vManage Release 20.4.1, you can create password policies using Cisco AAA on Cisco vEdge devices. We recommend configuring a password policy to ensure that all users or users of a specific group are prompted to use strong passwords. You can customize the password policy to meet the requirements of your organization.



Note You can only configure password policies for Cisco AAA using device CLI templates.

You can configure the following parameters:

password-policy min-password-length <i>length</i>	The minimum allowed length of a password. You can specify between 8 to 32 characters.
password-policy num-lower-case-characters <i>number-of-lower-case-characters</i>	The minimum number of lower case characters. You can specify between 1 to 128 characters.
password-policy num-numeric-characters <i>number-of-numeric-characters</i>	The minimum number of numeric characters. You can specify between 1 to 128 characters.
password-policy num-special-characters <i>number-of-special-characters</i>	The minimum number of special characters. You can specify between 1 to 128 characters.
password-policy num-upper-case-characters <i>number-of-upper-case-characters</i>	The minimum number of upper case characters. You can specify between 1 to 128 characters.

Configure Password Policies Using Cisco vManage

Table 55: Feature History

Feature Name	Release Information	Description
Support for Password Policies using Cisco AAA	Cisco SD-WAN Release 20.4.1 Cisco vManage Release 20.4.1	This feature allows you to create password policies for Cisco AAA. Password policies ensure that your users use strong passwords and can be customized based on your requirements. To configure password policies, push the <code>password-policy</code> commands to your device using Cisco SD-WAN Manager device CLI templates. For more information on the <code>password-policy</code> commands, see the aaa command reference page .

Configure password policies for Cisco AAA by doing the following:

1. Navigate to **Configuration > Templates**.
2. Click **Device Templates**.



Note In Cisco vManage Release 20.7.x and earlier releases, **Device Templates** is titled **Device**.

3. Click **Create Template**.
4. Click **CLI Template**.
5. From the **Device Model** drop-down list, choose your Cisco vEdge device.
6. Enter a **Template Name**.

7. Enter a **Description**.
8. (Optional) From the **Load Running config from reachable device**: drop-down list, choose a device from which to load the running configuration.
9. Enter or append the password policy configuration.
For more information on the `password-policy` commands, see the [aaa command reference page](#).
10. Click **Add**.
The device templates page appears.
11. Attach the templates to your devices as described in [Attach a Device Template to Devices](#).

Configuring IEEE 802.1X and IEEE 802.11i Authentication

IEEE 802.1X is a port-based network access control (PNAC) protocol that prevents unauthorized network devices from gaining access to wired networks (WANs), by providing authentication for devices that want to connect to a WAN.

IEEE 802.11i prevents unauthorized network devices from gaining access to wireless networks (WLANs). 802.11i implements WiFi Protected Access II (WPA2) to provide authentication for devices that want to connect to a WLAN on a Cisco vEdge 100wm device.

A RADIUS authentication server must authenticate each client connected to a port before that client can access any services offered by network.

This section describes how to configure RADIUS servers to use for 802.1X and 802.11i authentication. It describes how to enable 802.1X on Cisco vEdge device interfaces to have the router act as an 802.1X authenticator, responsible for authorizing or denying access to network devices on a WAN.

It also describes how to enable 802.11i on Cisco vEdge 100wm device routers to control access to WLANs.

It describes how to enable IEEE 802.1X and AAA on a port, and how to enable IEEE 802.1X RADIUS accounting.

Configure RADIUS Authentication Servers

Authentication services for IEEE 802.1X and IEEE 802.11i are provided by RADIUS authentication servers. You configure the RADIUS servers to use for 802.1X and 802.11i authentication on a system-wide basis:

```
vEdge(config)# system radius
vEdge(config-radius)# server ip-address
```

Specify the IP address of the RADIUS server. You can configure one or two RADIUS servers to perform 802.1X and 802.11i authentication. (Note that for AAA authentication, you can configure up to eight RADIUS servers.)

For each RADIUS server, you can configure a number of optional parameters.

You can configure the VPN through which the RADIUS server is reachable and the router interface to use to reach the server:

```
vEdge(config-server)# vpn vpn-id
vEdge(config-server)# source-interface interface-name
```

If you configure two RADIUS servers, they must both be in the same VPN, and they must both be reachable using the same source interface.

You must configure a tag to identify the RADIUS server:

```
vEdge(config-server)# tag tag
```

The tag can be from 4 through 16 characters. You use this tag when configuring the RADIUS servers to use with IEEE 802.1X authentication and with IEEE 802.11i WPA enterprise authentication.

For authentication between the router and the RADIUS server, you can authenticate and encrypt packets sent between the Cisco vEdge device and the RADIUS server, and you can configure a destination port for authentication requests. To authenticate and encrypt packets, configure a key:

```
vEdge(config-server)# secret-key password
```

Enter the password as clear text, which is immediately encrypted, or as an AES 128-bit encrypted key. The key must match the AES encryption key used on the RADIUS server.

By default, UDP port 1812 is used as the destination port on the RADIUS server to use for authentication requests. You can change the port number to a number from 1 through 65535. To disable authentication, set the port number to 0.

```
vEdge(config-server)# auth-port number
```

You can set the priority of a RADIUS server, to choose which one to use first when performing 802.1X authentication:

```
vEdge(config-server)# priority number
```

The priority can be a value from 0 through 7. The server with the lower priority number is given priority. If you do not include this command in the RADIUS server configuration, the priority is determined by the order in which you enter the IP addresses in the **system radius server** command.

By default, accounting is enabled for 802.1X and 802.11i interfaces. Accounting information is sent to UDP port 1813 on the RADIUS server. To change this port:

```
vEdge(config-server)# acct-port number
```

The port number can be from 1 through 65535.

Configure IEEE 802.1X Port Security

To enable basic 802.1X port security on an interface, configure it and at least one RADIUS server to use for 802.1X authentication. The 802.1X interface must be in VPN 0.

```
vEdge(config)# vpn 0  
interface interface-name  
vEdge(config-interface)# dot1x  
vEdge(config-dot1x)# radius-servers tag
```

For 802.1X authentication to work, you must also configure the same interface under an untagged bridge:

```
vEdge(config)# bridge number  
vEdge(config)# interface interface-name
```

The interface name in the **vpn 0 interface** and **bridge interface** commands must be the same. Do not configure a VLAN ID for this bridge so that it remains untagged.

You can enable 802.1X on a maximum of four wired physical interfaces. The interface cannot also be configured as a tunnel interface.

Configure the tags associated with one or two RADIUS servers to use for 802.1X client authentication and accounting. (You configure the tags with the **system radius server tag** command.) If you specify tags for two RADIUS servers, they must both be reachable in the same VPN. If you do not configure a priority value when you configure the RADIUS server with the **system radius server priority** command, the order in which you list the IP addresses is the order in which the RADIUS servers are tried.

Enable RADIUS Accounting

By default, the Cisco vEdge device never sends interim accounting updates to the 802.1X RADIUS accounting server. Accounting updates are sent only when the 802.1X session ends.

To enable the sending of interim accounting updates, configure the interval at which to send the updates:

```
vEdge(config-dot1x) # accounting-interval seconds
```

The time can be from 0 through 7200 seconds.

Enable MAC Authentication Bypass

IEEE 802.1X authentication is accomplished through an exchange of Extensible Authentication Protocol (EAP) packets. After 802.1X-compliant clients respond to the EAP packets, they can be authenticated and granted access to the network. Enabling MAC authentication bypass (MAB) provides a mechanism to allow non-802.1X-compliant clients to be authenticated and granted access to the network.

The Cisco vEdge device determines that a device is non-802.1X-compliant clients when the 802.1X authentication process times out while waiting for an EAPOL response from the client.

To enable MAC authentication bypass for an 802.1X interface on the Cisco vEdge device :

```
vEdge(config) # vpn 0 interface interface-name dot1x
vEdge(config-dot1x) # mac-authentication-bypass
```

With this configuration, the Cisco vEdge device authenticates non-802.1X-compliant clients using the configured RADIUS servers. The RADIUS server must be configured with the MAC addresses of non-802.1X-compliant clients that are allowed to access the network.

To enable MAB on the RADIUS server:

```
vEdge(config-dot1x) # mac-authentication-bypass server
```

To allow authentication to be performed for one or more non-802.1X-compliant clients before performing an authentication check with the RADIUS server, list their MAC addresses in the following command:

```
vEdge(config-dot1x) # mac-authentication-bypass allow mac-addresses
```

You can configure up to eight MAC addresses for MAC authentication bypass. For these devices, the Cisco vEdge device grants immediate network access based on their MAC addresses, and then sends a request to the RADIUS server to authenticate the devices.

Configure VLANs for Authenticated and Unauthenticated Clients

For clients that cannot be authenticated but that you want to provide limited network services to, you create VLANs to handle network access for these clients. You also create VLANs to handle authenticated clients.

You can create the following kinds of VLAN:

- Guest VLAN—Provide limited services to non-802.1X-compliant clients.
- Authentication Reject VLAN—Provide limited services to 802.1X-compliant clients that failed RADIUS authentication. An authentication-reject VLAN is similar to a restricted VLAN.

- Authentication Fail VLAN—Provide network access when RADIUS authentication or the RADIUS server fails. An authentication-fail VLAN is similar to a critical VLAN.
- Default VLAN—Provide network access to 802.1X-compliant clients that are successfully authenticated by the RADIUS server. If you do not configure a default VLAN on the Cisco vEdge device, successfully authenticated clients are placed into VLAN 0, which is the VLAN associated with an untagged bridge.

To configure the VLANs for authenticated and unauthenticated clients, first create the VLAN in a bridging domain, and then create the 802.1X VLANs for the unauthenticated clients by associating the bridging domain VLAN with an 802.1X VLAN.

To create the VLAN, configure a bridging domain to contain the VLAN:

```
vEdge(config)# bridge bridge-id
vEdge(config-bridge)# name text
vEdge(config-bridge)# vlan vlan-id
vEdge(config-bridge)# interface interface-name
vEdge(config-interface)# no shutdown
```

The bridging domain identifier is a number from 1 through 63. A best practice is to have the bridge domain ID be the same as the VLAN number.

The name is optional, but it is recommended that you configure a name that identifies the 802.1X VLAN type, such as Guest-VLAN and Default-VLAN.

The VLAN number can be from 1 through 4095. This is the number that you associate with an 802.1X VLAN.

The interface name is the interface that is running 802.1X.

Then configure the 802.1X VLANs to handle unauthenticated clients.

A guest VLAN provides limited services to non-802.1X-compliant clients, and it can be used to allow clients to download 802.1X client software. An interface running 802.1X assigns clients to a guest VLAN when the interface does not receive a response to EAP request/identity packets that it has sent to the client, or when the client does not send EAPOL packets and MAC authentication bypass is not enabled. To configure a guest VLAN:

```
vEdge(config)# vpn 0 interface interface-name interface dot1x
vEdge(config-dot1x)# guest-vlan vlan-id
```

The VLAN number must match one of the VLANs you configured in a bridging domain. A best practice is to have the VLAN number be the same as the bridge domain ID.

An authentication-reject VLAN provides limited services to 802.1X-compliant clients that have failed RADIUS authentication. To configure an authentication-reject VLAN:

```
vEdge(config-dot1x)# auth-reject-vlan vlan-id
```

The VLAN number must match one of the VLANs you configure in a bridging domain. A best practice is to have the VLAN number be the same as the bridge domain ID.

When the RADIUS authentication server is not available, 802.1X-compliant clients attempting to authenticate are placed in an authentication-fail VLAN if it is configured. If this VLAN is not configured, the authentication request is eventually dropped. To configure the authentication-fail VLAN:

```
vEdge(config-dot1x)# auth-fail-vlan vlan-id
```

The VLAN number must match one of the VLANs you configure in a bridging domain. A best practice is to have the VLAN number be the same as the bridge domain ID.

The following configuration snippet illustrates the interrelationship between the 802.1X configuration and the bridging domain configuration. This snippet shows that the bridging domain numbers match the VLAN

numbers, which is a recommended best practice. Also, the bridging domain name identifies the type of 802.1X VLAN.

```

system
...
radius
server 10.1.15.150
    tag freerad1
    source-interface ge0/0
    secret-key $4$L3rwZmsIic8zj4BgLEFXKw==
    priority 1
exit
server 10.20.24.150
    auth-port 2000
    acct-port 2001
    tag freerad2
    source-interface ge0/4
    secret-key $4$L3rwZmsIic8zj4BgLEFXKw==
    priority 2
exit
!
!
bridge 1
name Untagged_bridge
interface ge0/5
    no native-vlan
    no shutdown
!
!
bridge 10
name Authorize_VLAN
vlan 10
interface ge0/5
    no native-vlan
    no shutdown
!
!
bridge 20
name Guest_VLAN
vlan 20
interface ge0/5
    no native-vlan
    no shutdown
!
!
bridge 30
name Critical_VLAN
vlan 30
interface ge0/5
    no native-vlan
    no shutdown
!
!
bridge 40
name Restricted_VLAN
vlan 40
interface ge0/5
    no native-vlan
    no shutdown
!
!
vpn 0
interface ge0/0
    ip address 10.1.15.15/24

```



```
tunnel-interface
 encapsulation ipsec
 ...
 !
no shutdown
!
interface ge0/1
 ip address 60.0.1.16/24
 no shutdown
!
interface ge0/2
 ip address 10.1.19.15/24
 no shutdown
!
interface ge0/4
 ip address 10.20.24.15/24
 no shutdown
!
interface ge0/5
 dot1x
  auth-reject-vlan 40
  auth-fail-vlan 30
  guest-vlan 20
  default-vlan 10
  radius-servers freerad1
!
no shutdown
!
interface ge0/7
 ip address 10.0.100.15/24
 no shutdown
!
!
vpn 1
 interface ge0/2.1
  ip address 10.2.19.15/24
  mtu 1496
  no shutdown
!
 interface irb1
  ip address 56.0.1.15/24
  mac-address 00:00:00:00:aa:01
  no shutdown
  dhcp-server
   address-pool 56.0.1.0/25
   offer-time 600
   lease-time 86400
   admin-state up
   options
    default-gateway 56.0.1.15
  !
!
!
vpn 10
 interface ge0/2.10
  ip address 10.10.19.15/24
  mtu 1496
  no shutdown
!
 interface irb10
  ip address 56.0.10.15/24
  mac-address 00:00:00:00:aa:10
  no shutdown
```

```

dhcp-server
  address-pool 56.0.10.0/25
  offer-time 600
  lease-time 86400
  admin-state up
  options
    default-gateway 56.0.10.15
  !
!
!
!
vpn 20
  interface ge0/2.20
    ip address 10.20.19.15/24
    mtu 1496
    no shutdown
  !
  interface irb20
    ip address 56.0.20.15/24
    mac-address 00:00:00:00:aa:20
    no shutdown
  !
!
vpn 30
  interface ge0/2.30
    ip address 10.30.19.15/24
    mtu 1496
    no shutdown
  !
  interface irb30
    ip address 56.0.30.15/24
    mac-address 00:00:00:00:aa:30
    no shutdown
  !
!
vpn 40
  interface ge0/2.40
    ip address 10.40.19.15/24
    mtu 1496
    no shutdown
  !
  interface irb40
    ip address 56.0.40.15/24
    mac-address 00:00:00:00:aa:40
    no shutdown
  !
!
vpn 512
  interface eth0
    ip dhcp-client
    no shutdown
  !
!

```

Configure Control Direction

To configure how the 802.1X interface handles traffic when the client is unauthorized, set the control direction:

```
vEdge(config-dot1x)# control-direction (in-and-out | in-only)
```

The direction can be one of the following:

- **in-and-out**—The 802.1X interface can both send packets to and receive packets from the authorized client. Bidirectional control is the default behavior.

- **in-only**—The 802.1X interface can send packets to the unauthorized client, but cannot receive packets from that client.

Configure Authentication with Wake on LAN

IEEE 802.1X authentication wake on LAN (WoL) allows dormant clients to be powered up when the Cisco vEdge device receives a type of Ethernet frame called the magic packet. Administrators can use wake on LAN when to connect to systems that have been powered down.

When a client that uses wake on LAN and that attaches through an 802.1X port powers off, the 802.1X port becomes unauthorized. The port can only receive and send EAPOL packets, and wake-on-LAN magic packets cannot reach the client. When the device is powered off, it is not authorized, and the switch port is not opened.

Without wake on LAN, when an 802.1X port is unauthorized, the router's 802.1X interface block traffic other than EAPOL packets coming from unauthorized clients.

When you enable wake on LAN on an 802.1X port, the Cisco vEdge device is able to send magic packets even if the 802.1X port is unauthorized.

To enable wake on LAN on an 802.1X interface, use the following command:

```
vEdge(config)# vpn 0 interface interface-name dot1x
vEdge(config-dot1x)# wake-on-lan
```

Configure 802.1X Host Mode

The host mode of an 802.1X interfaces determines whether the interface grants access to a single client or to multiple clients. Three host modes are available:

- **Single-host mode**—The 802.1X interface grants access only to the first authenticated client. All other clients attempting access are denied and dropped.
- **Multiple-host mode**—A single 802.1X interface grants access to multiple clients. In this mode, only one of the attached clients must be authorized for the interface to grant access to all clients. If the interface becomes unauthorized, the Cisco vEdge device denies network access to all the attached clients.
- **Multiple-authentication mode**—A single 802.1X interface grants access to multiple authenticated clients on data VLANs.

To configure the host mode of the 802.1X interface, use the following command:

```
vEdge(config)# vpn 0 interface interface-name dot1x
vEdge(config-dot1x)# host-mode (multi-auth | multi-host | single-host)
```

Set the Timeout for Inactive Clients

By default, when a client has been inactive on the network for 1 hour, its authentication is revoked, and the client is timed out. To change the timeout interval, use the following command:

```
vEdge(config)# vpn 0 interface interface-name dot1x
vEdge(config-dot1x)# timeout inactivity minutes
```

The timeout interval can be from 0 through 1440 minutes (24 hours).

Enable Periodic Client Reauthentication

By default, once a client session is authenticated, that session remains functional indefinitely. To enable the periodic reauthentication of 802.1X clients, configure the number of minutes between reauthentication attempts:

```
vEdge(config)# vpn 0 interface interface-name dot1x
vEdge(config-dot1x)# reauthentication minutes
```

The time can be from 0 through 1440 minutes (24 hours)

Configure Dynamic Authorization Service for RADIUS Change of Authorization

Dynamic authorization service (DAS) allows an 802.1X interface on a Cisco vEdge device to accept change of authorization (CoA) requests from a RADIUS or other authentication server and to act on the requests. The Cisco Catalyst SD-WAN implementation of DAS supports disconnect packets, which immediately terminate user sessions, and reauthentication CoA requests, which modify session authorization attributes.

DAS, defined in RFC 5176, is an extension to RADIUS that allows the RADIUS server to dynamically change 802.1X session information without requiring the Cisco vEdge device to initiate the change request. When you enable DAS on the Cisco vEdge device, the router opens a socket to listen for CoA requests from the RADIUS server. If the network administrator of a RADIUS server modifies the authentication of an 802.1X client, the RADIUS server sends a CoA request to inform the router about the change of authorization. When the router receives the CoA request, it processes the requested change.

To enable DAS for an 802.1X interface, you configure information about the RADIUS server from which the interface can accept CoA requests. In the context of configuring DAS, the Cisco vEdge device is the server and the RADIUS server (or other authentication server) is the client.

To configure the RADIUS server from which to accept CoA requests, configure the server's IP address and the password that the RADIUS server uses to access the router's 802.1X interface:

```
vEdge(config)# vpn 0 interface interface-name dot1x
vEdge(config-dot1x)# das
vEdge(config-das)# client ip-address
vEdge(config-das)# secret-key password
```

You can configure the VPN through which the RADIUS server is reachable:

```
vEdge(config-das)# vpn vpn-id
```

By default, the 802.1X interface uses UDP port 3799 to listen for CoA request from the RADIUS server. You can change the port number:

```
vEdge(config-das)# port port-number
```

The port number can be a value from 1 through 65535. If you configure DAS on multiple 802.1X interfaces on a Cisco vEdge device, you must configure each interface to use a different UDP port.

By default, the CoA requests that the Cisco vEdge device receives from the DAS client are all honored, regardless of when the router receives them. To have the router handle CoA within a specified time, you require that the DAS client timestamp all CoA requests:

```
vEdge(config-das)# require-timestamp
```

With this configuration, the Cisco vEdge device processes only CoA requests that include an event timestamp. Non-timestamped CoA requests are dropped immediately.

When timestamping is configured, both the Cisco vEdge device and the RADIUS server check that the timestamp in the CoA request is current and within a specific time window. The default time window is 300 seconds (5 minutes). This behavior means that if the DAS timestamps a CoA at 15:00 and the router receives it at 15:04, the router honors the request. However, if the router receives the request at 15:10, the router drops the CoA request. You can change the time window to a time from 0 through 1000 seconds:

```
vEdge(config-das)# time-window seconds
```

Configure RADIUS Authentication and Accounting Attributes

For IEEE 802.1X authentication and accounting, the Cisco vEdge device, acting as a network access server (NAS), sends RADIUS attribute–value (AV) pairs to the RADIUS server. These AV pairs are defined in RFC 2865, RADIUS, RFC 2866, RADIUS Accounting, and RFC 2869, RADIUS Extensions. The AV pairs are placed in the Attributes field of the RADIUS packet.

By default, when you enable IEEE 802.1X port security, the following authentication attributes are included in messages sent to the RADIUS server:

Attribute Number	Attribute Name	Description
1	User-Name	Name of the user to be authenticated.
5	NAS-Port	Physical port number on the Cisco vEdge device that is authenticating the user.
12	Framed-MTU	Maximum MTU configured for the user.
30	Called-Station-Id	Phone number that the user called, using dialed number identification (DNIS) or similar technology used to access the RADIUS server.
31	Calling-Station-Id	Phone number that the call came in to the server, using automatic number identification (ANI) or similar technology.
44	Acct-Session-Id	Unique session identifier.
61	NAS-Port-Type	Type of physical port on the Cisco vEdge device that is authenticating the user.
77	Connect-Info	Nature of the user's connection.
79	EAP-Message	Encapsulate Extended Access Protocol (EAP) packets, to allow the Cisco vEdge device to authenticate dial-in users via EAP without having to run EAP.
80	Message-Authenticator	Sign RADIUS Access-Requests to prevent these requests from being spoofed by ARAP, CHAP, or EAP.

When you enable RADIUS accounting, the following accounting attributes are included, by default, in messages sent to the RADIUS server:

Attribute Number	Attribute Name	Description
1	User-Name	Name of the user to be authenticated.
5	NAS-Port	Physical port number on the Cisco vEdge device that is authenticating the user.
30	Called-Station-Id	Phone number that the user called, using dialed number identification (DNIS) or similar technology used to access the RADIUS server.

Attribute Number	Attribute Name	Description
31	Calling-Station-Id	Phone number that the call came in to the server, using automatic number identification (ANI) or similar technology.
40	Acct-Status-Type	Mark the beginning and end of an accounting request.
44	Acct-Session-Id	Unique accounting identifier used to match the start and stop records in a log file.
45	Acct-Authentic	How the user was authenticated.
61	NAS-Port-Type	Type of physical port on the Cisco vEdge device that is authenticating the user.
77	Connect-Info	Nature of the user's connection.

Several configuration commands allow you to add additional attribute information to RADIUS packets.

To include the NAS-IP-Address (attribute 4) in messages sent to the RADIUS server to indicate the IP address of the Cisco vEdge device that is acting as a NAS server:

```
vEdge(config-dot1x) nas-ip-address ip-address
```

To include the NAS-Identifier (attribute 32) in messages sent to the RADIUS server, use the following command:

```
vEdge(config-dot1x) # nas-identifier string
```

The NAS identifier is a unique string from 1 through 255 characters long that identifies the Cisco vEdge device that is acting as a NAS server.

To include a RADIUS authentication or accounting attribute of your choice in messages sent to the RADIUS server, use the following commands:

```
vEdge(config-dot1x) # auth-req-attr attribute-number (integer integer | octet
octet | string string)
vEdge(config-dot1x) # acct-req-attr attribute-number (integer integer | octet
octet | string
string)
```

Specify the desired value of the attribute as an integer, octet value, or string, depending on the attribute. For example, to set the Service-Type attribute to be authenticate-only:

```
vEdge(config-dot1x) # auth-req-attr 6 integer 8
```

Configure IEEE 802.11i Authentication

For Cisco vEdge device that support wireless LANs (WLANs), you can configure the router to support either a 2.4-GHz or 5-GHz radio frequency. Then, you segment the WLAN into multiple broadcast domains, which are called virtual access points, or VAPs. Users who connect to a VAP can be unauthenticated, or you can configure IEEE 802.11i authentication for each VAP.

For information about configuring the WLAN interface itself, see *Configuring WLAN Interfaces*.

To enable user authentication on the WLAN, you create a VAP on the desired radio frequency and then you configure Wi-Fi protected access (WPA) or WPA2 data protection and network access control for the VAP. WPA authenticates individual users on the WLAN using a username and password. WPA uses the Temporal

Key Integrity Protocol (TKIP), which is based on the RC4 cipher. WPA2 implements the NIST FIPS 140-2–compliant AES encryption algorithm along with IEEE 802.1X-based authentication, to enhance user access security over WPA. WPA2 uses the Counter Mode Cipher Block Chaining Message Authentication Code Protocol (CCMP), which is based on the AES cipher. Authentication is done either using preshared keys or through RADIUS authentication.

To enable personal authentication, which requires users to enter a password to connect to the WLAN, configure the authentication and password:

```
vEdge(config)# wlan frequency
vEdge(config-wlan)# interface vap number
vEdge(config-vap)# no shutdown
vEdge(config-vap)# data-security (wpa-personal | wpa/wpa2-personal | wpa2-personal)
vEdge(config-vap)# wpa-personal-key password
```

For the security, configure either WPA, WPA2, or both (WPA/WPA2). Enter the password either as clear text or an AES-encrypted key.

For each VAP, you can customize the security mode to control wireless client access.

To enable enterprise WPA security, configure the authentication and the RADIUS server to perform the authentication:

```
vEdge(config-vap)# data-security (wpa-enterprise | wpa/wpa2-enterprise | wpa2-enterprise)
vEdge(config-vap)# radius-servers tag
```

For the security, configure either WPA, WPA2, or both (WPA/WPA2). Enter the password either as clear text or an AES-encrypted key.

In the **radius-servers** command, enter the tags associated with one or two RADIUS servers to use for 802.11i authentication. (You configure the tags with the **system radius server tag** command.) If you specify tags for two RADIUS servers, they must both be reachable in the same VPN. If you do not configure a priority value when you configure the RADIUS server with the **system radius server priority** command, the order in which you list the IP addresses is the order in which the RADIUS servers are tried.

By default, management frames sent on the WLAN are not encrypted. For each VAP, you can configure the encryption to be optional or required:

```
vEdge(config-vap)# mgmt-security (none | optional | required)
```




CHAPTER 5

Role-Based Access Control (Cisco IOS XE Catalyst SD-WAN Release 17.12.x and Earlier)

Table 56: Feature History

Feature Name	Release Information	Description
Role-Based Access Control By Resource Group	Cisco SD-WAN Release 20.5.1 Cisco vManage Release 20.5.1	This feature introduces role-based access control (RBAC) based on sites or resource groups. It is a method of authorizing system access for users based on a combination of user groups and resource groups. For large Cisco Catalyst SD-WAN deployments across multiple geographical locations, this feature helps you to split the network administration among different regional administrators.
RBAC for Policies	Cisco vManage Release 20.6.1 Cisco SD-WAN Release 20.6.1	This feature allows you to create users and user groups with required read and write permissions for Cisco SD-WAN Manager policies. RBAC for policies provides users with the access to all the details of policies to help maximize the operational efficiency. It makes it easier to meet configuration requirements and guarantees that authorized users on the system are only given access to what they need.
Co-Management: Granular Role-Based Access Control for Feature Templates	Cisco vManage Release 20.7.1	This feature introduces greater granularity in assigning RBAC permissions for template use. This enables you to give a tenant self-management of network configuration tasks. Network administrators and managed service providers can use this feature to assign permissions to their end customers.

Feature Name	Release Information	Description
Co-Management: Improved Granular Configuration Task Permissions	Cisco vManage Release 20.9.1	To enable a user to self-manage specific configuration tasks, you can assign the user permissions to perform specific configuration tasks while excluding other tasks. This feature introduces numerous new permission options, enabling fine granularity in determining which configuration task permissions to provide to a user. .
RBAC for Security Operations and Network Operations Default User Groups	Cisco vManage Release 20.9.1	This feature provides the following default user groups: • network_operations user group for non-security policies • security_operations user group for security policies RBAC for policies allows you to create users and user groups with the required read and write permissions for security and non-security policies. Users can perform configuration and monitoring actions only for the authorized policy type.
Co-Management: Improved Granular Configuration for Resource group features	Cisco vManage Release 20.11.1	To enable a user to self-manage specific configuration tasks, you can assign the user permissions to perform specific configuration tasks while excluding other tasks. This feature introduces new permission options for the following configuration groups and feature profiles. • AppQoS under other feature profile • GPS under transport feature profile • Cisco VPN Interface GRE under WAN/LAN profile. • Cisco VPN Interface IPsec under WAN profile. • Cisco Multicast under LAN profile. • UCSE under other feature profile. • IPv4 Tracker and Tracker Group under transport and service feature profiles. • IPv6 DIA Tracker and Tracker Group, under transport feature profile.

Feature Name	Release Information	Description
Assigning Roles Locally for SSO-Authenticated Users	Cisco vManage Release 20.11.1	If you are using an identity provider, such as Okta, for security assertion markup language (SAML)-based single sign-on (SSO), then in most use cases, you define user roles through the identity provider. This feature enables you to assign user groups locally in Cisco SD-WAN Manager, in case no roles are defined for the user by the identity provider.

- [Information About RBAC, on page 123](#)
- [Restrictions for RBAC, on page 137](#)
- [Use Cases for RBAC, on page 137](#)
- [Configure RBAC, on page 137](#)
- [Configure RBAC Using the CLI, on page 168](#)
- [Verify RBAC, on page 170](#)
- [Monitor RBAC, on page 170](#)

Information About RBAC

Role-Based Access Control by VPN

Role-based access control (RBAC) is the process of restricting user access to network configurations and resources. In RBAC, users are assigned roles depending on the resources they need access to. The RBAC by VPN feature helps you to manage and control access to your network based on the VPNs. It involves setting permissions and privileges to enable access to authorized users.

RBAC by VPN

Role-based access by VPN allows a network administrator to define VPN groups with one or more network segments. The network administrator can associate a user with a VPN group that restricts user access to devices in the network and features of Cisco SD-WAN Manager.

RBAC by VPN provides the following restricted access to users configured with a VPN group:

- Access to VPN Dashboard
- Monitor devices, network, and application status via VPN dashboard
- VPN dashboard information restricted to devices with segments in the VPN group
- Monitor option restricted to devices with segments in the VPN group
- Interface monitoring on each device restricted to interfaces of segments in the VPN group

VPN Dashboard Overview

Users configured with VPN group can access only the VPN Dashboard, and it is read-only access. User with Admin access can create the VPN groups and has access to both Admin Dashboard and VPN Dashboard(s). Admin user can access these dashboards by choosing **Dashboard** from the Cisco SD-WAN Manager menu.

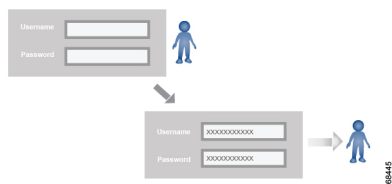
Role-Based Access with AAA

The Cisco Catalyst SD-WAN AAA software implements role-based access to control the authorization permissions for users on Cisco vEdge devices. Role-based access consists of three components:

- Users are those who are allowed to log in to a Cisco vEdge device.
- User groups are collections of users.
- Privileges are associated with each group. They define the commands that the group's users are authorized to issue.

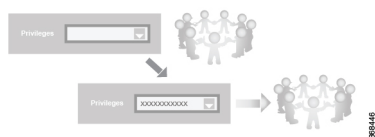
Users and User Groups

All users who are permitted to perform operations on a Cisco vEdge device must have a login account. For the login account, you configure a username and a password on the device itself. These allow the user to log in to that device. A username and password must be configured on each device that a user is allowed to access.

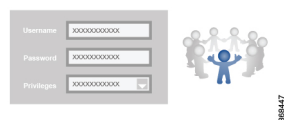


The Cisco Catalyst SD-WAN software provides one standard username, **admin**, which is a user who has full administrative privileges, similar to a UNIX superuser. By default, the **admin** username password is **admin**. You cannot delete or modify this username, but you can and should change the default password.

User groups pool together users who have common roles, or privileges, on the Cisco vEdge device. As part of configuring the login account information, you specify which user group or groups that user is a member of. You do not need to specify a group for the **admin** user, because this user is automatically in the user group **netadmin** and is permitted to perform all operations on the Cisco vEdge device.



The user group itself is where you configure the privileges associated with that group. These privileges correspond to the specific commands that the user is permitted to execute, effectively defining the role-based access to the Cisco Catalyst SD-WAN software elements.



The Cisco Catalyst SD-WAN software provides the following standard user groups:

- **basic**: The basic group is a configurable group and can be used for any users and privilege levels. This group is designed to include users who have permission to both view and modify information on the device.
- **operator**: The operator group is also a configurable group and can be used for any users and privilege levels. This group is designed to include users who have permission only to view information.
- **netadmin**: The netadmin group is a non-configurable group. By default, this group includes the **admin** user. You can add other users to this group. Users in this group are permitted to perform all operations on the device.
- Minimum supported release: Cisco vManage Release 20.9.1

network_operations: The **network_operations** group is a non-configurable group. Users in this group can perform all non-security-policy operations on the device and only view security policy information. For example, users can create or modify template configurations, manage disaster recovery, and create non-security policies such as application aware routing policy or CFlowD policy.

- Minimum supported release: Cisco vManage Release 20.9.1

security_operations: The **security_operations** group is a non-configurable group. Users in this group can perform all security operations on the device and only view non-security-policy information. For example, users can manage umbrella keys, licensing, IPS signatures auto update, TLS/SSL proxy settings, and so on.

Users of the **network_operations** group are authorized to apply policies to a device, revoke applied policies, and edit device templates. Users of the **security_operations** group require **network_operations** users to intervene on day-0 to deploy security policy on a device and on day-N to remove a deployed security policy. However, after a security policy is deployed on a device, **security_operations** users can modify the security policy without needing the **network_operations** users to intervene.

**Note**

Only admin users can view running and local configuration. Users associated with predefined operator user group do not have access to the running and local configurations. The predefined user group operator has only read access for the template configuration. If you need only a subset of admin user privileges, then you need to create a new user group with the selected features from the features list with both read and write access and associate the group with the custom user.

Privileges for Role-Based Access

Role-based access privileges are arranged into five categories, which are called *tasks*:

- Interface—Privileges for controlling the interfaces on the Cisco vEdge device.
- Policy—Privileges for controlling control plane policy, OMP, and data plane policy.
- Routing—Privileges for controlling the routing protocols, including BFD, BGP, OMP, and OSPF.
- Security—Privileges for controlling the security of the device, including installing software and certificates. Only users belonging to the **netadmin** group can install software on the system.
- System—General systemwide privileges.

The tables in the following sections detail the AAA authorization rules for users and user groups. These authorization rules apply to commands issued from the CLI and to those issued from Netconf.

User Authorization Rules for Operational Commands

The user authorization rules for operational commands are based simply on the username. Any user who is allowed to log in to the Cisco vEdge device can execute most operational commands. However, only the **admin** user can issue commands that affect the fundamental operation of the device, such as installing and upgrading the software and shutting down the device.

Note that any user can issue the **config** command to enter configuration mode, and once in configuration mode, they are allowed to issue any general configuration command. Also, any user is allowed to configure their password by issuing the **system aaa user self password password** command and then committing that configuration change. For the actual commands that configure device operation, authorization is defined according to user group membership. See User Group Authorization Rules for Configuration Commands.

The following tables lists the AAA authorization rules for general CLI commands. All the commands are operational commands except as noted. Also, some commands available to the "admin" user are available only if that user is in the "netadmin" user group.

CLI Command	Any User	Admin User
clear history	X	X
commit confirm	X	X
complete-on-space	X	X
config	X	X
exit	X	X
file	X	X
help	X	X
[no] history	X	X
idle-timeout	X	X
job	X	X
logout	—	X (users in netadmin group only)
monitor	X	X
nslookup	X	X
paginate	X	X
ping	X (users in netadmin group only)	X (users in netadmin group only)
poweroff	—	X(users in netadmin group only)
prompt1	X	X

CLI Command	Any User	Admin User
prompt2	X	X
quit	X	X
reboot	—	X (users in netadmin group only)
request aaa request admin-tech request firmware request interface-reset request nms request reset request software	—	X (users in netadmin group only)
request execute request download request upload	X	X
request (everything else)	—	X
rollback (configuration mode command)	—	X (users in netadmin group only)
screen-length	X	X
screen-width	X	X
show cli	X	X
show configuration commit list	X	X
show history	X	X
show jobs	X	X
show parser dump	X	X
show running-config	X	X
show users	X	X
system aaa user <i>self</i> password <i>password</i> (configuration mode command) (Note: A user cannot delete themselves)		
tcpdump	X	X
timestamp	X	X
tools ip-route	X	X
tools netstat	X	X
tools nping	X	X
traceroute	X	X

CLI Command	Any User	Admin User
vshell	X (The availability of vshell command is unavailable to all users that are not in netadmin group in Cisco vManage Release 20.9.5.)	X (The vshell AAA authorized access is limited only to users that are in netadmin group.)

User Group Authorization Rules for Operational Commands

The following table lists the user group authorization roles for operational commands.

Operational Command	Interface	Policy	Routing	Security	System
clear app		X			
clear app-route		X			
clear arp	X				
clear bfd			X		X
clear bgp			X		X
clear bridge	X				
clear cellular	X				
clear control				X	
clear crash					X
clear dhcp					X
clear dns					X
clear igmp			X		
clear installed-certificates				X	
clear interface	X				
clear ip			X		
clear notification					X
clear omp			X		
clear orchestrator				X	
clear ospf			X		
clear pim			X		

Operational Command	Interface	Policy	Routing	Security	System
clear policy		X			
clear pppoe	X				
clear system					X
clear tunnel				X	
clear wlan	X				
clear ztp				X	X
clock					X
debug bgp			X		
debug cellular	X				
debug cflowd		X			
debug chmgr					X
debug config-mgr					X
debug dhcp-client					X
debug dhcp-helper					X
debug dhcp-server					X
debug fpm		X			
debug ftm					X
debug igmp			X		
debug netconf					X
debug omp			X		
debug ospf			X		
debug pim			X		
debug resolver			X		
debug snmp					X
debug sysmgr					X
debug transport					X
debug ttm					X
debug vdaemon				X	X

Operational Command	Interface	Policy	Routing	Security	System
debug vrrp				X	
debug wlan	X				
request certificate				X	
request control-tunnel				X	
request controller				X	
request controller-upload				X	
request csr				X	
request device				X	
request device-upload				X	
request on-vbond-controller				X	
request port-hop				X	
request root-cert-chain				X	
request security				X	
request vedge				X	
request vedge-upload				X	
request vsmart-upload				X	
show aaa					X
show app		X			
show app-route		X			
show arp	X				
show bfd			X		X
show bgp			X		
show boot-partition					X
show bridge	X				
show cellular	X				
show certificate				X	
show clock					X

Operational Command	Interface	Policy	Routing	Security	System
show control				X	X
show crash					X
show debugs—same as debug commands					
show dhcp					X
show external-nat				X	X
show hardware					X
show igmp			X		
show interface	X				
show ip			X		X
show ipsec				X	
show licenses					X
show logging					X
show multicast			X		
show nms-server					X
show notification					X
show ntp					X
show omp		X	X		X
show orchestrator				X	
show ospf			X		
show pim			X		
show policer		X			
show policy		X			
show ppp	X				
show pppoe	X				
show reboot					X
show security-info				X	
show software					X

Operational Command	Interface	Policy	Routing	Security	System
show system					X
show transport					X
show tunnel				X	
show uptime					X
show users					X
show version					X
show vrrp	X				
show wlan	X				
show ztp				X	

User Group Authorization Rules for Configuration Commands

The following table lists the user group authorization rules for configuration commands.

Configuration Command	Interface	Policy	Routing	Security	System
apply-policy		X			
banner					X
bfd			X		X
bridge	X				
omp		X	X		X
policy		X			
security				X	X
snmp					X
system					X
vpn interface	X				
vpn ip			X		
vpn router			X		
vpn service			X		
vpn (everything else, including creating, deleting, and naming)					X

Configuration Command	Interface	Policy	Routing	Security	System
wlan	X				

RBAC By Resource Group Overview

Minimum supported releases: Cisco IOS XE Catalyst SD-WAN Release 17.5.1a and Cisco vManage Release 20.5.1

RBAC by resource groups is a method of restricting or authorizing system access for users based on user groups and resource groups. A user group defines the privileges of a user in the system and the resource group defines the organizations (domains) to which a user is allowed access. Because users are not directly assigned privileges, management of individual user privileges is simply a matter of assigning the appropriate user and resource groups.

For large Cisco Catalyst SD-WAN deployments across multiple geographical locations, you can split the network administration among different regional administrators.

Based on the user groups and resources groups to which network administrators are assigned, we can broadly classify them as Global Administrators and Regional Administrators. Global administrators have access to resources in every resource group and have full read-write privileges for all the features. Regional Administrators group have full read-write privileges for all the features, but the resources they can access is controlled by the resource groups to which they are assigned.

Global Admin

User accounts in the global resource group have access to all resources. A global admin is responsible for overseeing the entire network, but not involved in the operations of the individual devices on a daily basis. The global admin can assign devices to their corresponding regions, assign the regional admin accounts, manage the controllers, maintain sharable and centralized configurations, and when necessary, operate on the individual devices.

Any user in a single tenant setup with netadmin privileges and also part of global resource group is considered as global admin. Default admin user on Cisco SD-WAN Manager is also a global-admin, and that user can assign more global-admins. Global resource group encompasses all the WAN edges, controllers in the single view.

Global admin can switch to view only a specific resource group and can create templates. Local resource group admins, also called regional admins can clone the global templates and reuse them within their resource groups.

Regional Admin

The regional admins are responsible for day-to-day operations (configuration, monitoring, onboarding, and so on) for devices in their corresponding regions. They should not have access to or visibility into devices outside of their region. The following user groups can be created:

- resource group admin – full read/write access to devices in the corresponding resource group, can troubleshoot, monitor, attach or detach templates for the WAN edges in their group
- resource group operator – read-only access to WAN edges within their resource group
- resource group basic – basic access

Resource group admins can create new templates and attach or detach to the WAN edges in their group. They can also copy global templates and re-use them.

Resource group decides which resources the user has access to. However, the level of access is controlled by the existing user group.

- If user is in **resource_group_a** and user group **resource_group_admin**, they have full read/write access to all resources in **resource_group_a**.
- If user is in **resource_group_a** and user group **resource_group_operator**, they have read only access to all resources in **resource_group_a**.
- If user is in **resource_group_a** and user group **resource_group_basic**, they have read only access to interface and system resources in **resource_group_a**.

Global Resource Group

Global group is a special system pre-defined resource group that has different access control rules.

- Users within this group are considered as global-admins, who can have full access to all resources (devices, templates and policies) in the system and they can manage the resource groups and assign resources and users to groups.
- All other users have read-only access to resources within this group.
- The system default admin account (or tenantadmin account in a multi-tenant setup) is always in this group. This privilege cannot be changed. However, the admin account may add/remove other user accounts to or from this group.

IdP (SSO)-Managed Group

An identity provider (IdP) is a service that stores and verifies user identity. IdPs typically work with single sign-on (SSO) providers to authenticate users. If a user is authenticated with a SSO service of an IdP, the group information is also provided and managed by the IDP. An IdP passes the information about the user, including the user name and all the group names, where the user belongs to. Cisco SD-WAN Manager matches the group names with the group names stored in the database to further distinguish if a particular group name passed from IdP is for user group or resource group or VPN group.

Multi-Tenancy Support

With Cisco Catalyst SD-WAN multitenancy, a service provider can manage multiple customers, called tenants, from Cisco SD-WAN Manager. The tenants share Cisco SD-WAN Manager instances, Cisco SD-WAN Validator, and Cisco SD-WAN Controller. The domain name of the service provider has subdomains for each tenant. Cisco SD-WAN Manager is deployed and configured by the service provider. The provider enables multitenancy and creates a Cisco SD-WAN Manager cluster to serve tenants. Only the provider can access a Cisco SD-WAN Manager instance through the SSH terminal.

Provider has the following features:

- resource group is not applicable as the provider manages only the controllers.
- when provider provisions a new tenant, the default user account for the tenant is tenantadmin.
- other user accounts created by the provider are included in the default global resource group.
- when a provider creates a template for a tenant, the template is included in to the global resource group.

RBAC for Policies Overview

Minimum supported releases: Cisco SD-WAN Release 20.6.1 and Cisco vManage Release 20.6.1

RBAC for policies allows a user or user group to have selective Read and Write (RW) access to Cisco SD-WAN Manager policies. For example,

- A user with RW access for Cflowd policy can only configure Cflowd policy, but cannot configure application-aware routing policy.
- A user with RW access for application aware routing policy can only configure application-aware routing policy, but cannot configure other policies.

This feature is only supported for centralized and localized policies, but not supported for security policies.

Information About Granular RBAC for Templates

Minimum supported release: Cisco vManage Release 20.7.1

When setting user group permissions, you can use the following template permissions to provide an RBAC user with a specific degree of access to different types of templates. This gives you control over the types of device configurations that an RBAC user can apply.

Permission	Description
CLI Add-On Template	Provides access to the CLI add-on feature template.
Device CLI Template	Provides access to the device CLI template.
SIG Template	Provides access to the SIG feature template and SIG credential template.
Other Feature Templates	Provides access to all feature templates except the SIG feature template, SIG credential template, and CLI add-on feature template.
Feature Profile	Provides access to all feature profiles.
Config Group	Provides access to all the configuration groups.

You can specify granular RBAC for each feature profile by expanding it. With the set permissions to the usergroup, ensure that you are able to access required feature profiles from **Templates > Configuration Groups**.

Single-Tenant and Multi-Tenant Scenarios

You can use granular RBAC for feature templates in single-tenant and multi-tenant Cisco SD-WAN Manager scenarios.

You can create user groups to assign specific permissions to a tenant's various teams, enabling teams to manage only specific network services without granting permission to use device CLI templates. It might be undesirable to give a tenant permission to apply device CLI templates, as the device CLI template can override any other template or device configuration.

For example, you can create a user group for a tenant's security operations group, giving them read/write access only to the SIG Template option, which would enable the security operations group to work on security configuration.

Information About Granular Configuration Task Permissions

From Cisco vManage Release 20.9.1, numerous user permission options are available, providing you fine granularity when assigning a user with permissions to manage specific configuration tasks related to configuration groups and feature profiles.

Information About Assigning Roles Locally to a User Defined by an Identity Provider

Minimum release: Cisco vManage Release 20.11.1

When you define users in an identity provider, such as Okta, for SAML SSO, one attribute that you can define for each user is the role.

When a user logs in to a Cisco SD-WAN Manager instance, Cisco SD-WAN Manager retrieves information about the user from the identity provider, including the user's role or roles. The roles defined in the identity provider map to user group permissions in Cisco SD-WAN Manager. Based on the roles of the user, Cisco SD-WAN Manager provides the user with the permissions defined by the corresponding user group.

You can assign roles locally (not depending on the identity provider) for a user profile that does not have a role defined in the identity provider.

If you have defined roles for a user through the identity provider and have also assigned user groups locally for the same user, the roles defined through the identity provider take priority.

The following table summarizes the ways to provide a user with specific permissions:

Using or Not Using an Identity Provider for SAML SSO	Roles Defined in the Identity Provider	How User Permissions Are Defined
Not using an identity provider	Not applicable	In Cisco SD-WAN Manager, assign a user to one or more user groups locally. This provides the user with the corresponding user group permissions.
Using an identity provider	Identity provider has one or more roles defined for the user.	Define roles for the user through the identity provider. Cisco SD-WAN Manager provides the user with the user group permissions corresponding to the roles.
	Identity provider does not have a role defined for the user.	Use the Remote User option when adding a user (Administration > Manage Users > Add User). See Add a User, on page 161 . In Cisco SD-WAN Manager, assign a user to one or more user groups locally. This provides the user with the corresponding user group permissions.

Benefits of RBAC

Benefits of Granular RBAC for Feature Templates

Minimum supported release: Cisco vManage Release 20.7.1

The permissions that you add for co-management are useful for providing detailed control over access to network configuration. They are useful when using Cisco Catalyst SD-WAN with tenants, enabling you to provide a tenant access to specific types of templates. This enables you to give the tenant self-management of network configuration tasks within the tenant's VPN.

For information about the permissions added for co-management, see [Information About Granular RBAC for Templates, on page 135](#).

Restrictions for RBAC

Restrictions for Granular RBAC for Feature Templates

Minimum supported release: Cisco vManage Release 20.7.1

- To use any of the template restriction options that are provided for RBAC for co-management, provide permissions for the **Template Configuration** option. If a specific user role does not have any permissions assigned in the **Template Configuration** option, the **Templates** menu does not appear for the user in Cisco SD-WAN Manager. See [Manage Users](#).
- To enable an RBAC user to apply templates to devices, provide **Write** permission to the **Template Deploy** option.

Use Cases for RBAC

Use Cases for Assigning Roles Locally to a User Defined by an Identity Provider

Minimum release: Cisco vManage Release 20.11.1

An organization uses the identity provider, Okta, to authenticate users logging in to Cisco SD-WAN Manager.

A user defined through the identity provider has not been assigned any roles. A network administrator with access to Cisco SD-WAN Manager, but no access to the identity provider, can locally assign the user to a specific user group to provide the user with specific permissions.

Configure RBAC

Manage Users

From the Cisco SD-WAN Manager menu, choose **Administration > Manage Users** to add, edit, view, or delete users and user groups.

Please note the following:

- Only a user logged in as the **admin** user or a user who has Manage Users write permission can add, edit, or delete users and user groups from Cisco SD-WAN Manager.

- Each user group can have read or write permission for the features listed in this section. Write permission includes Read permission.
- All user groups, regardless of the read or write permissions selected, can view the information displayed in the Cisco SD-WAN Manager Dashboard.

Table 57: User Group Permissions for Different Device Types

Permissions	See This Section
User group permissions related to Cisco IOS XE Catalyst SD-WAN device configuration.	User Group Permissions: Cisco IOS XE Catalyst SD-WAN Devices
User group permissions related to Cisco Catalyst Wireless Gateway device configuration.	User Group Permissions: Cisco Catalyst Wireless Gateway Devices

User Group Permissions: Cisco IOS XE Catalyst SD-WAN device

Table 58: User Group Permissions: Cisco IOS XE Catalyst SD-WAN devices

Feature	Read Permission	Write Permission
Alarms	Set alarm filters and view the alarms generated on the devices on the Monitor > Logs > Alarms page. Cisco vManage Release 20.6.x and earlier: Set alarm filters and view the alarms generated on the devices on the Monitor > Alarms page.	No additional permissions.
Audit Log	Set audit log filters and view a log of all the activities on the devices on the Monitor > Logs > Alarms page and the Monitor > Logs > Audit Log page. Cisco vManage Release 20.6.x and earlier: Set audit log filters and view a log of all the activities on the devices on the Monitor > Alarms page and the Monitor > Audit Log page.	No additional permissions.

Feature	Read Permission	Write Permission
Certificates	View a list of the devices in the overlay network under Configuration > Certificates > WAN Edge List. View a certificate signing request (CSR) and certificate on the Configuration > Certificates > Controllers window. Note Starting from Cisco IOS XE Catalyst SD-WAN Release 17.13.1a, the Controllers tab is renamed as the Control Components tab to stay consistent with Cisco Catalyst SD-WAN rebranding.	Validate and invalidate a device, stage a device, and send the serial number of valid controller devices to the Cisco Catalyst SD-WAN Validator on the Configuration > Certificates > WAN Edge List window. Generate a CSR, install a signed certificate, reset the RSA key pair, and invalidate a controller device on the Configuration > Certificates > Controllers window. Note Starting from Cisco IOS XE Catalyst SD-WAN Release 17.13.1a, the Controllers tab is renamed as the Control Components tab to stay consistent with Cisco Catalyst SD-WAN rebranding.
CLI Add-On Template (Minimum supported release: Cisco vManage Release 20.7.1)	View the CLI add-on feature template on the Configuration > Templates window. Note This operation requires read permission for Template Configuration.	Create, edit, delete, and copy a CLI add-on feature template on the Configuration > Templates window. Note These operations require write permission for Template Configuration. Note For information about this option, see Information About Granular RBAC for Feature Templates
Cloud OnRamp	View the cloud applications on the Configuration > Cloud OnRamp for SaaS and Configuration > Cloud OnRamp for IaaS window.	No additional permissions.

Feature	Read Permission	Write Permission
Cluster	View information about the services running on Cisco SD-WAN Manager, a list of devices connected to a Cisco SD-WAN Manager server, and the services that are available and running on all the Cisco SD-WAN Manager servers in the cluster on the Administration > Cluster Management window.	Change the IP address of the current Cisco SD-WAN Manager, add a Cisco SD-WAN Manager server to the cluster, configure the statistics database, edit, and remove a Cisco SD-WAN Manager server from the cluster on the Administration > Cluster Management window.
Colocation	View the cloud applications on the Configuration > Cloud OnRamp for Colocation window.	No additional permissions.
Config Group > Device > Deploy (Minimum supported release: Cisco vManage Release 20.9.1)	This permission does not provide any functionality.	Deploy a configuration onto Cisco IOS XE Catalyst SD-WAN devices. Note To edit an existing feature configuration requires write permission for Template Configuration . For more details on deploying devices, see Deploy Devices .
Device CLI Template (Minimum supported release: Cisco vManage Release 20.7.1)	View the device CLI template on the Configuration > Templates window. Note This operation requires read permission for Template Configuration .	Create, edit, delete, and copy a device CLI template on the Configuration > Templates window. Note These operations require write permission for Template Configuration . Note For information about this option, see Information About Granular RBAC for Feature Templates

Feature	Read Permission	Write Permission
Device Inventory	View the running and local configuration of devices, a log of template activities, and the status of attaching configuration templates to devices on the Configuration > Devices > WAN Edge List window. View the running and local configuration of the devices and the status of attaching configuration templates to controller devices on the Configuration > Devices > Controllers window. Note Starting from Cisco IOS XE Catalyst SD-WAN Release 17.13.1a, the Controllers tab is renamed as the Control Components tab to stay consistent with Cisco Catalyst SD-WAN rebranding.	Upload a device's authorized serial number file to Cisco SD-WAN Manager, toggle a device from Cisco SD-WAN Manager configuration mode to CLI mode, copy a device configuration, and delete the device from the network on the Configuration > Devices > WAN Edge List window. Add and delete controller devices from the overlay network, and edit the IP address and login credentials of a controller device on the Configuration > Devices > Controllers window. Note Starting from Cisco IOS XE Catalyst SD-WAN Release 17.13.1a, the Controllers tab is renamed as the Control Components tab to stay consistent with Cisco Catalyst SD-WAN rebranding.

Feature	Read Permission	Write Permission
Device Monitoring	View the geographic location of the devices on the Monitor > Geography window. View events that have occurred on the devices on the Monitor > Logs > Events page. Cisco vManage Release 20.6.x and earlier: View events that have occurred on the devices on the Monitor > Events page. View a list of devices in the network, along with device status summary, SD-WAN Application Intelligence Engine (SAIE) and Cflowd flow information, transport location (TLOC) loss, latency, and jitter information, control and tunnel connections, system status, and events on the Monitor > Devices page (only when a device is selected). Note In Cisco vManage Release 20.7.x and earlier releases, the SAIE flow is called the deep packet inspection (DPI) flow. Cisco vManage Release 20.6.x and earlier: Device information is available in the Monitor > Network page.	Ping a device, run a traceroute, and analyze the traffic path for an IP packet on the Monitor > Devices page (only when a device is selected). Note These operations require read and write permissions for Device Monitoring.
Device Reboot	View the list of devices on which the reboot operation can be performed on the Maintenance > Device Reboot window.	Reboot one or more devices on the Maintenance > Device Reboot window.
Disaster Recovery	View information about active and standby clusters running on Cisco SD-WAN Manager on the Administration > Disaster Recovery window.	No additional permissions.

Feature	Read Permission	Write Permission
Events	View the geographic location of the devices on the Monitor > Logs > Events page. View the geographic location of the devices on the Monitor > Events page.	Ping a device, run a traceroute, and analyze the traffic path for an IP packet on the Monitor > Logs > Events page (only when a device is selected).
Feature Profile > Other > Thousandeyes (Minimum supported release: Cisco vManage Release 20.9.1)	View the ThousandEyes settings on the Configuration > Templates > (View configuration group) page, in the Other Profile section. Note This operation requires read permission for Template Configuration .	Create, edit, and delete the ThousandEyes settings on the Configuration > Templates > (Add or edit configuration group) page, in the Other Profile section. Note These operations require write permission for Template Configuration .
Feature Profile > Service > Dhcp (Minimum supported release: Cisco vManage Release 20.9.1)	View the DHCP settings on the Configuration > Templates > (View configuration group) page, in the Service Profile section. Note This operation requires read permission for Template Configuration .	Create, edit, and delete the DHCP settings on the Configuration > Templates > (Add or edit configuration group) page, in the Service Profile section. Note These operations require write permission for Template Configuration .
Feature Profile > Service > Lan/Vpn (Minimum supported release: Cisco vManage Release 20.9.1)	View the LAN/VPN settings on the Configuration > Templates > (View configuration group) page, in the Service Profile section. Note This operation requires read permission for Template Configuration .	Create, edit, and delete the LAN/VPN settings on the Configuration > Templates > (Add or edit configuration group) page, in the Service Profile section. Note These operations require write permission for Template Configuration .
Feature Profile > Service > Lan/Vpn/Interface/Ethernet (Minimum supported release: Cisco vManage Release 20.9.1)	View the Ethernet Interface settings on the Configuration > Templates > (View configuration group) page, in the Service Profile section. Note This operation requires read permission for Template Configuration .	Create, edit, and delete the Ethernet Interface settings on the Configuration > Templates > (Add or edit configuration group) page, in the Service Profile section. Note These operations require write permission for Template Configuration .

Feature	Read Permission	Write Permission
Feature Profile > Service > Lan/Vpn/Interface/Svi (Minimum supported release: Cisco vManage Release 20.9.1)	View the SVI Interface settings on the Configuration > Templates > (View configuration group) page, in the Service Profile section. Note This operation requires read permission for Template Configuration .	Create, edit, and delete the SVI Interface settings on the Configuration > Templates > (Add or edit configuration group) page, in the Service Profile section. Note These operations require write permission for Template Configuration .
Feature Profile > Service > Routing/Bgp (Minimum supported release: Cisco vManage Release 20.9.1)	View the Routing/BGP settings on the Configuration > Templates > (View configuration group) page, in the Service Profile section. Note This operation requires read permission for Template Configuration .	Create, edit, and delete the Routing/BGP settings on the Configuration > Templates > (Add or edit configuration group) page, in the Service Profile section. Note These operations require write permission for Template Configuration .
Feature Profile > Service > Routing/Ospf (Minimum supported release: Cisco vManage Release 20.9.1)	View the Routing/OSPF settings on the Configuration > Templates > (View configuration group) page, in the Service Profile section. Note This operation requires read permission for Template Configuration .	Create, edit, and delete the Routing/OSPF settings on the Configuration > Templates > (Add or edit configuration group) page, in the Service Profile section. Note These operations require write permission for Template Configuration .
Feature Profile > Service > Switchport (Minimum supported release: Cisco vManage Release 20.9.1)	View the Switchport settings on the Configuration > Templates > (View configuration group) page, in the Service Profile section. Note This operation requires read permission for Template Configuration .	Create, edit, and delete the Switchport settings on the Configuration > Templates > (Add or edit configuration group) page, in the Service Profile section. Note These operations require write permission for Template Configuration .

Feature	Read Permission	Write Permission
Feature Profile > Service > Wirelesslan (Minimum supported release: Cisco vManage Release 20.9.1)	View the Wireless LAN settings on the Configuration > Templates > (View configuration group) page, in the Service Profile section. Note This operation requires read permission for Template Configuration .	Create, edit, and delete the Wireless LAN settings on the Configuration > Templates > (Add or edit configuration group) page, in the Service Profile section. Note These operations require write permission for Template Configuration .
Feature Profile > System > Interface/Ethernet > Aaa (Minimum supported release: Cisco vManage Release 20.9.1)	View the AAA settings on the Configuration > Templates > (View configuration group) page, in the System Profile section. Note This operation requires read permission for Template Configuration .	Create, edit, and delete the AAA settings on the Configuration > Templates > (Add or edit configuration group) page, in the System Profile section. Note These operations require write permission for Template Configuration .
Feature Profile > System > Interface/Ethernet > Banner (Minimum supported release: Cisco vManage Release 20.9.1)	View the Banner settings on the Configuration > Templates > (View configuration group) page, in the System Profile section. Note This operation requires read permission for Template Configuration .	Create, edit, and delete the Banner settings on the Configuration > Templates > (Add or edit configuration group) page, in the System Profile section. Note These operations require write permission for Template Configuration .
Feature Profile > System > Basic (Minimum supported release: Cisco vManage Release 20.9.1)	View the Basic settings on the Configuration > Templates > (View configuration group) page, in the System Profile section. Note This operation requires read permission for Template Configuration .	Create, edit, and delete the Basic settings on the Configuration > Templates > (Add or edit configuration group) page, in the System Profile section. Note These operations require write permission for Template Configuration .

Feature	Read Permission	Write Permission
Feature Profile > System > Bfd (Minimum supported release: Cisco vManage Release 20.9.1)	View the BFD settings on the Configuration > Templates > (View configuration group) page, in the System Profile section. Note This operation requires read permission for Template Configuration .	Create, edit, and delete the BFD settings on the Configuration > Templates > (Add or edit configuration group) page, in the System Profile section. Note These operations require write permission for Template Configuration .
Feature Profile > System > Global (Minimum supported release: Cisco vManage Release 20.9.1)	View the Global settings on the Configuration > Templates > (View configuration group) page, in the System Profile section. Note This operation requires read permission for Template Configuration .	Create, edit, and delete the Global settings on the Configuration > Templates > (Add or edit configuration group) page, in the System Profile section. Note These operations require write permission for Template Configuration .
Feature Profile > System > Logging (Minimum supported release: Cisco vManage Release 20.9.1)	View the Logging settings on the Configuration > Templates > (View configuration group) page, in the System Profile section. Note This operation requires read permission for Template Configuration .	Create, edit, and delete the Logging settings on the Configuration > Templates > (Add or edit configuration group) page, in the System Profile section. Note These operations require write permission for Template Configuration .
Feature Profile > System > Ntp (Minimum supported release: Cisco vManage Release 20.9.1)	View the NTP settings on the Configuration > Templates > (View configuration group) page, in the System Profile section. Note This operation requires read permission for Template Configuration .	Create, edit, and delete the NTP settings on the Configuration > Templates > (Add or edit configuration group) page, in the System Profile section. Note These operations require write permission for Template Configuration .

Feature	Read Permission	Write Permission
Feature Profile > System > Omp (Minimum supported release: Cisco vManage Release 20.9.1)	View the OMP settings on the Configuration > Templates > (View configuration group) page, in the System Profile section. Note This operation requires read permission for Template Configuration .	Create, edit, and delete the OMP settings on the Configuration > Templates > (Add or edit configuration group) page, in the System Profile section. Note These operations require write permission for Template Configuration .
Feature Profile > System > Snmp (Minimum supported release: Cisco vManage Release 20.9.1)	View the SNMP settings on the Configuration > Templates > (View configuration group) page, in the System Profile section. Note This operation requires read permission for Template Configuration .	Create, edit, and delete the SNMP settings on the Configuration > Templates > (Add or edit configuration group) page, in the System Profile section. Note These operations require write permission for Template Configuration .
Feature Profile > Transport > Cellular Controller (Minimum supported release: Cisco vManage Release 20.9.1)	View the Cellular Controller settings on the Configuration > Templates > (View a configuration group) page, in the Transport & Management Profile section. Note This operation requires read permission for Template Configuration .	Create, edit, and delete the Cellular Controller settings on the Configuration > Templates > (Add or edit a configuration group) page, in the Transport & Management Profile section. Note These operations require write permission for Template Configuration .
Feature Profile > Transport > Cellular Profile (Minimum supported release: Cisco vManage Release 20.9.1)	View the Cellular Profile settings on the Configuration > Templates > (View a configuration group) page, in the Transport & Management Profile section. Note This operation requires read permission for Template Configuration .	Create, edit, and delete the Cellular Profile settings on the Configuration > Templates > (Add or edit a configuration group) page, in the Transport & Management Profile section. Note These operations require write permission for Template Configuration .

Feature	Read Permission	Write Permission
Feature Profile > Transport > Management/Vpn (Minimum supported release: Cisco vManage Release 20.9.1)	View the Management VPN settings on the Configuration > Templates > (View configuration group) page, in the Transport & Management Profile section. Note This operation requires read permission for Template Configuration .	Create, edit, and delete the Management VPN settings on the Configuration > Templates > (Add or edit a configuration group) page, in the Transport & Management Profile section. Note These operations require write permission for Template Configuration .
Feature Profile > Transport > Management/Vpn/Interface/Ethernet (Minimum supported release: Cisco vManage Release 20.9.1)	View the Management Ethernet Interface settings on the Configuration > Templates > (View configuration group) page, in the Transport & Management Profile section. Note This operation requires read permission for Template Configuration .	Create, edit, and delete the Management VPN and Management Internet Interface settings on the Configuration > Templates > (Add or edit a configuration group) page, in the Transport & Management Profile section. Note These operations require write permission for Template Configuration .
Feature Profile > Transport > Routing/Bgp (Minimum supported release: Cisco vManage Release 20.9.1)	View the BGP Routing settings on the Configuration > Templates > (View configuration group) page, in the Transport & Management Profile section. Note This operation requires read permission for Template Configuration .	Create, edit, and delete the BGP Routing settings on the Configuration > Templates > (Add or edit a configuration group) page, in the Transport & Management Profile section. Note These operations require write permission for Template Configuration .
Feature Profile > Transport > Tracker (Minimum supported release: Cisco vManage Release 20.9.1)	View the Tracker settings on the Configuration > Templates > (View configuration group) page, in the Transport & Management Profile section. Note This operation requires read permission for Template Configuration .	Create, edit, and delete the Tracker settings on the Configuration > Templates > (Add or edit a configuration group) page, in the Transport & Management Profile section. Note These operations require write permission for Template Configuration .

Feature	Read Permission	Write Permission
Feature Profile > Transport > Wan/Vpn (Minimum supported release: Cisco vManage Release 20.9.1)	View the Wan/Vpn settings on the Configuration > Templates > (View configuration group) page, in the Transport & Management Profile section. Note This operation requires read permission for Template Configuration .	Create, edit, and delete the Wan/Vpn settings on the Configuration > Templates > (Add or edit a configuration group) page, in the Transport & Management Profile section. Note These operations require write permission for Template Configuration .
Feature Profile > Transport > Wan/Vpn/Interface/Cellular (Minimum supported release: Cisco vManage Release 20.9.1)	View the Wan/Vpn/Interface/Cellular settings on the Configuration > Templates > (View configuration group) page, in the Transport & Management Profile section. Note This operation requires read permission for Template Configuration .	Create, edit, and delete the Wan/Vpn/Interface/Cellular settings on the Configuration > Templates > (Add or edit a configuration group) page, in the Transport & Management Profile section. Note These operations require write permission for Template Configuration .
Feature Profile > Transport > Wan/Vpn/Interface/Ethernet (Minimum supported release: Cisco vManage Release 20.9.1)	View the Wan/Vpn/Interface/Ethernet settings on the Configuration > Templates > (View configuration group) page, in the Transport & Management Profile section. Note This operation requires read permission for Template Configuration .	Create, edit, and delete the Wan/Vpn/Interface/Ethernet settings on the Configuration > Templates > (Add or edit a configuration group) page, in the Transport & Management Profile section. Note These operations require write permission for Template Configuration .
Integration Management	View information about controllers running on Cisco SD-WAN Manager, on the Administration > Integration Management window.	No additional permissions.

Feature	Read Permission	Write Permission
License Management	View license information of devices running on Cisco SD-WAN Manager, on the Administration > License Management window.	On the Administration > License Management page, configure use of a Cisco Smart Account, choose licenses to manage, and synchronize license information between Cisco SD-WAN Manager and the license server.
Interface	View information about the interfaces on a device on the Monitor > Devices > Interface page. Cisco vManage Release 20.6.x and earlier: View information about the interfaces on a device on the Monitor > Network > Interface page	Edit Chart Options to select the type of data to display, and edit the time period for which to display data on the Monitor > Devices > Interface page.
Application Monitoring (Minimum supported release: Cisco Catalyst SD-WAN Manager Release 20.12.1)	View the application health of the devices on the Monitor > Applications window.	View the application health of the devices on the Monitor > Applications window.
Manage Users	View users and user groups on the Administration > Manage Users window.	Add, edit, and delete users and user groups from Cisco SD-WAN Manager, and edit user group privileges on the Administration > Manage Users window.

Feature	Read Permission	Write Permission
Other Feature Templates (Minimum supported release: Cisco vManage Release 20.7.1)	View all feature templates except the SIG feature template, SIG credential template, and CLI add-on feature template on the Configuration > Templates window. Note This operation requires read permission for Template Configuration. Note To check the mutual authentication option, you need read permission for certificates. (Minimum supported release: Cisco Catalyst SD-WAN Manager Release 20.12.1)	Create, edit, delete, and copy all feature templates except the SIG feature template, SIG credential template, and CLI add-on feature template on the Configuration > Templates window. Note These operations require write permission for Template Configuration. Note For information about this option, see Information About Granular RBAC for Feature Templates Note To check the mutual authentication option, you need write permission for certificates. (Minimum supported release: Cisco Catalyst SD-WAN Manager Release 20.12.1)
Policy	View the common policies for all Cisco Catalyst SD-WAN Controllers or devices in the network on the Configuration > Policies window.	Create, edit, and delete the common policies for all Cisco Catalyst SD-WAN Controllers or devices in the network on the Configuration > Policies window.
Policy Configuration	View the list of policies created and details about them on the Configuration > Policies window.	Create, edit, and delete the common policies for all the Cisco Catalyst SD-WAN Controllers and devices in the network on the Configuration > Policies window.
Policy Deploy	View the current status of the Cisco Catalyst SD-WAN Controllers to which a policy is being applied on the Configuration > Policies window.	Activate and deactivate the common policies for all Cisco SD-WAN Manager servers in the network on the Configuration > Policies window.

Feature	Read Permission	Write Permission
RBAC VPN	View the VPN groups and segments based on roles on the Monitor > VPN page. Cisco vManage Release 20.6.x and earlier: View the VPN groups and segments based on roles on the Dashboard > VPN Dashboard page.	Add, edit, and delete VPNs and VPN groups from Cisco SD-WAN Manager, and edit VPN group privileges on the Administration > VPN Groups window.
Routing	View real-time routing information for a device on the Monitor > Devices > Real-Time page. Cisco vManage Release 20.6.x and earlier: View real-time routing information for a device on the Monitor > Network > Real-Time page.	Add command filters to speed up the display of information on the Monitor > Devices > Real-Time page.
Security	View the current status of the Cisco Catalyst SD-WAN Controllers to which a security policy is being applied on the Configuration > Security window.	Activate and deactivate the security policies for all Cisco SD-WAN Manager servers in the network on the Configuration > Security window.
Security Policy Configuration	Activate and deactivate the common policies for all Cisco SD-WAN Manager servers in the network on the Configuration > Security > Add Security Policy window.	Activate and deactivate the security policies for all Cisco SD-WAN Manager servers in the network on the Configuration > Security > Add Security Policy window.
Session Management	View user sessions on the Administration > Manage Users > User Sessions window.	Add, edit, and delete users and user groups from Cisco SD-WAN Manager, and edit user sessions on the Administration > Manage Users > User Sessions window.

Feature	Read Permission	Write Permission
Settings	View the organization name, Cisco Catalyst SD-WAN Validator DNS or IP address, certificate authorization settings, software version enforced on a device, custom banner on the Cisco SD-WAN Manager login page, and the current settings for collecting statistics on the Administration > Settings window.	Edit the organization name, Cisco Catalyst SD-WAN Validator DNS or IP address, certificate authorization settings, software version enforced on a device, custom banner on the Cisco SD-WAN Manager login page, current settings for collecting statistics, generate a certificate signing request (CSR) for a web server certificate, and install a certificate on the Administration > Settings window.
SIG Template (Minimum supported release: Cisco vManage Release 20.7.1)	View the SIG feature template and SIG credential template on the Configuration > Templates window. Note This operation requires read permission for Template Configuration .	Create, edit, delete, and copy a SIG feature template and SIG credential template on the Configuration > Templates window. Note These operations require write permission for Template Configuration . Note For information about this option, see Information About Granular RBAC for Feature Templates
SIG Tunnels (Minimum supported release: Cisco vManage Release 17.12)	View information about the SIG tunnels on the Monitor > Tunnels > SIG Tunnels page.	View information about the SIG tunnels on the Monitor > Tunnels > SIG Tunnels page.
Software Upgrade	View a list of devices, the custom banner on Cisco SD-WAN Manager on which a software upgrade can be performed, and the current software version running on a device on the Maintenance > Software Upgrade window.	Upload new software images on devices, upgrade, activate, and delete a software image on a device, and set a software image to be the default image on devices on the Maintenance > Software Upgrade window.

Feature	Read Permission	Write Permission
System	View system-wide parameters configured using Cisco SD-WAN Manager templates on the Configuration > Templates > Device Templates window. Note In Cisco vManage Release 20.7.x and earlier releases, Device Templates is called Device .	Configure system-wide parameters using Cisco SD-WAN Manager templates on the Configuration > Templates > Device Templates window. Note In Cisco vManage Release 20.7.x and earlier releases, Device Templates is called Device .
Template Configuration	View feature and device templates on the Configuration > Templates window.	Create, edit, delete, and copy a feature or device template on the Configuration > Templates window. Note Beginning with Cisco vManage Release 20.7.1, to create, edit, or delete a template that is already attached to a device, the user requires write permission for the Template Deploy option.
Template Deploy	View the devices attached to a device template on the Configuration > Templates window.	Attach a device to a device template on the Configuration > Templates window.
Tools	Use the admin tech command to collect the system status information for a device on the Tools > Operational Commands window.	Use the admin tech command to collect the system status information for a device, and use the interface reset command to shut down and then restart an interface on a device in a single operation on the Tools > Operational Commands window. Rediscover the network to locate new devices and synchronize them with Cisco SD-WAN Manager on the Tools > Operational Commands window. Establish an SSH session to the devices and issue CLI commands on the Tools > Operational Commands window.
vAnalytics	Launch Cisco SD-WAN Analytics on > vAnalytics window.	No additional permissions.

Feature	Read Permission	Write Permission
Workflows	Launch workflow library from > Workflows window.	No additional permissions.
Config Group > Device > Deploy (Minimum supported release: Cisco vManage Release 20.11.1)	View the devices associated to a configuration group on the Configuration > Templates > Edit Configuration Group > Associated Devices window.	Deploy a configuration onto Cisco IOS XE Catalyst SD-WAN devices. Note To edit an existing feature configuration requires write permission for Template Configuration . For more details on deploying devices, see Deploy Devices .
Feature Profile > Transport > IPv4 Tracker and Tracker Group (Minimum supported release: Cisco vManage Release 20.11.1)	View the IPv4 Tracker and Tracker Group settings on the Configuration > Templates > (View configuration group) page, in the Transport & Management Profile section. Note This operation requires read permission for Template Configuration .	Create, edit, and delete the IPv4 Tracker and Tracker Group settings on the Configuration > Templates > (Add or edit a configuration group) page, in the Transport & Management Profile section. Note These operations require write permission for Template Configuration .
Feature Profile > Transport > IPv6 Tracker and Tracker Group (Minimum supported release: Cisco vManage Release 20.11.1)	View the IPv6 Tracker and Tracker Group settings on the Configuration > Templates > (View configuration group) page, in the Transport & Management Profile section. Note This operation requires read permission for Template Configuration .	Create, edit, and delete the IPv6 Tracker and Tracker Group settings on the Configuration > Templates > (Add or edit a configuration group) page, in the Transport & Management Profile section. Note These operations require write permission for Template Configuration .

Feature	Read Permission	Write Permission
Feature Profile > Transport > Gps (Minimum supported release: Cisco vManage Release 20.11.1)	View the GPS settings on the Configuration > Templates > (View configuration group) page, in the Transport & Management Profile section. Note This operation requires read permission for Template Configuration .	Create, edit, and delete the Gps settings on the Configuration > Templates > (Add or edit a configuration group) page, in the Transport & Management Profile section. Note These operations require write permission for Template Configuration .
Feature Profile > Other > APPQoE (Minimum supported release: Cisco vManage Release 20.11.1)	View the APPQoE settings on the Configuration > Templates > (View configuration group) page, in the Other section. Note This operation requires read permission for Template Configuration .	Create, edit, and delete the APPQoE settings on the Configuration > Templates > (Add or edit a configuration group) page, in the Other section. Note These operations require write permission for Template Configuration .
Feature Profile > Other > UCSE (Minimum supported release: Cisco vManage Release 20.11.1)	View the UCSE settings on the Configuration > Templates > (View configuration group) page, in the Other section. Note This operation requires read permission for Template Configuration .	Create, edit, and delete the UCSE settings on the Configuration > Templates > (Add or edit a configuration group) page, in the Other section. Note These operations require write permission for Template Configuration .
Feature Profile > Wan Profile > Cisco VPN Interface IPSec (Minimum supported release: Cisco vManage Release 20.11.1)	View the Cisco VPN Interface IPSec settings on the Configuration > Templates > (View configuration group) page, in the Wan Profile section. Note This operation requires read permission for Template Configuration .	Create, edit, and delete the Cisco VPN Interface IPSec settings on the Configuration > Templates > (Add or edit a configuration group) page, in the Wan Profile section. Note These operations require write permission for Template Configuration .

Feature	Read Permission	Write Permission
Feature Profile > Wan/Lan Profile > Cisco VPN Interface GRE (Minimum supported release: Cisco vManage Release 20.11.1)	View the Cisco VPN Interface GRE settings on the Configuration > Templates > (View configuration group) page, in the Wan/Lan Profile section. Note This operation requires read permission for Template Configuration .	Create, edit, and delete the Cisco VPN Interface GRE settings on the Configuration > Templates > (Add or edit a configuration group) page, in the Wan/Lan Profile section. Note These operations require write permission for Template Configuration .
Feature Profile > Lan Profile > Cisco Multicast (Minimum supported release: Cisco vManage Release 20.11.1)	View the Cisco Multicast settings on the Configuration > Templates > (View configuration group) page, in the Lan Profile section. Note This operation requires read permission for Template Configuration .	Create, edit, and delete the Cisco Multicast settings on the Configuration > Templates > (Add or edit a configuration group) page, in the Lan Profile section. Note These operations require write permission for Template Configuration .

**Note**

To create Service, System and Transport feature profiles using configuration groups, you need to provide read and write permissions on the following features to access each configuration group.

- **Feature Profile > System**
- **Feature Profile > System > AAA**
- **Feature Profile > System > BFD**
- **Feature Profile > System > Banner**
- **Feature Profile > System > Basic**
- **Feature Profile > System > Logging**
- **Feature Profile > System > NTP**
- **Feature Profile > System > OMP**
- **Feature Profile > System > SNMP**
- **Feature Profile > Service**
- **Feature Profile > Service > BFD**
- **Feature Profile > Service > LAN/VPN**
- **Feature Profile > Service > LAN/VPN/Interface/Ethernet**
- **Feature Profile > Service > Routing/BGP**
- **Feature Profile > Service > Routing/OSPF**
- **Feature Profile > Service > Routing/DHCP**
- **Feature Profile > Service > Routing/Multicast**
- **Feature Profile > Transport**
- **Feature Profile > Transport > Routing/BGP**
- **Feature Profile > Transport > WAN/VPN**
- **Feature Profile > Transport > WAN/VPN/Interface/Ethernet**

For more details on configuring features using Configuration Groups, see [Feature Management](#).

User Group Permissions: Cisco Catalyst Wireless Gateway Devices

Table 59: User Group Permissions: Cisco Catalyst Wireless Gateway Devices

Feature	Read Permission	Write Permission
Feature Profile > Teleworker > Basic (Minimum supported release: Cisco vManage Release 20.9.1)	View the basic settings on the Configuration > Templates > (View mobility configuration group) page, in the Global Profile section. Note This operation requires read permission for Template Configuration .	Configure the basic settings on the Configuration > Templates > (Add or edit mobility configuration group) page, in the Global Profile section. Note This operation requires write permission for Template Configuration .
Feature Profile > Teleworker > Cellular (Minimum supported release: Cisco vManage Release 20.9.1)	View the cellular network settings on the Configuration > Templates > (View mobility configuration group) page, in the Global Profile section. Note This operation requires read permission for Template Configuration .	Configure the cellular network settings on the Configuration > Templates > (Add or edit mobility configuration group) page, in the Global Profile section. Note This operation requires write permission for Template Configuration .
Feature Profile > Teleworker > Ethernet (Minimum supported release: Cisco vManage Release 20.9.1)	View the ethernet settings on the Configuration > Templates > (View mobility configuration group) page, in the Global Profile section. Note This operation requires read permission for Template Configuration .	Configure the ethernet settings on the Configuration > Templates > (Add or edit mobility configuration group) page, in the Global Profile section. Note This operation requires write permission for Template Configuration .
Feature Profile > Teleworker > NetworkProtocol (Minimum supported release: Cisco vManage Release 20.9.1)	View the network protocol settings on the Configuration > Templates > (View mobility configuration group) page, in the Global Profile section. Note This operation requires read permission for Template Configuration .	Configure the network protocol settings on the Configuration > Templates > (Add or edit mobility configuration group) page, in the Global Profile section. Note This operation requires write permission for Template Configuration .

Feature	Read Permission	Write Permission
Feature Profile > Teleworker > SecurityPolicy (Minimum supported release: Cisco vManage Release 20.9.1)	View the security policy settings on the Configuration > Templates > (View mobility configuration group) page, in the Global Profile section. Note This operation requires read permission for Template Configuration .	Configure the security policy settings on the Configuration > Templates > (Add or edit mobility configuration group) page, in the Global Profile section. Note This operation requires write permission for Template Configuration .
Feature Profile > Teleworker > Vpn (Minimum supported release: Cisco vManage Release 20.9.1)	View the VPN settings on the Configuration > Templates > (View mobility configuration group) page, in the Global Profile section. Note This operation requires read permission for Template Configuration .	Configure the VPN settings on the Configuration > Templates > (Add or edit mobility configuration group) page, in the Global Profile section. Note This operation requires write permission for Template Configuration .
Feature Profile > Teleworker > Wifi (Minimum supported release: Cisco vManage Release 20.9.1)	View the Wi-Fi settings on the Configuration > Templates > (View mobility configuration group) page, in the Global Profile section. Note This operation requires read permission for Template Configuration .	Configure the Wi-Fi settings on the Configuration > Templates > (Add or edit mobility configuration group) page, in the Global Profile section. Note This operation requires write permission for Template Configuration .

RBAC User Group in a Multitenant Environment

The following is the list of user group permissions for role-based access control (RBAC) in a multitenant environment:

- R stands for read permission.
- W stands for write permission.

Table 60: RBAC User Group in Multitenant Environment

Feature	Provider Admin	Provider Operator	Tenant Admin	Tenant Operator
Cloud OnRamp	RW	R	RW	R
Colocation	RW	R	RW	R

Feature	Provider Admin	Provider Operator	Tenant Admin	Tenant Operator
RBAC VPN	RW	R	RW	R
Security	RW	R	RW	R
Security Policy Configuration	RW	R	RW	R
vAnalytics	RW	R	RW	R

Add a User

1. From the Cisco SD-WAN Manager menu, choose **Administration > Manage Users**.
2. By default **Users** is selected. The table displays the list of users configured in the device.
3. To edit, delete, or change password for an existing user, click ... and click **Edit**, **Delete**, or **Change Password** respectively.
4. To add a new user, click **Add User**.
5. Add **Full Name**, **Username**, **Password**, and **Confirm Password** details.
6. If you are using an identity provider, such as Okta, for security assertion markup language (SAML)-based single sign-on (SSO), then in most use cases, you define user roles through the identity provider. If no roles are defined for the user through the identity provider, you can enable the **Remote User** option and assign user groups locally in Cisco SD-WAN Manager. Assigning user groups locally provides an alternate method for assigning the user with permissions.

If you enable this option, enter an email address for the user.

If you have defined roles for a user through the identity provider and have also assigned user groups locally for the same user, the roles defined through the identity provider take priority.



Note This option is available from Cisco vManage Release 20.11.1.

7. In the **User Groups** drop-down list, select the user group where you want to add a user.
8. In the **Resource Group** drop-down list, select the resource group.



Note This field is available from Cisco SD-WAN Release 20.5.1.

9. Click **Add**.

Delete a User

If a user no longer needs access to devices, you can delete the user. Deleting a user does not log out the user if the user is logged in.

To delete a user:

1. From the Cisco SD-WAN Manager menu, choose **Administration > Manage Users**.
2. For the user you wish to delete, click ..., and click **Delete**.
3. To confirm the deletion of the user, click **OK**.

Edit User Details

You can update login information for a user, and add or remove a user from a user group. If you edit the details of a user who is logged in, the changes take effect after the user logs out.

To edit user details:

1. From the Cisco SD-WAN Manager menu, choose **Administration > Manage Users**.
2. For the user you wish to edit, click ..., and click **Edit**.
3. Edit the user details.

You can also add or remove the user from user groups.

4. Click **Update**.

Change a User Password

You can update passwords for users, as needed. We recommend that you use strong passwords.

Before You Begin

If you are changing the password for an admin user, detach device templates from all Cisco SD-WAN Manager instances in the cluster before you perform this procedure. You can reattach the device templates after you complete this procedure.

To change a password for a user:

1. From the Cisco SD-WAN Manager menu, choose **Administration > Manage Users**.
2. For the user you wish to change the password, click ... and click **Change Password**.
3. Enter the new password, and then confirm it.



Note Note that the user, if logged in, is logged out.

4. Click **Done**.

Check Users Logged In to a Device Using SSH Sessions

1. From the Cisco SD-WAN Manager menu, choose **Monitor > Devices**.

Cisco vManage Release 20.6.x and earlier: From the Cisco SD-WAN Manager menu, choose **Monitor > Network**.

2. Select the device you want to use under the **Hostname** column.
3. Click **Real Time**.
4. From **Device Options**, choose **AAA users** for Cisco IOS XE Catalyst SD-WAN devices or **Users** for Cisco vEdge devices.

A list of users logged in to this device is displayed.

Check Users Logged In to a Device Using HTTP Sessions

1. From the Cisco SD-WAN Manager menu, choose **Administration > Manage Users**.
2. Click **User Sessions**.

A list of all the active HTTP sessions within Cisco SD-WAN Manager is displayed, including, username, domain, source IP address, and so on.

Manage a User Group

Users are placed in groups, which define the specific configuration and operational commands that the users are authorized to view and modify. A single user can be in one or more groups. Cisco Catalyst SD-WAN software provides standard user groups, and you can create custom user groups, as needed:

- **basic**: Includes users who have permission to view interface and system information.
- **netadmin**: Includes the admin user, by default, who can perform all operations on the Cisco SD-WAN Manager. You can add other users to this group.
- **operator**: Includes users who have permission only to view information.
- Minimum supported release: Cisco vManage Release 20.9.1
- **network_operations**: Includes users who can perform non-security operations on Cisco SD-WAN Manager, such as viewing and modifying non-security policies, attaching and detaching device templates, and monitoring non-security data.
- Minimum supported release: Cisco vManage Release 20.9.1
- **security_operations**: Includes users who can perform security operations on Cisco SD-WAN Manager, such as viewing and modifying security policies, and monitoring security data.

Note: All user groups, regardless of the read or write permissions selected, can view the information displayed on the Cisco SD-WAN Manager Dashboard screen.

Delete a User Group

You can delete a user group when it is no longer needed. For example, you might delete a user group that you created for a specific project when that project ends.

1. From the Cisco SD-WAN Manager menu, choose **Administration > Manage Users**.
2. Click **User Groups**.
3. Click the name of the user group you wish to delete.



Note You cannot delete any of the default user groups—basic, netadmin, operator, network_operations, and security_operations.

4. Click **Trash** icon.
5. To confirm the deletion of the user group, click **OK**.

Edit User Group Privileges

You can edit group privileges for an existing user group. This procedure lets you change configured feature read and write permissions for the user group needed.

1. From the Cisco SD-WAN Manager menu, choose **Administration > Manage Users**.
2. Click **User Groups**.
3. Select the name of the user group whose privileges you wish to edit.



Note You cannot edit privileges for the any of the default user groups—basic, netadmin, operator, network_operations, and security_operations.

4. Click **Edit**, and edit privileges as needed.
5. Click **Save**.

If an **admin** user changes the privileges of a user by changing their group, and if that user is currently logged in to the device, the user is logged out and must log back in again.

Create User Groups

1. From the Cisco SD-WAN Manager menu, choose **Administration > Manage Users**.
2. Click **User Groups**.
3. Click **Add User Group**.
4. Enter **User Group Name**.
5. Select the **Read** or **Write** check box against feature that you want to assign to a user group.
6. Click **Add**.
7. You can view the new user group in the left navigation path. Click **Edit** to edit the existing read or write rules.
8. Click **Save**.

Configure and Manage VPN Segments

To configure VPN Segments:

1. From the Cisco SD-WAN Manager menu, choose **Administration > VPN Segments**. A web page displays the list of segments that are configured.
2. To edit or delete an existing segment, click ..., and click **Edit** or **Delete**.
3. To add new segment, click **Add Segment**.
4. Enter the name of the segment in the **Segment Name** field.
5. Enter the number of VPNs you want to configure in **VPN Number** field.
6. To add a new segment, click **Add**.

Configure and Manage VPN Groups

To configure VPN Groups:

1. From the Cisco SD-WAN Manager menu, choose **Administration > VPN Groups**. A web page displays the list of segments that are configured.
2. To edit or delete a VPN group, click ..., and click **Edit** or **Delete**.
3. To view the existing VPN in the dashboard, click ..., and click **View Dashboard**. The **VPN Dashboard** displays the device details of the VPN device configured.
4. To add new VPN group, click **Add Group**.
5. From **Create VPN Group**, enter VPN group name in the **VPN Group Name** field.
6. Enter a brief description of the VPN in the **Description** field.
7. Check **Enable User Group access** check box and enter the user group name.
8. From **Assign Segment**, click **Add Segment** drop-down list to add new or existing segment to the VPN group.
9. Enter the **Segment Name** and **VPN Number** in the respective fields.
10. To add the configure VPN group to a device, click **Add**.

Managing Resource Groups

Minimum supported releases: Cisco IOS XE Catalyst SD-WAN Release 17.5.1a and Cisco vManage Release 20.5.1

To configure Resource Groups:

1. From the Cisco SD-WAN Manager menu, choose **Administration > Resource Groups**. The table displays a list of resource groups that are configured in Cisco SD-WAN Manager.
2. To edit or delete a resource group, click ..., and click **Edit** or **Delete**.
3. To add new resource group, click **Add Resource Group**.
4. Enter **Resource Group Name** and the **Description**.
5. Under **Site ID**, enter **Range** or **Select ID(S)** from the drop-down list to include in the resource group.

6. To add the resource group to a device, click **Add**.

To add Users:

1. From the Cisco SD-WAN Manager menu, choose **Administration > Manage Users**. The Manage Users screen appears.
2. By default **Users** is selected. The table displays the list of users configured in the device.
3. To edit, delete, or change password for an existing user, click **...**, and click **Edit**, **Delete**, or **Change Password** respectively.
4. To add a new user, click **Add User**.
5. Add **Full Name**, **Username**, **Password**, and **Confirm Password** details.
6. From the **User Groups** drop-down list, select the user group where you want to add a user.
7. From the **Resource Group** drop-down list, select the resource group.



Note This field is available from Cisco SD-WAN Release 20.5.1.

8. Click **Add**.

Workflow to Configure RBAC for Policies

Minimum supported releases: Cisco SD-WAN Release 20.6.1 and Cisco vManage Release 20.6.1

To configure RBAC for policies, use the following workflow:

1. Create user groups with required Read or Write (R/W) access to selected control or data policies. For details on creating user groups, refer [Create User Groups](#).
2. Create users and assign them to required user groups. Refer [Create Users](#).
3. Create or modify or view policy configurations as required. For information about configuring policies, see [Configure Centralized Policies Using Cisco vManage](#).

Modify Policy Configurations

Minimum supported releases: Cisco SD-WAN Release 20.6.1 and Cisco vManage Release 20.6.1

1. Login to Cisco SD-WAN Manager with the new user details.
2. You can modify or update the configurations based on the requirement.

When you login to Cisco SD-WAN Manager with new user details, you can view only the user group components that are assigned to you. For more details on configuring policies, see [Policies Configuration Guide for vEdge Routers](#)

Assign Users to Configure RBAC for Policies

Minimum supported releases: Cisco SD-WAN Release 20.6.1 and Cisco vManage Release 20.6.1

To Assign User to Create or Modify a CFlowd Data Policy

To create a CFlowd user group:

1. From Cisco SD-WAN Manager, choose **Administration > Manage Users**.
2. Click **User Groups** and **Add User Group**.
3. Enter **User Group Name**.
For example, cflowd-policy-only.
4. Check the Read or Write check box against the CFlowD Policy feature that you want to assign to a user group.
5. Click **Add**.
6. You can view the new user group in the left navigation path. Click **Edit** to edit the existing read or write rules.
7. Click **Save**.

To create a CFlowd user:

1. In Cisco SD-WAN Manager, choose **Administration > Manage Users**.
2. Click **Users**.
3. Click **Add User**.
4. In the Add New User page, enter **Full Name**, **Username**, **Password**, and **Confirm Password** details.
5. Choose **cflowd-policy-only** from the **User Groups** drop-down.
Allow the **Resource Group** to select the default resource group.
6. Click **Add**. You can view the new user in the Users window.
7. To edit the existing read or write rules for a user, click **Edit**.

To modify a Cflowd policy:

1. Login to Cisco SD-WAN Manager with the new user credentials.
You can view access only to CFlowd Policies as your login is assigned to **cflowd-policy-only** user group.
2. You can create, modify, or update the configurations based on the requirement.

Configure Granular RBAC for Feature Templates

Minimum supported release: Cisco vManage Release 20.7.1

To configure specific template access, create a user group and assign the read and write permissions using the permission types described in Information About RBAC for Co-Management. The permission options for limiting template access appear with the other permission options that you choose when adding a user group.

For information about granular RBAC for feature templates, see [Information About Granular RBAC for Templates](#), on page 135.

For information about adding a user group, see [Create User Groups](#).

For a list of permission types and descriptions, see [Manage Users](#).

Configure RBAC Using the CLI

Configure Users Using CLI

You can use the CLI to configure user credentials on each device. This way, you can create additional users and give them access to specific devices. The credentials that you create for a user by using the CLI can be different from the Cisco SD-WAN Manager credentials for the user. In addition, you can create different credentials for a user on each device. All users with the **netadmin** privilege can create a new user.

To create a user account, configure the username and password, and place the user in a group:

```
vEdge(config)# system aaa
vEdge(config)# user username password password
vEdge(config-aaa)# group group-name
```

The Username can be 1 to 128 characters long, and it must start with a letter. The name can contain only lowercase letters, the digits 0 through 9, hyphens (-), underscores (_), and periods (.). The name cannot contain any uppercase letters. Because some usernames are reserved, you cannot configure them. For a list of reserved usernames, see the **aaa** configuration command in the Cisco Catalyst SD-WAN Command Reference Guide.

The Password is the password for a user. Each username must have a password, and users are allowed to change their own password. The CLI immediately encrypts the string and does not display a readable version of the password. When a user logs in to a Cisco vEdge device, they have five chances to enter the correct password. After the fifth incorrect attempt, the user is locked out of the device, and must wait for 15 minutes before attempting to log in again.



Note Enclose any user passwords that contain the special character ! in double quotation marks (“”). If a double quotation is not included for the entire password, the config database (?) treats the special character as a space and ignores the rest of the password.

For example, if the password is C!sc0, use “C!sc0”.

Group name is the name of a standard Cisco Catalyst SD-WAN group (**basic**, **netadmin**, or **operator**) or of a group configured with the **usergroup** command (discussed below). If an admin user changes the permission of a user by changing their group, and if that user is currently logged in to the device, the user is logged out and must log back in again.

The factory-default password for the admin username is admin. We strongly recommend that you modify this password the first time you configure a Cisco vEdge device :

```
vEdge(config)# system aaa admin password password
```

Configure the password as an ASCII string. The CLI immediately encrypts the string and does not display a readable version of the password, for example:

```
vEdge(config-user-admin)# show config
system
aaa
user admin
password $1$xULc8yYH$k71cTjvKESmeIGgImNDaC.
!
```



```

user eve
password $1$8z3q4qoU$F6DMBr9vPBF0s/sl45ax5.
group basic
!
!
!

```

If you are using RADIUS to perform AAA authentication, you can configure a specific RADIUS server to verify the password:

```
vEdge(config)# system aaa radius-servers tag
```

The tag is a string that you defined with the **radius server tag** command, as described in the Cisco Catalyst SD-WAN Command Reference Guide.

Creating Groups Using CLI

The Cisco Catalyst SD-WAN software provides default user groups: **basic**, **netadmin**, **operator**, **network_operations**, and **security_operations**. The username **admin** is automatically placed in the **netadmin** usergroup.

If needed, you can create additional custom groups and configure privilege roles that the group members have. To create a custom group with specific authorization, configure the group name and privileges:

```
vEdge(config)# system aaa usergroup group-name task privilege
```

group-name can be 1 to 128 characters long, and it must start with a letter. The name can contain only lowercase letters, the digits 0 through 9, hyphens (-), underscores (_), and periods (.). The name cannot contain any uppercase letters. Some group names are reserved, so you cannot configure them. For a list of them, see the aaa configuration command.

If a remote RADIUS or TACACS+ server validates authentication but does not specify a user group, the user is placed into the user group **basic**. If a remote server validates authentication and specifies a user group (say, X) using VSA Cisco SD-WAN-Group-Name, the user is placed into that user group only. However, if that user is also configured locally and belongs to a user group (say, Y), the user is placed into both the groups (X and Y).

In the **task** option, list the privilege roles that the group members have. The role can be one or more of the following: **interface**, **policy**, **routing**, **security**, and **system**.

In the following example, the **basic** user group has full access to the **system** and **interface** portions of the configuration and operational commands, and the **operator** user group can use all operational commands but can make no modifications to the configuration:

```

vEdge# show running-config system aaa
system
aaa
  usergroup basic
    task system read write
    task interface read write
  !
  usergroup operator
    task system read
    task interface read
    task policy read
    task routing read
    task security read
  !
  user admin
    password $1$tokPB7tf$VchR2JI9Sw1/dqgkqup9S.
  !

```

!

!

Verify RBAC

Verify Granular RBAC Permissions

Minimum supported release: Cisco vManage Release 20.7.1

Use this procedure to verify the permissions that you have configured for a user group.

1. From the Cisco SD-WAN Manager menu, choose **Administration > Manage Users**.
2. Click **User Groups**.
3. In the pane that displays the user groups, select a user group to display the read and write permissions assigned to the user group.
4. Scroll to the permissions that control template access to verify your configuration for the user group.

Monitor RBAC

Monitor devices for VPN Groups

To monitor devices:

1. From the Cisco SD-WAN Manager menu, choose **Monitor > Network**.
2. Click **WAN - Edge**.
3. Select the **VPN Group** and **VPN Segment** for which you want to monitor the network.
A web page displays the list of VPN groups and segments that are configured to a device.



CHAPTER 6

Configure Devices

You can create and store configurations for all devices—the Cisco SD-WAN Manager systems themselves, Cisco Catalyst SD-WAN Controllers, Cisco Catalyst SD-WAN Validators, and routers— by using Cisco SD-WAN Manager. When the devices start up, they contact Cisco SD-WAN Manager, which then downloads the device configuration to the device. (A device that is starting up first contacts the Cisco Catalyst SD-WAN Validator, which validates the device and then sends it the IP address of Cisco SD-WAN Manager.)

The general procedure for creating configuration for all devices is the same. This section provides a high-level description of the configuration procedure. It also describes the prerequisite steps that must be performed before you can create configurations and configure devices in the overlay network.

- [Device Configuration Workflow, on page 171](#)
- [Feature Templates, on page 172](#)
- [Device Templates, on page 172](#)
- [Template Variables, on page 173](#)
- [Configuration Prerequisites, on page 173](#)
- [Create a Device Template from Feature Templates, on page 174](#)
- [Default Device Templates, on page 189](#)
- [Configuring Devices using Cisco SD-WAN Manager, on page 190](#)

Device Configuration Workflow

Devices in the overlay network that are managed by Cisco SD-WAN Manager must be configured from Cisco SD-WAN Manager. The basic configuration procedure is straightforward:

1. Create feature templates.
 - a. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
 - b. Click **Feature Templates**, and click **Add Templates**.



Note In Cisco vManage Release 20.7.x and earlier releases, **Feature Templates** is titled **Feature**.

2. Create device templates.
 - a. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.

- b. Click **Device Templates**, and click **Create Templates**.



Note In Cisco vManage Release 20.7.x and earlier releases, **Device Templates** is titled **Device**.

3. Attach device templates to individual devices.
 - a. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
 - b. Click **Device Templates**, and choose a template.



Note In Cisco vManage Release 20.7.x and earlier releases, **Device Templates** is titled **Device**.

- c. Click ..., and select **Attach Devices**.

Feature Templates

Feature templates are the building blocks of complete configuration for a device. For each feature that you can enable on a device, Cisco SD-WAN Manager provides a template form that you fill out. The form allows you to set the values for all configurable parameters for that feature.

Because device configurations vary for different device types and the different types of routers, feature templates are specific to the type of device.

Some features are mandatory for device operation, so creating templates for these features is required. Also for the same feature, you can create multiple templates for the same device type.



Note In releases prior to Cisco SD-WAN Release 20.7.1, if you enter < or > special characters in a Cisco SD-WAN Manager feature template definition or description, Cisco SD-WAN Manager generates a 500 exception error while attempting to preview a Cisco SD-WAN Manager feature template.

Starting from Cisco SD-WAN Release 20.7.1, if you enter < or > special characters in a Cisco SD-WAN Manager feature template definition or description, the special characters are converted to their HTML equivalents, **<** and **>**. This applies to all feature templates. You no longer receive a 500 exception error when previewing a Cisco SD-WAN Manager feature template.

Device Templates

You create and store configurations for all devices—the Cisco SD-WAN Manager systems themselves, Cisco Catalyst SD-WAN Controllers, Cisco Catalyst SD-WAN Validators, and routers—by using Cisco SD-WAN Manager. When the devices start up, they contact Cisco SD-WAN Manager, which then downloads the device configuration to the device. (A device that is starting up first contacts the Cisco Catalyst SD-WAN Validator, which validates the device and then sends it the IP address of Cisco SD-WAN Manager.)

Device templates contain complete operational configuration for a device. You create device templates by consolidating individual feature templates.

Each device template is specific for a type of device. For each device type, if multiple devices have the same configuration, you can use the same device template for them. For example, many of the routers in a network might have the same basic configuration, so you can configure them with the same templates. (You specify the differences in the templates using configuration variables, which are discussed below.) If the configurations for the same type of devices are different, you create separate device templates.

You can also create a device template by entering a CLI text-style configuration directly on Cisco SD-WAN Manager. Typically, you upload a text file containing the configuration text (or cut the configuration text from a text file and paste it into Cisco SD-WAN Manager). You can also directly type the configuration text into Cisco SD-WAN Manager.

From Cisco vManage Release 20.5.1, device variable page shows text area instead of text input field to configure CLI device template for the ease of configuration.

Template Variables

Within a feature template, some configuration commands and command options are identical across all device types. Others—such as a device system IP address, its geographic latitude and longitude, the timezone, and the overlay network site identifier—are variable, changing from device to device. When you attach the device template to a device, you are prompted to enter actual values for these command variables. You can do this either manually, by typing the values for each variable and for each device, or you can upload an Excel file in CSV format that contains the values for each device.

Configuration Prerequisites

Security Prerequisites

Before you can configure any device in the network, that device must be validated and authenticated so that Cisco SD-WAN Manager systems, Cisco Catalyst SD-WAN Controllers, and Cisco Catalyst SD-WAN Validators recognize it as being allowed in the overlay network.

To validate and authenticate the controllers in the overlay network—Cisco SD-WAN Manager, Cisco SD-WAN Controller, and Cisco Catalyst SD-WAN Validators—a signed certificate must be installed on these devices.

To validate and authenticate the routers, you receive an authorized serial number file from Cisco, which lists the serial and chassis numbers for all the routers allowed in your network. Then, you upload the serial number file to Cisco SD-WAN Manager.

Variables Spreadsheet

The feature templates that you create most likely contain variables. To have Cisco SD-WAN Manager populate the variables with actual values when you attach a device template to a device, create an Excel file that lists the variable values for each device and save the file in CSV format.

In the spreadsheet, the header row contains the variable name and each row after that corresponds to a device, defining the values of the variables. The first three columns in the spreadsheet must be the following, in this order:

- **csv-deviceId**—Serial number of the device (used to uniquely identify the device). For routers, you receive the serial numbers in the authorized serial number file sent to you from Cisco. For other devices, the serial number is included in the signed certificate you receive from Symantec or from your root CA.
- **csv-deviceIP**—System IP address of the device (used to populate the **system ip address** command).
- **csv-host-name**—Hostname of the device (used to populate the **system hostname** command).

You can create a single spreadsheet for all devices in the overlay network—Cisco Catalyst SD-WAN Controllers, Cisco Catalyst SD-WAN Validators, and routers. You do not need to specify values for all variables for all devices.

Create a Device Template from Feature Templates

Device templates define a device's complete operational configuration. A device template consists of a number of feature templates. Each feature template defines the configuration for a particular Cisco Catalyst SD-WAN software feature. Some feature templates are mandatory, indicated with an asterisk (*), and some are optional. Each mandatory feature template, and some of the optional ones, have a factory-default template. For software features that have a factory-default template, you can use either the factory-default template (named `Factory_Default_feature-name_Template`) or you can create a custom feature template.

Create a Device Template from Feature Templates

To create a device template:

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Device Templates**.



Note In Cisco vManage Release 20.7.x and earlier releases, **Device Templates** is titled **Device**.

3. Click the **Create Template** drop-down list, and select **From Feature Template**.
4. From the **Device Model** drop-down list, select the type of device for which you wish to create the template.

Cisco SD-WAN Manager displays all the feature templates for that device type. The required feature templates are indicated with an asterisk (*), and the remaining templates are optional. The factory-default template for each feature is selected by default.

5. In the **Template Name** field, enter a name for the device template.
This field is mandatory and can contain only uppercase and lowercase letters, the digits 0 through 9, hyphens (-), and underscores (_). It cannot contain spaces or any other characters.
6. In the **Description** field, enter a description for the device template.
This field is mandatory, and it can contain any characters and spaces.
7. To view the factory-default configuration for a feature template, select the desired feature template and click **View Template**.
8. Click **Cancel** to return to the **Configuration Template** screen.

9. To create a custom template for a feature, select the desired factory-default feature template and click **Create Template**. The template form is displayed.
This form contains fields for naming the template and defining the feature parameters.
10. In the **Template Name** field, enter a name for the feature template.
This field is mandatory and can contain only uppercase and lowercase letters, the digits 0 through 9, hyphens (-), and underscores (_). It cannot contain spaces or any other characters.
11. In the **Description** field, enter a description for the feature template.
This field is mandatory, and it can contain any characters and spaces.
12. For each field, enter the desired value. You may need to click a tab or the plus sign (+) to display additional fields.
13. When you first open a feature template, for each parameter that has a default value, the scope is set to Default (indicated by a check mark), and the default setting or value is shown. To change the default or to enter a value, click the scope drop-down list of the parameter field and select one of the following:

Table 61:

Parameter Scope	Scope Description
Device Specific (indicated by a host icon)	Use a device-specific value for the parameter. For device-specific parameters, you cannot enter a value in the feature template. You enter the value when you attach a device to a device template. When you click Device Specific, the Enter Key box opens. This box displays a key, which is a unique string that identifies the parameter in a CSV file that you create. This file is an Excel spreadsheet that contains one column for each key. The header row contains the key names (one key per column), and each row after that corresponds to a device and defines the values of the keys for that device. You upload the CSV file when you attach a device to a device template. For more information, see Use Variable Values in Configuration Templates. To change the default key, type a new string and move the cursor out of the Enter Key box. Examples of device-specific parameters are system IP address, hostname, GPS location, and site ID.
Global (indicated by a globe icon)	Enter a value for the parameter, and apply that value to all devices. Examples of parameters that you might apply globally to a group of devices are DNS server, syslog server, and interface MTUs.

14. For some groups of parameters, you can mark the entire group as device-specific. To do this, check the **Mark as Optional Row** check box.
These parameters are then grayed out so that you cannot enter a value for them in the feature template. You enter the value or values when you attach a device to a device template.
15. Click **Save**.

16. Repeat Steps 6 through 13 to create a custom template for each additional software feature. For details on creating specific feature templates, see the templates listed in **Available Feature Templates**.
17. Click **Create**. The new configuration template is displayed in the Device Template table.

The Feature Templates column shows the number of feature templates that are included in the device template, and the Type column shows "Feature" to indicate that the device template was created from a collection of feature templates.

Another way to create device templates from feature templates is to first create one or more custom feature templates and then create device templates. You can create multiple feature templates for the same feature. For a list of feature templates, see **Available Feature Templates**.

1. Click **Feature**.
2. Click **Add Template**.
3. From **Select Devices**, select the type of device for which you wish to create a template.

You can create a single feature template for features that are available on multiple device types. You must, however, create separate feature templates for software features that are available only on the device type you are configuring.
4. Select the feature template. The template form is displayed.

This form contains fields for naming the template and fields for defining the required parameters. If the feature has optional parameters, then the template form shows a plus sign (+) after the required parameters.
5. In the **Template Name** field, enter a name for the feature template.

This field is mandatory and can contain only uppercase and lowercase letters, the digits 0 through 9, hyphens (-), and underscores (_). It cannot contain spaces or any other characters.
6. In the **Description** field, enter a description for the feature template.

This field is mandatory, and it can contain any characters and spaces.
7. For each required parameter, choose the desired value, and if applicable, select the scope of the parameter. Select the scope from the drop-down list of each parameter's value box.
8. Click the plus sign (+) from the required parameters to set the values of optional parameters.
9. Click **Save**.
10. Repeat Steps 2 to 9 for each additional feature template you wish to create.
11. Click **Device**.
12. Click the **Create Template** drop-down list and select **From Feature Template**.
13. From the **Device Model** drop-down list, select the type of device for which you wish to create the device template.

Cisco SD-WAN Manager displays the feature templates for the device type you selected. The required feature templates are indicated with an asterisk (*). The remaining templates are optional.
14. In the **Template Name** field, enter a name for the device template.

This field is mandatory and can contain only uppercase and lowercase letters, the digits 0 through 9, hyphens (-), and underscores (_). It cannot contain spaces or any other characters.

15. In the **Description** field, enter a description for the device template.
This field is mandatory, and it can contain any characters and spaces.
16. To view the factory-default configuration for a feature template, select the desired feature template and click **View Template**.
17. Click **Cancel** to return to the **Configuration Template** screen.
18. To use the factory-default configuration, click **Create** to create the device template. The new device template is displayed in the **Device Template** table. The Feature Templates column shows the number of feature templates that are included in the device template, and the Type column shows "Feature" to indicate that the device template was created from a collection of feature templates.
19. To modify the factory-default configuration, select the feature template for which you do not wish to use the factory-default template. From the drop-down list of available feature templates, select a feature template that you created.
20. Repeat Step 19 for each factory-default feature template you wish to modify.
21. Click **Create**. The new configuration template is displayed in the **Device Template** table.
The Feature Templates column shows the number of feature templates that are included in the device template, and the Type column shows "Feature" to indicate that the device template was created from a collection of feature templates.

Create a Device CLI Template

To create a device template by entering a CLI text-style configuration directly on the Cisco SD-WAN Manager:

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Device Templates**.



Note In Cisco vManage Release 20.7.x and earlier releases, **Device Templates** is titled **Device**.

3. Click the **Create Template** drop-down list and select **CLI Template**.
4. From the **Device Type** drop-down list, select the type of device for which you wish to create the template.
5. In the **Template Name** field, enter a name for the device template.
This field is mandatory and can contain only uppercase and lowercase letters, the digits 0 through 9, hyphens (–), and underscores (_). It cannot contain spaces or any other characters.
6. In the **Description** field, enter a description for the device template.
This field is mandatory, and it can contain any characters and spaces.
7. In the CLI Configuration box, enter the configuration either by typing it, cutting and pasting it, or uploading a file.
8. To convert an actual configuration value to a variable, select the value and click **Create Variable**. Enter the variable name, and click **Create Variable**. You can also type the variable name directly, in the format `{{variable-name}}`; for example, `{{hostname}}`.

9. Click **Add**. The new device template is displayed in the Device Template table.

The **Feature Templates** column shows the number of feature templates that are included in the device template, and the Type column shows "CLI" to indicate that the device template was created from CLI text.

Manage Device Templates

Edit a Device Template

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Device Templates** or **Feature Templates**, and select a template.



Note In Cisco vManage Release 20.7.x and earlier releases, **Device Templates** is titled **Device**, and **Feature Templates** is titled **Feature**.

3. Click ..., and click **Edit**.

You cannot change the name of a device or feature template when that is attached to a device.



Note You can edit templates simultaneously from one or more Cisco SD-WAN Manager servers. For simultaneous template edit operations, the following rules apply:

- You cannot edit the same device or feature template simultaneously.
- When you are editing a device template, all other feature templates attached to that device template are locked and you cannot perform any edit operations on them.
- When you are editing a feature template that is attached to a device template, that device template as well as all other feature templates attached to it are locked and you cannot perform any edit operations on them.

Delete a Template

Deleting a template does not remove the associated configuration from devices.

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Device Templates** or **Feature Templates**, and select a template.



Note In Cisco vManage Release 20.7.x and earlier releases, **Device Templates** is titled **Device**, and **Feature Templates** is titled **Feature**.

3. Click ..., and click **Delete**.
4. To confirm the deletion of the template, click **OK**.

Copy a Template

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Device Templates** or **Feature Templates**, and select a template.



Note In Cisco vManage Release 20.7.x and earlier releases, **Device Templates** is titled **Device**, and **Feature Templates** is titled **Feature**.

3. Click ..., and click **Copy**.
4. Enter a new template name and description.
5. Click **Copy**.

Edit a CLI Device Template

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Device Templates**, and select a template.



Note In Cisco vManage Release 20.7.x and earlier releases, **Device Templates** is titled **Device**.

3. Click ..., and click **Edit**.
4. Under **Device CLI Template**, edit the template.
5. Click **Update**.

Use Variable Values in Configuration Templates

An overlay network might have multiple devices of the same type that have nearly identical configurations. This situation most commonly occurs with routers when the routers that are located in multiple stores or branch locations provide identical services, but each individual router has its own hostname, IP address, GPS location, and other site-specific properties, such as BGP neighbors. This situation also occurs in a network with redundant controller devices, such as Cisco Catalyst SD-WAN Controllers, which must all be configured with identical policies, and Cisco SD-WAN Manager systems. Again, each controller has its own individual parameters, such as hostname and IP address.

To simplify the configuration process for these devices, you can create a single configuration template that contains both static configuration values and variable values. The static values are common across all the devices, and the variable values apply only to an individual device. You provide the actual values for the variables when you attach the individual device to the device configuration template.

You can configure a variable value for a parameter in a feature configuration template in two ways:

- Select the parameter scope to be **Device Specific**—For an individual configuration parameter, select **Device Specific** to mark the parameter as a variable. Each variable must be identified by a unique text string, which is called a *key*. When you select **Device Specific**, an Enter Key box opens and displays the

default key. You can use the default key, or you can change it by typing a new string and then moving the cursor out of the Enter Key box.

- Mark a group of related parameters as optional—For some features in some feature configuration templates, you can mark the entire feature as optional. To mark the feature in this way, click Mark as Optional Row in a section of a feature configuration template. The variable parameters are then dimmed, and you cannot configure values for them in the feature configuration template.

You enter the device-specific values for the variables when you attach the device to the configuration, in one of the following ways:

- From a file—When you are attaching a template to a device, you load a file to Cisco SD-WAN Manager. This is an Excel file in CSV format that lists all the variables and defines the variable's value for each device.
- Manually—When you attach a device template to a device, the Cisco SD-WAN Manager prompts you for the values for each of device-specific parameters, and you type in the value for each parameter.



Note Cisco Catalyst SD-WAN supports up to 500 variables in a template push operation.

Use a File for Variable Parameters

To load device-specific variable values from a file, you create a template variables file. This file is an Excel file in CSV format that lists all the variables in your the configurations of your devices and defines the values for each variable. You create this file offline and then import it into Cisco SD-WAN Manager server when you attach a device configuration to one or more devices in the overlay network.

We recommend that you create a template variables CSV file when your overlay network has more than a small number of Cisco vEdge devices.

CSV File Format

The CSV file is an Excel spreadsheet that contains one column for each variable that is required for the configuration of a device. The header row contains the variable names (one variable per column), and each row after that corresponds to a device and defines the values of the variables for that device.

You can create a single spreadsheet for all devices in the overlay network—Cisco vEdge devices, Cisco SD-WAN Manager systems, Cisco Catalyst SD-WAN Controllers, and Cisco Catalyst SD-WAN Validators—or you can create one spreadsheet for each device type. The system determines the device type from its serial number.

In the spreadsheet, for each device type and for each individual device, you specify values only for the required variables. When you do not need to specify a value for a variable, simply leave that cell blank.

The first three columns in the spreadsheet must be the following items and must be in the order shown:

Column	Column Heading	Description
1	csv-deviceId	Serial number of the device (used to uniquely identify the device). For Cisco vEdge devices, you receive the serial numbers in the authorized serial number file sent to you from Cisco. For other devices, the serial number is included in the signed certificate you receive from Symantec or from your root CA.
2	csv-deviceIP	System IP address of the device (used to populate the system ip address command).
3	csv-host-name	Hostname of the device (used to populate the system hostname command).

The headings for the remaining columns must be unique variable keys that are defined in the Enter Key box of a feature configuration template. These remaining columns can be in any order.

Generate a Skeleton CSV File

You can create a template variables CSV file manually, with the format described in the previous section, or you can have Cisco SD-WAN Manager generate a skeleton CSV file that contains all the required columns and column headings. This generated CSV file has one row for each Cisco device type, and it has the column headings for each of the variables that are required by all the feature templates included in the device configuration. The column heading text corresponds to the key string that identifies a device-specific parameter. Then you populate the rows with values for each variable.

To have Cisco SD-WAN Manager generate a skeleton CSV file:

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Feature Templates**, and click **Add Template**.



Note In Cisco vManage Release 20.7.x and earlier releases, **Feature Templates** is titled **Feature**.

3. Create the required feature templates for one Cisco vEdge device router, one Cisco Catalyst SD-WAN Controller, one Cisco SD-WAN Manager system, and one Cisco Catalyst SD-WAN Validator.

In each feature template:

- a. For fields that have default values, verify that you want to use that value for all devices. If you do not want to use the default, change the scope to **Global** or **Device-specific**.
 - b. For fields that apply to all devices, select the **Global** icon next to the field and set the desired global values.
 - c. For fields that are device specific, select the **Device-specific** icon next to the field and leave the field blank.
4. For each Cisco device type, create a device template.

5. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
6. Click **Device Templates**, and select the desired device template from the template list table.



Note In Cisco vManage Release 20.7.x and earlier releases, **Device Templates** is titled **Device**.

7. Click ..., and click **Export CSV**.
8. Repeat Steps 7 and 8 for each device template.

Edit the exported CSV file, adding at a minimum the device serial number, device system IP address, and device hostname for each device in the overlay network. Then add values for desired device-specific variables for each device. Note that variable names cannot contain forward slashes (/), backwards slashes (\), or parentheses (()).

If desired, you can combine the CSV files into a single file.

Import a CSV File

To use the device-specific variable values in the CSV file, import the file when you are attaching a device template to the Viptela device:

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Device Templates**.



Note In Cisco vManage Release 20.7.x and earlier releases, **Device Templates** is titled **Device**.

3. For the desired template, click ..., and select **Attach Devices**.
4. In the **Attach Devices** dialog box, select the desired devices in **Available Devices** and click the arrow to move them to **Selected Devices**.
5. Click **Attach**.
6. Click the Up arrow. The Upload CSV File box displays.
7. Choose the CSV file to upload, and click **Upload**.

During the attachment process, click Import file to load the Excel file. If Cisco SD-WAN Manager detects duplicate system IP addresses for devices in the overlay network, it displays a warning message or a pop-up window. You must correct the system IP addresses to remove any duplicates before you can continue the process of attaching device templates to Viptela devices.

Manually Enter Values for Device-Specific Variables and for Optional Rows

For parameters in a feature template that you configure as device-specific, when you attach a device template to a device, Cisco SD-WAN Manager prompts you for the values to use for these parameters. Entering device-specific values in this manner is useful in test or POC networks, or if you are deploying a small network. This method generally does not scale well for larger networks.

For situations in which the configuration for many devices is identical except for a few parameters, in the feature configuration template, you can specify that the parameter be an optional row in the configuration. By selecting optional row, the feature template automatically marks the parameters as device-specific, and these parameters are dimmed so that you cannot set them in the template. You do not have to individually mark the parameters as device specific. Then, when you attach a device template to a device, Cisco SD-WAN Manager prompts you for the values to use for these parameters. Using optional rows to enter device-specific values is useful when a group of many Cisco vEdge devices provide identical services at their branch or site, but individual routers have their own hostname, IP address, GPS location, and other site or store properties, such as BGP neighbors.

Optional rows are available for some parameters in some feature configuration templates. To treat a parameter or set of parameters as an optional row, click the **Mark as Optional Row** box. For these types of parameters, the feature configuration template has a table listing all the configured parameters. The Optional column indicates which are optional rows,

To manually enter values for device-specific variables or for variables in optional rows when you attach the template to a device:

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Device Templates**, and select the desired device template.



Note In Cisco vManage Release 20.7.x and earlier releases, **Device Templates** is titled **Device**.

3. Click ..., and click **Attach Devices**. The **Attach Devices** dialog box opens.
4. Choose one or more devices from **Available Devices** and move them to **Selected Devices**.
5. Click **Attach**.
6. In the **Chassis Number** list, select the desired device.
7. Click ..., and click **Edit Device Template**. The **Update Device Template** dialog box opens.
8. Enter values for the optional parameters. When you are using optional rows, if you do not want to include the parameter for the specific device, do not specify a value.
9. Click **Update**.
10. Click **Next**.

If any devices have the same system IP address, a dialog box appears or an error message is displayed when you click **Next**. Modify the system IP addresses so that there are no duplicates, and click **Save**. Then click **Next** again.



Note You need to shut down the OMP on the device, before changing the system-ip on the device.

11. In the left pane, select the device. The right pane displays the device configuration and the **Config Preview** tab in the upper right corner is selected.
12. Click **Config Diff** to preview the differences between this configuration and the configuration currently running on the device, if applicable. To edit the variable values entered in the previous screen, click **Back**.

13. Click **Configure Devices** to push the configuration to the devices.

The Status column displays whether the configuration was successfully pushed. Click the **right angle bracket** to the left of the row to display details of the push operation.

View Device Templates

View a Template

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Device Templates** or **Feature Templates**, and select a template you wish to view.



Note In Cisco vManage Release 20.7.x and earlier releases, **Device Templates** is titled **Device**, and **Feature Templates** is titled **Feature**.

3. Click ..., and then click **View**.

View Device Templates Attached to a Feature Template

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Feature Templates**, and select a template you wish to view.



Note In Cisco vManage Release 20.7.x and earlier releases, **Feature Templates** is titled **Feature**.

3. Click ..., and click **Show Attached Device Templates**.

Device Templates dialog box opens, displaying the names of the device templates to which the feature template is attached.

View Devices Attached to a Device Template

For a device template that you created from feature templates:

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Device Templates**, and select a template you wish to view.



Note In Cisco vManage Release 20.7.x and earlier releases, **Device Templates** is titled **Device**.

3. Click ..., and click **Attach Devices**.
4. From **Attach Devices**, click **Attached Devices**.

For a device template that you created from a CLI template:

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Device Templates**, and select a template you wish to view.



Note In Cisco vManage Release 20.7.x and earlier releases, **Device Templates** is titled **Device**.

3. Click ..., and then click **Show Attached Devices**.

Attach and Detach a Device Template

To configure a device on the network, you attach a device template to the device. You can attach only one device template to a device, so the template—whether you created it by consolidating individual feature templates or by entering a CLI text-style configuration—must contain the complete configuration for the device. You cannot mix and match feature templates and CLI-style configurations.

On Cisco vEdge devices in the overlay network, you can perform the same operations, in parallel, from one or more Cisco SD-WAN Manager servers. You can perform the following template operations in parallel:

- Attach a device template to devices
- Detach a device template from a device
- Change the variable values for a device template that has devices attached to it

For template operations, the following rules apply:

- When a device template is already attached to a device, you can modify one of its feature templates. Then when you click **Update > Configure Devices**, all other template operations—including attach devices, detach devices, and edit device values—are locked on all Cisco SD-WAN Manager servers until the update operation completes. This means that a user on another Cisco SD-WAN Manager server cannot perform any template operations until the update completes.
- You can perform the attach and detach device template operations on different devices, from one or more Cisco SD-WAN Manager servers, at the same time. However, if any one of these operations is in progress on one Cisco SD-WAN Manager server, you cannot edit any feature templates on any of the servers until the attach or detach operation completes.



Note You need to recreate the feature templates as the templates created prior to Cisco vManage Release 20.5 fails when attached to the device.

If the device being configured is present and operational on the network, the configuration is sent to the device immediately and takes effect immediately. If the device has not yet joined the network, the pushing of the configuration to the device is scheduled. When the device joins the network, Cisco SD-WAN Manager pushes the configuration immediately after it learns that the device is present in the network.

Attach a Device Template to Devices

You can attach the same templates to multiple devices, and you can do so simultaneously, in a single operation.

To attach a device template to one or more devices:

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Device Templates** and select the desired template.



Note In Cisco vManage Release 20.7.x and earlier releases, **Device Templates** is titled **Device**.

3. Click ..., and click **Attach Devices**. The **Attach Devices** dialog box opens with the **Select Devices** tab selected
4. In the **Available Devices** column on the left, select a group and search for one or more devices, select a device from the list, or click **Select All**.
5. Click the arrow pointing right to move the device to the **Selected Devices** column on the right.
6. Click **Attach**.
7. If the template contains variables, enter the missing variable values for each device you selected in one of the following ways:
 - Enter the values manually for each device either in the table column or by clicking ... and **Edit Device Template**. When you are using optional rows, if you do not want to include the parameter for the specific device, do not specify a value.
 - Click **Import File** to upload a CSV file that lists all the variables and defines each variable's value for each device.
8. Click **Update**
9. Click **Next**.
If any devices have the same system IP address, a dialog box appears or an error message is displayed when you click **Next**. Modify the system IP addresses so that there are no duplicates, and click **Save**. Then click **Next** again.
10. In the left pane, select the device, to preview the configuration that is ready to be pushed to the device. The right pane displays the device's configuration and the **Config Preview** tab is selected. Click the **Config Diff** tab to view the differences between this configuration and the configuration currently running on the device, if applicable. Click the **Back** button to edit the variable values entered in the previous screen.
11. If you are attaching a Cisco vEdge device, click **Configure Device Rollback Timer** to configure the time interval at which the device rolls back to its previous configuration if the router loses its control connection to the overlay network. The **Configure Device Rollback Time** dialog box is displayed.
 - a. From the **Devices** drop-down list, select a device.
 - b. To enable the rollback timer, in the **Set Rollback slider**, drag the slider to the left to enable the rollback timer. When you do this, the slider changes in color from gray to green.
 - c. To disable the rollback timer, click the **Enable Rollback** slider. When you disable the timer, the Password field dialog box opens. Enter the password that you used to log in to Cisco SD-WAN Manager.
 - d. In the **Device Rollback Time slider**, drag the slider to the desired value. The default time is 5 minutes. You can configure a time from 6 to 15 minutes.

- e. To exclude a device from the rollback timer setting, click **Add Exception** and select the devices to exclude.
 - f. The table at the bottom of the **Configure Device Rollback Time** dialog box lists all the devices to which you are attaching the template and their rollback time. To delete a configured rollback time, click the **Trash** icon from the device name.
 - g. Click **Save**.
12. Click **Configure Devices** to push the configuration to the devices. The **Status** column displays whether the configuration was successfully pushed. Click the right angle bracket to display details of the push operation.

Export a Variables Spreadsheet in CSV Format for a Template

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Device Templates** and select the desired template.



Note In Cisco vManage Release 20.7.x and earlier releases, **Device Templates** is titled **Device**.

3. Click ..., and click **Export CSV**.

Determine Why a Device Rejects a Template

When you attach a template to a device using the screen, the device might reject the template. One reason that this may occur is because the device template contains incorrect variable values. When a device rejects a template, it reverts to the previous configuration.

To determine why the device rejected the template:

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Device Templates**.



Note In Cisco vManage Release 20.7.x and earlier releases, **Device Templates** is titled **Device**.

3. Locate the device. The **Template Status** column indicates why the device rejected the template.

Change the Device Rollback Timer

By default, when you attach a Cisco vEdge device to a configuration template, if the router is unable to successfully start after 5 minutes, it returns to, or rolls back to, the previous configuration. For a configuration that you have created from the CLI, you can change the device's rollback timer:

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Device Templates**, and choose a device template.



Note In Cisco vManage Release 20.7.x and earlier releases, **Device Templates** is titled **Device**.

3. Click ..., and click **Change Device Values**.
The right pane displays the device's configuration, and the **Config Preview** tab is selected.
4. In the left pane, click the name of a device.
5. Click **Configure Device Rollback Timer**. The **Configure Device Rollback Time** pop up page is displayed.
6. From the **Devices** drop-down list, select a device.
7. To enable the rollback timer, in the **Set Rollback slider** drag the slider to the left to enable the rollback timer. When you do this, the slider changes in color from gray to green.
8. To disable the rollback timer, click **Enable Rollback slider**. When you disable the timer, the **Password** field dialog box appears. Enter the password that you used to log in to Cisco SD-WAN Manager.
9. In the **Device Rollback Time** slider, drag the slider to the desired value. The default time is 5 minutes. You can configure a time from 6 to 15 minutes.
10. To exclude a device from the rollback timer setting, click **Add Exception** and select the devices to exclude.
11. The table of the **Configure Device Rollback Time** dialog box lists all the devices to which you are attaching the template and their rollback time. To delete a configured rollback time, click the **Trash** icon of the device name.
12. Click **Save**.
13. Click **Configure Devices** to push the configuration to the devices. The **Status** column displays whether the configuration was successfully pushed. Click (+) to display details of the push operation.

Preview Device Configuration and View Configuration Differences

For a configuration that you have created from the CLI:

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Device Templates**, and choose the desired device template.



Note In Cisco vManage Release 20.7.x and earlier releases, **Device Templates** is titled **Device**.

3. Click ..., and click **Change Device Values**.
The right pane displays the device's configuration, and **Config Preview** is selected.
4. Click the name of a device.
5. Click **Config Diff** to view the differences between this configuration and the configuration currently running on the device, if applicable. Click **Back** to edit the variable values entered in the previous screen.

- Click **Configure Devices** to push the configuration to the devices. The Status column displays whether the configuration was successfully pushed. Click the right angle bracket to display details of the push operation.

Change Variable Values for a Device

For a configuration that you have created from device configuration templates, if the templates contain variables, Cisco SD-WAN Manager can automatically populate the variables with actual values when you attach the templates to the devices. To do this, you create an Excel file that lists the variable values for each device and save the file in CSV format. You can also enter values for these variables manually.

After you have pushed the configuration to a device, you can change the value assigned to any variable:

- From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
- Click **Device Templates**, and choose the desired device template.



Note In Cisco vManage Release 20.7.x and earlier releases, **Device Templates** is titled **Device**.

- Click ..., and click **Change Device Values**.
The screen displays a table of all the devices that are attached to that device template.
- For the desired device, click ..., and click **Edit Device Template**.
- In the **Update Device Template** dialog box, enter values for the items in the variable list.
- Click **Update**.
- Click **Next**.
- Click **Configure Devices** to push the configuration to the device. The Status column displays if the configuration was successfully pushed or not. Click the right angle bracket to display the details of the push operation.

Default Device Templates

Table 62: Feature History

Feature Name	Release Information	Description
Default Device Templates	Cisco SD-WAN Release 20.1.1	A default device template provides basic information that you can use to bring up devices in a deployment quickly. This feature is supported on the Cisco Cloud Services Router 1000V Series, Cisco C1111-8PLTELA Integrated Services Routers, and Cisco 4331 Integrated Services Routers.

A default device template provides basic information that you can use to bring up devices in a deployment. It provides a way for you to quickly provision devices with the minimum information that they need to operate in your network.

You cannot directly edit or update information in a device default template, but you can copy the template and then edit the copy.

To use a default device template:

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Device Templates**.



Note In Cisco vManage Release 20.7.x and earlier releases, **Device Templates** is titled **Device**.

3. From the **Template Type** drop-down list, select **Default**.

A list of default device templates displays.

4. Perform any of these actions:

- To attach a default device template to devices, click ..., and select **Attach Devices**.

In the **Attach Devices** dialog box, select the devices that you want attach, and then click **Attach**.

- To view the configuration settings for a default device template, click ..., and choose **View**.

- To copy a default device template, click ..., and choose **View**.

In the **Template Copy** dialog box, enter a unique name and a description for the copy that you are creating, and then click **Copy**.

The copied version becomes a feature template that you can edit.

- To create an Excel file in CSV format that contains device-specific settings from a device template, click ..., and choose **Export CSV**. Use the dialog box that displays to open or save the CSV file.

You can use this CSV file as a reference for device-specific settings when you create other device templates.

Configuring Devices using Cisco SD-WAN Manager

Use the **Devices** screen to add and delete devices, toggle the mode of a device between CLI and Cisco SD-WAN Manager, upload the WAN Edge Serial number file, export bootstrap configuration and, and perform other device-related tasks.

The screenshot shows the Cisco vManage interface. The left sidebar contains a menu with options: Dashboard, Monitor, Configuration (selected), Devices, Certificates, Templates, Policies, Security, CloudExpress, Cloud onRamp, Tools, Maintenance, Administration, and vAnalytics. The main content area is titled 'CONFIGURATION | DEVICES' and shows the 'WAN Edge List' tab. Above the table are buttons for 'Change Mode', 'Upload WAN Edge List', 'Export Bootstrap Configuration', and 'Sync Smart Account'. The table has columns: State, Device Model, Chassis Number, Serial No./Token, Hostname, System IP, and Site ID. There are 8 rows of data.

State	Device Model	Chassis Number	Serial No./Token	Hostname	System IP	Site ID
✓	vEdge Cloud	f3967a34-d454-49cd-8494-05c73d88c9...	12345703	vm11	172.16.255.21	100
✓	vEdge Cloud	537fcec5-00f8-4062-a894-9db7d11f4fd4	12345715	vm1	172.16.255.11	100
✓	vEdge Cloud	3de2abff-0251-4718-b4d2-023d2d39b9...	12345711	vm4	172.16.255.14	400
✓	ISR4331	ISR4331-FD02106254L	0153B8A8	ISR4331-SDWAN-2	172.16.255.129	1800
✓	CSR1000v	CSR-e109d8c2-7541-40d3-b3f7-362116...	12345607	CSR-cEdge2	172.16.255.134	1900
✗	CSR1000v	CSR-97a0fe05-a03e-4a1c-afb3-4f9e714...	1234560A	CSR-cEdge1	172.16.255.130	1600
✓	ISR4221	ISR4221/K9-FOC22034WR7	0254FAFB	ISR4221	172.16.255.139	2000
✓	C1111-8P	C1111-8P-FOC215124MH	0160C45F	C1111-8P	172.16.255.138	2001

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Change Configuration Modes

A device can be in either of these configuration modes:

- Cisco SD-WAN Manager mode—A template is attached to the device and you cannot change the configuration on the device by using the CLI.
- CLI mode—No template is attached to the device and the device can be configured locally by using the CLI.

When you attach a template to a device from Cisco SD-WAN Manager, it puts the device in Cisco SD-WAN Manager mode. You can change the device back to CLI mode if needed to make local changes to its configuration.

To toggle a router from Cisco SD-WAN Manager mode to CLI mode:

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Devices**.
2. Click **WAN Edge List**, and select a device.
3. Click the **Change Mode** drop-down list and select **CLI mode**.



Note

Starting from Cisco IOS XE SD-WAN Release 17.11.1a, click the ... icon adjacent to the device that you want to change from Cisco SD-WAN Manager mode to the CLI mode and click **Config Lock (Provision Device)**.

The **Config Lock (Provision Device)** option appears only if a template is attached to the device or if a configuration group is deployed to the device.

An SSH window opens. To log in to the device, enter a username and password. You can then issue CLI commands to configure or monitor the device.

To toggle a controller device from Cisco SD-WAN Manager mode to CLI mode:

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Devices**.
2. Click **Controllers**, and select a device.



Note Starting from Cisco IOS XE Catalyst SD-WAN Release 17.13.1a, the **Controllers** tab is renamed as the **Control Components** tab to stay consistent with Cisco Catalyst SD-WAN rebranding.

3. Click the **Change Mode** drop-down list.
4. Select **CLI mode** and then select the device type. The **Change Mode - CLI** window opens.
5. From the **Manager mode** pane, select the device and click the right arrow to move the device to the **CLI mode** pane.
6. Click **Update to CLI Mode**.

An SSH window opens. To log in to the device, enter a username and password. You can then issue CLI commands to configure or monitor the device.



Note Starting from Cisco IOS XE SD-WAN Release 17.11.1a, click the ... icon adjacent to the device that you want to change from Cisco SD-WAN Manager mode to the CLI mode and click **Config Lock (Provision Device)**. The **Config Lock (Provision Device)** option appears only if a template is attached to the device or if a configuration group is deployed to the device.

Upload WAN Edge Router Authorized Serial Number File

The WAN eEdge router authorized serial number file contains the chassis number and the certificate serial numbers of all valid Cisco vEdge devices in the overlay network. You retrieve a serial number file from the Cisco Plug-and-Play (PnP) portal and upload it to Cisco SD-WAN Manager. (For more information about Cisco PnP, see [Cisco Plug and Play Support Guide for Cisco Catalyst SD-WAN Products](#).) From Cisco SD-WAN Manager, you send the file to the controllers in the network. This file is required to allow the Cisco Catalyst SD-WAN overlay network components to validate and authenticate each other and to allow the overlay network to become operational.

To upload the WAN edge router authorized serial number file to Cisco SD-WAN Manager and then download it to controllers in the network:

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Devices**.
2. Click **WAN Edge List**, and click **Upload WAN Edge List**.

The Quick Connect workflow opens, enabling you to upload the serial number file. See [Quick Connect Workflow](#) for details.

3. (This step applies only for releases earlier than Cisco Catalyst SD-WAN Manager Release 20.14.1.) Under **Upload WAN Edge List** screen:

- a. Click **Choose File** and select the WAN edge router authorized serial number file you received from Cisco PnP.
- b. To automatically validate the routers and send their chassis and serial numbers to the controllers, ensure that the **Validate the uploaded vEdge List and send to controllers** check box is selected. If you do not select this option, you must individually validate each router in **Configuration > Certificates > WAN Edge List**.
- c. Click **Upload**.

A list of routers in the network is displayed in the router table, with details about each router.

Starting from Cisco vManage Release 20.9.2, you can monitor the newly added WAN Edge devices in the **Monitor > Devices** page.

Upload WAN Edge Router Serial Numbers from Cisco Smart Account

To allow Cisco Catalyst SD-WAN overlay network components to validate and authenticate each other and to allow the overlay network to become operational, Cisco Catalyst SD-WAN requires chassis numbers of all valid Cisco vEdge devices in the overlay network.

In addition, certificate serial numbers, are required for all devices.

To upload the WAN edge router authorized serial numbers from a Cisco Smart account to Cisco SD-WAN Manager and then download it to all the controllers in the overlay network:

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Devices**.
2. Click **WAN Edge List**, and click **Sync Smart Account**.
3. In the **Sync Smart Account** window:
 - a. Enter the **Username** and **Password** for your Smart account.
 - b. To automatically validate the routers and send their chassis and serial numbers to the controllers, check the **Validate the Uploaded WAN Edge List and Send to Controllers** check box. If you do not select this option, you must individually validate each router in **Configuration > Certificates > WAN Edge List**.
 - c. Click **Sync**.

A list of routers in the network is displayed in the router table, with details about each router.

Starting from Cisco vManage Release 20.9.2, you can monitor the newly added WAN Edge devices in the **Monitor > Devices** page.

Generate Bootstrap Configuration for a vEdge Cloud Router

For vEdge Cloud routers, you need to generate a bootstrap configuration file that you use when you create vEdge cloud VM instances.

To generate and download a bootstrap configuration for one or more vEdge Cloud routers:

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Devices**.
2. Click **WAN Edge List**, and click **Export Bootstrap Configuration**.

3. In the **Export Bootstrap Configuration** window, in the **Bootstrap Configuration** field, click **Cloud-Init** or **Encoded String**, depending the Hypervisor you are using to bring up the vEdge Cloud router.
4. Select the devices to configure from the **Available Devices** pane, or click **Select All** to select all devices.
5. Click the right arrow to move the devices to the **Selected Devices** pane.
6. Click **Generate Configuration**. The configurations are downloaded to the vManage NMS.
7. Provision the vEdge Cloud router instance in AWS, KVM, or ESXi with the bootstrap configuration. By default, ge0/0 is the device's tunnel interface and is a DHCP client. To use an interface other than ge0/0 as the tunnel interface or to use a static IP as the IP address, reconfigure the device through the CLI. For more information about configuring interfaces, see **Configure Network Interfaces**.

After you provision the vEdge Cloud router instance, vManage NMS installs a certificate on the device and the device's token changes to a serial number. After the device's control connections to vManage NMS come up, any templates attached to the device are automatically pushed to the device.

Export Device Data in CSV Format

In an overlay network, you might have multiple devices of the same type that have identical or effectively identical configurations. For example, in a network with redundant Cisco Catalyst SD-WAN Controllers, each controller must be configured with identical policies. Another example is a network with Cisco vEdge devices at multiple sites, where each Cisco vEdge device is providing identical services at each site.

Because the configurations for these devices are essentially identical, you can create one set of feature templates, which you then consolidate into one device template that you use to configure all the devices. You can create an Excel file in CSV format that lists the variables and defines each device specific variable value for each device. Then you can load the file when you attach a device template to a device.

To export data for all devices to a file in CSV format, click the Export icon. This icon, which is a downward-pointing arrow, is located to the right of the filter criteria both in the WAN Edge List and in the Controllers tab.



Note Starting from Cisco IOS XE Cisco IOS XE Catalyst SD-WAN Release 17.13.1a, the **Controllers** tab is renamed as the **Control Components** tab to stay consistent with Cisco Catalyst SD-WAN rebranding.

Cisco SD-WAN Manager downloads all data from the device table to an Excel file in CSV format.

View and Copy Device Configuration

View a Device's Running Configuration

Running configuration is configuration information that Cisco SD-WAN Manager obtains from the memory of a device. This information can be useful for troubleshooting.

To view a device's running configuration:

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Devices**.
2. Click **WAN Edge List** or **Controllers**, and select the device.



Note Starting from Cisco IOS XE Catalyst SD-WAN Release 17.13.1a, the **Controllers** tab is renamed as the **Control Components** tab to stay consistent with Cisco Catalyst SD-WAN rebranding.

3. Click ..., and click **Running Configuration**.

View a Device's Local Configuration

Local configuration is configuration that Cisco SD-WAN Manager has stored for a device. This information can be useful for troubleshooting or for determining how to access a device if, for example, a device is not reachable from Cisco SD-WAN Manager.

To view a device's local configuration created using Configuration ► Templates:

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Devices**.
2. Click **WAN Edge List** or **Controllers**, and select the device.



Note Starting from Cisco IOS XE Catalyst SD-WAN Release 17.13.1a, the **Controllers** tab is renamed as the **Control Components** tab to stay consistent with Cisco Catalyst SD-WAN rebranding.

3. Click ..., and click **Local Configuration**.

Copy Router Configuration

When you are replacing one router at a site with another router, you copy the old router's configuration to the new router. Then you remove the old router from the network and add the new one.

To copy the configuration from the old router to the new router:

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Certificates**.
2. Mark the new Cisco vEdge device as invalid.
3. From the Cisco SD-WAN Manager menu, choose **Configuration > Devices**.
4. Under **WAN Edge List**, select the old router.
5. Click ..., and click **Copy Configuration**.
6. In the **Copy Configuration** window, select the new router.
7. To confirm the copy of the configuration, click **Update**.

After you have copied the configuration to the new router, you can add the new router to the network. First, delete the old router from the network, as described below. Then add the new router to the network:

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Certificates**.
2. Mark the new router as valid.
3. Click **Send to Controller**.

Delete a WAN Edge Router

Delete a router if you need to remove it from your deployment. Doing so removes from the WAN edge router serial number list any of the following items that are stored for the router:

- Chassis number
- Certificate serial number
- Subject SUDI serial number



Note Deleting a router also permanently removes the router configuration from Cisco SD-WAN Manager.

To delete a router:

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Certificates**.
2. Mark the WAN Edge router as invalid.
3. From the Cisco SD-WAN Manager menu, choose **Configuration > Devices**.
4. Click **WAN Edge List**, and select the router.
5. Click ..., and click **Delete WAN Edge**.
6. To confirm deletion of the device, click **OK**.
7. From the Cisco SD-WAN Manager menu, choose **Configuration > Certificates**.
8. Click **Send to Controller**.

Decommission a Cloud Router

Decommissioning a cloud router (such as a vEdge Cloud router) removes the device's serial number from Cisco SD-WAN Manager and generates a new token for the device. To do so:

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Devices**.
2. Click **WAN Edge List**, and select a cloud router.
3. Click ..., and click **Decommission WAN Edge**.
4. To confirm the decommissioning of the router, click **OK**.

From Cisco Catalyst SD-WAN Manager Release 20.15.1, the process to decommission a WAN edge router has been modified. The scenarios below highlight the updates.

Table 63: Updates to Deleting a WAN Edge Router

Scenario	Action
Decommission a compromised device	Click Delete WAN Edge

Scenario	Action
Device is reachable	Perform a mandatory configuration unlock before proceeding with the decommissioning process.
Device is unreachable	The device will be unlocked after certain time when the device is unreachable. If you no longer have the device onboarded, with no Cisco SD-WAN Manager visibility, you can hold the power button for 5–10 seconds for a config reset, or 10-20 seconds for a software reset. If you have console/terminal access, you can run the request config reset command. Reuse of UUID is only possible after decommissioning a device. In this case, the Decommission option is not available since the device is offline.

**Note**

- The **Decommission WAN Edge** feature applies only to cloud WAN edge devices and retains the cloud WAN Edge's UUID generated on the PnP Portal.
- Physical devices do not support the **Decommission WAN Edge** functionality.

View Template Log and Device Bringup

View Log of Template Activities

A log of template activities contains information that relates to creating, editing, and deleting configuration templates, and the status of attaching configuration templates to devices. This information can be useful for troubleshooting.

To view a log of template activities:

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Devices**.
2. Click **WAN Edge List** or **Controllers**, and select the device.

**Note**

Starting from Cisco IOS XE Cisco IOS XE Catalyst SD-WAN Release 17.13.1a, the Controllers tab is renamed as the Control Components tab to stay consistent with Cisco Catalyst SD-WAN rebranding.

3. Click ..., and click **Template Log**.

View Status of Device Bringup

You can view the status of the operations involved in bringing a router or controller up in the overlay network. This information can help you monitor these operations.

To view the status of a device bringup:

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Devices**.
2. Click **WAN Edge List** or **Controllers**, and select the device.



Note Starting from Cisco IOS XE Cisco IOS XE Catalyst SD-WAN Release 17.13.1a, the **Controllers** tab is renamed as the **Control Components** tab to stay consistent with Cisco Catalyst SD-WAN rebranding.

3. Click ..., and click **Device Bring Up**.

Add a Cisco SD-WAN Validator

A Cisco Catalyst SD-WAN Validator automatically orchestrates connectivity between Cisco vEdge devices and Cisco SD-WAN Manager. If any Cisco vEdge device or Cisco Catalyst SD-WAN Controller is behind a NAT, the Cisco Catalyst SD-WAN Validator also serves as an initial NAT-traversal orchestrator. To add a Cisco Catalyst SD-WAN Validator:

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Devices**.
2. Click **Controllers**.



Note Starting from Cisco IOS XE Catalyst SD-WAN Release 17.13.1a, the **Controllers** tab is renamed as the **Control Components** tab to stay consistent with Cisco Catalyst SD-WAN.

3. Click **Add Validator**.
4. In the **Add Validator** window:
 - a. Enter **Validator Management IP Address** of the Cisco SD-WAN Validator.
 - b. Enter the **Username** and **Password** to access the Cisco SD-WAN Validator.
 - c. To allow the certificate-generation process to occur automatically, check the **Generate CSR** check box.
 - d. Click **Add**.
5. Repeat Steps 2, 3 and 4 to add additional Cisco Catalyst SD-WAN Validators.

The new Cisco Catalyst SD-WAN Validator is added to the list of controllers in the Controllers screen.

Configure Cisco SD-WAN Controllers

Add a Cisco SD-WAN Controller

After the Cisco Catalyst SD-WAN Validator authenticates Cisco vEdge devices, the Cisco Catalyst SD-WAN Validator provides Cisco vEdge devices information that they need to connect to the Cisco Catalyst SD-WAN Controller. A Cisco Catalyst SD-WAN Controller controls the flow of data traffic throughout the network via data and app-route policies. To configure Cisco Catalyst SD-WAN Controllers:

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Devices**.
2. Click **Controllers**.



Note Cisco IOS XE Catalyst SD-WAN Release Cisco IOS XE Catalyst SD-WAN Release 17.13.1a, the **Controllers** tab is renamed as the **Control Components** tab to stay consistent with Cisco Catalyst SD-WAN rebranding.

3. Click **Add Controller**.
4. In the **Add Controller** window:
 - a. Enter the system IP address of the Cisco Catalyst SD-WAN Controller.
 - b. Enter the username and password to access the Cisco Catalyst SD-WAN Controller.
 - c. Select the protocol to use for control-plane connections. The default is DTLS. The DTLS (Datagram Transport Layer Security) protocol is designed to provide security for UDP communications.
 - d. If you select TLS, enter the port number to use for TLS connections. The default is 23456.
The TLS (Transport Socket Layer) protocol that provides communications security over a network.
 - e. Check the **Generate CSR** check box to allow the certificate-generation process to occur automatically.
 - f. Click **Add**.
5. Repeat Steps 2, 3 and 4 to add additional Cisco Catalyst SD-WAN Controllers. Cisco SD-WAN Manager can support up to 20 Cisco Catalyst SD-WAN Controllers in the network.

The new Cisco Catalyst SD-WAN Controller is added to the list of controllers in the Controllers screen.

Edit Controller Details

Editing controller details lets you update the IP address and login credentials of a controller device. To edit controller details:

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Devices**.
2. Click **Controllers**, and select the controller.



Note Cisco IOS XE Catalyst SD-WAN Release Cisco IOS XE Catalyst SD-WAN Release 17.13.1a, the **Controllers** tab is renamed as the **Control Components** tab to stay consistent with Cisco Catalyst SD-WAN rebranding.

3. Click **...**, and click **Edit**.

4. In the **Edit** window, edit the IP address and the login credentials.
5. Click **Save**.

Delete a Controller

Deleting a controller removes it from the overlay. Delete a controller if you are replacing it or if you no longer need it in your network.

To delete a controller:

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Devices**.
2. Click **Controllers**, and select the controller.



Note Cisco IOS XE Catalyst SD-WAN Release Cisco IOS XE Catalyst SD-WAN Release 17.13.1a, the **Controllers** tab is renamed as the **Control Components** tab to stay consistent with Cisco Catalyst SD-WAN rebranding.

3. Click ..., and click **Invalidate**.
4. To confirm the removal of the device and all its control connections, click **OK**.

Configure Reverse Proxy on Controllers

To configure reverse proxy on an individual Cisco SD-WAN Manager and Cisco Catalyst SD-WAN Controller:

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Devices**.
2. Click **Controllers**, and select the controller.



Note Cisco IOS XE Catalyst SD-WAN Release Cisco IOS XE Catalyst SD-WAN Release 17.13.1a, the **Controllers** tab is renamed as the **Control Components** tab to stay consistent with Cisco Catalyst SD-WAN rebranding.

3. Click ..., and click **Add Reverse Proxy**.
The **Add Reverse Proxy** dialog box is displayed.
4. Click **Add Reverse Proxy**.
5. Configure the private IP address and port number for the device. The private IP address is the IP address of the transport interface in VPN 0. The default port number is 12346. This is the port used to establish the connections that handle control and traffic in the overlay network.
6. Configure the proxy IP address and port number for the device, to create the mapping between the private and public IP addresses and port numbers.
7. If the Cisco SD-WAN Manager NMS or Cisco Catalyst SD-WAN Controller has multiple cores, repeat Steps 5 and 6 for each core.
8. Click **Add**.

To enable reverse proxy in the overlay network, from the Cisco SD-WAN Manager menu, choose **Administration > Settings**. Then **Proxy**. Go to **Reverse Proxy**, and enable **Reverse Proxy**. Click **Save**.

Configure UCSE Using a Configuration Group

Before you begin

On the **Configuration > Configuration Groups** page, choose **SD-WAN** as the solution type.

Procedure

Step 1 From the Cisco SD-WAN Manager menu, choose **Configuration > Configuration Groups**.

Step 2 Create and configure a UCSE feature in Other profile.

- a. Configure parameter scope.

Table 64: Parameter

Parameter Scope	Scope Description
Global (Indicated by a globe icon)	Enter a value for the parameter and apply that value to all devices. Examples of parameters that you might apply globally to a group of devices are DNS server, syslog server, and interface MTUs.
Device Specific (Indicated by a host icon)	Use a device-specific value for the parameter. Choose Device Specific to provide a value for the key in the Enter Key field. The key is a unique string that helps identify the parameter. To change the default key, type a new string in the Enter Key field. Examples of device-specific parameters are system IP address, host name, GPS location, and site ID.
Default (indicated by a check mark)	The default value is shown for parameters that have a default setting.

- b. Configure options for the UCSE feature.

Table 65: Settings

Field	Description
Type	Choose a feature from the drop-down list.
Feature Name*	Enter a name for the feature. The name can be up to 128 characters and can contain only alphanumeric characters.
Description	Enter a description of the feature. The description can be up to 2048 characters and can contain only alphanumeric characters.

- c. Configure basic settings.

Table 66: Basic Configuration

Field	Description
Bay*	Specify the number for the SAS drive bays. The input value must be an integer.
Slot*	Specify the slot numbers for the mezzanine adapters. The input value must be an integer.

- d. Configure IMC.

Table 67: IMC

Field	Description
Access Port	Configure the interface as an access port. You can configure only one VLAN on an access port, and the port can carry traffic for only one VLAN. Not all hardware models have a dedicated access port. See the release notes for your Cisco Catalyst SD-WAN release for the supported hardware. Available options: • Dedicated • Shared Configure the appropriate port (GE or TE) based on the hardware module.
IPv4 Address*	Provide the UCS-E management port address.
Default Gateway*	Gateway tracking determine, for static routes, whether the next hop is reachable before adding that route to the device's route table. Default: Enabled.
VLAN ID	Provide the VLAN number, which can be a value from 1 through 4094.
Assign Priority	Assign the priority.

- e. Configure advanced settings.

Table 68: Advanced Configuration

Field	Description
Interface Name*	Specify the name of the interface.
Layer	Specify the layer details necessary for traffic exchange between different VLANs.
UCSE Interface VPN	Specify the details of the UCS-E interface VPN.
IPv4 Address	Provide the UCS-E management port address.

What to do next

Also see [Deploy a configuration group](#).



CHAPTER 7

Network Hierarchy and Resource Management

Table 69: Feature History

Feature Name	Release Information	Description
Network Hierarchy and Resource Management	Cisco SD-WAN Release 20.9.1 Cisco vManage Release 20.9.1	This feature enables you to create a network hierarchy in Cisco SD-WAN Manager to represent the geographical locations of your network. The network hierarchy and the associated resource IDs, including region IDs and site IDs, help you apply configuration settings to a device. In addition, the introduction of the resource manager in Cisco SD-WAN Manager automatically manages these resource IDs, thereby simplifying the overall user experience of Cisco Catalyst SD-WAN. Note that you can create a region only if you enable the Multi-Region Fabric option in Cisco SD-WAN Manager.
Network Hierarchy and Resource Management (Phase II)	Cisco IOS XE Catalyst SD-WAN Release 17.10.1a Cisco vManage Release 20.10.1	The following enhancements are introduced in the Network Hierarchy and Resource Management feature. • Creation of a system IP pool on the Configuration > Network Hierarchy page • Automatic assignment of site ID, system IP, and hostname to a device in the Quick Connect workflow • Display of detailed information on the Configuration > Network Hierarchy page, including site ID pool, region ID pool, and the list of devices associated with a site
Support for Software Defined Remote Access Pools	Cisco IOS XE Catalyst SD-WAN Release 17.11.1a Cisco vManage Release 20.11.1	Remote access refers to enabling secure access to an organization's network from devices at remote locations. The resource pool manager manages the IPv4 and IPv6 private IP address pools for Cisco Catalyst SD-WAN remote access devices. You can create a software defined remote access pool using the Configuration > Network Hierarchy page.

Feature Name	Release Information	Description
Support for Traffic Flow Collectors	Cisco IOS XE Catalyst SD-WAN Release 17.13.1a Cisco Catalyst SD-WAN Manager Release 20.13.1	This feature enables you to configure traffic flow collectors such as the Cflowd server and security logging server. Cflowd monitors service side traffic flowing through devices in the overlay network and exports flow information to the collector. Enable security logging and configure servers for high-speed logging (HSL) and collecting external syslogs. You can configure the traffic flow collectors by navigating to Configuration > Network Hierarchy > Collectors.

- [Information About Network Hierarchy and Resource Management, on page 206](#)
- [Supported Devices for Network Hierarchy and Resource Management, on page 207](#)
- [Restrictions for Network Hierarchy and Resource Management, on page 207](#)
- [Manage a Network Hierarchy, on page 208](#)
- [Assign Resource IDs to Devices, on page 217](#)
- [Configure Collectors in a Network Hierarchy, on page 220](#)

Information About Network Hierarchy and Resource Management

Overview of Network Hierarchy

You can create a network hierarchy in Cisco SD-WAN Manager to represent the geographical locations of your network. Your network hierarchy can contain three types of nodes—regions, areas, and sites. The resource IDs assigned to the nodes help you identify where to apply configuration settings later.

By default, there is one node called global in the network hierarchy.

The network hierarchy has a predetermined hierarchy with three types of nodes:

- **Region:** It represents a region in a multiregion fabric-based Cisco Catalyst SD-WAN deployment. The Multi-Region Fabric feature provides the option to divide the architecture of the Cisco Catalyst SD-WAN overlay network into multiple regional networks that operate distinctly from one another, and a central core-region network for managing inter-regional traffic.

You can create a region only if you enable the **Multi-Region Fabric** option in Cisco SD-WAN Manager. For complete information about the Multi-Region Fabric feature, see the [Cisco Catalyst SD-WAN Multi-Region Fabric \(also Hierarchical SD-WAN\) Configuration Guide](#).

- **Group/Area:** A group, also called an area, is a logical grouping of nodes in a network hierarchy. You can group sites, regions, other areas, or any combination of these into an area.
- **Site:** A site is the lowest level of node or the leaf node in a network hierarchy. You cannot create a child node under a site. You can only associate devices to a site.

For complete information about creating and managing different nodes in a network hierarchy, see [Manage a Network Hierarchy](#).

Overview of Resource Management

The resource manager in Cisco SD-WAN Manager manages the resource IDs, that is, region IDs and site IDs. It automatically generates a region ID for a region that you create on the **Configuration > Network Hierarchy** page. Similarly, it generates a site ID for a site if you do not specify it.

You can assign a site ID and a region ID to a device. For complete information about assigning resource IDs to devices, see [Assign Resource IDs to Devices](#).

If you upgrade from an earlier version of Cisco SD-WAN Manager to Cisco vManage Release 20.9.1, the resource manager in Cisco SD-WAN Manager automatically creates sites based on the site IDs of the existing devices in your setup. Sites are named as SITE_<id>. Cisco SD-WAN Manager displays these sites under the global node on the **Network Hierarchy** page. It also associates the existing devices with their sites in the network hierarchy.

Benefits of Network Hierarchy and Resource Management

- Automates the management of regions and sites.
- Saves the manual effort in an upgrade scenario when Cisco SD-WAN Manager discovers all your existing sites and displays them in the network hierarchy.
- Simplifies the onboarding and configuration of devices.
- Monitors and collects information about traffic flow.

Supported Devices for Network Hierarchy and Resource Management

This feature is supported on Cisco IOS XE Catalyst SD-WAN devices and Cisco vEdge devices.

Restrictions for Network Hierarchy and Resource Management

- You can delete a node only if it does not have any child node. For example, you can delete a site only if no devices are associated with it.
- A site is the lowest level of a node or the leaf node in a network hierarchy. You cannot create a child node under a site.
- You cannot create more than one region node between the global node and a site node.
- You cannot create a region in a multitenant deployment.
- The maximum combined number of regions and secondary regions is 63 (region ID numbers 1 through 63).

Manage a Network Hierarchy

The Network Hierarchy and Resource Management feature enables you to do the following:

- Create a region
- Create an area
- Create, edit, and delete a site

Create a Region in a Network Hierarchy

Before You Begin

(For Cisco Catalyst SD-WAN Manager Release 20.12.x or earlier) Ensure that the **Multi-Region Fabric** option in Cisco SD-WAN Manager is enabled. See [Enable Multi-Region Fabric](#) in the *Cisco Catalyst SD-WAN Multi-Region Fabric Configuration Guide*.

From Cisco Catalyst SD-WAN Manager Release 20.13.1, configuring regions is enabled by default. It does not require enabling Multi-Region Fabric.

Create a Region, Cisco Catalyst SD-WAN Manager Release 20.13.1 and Later

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Network Hierarchy**.
2. Click ... adjacent to **Global** in the left pane and choose **Add Node**.
3. Do one of the following:
 - If Multi-Region Fabric is not enabled:

In the **Add Node** pop-up window, check the **Behave as SDWAN Region** checkbox.

If you do not check this checkbox, this procedure creates a new group within the network hierarchy instead of a region.
 - If Multi-Region Fabric is enabled:

In the **Add Node** pop-up window, choose **Region**.
4. Configure the following:

Field	Description
Name	Name for the region. The name must be unique and can contain only letters, the digits 0 through 9, hyphens (-), underscores (_), and periods (.).
Description	Description of the region.
Parent drop-down list	Choose a parent node.

5. Click **Add**.

The new region appears in the left pane.

6. (Optional) You can click a region name or a secondary region name in the left pane to display the automatically assigned region ID number. The region ID number appears above the table in the right pane. The maximum combined number of regions and secondary regions is 63 (region ID numbers 1 through 63).

Create a Region, Cisco Catalyst SD-WAN Manager Release 20.12.x or Earlier

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Network Hierarchy**.
2. Click ... adjacent to a node (global or area) in the left pane and choose **Add MRF Region**.



Note In Cisco vManage Release 20.9.x, you can also use the **Add Node** option to add a region.

3. In the **Name** field, enter a name for the region. The name must be unique and can contain only letters, the digits 0 through 9, hyphens (-), underscores (_), and periods (.).
4. In the **Description** field, enter a description of the region.
5. From the **Parent** drop-down list, choose a parent node.
6. Click **Add**.

Create a Subregion in a Network Hierarchy



Note From Cisco IOS XE Catalyst SD-WAN Release 17.15.1a and Cisco Catalyst SD-WAN Control Components Release 20.15.1, configuration of this feature is supported only through API.

Before You Begin

Minimum supported release: Cisco Catalyst SD-WAN Manager Release 20.13.1

- From Cisco Catalyst SD-WAN Manager Release 20.13.1, configuring subregions is enabled by default. It does not require enabling Multi-Region Fabric.
- Create a region before creating a subregion. See [Create a Region in a Network Hierarchy, on page 208](#).
- For the maximum combined number of regions and secondary regions, see [Restrictions for Network Hierarchy and Resource Management, on page 207](#).

Create a Subregion

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Network Hierarchy**.
2. Click ... adjacent to a region in the left pane and choose **Add MRF Sub Region**.
3. In the **Add Sub-Region** pop-up window, configure the following:

Field	Description
Name	Name for the region. The name must be unique and can contain only letters, the digits 0 through 9, hyphens (-), underscores (_), and periods (.).
Description	Description of the region.
Parent	This field is automatically populated with the region to which you are adding the subregion, and is not configurable.

- Click **Add**.

The new subregion appears in the left pane.

Create a Secondary Region in a Network Hierarchy



Note From Cisco IOS XE Catalyst SD-WAN Release 17.15.1a and Cisco Catalyst SD-WAN Control Components Release 20.15.1, configuration of this feature is supported only through API.

Before You Begin

Minimum supported release: Cisco Catalyst SD-WAN Manager Release 20.13.1

- Create a region before creating a subregion. See [Create a Region in a Network Hierarchy, on page 208](#).
- For the maximum combined number of regions and secondary regions, see [Restrictions for Network Hierarchy and Resource Management, on page 207](#).

Create a Secondary Region

- From the Cisco SD-WAN Manager menu, choose **Configuration > Network Hierarchy**.
- Click ... adjacent to **Global** in the left pane and choose **Add Node**.
- In the **Add Node** pop-up window, click **Secondary Region**.
- Configure the following:

Field	Description
Name	Name for the region. The name must be unique and can contain only letters, the digits 0 through 9, hyphens (-), underscores (_), and periods (.).
Description	Description of the region.
Parent	This field shows Secondary Regions , and is not configurable.

- Click **Add**.

The new secondary region appears in the left pane, in the **Secondary Regions** section.

6. (Optional) You can click a region name or a secondary region name in the left pane to display the automatically assigned region ID number. The region ID number appears above the table in the right pane. The maximum combined number of regions and secondary regions is 63 (region ID numbers 1 through 63).

Create a Group in a Network Hierarchy

Before You Begin

In Cisco Catalyst SD-WAN Manager Release 20.12.x and earlier, a group is called an area.

Create a Group

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Network Hierarchy**.
2. Click ... adjacent to a node (global, region, or group) in the left pane and choose **Add Node**.
3. In the **Add Node** pop-up window, in the **Type** field, choose **Group**.
(In Cisco Catalyst SD-WAN Manager Release 20.12.x and earlier, choose **Add Area**.)



Note In Cisco vManage Release 20.9.x, you can also use the **Add Node** option to add an area.

4. In the **Name** field, enter a name for the group. The name must be unique and can contain only letters, the digits 0 through 9, hyphens (-), underscores (_), and periods (.).
5. In the **Description** field, enter a description of the group.
6. From the **Parent** drop-down list, choose a parent node.
7. Click **Add**.

Create a Site in a Network Hierarchy

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Network Hierarchy**.
2. Click ... adjacent to a node (global, region, or area) in the left pane and choose **Add Site**.



Note In Cisco vManage Release 20.9.x, you can also use the **Add Node** option to add a site.

3. In the **Name** field, enter a name for the site. The name must be unique and can contain only letters, the digits 0 through 9, hyphens (-), underscores (_), and periods (.).
4. In the **Description** field, enter a description of the site.
5. From the **Parent** drop-down list, choose a parent node.

6. In the **Site ID** field, enter a site ID.
If you do not enter the site ID, Cisco SD-WAN Manager generates a site ID for the site.
7. In the **Address** field, enter the address.
When you enter an address, latitude and longitude fields are auto populated.
8. In the **Latitude** field, enter the latitude of the site.
9. In the **Longitude** field, enter the longitude of the site.
10. Click **Add**.



Note The **Address**, **Latitude**, and **Longitude** fields are included starting from Cisco IOS XE Catalyst SD-WAN Release 17.18.1a. When you type in the address, the latitude and longitude fields are auto-populated.

Conversely, you can also enter your own latitude and longitude. If a corresponding location is found, then the address field will be auto-populated. If you do not want to provide address data for your site, then you can check the **Undisclosed Address** check box.

Field	Description
Name	Enter a name for the site. The name must be unique and can contain only letters, the digits 0 through 9, hyphens (-), underscores (_), and periods (.).
Description	Enter a description of the site.
Parent	From the drop-down list, choose a parent node.
Site ID	Enter a site ID. If you do not enter the site ID, Cisco SD-WAN Manager generates a site ID for the site.
Address	Enter the address. When you enter an address, latitude and longitude fields are auto populated.
Latitude	Specifies latitude of the site. When you enter the latitude of the site, the address field is auto populated if the location of the site is found.
Longitude	Specifies longitude of the site. When you enter the longitude of the site, the address field is auto populated if the location of the site is found.

Edit a WAN Region

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Network Hierarchy**.
2. Click ... adjacent to the region name and choose **Edit WAN Region**.

In releases earlier than Cisco Catalyst SD-WAN Manager Release 20.15.1, choose **Edit MRF Region**.

3. Edit the options as needed. You can edit the name, description, and parent of the region.
4. Click **Save**.

Delete a WAN Region

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Network Hierarchy**.
2. Click ... adjacent to the region name and choose **Delete WAN Region**.

In releases earlier than Cisco Catalyst SD-WAN Manager Release 20.15.1, choose **Delete MRF Region**.

3. In the confirmation dialog box, click **Yes**.

Edit a Group

Before You Begin

In Cisco Catalyst SD-WAN Manager Release 20.12.x and earlier, a group is called an area.

Edit a Group

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Network Hierarchy**.
2. Click ... adjacent to the group name and choose **Edit Group**.
(In Cisco Catalyst SD-WAN Manager Release 20.12.x and earlier, choose **Edit Area**.)
3. Edit the options as needed. You can edit the name, description, and parent of the group.
4. Click **Save**.

Delete a Group

Before You Begin

In Cisco Catalyst SD-WAN Manager Release 20.12.x and earlier, a group is called an area.

Delete a Group

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Network Hierarchy**.
2. Click ... adjacent to the group name and choose **Delete Group**.
(In Cisco Catalyst SD-WAN Manager Release 20.12.x and earlier, choose **Delete Area**.)
3. In the confirmation dialog box, click **Yes**.

Edit a Site

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Network Hierarchy**.
2. Click ... adjacent to the site name and choose **Edit Site**.
3. Edit the options as needed. You can edit only the name, description, and parent of the site.
4. Click **Save**.



Note After reassigning a site to a different region in the **Network Hierarchy Manager**, follow the procedures in the sections [Assign a Site ID to a Device](#) and [Assign a Region ID to a Device](#) to ensure configuration updates are properly applied to the affected device.

Delete a Site

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Network Hierarchy**.
2. Click ... adjacent to the site name and choose **Delete Site**.
3. In the confirmation dialog box, click **Yes**.

Create a System IP Pool

Minimum releases: Cisco IOS XE Catalyst SD-WAN Release 17.10.1a, Cisco vManage Release 20.10.1

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Network Hierarchy**.
The page displays the site pool and region pool for the Global node.
2. Click **Pools**.
3. Click **Add Pool**.
4. In the **Pool Name** field, enter a name for the pool.
5. In the **Pool Description** field, enter a description of the pool.
6. From the **Pool Type** drop-down list, choose **System IP**.
7. In the **IP Subnet*** field, enter an IP address.
8. In the **Prefix Length*** field, enter the prefix length of the system IP pool.
9. Click **Add**.



Note You can create only one system IP pool. If you want to make any changes to the pool, you must edit the existing pool.

Edit a System IP Pool

Minimum releases: Cisco IOS XE Catalyst SD-WAN Release 17.10.1a, Cisco vManage Release 20.10.1

1. From the Cisco SD-WAN Manager menu, choose **Configuration** > **Network Hierarchy**.

The page displays the site pool and region pool for the Global node. The system IP pool is also displayed if you have already created it.

2. Click ... adjacent to the system IP name and choose **Edit**.
3. Edit the options as needed.



Note You can only expand the pool range and cannot enter a lower IP address than the already specified IP address.

4. Click **Save**.

Create a Remote Access Pool

Minimum supported release: Cisco vManage Release 20.11.1

The resource pool manager supports creation of IPv4 and IPv6 private IP pools for Cisco Catalyst SD-WAN remote access devices. In the remote access configuration you can select the remote access private IP Pool by defining the number of IP addresses.

For more information on Software Defined Remote Access, see [Cisco Catalyst SD-WAN Remote Access](#).

1. From the Cisco SD-WAN Manager menu, choose **Configuration** > **Network Hierarchy**.

The page displays the site pool and region pool for the Global node.

2. Click **Add Pool**.
3. In the **Pool Name** field, enter a name for the pool.
4. In the **Pool Description** field, enter a description of the pool.
5. From the **Pool Type** drop-down list, choose **Remote Access**.
6. Choose the **IP Type** by clicking the radio button next to **IPv4** or **IPv6**.
7. In the **IP Subnet** field, enter an IP subnet.
8. In the **Prefix Length** field, enter the prefix length of the remote access pool.
9. Click **Add**.

Edit a Remote Access Pool

Minimum supported release: Cisco vManage Release 20.11.1

You can edit a remote access pool only when you want to expand the pool range.

1. From the Cisco SD-WAN Manager menu, choose **Configuration** > **Network Hierarchy**.

The page displays the site pool and region pool for the Global node. The remote access pool is also displayed if you have already created it.

2. Click ... adjacent to the remote access pool name and choose **Edit**.
3. Edit the options as needed.



Note When you edit a remote access pool, the new pool range cannot be less than the existing pool range

4. Click **Save**.

Create an IP Pool for ThousandEyes

SUMMARY STEPS

1. From the menu, choose **Configuration > Network Hierarchy**.
2. Examine the audit logs to determine if the ThousandEyes test has been posted or failed.
3. Click **Pools**.
4. Click **Add Pool**.
5. In the **Pool Name** field, enter a name for the pool.
6. In the **Pool Description** field, enter a description of the pool.
7. From the **Pool Type** drop-down list, choose **ThousadEyes**.
8. In the **IP Subnet*** field, enter an IP address.
9. In the **Prefix Length*** field, enter the prefix length of the system IP pool.
10. Click **Add**.

DETAILED STEPS

Procedure

	Command or Action	Purpose
Step 1	From the menu, choose Configuration > Network Hierarchy .	The page displays the site pool and region pool for the Global node.
Step 2	Examine the audit logs to determine if the ThousandEyes test has been posted or failed.	
Step 3	Click Pools .	
Step 4	Click Add Pool .	
Step 5	In the Pool Name field, enter a name for the pool.	
Step 6	In the Pool Description field, enter a description of the pool.	
Step 7	From the Pool Type drop-down list, choose ThousadEyes .	
Step 8	In the IP Subnet* field, enter an IP address.	

	Command or Action	Purpose
Step 9	In the Prefix Length* field, enter the prefix length of the system IP pool.	
Step 10	Click Add .	

Delete a Pool

Minimum supported release: Cisco vManage Release 20.11.1

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Network Hierarchy**.
2. In the Global page, click ... adjacent to the pool name and choose **Delete**.
3. In the confirmation dialog box, click **Yes**.

**Note**

You can delete a pool only when the pool resources are not in use.

Assign Resource IDs to Devices

The Network Hierarchy and Resource Management feature enables you to do the following:

- Assign a site ID to a device
- Assign a region ID to a device

Assign a Site ID to a Device

You can assign a site ID to a device using one of the following ways.

Use the Quick Connect Workflow

1. From the Cisco SD-WAN Manager menu, choose **Workflows > Workflow Library**.
2. Start the **Quick Connect** workflow.
3. Follow the instructions provided in the workflow.
4. On the **Add and Review Device Configuration** page, enter the site ID of the device.

**Note**

- You can use any of the existing site IDs that are available in the network hierarchy or enter a new site ID. If you enter a new site ID without creating a node in the network hierarchy, the site is automatically created and listed on the **Configuration > Network Hierarchy** page.
- (Minimum releases: Cisco IOS XE Catalyst SD-WAN Release 17.10.1a, Cisco vManage Release 20.10.1)
If you want Cisco SD-WAN Manager to automatically generate a site ID for the device, do not make any change to the default value, **AUTO**.

Use a Template

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Devices > WAN Edge List**.
2. Check if a device is attached to a device template.
3. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates > Feature Templates**.
4. Click ... adjacent to the System feature template and choose **Edit**.
5. Click the **Basic Configuration** tab and set the scope of the **Site ID** field to **Global** and enter the site ID.
6. Click **Update**.
7. Click **Configure Devices** to push the configuration to the device.

In Step 5, if you set the scope of the **Site ID** field to **Device Specific**, do the following:

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates > Device Templates**.
2. Click ... adjacent to the device template and choose **Edit Device Template**.
3. In the **Site ID** field, enter the site ID.

You can use any of the existing site IDs that are available in the network hierarchy or enter a new site ID. If you enter a new site ID without creating a node in the network hierarchy, the site is automatically created and listed on the **Configuration > Network Hierarchy** page.

4. Click **Update**.
5. Click **Configure Devices** to push the configuration to the device.

Use a Configuration Group

The configuration group flow is applicable only for the Cisco IOS XE Catalyst SD-WAN devices.

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates > Configuration Groups**.
2. Click ... adjacent to the configuration group name and choose **Edit**.
3. Click **Associated Devices**.
4. Choose a device that is associated with the configuration group and click **Deploy**.
The **Deploy Configuration Group** workflow starts.
5. Follow the instructions provided in the workflow.

6. On the **Add and Review Device Configuration** page, enter the site ID of the device.

You can use any of the existing site IDs that are available in the network hierarchy or enter a new site ID. If you enter a new site ID without creating a node in the network hierarchy, the site is automatically created and listed on the **Configuration > Network Hierarchy** page.

Assign a Region ID to a Device

Before You Begin

- Have access to the **Multi-Region Fabric** feature.
- Ensure that the region is available in the network hierarchy.

Assign a Region ID

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Devices > WAN Edge List**.
2. Check if the corresponding device is attached to a device template.
3. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates > Feature Templates**.
4. Click ... adjacent to the System feature template and choose **Edit**.
5. Click the **Basic Configuration** tab and set the scope of the **Region ID** field to **Global** and enter the region ID.

You can use any of the existing region IDs that are available in the network hierarchy. If the specified region ID is not available in the network hierarchy, the template push operation to the devices fails.

6. Click **Update**.
7. Click **Configure Devices** to push the configuration to the device.

In Step 5, if you set the scope of the **Region ID** field to **Device Specific**, do the following:

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates > Device Templates**.
2. Click ... adjacent to the device template and choose **Edit Device Template**.
3. In the **Region ID** field, enter the region ID.
4. Click **Update**.
5. Click **Configure Devices** to push the configuration to the device.

Assign a System IP to a Device

Minimum releases: Cisco IOS XE Catalyst SD-WAN Release 17.10.1a, Cisco vManage Release 20.10.1

1. From the Cisco SD-WAN Manager menu, choose **Workflows > Workflow Library**.
2. Start the **Quick Connect** workflow.
3. Follow the instructions provided in the workflow.

4. On the **Add and Review Device Configuration** page, enter the system IP of the device. If you want Cisco SD-WAN Manager to automatically generate a system IP for the device, do not make any change to the default value, **AUTO**.

Assign a Hostname to a Device

Minimum releases: Cisco IOS XE Catalyst SD-WAN Release 17.10.1a, Cisco vManage Release 20.10.1

1. From the Cisco SD-WAN Manager menu, choose **Workflows > Workflow Library**.
2. Start the **Quick Connect** workflow.
3. Follow the instructions provided in the workflow.
4. On the **Add and Review Device Configuration** page, enter the hostname of the device. If you want Cisco SD-WAN Manager to automatically generate a hostname for the device, do not make any change to the default value, **AUTO**.

Configure Collectors in a Network Hierarchy

Minimum supported releases: Cisco IOS XE Catalyst SD-WAN Release 17.13.1a, Cisco Catalyst SD-WAN Manager Release 20.13.1.

Configure Cflowd and security logging servers that help monitor traffic flow and collect information about service-side traffic.

Information About Collectors

Minimum supported releases: Cisco IOS XE Catalyst SD-WAN Release 17.13.1a, Cisco Catalyst SD-WAN Manager Release 20.13.1.

Collectors process traffic flowing through routers in the overlay network and export flow information to a server. The collectors maintain information about the flow and data that is extracted from the IP headers of the packets in the traffic flow.

You can configure the location of cflowd collectors, how often sets of sampled flows are sent to the collectors, and how often the samples are sent to the collectors (on Cisco SD-WAN Controllers only). You can configure a maximum of four cflowd collectors per Cisco IOS XE Catalyst SD-WAN Device. To have a cflowd configuration take effect, apply it with the appropriate data policy.

Configure Cflowd

Minimum supported releases: Cisco IOS XE Catalyst SD-WAN Release 17.13.1a, Cisco Catalyst SD-WAN Manager Release 20.13.1.

Before You Begin

You can configure the location of cflowd collectors, how often sets of sampled flows are sent to the collectors, and how often the samples are sent to the collectors (on Cisco SD-WAN Controllers only). You can configure

a maximum of four cflowd collectors per Cisco IOS XE Catalyst SD-WAN device. To have a cflowd configuration take effect, apply it with the appropriate data policy.

Ensure that you specify the granular role-based access control (RBAC) for Cflowd and policy groups. With specific permissions to the user group, ensure that you are able to access policy groups from **Configuration > Policy Groups**. For more information about configuring RBAC for policy groups, see Configure RBAC for policy groups in [Prerequisites for Policy Groups](#).

1. From the Cisco SD-WAN Manager menu, choose **Administration > Users and Access > Roles**.
2. Click **Edit** next to existing roles or click **Add Role** to create a new role.
3. Choose the desired permission for the **Cflowd** feature under **Network Settings** and click **Update**.

Configure Cflowd

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Network Hierarchy > Collectors**.
2. Enable Cflowd and configure the values in the following table for the collector server:

Field	Description
Add Collector Server	
VPN ID	VPN ID of the server. Range: 1 to 65525, excluding 512. For details see the VRF range behavior change described here .
IPv4/IPv6 Address	IPv4 or IPv6 address of the collector server.
UDP Port	UDP port number of the collector server. Range: 1024 through 65535
Export Spreading	Toggle to enable or disable the export spreading configuration.
BFD Metrics Exporting	Toggle to enable or disable Bidirectional Forwarding Detection (BFD) metrics.
Exporting Interval	Interval in seconds for sending BFD metrics. Exporting Interval appears if you have enabled BFD Metrics Exporting . The default BFD export interval is 600 seconds.
Advanced Settings	
Active Flow Timeout (Seconds)	Active flow timeout value. Range: 30 through 3600 Default: 600 seconds.
Inactive Flow Timeout (Seconds)	Inactive flow timeout value. Range: 1 through 3600 Default: 60 seconds.

Field	Description
Flow Refresh Time (Seconds)	Flow refresh time in seconds. Range: 60 through 86400 seconds. Default: 600 seconds.
Sampling Rate	Sample duration in seconds. Range: 1 through 65536. Default: 1 second.
Collect TLOC Loopback	Enable to collect information about the TLOC loopback.
Protocol	Traffic protocol type to apply the collector to. The options are: IPv4 , IPv6 , or both . The default protocol is IPv4 .
TOS	Type of field in the IPv4 header.
Re-marked DSCP	Traffic output of the router's data policy.

You can configure up to four collector servers.

3. Click **Save**.

The Cflowd settings that you configure are applied to the application priority and SLA policy when the policy is deployed to Cisco Catalyst SD-WAN devices. You can monitor application traffic flow over IPv4, IPv6, or both network addresses. For more information about configuring additional settings, see **Monitor traffic flow** in [Application Priority and SLA](#).

Configure Security Logging

Minimum supported releases: Cisco IOS XE Catalyst SD-WAN Release 17.13.1a, Cisco Catalyst SD-WAN Manager Release 20.13.1.

Configure Security Logging

You can set up security logging for Cisco IOS XE Catalyst SD-WAN devices by configuring the location of the destination IP address of the log server. You can configure up to four destination servers along with the source interface to collect the syslogs for High Speed Logging (HSL). The IP address for the destination server can be IPv4, IPv6, or both. For more information about configuring HSL, see [Configure Firewall High-Speed Logging Using the CLI Template](#). You can configure the external syslog server to export UTD logs. For more information about UTD logging, see [Create Unified Security Policy Summary](#) page.

Before You Begin

Ensure that you specify the granular role-based access control (RBAC) for security logging. Ensure that you are able to access policy groups from **Configuration > Policy Groups** by configuring specific permissions to the user group. For more information about configuring RBAC for policy groups, see "Configure RBAC for policy groups" in [Prerequisites for Policy Groups](#).

1. From the Cisco SD-WAN Manager menu, choose **Administration > Users and Access > Roles**.

2. Click **Edit** adjacent to existing roles or click **Add Role** to create a new role.
3. Choose the permission you wish to configure for the **Security Logging** feature under **Network Settings** and click **Update**.

Configure Security Logging

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Network Hierarchy > Collectors**.
2. Enable **Security Logging** and configure the values in the following table for the high-speed logging and external syslog servers:

Field	Description
High Speed Logging	Configure the following values for the high-speed logging server: • VPN: VPN name of the high-speed logging server. The VPNs available in the drop-down list are ones that are previously configured in the configuration groups in Cisco SD-WAN Manager. • Server IP: IPv4 or IPv6 address of the log collector server. • Port: Port number on which the log collector server is listening for incoming packets.
External Syslog Server	Configure the following values for the external syslog server: • VPN: VPN name of the external syslog server. The VPNs available in the drop-down list are ones that are previously configured in the configuration groups in Cisco SD-WAN Manager. • Server IP: IPv4 or IPv6 address of the external syslog server.

You can configure up to four high-speed logging servers.



Note

Starting from Cisco IOS XE Catalyst SD-WAN Release 17.16.1a and Cisco Catalyst SD-WAN Manager Release 20.16.1, server labels (**Server 1**, **Server 2**, **Server 3**, and **Server 4**) are added to the high-speed logging server and the syslog server.

For device-specific server settings, add the associated source interface in the **Additional Settings** of the NGFW policy. For more information about configuring additional settings in the policy groups, see [Configure NGFW Additional Settings](#) section.

For Global server settings, define the global NHM values. For more information about defining global values on NHM, see [Network Hierarchy and Resource Management](#).

3. Click **Save**.

The security logging settings that you configure are applied along with the embedded security policy when the policy is deployed to Cisco Catalyst SD-WAN devices. For more information about configuring the embedded security policy, see [Configure Embedded Security](#).



CHAPTER 8

Configure Network Interfaces

In the Cisco Catalyst SD-WAN overlay network design, interfaces are associated with VPNs. The interfaces that participate in a VPN are configured and enabled in that VPN. Each interface can be present only in a single VPN.

At a high level, for an interface to be operational, you must configure an IP address for the interface and mark it as operational (**no shutdown**). In practice, you always configure additional parameters for each interface.

You can configure up to 512 interfaces on a Cisco vEdge device. This number includes physical interfaces, loopback interfaces, and subinterfaces.



Note To maximize the efficiency of the load-balancing among Cisco Catalyst SD-WAN Controllers, use sequential numbers when assigning system IP addresses to the Cisco vEdge devices in the domain. Example of a sequential numbering schemes is 172.16.1.1, 172.16.1.2, 172.16.1.3, and so on.



Note Ensure that any network interface configured on a device has a unique IP address. If the IP address of the interface conflicts with the system IP address of Cisco SD-WAN Manager instance, it can break the NETCONF session and lead Cisco SD-WAN Manager to read the device as offline.



Note If you try to configure an interface or sub-interface beyond the supported limit, the device generates a notification to Cisco SD-WAN Manager.

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Configure VPN

VPN

Use the VPN template for all Cisco Catalyst SD-WAN devices running the Cisco Catalyst SD-WAN software.

To configure VPNs using Cisco SD-WAN Manager templates, follow this general workflow:

1. Create VPN feature templates to configure VPN parameters. You create a separate VPN feature template for each VPN. For example, create one feature template for VPN 0, a second for VPN 1, and a third for VPN 512.

For Cisco SD-WAN Manager Network Management Systems and Cisco Catalyst SD-WAN Controllers, you can configure only VPNs 0 and 512. Create templates for these VPNs only if you want to modify the default settings for the VPN. For Cisco vEdge devices, you can create templates for these two VPNs and for additional VPN feature templates to segment service-side user networks.

- **VPN 0—Transport VPN**, which carries control traffic via the configured WAN transport interfaces. Initially, VPN 0 contains all of a device's interfaces except for the management interface, and all interfaces are disabled.
- **VPN 512—Management VPN**, which carries out-of-band network management traffic among the Cisco vEdge devices in the overlay network. The interface used for management traffic resides in VPN 512. By default, VPN 512 is configured and enabled on all Cisco vEdge devices except for Cisco vEdge 100. For controller devices, by default, VPN 512 is not configured.

- **VPNs 1–511, 513–65525—Service VPNs**, for service-side data traffic on Cisco vEdge devices. For details, see the VRF range behavior change described [here](#).

2. Create interface feature templates to configure the interfaces in the VPN.

Create a VPN Template



Note You can configure a static route through the VPN template.

Procedure

- Step 1** From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
- Step 2** Click **Device Templates**, and click **Create Template**.

Note
In Cisco vManage Release 20.7.x and earlier releases **Device Templates** is called **Device**.
- Step 3** From the **Create Template** drop-down list, choose **From Feature Template**.
- Step 4** From the **Device Model** drop-down list, choose the type of device for which you wish to create the template.
- Step 5** To create a template for VPN 0 or VPN 512:
 - a. Click **Transport & Management VPN**, or scroll to the **Transport & Management VPN** section.
 - b. From the VPN 0 or VPN 512 drop-down list, click **Create Template**. The VPN template form appears.
The form contains fields for naming the template, and fields for defining VPN parameters.
- Step 6** To create a template for VPNs 1 through 511, and 513 through 65530: Create a template for VPNs. Range: 1 to 65525, excluding 512. For details see the VRF range behavior change described [here](#).
 - a. Click **Service VPN**, or scroll to the **Service VPN** section.
 - b. Click the **Service VPN** drop-down list.
 - c. From the **VPN** drop-down list, click **Create Template**. The VPN template form displays.
The form contains fields for naming the template, and fields for defining VPN parameters.
- Step 7** In **Template Name**, enter a name for the template. The name can be up to 128 characters and can contain only alphanumeric characters.
- Step 8** In **Template Description**, enter a description of the template. The description can be up to 2048 characters and can contain only alphanumeric characters.

Configure Service VPN Using Configuration Groups

This feature helps you configure a service VPN (Range: 1 to 65525, excluding 512. For details see the VRF range behavior change described [here](#).) or the LAN VPN.

The following tables describes the options for configuring the Service VPN feature.

Before you begin

On the **Configuration > Configuration Groups** page, choose **SD-WAN** as the solution type.

Procedure

Step 1 From the Cisco SD-WAN Manager menu, choose **Configuration > Configuration Groups**.

Step 2 Create and configure Service VPN in a Service profile.

a) Enter the basic configuration information.

Table 70: Basic Configuration

Field	Description
VPN*	Enter the numeric identifier of the VPN.
Name*	Enter a name for the VPN.
OMP Admin Distance IPv4	Administrative distance for OMP routes. The Cisco SD-WAN Controllers learn the topology of the overlay network and the services available in the network using OMP routes. The distance can be a value between 1–255.
OMP Admin Distance IPv6	Administrative distance for OMP routes. The Cisco SD-WAN Controllers learn the topology of the overlay network and the services available in the network using OMP routes. The distance can be a value between 1–255.

b) Enter DNS information.

Table 71: DNS

Field	Description
Add DNS IPv4	
Primary DNS Address (IPv4)	Enter the IP address of the primary IPv4 DNS server in this VPN.
Secondary DNS Address (IPv4)	Enter the IP address of a secondary IPv4 DNS server in this VPN.
Add DNS IPv6	
Primary DNS Address (IPv6)	Enter the IP address of the primary IPv6 DNS server in this VPN.
Secondary DNS Address (IPv6)	Enter the IP address of a secondary IPv6 DNS server in this VPN.

c) Enter host hopping information.

Table 72: Host Mapping

Field	Description
Add New Host Mapping	
Hostname*	Enter the hostname of the DNS server. The name can be up to 128 characters.
List of IP*	Enter up to eight IP addresses to associate with the hostname. Separate the entries with commas.

- d) Enter advertise OMP information.

Table 73: Advertise OMP

Field	Description
Add OMP Advertise IPv4	
Protocol	Choose a protocol to configure route advertisements to OMP, for this VPN: • bgp • ospf • ospfv3 • connected • static • network • aggregate Applied to Region: (Minimum supported release: Cisco Catalyst SD-WAN Manager Release 20.13.1) In a Multi-Region Fabric scenario, route aggregation is a method for reducing the number of entries that routers in a network must maintain in routing tables, for better scaling. Choose core, access, or core-and-access, to apply route aggregation only to access regions, the core region, or both. This option is applicable only to a Multi-Region Fabric border router, not an edge router or a transport gateway. • eigrp • lisp • isis
Select Route Policy	Enter the name of the route policy. Route policy is not supported in Cisco vManage Release 20.9.1.
Add OMP Advertise IPv6	

Field	Description
Protocol	Note Advertising IPv6 OMP routes as network statements is not supported. This applies when using the Service VPN feature in a configuration group, and applies also when using a Cisco VPN feature template. You can configure to advertise: • IPv6 routes by BGP and OSPF protocols • Connected routes, static routes, and aggregate routes The reason for the lack of support is that the Service VPN feature and the Cisco VPN feature template both use the advertise network prefix command, which does not fully support IPv6 addresses. Choose a protocol to configure route advertisements to OMP, for this VPN: • BGP • OSPF • Connected • Static • Network • Aggregate Applied to Region: (Minimum supported release: Cisco Catalyst SD-WAN Manager Release 20.13.1) In a Multi-Region Fabric scenario, route aggregation is a method for reducing the number of entries that routers in a network must maintain in routing tables, for better scaling. Choose core, access, or core-and-access, to apply route aggregation only to access regions, the core region, or both. This option is applicable only to a Multi-Region Fabric border router, not an edge router or a transport gateway.
Select Route Policy	Enter the name of the route policy. Route policy is not supported in Cisco vManage Release 20.9.1.
Protocol Sub Type	When you choose the OSPF protocol, specify the sub type as external.

e) Enter route information.

Table 74: Route

Field	Description
Add IPv4 Static Route	
Network Address*	Enter the IPv4 address or prefix, in decimal four-point-dotted notation, and the prefix length of the IPv4 static route to configure in the VPN.

Field	Description
Subnet Mask*	Enter the subnet mask.
Next Hop/Null 0/VPN/DHCP	Choose one of the following options to configure the next hop to reach the static route: • Next Hop: When you choose this option, the IPv4 Route Gateway Next Hop field appears. Enable this option to add the next hop. You can add a hop with and without a tracker. When you click Add Next Hop, the following fields appear: • Address*: Enter the next-hop IPv4 address. • Administrative Distance*: Enter the administrative distance for the route. When you click Add Next Hop with Tracker, the following fields appear: • Address*: Enter the next-hop IPv4 address. • Administrative Distance*: Enter the administrative distance for the route. • Tracker*: Enter the name of the gateway tracker to determine whether the next hop is reachable before adding that route to the route table of the device. • Null 0: When you choose this option, the following field appears: • IPv4 Route Null 0*: Enable this option to set the next hop to be the null interface. All packets sent to this interface are dropped without sending any ICMP messages. • VPN: When you choose this option, the following field appears: • IPv4 Route VPN*: Selects VPN as the gateway to direct packets to the transport VPN. • DHCP: When you choose this option, the following field appears: • IPv4 Route Gateway DHCP*: Assigns a static route for the default next-hop router when the DHCP server is accessed for an IP address.
Add BGP Routing	Choose a BGP route.
Add OSPF Routing	Choose an OSPF route.
Add IPv6 Static Route	
Prefix*	Enter the IPv6 address or prefix, in decimal four-point-dotted notation, and the prefix length of the IPv6 static route to configure in the VPN.

Field	Description
Next Hop/Null 0/NAT	Choose one of the following options to configure the next hop to reach the static route: • Next Hop: When you choose this option and click Add Next Hop, the following fields appear: • Address*: Enter the next-hop IPv6 address. • Administrative distance*: Enter the administrative distance for the route. • Null 0: When you choose this option, the following field appears: • IPv6 Route Null 0*: Enable this option to set the next hop to be the null interface. All packets sent to this interface are dropped without sending any ICMP messages. • NAT: When you choose this option, the following field appears: • IPv6 NAT*: Choose NAT64 or NAT66. • Interface: When you choose this option, the following fields appear: • Interface Name: Choose IPv6 interface name for the IPsec tunnel. • Next Hop: Enter the IPv6 address and the administrative distance for the next hop.

f) Enter service information.

Table 75: Service

Field	Description
Add Service	
Service Type	Choose a service available at the local site and in the VPN. Values: FW, IDS, IDP, netsvc1, netsvc2, netsvc3, netsvc4, TE, SIG
IPv4 Addresses (Maximum: 4)*	Enter up to four IP address, separated by commas. The service is advertised to the Cisco SD-WAN Controller only if one of the addresses can be resolved locally, at the local site, not via routes learned through OMP. You can configure up to four IP addresses.
Tracking*	Cisco Catalyst SD-WAN tests each service device periodically to check whether it is operational. Tracking saves the results of the periodic tests in a service log. Tracking is enabled by default.

g) Enter service route information.

Table 76: Service Route

Field	Description
Add Service Route	
Prefix*	Enter the IP address or prefix. For Umbrella SIG, use any RFC 1918 subnet for Service IP addresses.
Service*	Configure routes pointing to any service. Values: FW , IDS , IDP , netsvc1 , netsvc2 , netsvc3 , netsvc4 .
VPN*	Destination VPN to resolve the prefix.

- h) Enter GRE route information.

Table 77: GRE Route

Field	Description
Add GRE Route	
Prefix*	Enter the IP address or prefix, in decimal four-part-dotted notation, and prefix length of the GRE-specific static route.
Interface*	Enter the name of one or two GRE tunnels to use to reach the service.
VPN*	Enter the number of the VPN to reach the service. This must be VPN 0.

- i) Enter IPSEC route information.

Table 78: IPSEC Route

Field	Description
Add ipSec Route	
Prefix*	Enter the IP address or prefix, in decimal four-part-dotted notation, and prefix length of the IPsec-specific static route.
Interface*	Enter the name of one or two IPsec tunnel interfaces. If you configure two interfaces, the first is the primary IPsec tunnel, and the second is the backup. All packets are sent only to the primary tunnel. If that tunnel fails, all packets are then sent to the secondary tunnel. If the primary tunnel comes back up, all traffic is moved back to the primary IPsec tunnel.

- j) Enter NAT information

Table 79: NAT

Field	Description
Nat Pool	

Field	Description
NatPool Name*	Enter a NAT pool number configured in the centralized data policy. The NAT pool name must be unique across VPNs and VRFs. You can configure up to 31 (1–32) NAT pools per router.
Prefix Length*	Enter the NAT pool prefix length.
Range Start*	Enter a starting IP address for the NAT pool.
Range End*	Enter a closing IP address for the NAT pool.
Overload*	Enable this option to configure per-port translation. If this option is disabled, only dynamic NAT is configured on the end device. Per-port NAT is not configured. Default: Enabled
Direction*	Choose the NAT direction.
Nat64 V4 Pool	
Nat64 V4 Pool Name*	Enter a NAT pool number configured in the centralized data policy. The NAT pool name must be unique across VPNs and VRFs. You can configure up to 31 (1–32) NAT pools per router.
Nat 64 V4 Pool Range Start*	Enter a starting IP address for the NAT pool.
Nat 64 V4 Pool Range End*	Enter a closing IP address for the NAT pool.
Overload*	Enable this option to configure per-port translation. If this option is disabled, only dynamic NAT is configured on the end device. Per-port NAT is not configured. Default: Disabled

k) Enter route leak information.

Table 80: Route leak from Global VPN

Field	Description
Route Protocol*	Choose a protocol to configure leak routes from global VPN to the service VPN that you are configuring: • static • connected • bgp • ospf
Select Route Policy	Choose a route policy from the drop-down list.
Redistribution (in service VPN)	

Field	Description
Protocol*	Choose a protocol from the available options to redistribute the leaked routes: • bgp • ospf • (Minimum supported release: Cisco Catalyst SD-WAN Manager Release 20.18.1) • eigrp
Select Route Policy	Choose a route policy from the drop-down list.

- 1) Enter route target information.

Table 81: Route Target

Field	Description
IPv4 Settings	
Import Route Target List: Route Target*	Configure a route target for IPv4 interfaces. It imports routing information from the target VPN extended community.
Export Route Target List: Route Target*	Configure a route target for IPv4 interfaces. It exports routing information to the target VPN extended community.
IPv6 Settings	
Import Route Target List: Route Target*	Configure a route target for IPv6 interfaces. It imports routing information from the target VPN extended community.
Export Route Target List: Route Target*	Configure a route target for IPv6 interfaces. It exports routing information to the target VPN extended community.

What to do next

Also see [Deploy a Configuration Group](#).

Changing the Scope for a Parameter Value

When you first open a feature template, for each parameter that has a default value, the scope is set to Default (a ) , and the default setting or value is shown. To change the default or to enter a value, click the scope drop-down list and select one of the following:

Parameter Name	Description
 Device Specific	Use a device-specific value for the parameter. For device-specific parameters, you cannot enter a value in the feature template. You enter the value when you attach a device to a device template. When you click Device Specific, the Enter Key box opens. This box displays a key, which is a unique string that identifies the parameter in a CSV file that you create. This file is an Excel spreadsheet that contains one column for each key. The header row contains the key names (one key per column), and each row after that corresponds to a device and defines the values of the keys for that device. You upload the CSV file when you attach a device to a device template. For more information, see Create a Template Variables Spreadsheet Note When you are using a CSV file for configuring device-specific variables in the device attach flow, ensure to fill all the mandatory fields before uploading. To change the default key, type a new string and move the cursor out of the Enter Key box. Examples of device-specific parameters are system IP address, hostname, GPS location, and site ID.
 Global	Enter a value for the parameter, and apply that value to all devices. Examples of parameters that you might apply globally to a group of devices are DNS server, syslog server, and interface MTUs.

Once you have created and named the template, enter the following values. Parameters marked with an asterisk are required.

Configure Basic VPN Parameters Using a Configuration Group

Before you begin

On the **Configuration > Configuration Groups** page, choose **SD-WAN** as the solution type.

Procedure

-
- Step 1** From the Cisco SD-WAN Manager menu, choose **Configuration > Configuration Groups**.
- Step 2** Create and configure a Transport VPN feature in Transport and Management profile.

a) Basic Configuration

Table 82: Basic Configuration

Field	Description
VPN	Enter the numeric identifier of the VPN.
Enhance ECMP Keying	Enable the use in the ECMP hash key of Layer 4 source and destination ports, in addition to the combination of the source IP address, destination IP address, protocol, and DSCP field, as the ECMP hash key. Default: Disabled

What to do next

Also see [Deploy a Configuration Group](#).

Configure Basic VPN Parameters

To configure basic VPN parameters, choose **Basic Configuration** and then configure the following parameters. Parameters marked with an asterisk are required to configure a VPN.

Parameter Name	Description
VPN	Enter the numeric identifier of the VPN. Range: 1 to 65525, excluding 512. For details see the VRF range behavior change described here . Range for Cisco vEdge devices: 0 through 65530
Name	Enter a name for the VPN.
Enhance ECMP keying	Click On to enable the use in the ECMP hash key of Layer 4 source and destination ports, in addition to the combination of the source IP address, destination IP address, protocol, and DSCP field, as the ECMP hash key. ECMP keying is Off by default.
Enable TCP Optimization Cisco vEdge devices only	Click On to enable TCP optimization for a service-side VPN (a VPN other than VPN 0 and VPN 512). TCP optimization fine-tunes TCP to decrease round-trip latency and improve throughput for TCP traffic.



Note To complete the configuration of the transport VPN on a router, you must configure at least one interface in VPN 0.

To save the feature template, click **Save**.

Configure DNS and Static Hostname Mapping Using a Configuration Group

Before you begin

On the **Configuration > Configuration Groups** page, choose **SD-WAN** as the solution type.

Procedure

Step 1 From the Cisco SD-WAN Manager menu, choose **Configuration > Configuration Groups**.

Step 2 Create and configure a Transport VPN feature in Transport and Management profile.

a) DNS

Table 83: DNS

Field	Description
Add DNS	
Primary DNS Address (IPv4)	Enter the IP address of the primary IPv4 DNS server in this VPN.
Secondary DNS Address (IPv4)	Enter the IP address of a secondary IPv4 DNS server in this VPN.
Add DNS IPv6	
Primary DNS Address (IPv6)	Enter the IP address of the primary IPv6 DNS server in this VPN.
Secondary DNS Address (IPv6)	Enter the IP address of a secondary IPv6 DNS server in this VPN.

b) Host Mapping

Table 84: Host Mapping

Field	Description
Add New Host Mapping	
Hostname*	Enter the hostname of the DNS server. The name can be up to 128 characters.
List of IP*	Enter up to 14 IP addresses to associate with the hostname. Separate the entries with commas.

What to do next

Also see [Deploy a Configuration Group](#).

Configure DNS and Static Hostname Mapping

To configure DNS addresses and static hostname mapping, click **DNS** and configure the following parameters:

Parameter Name	Options	Description
Primary DNS Address		Click either IPv4 or IPv6 , and enter the IP address of the primary DNS server in this VPN.
New DNS Address		Click New DNS Address and enter the IP address of a secondary DNS server in this VPN. This field appears only if you have specified a primary DNS address.
	Mark as Optional Row	Check the Mark as Optional Row check box to mark this configuration as device-specific. To include this configuration for a device, enter the requested variable values when you attach a device template to a device, or create a template variables spreadsheet to apply the variables.
	Hostname	Enter the hostname of the DNS server. The name can be up to 128 characters.
	List of IP Addresses	Enter up to eight IP addresses to associate with the hostname. Separate the entries with commas.
To save the DNS server configuration, click Add .		

To save the feature template, click **Save**.

CLI Equivalent

```
vpn vpn-id
  dns ip-address (primary | secondary)
  host hostname ip ip-address
```

Configure Interfaces in the WAN Transport VPN (VPN 0)

This topic describes how to configure the general properties of WAN transport and service-side network interfaces. For information about how to configure specific interface types and properties—including cellular interfaces, DHCP, PPPoE, VRRP, and WLAN interfaces.

VPN 0 is the WAN transport VPN. This VPN handles all control plane traffic, which is carried over OMP sessions, in the overlay network. For a Cisco vEdge device to participate in the overlay network, at least one interface must be configured in VPN 0, and at least one interface must connect to a WAN transport network, such as the Internet or an MPLS or a metro Ethernet network. This WAN transport interface is referred to as a tunnel interface. At a minimum, for this interface, you must configure an IP address, enable the interface, and set it to be a tunnel interface.

To configure a tunnel interface on a Cisco Catalyst SD-WAN Controller or a Cisco SD-WAN Manager, you create an interface in VPN 0, assign an IP address or configure the interface to receive an IP address from DHCP, and mark it as a tunnel interface. The IP address can be either an IPv4 or IPv6 address. To enable dual stack, configure both address types. You can optionally associate a color with the tunnel.



Note You can configure IPv6 addresses only on transport interfaces in VPN 0 and but not supported in VPN 512.

```
vSmart/vManage(config)# vpn 0
vSmart/vManage(config-vpn-0)# interface interface-name
```

```
vSmart/vManage(config-interface)# [ip address prefix / length | ip dhcp-client [dhcp-distance
number]
vSmart/vManage(config-interface)# [ipv6 address prefix / length | ipv6 dhcp-client
[dhcp-distance number] [dhcp-rapid-commit]
vSmart/vManage(config-interface)# no shutdown
vSmart/vManage(config-interface)# tunnel-interface
vSmart/vManage(config-tunnel-interface)# color color
vSmart/vManage(config-tunnel-interface)# [no] allow-service service
```

Tunnel interfaces on Cisco vEdge devices must have an IP address, a color, and an encapsulation type. The IP address can be either an IPv4 or IPv6 address. To enable dual stack in releases before Cisco SD-WAN Release 20.3.2, configure both address types.

To use dual stack with Cisco vEdge devices from Cisco SD-WAN Release 20.3.2, configure all controllers with both IPv4 and IPv6 addresses. In addition, configure DNS for the Cisco Catalyst SD-WAN Validator interface to resolve IPv4 and IPv6 address types so that controllers can reach the Cisco Catalyst SD-WAN Validator through either IP address type.



Note Starting from Cisco vManage Release 20.6.1, in case of a dual-stack configuration, if an IPv4 address or the fully qualified domain name (FQDN) is not available, but an IPv6 address is available, then the IPv6 address is used to connect to the Cisco Catalyst SD-WAN Validator.

For the tunnel interface, you can configure a static IPv4 or IPv6 address, or you can configure the interface to receive its address from a DHCP server. To enable dual stack, configure both an IPv4 and an IPv6 address on the tunnel interface.

From Cisco SD-WAN Release 20.3.2, Cisco vEdge devices do not support dual stack on the same TLOC or interface. Only one address type can be provisioned for a TLOC or interface. Using a second address type requires a second TLOC or interface on which it can be provisioned.

```
vEdge(config)# vpn 0
vEdge(config-vpn-0)# interface interface-name
vEdge(config-interface)# [ip address prefix / length | ip dhcp-client [dhcp-distance number]
vEdge(config-interface)# [ipv6 address prefix / length | ipv6 dhcp-client [dhcp-distance
number] [dhcp-rapid-commit]
vEdge(config-interface)# no shutdown
vEdge(config-interface)# tunnel-interface
vEdge(config-tunnel-interface)# color color [restrict]
vEdge(config-tunnel-interface)# encapsulation (gre | ipsec)
vEdge(config-tunnel-interface)# [no] allow-service service
```

On Cisco Catalyst SD-WAN Controllers and Cisco Catalyst SD-WAN Controller NMSs, *interface-name* can be either **eth number** or **loopback number**. Because Cisco Catalyst SD-WAN Controllers and Cisco Catalyst SD-WAN Controller NMSs participate only in the overlay network's control plane, the VPNs that you can configure on these devices are VPN 0 and VPN 512. Hence, all interfaces are present only on these VPNs.

On Cisco vEdge devices, *interface-name* can be **ge slot/port**, **gre number**, **ipsec number**, **loopback string**, **natpool number**, or **ppp number**.

To enable the interface, include the **no shutdown** command.

Color is a Cisco Catalyst SD-WAN software construct that identifies the transport tunnel. It can be **3g**, **biz-internet**, **blue**, **bronze**, **custom1**, **custom2**, **custom3**, **default**, **gold**, **green**, **lte**, **metro-ethernet**, **mpls**, **private1** through **private6**, **public-internet**, **red**, and **silver**. The colors **metro-ethernet**, **mpls**, and **private1** through **private6** are referred to as *private colors*, because they use private addresses to connect to the remote

side Cisco vEdge device in a private network. You can use these colors in a public network provided that there is no NAT device between the local and remote Cisco vEdge devices.

To limit the remote TLOCs that the local TLOC can establish BFD sessions with, mark the TLOC with the **restrict** option. When a TLOC is marked as restricted, a TLOC on the local router establishes tunnel connections with a remote TLOC only if the remote TLOC has the same color.



Note When a WAN edge device is configured with two IPv6 TLOCs, one with static default route and the other one with IPv6 address autoconfig default which is the IPv6 neighbor discovery default route, the IPv6 neighbor discovery default route is not installed in the routing table. In this case, the IPv6 TLOC with IPv6 neighbor discovery default route does not work.

For IPv6 TLOC with IPv6 neighbor discovery default route to work, you can configure the static route for TLOC with IPv6 neighbor discovery to overwrite the IPv6 neighbor discovery default route and ensure that both the static routes are installed into the routing table. You can also use the IPv6 neighbor discovery default route on all interfaces.

On a Cisco Catalyst SD-WAN Controller or Cisco Catalyst SD-WAN Controller NMS, you can configure one tunnel interface. On a Cisco vEdge device, you can configure up to eight tunnel interfaces.

This means that each Cisco vEdge device can have up to eight TLOCs.

On Cisco vEdge devices, you must configure the tunnel encapsulation. The encapsulation can be either IPsec or GRE. For IPsec encapsulation, the default MTU is 1442 bytes, and for GRE it is 1468 bytes. These values are a function of overhead required for BFD path MTU discovery, which is enabled by default on all TLOCs. (For more information, see *Configuring Control Plane and Data Plane High Availability Parameters*.) You can configure both IPsec and GRE encapsulation by including two **encapsulation** commands under the same **tunnel-interface** command. On the remote Cisco vEdge device, you must configure the same tunnel encapsulation type or types so that the two routers can exchange data traffic. Data transmitted out of an IPsec tunnel can be received only by an IPsec tunnel, and data sent on a GRE tunnel can be received only by a GRE tunnel. The Cisco Catalyst SD-WAN software automatically selects the correct tunnel on the destination Cisco vEdge device.

A tunnel interface allows only DTLS, TLS, and, for Cisco vEdge devices, IPsec traffic to pass through the tunnel. To allow additional traffic to pass without having to create explicit policies or access lists, enable them by including one **allow-service** command for each service. You can also explicitly disallow services by including the **no allow-service** command. Note that services affect only physical interfaces. You can allow or disallow these services on a tunnel interface:

Service	Cisco vEdge device	Cisco Catalyst SD-WAN Controller	Cisco Catalyst SD-WAN Controller
all (Overrides any commands that allow or disallow individual services)	X	X	X
bgp	X	—	—
dhcp (for DHCPv4 and DHCPv6)	X	—	—
dns	X	—	—
https	—	X	—

Service	Cisco vEdge device	Cisco Catalyst SD-WAN Controller	Cisco Catalyst SD-WAN Controller
icmp	X	X	X
netconf	—	X	—
ntp	X	—	—
ospf	X	—	—
sshd	X	X	X
stun	X	X	X

The **allow-service stun** command pertains to allowing or disallowing a Cisco vEdge device to generate requests to a generic STUN server so that the device can determine whether it is behind a NAT and, if so, what kind of NAT it is and what the device's public IP address and public port number are. On a Cisco vEdge device that is behind a NAT, you can also have tunnel interface to discover its public IP address and port number from the Cisco Catalyst SD-WAN Validator.

```
vEdge(config-tunnel-interface)# vbond-as-stun-server
```

With this configuration, the Cisco vEdge device uses the Cisco Catalyst SD-WAN Validator as a STUN server, so the router can determine its public IP address and public port number. (With this configuration, the router cannot learn the type of NAT that it is behind.) No overlay network control traffic is sent and no keys are exchanged over tunnel interface configured to the the Cisco Catalyst SD-WAN Validator as a STUN server. However, BFD does come up on the tunnel, and data traffic can be sent on it. Because no control traffic is sent over a tunnel interface that is configured to use the Cisco Catalyst SD-WAN Validator as a STUN server, you must configure at least one other tunnel interface on the Cisco vEdge device so that it can exchange control traffic with the Cisco Catalyst SD-WAN Controller and the Cisco Catalyst SD-WAN Controller NMS.

You can log the headers of all packets that are dropped because they do not match a service configured with an **allow-service** command. You can use these logs for security purposes, for example, to monitor the flows that are being directed to a WAN interface and to determine, in the case of a DDoS attack, which IP addresses to block.

```
vEdge(config)# policy implicit-acl-logging
```

When you enable implicit ACL logging, by default, the headers of all dropped packets are logged. It is recommended that you configure a limit to the number of packets logged with the **policy log-frequency** configuration command.

On a Cisco vEdge device, services that you configure on a tunnel interface act as implicit access lists (ACLs). If you apply a localized data policy on a tunnel interface by configuring an ACL with the **policy access-list** command, this ACL is an explicit ACL. For information about how packets matching both implicit and explicit ACLs are handled, see [Configuring Localized Data Policy for IPv4](#) or [Configuring Localized Data Policy for IPv6](#).

For each transport tunnel on a vEdge router and for each encapsulation type on a single transport tunnel, the Cisco SD-WAN software creates a TLOC, which consists of the router's system IP address, the color, and the encapsulation. The OMP session running on the tunnel sends the TLOC, as a TLOC route, to the Cisco Catalyst SD-WAN Controller, which uses it to determine the overlay network topology and to determine the best paths for data traffic across the overlay network.

To display information about interfaces in the WAN transport VPN that are configured with IPv4 addresses, use the **show interface** command. For example:

```
vEdge# show interface vpn 0
```

VPN	INTERFACE	IP ADDRESS	IF ADMIN STATUS	IF OPER STATUS	ENCAP TYPE	PORT TYPE	MTU	HWADDR	SPEED MBPS	DUPLEX	TCP MSS ADJUST	UPTIME	RX PACKETS	TX PACKETS
0	ge0/1	10.0.5.21/24	Up	Up	null	transport	1500	00:0c:29:6c:30:c1	10	full	0	0:04:03:41	260025	260145
0	ge0/2	-	Down	Up	null	service	1500	00:0c:29:6c:30:cb	10	full	0	0:04:03:41	3506	1
0	ge0/3	-	Down	Up	null	service	1500	00:0c:29:6c:30:d5	10	full	0	0:04:03:41	260	1
0	ge0/4	-	Down	Up	null	service	1500	00:0c:29:6c:30:df	10	full	0	0:04:03:41	260	1
0	ge0/5	-	Down	Up	null	service	1500	00:0c:29:6c:30:e9	10	full	0	0:04:03:41	260	1
0	ge0/6	10.0.7.21/24	Up	Up	null	service	1500	00:0c:29:6c:30:f3	10	full	0	0:04:03:41	265	2
0	ge0/7	10.0.100.21/24	Up	Up	null	service	1500	00:0c:29:6c:30:fd	10	full	0	0:04:03:41	278	2
0	system	172.16.255.21/32	Up	Up	null	loopback	1500	00:00:00:00:00:00	10	full	0	0:04:03:37	0	0

To display information for interfaces configured with IPv6 addresses, use the **show ipv6 interface** command. For example:

```
vEdge# show ipv6 interface vpn 0
```

				IF	IF						TCP								
AF				ADMIN	OPER	ENCAP						SPEED	MSS			RX	TX		
VPN	INTERFACE	TYPE	IPV6 ADDRESS	STATUS	STATUS	TYPE	PORT	TYPE	MTU	HWADDR	MBPS	DUPLEX	ADJUST	UPTIME	PACKETS	PACKETS	LINK	LOCAL ADDRESS	
0	ge0/1	ipv6	2001::a00:1a0b/120	Up	Up	null	service	1500	00:0c:29:ab:b7:62	1000	full	1420	0:01:30:00	2	6		fe80::20c:29ff:feab:b762/64		
0	ge0/2	ipv6	2001::a00:50b/120	Up	Up	null	service	1500	00:0c:29:ab:b7:6c	1000	full	1420	0:01:30:00	21	5		fe80::20c:29ff:feab:b76c/64		
0	ge0/3	ipv6	fd00:1234::/16	Up	Up	null	service	1500	00:0c:29:ab:b7:76	1000	full	1420	0:01:08:33	0	8		fe80::20c:29ff:feab:b776/64		
0	ge0/4	ipv6	-	Up	Up	null	service	1500	00:0c:29:ab:b7:80	1000	full	1420	0:01:30:00	18	5		fe80::20c:29ff:feab:b780/64		
0	ge0/5	ipv6	-	Down	Up	null	service	1500	00:0c:29:ab:b7:8a	1000	full	1420	0:01:44:19	1	1		fe80::20c:29ff:feab:b78a/64		
0	ge0/6	ipv6	-	Down	Up	null	service	1500	00:0c:29:ab:b7:94	1000	full	1420	0:01:44:19	0	1		fe80::20c:29ff:feab:b794/64		
0	ge0/7	ipv6	-	Up	Up	null	service	1500	00:0c:29:ab:b7:9e	1000	full	1420	0:01:43:02	55	5		fe80::20c:29ff:feab:b79e/64		
0	system	ipv6	-	Up	Up	null	loopback	1500	00:00:00:00:00:00	10	full	1420	0:01:29:31	0	0		-		
0	loopback1	ipv6	2001::a00:6501/128	Up	Up	null	transport	1500	00:00:00:00:00:00	10	full	1420	0:03:49:09	0	0		-		
0	loopback2	ipv6	2001::a00:6502/128	Up	Up	null	transport	1500	00:00:00:00:00:00	10	full	1420	0:03:49:05	0	0		-		
0	loopback3	ipv6	2001::a00:6503/128	Up	Up	null	transport	1500	00:00:00:00:00:00	10	full	1420	0:03:49:01	0	0		-		
0	loopback4	ipv6	2001::a00:6504/128	Up	Up	null	transport	1500	00:00:00:00:00:00	10	full	1420	0:03:48:54	0	0		-		

In the command output, a port type of "transport" indicates that the interface is configured as a tunnel interface, and a port type of "service" indicates that the interface is not configured as a tunnel interface and can be used for data plane traffic. The port type for the system IP address interface is "loopback".

Configure Other WAN Interface Properties

You can modify the distribution of data traffic across transport tunnels by applying a data policy in which the action sets TLOC attributes (IP address, color, and encapsulation) to apply to matching data packets. For more information, see the Configuring Centralized Data Policy.

Extend the WAN Transport VPN

When two Cisco vEdge devices are collocated at a physical site that has only one WAN circuit, you can configure the Cisco vEdge device that is not connected to the circuit to be able to establish WAN transport tunnels through the other router's TLOCs. In this way, you extend the WAN transport VPN so that both routers can establish tunnel interfaces, and hence can establish independent TLOCs, in the overlay network. (Note that you can configure the two routers themselves with different site identifiers).

The following figure illustrates a site with two Cisco vEdge devices. Cisco vEdge device-1 terminates one WAN circuit from the Internet and the second Cisco vEdge device-2 terminates the private MPLS network. Each router has one TLOC. You can configure Cisco vEdge device-2 to extend its WAN transport VPN to Cisco vEdge device1 so that Cisco vEdge device-1 can participate independently in the overlay network. You can also make a similar configuration for vEdge1 so that the WAN transport can be extended from Cisco vEdge device1 to Cisco vEdge device2.



When you extend the WAN transport VPN, no BFD sessions are established between the two collocated vEdge routers.

You cannot configure TLOC extensions on cellular (LTE) interfaces.

To extend the WAN transport VPN, you configure the interface between the two routers:

- For the router that is not connected to the circuit, you configure a standard tunnel interface in VPN 0.
- For the router that is physically connected to the WAN or private transport, you associate the physical interface that connects to the circuit, configuring this in VPN 0 but not in a tunnel interface.

To configure the non-connected router (Cisco vEdge device-1 in the figure above), create a tunnel interface in VPN 0 on the physical interface to the connected router.

```
vEdge-1 (config-vpn-0) # interface ge slot/ port
vEdge-1 (config-interface) # ip address prefix / length
vEdge-1 (config-interface) # no shutdown
vEdge-1 (config-interface) # mtu number
vEdge-1 (config-interface) # tunnel-interface
vEdge-1 (config-tunnel-interface) # color color
```

For the router connected to the WAN or private transport (Cisco vEdge device-2 in the figure above), configure the interface that connects to the non-connected router, again in VPN 0:

```
vEdge-2 (config-vpn-0) # interface ge slot/port
vEdge-2 (config-interface) # ip address prefix / length
vEdge-2 (config-interface) # tloc-extension geslot / port
vEdge-2 (config-interface) # no shutdown
vEdge-2 (config-interface) # mtu number
```

The physical interface in the **interface** command is the one that connects to the other router.

The **tloc-extension** command creates the binding between the non-connected router and the WAN or private network. In this command, you specify the physical interface that connects to the WAN or private network circuit.

If the circuit connects to a public network:

- Configure a NAT on the public-network-facing interface on the Cisco vEdge device. The NAT configuration is required because the two Cisco vEdge devices are sharing the same transport tunnel.
- Configure a static route on the non-connected router to the TLOC-extended interface on the router connected to the public network.

If the circuit connects to a private network, such as an MPLS network:

- Enable routing on the non-connected router so that the interface on the non-connected router is advertised into the private network.
- Depending on the routing protocol you are using, enable either OSPF or BGP service on the non-connected router interface so that routing between the non-connected and the connected routers comes up. To do this, use the **allow-service** command.

You cannot extend a TLOC configured on a loopback interface, that is, when you use a loopback interface to connect to the public or private network. You can extend a TLOC only on a physical interface.

If one of the routers is connected to two WAN transports (such as the Internet and an MPLS network), create subinterfaces between the two routers, creating the tunnel on the subinterface. The subinterfaces on the two routers must be in the same subnet. Because you are using a subinterface, the interface's MTU must be at least 4 bytes less than the physical MTU.

Here is a sample configuration that corresponds to the figure shown above. Because the router Cisco vEdge device-2 connects to two transports, we create subinterfaces between the Cisco vEdge device-1 and Cisco vEdge device-2 routers. One subinterface binds to the Internet circuit, and the second one binds to the MPLS connection.

```
vEdge-1# show running-config vpn 0
interface ge0/2.101
  ip address 192.168.19.15/24
  mtu 1496
  tunnel-interface
    color lte
  ...
  !
  no shutdown
!
interface ge0/2.102
  ip address 192.168.20.15/24
  mtu 1496
  tunnel-interface
    color mpls
  ...
  !
  no shutdown
!
ip route 0.0.0.0/0 192.168.19.16
vEdge-2# show running-config vpn 0
interface ge0/0
  ip address 172.16.255.2
  tunnel-interface
    color lte
  ...
  !
  no shutdown
!
interface ge0/3
  ip address 172.16.255.16
  tunnel-interface
    color mpls
  ...
  !
  no shutdown
!
interface ge0/2.101
  ip address 192.168.19.16/24
  mtu 1496
  tloc-extension ge0/0
  no shutdown
!
interface ge0/2.102
  ip address 192.168.20.16/24
  mtu 1496
  tloc-extension ge0/3
  no shutdown
!
```

For this example configuration, Cisco vEdge device-1 establishes two control connections to each Cisco Catalyst SD-WAN Controller in the overlay network—one connection for the LTE tunnel and the second for the MPLS tunnel. These control connections are separate and independent from those established on Cisco

vEdge device-2. The following output shows the control connections on vEdge-1 in a network with two Cisco Catalyst SD-WAN Controllers:

```
vEdge-1# show control connections
```

PEER TYPE	PEER PROTOCOL	PEER SYSTEM IP	SITE ID	DOMAIN ID	PEER PRIVATE IP	PEER PRIVATE PORT	PEER PUBLIC IP	PEER PUBLIC PORT	LOCAL COLOR	STATE	UPTIME	CONTROLLER GROUP NAME
vsmart	dtls	172.16.255.19	100	1	10.0.5.19	12346	10.0.5.19	12346	lte	up	0:00:18:43	default
vsmart	dtls	172.16.255.19	100	1	10.0.5.19	12346	10.0.5.19	12346	mpls	up	0:00:18:32	default
vsmart	dtls	172.16.255.20	200	1	10.0.12.20	12346	10.0.12.20	12346	lte	up	0:00:18:38	default
vsmart	dtls	172.16.255.20	200	1	10.0.12.20	12346	10.0.12.20	12346	mpls	up	0:00:18:27	default

You can verify that the two Cisco vEdge devices have established no BFD sessions between them. On Cisco vEdge device-1, we see no BFD sessions to Cisco vEdge device-2 (system IP address 172.16.255.16):

```
vEdge-1# show bfd sessions
```

SYSTEM IP	SITE ID	STATE	SOURCE TLOC COLOR	REMOTE TLOC COLOR	SOURCE IP	DST PUBLIC IP	DST PUBLIC PORT	ENCAP	DETECT MULTIPLIER	TX INTERVAL(msec)	UPTIME	TRANSITIONS
172.16.255.11	100	up	lte	lte	192.168.19.15	10.0.101.1	12346	ipsec	20	1000	0:00:20:26	0
172.16.255.11	100	up	lte	3g	192.168.19.15	10.0.101.2	12346	ipsec	20	1000	0:00:20:26	0
172.16.255.11	100	up	lte	gold	192.168.19.15	10.0.101.3	12346	ipsec	20	1000	0:00:20:26	0
172.16.255.11	100	up	lte	red	192.168.19.15	10.0.101.4	12346	ipsec	20	1000	0:00:20:26	0
172.16.255.11	100	up	mpls	lte	192.168.20.15	10.0.101.1	12346	ipsec	20	1000	0:00:20:26	0
172.16.255.11	100	up	mpls	3g	192.168.20.15	10.0.101.2	12346	ipsec	20	1000	0:00:20:26	0
172.16.255.11	100	up	mpls	gold	192.168.20.15	10.0.101.3	12346	ipsec	20	1000	0:00:20:26	0
172.16.255.11	100	up	mpls	red	192.168.20.15	10.0.101.4	12346	ipsec	20	1000	0:00:20:26	0
172.16.255.14	400	up	lte	lte	192.168.19.15	10.1.14.14	12360	ipsec	20	1000	0:00:20:26	0
172.16.255.14	400	up	mpls	lte	192.168.20.15	10.1.14.14	12360	ipsec	20	1000	0:00:20:26	0
172.16.255.21	100	up	lte	lte	192.168.19.15	10.0.111.1	12346	ipsec	20	1000	0:00:20:26	0
172.16.255.21	100	up	lte	3g	192.168.19.15	10.0.111.2	12346	ipsec	20	1000	0:00:20:26	0
172.16.255.21	100	up	mpls	lte	192.168.20.15	10.0.111.1	12346	ipsec	20	1000	0:00:20:26	0
172.16.255.21	100	up	mpls	3g	192.168.20.15	10.0.111.2	12346	ipsec	20	1000	0:00:20:26	0

Configure GRE Interfaces and Advertise Services to Them

When a service, such as a firewall, is available on a device that supports only GRE tunnels, you can configure a GRE tunnel on the vEdge router to connect to the remote device.

You then advertise that the service is available via a GRE tunnel, and you direct the appropriate traffic to the tunnel either by creating centralized data policy or by configuring GRE-specific static routes.

Create a GRE tunnel by configuring a GRE interface. GRE interfaces are logical interfaces, and you configure them just like any other physical interface. A GRE interface is a logical interface, you must bind it to a physical interface or a PPPoE interface, as described below.

To configure a GRE tunnel interface to a remote device that is reachable through a transport network, configure the tunnel in VPN 0:

```
vEdge(config)# vpn 0 interface gre number
vEdge(config-interface-gre)# (tunnel-source ip-address | tunnel-source-interface
interface-name)
vEdge(config-interface-gre)# tunnel-destination ip-address
vEdge(config-interface-gre)# no shutdown
```

The GRE interface has a name in the format **gre number**, where *number* can be from 1 through 255.

To configure the source of the GRE tunnel on the local device, you can specify either the IP address of the physical interface or PPPoE interface (in the **tunnel-source** command) or the name of the physical interface or PPPoE interface (in the **tunnel-source-interface** command). Ensure that the physical interface is configured in the same VPN in which the GRE interface is located.

To configure the destination of the GRE tunnel, specify the IP address of the remote device in the **tunnel-destination** command.

The combination of a source address (or source interface name) and a destination address defines a single GRE tunnel. Only one GRE tunnel can exist that uses a specific source address (or interface name) and destination address pair.

You can optionally configure an IP address for the GRE tunnel itself:

```
vEdge(config-interface-gre)# ip address ip-address
```

Because GRE tunnels are stateless, the only way for the local router to determine whether the remote end of the tunnel is up, is to periodically send keepalive messages over the tunnel. The keepalive packets are looped back to the sender, and receipt of these packets by the local router indicates that the remote GRE device is up. By default, the GRE interface sends keepalive packets every 10 seconds, and if it receives no response, retries 3 times before declaring the remote device to be down. You can modify these default values with the **keepalive** command:

```
vEdge(config-interface-gre)# keepalive seconds retries
```

The keepalive interval can be from 0 through 65535 seconds, and the number of retries can be from 0 through 255. If you configure an IP address for the GRE interface, that IP address generates the keepalive messages.

If the vEdge router sits behind a NAT and you have configured GRE encapsulation, you must disable keepalives, with a **keepalive 0 0** command. (Note that you cannot disable keepalives by issuing a **no keepalive** command. This command returns the keepalive to its default settings of sending a keepalive packet every 10 seconds and retrying 3 times before declaring the remote device down.)

For GRE interfaces, you can configure only the following additional interface properties:

```
vEdge(config-interface-gre)# access-list acl-name
vEdge(config-interface-gre)# block-non-source-ip
vEdge(config-interface-gre)# clear-dont-fragment
vEdge(config-interface-gre)# description text
vEdge(config-interface-gre)# mtu bytes
vEdge(config-interface-gre)# policer policer-name
vEdge(config-interface-gre)# rewrite-rule rule-name
vEdge(config-interface-gre)# tcp-mss-adjust
```

GRE interfaces do not support cFlowd traffic monitoring.

You can configure one or two GRE interfaces per service. When you configure two, the first interface is the primary GRE tunnel, and the second is the backup tunnel. All packets are sent only to the primary tunnel. If that tunnel fails, all packets are then sent to the secondary tunnel. If the primary tunnel comes back up, all traffic is moved back to the primary GRE tunnel.

You direct data traffic from the service VPN to the GRE tunnel in one of two ways: either with a GRE-specific static route or with a centralized data policy.

To create a GRE-specific static route in the service VPN (a VPN other than VPN 0 or VPN 512), use the **ip gre-route** command:

```
vEdge(config-vpn)# ip gre-route prefix vpn 0 interface gre number [gre number2]
```

This GRE-specific static route directs traffic from the specified prefix to the primary GRE interface, and optionally to the secondary GRE interface, in VPN 0. The OMP administrative distance of a GRE-specific static route is 5, and the admin distance for a regular static route (configured with the **ip route** command) is 1. For more information, see *Unicast Overlay Routing Overview*.

To direct the data traffic to the GRE tunnel using a centralized data policy is a two-part process: you advertise the service in the service VPN, and then you create a centralized data policy on the Cisco Catalyst SD-WAN Controller to forward matching traffic to that service.

To advertise the service, include the **service** command in the service VPN (a VPN other than VPN 0 or VPN 512):

```
vEdge(config-vpn)# service service-name interface gre number [gre number2]
```

The service name can be **FW**, **IDP**, **IDS**, or **TE**, or a custom service name **netsvc1** through **netsvc4**. For more information on service-names, see Service Chaining. The interface is the GRE interface in VPN 0 that is used to reach the service. If you have configured a primary and a backup GRE tunnel, list the two GRE interfaces (**gre number1 gre number2**) in the **service** command. Once you have configured a service as a reachable GRE interface, you cannot delete the GRE interface from the configuration. To delete the GRE interface, you must first delete the service. You can, however, reconfigure the service itself, by modifying the **service** command.

Then, create a data policy on the Cisco Catalyst SD-WAN Controller that applies to the service VPN. In the action portion of the data policy, you must explicitly configure the policy to service the packets destined for the GRE tunnel. To do this, include the **local** option in the **set service** command:

```
vSmart(config-policy-data-policy-vpn-list-vpn-sequence) # action accept
vSmart(config-action-accept) # set service service-name local
```

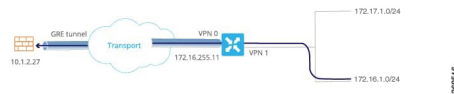
If the GRE tunnel used to reach the service is down, packet routing falls back to using standard routing. To drop packets when a GRE tunnel to the service is unreachable, add the **restrict** option:

```
vSmart(config-policy-data-policy-vpn-list-vpn-sequence) # action accept
vSmart(config-action-accept) # set service service-name local restrict
```

To monitor GRE tunnels and their traffic, use the following commands:

- **show interface** —List data traffic transmitted and received on GRE tunnels.
- **show tunnel gre-keepalives** —List GRE keepalive traffic transmitted and received on GRE tunnels.
- **show tunnel statistics** —List both data and keepalive traffic transmitted and received on GRE tunnels.

The following figure illustrates an example of configuring a GRE tunnel in VPN 0, to allow traffic to be redirected to a service that is not located at the same site as the vEdge router. In this example, local traffic is directed to the GRE tunnel using a centralized data policy, which is configured on the Cisco Catalyst SD-WAN Controller.



The configuration looks like this:

```
vEdge# show running-config vpn 0
vpn 0
  interface gre1
    ip address 172.16.111.11/24
    keepalive 60 10
    tunnel-source 172.16.255.11
    tunnel-destination 10.1.2.27
    no shutdown
  !
!
vEdge# show running-config vpn 1 service
vpn 1
  service FW interface gre1

vSmart# show running-config policy
policy
  lists
    prefix-list for-firewall
      ip-prefix 172.16.1.0/24
    site-list my-site
      site-id 100
    vpn-list for-vpn-1
```



```

    vpn 1
    data-policy to-gre-tunnel
    vpn-list for-vpn-1
    sequence 10
    match
    source-data-prefix-list for-firewall
    action accept
    set service FW local
  apply-policy site-list my-site
  data-policy to-gre-tunnel from-service

```

Here is an example of the same configuring using a GRE-specific static route to direct data traffic from VPN 1 into the GRE tunnels:

```

vEdge# show running-config
vpn 0
  interface gre1
    ip address 172.16.111.11/24
    keepalive 60 10
    tunnel-source 172.16.255.11
    tunnel-destination 10.1.2.27
    no shutdown
  !
!
vpn 1
  ip gre-route 172.16.1.0/24 vpn 0 interface gre1

```

The **show interface** command displays the GRE interface in VPN 0:

vEdge# **show interface vpn 0**

VPN	INTERFACE	IP ADDRESS	IF ADMIN STATUS	IF OPER STATUS	ENCAP TYPE	PORT TYPE	MTU	HWADDR	SPEED MBPS	DUPLEX	TCP MSS ADJUST	UPTIME	RX PACKETS	TX PACKETS
0	gre1	172.16.111.11/24	Up	Down	null	service	1500	0a:00:05:0b:00:00	-	-	1420	-	0	0
0	ge0/1	10.0.26.11/24	Up	Up	null	service	1500	00:0c:29:ab:b7:62	10	full	1420	0:03:35:14	89	5
0	ge0/2	10.0.5.11/24	Up	Up	null	transport	1500	00:0c:29:ab:b7:6c	10	full	1420	0:03:35:14	9353	18563
0	ge0/3	-	Down	Up	null	service	1500	00:0c:29:ab:b7:76	10	full	1420	0:03:57:52	99	0
0	ge0/4	10.0.7.11/24	Up	Up	null	service	1500	00:0c:29:ab:b7:80	10	full	1420	0:03:35:14	89	5
0	ge0/5	-	Down	Up	null	service	1500	00:0c:29:ab:b7:8a	10	full	1420	0:03:57:52	97	0
0	ge0/6	-	Down	Up	null	service	1500	00:0c:29:ab:b7:94	10	full	1420	0:03:57:52	85	0
0	ge0/7	10.0.100.11/24	Up	Up	null	service	1500	00:0c:29:ab:b7:9e	10	full	1420	0:03:56:30	3146	2402
0	system	172.16.255.11/32	Up	Up	null	loopback	1500	00:00:00:00:00:00	10	full	1420	0:03:34:15	0	0

You can also view the GRE tunnel information:

vEdge# **show tunnel gre-keepalives**

VPN	IF NAME	SOURCE IP	DEST IP	ADMIN STATE	OPER STATE	KA ENABLED	REMOTE TX PACKETS	REMOTE RX PACKETS	TX PACKETS	RX PACKETS	TX ERRORS	RX ERRORS	TRANSITIONS
0	gre1	10.0.5.11	10.1.2.27	up	down	true	0	0	442	0	0	0	0

vEdge# **show tunnel statistics**

```

tunnel statistics gre 10.0.5.11 10.1.2.27 0 0
  tunnel-mtu      1460
  tx_pkts         451
  tx_octets       54120
  rx_pkts         0
  rx_octets       0
  tcp-mss-adjust  1380

```

Configure the System Interface

For each Cisco vEdge device, you configure a system interface with the **system system-ip** command. The system interface's IP address is a persistent address that identifies the Cisco vEdge device. It is similar to a router ID on a regular router, which is the address used to identify the router from which packets originated.

```
vEdge(config)# system system-ip ipv4-address
```

Specify the system IP address as an IPv4 address in decimal four-part dotted notation. Specify just the address; the prefix length (/32) is implicit.

The system IP address can be any IPv4 address except for 0.0.0.0/8, 127.0.0.0/8, and 224.0.0.0/4, and 240.0.0.0/4 and later. Each device in the overlay network must have a unique system IP address. You cannot use this same address for another interface in VPN 0.

The system interface is placed in VPN 0, as a loopback interface named **system**. Note that this is not the same as a loopback address that you configure for an interface.

To display information about the system interface, use the **show interface** command. For example:

```
vEdge# show running-config system system-ip
system
 system-ip 172.16.255.11
!
vEdge# show interface vpn 0
```

VPN	INTERFACE	IP ADDRESS	IF ADMIN STATUS	IF OPER STATUS	ENCAP TYPE	PORT TYPE	MTU	HWADDR	SPEED MBPS	DUPLEX	TCP MSS ADJUST	UPTIME	RX PACKETS	TX PACKETS
0	ge0/1	10.0.26.11/24	Up	Up	null	service	1500	00:0c:29:ab:b7:62	1000	full	1420	0:10:32:16	1606	8
0	ge0/2	10.0.5.11/24	Up	Up	null	transport	1500	00:0c:29:ab:b7:6c	1000	full	1420	0:10:32:16	307113	303457
0	ge0/3	-	Down	Up	null	service	1500	00:0c:29:ab:b7:76	1000	full	1420	0:10:47:49	1608	0
0	ge0/4	10.0.7.11/24	Up	Up	null	service	1500	00:0c:29:ab:b7:80	1000	full	1420	0:10:32:16	1612	8
0	ge0/5	-	Down	Up	null	service	1500	00:0c:29:ab:b7:8a	1000	full	1420	0:10:47:49	1621	0
0	ge0/6	-	Down	Up	null	service	1500	00:0c:29:ab:b7:94	1000	full	1420	0:10:47:49	1600	0
0	ge0/7	10.0.100.11/24	Up	Up	null	service	1500	00:0c:29:ab:b7:9e	1000	full	1420	0:10:47:31	3128	1165
0	system	172.16.255.11/32	Up	Up	null	loopback	1500	00:00:00:00:00:00	10	full	1420	0:10:31:58	0	0

The system IP address is used as one of the attributes of the OMP TLOC. Each TLOC is uniquely identified by a 3-tuple comprising the system IP address, a color, and an encapsulation. To display TLOC information, use the **show omp tlocs** command.

For device management purposes, it is recommended as a best practice that you also configure the same system IP address on a loopback interface that is located in a service-side VPN that is an appropriate VPN for management purposes. You use a loopback interface because it is always reachable when the router is operational and when the overlay network is up. If you were to configure the system IP address on a physical interface, both the router and the interface would have to be up for the router to be reachable. You use a service-side VPN because it is reachable from the data center. Service-side VPNs are VPNs other than VPN 0 (the WAN transport VPN) and VPN 512 (the management VPN), and they are used to route data traffic.

Here is an example of configuring the system IP address on a loopback interface in VPN 1:

```
vEdge# config
Entering configuration mode terminal
vEdge(config)# vpn 1
vEdge(config-vpn-1)# interface loopback0 ip address 172.16.255.11/32
vEdge(config-vpn-1)# no shutdown
vEdge(config-interface-loopback0)# commit and-quit
Commit complete.
vEdge# show interface
```

VPN	INTERFACE	IP ADDRESS	IF ADMIN STATUS	IF OPER STATUS	ENCAP TYPE	PORT TYPE	MTU	HWADDR	SPEED MBPS	DUPLEX	TCP MSS ADJUST	UPTIME	RX PACKETS	TX PACKETS
0	ge0/1	10.0.26.11/24	Up	Up	null	service	1500	00:0c:29:ab:b7:62	1000	full	1420	0:10:27:33	1597	8
0	ge0/2	10.0.5.11/24	Up	Up	null	transport	1500	00:0c:29:ab:b7:6c	1000	full	1420	0:10:27:33	304819	301173
0	ge0/3	-	Down	Up	null	service	1500	00:0c:29:ab:b7:76	1000	full	1420	0:10:43:07	1599	0
0	ge0/4	10.0.7.11/24	Up	Up	null	service	1500	00:0c:29:ab:b7:80	1000	full	1420	0:10:27:33	1603	8

```

0   ge0/5   -           Down   Up      null   service 1500 00:0c:29:ab:b7:8a 1000 full 1420 0:10:43:07 1612 0
0   ge0/6   -           Down   Up      null   service 1500 00:0c:29:ab:b7:94 1000 full 1420 0:10:43:07 1591 0
0   ge0/7   10.0.100.11/24 Up     Up      null   service 1500 00:0c:29:ab:b7:9e 1000 full 1420 0:10:42:48 3118 1164
0   system  172.16.255.11/32 Up     Up      null   loopback 1500 00:00:00:00:00:00 10   full 1420 0:10:27:15 0 0
1   ge0/0   10.2.2.11/24   Up     Up      null   service 1500 00:0c:29:ab:b7:58 1000 full 1420 0:10:27:30 5734 4204
1   loopback0 172.16.255.11/32 Up     Up      null   service 1500 00:00:00:00:00:00 10   full 1420 0:00:00:28 0 0
512 eth0     10.0.1.11/24   Up     Up      null   service 1500 00:50:56:00:01:0b 1000 full 0 0:10:43:03 20801 14368

```

Configure Control Plane High Availability

A highly available Cisco Catalyst SD-WAN network contains two or more Cisco Catalyst SD-WAN Controllers in each domain. A Cisco Catalyst SD-WAN domain can have up to eight Cisco Catalyst SD-WAN Controllers, and each Cisco vEdge device, by default, connects to two of them. You change this value on a per-tunnel basis:

```
vEdge(config-tunnel-interface)# max-controllers number
```

When the number of Cisco Catalyst SD-WAN Controllers in a domain is greater than the maximum number of controllers that a domain's Cisco vEdge devices are allowed to connect to, the Cisco SD-WAN software load-balances the connections among the available Cisco Catalyst SD-WAN Controllers.

Configure Other Interfaces

Configure Interfaces in the Management (VPN 512)

On all Cisco Catalyst SD-WAN devices, VPN 512 is used for out-of-band management, by default as part of the factory-default configuration. On Cisco vEdge devices the interface type for management interfaces is **mgmt**, and the initial address for the interface is 192.168.1.1.

```

vEdge# show running-config vpn 512
vpn 512
  interface mgmt0
    ip dhcp-client
    no shutdown
  !
!

```

To display information about the configured management interfaces, use the **show interface** command. For example:

```

vEdge# show interface vpn 512

```

VPN	INTERFACE	IP ADDRESS	IF ADMIN STATUS	IF OPER STATUS	ENCAP TYPE	PORT TYPE	MTU	HWADDR	SPEED MBPS	DUPLEX	TCP MSS ADJUST	UPTIME	RX PACKETS	TX PACKETS
512	mgmt0	192.168.1.1/24	Up	Up	null	service	1500	00:50:56:00:01:1f	1000	full	0	0:04:08:01	1131	608



Note VPN 512 is not advertised in the overlay. It is local to the device. If you need a management VPN that is reachable through the overlay, create a VPN with a number other than 512.

Configure Service-Side Interfaces for Carrying Data Traffic

On Cisco vEdge devices, the VPNs other than 0 and 512 are service-side VPNs, and the interfaces in these VPNs connect the router to service-side LANs and WLANs. These interfaces are the interfaces that carry data

traffic between vEdge routers and sites across the overlay network. At a minimum, for these interfaces, you must configure an IPv4 address, and you must enable the interface:

```
vEdge(config)# vpn vpn-id
vEdge(config-vpn)# interface ge slot / port
vEdge(config-interface)# ip address prefix/length
vEdge(config-interface)# no shutdown
```

For service-side interfaces, you can configure up to four secondary IP addresses.

```
vEdge(config)# vpn vpn-id
vEdge(config-vpn)# interface ge slot/port
vEdge(config-interface)# ip secondary-address ipv4-address
```

To display information about the configured data traffic interfaces, use the **show interface** command.

```
vEdge# show interface vpn 1
```

VPN	INTERFACE	IP ADDRESS	IF ADMIN STATUS	IF OPER STATUS	ENCAP TYPE	PORT TYPE	MTU	HWADDR	SPEED MBPS	DUPLEX	TCP MSS ADJUST	UPTIME	RX PACKETS	TX PACKETS
1	ge0/1	10.192.1.1/28	Up	Up	null	service	1500	00:0c:bd:05:f0:84	100	full	0	1:05:44:07	399	331
1	loopback1	10.255.1.1/32	Up	Up	null	service	1500	00:00:00:00:00:00	10	full	0	1:05:44:07	0	0

For some protocols, you specify an interface as part of the protocol's configuration. In these cases, the interface used by the protocol must be the same as one of the interfaces configured in the VPN. As example is OSPF, where you place interfaces in OSPF areas. In this example, the interface **ge0/0** is configured in VPN 1, and this interface is configured to be in the OSPF backbone area:

```
vEdge# show running-config vpn 1
vpn 1
router
ospf
router-id 172.16.255.21
timers spf 200 1000 10000
redistribute static
redistribute omp
area 0
interface ge0/0
exit
exit
!
!
interface ge0/0
ip address 10.2.3.21/24
no shutdown
!
!
```

Configure Loopback Interfaces

Use the interface name format **loopback string**, where *string* can be any alphanumeric value and can include underscores (_) and hyphens (-). The total interface name, including the string "loopback", can be a maximum of 16 characters long. (Note that because of the flexibility of interface naming in the CLI, the interfaces **lo0** and **loopback0** are parsed as different strings and as such are not interchangeable. For the CLI to recognize as interface as a loopback interface, its name must start with the full string **loopback**.)

One special use of loopback interfaces is to configure data traffic exchange across private WANs, such as MPLS or metro Ethernet networks. To allow a router that is behind a private network to communicate directly over the private WAN with other edge routers, you direct data traffic to a loopback interface that is configured as a tunnel interface rather than to an actual physical WAN interface.

Configure Interface Properties

Set the Interface Speed

When a Cisco vEdge device comes up, the Cisco Catalyst SD-WAN software autodetects the SFPs present in the router and sets the interface speed accordingly. The software then negotiates the interface speed with the device at the remote end of the connection to establish the actual speed of the interface. To display the hardware present in the router, use the **show hardware inventory** command:

```
vEdge# show hardware inventory
```

HW TYPE	HW DEV INDEX	VERSION	PART NUMBER	SERIAL NUMBER	DESCRIPTION
Chassis	0	3.1	vEdge-1000	110D145130001	vEdge-1000
CPU	0	None	None	None	Quad-Core Octeon-II
DRAM	0	None	None	None	2048 MB DDR3
Flash	0	None	None	None	nor Flash - 16.00 MB
eMMC	0	None	None	None	eMMC - 7.31 GB
PIM	0	None	ge-fixed-8	None	8x 1GE Fixed Module
Transceiver	0	A	FCLF-8521-3	PQD3FHL	Port 0/0, Type 0x8 (Copper), Vendor FINISAR CORP.
Transceiver	1	PB	1GBT-SFP05	0000000687	Port 0/1, Type 0x8 (Copper), Vendor BEL-FUSE
FanTray	0	None	None	None	Fixed Fan Tray - 2 Fans

To display the actual speed of each interface, use the **show interface** command. Here, interface **ge0/0**, which connects to the WAN cloud, is running at 1000 Mbps (1Gbps; it is the 1GE PIM highlighted in the output above), and interface **ge0/1**, which connects to a device at the local site, has negotiated a speed of 100 Mbps.

```
vEdge# show interface
```

VPN	INTERFACE	IP ADDRESS	IF ADMIN STATUS	IF OPER STATUS	ENCAP TYPE	PORT TYPE	MTU	HWADDR	SPEED MBPS	DUPLEX	TCP MSS ADJUST	UPTIME	RX PACKETS	TX PACKETS
0	ge0/0	192.168.1.4/24	Up	Up	null	transport	1500	00:0c:bd:05:f0:83	1000	full	1300	0:06:10:59	2176305	2168760
0	ge0/2	-	Down	Down	null	service	1500	00:0c:bd:05:f0:81	-	-	0	-	0	0
0	ge0/3	-	Down	Down	null	service	1500	00:0c:bd:05:f0:82	-	-	0	-	0	0
0	ge0/4	-	Down	Down	null	service	1500	00:0c:bd:05:f0:87	-	-	0	-	0	0
0	ge0/5	-	Down	Down	null	service	1500	00:0c:bd:05:f0:88	-	-	0	-	0	0
0	ge0/6	-	Down	Down	null	service	1500	00:0c:bd:05:f0:85	-	-	0	-	0	0
0	ge0/7	-	Down	Down	null	service	1500	00:0c:bd:05:f0:86	-	-	0	-	0	0
0	system	10.255.1.1/32	Up	Up	null	loopback	1500	00:00:00:00:00:00	10	full	0	0:06:11:15	0	0
1	ge0/1	10.192.1.1/28	Up	Up	null	service	1500	00:0c:bd:05:f0:84	100	full	0	0:06:10:59	87	67
1	loopback1	10.255.1.1/32	Up	Up	null	service	1500	00:00:00:00:00:00	10	full	0	0:06:10:59	0	0
2	loopback0	10.192.1.2/32	Up	Up	null	service	1500	00:00:00:00:00:00	10	full	0	0:06:10:59	0	0
512	mgmt0	-	Up	Down	null	mgmt	1500	00:0c:bd:05:f0:80	-	-	0	-	0	0

For non-physical interfaces, such as those for the system IP address and loopback interfaces, the interface speed is set by default to 10 Mbps.

To override the speed negotiated by the two devices on the interface, disable autonegotiation and configure the desired speed:

```
vEdge(config-vpn)# interface interface-name no autonegotiate
vEdge(config-vpn)# interface interface-name speed (10 | 100)
```

For Cisco Catalyst SD-WAN Controllers and Cisco SD-WAN Manager systems, the initial interface speeds are 1000 Mbps, and the operating speed is negotiated with the device at the remote end of the interface. The controller interface speed may vary depending upon the virtualization platform, the NIC used, and the drivers that are present in the software.

Set the Interface MTU

By default, all interfaces have an MTU of 1500 bytes. You can modify this on an interface:

```
vEdge(config-vpn) # interface interface-name mtu bytes
```

The MTU can range from 576 through 2000 bytes.

To display an interface's MTU, use the **show interface** command.

For Cisco Catalyst SD-WAN Validator, Cisco SD-WAN Manager, and Cisco Catalyst SD-WAN Controller devices, you can configure interfaces to use ICMP to perform path MTU (PMTU) discovery. When PMTU discovery is enabled, the device to automatically negotiates the largest MTU size that the interface supports in an attempt to minimize or eliminate packet fragmentation:

```
vEdge(config-vpn) # interface interface-name pmtu
```

On Cisco vEdge device, the Cisco Catalyst SD-WAN BFD software automatically performs PMTU discovery on each transport connection (that is, for each TLOC, or color). BFD PMTU discovery is enabled by default, and it is recommended that you use it and not disable it. To explicitly configure BFD to perform PMTU discovery, use the **bfd color pmtu-discovery** configuration command. However, you can choose to instead use ICMP to perform PMTU discovery:

```
vEdge(config-vpn) # interface interface-name pmtu
```

BFD is a data plane protocol and so does not run on Cisco Catalyst SD-WAN Validator, Cisco SD-WAN Manager, and Cisco Catalyst SD-WAN Controller devices.



Note If you set an MTU on Cisco vEdge hardware device, when a packet whose size is larger than the MTU is received, the vEdge interface drops the packet. This is true, if the "Do Not Fragment" bit is set or not. However, this behavior is not true for vEdge Cloud devices.



Note From Cisco SD-WAN release 20.5 and later releases, PMTU discovery on Cisco vEdge devices is enabled for asymmetric networks. PMTU is calculated based on the egress path MTU.

Configure Bandwidth Reference Values for Network Interfaces

Table 85: Feature History

Feature Name	Release Information	Description
Upstream and Downstream Bandwidth Reference Values	Cisco IOS XE Catalyst SD-WAN Release 17.16.1a Cisco Catalyst SD-WAN Manager Release 20.16.1	Use the upstream and downstream bandwidth reference values to govern how Cisco SD-WAN Manager displays interface utilization percentages in charts. The values also act as configurable thresholds that trigger interface-bw events when a network interface's utilization exceeds a defined point.

Information About Bandwidth Utilization Reference Values

Minimum supported releases: Cisco IOS XE Catalyst SD-WAN Release 17.16.1a and Cisco Catalyst SD-WAN Manager Release 20.16.1

You can configure bandwidth utilization reference values for each interface. These reference values are called upstream bandwidth and downstream bandwidth. You can configure different reference values for upstream (egress) traffic and downstream (ingress) traffic.

These values act as reference values for calculating bandwidth utilization and generating notifications. Cisco SD-WAN Manager uses these reference values to display bandwidth utilization percentages in charts.

Information About Generating Notifications

To receive event notifications when the traffic bandwidth exceeds 85% utilization, configure reference values for both upstream (transmitted) and downstream (received) traffic. These values act as thresholds for generating the interface-bw events. The range is from 1 and 2,147,483,647 kbps.

The device samples the interface traffic each 10 seconds. If the received or transmitted bandwidth exceeds 85 percent of the configured value in 85 percent of the sampled intervals in a continuous 5-minute period, the device generates an event notification. After the first event notification is generated, sampling continues at the same frequency, but notifications are rate-limited to once per hour. A second event notification is sent, and subsequent notifications are sent, if the bandwidth exceeds 85 percent of the value in 85 percent of the 10-second sampling intervals over the next 1-hour period. If, after 1 hour, the device does not send another event notification, the interval reverts to 5 minutes.

The upstream bandwidth and downstream bandwidth settings are solely for monitoring purposes and do not impose a bandwidth limit on the traffic. For example, in some network configurations, the full bandwidth of an interface may not be available. To ensure that utilization values reflect the available network bandwidth, set bandwidth utilization reference values lower than an interface's full speed.

You can verify the configured upstream and downstream bandwidth values using the **show interface detail** command, which will display the upstream bandwidth (tx-kbps) and downstream bandwidth (rx-kbps) fields showing current usage.

You can monitor transport circuit bandwidth on Cisco IOS XE Catalyst SD-WAN devices and on Cisco SD-WAN Manager.

Information About Monitoring Bandwidth Utilization with Interface Charts

Cisco SD-WAN Manager provides a chart showing bandwidth utilization for each interface of a device. To view the chart, from the Cisco SD-WAN Manager menu, choose **Monitor > Devices**, click a device and click **Interface**.

Configuring upstream or downstream reference values affects how the chart displays the percentages:

- No bandwidth utilization reference values configured: By default, devices calculate the bandwidth utilization value according to the interface speed of the connection.
- Bandwidth utilization reference values configured: If you configure bandwidth utilization reference values, devices calculate the bandwidth utilization value as a percentage of the reference point.

For example, if you configure upstream bandwidth and downstream bandwidth values of 500 megabits per second, and if the downstream utilization is 500 megabits per second, the device reports downstream utilization as 100%.

Devices limit the calculated utilization value to 100% even if the traffic utilization is more than 100% of the configured reference value.

Restrictions for Upstream and Downstream Bandwidth Utilization

- Cisco SD-WAN Manager supports bandwidth data rate and utilization statistics only for a primary network interface, not a subinterface or loopback interface. You can still configure upstream and downstream bandwidth under a subinterface or loopback interface as a reference value or for per-tunnel QoS functionality.
- To configure upstream and downstream bandwidth references for service-side VPN interfaces, you can only use a CLI template.

Configure Upstream and Downstream Bandwidth Reference Values Using a Feature Template

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Feature Templates**.
3. Do one of these:
 - Click **Add Template**, choose a device, and create a Cisco VPN Interface Ethernet template.
 - Edit an existing Cisco VPN Interface Ethernet template.
4. In the **Basic Configuration** section, enter the upstream and downstream bandwidth reference values.
5. Click **Update** to save the template.

Configure Upstream and Downstream Bandwidth Reference Values Using a CLI Add-On Template

For more information about using CLI templates, see [CLI Add-On Feature Templates](#) and [CLI Templates](#).



Note By default, CLI templates execute commands in global config mode.

This section provides example CLI configurations to configure upstream and downstream bandwidth reference values in Cisco SD-WAN Manager.

1. Enter the SD-WAN configuration mode.
sdwan
2. Enter interface configuration mode.
interface *interface-name*
3. Define the upstream or downstream bandwidth reference value in kbps.
bandwidth-upstream *upstream-value*
bandwidth-downstream *downstream-value*

Here is a complete configuration example for configuring upstream and downstream bandwidth reference values:


```
sdwan
!
interface GigabitEthernet1
bandwidth-upstream 10000
bandwidth-downstream 10000
```

Configure Upstream and Downstream Bandwidth Reference Values Using Configuration Groups

Minimum supported releases: Cisco IOS XE Catalyst SD-WAN Release 17.16.1a and Cisco Catalyst SD-WAN Manager Release 20.16.1

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Configuration Groups**.
2. Under **Transport & Management Profile**, click ... adjacent to the transport profile, and choose **Edit**.
3. Click the edit icon adjacent to Ethernet Interface.
4. In the **Basic Configuration** section, enter the upstream and downstream bandwidth reference values.
5. Click **Save**.

Monitor Upstream and Downstream Bandwidth Events Using the Cisco SD-WAN Manager

1. From the Cisco SD-WAN Manager menu, choose **Monitor > Logs > Events**.
2. View the details of upstream and downstream bandwidth values in the device configuration section. The default name for interface bandwidth-related events is **interface-bw**.

Monitor Upstream and Downstream Bandwidth Reference Values Using the CLI

Monitor Interface Alarms

Use the **show platform software sdwan interface-alarm summary** command to display the bandwidth reference values for an interface in the alarm summary. This sample output shows the summary details of a GigabitEthernet interface alarm.

See [Interface Alarm Summary](#), and [Base and High Intervals](#) for more information.

```
device# show platform software sdwan interface-alarm summary
===== Interface Alarm Summary =====
Interface Name      Upstream (kbps)  Downstream (kbps)  Base Interval(s)  High Interval(s)
GigabitEthernet1    10000            10000              300               3600
```

Monitor Bandwidth Utilization Using the Cisco SD-WAN Manager

1. From the Cisco SD-WAN Manager menu, choose **Monitor > Devices**.
2. Click the **Device** tab.
3. Click the hostname of the device you want to monitor.
4. From the **Applications** section, choose **Interface**.
5. Click the **Chart Options** drop-down list and choose **Utilization**.
6. Hover over the real-time or historical data points to view the utilization percentage.

Enable DHCP Server using Cisco SD-WAN Manager

Use the DHCP-Server template for all Cisco Catalyst SD-WANs.

You enable DHCP server functionality on a Cisco Catalyst SD-WAN device interface so it can assign IP addresses to hosts in the service-side network.

To configure a Cisco Catalyst SD-WAN device to act as a DHCP server using Cisco SD-WAN Manager templates:

1. Create a DHCP-Server feature template to configure DHCP server parameters, as described in this topic.
2. Create one or more interface feature templates, as described in the VPN-Interface-Ethernet and the VPN-Interface-PPP-Ethernet help topics.
3. Create a VPN feature template to configure VPN parameters. See the VPN help topic.

To configure a Cisco vEdge device interface to be a DHCP helper so that it forwards broadcast DHCP requests that it receives from DHCP servers, in the DHCP Helper field of the applicable interfaces template, enter the addresses of the DHCP servers.

Navigate to the Template Screen and Name the Template

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Device Templates**, and then click **Create Template**.



Note In Cisco vManage Release 20.7.x and earlier releases, **Device Templates** is titled **Device**.

3. From the **Create Template** drop-down list, choose **From Feature Template**.
4. From the **Device Model** drop-down list, select the type of device for which you are creating the template.
5. Click **Service VPN** or scroll to the **Service VPN** section.
6. Click **Service VPN** drop-down list.
7. From **Additional VPN Templates**, click **VPN Interface**.
8. From the **Sub-Templates** drop-down list, choose **DHCP Server**.
9. From the **DHCP Server** drop-down list, click **Create Template**. The DHCP-Server template form is displayed.

This form contains fields for naming the template, and fields for defining the DHCP Server parameters.

10. In **Template Name**, enter a name for the template.
The name can be up to 128 characters and can contain only alphanumeric characters.
11. In **Template Description**, enter a description of the template.
The description can be up to 2048 characters and can contain only alphanumeric characters.

When you first open a feature template, for each parameter that has a default value, the scope is set to Default (indicated by a check mark), and the default setting or value is shown. To change the default or to enter a value, click the **Scope** drop-down list.

Minimum DHCP Server Configuration

To configure DHCP server functionality, select **Basic Configuration** and configure the following parameters. Parameters marked with an asterisk as required to configure DHCP servers.

Table 86:

Parameter Name	Description
Address Pool*	Enter the IPv4 prefix range, in the format <i>prefix/length</i> , for the pool of addresses in the service-side network for which the router interface acts as DHCP server.
Exclude Addresses	Enter one or more IP addresses to exclude from the DHCP address pool. To specify multiple individual addresses, list them separated by a comma. To specify a range of addresses, separate them with a hyphen.
Maximum Leases	Specify the number of IP addresses that can be assigned on this interface. <i>Range:</i> 0 through 4294967295
Lease Time	Specify how long a DHCP-assigned IP address is valid. <i>Range:</i> 0 through 4294967295 seconds
Offer Time	Specify how long the IP address offered to a DHCP client is reserved for that client. By default, an offered IP address is reserved indefinitely, until the DHCP server runs out of addresses. At that point, the address is offered to another client. <i>Range:</i> 0 through 4294967295 seconds <i>Default:</i> 600 seconds
Administrative State	Select Up to enable or Down to disable the DHCP functionality on the interface. By default, DHCP server functionality is disabled on an interface.

To save the feature template, click **Save**.

```
vpn vpn-id
interface geslot/port
dhcp-server address-pool prefix/length admin-state (down | up)
    exclude ip-address
    lease-time seconds
    max-leases number
    offer-time minutes
```

Configure Static Leases

To configure a static lease to assign a static IP address to a client device on the service-side network, click **Static Lease**, and click **Add New Static Lease** and configure the following parameters:

Table 87:

Parameter Name	Description
MAC Address	Enter the MAC address of the client to which the static IP address is being assigned.

Parameter Name	Description
IP Address	Enter the static IP address to assign to the client.
Hostname	Enter the hostname of the client device.

To edit a static lease, click **pencil** icon.

To remove a static lease, click **trash** icon.

To save the feature template, click **Save**.

CLI equivalent:

```
vpn vpn-id
interface geslot/port
dhcp-server static-lease mac-address ip ip-address host-name hostname
```

Configure Advanced Options

To configure a advanced DHCP server options, click **Advanced** and then configure the following parameters:

Table 88:

Parameter Name	Description
Interface MTU	Specify the maximum MTU size of packets on the interface. <i>Range:</i> 68 to 65535 bytes
Domain Name	Specify the domain name that the DHCP client uses to resolve hostnames.
Default Gateway	Enter the IP address of a default gateway in the service-side network.
DNS Servers	Enter one or more IP address for a DNS server in the service-side network. Separate multiple entries with a comma. You can specify up to eight addresses.
TFTP Servers	Enter the IP address of a TFTP server in the service-side network. You can specify one or two addresses. If two, separate them with a comma.

To save the feature template, click **Save**.

CLI equivalent:

```
vpn vpn-id
interface geslot/port
dhcp-server options
    default-gateway ip-address
    dns-servers ip-address
    domain-name domain-name
    interface-mtu mtu
    tftp-servers ip-address
```

Release Information

Introduced in Cisco SD-WAN Manager in Release 15.2.

Configure DHCP Using CLI

When you configure a tunnel interface on a Cisco vEdge device, a number of services are enabled by default on that interface, including DHCP.

A Cisco vEdge device can act as a DHCP server for the service-side network to which it is connected, and it can also act as a DHCP helper, forwarding requests for IP addresses from devices in the service-side network to a DHCP server that is in a different subnet on the service side of the Cisco vEdge device.

Enable DHCP on the WAN Interface

On a Cisco vEdge device's WAN interface—the interface configured as a tunnel interface in VPN 0, the transport VPN—DHCP is enabled by default. You can see this by using the **details** filter with the **show running-config** command. This command also shows that the DNS and ICMP services are enabled by default.

```
vm1# show running-config vpn 0 interface ge0/2 tunnel-interface | details
vpn 0
  interface ge0/2
    tunnel-interface
      encapsulation ipsec weight 1
      color lte
      control-connections
      carrier default
      no allow-service all
      no allow-service bgp
      allow-service dhcp
      allow-service dns
      allow-service icmp
      no allow-service ospf
      no allow-service sshd
      no allow-service ntp
      no allow-service stun
    !
  !
!
```

Enabling DHCP on the router's WAN interface allows the device that actually connects the router to the transport network (such as a DSL router) to dynamically assign a DHCP address to the Cisco vEdge device. The DHCP service in VPN 0 affects the transport-side network.

Configure Cisco vEdge Device as a DHCP Server

One or more service-side interfaces on Cisco vEdge device can act as a DHCP server, assigning IP addresses to hosts in the service-side network. To do this, configure this function on the interface that connects the Cisco vEdge device to the local site's network. At a minimum, you must configure the pool of IP addresses available for assigning to hosts:

```
vEdge(config-vpn)# interface ge slot / port dhcp-serveraddress-pool ip-address / prefix
vEdge(config-dhcp-server)#
```

You can exclude IP addresses that fall within the range of the DHCP address pool:

```
vEdge(config-dhcp-server)#exclude ip-address
```

To specify multiple individual addresses, list them in a single **exclude** command, separated by a space (for example, **exclude 10.1.1.1 10.2.2.2 10.3.3.3**). To specify a range of addresses, separate them with a hyphen (for example, **exclude 10.255.1.1-10.255.1.10**).

You can also statically assign IP addresses to a host:

```
vEdge(config-dhcp-server)# static-lease mac-address ip ip-address
```

By default, the DHCP server on a single interface can assign 254 DHCP leases, and each lease is valid for 24 hours. The offer of an IP address is valid indefinitely, until that DHCP server runs out of addresses to offer. You can modify these values:

```
vEdge(config-dhcp-server) # max-leases number
vEdge(config-dhcp-server) # lease-time seconds
vEdge(config-dhcp-server) # offer-time seconds
```

These values can range from 0 through ($2^{32} - 1$).

The Cisco Catalyst SD-WAN software supports DHCP server options that allow you to configure the IP addresses of a default gateway, DNS server, and TFTP server in the service-side network and the network mask of the service-side network:

```
vEdge(config-dhcp-server) # options default-gateway ip-address
vEdge(config-dhcp-server) # options dns-servers ip-address
vEdge(config-dhcp-server) # options domain-name domain-name
vEdge(config-dhcp-server) # options interface-mtu mtu
vEdge(config-dhcp-server) # options tftp-servers ip-address
vEdge(config-dhcp-server) # options option-code 43 ascii | hex
vEdge(config-dhcp-server) # options option-code 191 ascii
```

Configure a Cisco vEdge Device as a DHCP Helper

One or more service-side interfaces on a Cisco vEdge device can be a DHCP helper. With this configuration, the interface forwards any broadcast BOOTP DHCP requests that it receives from hosts on the service-side network to the DHCP server or servers specified by the configured IP helper address (or addresses) and returns the assigned IP address to the requester.

When the DHCP server at the Cisco vEdge device's local site is on a different segment than the devices connected to the Cisco vEdge device or than the Cisco vEdge device itself. When configured as a DHCP helper, the Cisco vEdge device interface forwards any broadcast BOOTP DHCP requests that it receives to the DHCP server specified by the configured IP helper address.

To configure an interface as a DHCP helper, configure the IP address of the DHCP server on the interface that connects to the local site's network:

```
vEdge(config-vpn) # interface ge slot/port dhcp-helper ip-address
```

You can configure up to four IP addresses, and you must enter the addresses in a single **dhcp-helper** command.

In Releases 17.2.2 and later, you can configure up to eight IP address. You must enter all the addresses in a single **dhcp-helper** command.

Configuring PPPoE

The Point-to-Point Protocol over Ethernet (PPPoE) connects multiple users over an Ethernet local area network to a remote site through common customer premises equipment. PPPoE is commonly used in a broadband aggregation, such as by digital subscriber line (DSL). PPPoE provides authentication with the CHAP or PAP protocol. In the Cisco Catalyst SD-WAN overlay network, Cisco Catalyst SD-WAN devices can run the PPPoE client. The PPPoE server component is not supported.

To configure PPPoE client on a Cisco Catalyst SD-WAN device, you create a PPP logical interface and link it to a physical interface. The PPPoE connection comes up when the physical interface comes up. You can link a PPP interface to only one physical interface on a Cisco Catalyst SD-WAN device, and you can link a physical interface to only one PPP interface. To enable more than one PPPoE interfaces on a Cisco Catalyst SD-WAN device, configure multiple PPP interfaces.

It is recommended that you configure quality of service (QoS) and shaping rate on a PPPoE-enabled physical interface, and not on the PPP interface.

PPPoE-enabled physical interfaces do not support:

- 802.1Q
- Subinterfaces
- NAT, PMTU, and tunnel interfaces. These are configured on the PPP interface and therefore not available on PPPoE-enabled interfaces.

The Cisco Catalyst SD-WAN implementation of PPPoE does not support the Compression Control Protocol (CCP) options, as defined in RFC 1962.



Note Follow these steps to replace a template configured with PPPoE as WAN interface with a regular interface in Dialer100:

1. Remove the IP address assigned to the dialer interface using the command:

```
no ip address <ip> <mask>
```

2. Add a new IP address for the dialer interface.

Configure PPPoE from Cisco SD-WAN Manager Templates

To use Cisco SD-WAN Manager templates to configure PPPoE on Cisco vEdge device, you create three feature templates and one device template:

- Create a VPN-Interface-PPP feature template to configure PPP parameters for the PPP virtual interface.
- Create a VPN-Interface-PPP-Ethernet feature template to configure a PPPoE-enabled interface.
- Optionally, create a VPN feature template to modify the default configuration of VPN 0.
- Create a device template that incorporates the VPN-Interface-PPP, VPN-Interface-PPP-Ethernet, and VPN feature templates.

Create a VPN-Interface-PPP feature template to configure PPP parameters for the PPP virtual interface:

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Feature Templates**, and click **Add Template**.



Note In Cisco vManage Release 20.7.x and earlier releases, **Feature Templates** is titled **Feature**.

3. Choose Cisco vEdge device Cloud or a router model.
4. Choose the **VPN-Interface-PPP** template.
5. In the template, configure the following parameters:

Table 89:

Parameter Field	Procedure
Template Name	Enter a name for the template. It can be up to 128 alphanumeric characters.
Description	Enter a description for the template. It can be up to 2048 alphanumeric characters.
Shutdown	Click No to enable the PPP virtual interface.
Interface Name	Enter the number of the PPP interface. It can be from 1 through 31.
Description (optional)	Enter a description for the PPP virtual interface.
Authentication Protocol	Select either CHAP or PAP to configure one authentication protocol, or select PAP and CHAP to configure both. For CHAP, enter the hostname and password provided by your ISP. For PAP, enter the username and password provided by your ISP. If you are configuring both PAP and CHAP, to use the same username and password for both, click Same Credentials for PAP and CHAP.
AC Name (optional)	Select the PPP tab, and in the AC Name field, enter the name of the the name of the access concentrator used by PPPoE to route connections to the Internet.
IP MTU	Click Advanced , and in the IP MTU field, ensure that the IP MTU is at least 8 bytes less than the MTU on the physical interface. The maximum MTU for a PPP interface is 1492 bytes. If the PPPoE server does not specify a maximum receive unit (MRU), the MTU value for the PPP interface is used as the MRU. Starting from Cisco vManage Release 20.9.1, there is 8 bytes overheads deduced based on the specified IP MTU value when configuration is pushed to the device.
Save	To save the feature template, click Save .

To create a VPN-Interface-PPP-Ethernet feature template to enable the PPPoE client on the physical interfaces:

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Feature Templates**, and click **Add Template**.



Note In Cisco vManage Release 20.7.x and earlier releases, **Feature Templates** is titled **Feature**.

3. Choose Cisco vEdge device Cloud or a router model.
4. Choose the **VPN-Interface-PPP-Ethernet** template.
5. In the template, configure the following parameters:

Parameter Field	Procedure
Template Name	Enter a name for the template. It can be up to 128 alphanumeric characters.
Description	Enter a description for the template. It can be up to 2048 alphanumeric characters.
Shutdown	Click No to enable the PPPoE-enabled interface.

Parameter Field	Procedure
Interface Name	Enter the name of the physical interface in VPN 0 to associate with the PPP interface.
Description (optional)	Enter a description for the PPPoE-enabled interface.
IP Configuration	Assign an IP address to the physical interface: - To use DHCP, select Dynamic. The default administrative distance of routes learned from DHCP is 1. - To configure the IP address directly, enter of the IPv4 address of the interface.
DHCP Helper (optional)	Enter up to four IP addresses for DHCP servers in the network.
Save	To save the feature template, click Save .

To create a VPN feature template to configure the PPPoE-enabled interface in VPN 0, the transport VPN:

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Feature Templates**, and click **Add Template**.



Note In Cisco vManage Release 20.7.x and earlier releases, **Feature Templates** is titled **Feature**.

3. Choose Cisco vEdge device Cloud or a router model.
4. Choose the **VPN** template.
5. In the template, configure the following parameters:

Parameter Field	Procedure
Template Name	Enter a name for the template. It can be up to 128 alphanumeric characters.
Description	Enter a description for the template. It can be up to 2048 alphanumeric characters.
VPN Identifier	Enter VPN identifier 0.
Name	Enter a name for the VPN.
Other interface parameters	Configure the desired interface properties.
Save	To save the feature template, click Save .

To create a device template that incorporates the VPN-Interface-PPP, VPN-Interface-PPP-Ethernet, and VPN feature templates:

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Device Templates**, and then click **Create Template**.



Note In Cisco vManage Release 20.7.x and earlier releases, **Device Templates** is titled **Device**.

3. From the **Create Template** drop-down list, choose **From Feature Template**.
4. From the **Device Model** drop-down list, choose the type of device for which you are creating the device template.

Cisco SD-WAN Manager displays the feature templates for the device type you selected. Required templates are indicated with an asterisk (*).
5. Enter a name and description for the device template. These fields are mandatory. The template name cannot contain special characters.
6. In **Transport & Management VPN**, under **VPN 0**, from the drop-down list of available templates, select the desired feature template. The list of available templates are the ones that you have previously created.
7. In **Additional VPN 0 Templates**, click the plus sign (+) next to **VPN Interface PPP**.
8. From **VPN-Interface-PPP** and **VPN-Interface-PPP-Ethernet** fields, select the feature templates to use.
9. To configure multiple PPPoE-enabled interfaces in VPN 0, click the plus sign (+) next to Sub-Templates.
10. To include additional feature templates in the device template, in the remaining sections, select the feature templates in turn, and from the drop-down list of available templates, select the desired template. The list of available templates are the ones that you have previously created. Ensure that you select templates for all mandatory feature templates and for any desired optional feature templates.
11. To create the device template, click **Create**.

To attach a device template to a device:

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Device Templates**.



Note In Cisco vManage Release 20.7.x and earlier releases, **Device Templates** is titled **Device**.

3. Choose a template.
4. Click ..., and click **Attach Device**.
5. Search for a device or select a device from the Available Device(s) column to the left.
6. Click the arrow pointing right to move the device to the Selected Device(s) column on the right.
7. Click **Attach**.

Configure PPPoE from the CLI

Table 90: Feature History

Feature Name	Release Information	Feature Description
Assign Static IP Address to PPP Interface.	Cisco SD-WAN Release 20.4.1 Cisco vManage Release 20.4.1	This feature enables you to assign a static IP address to a PPP interface and configure PPP interface echo requests.

To use the CLI to configure PPPoE on Cisco vEdge devices:

1. Create a PPP interface. The interface number can be from 1 through 31.

```
vEdge(config-vpn) # interface ppp number
```
2. Configure an authentication method for PPPoE and authentication credentials. You can configure both CHAP and PAP authentication on the same PPP interface. The software tries both methods and uses the first one that succeeds.

```
vEdge(config-interface-ppp) # ppp authentication chap hostname name password password
```

```
vEdge(config-interface-ppp) # ppp authentication pap password password sent-username username
```
3. Enable the PPP interface to be operationally up:

```
vEdge(config-interface-ppp) # no shutdown
```
4. Configure the MTU of the PPP interface. The maximum MTU for a PPP interface is 1492 bytes. If maximum receive unit (MRU) is not specified by the PPPoE server, the MTU value for the PPP interface is used as the MRU.

```
vEdge(config-interface-ppp) # mtu bytes
```
5. Configure a tunnel interface for the PPP interface:

```
vEdge(config-interface-ppp) # tunnel-interface color color
```
6. Optionally, configure the name of the access concentrator used by PPPoE to route connections to the internet:

```
vEdge(config-interface-ppp) # ppp ac-name name
```
7. Link a physical Gigabit Ethernet interface in VPN 0 to the PPP interface:

```
vEdge(config-vpn) # interface slot|port
```

```
vEdge(config-interface-ge) # pppoe-client ppp-interface ppp number
```
8. Enable the physical Gigabit Ethernet interface to be operationally up:

```
vEdge(config-interface-ge) # no shutdown
```
9. Optionally, configure a static IP address for the PPP interface:

```
vEdge(config-vpn) # interface ppp
```

```
vEdge(config-interface-ppp) # ppp local-ip ipv4-address
```
10. Optionally, configure the number of consecutive echo requests after which the PPP interface terminates if no responses are received:

```
vEdge(config-interface-ppp) # ppp lcp-echo-failure number
```

11. Optionally, configure the number of seconds between echo requests that the PPP interface sends:

```
vEdge(config-interface-ppp)# ppp lcp-echo-interval seconds
```

Here is an example of a PPPoE configuration:

```
vEdge# show running-config vpn 0
vpn 0
  interface ge0/1
    pppoe-client ppp-interface ppp10
    no shutdown
  !
  interface ppp10
    ppp authentication chap
    hostname branch100@corp.bank.myisp.net
    password $4$OHHjdmsC6M8zj4BgLEFXKw==
  !
  ppp ac-name ac_name
  ppp local-ip 10.1.5.15
  ppp lcp-echo-failure 5
  ppp lcp-echo-interval 25
  tunnel-interface
    encapsulation ipsec
    color gold
    no allow-service all
    no allow-service bgp
    allow-service dhcp
    allow-service dns
    allow-service icmp
    no allow-service ospf
    no allow-service sshd
    no allow-service ntp
    no allow-service stun
  !
  mtu 1492
  no shutdown
!
```

To view existing PPP interfaces, use the **show ppp interface** command.

```
vEdge# show ppp interface
```

VPN	IFNAME	PPPOE INTERFACE	INTERFACE IP	GATEWAY IP	PRIMARY DNS	SECONDARY DNS	MTU
0	ppp10	ge0/1	10.0.0.11	10.255.255.254	10.8.8.8	10.8.4.4	1150

To view PPPoE session information, use the **show pppoe session** command.

```
vEdge# show pppoe session
```

VPN	IFNAME	SESSION ID	SERVER MAC	LOCAL MAC	PPP INTERFACE	SERVICE AC NAME	SERVICE NAME
0	ge0/1	1	00:0c:29:2e:20:1a	00:0c:29:be:27:f5	ppp1	branch100	-
0	ge0/3	1	00:0c:29:2e:20:24	00:0c:29:be:27:13	ppp2	branch100	-

Configure PPPoE Over ATM

Table 91: Feature History

Feature Name	Release Information	Description
Configure PPPoE over ATM	Cisco IOS XE Catalyst SD-WAN Release 17.4.1a Cisco vManage Release 20.4.1	This feature provides support for configuring PPPoEoA on Cisco IOS XE Catalyst SD-WAN devices. PPPoEoA uses AAL5MUX encapsulation which delivers better efficiency compared to other encapsulation methods.

You can configure PPPoE over ATM interfaces (PPPoEoA) on Cisco IOS XE Catalyst SD-WAN devices that support ADSL. PPPoEoA uses ATM Adaptation Layer 5 Multiplexed Encapsulation (AAL5MUX) encapsulation to carry PPPoE over ATM permanent virtual circuits (PVCs), providing efficiency gain over AAL5 LLC/SNAP encapsulation.

PPPoEoA over AAL5MUX reduces Subnetwork Access Protocol (SNAP) encapsulation bandwidth usage, using multiplexed (MUX) encapsulation to reduce the number of cells needed to carry voice packets. Deploying the PPPoEoA over ATM AAL5MUX feature in a VoIP environment results in improved throughput and bandwidth usage.

Supported Platforms for PPPoE Over ATM

The following platforms support PPPoE over ATM:

- Cisco 1100 4G/6G Series Integrated Services routers.
- Cisco1100 Series Integrated Service routers.
- Cisco1109 Series Integrated Service routers.
- Cisco111x Series Integrated Service routers.
- Cisco1111x Series Integrated Service routers.
- Cisco1120 Series Integrated Service routers.
- Cisco1160 Series Integrated Service routers.

Configure PPPoE Over ATM using Cisco SD-WAN Manager

You can configure PPPoE using in Cisco SD-WAN Manager using the device CLI template.

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. From **Device Templates**, click **Create Template**.



Note In Cisco vManage Release 20.7.x and earlier releases, **Device Templates** is titled **Device**.

3. From the **Create Template** drop-down list, select **CLI Template**.
4. From the **Device Model** drop-down list, select the type of device for which you are creating the template.
5. In **Template Name**, enter a name for the template. The name can be up to 128 characters and can contain only alphanumeric characters.
6. In **Template Description**, enter a description of the template. The description can be up to 2048 characters and can contain only alphanumeric characters.
7. Choose **Device configuration**. Using this option, you can provide IOS-XE configuration commands that appear in the output of the `show sdwan running-config` command.
8. (Optional) To load the running config of a connected device, select it from the Load Running config from reachable device list and click **Search**.
9. In **CLI Configuration**, enter the configuration either by typing it, cutting and pasting it, or uploading a file. The configuration for PPPoEoA is available in the [Configure PPPoEoA on the CLI](#) section.
10. To convert an actual configuration value to a variable, select the value and click **Create Variable**. Enter the variable name, and click **Create Variable**. You can also type the variable name directly, in the format `{{variable-name}}`; for example, `{{hostname}}`.
11. Click **Add**. The new device template is displayed in the Device Template table. The **Type** column shows **CLI** to indicate that the device template was created from CLI text.

Configure PPPoE Over ATM on the CLI

This section provides example CLI configurations to configure PPoE over ATM on the CLI.

```
Device(config)# interface atm number
Device(config)# no ip address
Device(config)# interface atm number point-to-point
Device(config)# no atm enable-ilmi-trap
Device(config)# encapsulation aal5mux pppoe-client
Device(config)# pppoe-client dial-pool-number number
Device(config)# interface Dialer dialer-rotary-group-number
Device(config)# mtu bytes
Device(config)# ip address negotiated
Device(config-if)# encapsulation encapsulation-type
Device(config)# load-interval seconds
Device(config)# dialer pool number
Device(config)# dialer-group group-number
Device(config)# ppp mtu adaptive
Device(config)# ppp chap hostname hostname
Device(config)# ppp chap password secret
Device(config)# ppp ipcp address required
Device(config)# ppp link reorders
```

Configuration Example for Configuring PPPoE Over ATM Interfaces

This example shows configuring PPPoE over ATM interfaces.

```
Device(config)# interface ATM0/1/0
Device(config)# no ip address
Device(config)# no atm enable-ilmi-trap
!
Device(config)# interface ATM0/1/0.10 point-to-point
Device(config)# no atm enable-ilmi-trap
Device(config)# cdp enable
Device(config)# pvc 22/62
Device(config)# uabr 1045
Device(config-if)# encapsulation aal5mux pppoe-client
Device(config)# pppoe-client dial-pool-number 120
!
!
Device(config)# interface Dialer 120
Device(config)# mtu 1492
Device(config)# ip address negotiated
Device(config)# ip nat outside
Device(config-if)# encapsulation ppp
Device(config)# load-interval 30
Device(config)# dialer pool 120
Device(config)# dialer-group 1
Device(config)# ppp mtu adaptive
Device(config)# ppp chap hostname test@cisco.com
Device(config)# ppp chap password 0 cisco
Device(config)# ppp ipcp address required
Device(config)# ppp link reorders
!
```

Configuring VRRP

Table 92: Feature History

Feature Name	Release Information	Description
Support for Multiple VRRP Groups on the Same LAN Interface or Sub-interface	Cisco SD-WAN Release 20.3.1 Cisco vManage Release 20.3.1	This feature increases support from one VRRP group per interface to five VRRP groups per interface. Multiple VRRP groups are useful for providing redundancy and for load balancing.



Note The x710 NIC must have the **t->system-> vrrp-advt-with-phymac** command configured, for VRRP to function.

The Virtual Router Redundancy Protocol (VRRP) is a LAN-side protocol that provides redundant gateway service for switches and other IP end stations. In the Cisco Catalyst SD-WAN software, you configure VRRP on an interface, and typically on a subinterface, within a VPN.

VRRP is only supported with service-side VPNs (VPN 0 and 512 reserved) and if sub-interfaces are used, then the VRRP physical interface must be configured in VPN 0.

```
vEdge(config-vpn-0)# interface ge- slot / port
vEdge(config-interface-ge)# no shutdown
```

For each VRRP interface (or subinterface), you assign an IP address and you place that interface in a VRRP group.

```
vEdge(config-vpn)# interface ge- slot / port . subinterface
vEdge(config-interface-ge)# ip address prefix / length
vEdge(config-interface-ge)# vrrp group-number
```

The group number identifies the virtual router. You can configure a maximum of 512 groups on a router. In a typical VRRP topology, two physical routers are configured to act as a single virtual router, so you configure the same group number on interfaces on both these routers.

For each virtual router ID, you must configure an IP address.

```
vEdge(config-vrrp)# ipv4 ip-address
```

Within each VRRP group, the router with the higher priority value is elected as primary VRRP. By default, each virtual router IP address has a default primary election priority of 100, so the router with the higher IP address is elected as primary. You can modify the priority value, setting it to a value from 1 through 254.

```
vEdge(config-vrrp)# priority number
```

The primary VRRP periodically sends advertisement messages, indicating that it is still operating. If backup routers miss three consecutive VRRP advertisements, they assume that the primary VRRP is down and elect a new primary VRRP. By default, these messages are sent every second. You can change the VRRP advertisement time to be a value from 1 through 3600 seconds.

```
vEdge(config-vrrp)# timer seconds
```

By default, VRRP uses the state of the interface on which it is running, to determine which router is the primary virtual router. This interface is on the service (LAN) side of the router. When the interface for the primary VRRP goes down, a new primary VRRP virtual router is elected based on the VRRP priority value. Because VRRP runs on a LAN interface, if a router loses all its WAN control connections, the LAN interface still indicates that it is up even though the router is functionally unable to participate in VRRP. To take WAN side connectivity into account for VRRP, you can configure one of the following:

- Track the Overlay Management Protocol (OMP) session running on the WAN connection when determining the primary VRRP virtual router.

```
vEdge(config-vrrp)# track-omp
```

If all OMP sessions are lost on the primary VRRP router, VRRP elects a new default gateway from among all the gateways that have one or more active OMP sessions even if the gateway chosen has a lower VRRP priority than the current primary VRRP router. With this option, VRRP failover occurs once the OMP state changes from up to down, which occurs when the OMP hold timer expires. Until the hold timer expires and a new primary VRRP is elected, all overlay traffic is dropped. When the OMP session recovers, the local VRRP interface claims itself as primary VRRP even before it learns and installs OMP routes from the Cisco Catalyst SD-WAN Controllers. Until the routers are learned, traffic is also dropped.

- Track both the OMP session and a list of remote prefixes. *list-name* is the name of a prefix list configured with the **policy lists prefix-list** command on the Cisco vEdge device :

```
vEdge(config-vrrp)# track-prefix-list list-name
```


If all OMP sessions are lost, VRRP failover occurs as described for the **track-omp** option. In addition, if reachability to all the prefixes in the list is lost, VRRP failover occurs immediately, without waiting for the OMP hold timer to expire, thus minimizing the amount of overlay traffic is dropped while the router determines the primary VRRP.

As discussed above, the IEEE 802.1Q protocol adds 4 bytes to each packet's length. Hence, for packets to be transmitted, either increase the MTU size on the physical interface in VPN 0 (the default MTU is 1500 bytes) or decrease the MTU size on the VRRP interface.

For devices running on Cisco IOS XE Catalyst SD-WAN Release 17.4.1a and later, adjusting the MTU size is not required, both the physical interface and sub interface can have the same MTU size.

Here is an example of configuring VRRP on redundant physical interfaces. For subinterface 2, vEdge1 is configured to act as the primary VRRP, and for subinterface 3, vEdge2 acts as the primary VRRP.

```
vEdge1# show running-config vpn 1
vpn 1
  interface ge0/6.2
    ip address 10.2.2.3/24
    mtu 1496
    no shutdown
    vrrp 2
      ipv4 10.2.2.1
      track-prefix-list vrrp-prefix-list1
  !
  !
  interface ge0/6.3
    ip address 10.2.3.5/24
    mtu 1496
    shutdown
    vrrp 3
      ipv4 10.2.3.11
      track-prefix-list vrrp-prefix-list1
  !
  !
  !
```

```
vEdge2# show running-config vpn 1
vpn 1
  interface ge0/1.2
    ip address 10.2.2.4/24
    mtu 1496
    no shutdown
    vrrp 2
      ipv4 10.2.2.1
      track-prefix-list vrrp-prefix-list2
  !
  !
  interface ge0/1.3
    ip address 10.2.3.6/24
    mtu 1496
    no shutdown
    vrrp 3
      ipv4 10.2.3.11
      track-prefix-list vrrp-prefix-list2
  !
  !
  !
```

```
vEdge1# show interface vpn 1
```

VPN	INTERFACE	RX IP ADDRESS ADJUST	TX PACKETS	IF ADMIN STATUS	IF OPER STATUS	ENCAP TYPE	PORT TYPE	MTU	HWADDR	SPEED MBPS	TCP MSS DUPLEX
1	ge0/6.2	10.2.2.3/24	Up	Up	vlan	service	1496	00:0c:29:ab:b7:94	10	full	0
	0:00:05:52	0	357								

```

1    ge0/6.3    10.2.3.5/24    Down    Down    vlan    service    1496    00:0c:29:ab:b7:94    -    -    0
-    -    0    0

```

```
vEdge1# show vrrp interfaces
```

VPN	IF NAME	ID	GROUP	TRACK VIRTUAL PREFIX LIST	PREFIX LIST	VIRTUAL MAC	PRIORITY	VRRP STATE	OMP STATE	ADVERTISEMENT TIMER	MASTER	
											DOWN TIMER	LAST
1	ge0/6.2	2		10.2.2.1	00:0c:29:ab:b7:94	100	master	down	1	3		
	2015-05-01T20:09:37+00:00			-	-							
	ge0/6.3	3		10.2.3.11	00:00:00:00:00:00	100	init	down	1	3		
	0000-00-00T00:00:00+00:00			-	-							

In the following example, Router-1 is the primary VRRP, because it has a higher priority value than Router 2:

```

Router-1# show running-config vpn 1
vpn 1
!
interface ge0/1.15
ip address 10.10.1.2/24
mtu 1496
no shutdown
vrrp 15
priority 110
track-omp
ipv4 10.20.23.1
!
!
!

```

```
Router-1# show vrrp vpn 1
```

VPN	IF NAME	ID	GROUP	TRACK VIRTUAL IP PREFIX LIST	PREFIX LIST	VIRTUAL MAC	PRIORITY	VRRP STATE	OMP STATE	ADVERTISEMENT TIMER	MASTER	
											DOWN TIMER	LAST
1	ge0/1.1	1		10.20.22.1	00:0c:bd:08:79:a4	100	backup	up	1	3		
	2016-01-13T03:10:55+00:00			-	-							
	ge0/1.5	5		10.20.22.193	00:0c:bd:08:79:a4	100	backup	up	1	3		
	2016-01-13T03:10:55+00:00			-	-							
	ge0/1.10	10		10.20.22.225	00:0c:bd:08:79:a4	100	backup	up	1	3		
	2016-01-13T03:10:55+00:00			-	-							
	ge0/1.15	15		10.20.23.1	00:0c:bd:08:79:a4	110	master	up	1	3		
	2016-01-13T03:10:56+00:00			-	-							
	ge0/1.20	20		10.20.24.1	00:0c:bd:08:79:a4	100	backup	up	1	3		
	2016-01-13T03:10:56+00:00			-	-							
	ge0/1.25	25		10.20.25.1	00:0c:bd:08:79:a4	110	master	up	1	3		
	2016-01-13T03:10:56+00:00			-	-							
	ge0/1.30	30		10.20.25.129	00:0c:bd:08:79:a4	100	backup	up	1	3		
	2016-01-13T03:10:56+00:00			-	-							

```
Router-1# show vrrp vpn 1 interfaces ge0/1.15 groups 15
```

GROUP	ID	PREFIX LIST	VIRTUAL IP	VIRTUAL MAC	PRIORITY	VRRP STATE	OMP STATE	ADVERTISEMENT TIMER	MASTER		
									DOWN TIMER	LAST	STATE CHANGE
1	10.20.33.1	00:0c:bd:08:79:a4	110	master	up	1	3				
	2016-01-13T03:10:56+00:00	-	-								

```

Router-2# show running-config vpn 1
vpn 1

```

```

!
interface ge0/1.15
 ip address 10.10.1.3/24
 mtu 1496
 no shutdown
 vrrp 15
 track-omp
 ipv4 10.20.23.1
!
!
!

```

```
Router-2# show vrrp vpn 1 interfaces groups
```

										MASTER
GROUP		TRACK	PREFIX				VRRP	OMP	ADVERTISEMENT	DOWN
IF NAME	ID	VIRTUAL IP	LIST	PRIORITY	STATE	STATE	TIMER	TIMER	LAST	
STATE	CHANGE	TIME	LIST	STATE						
ge0/1.1	1	10.20.32.1	00:0c:bd:08:2b:a5	110	master	up	1	3		
2016-01-13T00:22:15+00:00	-	-	-							
ge0/1.5	5	10.20.32.193	00:0c:bd:08:2b:a5	110	master	up	1	3		
2016-01-13T00:22:15+00:00	-	-	-							
ge0/1.10	10	10.20.32.225	00:0c:bd:08:2b:a5	110	master	up	1	3		
2016-01-13T00:22:15+00:00	-	-	-							
ge0/1.15	15	10.20.33.1	00:0c:bd:08:2b:a5	100	backup	up	1	3		
2016-01-13T03:10:56+00:00	-	-	-							
ge0/1.20	20	10.20.34.1	00:0c:bd:08:2b:a5	110	master	up	1	3		
2016-01-13T00:22:16+00:00	-	-	-							
ge0/1.25	25	10.20.35.1	00:0c:bd:08:2b:a5	100	backup	up	1	3		
2016-01-13T03:10:56+00:00	-	-	-							
ge0/1.30	30	10.20.35.129	00:0c:bd:08:2b:a5	100	master	up	1	3		
2016-01-13T00:22:16+00:00	-	-	-							

```
Router-2# show vrrp vpn 100 interfaces groups 15
```

										MASTER
GROUP		TRACK	PREFIX				VRRP	OMP	ADVERTISEMENT	DOWN
IF NAME	ID	VIRTUAL IP	LIST	PRIORITY	STATE	STATE	TIMER	TIMER	LAST	
STATE	CHANGE	TIME	LIST	STATE						
ge0/0.15	15	10.20.33.1	00:0c:bd:08:2b:a5	100	backup	up	1	3		
2016-01-13T03:10:56+00:00	-	-	-							

Multiple VRRP Groups on One Interface

Cisco SD-WAN supports configuring multiple VRRP groups on an interface. A use case for configuring this is where primary and secondary IP addresses have been assigned to a single interface. On one interface, you can configure:

- One primary IP address
- Up to four secondary IP addresses

To support each of these IP addresses, you can configure up to 5 VRRP groups (each with a unique group ID) on an interface, subinterface, or integrated routing and bridging (IRB) interface that supports VRRP groups.

The following is an example of configuring 5 VRRP groups on 1 interface.

```

vpn 2
interface ge0/4.2
 ip address 10.0.1.10/24
 ip secondary-address 10.0.2.10/24
 ip secondary-address 10.0.3.10/24

```

```

ip secondary-address 10.0.4.10/24
mtu 1496
no shutdown
vrrp 1
  priority 101
  ipv4 10.0.1.1
!
vrrp 2
  ipv4 10.0.1.2
!
vrrp 3
  priority 101
  ipv4 10.0.2.1
!
vrrp 4
  ipv4 10.0.3.1
!
vrrp 5
  ipv4 10.0.4.1
!
!
!

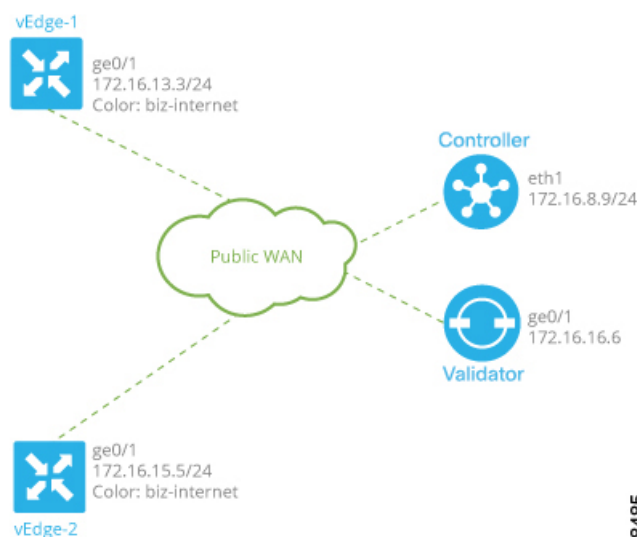
```

Network Interface Configuration Examples for Cisco vEdge Devices

This topic provides examples of configuring interfaces on Cisco vEdge devices to allow the flow of data traffic across both public and private WAN transport networks.

Connect to a Public WAN

This example shows a basic configuration for two connected to the same public WAN network (such as the Internet). The Cisco Catalyst SD-WAN Controller and Cisco Catalyst SD-WAN Validator are also connected to the public WAN network, and the Cisco Catalyst SD-WAN Controller is able to reach all destinations on the public WAN.



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For Cisco vEdge device-1, the interface ge0/1 connects to the public WAN, so it is the interface that is configured as a tunnel interface. The tunnel has a color of biz-internet, and the encapsulation used for data traffic is IPsec. The Cisco SD-WAN software creates a single TLOC for this interface, comprising the interface's IP address, color, and encapsulation, and the TLOC is sent to the Cisco Catalyst SD-WAN Controller over the OMP session running on the tunnel. The configuration also includes a default route to ensure that the router can reach the Cisco Catalyst SD-WAN Validator and Cisco Catalyst SD-WAN Controller.

```
vpn 0
  interface ge0/1
    ip address 172.16.13.3/24
    tunnel-interface
      encapsulation ipsec
      color biz-internet
      allow-service dhcp
      allow-service dns
      allow-service icmp
      no allow-service sshd
      no allow-service ntp
      no allow-service stun
    !
    no shutdown
  !
  ip route 0.0.0.0/0 0.0.0.0
!
```

The configuration for Cisco vEdge device-2 is similar to that for Cisco vEdge device-1:

```
vpn 0
  interface ge0/1
    ip address 172.16.15.5/24
    tunnel-interface
      encapsulation ipsec
      color biz-internet
      allow-service dhcp
      allow-service dns
      allow-service icmp
      no allow-service sshd
      no allow-service ntp
      no allow-service stun
    !
    no shutdown
  !
  ip route 0.0.0.0/0 0.0.0.0
!
```

On the Cisco Catalyst SD-WAN Controller and Cisco Catalyst SD-WAN Validator, you configure a tunnel interface and default IP route to reach the WAN transport. For the tunnel, color has no meaning because these devices have no TLOCs.

```
vpn 0
  interface eth1
    ip address 172.16.8.9/24
    tunnel-interface
    !
    no shutdown
  !
  ip route 0.0.0.0/0 0.0.0.0
!

vpn 0
  interface ge0/1
    ip address 172.16.16.6/24
    tunnel-interface
    !
```

```

    no shutdown
    !
    ip route 0.0.0.0/0 0.0.0.0
    !

```

Use the **show interface** command to check that the interfaces are operational and that the tunnel connections have been established. In the Port Type column, tunnel connections are marked as "transport".

```
vEdge-1# show interface vpn 0
```

			IF							TCP			
			ADMIN	OPER	ENCAP				SPEED		MSS		
RX	TX												
VPN	INTERFACE	IP ADDRESS	STATUS	STATUS	TYPE	PORT	TYPE	MTU	HWADDR	MBPS	DUPLEX	ADJUST	UPTIME
	PACKETS	PACKETS											
0	ge0/0	172.16.13.3/24	Up	Up	null	transport		1500	00:0c:29:7d:1e:fe	10	full	0	0:02:26:20
88358	88202												
0	ge0/1	10.1.17.15/24	Up	Up	null	service		1500	00:0c:29:7d:1e:08	10	full	0	0:02:26:20
217	1												
0	ge0/2	-	Down	Up	null	service		1500	00:0c:29:7d:1e:12	10	full	0	0:02:26:20
217	0												
0	ge0/3	10.0.20.15/24	Up	Up	null	service		1500	00:0c:29:7d:1e:1c	10	full	0	0:02:26:20
218	1												
0	ge0/6	172.17.1.15/24	Up	Up	null	service		1500	00:0c:29:7d:1e:3a	10	full	0	0:02:26:20
217	1												
0	ge0/7	10.0.100.15/24	Up	Up	null	service		1500	00:0c:29:7d:1e:44	10	full	0	0:02:25:02
850	550												
0	system	172.16.255.3/32	Up	Up	null	loopback		1500	00:00:00:00:00:00	10	full	0	0:02:13:31
0	0												

Use the **show control connections** command to check that the Cisco vEdge device has a DTLS or TLS session established to the Cisco Catalyst SD-WAN Controller.

```
vEdge-1# show control connections
```

			PEER		PEER		PEER			
PEER	PEER	PEER	SITE	DOMAIN	PEER	PRIVATE	PEER	PUBLIC		
TYPE	PROTOCOL	SYSTEM IP	ID	ID	PRIVATE IP	PORT	PUBLIC IP	PORT	LOCAL	COLOR
STATE		UPTIME								
vsmart	dtls	172.16.255.19	100	1	10.0.5.19	12346	10.0.5.19	12346	biz-internet	
up		0:02:13:13								
vsmart	dtls	172.16.255.20	200	1	10.0.12.20	12346	10.0.12.20	12346	biz-internet	
up		0:02:13:13								

Use the **show bfd sessions** command to display information about the BFD sessions that have been established between the local Cisco vEdge device and remote routers:

```
vEdge-1# show bfd sessions
```

		SOURCE TLOC		REMOTE TLOC		DST PUBLIC	
DST PUBLIC	DETECT	TX					
SYSTEM IP	SITE ID	STATE	COLOR	COLOR	SOURCE IP	IP	
PORT	ENCAP	MULTIPLIER	INTERVAL (msec)	UPTIME	TRANSITIONS		
172.16.255.11	100	up	biz-internet	biz-internet	10.1.15.15	10.0.5.11	
12346	ipsec	20	1000	0:02:24:59	1		
172.16.255.14	400	up	biz-internet	biz-internet	10.1.15.15	10.1.14.14	
12360	ipsec	20	1000	0:02:24:59	1		
172.16.255.16	600	up	biz-internet	biz-internet	10.1.15.15	10.1.16.16	
12346	ipsec	20	1000	0:02:24:59	1		
172.16.255.21	100	up	biz-internet	biz-internet	10.1.15.15	10.0.5.21	
12346	ipsec	20	1000	0:02:24:59	1		

Use the **show omp tlocs** command to list the TLOCs that the local router has learned from the Cisco Catalyst SD-WAN Controller:

```

vEdge-1# show omp tlocs
C -> chosen
I -> installed

```

Red -> redistributed
 Rej -> rejected
 L -> looped
 R -> resolved
 S -> stale
 Ext -> extranet
 Inv -> invalid

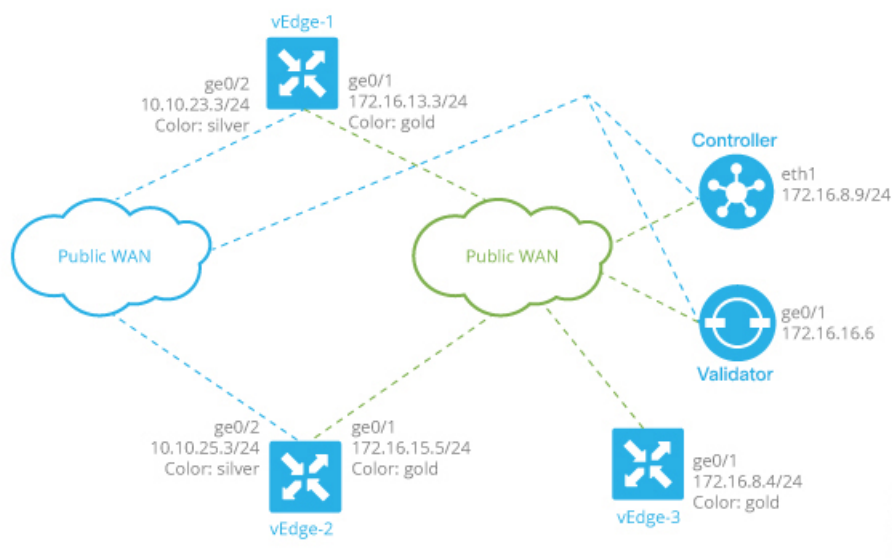
ADDRESS		PRIVATE		BFD	ENCAP	FROM PEER	STATUS	PUBLIC	
FAMILY	TLOC IP	IP	PORT	COLOR STATUS				IP	PORT
ipv4	172.16.255.11		12346	biz-internet up	ipsec	172.16.255.19	C,I,R	10.0.5.11	12346
10.0.5.11						172.16.255.20	C,R	10.0.5.11	12346
10.0.5.11			12346	up					
	172.16.255.14			biz-internet	ipsec	172.16.255.19	C,I,R	10.1.14.14	12360
10.1.14.14			12360	up		172.16.255.20	C,R	10.1.14.14	12360
10.1.14.14			12360	up					
	172.16.255.16			biz-internet	ipsec	172.16.255.19	C,I,R	10.1.16.16	12346
10.1.16.16			12346	up		172.16.255.20	C,R	10.1.16.16	12346
10.1.16.16			12346	up					
	172.16.255.21			biz-internet	ipsec	172.16.255.19	C,I,R	10.0.5.21	12346
10.0.5.21			12346	up		172.16.255.20	C,R	10.0.5.21	12346
10.0.5.21			12346	up					

Connect to Two Public WANs

In this example, two Cisco vEdge devices at two different sites connect to two public WANs, and hence each router has two tunnel connections. To direct traffic to the two different WANs, each tunnel interface is assigned a different color (here, **silver** and **gold**). Because each router has two tunnels, each router has two TLOCs.

A third router at a third site, vEdge-3, connects only to one of the public WANs.

The Cisco Catalyst SD-WAN Controller and Cisco Catalyst SD-WAN Validator are connected to one of the public WAN networks. (In reality, it does not matter which of the two networks they are connected to, nor does it matter whether the two devices are connected to the same network). The Cisco Catalyst SD-WAN Controller is able to reach all destinations on the public WAN. To ensure that the Cisco Catalyst SD-WAN Validator is accessible via each transport tunnel on the routers, a default route is configured for each interface. In our example, we configure a static default route, but you can also use DHCP.



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The configurations for vEdge-1 and vEdge-2 are similar. We configure two tunnel interfaces, one with color **silver** and the other with color **gold**, and we configure static default routes for both tunnel interfaces. Here is the configuration for vEdge-1:

```
vpn 0
  interface ge0/1
    ip address 172.16.13.3/24
    tunnel-interface
      encapsulation ipsec
      color silver
    !
    no shutdown
  !
  interface ge0/2
    ip address 10.10.23.3/24
    tunnel-interface
      encapsulation ipsec
      color gold
    !
    no shutdown
  !
  ip route 0.0.0.0/0 0.0.0.0
```

The configuration for vEdge-2 is similar:

```
vpn 0
  interface ge0/1
    ip address 172.16.15.5/24
    tunnel-interface
      encapsulation ipsec
      color silver
    !
    no shutdown
  !
  interface ge0/2
    ip address 10.10.25.3/24
    tunnel-interface
      encapsulation ipsec
      color gold
    !
    no shutdown
```



```
!
ip route 0.0.0.0/0 0.0.0.0
```

The third router, vEdge-3, connects only to one of the public WAN networks, and its tunnel interface is assigned the color "gold":

```
vpn 0
  interface ge0/1
    ip address 172.16.8.4/24
    tunnel-interface
      encapsulation ipsec
      color gold
    !
    no shutdown
  !
  ip route 0.0.0.0/0 0.0.0.0
```

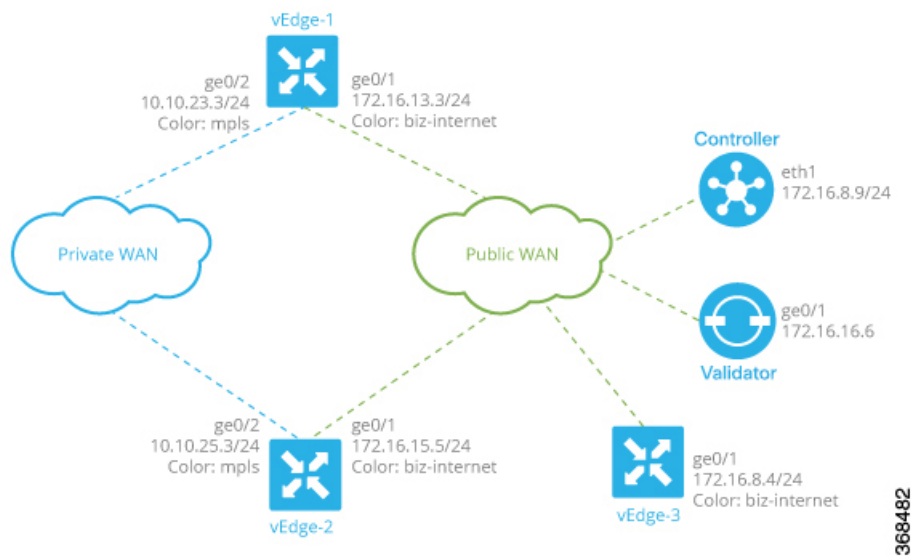
On the Cisco Catalyst SD-WAN Controller and Cisco Catalyst SD-WAN Validator, you configure a tunnel interface and default IP route to reach the WAN transport. For the tunnel, color has no meaning because these devices have no TLOCs.

```
vpn 0
  interface eth1
    ip address 172.16.8.9/24
    tunnel-interface
    !
    no shutdown
  ip route 0.0.0.0/0 0.0.0.0

vpn 0
  interface ge0/1
    ip address 172.16.16.6/24
    tunnel-interface
    !
    no shutdown
  ip route 0.0.0.0/0 0.0.0.0
```

Connect to Public and Private WANs, with Separation of Network Traffic

In this example, two Cisco vEdge devices at two different sites each connect to the same public WAN (here, the Internet) and the same private WAN (here, an MPLS network). We want to separate the MPLS network completely so that it is not reachable by the Internet. The Cisco Catalyst SD-WAN Controller and Cisco Catalyst SD-WAN Validator are hosted in the provider's cloud, which is reachable only via the Internet. A third Cisco vEdge device at a third site connects only to the public WAN (Internet).



In this example topology, we need to ensure the following:

- Complete traffic separation exists between private-WAN (MPLS) traffic and public-WAN (Internet) traffic.
- Each site (that is, each Cisco vEdge device) must have a connection to the Internet, because this is the only way that the overlay network can come up.

To maintain complete separation between the public and private networks so that all MPLS traffic stays within the MPLS network, and so that only public traffic passes over the Internet, we create two overlays, one for the private MPLS WAN and the second for the public Internet. For the private overlay, we want to create data traffic tunnels (which run IPsec and BFD sessions) between private-WAN TLOCs, and for the public overlay we want to create these tunnel connections between Internet TLOCs. To make sure that no data traffic tunnels are established between private-WAN TLOCs and Internet TLOCs, or vice versa, we associate the **restrict** attribute with the color on the private-WAN TLOCs. When a TLOC is marked as restricted, a TLOC on the local router establishes tunnel connections with a remote TLOC only if the remote TLOC has the same color. Put another way, BFD sessions come up between two private-WAN TLOCs and they come up between two public-WAN TLOCs, but they do not come up between an MPLS TLOC and an Internet TLOC.

Each site must have a connection to the public (Internet) WAN so that the overlay network can come up. In this topology, the Cisco Catalyst SD-WAN Controller and Cisco Catalyst SD-WAN Validator are reachable only via the Internet, but the MPLS network is completely isolated from the Internet. This means that if a Cisco vEdge device were to connect just to the MPLS network, it would never be able to discover the Cisco Catalyst SD-WAN Controller and Cisco Catalyst SD-WAN Validators and so would never be able to establish control connections in the overlay network. In order for a Cisco vEdge device in the MPLS network to participate in overlay routing, it must have at least one tunnel connection, or more specifically, one TLOC, to the Internet WAN. (Up to seven TLOCs can be configured on each Cisco vEdge device). The overlay network routes that the router learns over the public-WAN tunnel connection populate the routing table on the Cisco vEdge device and allow the router and all its interfaces and TLOCs to participate in the overlay network.

By default, all tunnel connections attempt to establish control connections in the overlay network. Because the MPLS tunnel connections are never going to be able to establish these connections to the Cisco Catalyst SD-WAN Controller or Cisco Catalyst SD-WAN Validators, we include the **max-control-connections 0** command in the configuration. While there is no harm in having the MPLS tunnels attempt to establish control

connections, these attempts will never succeed, so disabling them saves resources on the Cisco vEdge device. Note that **max-control-connections 0** command works only when there is no NAT device between the Cisco vEdge device and the PE router in the private WAN.

Connectivity to sites in the private MPLS WAN is possible only by enabling service-side routing.

Here is the configuration for the tunnel interfaces on vEdge-1. This snippet does not include the service-side routing configuration.

```
vpn 0
  interface ge0/1
    ip address 172.16.13.3/24
    tunnel-interface
      encapsulation ipsec
      color biz-internet
    !
    no shutdown
  !
  interface ge0/2
    ip address 10.10.23.3/24
    tunnel-interface
      encapsulation ipsec
      color mpls restrict
      max-control-connections 0
    !
    no shutdown
  !
  ip route 0.0.0.0/0 0.0.0.0
```

The configuration on vEdge-2 is quite similar:

```
vpn 0
  interface ge0/1
    ip address 172.16.15.5/24
    tunnel-interface
      encapsulation ipsec
      color biz-internet
    !
    no shutdown
  !
  interface ge0/2
    ip address 10.10.25.3/24
    tunnel-interface
      encapsulation ipsec
      color mpls restrict
      max-control-connections 0
    !
    no shutdown
  !
  ip route 0.0.0.0/0 0.0.0.0
!
```

The vEdge-3 router connects only to the public Internet WAN:

```
vpn 0
  interface ge0/1
    ip address 172.16.8.4/24
    tunnel-interface
      encapsulation ipsec
      color biz-internet
    !
    no shutdown
  !
  ip route 0.0.0.0/0 0.0.0.0
!
```

On the Cisco Catalyst SD-WAN Controller and Cisco Catalyst SD-WAN Validator, you configure a tunnel interface and default IP route to reach the WAN transport. For the tunnel, color has no meaning because these devices have no TLOCs.

```

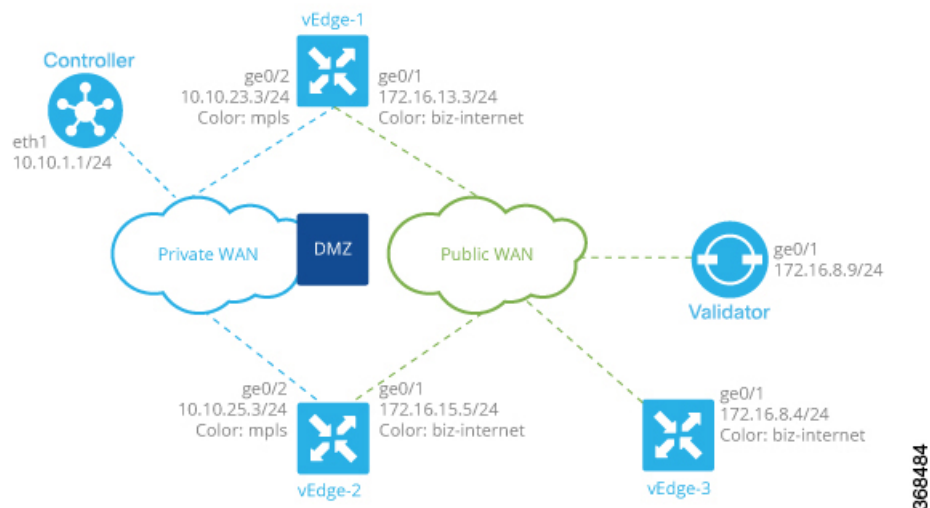
vpn 0
  interface eth1
    ip address 172.16.8.9/24
    tunnel-interface
    !
    no shutdown
  !
  ip route 0.0.0.0/0 0.0.0.0
!

vpn 0
  interface ge0/1
    ip address 172.16.16.6/24
    tunnel-interface
    !
    no shutdown
  !
  ip route 0.0.0.0/0 0.0.0.0
!

```

Connect to Public and Private WANs, with Ubiquitous Connectivity to Both WANs

This example is a variant of the previous example. We still have two Cisco vEdge devices at two different sites each connect to the same public WAN (here, the Internet) and the same private WAN (here, an MPLS network). However, now we want sites on the MPLS network and the Internet to be able to exchange data traffic. This topology requires a single overlay over both the public and private WANs. Control connections are present over both transports, and we want IPsec tunnel connections running BFD sessions to exist from private-WAN TLOCs to private-WAN TLOCs, from Internet TLOCs to Internet TLOCs, from private-WAN TLOCs to Internet TLOCs, and from Internet TLOCs to private-WAN TLOCs. This full possibility of TLOCs allows the establishment of a ubiquitous data plane in the overlay network.



For this configuration to work, the Cisco Catalyst SD-WAN Validator must be reachable over both WAN transports. Because it is on the public WAN (that is, on the Internet), there needs to be connectivity from the private WAN to the Internet. This could be provided via a DMZ, as shown in the figure above. The Cisco Catalyst SD-WAN Controller can be either on the public or the private LAN. If there are multiple controllers, some can be on public LAN and others on private LAN.

On each Cisco vEdge device, you configure private-WAN TLOCs, assigning a private color (**metro-ethernet**, **mpls**, or **private1** through **private6**) to the tunnel interface. You also configure public TLOCs, assigning any other color (or you can leave the color as **default**). Each Cisco vEdge device needs two routes to reach the Cisco Catalyst SD-WAN Validator, one via the private WAN and one via the public WAN.

With such a configuration:

- Control connections are established over each WAN transport.
- BFD/IPsec comes up between all TLOCs (if no policy is configured to change this).
- A given site can be dual-homed to both WAN transports or single-homed to either one.

Here is an example of the configuration on one of the Cisco vEdge devices, vEdge-1:

```
vpn 0
 interface ge0/1
   description "Connection to public WAN"
   ip address 172.16.31.3/24
   tunnel-interface
     encapsulation ipsec
     color biz-internet
   !
   no shutdown
 !
 interface ge0/2
   description "Connection to private WAN"
   ip address 10.10.23.3/24
   tunnel-interface
     encapsulation ipsec
     color mpls
   !
   no shutdown
 !
 ip route 0.0.0.0/0 0.0.0.0
 !
```

The **show control connections** command lists two DTLS sessions to the Cisco Catalyst SD-WAN Controller, one from the public tunnel (color of **biz-internet**) and one from the private tunnel (color of **mpls**):

```
vEdge-1# show control connections
```

						PEER	
PEER	PEER	PEER		SITE	DOMAIN	PEER	PRIVATE
PUBLIC							PEER
TYPE	PROTOCOL	SYSTEM	IP	ID	ID	PRIVATE IP	PUBLIC IP
PORT	LOCAL	COLOR	STATE	UPTIME			
vsmart	dtls	10.255.1.9		900	1	172.16.8.2	172.16.8.2
12346	mpls	up			0:01:41:17		
vsmart	dtls	10.255.1.9		900	1	172.16.8.2	172.16.8.2
12346	biz-internet	up			0:01:41:33		

The **show bfd sessions** command output shows that vEdge-1 has separate tunnel connections that are running separate BFD sessions for each color:

```
vEdge-1# show bfd sessions
```

DST PUBLIC		DETECT		SOURCE TLOC		REMOTE TLOC		DST PUBLIC	
SYSTEM	IP	SITE	ID	TX	COLOR	COLOR	SOURCE IP	IP	
PORT	ENCAP	MULTIPLIER	STATE	INTERVAL	(msec)	UPTIME	TRANSITIONS		
10.255.1.5	500	up		mpls		biz-internet	10.10.23.3	172.16.51.5	
12346	ipsec	3		1000		0:06:07:19	1		
10.255.1.5	500	up		biz-internet		biz-internet	172.16.31.3	172.16.51.5	
12360	ipsec	3		1000		0:06:07:19	1		
10.255.1.6	600	up		mpls		biz-internet	10.10.23.3	172.16.16.6	

```

12346      ipsec 3      1000      0:06:07:19      1
10.255.1.6      600      up      biz-internet      biz-internet      172.16.31.3      172.16.16.6
12346      ipsec 3      1000      0:06:07:19      1

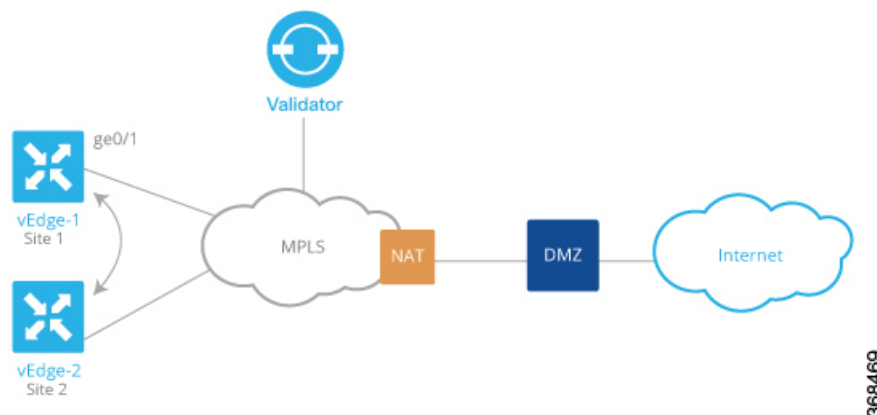
```

Exchange Data Traffic within a Single Private WAN

When the Cisco vEdge device is connected to a private WAN network, such as an MPLS or a metro Ethernet network, and when the carrier hosting the private network does not advertise the router's IP address, remote Cisco vEdge devices on the same private network but at different sites can never learn how to reach that router and hence are not able to exchange data traffic with it by going only through the private network. Instead, the remote routers must route data traffic through a local NAT and over the Internet to a Cisco Catalyst SD-WAN Validator, which then provides routing information to direct the traffic to its destination. This process can add significant overhead to data traffic exchange, because the Cisco Catalyst SD-WAN Validator may physically be located at a different site or a long distance from the two Cisco vEdge devices and because it may be situated behind a DMZ.

To allow Cisco vEdge devices at different overlay network sites on the private network to exchange data traffic directly using their private IP addresses, you configure their WAN interfaces to have one of eight private colors, **metro-ethernet**, **mpls**, and **private1** through **private6**. Of these four colors, the WAN interfaces on the Cisco vEdge devices must be marked with the same color so that they can exchange data traffic.

To illustrate the exchange of data traffic across private WANs, let's look at a simple topology in which two Cisco vEdge devices are both connected to the same private WAN. The following figure shows that these two Cisco vEdge devices are connected to the same private MPLS network. The vEdge-1 router is located at Site 1, and vEdge-2 is at Site 2. Both routers are directly connected to PE routers in the carrier's MPLS cloud, and you want both routers to be able to communicate using their private IP addresses.



This topology requires a special configuration to allow traffic exchange using private IP addresses because:

- The Cisco vEdge devices are in different sites; that is, they are configured with different site IDs.
- The Cisco vEdge devices are directly connected to the PE routers in the carrier's MPLS cloud.
- The MPLS carrier does not advertise the link between the Cisco vEdge device and its PE router.

To be clear, if the situation were one of the following, no special configuration would be required:

- vEdge-1 and vEdge-2 are configured with the same site ID.
- vEdge-1 and vEdge-2 are in different sites, and the Cisco vEdge device connects to a CE router that, in turn, connects to the MPLS cloud.

- vEdge-1 and vEdge-2 are in different sites, the Cisco vEdge device connects to the PE router in the MPLS cloud, and the private network carrier advertises the link between the Cisco vEdge device and the PE router in the MPLS cloud.
- vEdge-1 and vEdge-2 are in different sites, and you want them to communicate using their public IP addresses.

In this topology, because the MPLS carrier does not advertise the link between the Cisco vEdge device and the PE router, you use a loopback interface on the each Cisco vEdge device to handle the data traffic instead of using the physical interface that connects to the WAN. Even though the loopback interface is a virtual interface, when you configure it on the Cisco vEdge device, it is treated like a physical interface: the loopback interface is a terminus for both a DTLS tunnel connection and an IPsec tunnel connection, and a TLOC is created for it.

This loopback interface acts as a transport interface, so you must configure it in VPN 0.

For the vEdge-1 and vEdge-2 routers to be able to communicate using their private IP addresses over the MPLS cloud, you set the color of their loopback interfaces to be the same and to one of eight special colors—**metro-ethernet**, **mpls**, and **private1** through **private6**.

Here is the configuration on vEdge-1:

```
vedge-1(config)# vpn 0
vedge-1(config-vpn-0)# interface loopback1
vedge-1(config-interface-loopback1)# ip address 172.16.255.25/32
vedge-1(config-interface-loopback1)# tunnel-interface
vedge-1(config-tunnel-interface)# color mpls
vedge-1(config-interface-tunnel-interface)# exit
vedge-1(config-tunnel-interface)# no shutdown
vedge-1(config-tunnel-interface)# commit and-quit
vedge-1# show running-config vpn 0
...
interface loopback1
 ip-address 172.16.255.25/32
 tunnel-interface
  color mpls
 !
 no shutdown
 !
```

On vEdge-2, you configure a loopback interface with the same tunnel interface color that you used for vEdge-1:

```
vedge-2# show running-config vpn 0
vpn 0
 interface loopback2
 ip address 172.17.255.26/32
 tunnel-interface
  color mpls
 no shutdown
 !
```

Use the **show interface** command to verify that the loopback interface is up and running. The output shows that the loopback interface is operating as a transport interface, so this is how you know that it is sending and receiving data traffic over the private network.

```
vedge-1# show interface
```

			IF	IF							
TCP			ADMIN	OPER	ENCAP					SPEED	
MSS		RX	TX								
VPN	INTERFACE	IP ADDRESS	STATUS	STATUS	TYPE	PORT	TYPE	MTU	HWADDR	MBPS	
DUPLEX	ADJUST	UPTIME	PACKETS	PACKETS							

```

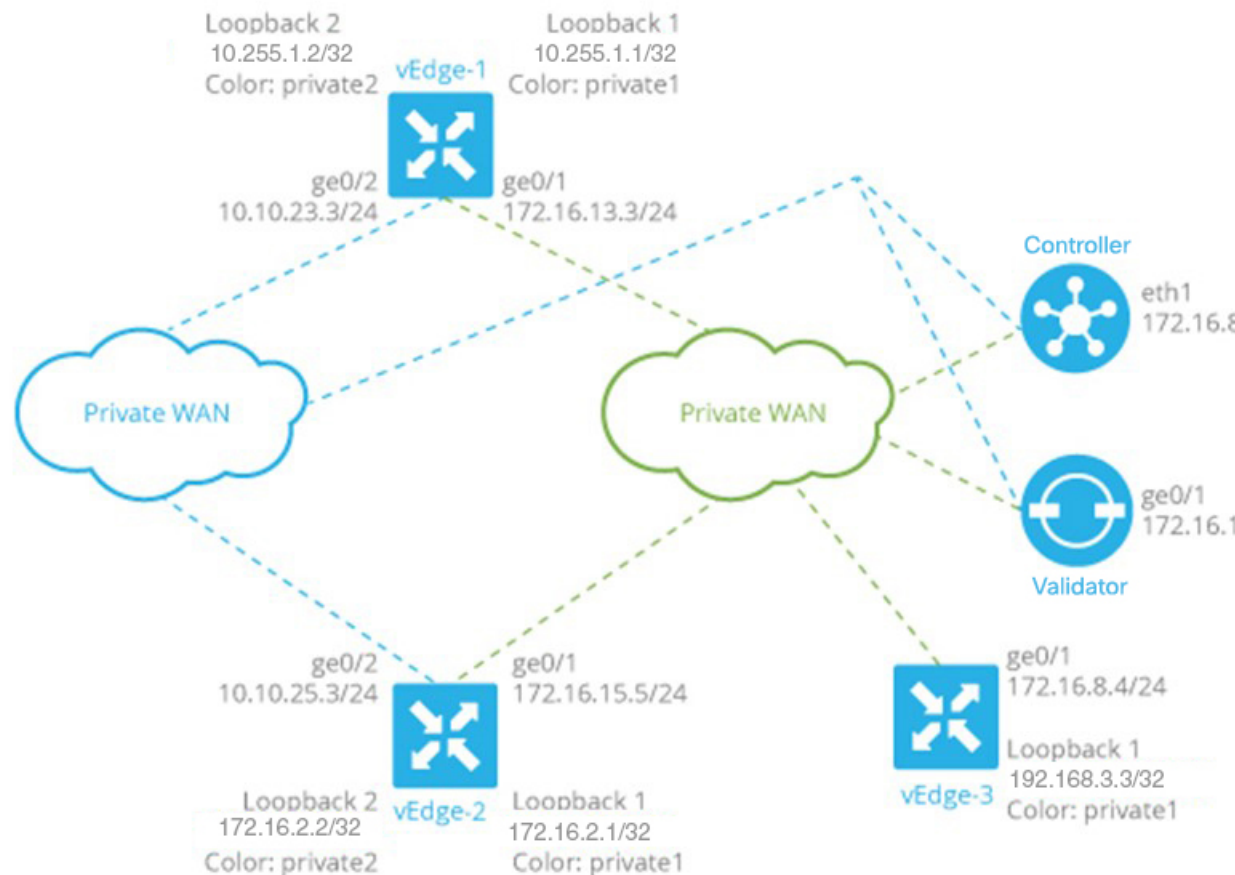
0   ge0/0      10.1.15.15/24      Up      Up      null  transport  1500  00:0c:29:7d:1e:fe  10    full
0   0          0:07:38:49  213199  243908
0   ge0/1      10.1.17.15/24      Up      Up      null  service    1500  00:0c:29:7d:1e:08  10    full
0   0          0:07:38:49  197     3
0   ge0/2      -                          Down    Down    null  service    1500  00:0c:29:7d:1e:12  -      -
0   0          -                          1
0   ge0/3      10.0.20.15/24      Up      Up      null  service    1500  00:0c:29:7d:1e:1c  10    full
0   0          0:07:38:49  221     27
0   ge0/6      172.17.1.15/24     Up      Up      null  service    1500  00:0c:29:7d:1e:3a  10    full
0   0          0:07:38:49  196     3
0   ge0/7      10.0.100.15/24     Up      Up      null  service    1500  00:0c:29:7d:1e:44  10    full
0   0          0:07:44:47  783     497
0   loopback1  172.16.255.25/32   Up      Up      null  transport  1500  00:00:00:00:00:00  10    full
0   0          0:00:00:20  0        0
0   system     172.16.255.15/32   Up      Up      null  loopback   1500  00:00:00:00:00:00  10    full
0   0          0:07:38:25  0        0
1   ge0/4      10.20.24.15/24     Up      Up      null  service    1500  00:0c:29:7d:1e:26  10    full
0   0          0:07:38:46  27594    27405
1   ge0/5      172.16.1.15/24     Up      Up      null  service    1500  00:0c:29:7d:1e:30  10    full
0   0          0:07:38:46  196     2
512 eth0       10.0.1.15/24       Up      Up      null  service    1500  00:50:56:00:01:05  1000  full
0   0          0:07:45:55  15053    10333

```

To allow Cisco vEdge device at different overlay network sites on the private network to exchange data traffic directly, you use a loopback interface on the each Cisco vEdge device to handle the data traffic instead of using the physical interface that connects to the WAN. You associate the same tag, called a carrier tag, with each loopback interface so that all the routers learn that they are on the same private WAN. Because the loopback interfaces are advertised across the overlay network, the vEdge routers are able to learn reachability information, and they can exchange data traffic over the private network. To allow the data traffic to actually be transmitted out the WAN interface, you bind the loopback interface to a physical WAN interface, specifically to the interface that connects to the private network. Remember that this is the interface that the private network does not advertise. However, it is still capable of transmitting data traffic.

Exchange Data Traffic between Two Private WANs

This example shows a topology with two different private networks, possibly the networks of two different network providers, and all the Cisco Catalyst SD-WAN devices are located somewhere on one or both of the private networks. Two Cisco vEdge devices are located at two different sites, and they both connect to both private networks. A third Cisco vEdge device connects to only one of the private WANs. The Cisco Catalyst SD-WAN Validator and Cisco Catalyst SD-WAN Controller both sit in one of the private WANs, perhaps in a data center, and they are reachable over both private WANs. For the Cisco vEdge devices to be able to establish control connections, the subnetworks where the Cisco Catalyst SD-WAN Validator and Cisco Catalyst SD-WAN Controller devices reside must be advertised into each private WAN. Each private WAN CPE router then advertises these subnets in its VRF, and each Cisco vEdge device learns those prefixes from each PE router that it is connected to.



Because both WANs are private, we need only a single overlay. In this overlay network, without policy, IPsec tunnels running BFD sessions exist from any TLOC connected to either transport network to any TLOC in the other transport as well as to any TLOC in the same WAN transport network.

As with the previous examples in this topic, it is possible to configure the tunnel interfaces on the routers' physical interfaces. If you do this, you also need to configure a routing protocol between the Cisco vEdge device at its peer PE router, and you need to configure access lists on the Cisco vEdge device to advertise all the routes in both private networks.

A simpler configuration option that avoids the need for access lists is to use loopback interfaces as the tunnel interfaces, and then bind each loopback interface to the physical interface that connects to the private network. Here, the loopback interfaces become the end points of the tunnel, and the TLOC connections in the overlay network run between loopback interfaces, not between physical interfaces. So in the figure shown above, on router vEdge-1, the tunnel connections originate at the Loopback1 and Loopback2 interfaces. This router has two TLOCs: {10.255.1.1, private1, ipsec} and {10.255.1.2, private2, ipsec}.

The WAN interfaces on the Cisco vEdge devices must run a routing protocol with their peer PE routers. The routing protocol must advertise the Cisco vEdge device's loopback addresses to both PE routers so that all Cisco vEdge devices on the two private networks can learn routes to each other. A simple way to advertise the loopback addresses is to redistribute routes learned from other (connected) interfaces on the same router. (You do this instead of creating access lists). If, for example, you are using OSPF, you can advertise the loopback addresses by including the **redistribute connected** command in the OSPF configuration. Looking at the figure above, the **ge0/2** interface on vEdge-1 needs to advertise both the Loopback1 and Loopback2

interfaces to the blue private WAN, and **ge0/1** must advertise also advertise both these loopback interfaces to the green private WAN.

With this configuration:

- The Cisco vEdge devices learn the routes to the Cisco Catalyst SD-WAN Validator and Cisco Catalyst SD-WAN Controller over each private WAN transport.
- The Cisco vEdge devices learn every other Cisco vEdge device's loopback address over each WAN transport network.
- The end points of the tunnel connections between each pair of Cisco vEdge devices are the loopback interfaces, not the physical (**ge**) interfaces.
- The overlay network has data plane connectivity between any TLOCs and has a control plane over both transport networks.

Here is the interface configuration for VPN 0 on vEdge-1. Highlighted are the commands that bind the loopback interfaces to their physical interfaces. Notice that the tunnel interfaces, and the basic tunnel interface properties (encapsulation and color), are configured on the loopback interfaces, not on the Gigabit Ethernet interfaces.

```
vpn 0
  interface loopback1
    ip address 10.255.1.2/32
    tunnel-interface
      encapsulation ipsec
      color private1
      bind ge0/1
    !
    no shutdown
  !
  interface loopback2
    ip address 10.255.1.1/32
    tunnel-interface
      encapsulation ipsec
      color private2
      bind ge0/2
    !
    no shutdown
  !
  interface ge0/1
    ip address 172.16.13.3/24
    no shutdown
  !
  interface ge0/2
    ip address 10.10.23.3/24
    no shutdown
  !
  ip route 0.0.0.0/0 0.0.0.0
```

The configuration for vEdge-2 is similar:

```
vpn 0
  interface loopback1
    ip address 172.16.2.1/32
    tunnel-interface
      encapsulation ipsec
      color private1
      bind ge0/1
    !
    no shutdown
  !
```

```

interface loopback2
 ip address 172.16.2.2/32
 tunnel-interface
 encapsulation ipsec
 color private2
 bind ge0/2
 !
 no shutdown
 !
interface ge0/1
 ip address 172.16.15.5/24
 no shutdown
 !
interface ge0/2
 ip address 10.10.25.3/24
 no shutdown
 !
ip route 0.0.0.0/0 0.0.0.0
!
```

The vEdge-3 router connects only to the green private WAN:

```

vpn 0
 interface loopback1
 ip address 192.168.3.3/32
 tunnel-interface
 encapsulation ipsec
 color private1
 bind ge0/1
 !
 no shutdown
 !
interface ge0/1
 ip address 172.16.8.4/24
 no shutdown
 !
ip route 0.0.0.0/0 0.0.0.0
!
```

On the Cisco Catalyst SD-WAN Controller and Cisco Catalyst SD-WAN Validator, you configure a tunnel interface and default IP route to reach the WAN transport. For the tunnel, color has no meaning because these devices have no TLOCs.

```

vpn 0
 interface eth1
 ip address 172.16.8.9/24
 tunnel-interface
 !
 no shutdown
 !
ip route 0.0.0.0/0 0.0.0.0
!

vpn 0
 interface ge0/1
 ip address 172.16.16.6/24
 tunnel-interface
 !
 no shutdown
 !
ip route 0.0.0.0/0 0.0.0.0
!
```

Connect to a WAN Using PPPoE

This example shows a Cisco vEdge device with a TLOC tunnel interface and an interface enabled for Point-to-Point Protocol over Ethernet (PPPoE). The PPP interface defines the authentication method and credentials and is linked to the PPPoE-enabled interface.



Here is the interface configuration for VPN 0:

```
vpn 0
interface ge0/1
no shutdown
!
tunnel-interface
encapsulation ipsec
color biz-internet
allow-service dhcp
allow-service dns
allow-service icmp
no allow-service sshd
no allow-service ntp
no allow-service stun
!
no shutdown
!
interface ge0/3
pppoe-client ppp-interface ppp10
no shutdown
!
interface ppp10
ppp authentication chap
hostname branch100@corp.bank.myisp.net
password $4$OHHjdmsC6M8zj4BgLEFXKw==
!
tunnel-interface
encapsulation ipsec
color gold
allow-service dhcp
allow-service dns
allow-service icmp
no allow-service sshd
no allow-service ntp
no allow-service stun
!
no shutdown
!
```

Use the **show ppp interface** command to view existing PPP interfaces:

```
vEdge# show ppp interface
```

VPN	IFNAME	PPPOE INTERFACE	INTERFACE IP	GATEWAY IP	PRIMARY DNS	SECONDARY DNS	MTU
0	ppp10	ge0/3	10.0.0.11	10.255.255.254	10.8.8.8	10.8.4.4	1150

Use the **show pppoe session** and **show pppoe statistics** commands to view information about PPPoE sessions:

```
vEdge# show pppoe session
```

SESSION				PPP		SERVICE	
VPN	IFNAME	ID	SERVER MAC	LOCAL MAC	INTERFACE	AC NAME	NAME
0	ge0/1	1	00:0c:29:2e:20:1a	00:0c:29:be:27:f5	ppp1	branch100	-
0	ge0/3	1	00:0c:29:2e:20:24	00:0c:29:be:27:13	ppp2	branch100	-

```
vEdge# show pppoe statistics
```

pppoe_tx_pkts	:	73
pppoe_rx_pkts	:	39
pppoe_tx_session_drops	:	0
pppoe_rx_session_drops	:	0
pppoe_inv_discovery_pkts	:	0
pppoe_ccp_pkts	:	12
pppoe_ipcp_pkts	:	16
pppoe_lcp_pkts	:	35
pppoe_padi_pkts	:	4
pppoe_pado_pkts	:	2
pppoe_padr_pkts	:	2
pppoe_pads_pkts	:	2
pppoe_padt_pkts	:	2

Configure VPN Ethernet Interface Using a Configuration Group

Before you begin

On the **Configuration > Configuration Groups** page, choose **SD-WAN** as the solution type.

Procedure

Step 1 From the Cisco SD-WAN Manager menu, choose **Configuration > Configuration Groups**.

Step 2 Create and configure a Transport VPN feature in Transport and Management profile.

Step 3 Create and configure Ethernet Interface feature in Transport VPN.

a) Configure basic VPN parameters.

Field	Description
Shutdown	Enable or disable the interface.
Interface Name*	Enter a name for the interface. Spell out the interface names completely (for example, GigabitEthernet0/0/0). Configure all the interfaces of the router, even if you are not using them, so that they are configured in the shutdown state and so that all default values for them are configured.
Description	Enter a description for the interface.

Field	Description
Auto Detect Bandwidth	Enable this option to automatically detect the bandwidth for WAN interfaces. The device detects the bandwidth by contacting an iPerf3 server to perform a speed test.
IPv4 Settings	Configure an IPv4 VPN interface. • Dynamic: Choose Dynamic to set the interface as a Dynamic Host Configuration Protocol (DHCP) client so that the interface receives its IP address from a DHCP server. • Static: Choose Static to enter an IP address that doesn't change.
Dynamic DHCP Distance	Enter an administrative distance value for routes learned from a DHCP server. This option is available when you choose Dynamic . Default: 1
IP Address	Enter a static IPv4 address. This option is available when you choose Static .
Subnet Mask	Enter the subnet mask. From Cisco Catalyst SD-WAN Control Components Release 20.15.1, if you enter a variable in the IP Address field, this field automatically uses a variable.
Configure Secondary IP Address	Enter up to four secondary IPv4 addresses for a service-side interface. • IP Address: Enter the IP address. • Subnet Mask: Enter the subnet mask.
DHCP Helper	To designate the interface as a DHCP helper on a router, enter up to eight IP addresses, separated by commas, for DHCP servers in the network. A DHCP helper interface forwards BOOTP (broadcast) DHCP requests that it receives from the specified DHCP servers.
IPv6 Settings	Configure an IPv6 VPN interface. • Dynamic: Choose Dynamic to set the interface as a Dynamic Host Configuration Protocol (DHCP) client so that the interface receives its IP address from a DHCP server. • Static: Choose Static to enter an IP address that doesn't change. • None
IPv6 Address Primary	Enter a static IPv6 address. This option is available when you choose Static .
Add Secondary Ipv6	
IP Address	Enter up to two secondary IPv6 addresses for a service-side interface.
Bandwidth Upstream	Enter upstream bandwidth reference value.
Bandwidth Downstream	Enter downstream bandwidth reference value.

b) Apply Access Lists and QoS Parameters

Field	Description
Adaptive QoS	To enable or disable adaptive QoS on an ethernet interface on the transport side.
Shaping Rate	Enter the shaping rate to control the maximum rate of traffic sent.
ACL	To define IPv4 and IPv6 ACL as ingress and egress.

c) Create a tunnel interface.

Field	Description
Tunnel Interface	Enable this option to create a tunnel interface.
Per-tunnel QoS	Enable this option to apply a Quality of Service (QoS) policy on individual tunnels.
Color	Choose a color for the TLOC.
Color Description	Minimum supported release: Cisco Catalyst SD-WAN Manager Release 20.18.1 Enter a description associated to the TLOC color.
Restrict	Enable this option to limit the remote TLOCs that the local TLOC can establish BFD sessions with. When a TLOC is marked as restricted, a TLOC on the local router establishes tunnel connections with a remote TLOC only if the remote TLOC has the same color.
Groups	Enter a group number. Range: 1 through 4294967295
Border	Enable this option to set the TLOC as a border TLOC.
Maximum Control Connections	Specify the maximum number of Cisco SD-WAN Controllers that the WAN tunnel interface can connect to. To have the tunnel establish no control connections, set the number to 0. Range: 0 through 100 Default: 2
Validator As Stun Server	Enable Session Traversal Utilities for NAT (STUN) to allow the tunnel interface to discover its public IP address and port number when the Cisco IOS XE Catalyst SD-WAN device is located behind a NAT.
Exclude Controller Group List	Set the identifiers of one or more Cisco SD-WAN Controller groups that this tunnel is not allowed to connect to. Range: 1 through 100

Field	Description
Manager Connection Preference	Set the preference for using a tunnel interface to exchange control traffic with Cisco SD-WAN Manager. Range: 0 through 8 Default: 5
Full Port Hop	Minimum release: Cisco IOS XE Catalyst SD-WAN Release 17.18.1a Enable full port hopping at the TLOC level to allow devices to establish connections with controllers by switching to the next port if the current port is blocked or non-functional. Default: Disabled
Port Hop	Enable port hopping. If port hopping is enabled globally, you can disable it on an individual TLOC (tunnel interface). Default: Enabled Starting from Cisco IOS XE Catalyst SD-WAN Release 17.18.1a, this field is deprecated. Instead use the Full Port Hop option. See the Full Port Hop field.
Low-Bandwidth Link	Enable this option to characterize the tunnel interface as a low-bandwidth link.
Tunnel TCP MSS	Specify the maximum segment size (MSS) of TCP SYN packets passing through the router. By default, the MSS is dynamically adjusted based on the interface or tunnel MTU such that TCP SYN packets are never fragmented. Range: 500 to 1460 bytes Default: None
Clear-Dont-Fragment	Enable this option to clear the Don't Fragment (DF) bit in the IPv4 packet header for packets being transmitted out the interface. When the DF bit is cleared, packets larger than the MTU of the interface are fragmented before being sent.
CTS SGT Propagation	Enable CTS SGT propagation on an interface.
Network Broadcast	Enable this option to accept and respond to network-prefix-directed broadcasts.

Field	Description
Allow Service	Allow or disallow the following services on the interface: • All • BGP • DHCP • NTP • SSH • DNS • ICMP • HTTPS • OSPF • STUN • SNMP • NETCONF • BFD
Encapsulation	

Field	Description
Encapsulation*	Choose an encapsulation type: • gre: Use GRE encapsulation on the tunnel interface. • ipsec: Use IPsec encapsulation on the tunnel interface. Note If you select both IPsec and GRE encapsulations, two TLOCs are created for the tunnel interface that have the same IP addresses and colors, but that differ by their encapsulation. When you choose gre, the following fields appear: • GRE Preference: Enter a preference value for directing traffic to the tunnel. A higher value is preferred over a lower value. Range: 0 through 4294967295 Default: 0 • GRE Weight: Enter a weight to use to balance traffic across multiple TLOCs. A higher value sends more traffic to the tunnel. Range: 1 through 255 Default: 1 When you choose ipsec, the following fields appear: • IPSEC Preference: Enter a preference value for directing traffic to the tunnel. A higher value is preferred over a lower value. Range: 0 through 4294967295 Default: 0 • IPSEC Weight: Enter a weight to use to balance traffic across multiple TLOCs. A higher value sends more traffic to the tunnel. Range: 1 through 255 Default: 1
Multi-Region Fabric Note These options appear only when Multi-Region Fabric is enabled.	
Connect to Core Region	(Minimum supported release: Cisco Catalyst SD-WAN Manager Release 20.13.1) (Applicable to a border router only) In a Multi-Region Fabric scenario, enable this option to specify how to use the Ethernet interface: • Share Interface with Access Region: Share the interface between the access region and core region. • Keep Exclusive to Core Region: Use the interface only for the core region.

Field	Description
Connect to Secondary Region	(Minimum supported release: Cisco Catalyst SD-WAN Manager Release 20.13.1) (Applicable to an edge router only) In a Multi-Region Fabric scenario, enable this option to specify how to use the Ethernet interface: • Share Interface with Access Region: Share the interface between the primary and secondary regions. • Keep Exclusive to Secondary Region: Use the interface only for the secondary region.

d) Configure an interface as a NAT device.

Field	Description
IPv4 Settings	
NAT	Enable this option to have the interface act as a NAT device.
NAT Type	Choose the NAT translation type for IPv4: • interface • pool • loopback Default: interface. It is supported for NAT64.
UDP Timeout	Specify when NAT translations over UDP sessions time out. Range: 1 through 8947 minutes Default: 1 minute
TCP Timeout	Specify when NAT translations over TCP sessions time out. Range: 1 through 8947 minutes Default: 60 minutes (1 hour)

Field	Description
Add Multiple NAT	Choose the NAT type: • Interface: This is the default value. • Pool: Configure the following: • Pool ID: Enter a NAT pool number configured in the centralized data policy. The NAT pool name must be unique across VPNs and VRFs. You can configure up to 31 (1–32) NAT pools per router. • Range Start: Enter a starting IP address for the NAT pool. • Range End: Enter a closing IP address for the NAT pool. • Prefix length: Specify the maximum number of source IP addresses that can be NATed in the NAT pool. • Overload: Enable this option to configure per-port translation. If this option is disabled, only dynamic NAT is configured on the end device. Per-port NAT is not configured. Default: Disabled • Loopback: Provide a value for the NAT inside source loopback interface.
Configure New Static NAT	Add a static NAT mapping
Source IP	Enter the source IP address to be translated.
Translate IP	Enter the translated source IP address.
Direction	Choose the direction in which to perform network address translation. • inside: Translates the IP address of packets that are coming from the service side of the device and that are destined for the transport side of the router. • outside: Translates the IP address of packets that are coming to the device from the transport side device and that are destined for a service-side device.
Source VPN	Enter the source VPN ID.
IPv6 Settings	
IPv6 NAT	Enable this option to have the interface act as a NAT device.
Select NAT	Choose NAT64 or NAT66. When you choose NAT66, the following fields appear: • Source Prefix: Enter the source IPv6 prefix. • Translated Source Prefix: Enter the translated source prefix. • Source VPN ID: Enter the source VPN ID. • Egress Interface: Enable this option to have the interface act as an egress interface.

- e) Add ARP table entries.

Field	Description
IP Address	Enter the IP address for the ARP entry in dotted decimal notation or as a fully qualified host name.
MAC Address	Enter the MAC address in colon-separated hexadecimal notation.

- f) Configure advanced properties.

Field	Description
Duplex	Specify whether the interface runs in full-duplex or half-duplex mode. Default: full
MAC Address	Specify a MAC address to associate with the interface, in colon-separated hexadecimal notation.
IP MTU	Specify the maximum MTU size of packets on the interface. Range: 576 through 9216 Default: 1500 bytes
Interface MTU	Enter the maximum transmission unit size for frames received and transmitted on the interface. Range: 1500 through 1518 (GigabitEthernet0), 1500 through 9216 (other GigabitEthernet) Default: 1500 bytes
TCP MSS	Specify the maximum segment size (MSS) of TCP SYN packets passing through the router. By default, the MSS is dynamically adjusted based on the interface or tunnel MTU such that TCP SYN packets are never fragmented. Range: 500 to 1460 bytes Default: None
Speed	Specify the speed of the interface, for use when the remote end of the connection does not support autonegotiation. Values: 10, 100, 1000, 2500, or 10000 Mbps
ARP Timeout	ARP timeout controls how long we maintain the ARP cache on a router. Specify how long it takes for a dynamically learned ARP entry to time out. Range: 0 through 2147483 seconds Default: 1200 seconds
Autonegotiate	Enable this option to turn on autonegotiation.

Field	Description
Media Type	Specify the physical media connection type on the interface. Choose one of the following: • auto-select: A connection is automatically selected. • rj45: Specifies an RJ-45 physical connection. • sfp: Specifies a small-form factor pluggable (SFP) physical connection for fiber media.
TLOC Extension	Enter the name of a physical interface on the same router that connects to the WAN transport. This configuration then binds this service-side interface to the WAN transport. A second router at the same site that itself has no direct connection to the WAN (generally because the site has only a single WAN connection) and that connects to this service-side interface is then provided with a connection to the WAN. Note TLOC extension over L3 is supported only for Cisco IOS XE Catalyst SD-WAN devices. If configuring TLOC extension over L3 for a Cisco IOS XE Catalyst SD-WAN device, enter the IP address of the L3 interface.
GRE tunnel source IP	Enter the IP address of the extended WAN interface.
XConnect	Enter the name of a physical interface on the same router that connects to the WAN transport.
Load Interval	Enter an interval value for interface load calculation.
IP Directed Broadcast	An IP directed broadcast is an IP packet whose destination address is a valid broadcast address for some IP subnet, but which originates from a node that is not itself part of that destination subnet. A device that is not directly connected to its destination subnet forwards an IP directed broadcast in the same way it would forward unicast IP packets destined to a host on that subnet. When a directed broadcast packet reaches a device that is directly connected to its destination subnet, that packet is broadcast on the destination subnet. The destination address in the IP header of the packet is rewritten to the configured IP broadcast address for the subnet, and the packet is sent as a link-layer broadcast. If directed broadcast is enabled for an interface, incoming IP packets whose addresses identify them as directed broadcasts intended for the subnet to which that interface is attached are broadcast on that subnet.
ICMP Redirect Disable	ICMP redirects are sent by a router to the sender of an IP packet when a packet is being routed sub-optimally. The ICMP redirect informs the sending host to forward subsequent packets to that same destination through a different gateway. By default, an interface allows ICMP redirect messages.

What to do next

Also see [Deploy a Configuration Group](#).

Configure Management VPN Using a Configuration Group

Before you begin

On the **Configuration > Configuration Groups** page, choose **SD-WAN** as the solution type.

Procedure

Step 1 From the Cisco SD-WAN Manager menu, choose **Configuration > Configuration Groups**.

Step 2 Create and configure a management VPN feature in a Transport and Management Profile.

- a. Configure management VPN.

Table 93: Management VPN

Field	Description
Type	Choose a feature from the drop-down list.
Feature Name*	Enter a name for the feature.
Description	Enter a description of the feature. The description can contain any characters and spaces.

- b. Configure basic settings.

Table 94: Basic Configuration

Field	Description
VPN	Management VPN carries out-of-band network management traffic among the Cisco IOS XE Catalyst SD-WAN devices in the overlay network. The interface used for management traffic resides in VPN 512. By default, VPN 512 is configured and enabled on all Cisco IOS XE Catalyst SD-WAN devices.
Name	Enter a name for the interface.

- c. Configure DNS.

Table 95: DNS

Field	Description
Add DNS	
Primary DNS Address (IPv4)	Enter the IPv4 address of the primary DNS server in this VPN.

Field	Description
Secondary DNS Address (IPv4)	Enter the IPv4 address of a secondary DNS server in this VPN.
Add DNS IPv6	
Primary DNS Address (IPv6)	Enter the IPv6 address of the primary DNS server in this VPN.
Secondary DNS Address (IPv6)	Enter the IPv6 address of a secondary DNS server in this VPN.

- d. Configure host mapping.

Table 96: Host Mapping

Field	Description
Add New Host Mapping	
Hostname*	Enter the hostname of the DNS server. The name can be up to 128 characters.
List of IP Address*	Enter IP addresses to associate with the hostname. Separate the entries with commas.

- e. Configure IPv4/IPv6 static route.

Table 97: IPv4/IPv6 Static Route

Field	Description
Add IPv4 Static Route	
IP Address*	Enter the IPv4 address or prefix, in decimal four-point-dotted notation, and the prefix length of the IPv4 static route to configure in the VPN.
Subnet Mask*	Enter the subnet mask.
Gateway*	Choose one of the following options to configure the next hop to reach the static route: • nextHop: When you choose this option and click Add Next Hop, the following fields appear: • Address*: Enter the next-hop IPv4 address. • Administrative distance*: Enter the administrative distance for the route. • dhcp • null0: When you choose this option, the following field appears: • Administrative distance: Enter the administrative distance for the route.
Add IPv6 Static Route	

Field	Description
Prefix*	Enter the IPv6 address or prefix, in decimal four-point-dotted notation, and the prefix length of the IPv6 static route to configure in the VPN.
Next Hop/Null 0/NAT	Choose one of the following options to configure the next hop to reach the static route: • Next Hop: When you choose this option and click Add Next Hop, the following fields appear: • Address*: Enter the next-hop IPv6 address. • Administrative distance*: Enter the administrative distance for the route. • Null 0: When you choose this option, the following field appears: • NULL0*: Enable this option to set the next hop to be the null interface. All packets sent to this interface are dropped without sending any ICMP messages. • NAT: When you choose this option, the following field appears: • IPv6 NAT: Choose NAT64 or NAT66.

What to do next

Also see [Deploy a configuration group](#).

Configure VPN Ethernet Interface

Procedure

Step 1 From the Cisco SD-WAN Manager menu, choose **Configuration** > **Templates**.

Step 2 Click **Device Templates**, and click **Create Template**.

Note

In Cisco vManage Release 20.7.x and earlier releases, **Device Templates** is titled **Device**.

Step 3 From the **Create Template** drop-down list, choose **From Feature Template**.

Step 4 From the **Device Model** drop-down list, choose the type of device for which you are creating the template.

Step 5 To create a template for VPN 0 or VPN 512:

- a. Click **Transport & Management VPN** or scroll to the **Transport & Management VPN** section.
- b. Under **Additional VPN 0 Templates**, click **VPN Interface**.

- c. From the **VPN Interface** drop-down list, click **Create Template**. The **VPN Interface Ethernet** template form displays.

This form contains fields for naming the template, and fields for defining the VPN Interface Ethernet parameters.

Step 6 To create a template for VPNs 1 through 511, and 513 through 65530:

- a. Click **Service VPN**, or scroll to the **Service VPN** section.
- b. Click the **Service VPN** drop-down list.
- c. Under **Additional VPN** templates, click **VPN Interface**.
- d. From the **VPN Interface** drop-down list, click **Create Template**. The VPN Interface Ethernet template form displays.

This form contains fields for naming the template, and fields for defining the VPN Interface Ethernet parameters.

Step 7 In **Template Name**, enter a name for the template. The name can be up to 128 characters and can contain only alphanumeric characters.

Step 8 In **Template Description**, enter a description of the template. The description can be up to 2048 characters and can contain only alphanumeric characters.

Configure Basic Interface Functionality

To configure basic interface functionality in a VPN, choose **Basic Configuration** and configure the following parameters:



Note Parameters marked with an asterisk are required to configure an interface.

Parameter Name	IPv4 or IPv6	Options	Description
Shutdown*			Click No to enable the interface.
Interface name*			Enter a name for the interface.
Description			Enter a description for the interface.
IPv4 / IPv6			Click IPv4 to configure an IPv4 VPN interface. Click IPv6 to configure an IPv6 interface.
Dynamic			Click Dynamic to set the interface as a Dynamic Host Configuration Protocol (DHCP) client, so that the interface receives its IP address from a DHCP server.
	Both	DHCP Distance	Optionally, enter an administrative distance value for routes learned from a DHCP server. Default is 1.
	IPv6	DHCP Rapid Commit	Optionally, configure the DHCP IPv6 local server to support DHCP Rapid Commit, to enable faster client configuration and confirmation in busy environments. Click On to enable DHCP rapid commit. Click Off to continue using the regular commit process.

Parameter Name	IPv4 or IPv6	Options	Description
Static	Click Static to enter an IP address that doesn't change.		
	IPv4	IPv4 Address	Enter a static IPv4 address.
	IPv6	IPv6 Address	Enter a static IPv6 address.
Secondary IP Address	IPv4	Click Add to enter up to four secondary IPv4 addresses for a service-side interface.	
IPv6 Address	IPv6	Click Add to enter up to two secondary IPv6 addresses for a service-side interface.	
DHCP Helper	Both	To designate the interface as a DHCP helper on a router, enter up to eight IP addresses, separated by commas, for DHCP servers in the network. A DHCP helper interface forwards BootP (broadcast) DHCP requests that it receives from the specified DHCP servers.	
Block Non-Source IP	Yes / No	Click Yes to have the interface forward traffic only if the source IP address of the traffic matches the interface's IP prefix range. Click No to allow other traffic.	
Bandwidth Upstream	For Cisco vEdge devices and vManage: For transmitted traffic, set the bandwidth above which to generate notifications. Range: 1 through (232 / 2) – 1 kbps		
Bandwidth Downstream	For Cisco vEdge devices and vManage: For received traffic, set the bandwidth above which to generate notifications. Range: 1 through (232 / 2) – 1 kbps		

To save the feature template, click **Save**.

CLI Equivalent

```

vpn vpn-id
  interface interface-name
    bandwidth-downstream kbps
    bandwidth-upstream kbps
    block-non-source-ip
    description text
    dhcp-helper ip-address
    (ip address ipv4-prefix/length | ip dhcp-client [dhcp-distance number])
    (ipv6 address ipv6-prefix/length | ipv6 dhcp-client [dhcp-distance number])
  [dhcp-rapid-commit]
  secondary-address ipv4-address
  [no] shutdown

```

Create a Tunnel Interface

On Cisco vEdge devices, you can configure up to eight tunnel interfaces. This means that each Cisco vEdge device router can have up to eight TLOCs. On Cisco Catalyst SD-WAN Controllers and Cisco SD-WAN Manager, you can configure one tunnel interface.

For the control plane to establish itself so that the overlay network can function, you must configure WAN transport interfaces in VPN 0. The WAN interface will enable the flow of tunnel traffic to the overlay. You can add other parameters shown in the table below only after you configure the WAN interface as a tunnel interface.

To configure a tunnel interface, select **Interface Tunnel** and configure the following parameters:

To configure additional tunnel interface parameters, click **Advanced Options**:

Parameter Name	Cisco vEdge devices Only	Description
GRE	Yes	Use GRE encapsulation on the tunnel interface. By default, GRE is disabled. If you select both IPsec and GRE encapsulations, two TLOCs are created for the tunnel interface that have the same IP addresses and colors, but that differ by their encapsulation.
IPsec	Yes	Use IPsec encapsulation on the tunnel interface. By default, IPsec is enabled. If you select both IPsec and GRE encapsulations, two TLOCs are created for the tunnel interface that have the same IP addresses and colors, but that differ by their encapsulation.
IPsec Preference	Yes	Specify a preference value for directing traffic to the tunnel. A higher value is preferred over a lower value. Range: 0 through 4294967295 Default: 0
IPsec Weight	Yes	Enter a weight to use to balance traffic across multiple TLOCs. A higher value sends more traffic to the tunnel. Range: 1 through 255 Default: 1
Carrier	No	Select the carrier name or private network identifier to associate with the tunnel. Values: carrier1, carrier2, carrier3, carrier4, carrier5, carrier6, carrier7, carrier8, default Default: default
Bind Loopback Tunnel	Yes	Enter the name of a physical interface to bind to a loopback interface.
Last-Resort Circuit	Yes	Select to use the tunnel interface as the circuit of last resort.

Parameter Name	Cisco vEdge devices Only	Description
NAT Refresh Interval	No	Enter the interval between NAT refresh packets sent on a DTLS or TLS WAN transport connection. Range: 1 through 60 seconds Default: 5 seconds
Hello Interval	No	Enter the interval between Hello packets sent on a DTLS or TLS WAN transport connection. Range: 100 through 10000 milliseconds Default: 1000 milliseconds (1 second)
Hello Tolerance	No	Enter the time to wait for a Hello packet on a DTLS or TLS WAN transport connection before declaring that transport tunnel to be down. Range: 12 through 60 seconds Default: 12 seconds

Configure Tunnel Interface CLI on vEdge Devices

```

vpn 0
  interface interface-name
    tunnel-interface
      allow-service service-name
      bind interface-name (on vEdge routers only)
      carrier carrier-name
      color color
      encapsulation (gre | ipsec) (on vEdge routers only)
      preference number
      weight number
      exclude-controller-group-list number (on vEdge routers only)
      hello-interval milliseconds
      hello-tolerance seconds
      last-resort-circuit (on vEdge routers only)
      low-bandwidth-link
      max-control-connections number (on vEdge routers only)
      nat-refresh-interval seconds
      vbond-as-stun-server
      vmanage-connection-preference number (on vEdge routers only)

```

Associate a Carrier Name with a Tunnel Interface

To associate a carrier name or private network identifier with a tunnel interface, use the **carrier** command. *carrier-name* can be **default** and **carrier1** through **carrier8**:

```

vEdge(config)# vpn 0
vEdge(config-vpn-0)# interface interface-name
vEdge(config-interface)# tunnel-interface
vEdge(config-tunnel-interface)# carrier carrier-name

```

Create Tunnel Groups

By default, WAN Edge routers try to build tunnels with all other TLOCs in the network, regardless of color. When the restrict option is used with the color designation under the tunnel configuration, the TLOC is restricted to only building tunnels to TLOCs of the same color. For more information on the restrict option see, [Configure Interfaces in the WAN Transport VPN\(VPN0\)](#).

The tunnel group feature is similar to the restrict option but gives more flexibility because once a tunnel group ID is assigned under a tunnel, only TLOCs with the same tunnel group IDs can form tunnels with each other irrespective of color.

If a TLOC is associated with a tunnel group ID, it continues to form tunnels with other TLOCs in the network that are not associated with any tunnel group IDs.



Note The restrict option can still be used in conjunction with this feature. If used, then an interface with a tunnel group ID and restrict option defined on an interface will only form a tunnel with other interfaces with the same tunnel group ID and color.

Configure Tunnel Groups on Cisco vEdge devices Using CLI

To configure tunnel groups on Cisco vEdge devices:

```
vEdge(config)# vpn 0
vEdge(config-vpn-0)# interface interface name
vEdge(config-interface-interface name)# tunnel-interface
vEdge(config-tunnel-interface)# group group-id
```

Limit Keepalive Traffic on a Tunnel Interface

By default, Cisco vEdge devices send a Hello packet once per second to determine whether the tunnel interface between two devices is still operational and to keep the tunnel alive. The combination of a hello interval and a hello tolerance determines how long to wait before declaring a DTLS or TLS tunnel to be down. The default hello interval is 1 second, and the default tolerance is 12 seconds. With these default values, if no Hello packet is received within 11 seconds, the tunnel is declared down at 12 seconds.

If the hello interval or the hello tolerance, or both, are different at the two ends of a DTLS or TLS tunnel, the tunnel chooses the interval and tolerance as follows:

- For a tunnel connection between two controller devices, the tunnel uses the lower hello interval and the higher tolerance interval for the connection between the two devices. (Controller devices are Cisco SD-WAN Validator, Cisco SD-WAN Manager, and Cisco SD-WAN Controller.) This choice is made in case one of the controllers has a slower WAN connection. The hello interval and tolerance times are chosen separately for each pair of controller devices.
- For a tunnel connection between a Cisco vEdge device and any controller device, the tunnel uses the hello interval and tolerance times configured on the router. This choice is made to minimize the amount of traffic sent over the tunnel, to allow for situations where the cost of a link is a function of the amount of traffic traversing the link. The hello interval and tolerance times are chosen separately for each tunnel between a Cisco vEdge device and a controller device.

To minimize the amount of keepalive traffic on a tunnel interface, increase the Hello packet interval and tolerance on the tunnel interface:

```
vEdge(config-tunnel-interface)# hello-interval milliseconds
vEdge(config-tunnel-interface)# hello-tolerance seconds
```

The default hello interval is 1000 milliseconds, and it can be a time in the range 100 through 600000 milliseconds (10 minutes). The default hello tolerance is 12 seconds, and it can be a time in the range 12 through 600 seconds (10 minutes). The hello tolerance interval must be at most one-half the OMP hold time. You can configure the OMP hold time with the **omp timers holdtime** command.

Configure Multiple Tunnel Interfaces on a vEdge Router

On a Cisco vEdge device, you can configure up to eight tunnel interfaces in the transport interface (VPN 0). This means that each Cisco vEdge device can have up to eight TLOCs.

When a Cisco vEdge device has multiple TLOCs, each TLOC is preferred equally and traffic to each TLOC is weighted equally, resulting in ECMP routing. ECMP routing is performed regardless of the encapsulation used on the transport tunnel, so if, for example, a router has one IPsec and one GRE tunnel, with ECMP traffic is forwarded equally between the two tunnels. You can change the traffic distribution by modifying the preference or the weight, or both, associated with a TLOC. (Note that you can also affect or change the traffic distribution by applying a policy on the interface that affects traffic flow.)

```
vEdge(config)# vpn 0
vEdge(config-vpn-0)# interface interface-name
vEdge(config-tunnel-interface) encapsulation (gre | ipsec)
vEdge(config-encapsulation)# preference number
vEdge(config-encapsulation)# weight number
```

The **preference** command controls the preference for directing inbound and outbound traffic to a tunnel. The preference can be a value from 0 through 4294967295 ($2^{32} - 1$), and the default value is 0. A higher value is preferred over a lower value.

When a Cisco vEdge device has two or more tunnels, if all the TLOCs have the same preference and no policy is applied that affects traffic flow, all the TLOCs are advertised into OMP. When the router transmits or receives traffic, it distributes traffic flows evenly among the tunnels, using ECMP.

When a Cisco vEdge device has two or more tunnels, if the TLOCs have different preferences and a policy is that affects traffic flow is not applied, all the TLOCs are advertised to Cisco Catalyst SD-WAN Controller via OMP for further processing based on the control policy applied on Cisco Catalyst SD-WAN Controller for the corresponding vEdge site-id. When the router transmits or receives traffic, it sends traffic to or receives traffic from only the TLOC with the highest preference. When there are three or more tunnels and two of them have the same preference, traffic flows are distributed evenly between these two tunnels.

A remote Cisco vEdge device trying to reach one of these prefixes selects which TLOC to use from the set of TLOCs that have been advertised. So, for example, if a remote router selects a GRE TLOC on the local router, the remote router must have its own GRE TLOC to be able to reach the prefix. If the remote router has no GRE TLOC, it is unable to reach the prefix. If the remote router has a single GRE TLOC, it selects that tunnel even if there is an IPsec TLOC with a higher preference. If the remote router has multiple GRE TLOCs, it selects from among them, choosing the one with the highest preference or using ECMP among GRE TLOCs with equal preference, regardless of whether there is an IPsec TLOC with a higher preference.

The **weight** command controls how traffic is balanced across multiple TLOCs that have equal preferences values. The weight can be a value from 1 through 255, and the default is 1. When the weight value is higher, the router sends more traffic to the TLOC. You typically set the weight based on the bandwidth of the TLOC. When a router has two or more TLOCs, all with the highest equal preference value, traffic distribution is

weighted according to the configured weight value. For example, if TLOC A has weight 10, and TLOC B has weight 1, and both TLOCs have the same preference value, then roughly 10 flows are sent out TLOC A for every 1 flow sent out TLOC B.

Configure an Interface as a NAT Device

For information on how to configure NAT, see the [Cisco Catalyst SD-WAN NAT Configuration Guide, Cisco IOS XE Catalyst SD-WAN Release 17.x](#).

Configure IPv4 NAT CLI Equivalent on vEdge

CLI Equivalent

```
vpn vpn-id
interface interface-name
  nat
    block-icmp-error
    refresh (bi-directional | outbound)
    respond-to-ping
    tcp-timeout minutes
    udp-timeout minutes
```

Configure NAT64 CLI Equivalent on Cisco vEdge Device

CLI Equivalent

```
interface interface-name
nat64 enable
  tcp-timeout minutes
  udp-timeout minutes
```

VPN Interface NAT Pool using Cisco SD-WAN Manager

Create NAT Pool Interfaces in a VPN

To create Network Address Translation (NAT) pools of IP addresses in VPNs, use the **VPN Interface NAT Pool** template for Cisco vEdge devices. To configure NAT pool interfaces in a VPN using Cisco SD-WAN Manager templates:

1. Create a **VPN Interface NAT Pool** template for Cisco vEdge devices to configure Ethernet interface parameters, as described in this article.
2. Create a VPN feature template to configure parameters for a service-side VPN.
3. Optionally, create a data policy to direct data traffic to a service-side NAT.

Create a VPN Interface NAT Pool Template

You can open a new **VPN Interface NATPool** template for Cisco vEdge devices from the VPN section of a device template.

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Feature Templates**.



Note In Cisco vManage Release 20.7.x and earlier releases, **Feature Templates** is titled **Feature**.

3. Click **Add Template**.
4. Select a device from the list.
5. From the **VPN** section, click **VPN Interface NATPool**.

The VPN Interface Ethernet template form displays. This form contains fields for naming the template, fields for defining the VPN Interface NAT Pool parameters.

1. In the required **Template Name** field, enter a name for the template.
The name can be up to 128 characters and can contain only alphanumeric characters.
2. In the optional **Template Description** field, enter a description of the template.
The description can be up to 2048 characters and can contain only alphanumeric characters.

Parameter Menus and Options

Parameter Menus and Options

When you first open a feature template, for each parameter that has a default value, the scope is set to Default (indicated by a ) and the default setting or value is shown. To change the default or to enter a value, click the scope drop-down to the left of the parameter field and select the appropriate option.

Configure a NAT Pool Interface

To configure a NAT pool interface, configure the following parameters. Parameters marked with an asterisk are required to configure the interface.

Basic Configuration

Enter the following basic configuration parameters:

Table 98:

Parameter Name	Description
Shutdown*	Yes Click No to enable the interface. No
Interface Name (1...31)*	Enter a number for the NAT pool interface to use for service-side NAT. For example, <i>natpool22</i> . Range: 1-31
Description	Enter a description for the interface.

Parameter Name	Description
IPv4 Address*	Enter the IPv4 address of the interface. The address length determines the number of NAT addresses that the router use at the same time. A Cisco vEdge device router can support a maximum of 250 NAT IP addresses.
Refresh Mode	Select how NAT mappings are refreshed:
bi-directional	Keep active the NAT mappings for inbound and outbound traffic.
outbound	Keep active the NAT mappings for outbound traffic. This is the default.
UDP Timeout	Enter the time when NAT translations over UDP sessions time out. <i>Default:</i> 1 minute Range: 1-65536 minutes
TCP Timeout	Enter the time when NAT translations over TCP sessions time out. <i>Default:</i> 60 minutes (1 hour) Range: 1-65536 minutes
Block ICMP	Select whether a Cisco vEdge device that is acting as a NAT device should receive inbound ICMP error messages. By default, the router blocks these error messages. Click Off to receive the ICMP error messages.
Direction	Select the direction in which the NAT interface performs address translation:
inside	Translate the source IP address of packets that are coming from the service side of the Cisco vEdge device and that are destined to transport side of the router. This is the default.
outside	Translate the source IP address of packets that are coming to the Cisco vEdge device from the transport side of the Cisco vEdge device and that are destined to a service-side device.
Overload	Click No to disable dynamic NAT. By default, dynamic NAT is enabled.

Configure a Tracker Interface

1. To create one or more tracker interfaces, click **Tracker**, and click **New Tracker**.
2. Select one or more interfaces to track the status of service interfaces.
3. To save the tracker interfaces, click **Add**.
4. To save the feature template, click **Save**.

NAT Pool Interface CLI Equivalent Commands on Cisco vEdge Devices

Use the following commands to configure NAT Pool interfaces on Cisco vEdge devices.

```
vpn vpn-id
  interface natpoolnumber
    ip address prefix/length
  nat
```

```

tracker tracker-name1
    tracker-name2, tracker-name3
direction (inside | outside)
[no] overload
refresh (bi-directional | outbound)
static source-ip ip-address1 translate-ip ip-address2 (inside | outside)
tcp-timeout minutes
udp-timeout minutes
[no] shutdown

```

Configure Port-Forwarding Rules

To create port-forwarding rules to allow requests from an external network to reach devices on the internal network:

1. Click **Port Forward**.
2. Click **New Port Forwarding Rule**, and configure the parameters. You can create up to 128 rules.
3. To save the rule, click **Add**.
4. To save the feature template, click **Save**.

Table 99:

Parameter Name	Values	Description
Port Start Range	Enter the starting port number. This number must be less than or equal to the ending port number.	
Port End Range	Enter the ending port number. To apply port forwarding to a single port, specify the same port number for the starting and ending numbers. When applying port forwarding to a range of ports, the range includes the two port numbers that you specify.	
Protocol	TCP UDP	Select the protocol to apply the port-forwarding rule to. To match the same ports for both TCP and UDP traffic, configure two rules.
VPN	0-65535	Private VPN in which the internal server resides.
Private IP	Enter an IP address to use within the firewall. A best practice is to specify the IP address of a service-side VPN.	

Port Forwarding CLI Equivalent for vEdge

```

vpn vpn-id
interface natpoolnumber
nat
port-forward port-start port-number1 port-end port-number2 proto (tcp | udp)
private-ip-address ip address private-vpn vpn-id

```

Static NAT CLI Equivalent Commands on Cisco vEdge Device

```
vpn vpn-id
  interface natpoolnumber
    nat
      port-forward port-start port-number1 port-end port-number2 proto (tcp | udp)
      private-ip-address ip address private-vpn vpn-id
```

Release Information

Introduced in Cisco SD-WAN Manager NMS Release 16.3. In Release 17.2.2, add support for tracker interface status. In Release 18.4, updated images; add support for multiple tracker interfaces.

Apply Access Lists and QoS Parameters

Quality of service (QoS) helps determine how a service will perform. By configuring QoS, enhance the performance of an application on the WAN. To configure a shaping rate for an interface and to apply a QoS map, a rewrite rule, access lists, and policers to a interface, click **ACL/QoS**, and configure the following parameters:

Parameter Name	Description
Shaping rate	Configure the aggregate traffic transmission rate on the interface to be less than line rate, in kilobits per second (kbps).
QoS Map	Specify the name of the QoS map to apply to packets being transmitted out the interface.
Rewrite Rule	Click On , and specify the name of the rewrite rule to apply on the interface.
Ingress ACL – IPv4	Click On , and specify the name of the access list to apply to IPv4 packets being received on the interface.
Egress ACL – IPv4	Click On , and specify the name of the access list to apply to IPv4 packets being transmitted on the interface.
Ingress ACL – IPv6	Click On , and specify the name of the access list to apply to IPv6 packets being received on the interface.
Egress ACL – IPv6	Click On , and specify the name of the access list to apply to IPv6 packets being transmitted on the interface.
Ingress Policer	Click On , and specify the name of the policer to apply to packets received on the interface.
Egress Policer	Click On , and specify the name of the policer to apply to packets being transmitted on the interface.

To save the feature template, click **Save**.

CLI Equivalent

```
vpn vpn-id
  interface interface-name
    access-list acl-list (in | out)
    policer policer-name (in |out)
    qos-map name
```

```
rewrite-rule name
shaping-rate name
```

Add ARP Table Entries

The Address Resolution Protocol (ARP) helps associate a link layer address (such as the MAC address of a device) to its assigned internet layer address. Configure a static ARP address when dynamic mapping is not functional. To configure static ARP table entries on the interface, select ARP. Then click **Add New ARP** and configure the following parameters:

Parameter Name	Description
IP Address	Enter the IP address for the ARP entry in dotted decimal notation or as a fully qualified host name.
MAC Address	Enter the MAC address in colon-separated hexadecimal notation.

To save the ARP configuration, click **Add**.

To save the feature template, click **Save**.

CLI Equivalent

```
vpn vpn-id
interface interface-name arp ip ip-address mac mac-address
```

Configuring VRRP

To have an interface run the Virtual Router Redundancy Protocol (VRRP), which allows multiple routers to share a common virtual IP address for default gateway redundancy, select the VRRP tab. Then click **Add New VRRP** and configure the following parameters:

Parameter Name	Description
Group ID	Enter the virtual router ID, which is a numeric identifier of the virtual router. You can configure a maximum of 24 groups. Range: 1 through 255
Priority	Enter the priority level of the router. The router with the highest priority is elected as primary VRRP router. If two routers have the same priority, the one with the higher IP address is elected as primary VRRP router. Range: 1 through 254 Default: 100

Parameter Name	Description
Timer (milliseconds)	Specify how often the primary VRRP router sends VRRP advertisement messages. If subordinate routers miss three consecutive VRRP advertisements, they elect a new primary VRRP routers. Range: 100 through 40950 milliseconds Default: 1000 msecs Note When the timer is 100 ms for the VRRP feature template on Cisco IOS XE Catalyst SD-WAN devices, the VRRP fails if the traffic is high on LAN interface. Use the 100 msec timer only if the Cisco IOS XE Catalyst SD-WAN device platform supports it, and if there are fewer tunnel groups.
Track OMP Track Prefix List	By default, VRRP uses of the state of the service (LAN) interface on which it is running to determine which router is the primary virtual router. if a router loses all its WAN control connections, the LAN interface still indicates that it is up even though the router is functionally unable to participate in VRRP. To take WAN side connectivity into account for VRRP, configure one of the following: Track OMP—Click On for VRRP to track the Overlay Management Protocol (OMP) session running on the WAN connection. If the primary VRRP router loses all its OMP sessions, VRRP elects a new default gateway from those that have at least one active OMP session. Note From Cisco IOS XE Catalyst SD-WAN Release 17.18.1a, enabling Track OMP changes the device CLI command from vrrp track omp shutdown to vrrp track omp decrement 10. Track Prefix List—Track both the OMP session and a list of remote prefixes, which is defined in a prefix list configured on the local router. If the primary VRRP router loses all its OMP sessions, VRRP failover occurs as described for the Track OMP option. In addition, if reachability to all of the prefixes in the list is lost, VRRP failover occurs immediately, without waiting for the OMP hold timer to expire, thus minimizing the amount of overlay traffic is dropped while the routers determine the primary VRRP router.
IP Address	Enter the IP address of the virtual router. This address must be different from the configured interface IP addresses of both the local router and the peer running VRRP.

CLI Equivalent

```

vpn vpn-id
  interface geslot/port[.subinterface]
    vrrp group-number
      ipv4 ip-address
      priority number
      timer seconds
      (track-omp | track-prefix-list list-name)

```

Configure a Prefix List for VRRP

You can configure prefix list tracking for VRRP using device and feature templates. To configure a prefix list, do the following:

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Policy**.
2. Click **Localized Policy**.
3. From the **Custom Options** drop-down list, click **Lists**.
4. Click **Prefix** from the left pane, and click **New Prefix List**.
5. In **Prefix List Name**, enter a name for the prefix list.
6. Choose **IPv4** as the **Internet Protocol**.
7. In **Add Prefix**, enter the prefix entries separated by commas.
8. Click **Add**.
9. Click **Next** and configure **Forwarding Classes/QoS**.
10. Click **Next** and configure **Access Control Lists**.
11. Click **Next** and in **Route Policy** pane, select a relevant route policy and click **...**, and click **Edit** to add the newly added prefix list.
12. From the **Match** pane, click **AS Path List** and in the **Address**, choose the newly added prefix list.
13. Click **Save Match and Actions**.
14. Click **Next** and enter the **Policy Name** and **Policy Description** in the **Policy Overview** screen.
15. Click **Save Policy**.

Configure a Prefix List for VRRP in the Device Template

To configure the Prefix List to the VRRP and the localized policy in the device template, do the following:

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Device Templates**.



Note In Cisco vManage Release 20.7.x and earlier releases, **Device Templates** is titled **Device**.

3. Select a relevant device template and click **...** and click **Edit** to edit the template details.
4. From **Policy**, select the policy with the newly added prefix list.
5. Click **Update**.
6. Click **Feature Templates**.
7. Select a relevant device template and click **...** and click **Edit** to edit the template details.
8. Click **VRRP**.

9. Select a relevant group ID and click the pen icon to associate the new prefix-list to the VRRP details.
10. Click the **Track Prefix List** drop-down list and enter the newly added prefix-list name.
11. Click **Save Changes**.
12. Click **Update** to save the changes.
13. Click **Device Templates** and select the policy with the newly added prefix list.
14. Click ... and click **Attach Devices**.
15. From **Available Devices**, double-click the relevant device to move it to **Selected Devices**, and then click **Attach**.

Configure Prefix List For VRRP Using Configuration Groups

Before you begin

On the **Configuration > Configuration Groups** page, choose **SD-WAN** as the solution type.

Procedure

Step 1 From the Cisco SD-WAN Manager menu, choose **Configuration > Configuration Groups**.

Step 2 Create and configure Prefix List for VRRP in a Policy Object Profile.

- a. Choose the **Prefix** policy object from the **Select Policy Object** drop-down list.
- b. Enter the **Prefix List Name**.
- c. In the **Internet Protocol** field, click **IPv4** or **IPv6**.
- d. Under **Add Prefix**, enter the prefix for the list. Optionally, click the **Choose a file** link to import a prefix list.
- e. Click **Save**.

The following table describe the options for configuring the prefix.

Table 100: Prefix List

Field	Description
Prefix List Name	Enter a name for the prefix list.
Internet Protocol	Specifies the internet protocol. The options are IPv4 and IPv6.

What to do next

Also see [Deploy a configuration group](#).

Configure Advanced Properties

To configure other interface properties, select the **Advanced** tab and configure the following parameters:

Parameter Name	Description
Duplex	Choose full or half to specify whether the interface runs in full-duplex or half-duplex mode. Default: full
MAC Address	Specify a MAC address to associate with the interface, in colon-separated hexadecimal notation.
IP MTU	Specify the maximum MTU size of packets on the interface. Range: 576 through 1804 Default: 1500 bytes
PMTU Discovery	Click On to enable path MTU discovery on the interface. PMTU determines the largest MTU size that the interface supports so that packet fragmentation does not occur.
Flow Control	Select a setting for bidirectional flow control, which is a mechanism for temporarily stopping the transmission of data on the interface. Values: autonet, both, egress, ingress, none Default: autoneg
TCP MSS	Specify the maximum segment size (MSS) of TCP SYN packets passing through the router. By default, the MSS is dynamically adjusted based on the interface or tunnel MTU such that TCP SYN packets are never fragmented. Range: 552 to 1460 bytes Default: None
Speed	Specify the speed of the interface for use when the remote end of the connection does not support autonegotiation. Values: 10, 100, 1000, or 10000 Mbps
Clear-Dont-Fragment	Click On to clear the Don't Fragment (DF) bit in the IPv4 packet header for packets being transmitted out the interface. When the DF bit is cleared, packets larger than that interface's MTU are fragmented before being sent. Note Clear-Dont-Fragment clears the DF bit when there is fragmentation needed and the DF bit is set. For packets not requiring fragmentation, the DF bit is not affected.
Static Ingress QoS	Specify a queue number to use for incoming traffic. Range: 0 through 7
ARP Timeout	Specify how long it takes for a dynamically learned ARP entry to time out. Range: 0 through 2678400 seconds (744 hours) Default: 1200 (20 minutes)

Parameter Name	Description
Autonegotiation	Click Off to turn autonegotiation off. By default, an interface runs in autonegotiation mode.
TLOC Extension	Enter the name of a physical interface on the same router that connects to the WAN transport. This configuration then binds this service-side interface to the WAN transport. A second router at the same site that itself has no direct connection to the WAN (generally because the site has only a single WAN connection) and that connects to this service-side interface is then provided with a connection to the WAN. Note that TLOC extension over L3 is only supported for Cisco IOS XE routers. If configuring TLOC extension over L3 for a Cisco IOS XE router, enter the IP address of the L3 interface.
Power over Ethernet	Click On to enable PoE on the interface.
ICMP Redirect	Click Disable to disable ICMP redirect messages on the interface. By default, an interface allows ICMP redirect messages.

To save the feature template, click **Save**.

CLI Equivalent

```

vpn vpn-id
  interface interface-name
    arp-timeout seconds (on vEdge routers only)
    [no] autonegotiate
    clear-dont-fragment
    duplex (full | half)
    flow-control control
    icmp-redirect-disable (on vEdge routers only)
    mac-address mac-address
    mtu bytes
    pmtu
    pppoe-client (on vEdge 100m and vEdge 100wm routers only)
    ppp-interface pppnumber
    speed speed
    static-ingress-qos number (on vEdge routers only)
    tcp-mss-adjust bytes
    tloc-extension interface-name (on vEdge routers only)
    tracker tracker-name (on vEdge routers only)

```

VPN Interface Bridge

Use the VPN Interface Bridge template for all Cisco vEdge device Cloud and Cisco vEdge devices.

Integrated routing and bridging (IRB) allows Cisco vEdge devices in different bridge domains to communicate with each other. To enable IRB, create logical IRB interfaces to connect a bridge domain to a VPN. The VPN provides the Layer 3 routing services necessary so that traffic can be exchanged between different VLANs. Each bridge domain can have a single IRB interface and can connect to a single VPN, and a single VPN can connect to multiple bridge domains on a Cisco vEdge device.

To configure a bridge interface using Cisco SD-WAN Manager templates:

1. Create a VPN Interface Bridge feature template to configure parameters for logical IRB interfaces, as described in this article.
2. Create a Bridge feature template for each bridging domain, to configure the bridging domain parameters. See the Bridge help topic.

Navigate to the Template Screen and Name the Template

1. From the Cisco SD-WAN Manager menu, choose **Configuration** > **Templates**.
2. Click **Device Templates**.



Note In Cisco vManage Release 20.7.x and earlier releases, **Device Templates** is titled **Device**.

3. From the **Create Template** drop-down list, select **From Feature Template**.
4. From the **Device Model** drop-down list, select the type of device for which you are creating the template.
5. Click **Service VPN** or scroll to the **Service VPN** section.
6. Click the **Service VPN** drop-down list.
7. From **Additional VPN Templates**, click **VPN Interface Bridge**.
8. From the **VPN Interface Bridge** drop-down list, click **Create Template**.
The VPN Interface Bridge template form is displayed. The top of the form contains fields for naming the template, and the bottom contains fields for defining VPN Interface Bridge parameters.
9. In **Template Name**, enter a name for the template. The name can be up to 128 characters and can contain only alphanumeric characters.
10. In **Template Description**, enter a description of the template. The description can be up to 2048 characters and can contain only alphanumeric characters.

When you first open a feature template, for each parameter that has a default value, the scope is set to Default (indicated by a check mark), and the default setting or value is shown. To change the default or to enter a value, click the scope drop-down to the left of the parameter field and select one of the following:

Table 101:

Parameter Scope	Scope Description
Device Specific (indicated by a host icon)	Use a device-specific value for the parameter. For device-specific parameters, you cannot enter a value in the feature template. You enter the value when you attach a Viptela device to a device template . When you click Device Specific, the Enter Key box opens. This box displays a key, which is a unique string that identifies the parameter in a CSV file that you create. This file is an Excel spreadsheet that contains one column for each key. The header row contains the key names (one key per column), and each row after that corresponds to a device and defines the values of the keys for that device. You upload the CSV file when you attach a Viptela device to a device template. For more information, see Create a Template Variables Spreadsheet . To change the default key, type a new string and move the cursor out of the Enter Key box. Examples of device-specific parameters are system IP address, hostname, GPS location, and site ID.
Global (indicated by a globe icon)	Enter a value for the parameter, and apply that value to all devices. Examples of parameters that you might apply globally to a group of devices are DNS server, syslog server, and interface MTUs.

Release Information

Introduced in Cisco SD-WAN Manager NMS in Release 15.3. In Release 18.2, add support for disabling ICMP redirect messages.

Create a Bridging Interface

To configure an interface to use for bridging servers, select **Basic Configuration** and configure the following parameters. Parameters marked with an asterisk are required to configure bridging.

Table 102:

Parameter Name	Description
Shutdown*	Click No to enable the interface.
Interface name*	Enter the name of the interface, in the format irb number . The IRB interface number can be from 1 through 63, and must be the same as the VPN identifier configured in the Bridge feature template for the bridging domain that the IRB is connected to.
Description	Enter a description for the interface.
IPv4 Address*	Enter the IPv4 address of the router.

Parameter Name	Description
DHCP Helper	Enter up to eight IP addresses for DHCP servers in the network, separated by commas, to have the interface be a DHCP helper. A DHCP helper interface forwards BOOTP (Broadcast) DHCP requests that it receives from the specified DHCP servers.
Block Non-Source IP	Click Yes to have the interface forward traffic only if the source IP address of the traffic matches the interface's IP prefix range.
Secondary IP Address (on Cisco vEdge devices)	Click Add to configure up to four secondary IPv4 addresses for a service-side interface.

To save the template, click **Save**.

CLI equivalent:

```
vpn vpn-id interface irbnumber description "text description" dhcp-helper ip-addresses
ip address prefix/length mac-address mac-address mtu bytes secondary-address ipv4-address
[no] shutdown tcp-mss-adjust bytes
```

Apply Access Lists

Apply Access Lists

To apply access lists to IRB interfaces, select the ACL tab and configure the following parameters. The ACL filter determines what is allowed in or out of a bridging domain:

Table 103:

Parameter Name	Description
Ingress ACL – IPv4	Click On , and specify the name of an IPv4 access list to packets being received on the interface.
Egress ACL– IPv4	Click On , and specify the name of an IPv4 access list to packets being transmitted on the interface.

To save the feature template, click **Save**.

CLI equivalent:

```
vpn vpn-id interface irbnumber access-list acl-name (in | out)
```

Configure VRRP

To have an interface run the Virtual Router Redundancy Protocol (VRRP), which allows multiple routers to share a common virtual IP address for default gateway redundancy, choose **VRRP**. Then click **Add New VRRP** and configure the following parameters:

Table 104:

Parameter Name	Description
Group ID	Enter the virtual router ID, which is a numeric identifier of the virtual router. You can configure a maximum of 24 groups. <i>Range:</i> 1 through 255
Priority	Enter the priority level of the router. There router with the highest priority is elected as primary VRRP router. If twoCisco vEdge devices have the same priority, the one with the higher IP address is elected as primary VRRP router. <i>Range:</i> 1 through 254 <i>Default:</i> 100
Timer (milliseconds)	Specify how often the primary VRRP router sends VRRP advertisement messages. If subordinate routers miss three consecutive VRRP advertisements, they elect a new primary VRRP router. <i>Range:</i> 100 through 40950 milliseconds <i>Default:</i> 1000 msec Note When the timer is 100 ms for the VRRP feature template on s, the VRRP fails if the traffic is high on LAN interface. Use the 100 msec timer only if the platform supports it, and if there are fewer tunnel groups.
Track OMP Track Prefix List	By default, VRRP uses of the state of the service (LAN) interface on which it is running to determine which Cisco vEdge device is the primary virtual router. if a Cisco vEdge device loses all its WAN control connections, the LAN interface still indicates that it is up even though the router is functionally unable to participate in VRRP. To take WAN side connectivity into account for VRRP, configure one of the following: Track OMP—Click On for VRRP to track the Overlay Management Protocol (OMP) session running on the WAN connection. If the primary VRRP router loses all its OMP sessions, VRRP elects a new default gateway from those that have at least one active OMP session. Track Prefix List—Track both the OMP session and a list of remote prefixes, which is defined in a prefix list configured on the local router. If the primary VRRP router loses all its OMP sessions, VRRP failover occurs as described for the Track OMP option. In addition, if reachability to all of the prefixes in the list is lost, VRRP failover occurs immediately, without waiting for the OMP hold timer to expire, thus minimizing the amount of overlay traffic is dropped while the Cisco vEdge devices determine the primary VRRP router.
IP Address	Enter the IP address of the virtual router. This address must be different from the configured interface IP addresses of both the local Cisco vEdge device and the peer running VRRP.

To save the VRRP configuration, click **Add**.

To save the feature template, click **Save**.

CLI equivalent:

```

vpn vpn-id
interface irbnumber[.subinterface]
  vrrp group-number
  ipv4 ip-address
  priority number
  timer seconds
  (track-omp | track-prefix-list list-name)

```

Add ARP Table Entries

To configure static Address Resolution Protocol (ARP) table entries on the interface, choose **ARP**. Then click **Add New ARP** and configure the following parameters:

Table 105:

Parameter Name	Description
IP Address	Enter the IP address for the ARP entry in dotted decimal notation or as a fully qualified host name.
MAC Address	Enter the MAC address in colon-separated hexadecimal notation.

To save the ARP configuration, click **Add**.

To save the feature template, click **Save**.

CLI equivalent:

```

vpn vpn-id interface irbnumber arp
  ip address ip-address mac mac-address

```

Configure Advanced Properties

To configure other interface properties, click **Advanced** and configure the following parameters:

Table 106:

Parameter Name	Description
MAC Address	MAC addresses can be static or dynamic. A static MAC address is manually configured as opposed to a dynamic MAC address that is one learned via an ARP request. You can configure a static MAC on a router's interface or indicate a static MAC that identifies a router's interface. Specify a MAC address to associate with the interface, in colon-separated hexadecimal notation.
IP MTU	Similar to MTU, IP MTU only affects IP packets. If an IP packet exceeds the IP MTU, then the packet will be fragmented. Specify the maximum MTU size of packets on the interface. <i>Range:</i> 576 through 1804 <i>Default:</i> 1500 bytes

Parameter Name	Description
TCP MSS	TCP MSS will affect any packet that contains an initial TCP header that flows through the router. When configured, TCP MSS will be examined against the MSS exchanged in the three-way handshake. The MSS in the header will be lowered if the configured setting is lower than what is in the header. If the header value is already lower, it will flow through unmodified. The end hosts will use the lower setting of the two hosts. If the TCP MSS is to be configured, it should be set it at 40 bytes lower than the minimum path MTU. Specify the maximum segment size (MSS) of TPC SYN packets passing through the Cisco vEdge device. By default, the MSS is dynamically adjusted based on the interface or tunnel MTU such that TCP SYN packets are never fragmented.<i>Range:</i> 552 to 1460 bytes<i>Default:</i> None
Clear-Dont-Fragment	Configure Clear-Dont-Fragment if there are packets arriving on an interface with the DF bit set. If these packets are larger than the MTU will allow, they are dropped. If you clear the df-bit, the packets will be fragmented and sent. Click On to clear the Dont Fragment (DF) bit in the IPv4 packet header for packets being transmitted out the interface. When the DF bit is cleared, packets larger than that interface's MTU are fragmented before being sent. Note Clear-Dont-Fragment clears the DF bit when there is fragmentation needed and the DF bit is set. For packets not requiring fragmentation, the DF bit is not affected.
ARP Timeout	ARP Timeout controls how long we maintain the ARP cache on a router. Specify how long it takes for a dynamically learned ARP entry to time out. <i>Range:</i> 0 through 2678400 seconds (744 hours)<i>Default:</i> 1200 seconds (20 minutes)
ICMP Redirect	ICMP Redirects are sent by a router to the sender of an IP packet when a packet is being routed sub-optimally. The ICMP Redirect informs the sending host to forward subsequent packets to that same destination through a different gateway. To disable ICMP redirect messages on the interface, click Disable. By default, an interface allows ICMP redirect messages.

To save the feature template, click **Save**.

CLI equivalent:

```
vpn vpn-id interface irbnumber arp-timeout seconds clear-dont-fragment
icmp-redirect-disable mac-address mac-address mtu bytes tcp-mss-adjust bytes
```

Configure DSL IPoE Using a Configuration Group

Before you begin

On the **Configuration > Configuration Groups** page, choose **SD-WAN** as the solution type.

Procedure

Step 1 From the Cisco SD-WAN Manager menu, choose **Configuration > Configuration Groups**.

Step 2 Create and configure the DSL IPoE parameters in a Transport and Management Profile.

- a. Configure basic configuration.

Table 107: Basic Configuration

Parameter Name	Description
Controller Slot*	Enter the slot number of the controller, in the following format: <i>slot/subslot/port</i> (for example, 0/2/0)
Controller Mode	Select the operating mode of the DSL controller from the drop-down list: • ADSL1: Use ITU G.992.1 Annex A full-rate mode, which provides a downstream rate of 1.3 Mbps and an upstream rate of 1.8 Mbps. • ADSL2: Use ITU G.992.3 Annex A, Annex L, and Annex M, which provides a downstream rate of 12 Mbps and an upstream rate of 1.3 Mbps. • ADSL2+: Use ITU G.992.5 Annex A and Annex M, which provides a downstream rate of 24 Mbps and an upstream rate of 3.3 Mbps. • ANSI: Operating in ADSL2/2+ mode, as defined in ITU G.991.1, G.992.3, and G.992.5, Annex A and Annex M, and in VDSL2 mode, as defined in ITU-T G.993.2. • VDSL2: Operate in VDSL2 mode, as defined in ITU-T G.993.2, which uses frequencies of up to 30 MHz to provide a downstream rate of 200 Mbps and an upstream rate of 100 Mbps.
SRA	Enabled by default. Click No to disable seamless rate adaptation on the interface. SRA adjusts the line rate based on current line conditions.

- b. Configure Ethernet.

Table 108: Ethernet

Parameter Name	Description
Ethernet Interface Name *	Enter the name of an ethernet interface. For IOS XE routers, you must spell out the interface names completely (for example, GigabitEthernet0/0/0).
Description	Enter a description for the interface.
VLAN ID	Enter the VLAN identifier of the Ethernet interface.

- c. Configure Tunnel.

Table 109: Tunnel

Parameter Name	Description
Tunnel Interface	
Per Tunnel QoS	Enable per tunnel QoS and choose from the following values to configure hub-to-spoke network topologies: • Spoke • Hub
Color	Select a color for the TLOC.
Color Description	Minimum supported release: Cisco Catalyst SD-WAN Manager Release 20.18.1 Enter a description associated to the TLOC color.
Groups	Enter the list of groups in the field.
Exclude Controller Group List	Set the Cisco SD-WAN Controllers that the tunnel interface is not allowed to connect to. Range: 0 through 100
Maximum Control Connections	Specify the maximum number of Cisco SD-WAN Controllers that the WAN tunnel interface can connect to. To have the tunnel establish no control connections, set the number to 0. Range: 0 through 8 Default: 2
Cisco SD-WAN Manager Connection Preference	Set the preference for using a tunnel interface to exchange control traffic with Cisco SD-WAN Manager. Range: 0 through 8 Default: 5
Tunnel TCP MSS	TCP MSS affects any packet that contains an initial TCP header that flows through the router. When configured, TCP MSS is examined against the MSS exchanged in the three-way handshake. The MSS in the header is lowered if the configured TCP MSS setting is lower than the MSS in the header. If the MSS header value is already lower than the TCP MSS, the packets flow through unmodified. The host at the end of the tunnel uses the lower setting of the two hosts. To configure TCP MSS, provide a value that is 40 bytes lower than the minimum path MTU. Specify the MSS of TPC SYN packets passing through the Cisco vEdge device. By default, the MSS is dynamically adjusted based on the interface or tunnel MTU such that TCP SYN packets are never fragmented. Range: 552 through 1460 bytes Default: None
Border	From the drop-down list, select Global . Click On to set TLOC as border TLOC.
Validator As Stun Server	Click On to enable Session Traversal Utilities for NAT (STUN) to allow the tunnel interface to discover its public IP address and port number when the router is located behind a NAT.

Parameter Name	Description
Full Port Hop	Minimum release: Cisco IOS XE Catalyst SD-WAN Release 17.18.1a Enable full port hopping at the TLOC level to allow devices to establish connections with controllers by switching to the next port if the current port is blocked or non-functional. Default: Disabled
Port Hop	From the drop-down list, select Global. Click Off to allow port hopping on tunnel interface. Default: On, which disallows port hopping on tunnel interface. Starting from Cisco IOS XE Catalyst SD-WAN Release 17.18.1a, this field is deprecated. Instead use the Full Port Hop option. See the Full Port Hop field.
Low-Bandwidth Link	Click On to set the tunnel interface as a low-bandwidth link. Default: Off
Clear-Dont-Fragment	Configure Clear-Dont-Fragment for packets that arrive at an interface that has Don't Fragment configured. If these packets are larger than what MTU allows, they are dropped. If you clear the Don't Fragment bit, the packets are fragmented and sent. Click On to clear the Dont Fragment bit in the IPv4 packet header for packets being transmitted out of the interface. When the Dont Fragment bit is cleared, the router fragments packets larger than the MTU of the interface before sending the packets. Note Clear-Dont-Fragment clears the Dont Fragment bit and the Dont Fragment bit is set. For packets not requiring fragmentation, the Dont Fragment bit is not affected.
Network Broadcast	From the drop-down list, select Global. Click On to accept and respond to network-prefix-directed broadcasts. Enable this parameter only if the Directed Broadcast is enabled on the LAN interface feature template. Default: Off
Carrier	From the drop-down list, select Global and select the carrier name or private network identifier to associate with the tunnel. Values: carrier1, carrier2, carrier3, carrier4, carrier5, carrier6, carrier7, carrier8, default. Default: default
Bind Loopback Tunnel	Enter the name of a physical interface to bind to a loopback interface. The interface name has the following format: ge <i>slot/port</i>
NAT Refresh Interval	Set the interval between NAT refresh packets sent on a DTLS or TLS WAN transport connection. Range: 1 through 60 seconds Default: 5 seconds

Parameter Name	Description
Hello Interval	Enter the interval between Hello packets sent on a DTLS or TLS WAN transport connection. Range: 100 through 10000 milliseconds Default: 1000 milliseconds (1 second)
Hello Tolerance	Enter the time to wait for a Hello packet on a DTLS or TLS WAN transport connection before declaring that transport tunnel to be down. Range: 12 through 60 seconds. Default: 12 seconds.
Last Resort Circuit	Select to use the tunnel interface as the circuit of last resort. Note It is assumed that an interface configured as a circuit of last resort is unavailable and is skipped while calculating the number of control connections. As a result, the cellular modem becomes dormant, and no traffic is sent over the circuit. When the configurations are activated on the edge device with cellular interfaces, all the interfaces begin the process of establishing control and BFD connections. When one or more of the primary interfaces establishes a BFD connection, the circuit of last resort shuts itself down. If the primary interfaces lose their connections to remote edges, the circuit of last resort activates itself, triggering a BFD TLOC Down alarm and a Control TLOC Down alarm on the edge device. The last resort interfaces are a backup circuit on edge device and are activated when all other transport links BFD sessions fail. In this mode, the radio interface is turned off, and no control or data connections exist over the cellular interface.
Allow Services	Click On or Off for each service to enable or disable the service on the cellular interface.
Encapsulation	

Parameter Name	Description
Encapsulation	Enable atleast one of the following encapsulation methods: • IPsec: Enter a value to set the preference for directing traffic to the tunnel. A higher value is preferred over a lower value. Range: 0 through 4294967295 Default: 0 • IPsec Preference: From the drop-down list, select Global and enter a value to set the preference for directing traffic to the tunnel. A higher value is preferred over a lower value. Range: 0 through 4294967295 Default: 0 • IPsec Weight: From the drop-down list, select Global and enter a value to set weight for balancing traffic across multiple TLOCs. A higher value sends more traffic to the tunnel. Range: 1 through 255 Default: 1 • GRE: Enter a value to set GRE preference for TLOC. Range: 0 through 4294967295 • GRE Preference: From the drop-down list, select Global and enter a value to set the preference for directing traffic to the tunnel. A higher value is preferred over a lower value. Range: 0 through 4294967295 Default: 0 • GRE Weight: From the drop-down list, select Global and enter a value to set weight for balancing traffic across multiple TLOCs. A higher value sends more traffic to the tunnel. Range: 1 through 255 Default: 1

d. Configure NAT.

Table 110: NAT

Parameter Name	Description
UDP Timeout (Minutes)	Specify when NAT translations over UDP sessions time out. Range: 1 through 65536 minutes Default: 1 minute

Parameter Name	Description
TCP Timeout (Minutes)	Specify when NAT translations over TCP sessions time out. Range: 1 through 65536 minutes Default: 60 minutes (1 hour)

e. Configure QoS.

Table 111: QoS

Parameter Name	Description
Adaptive QoS	Enter adaptive QoS parameters. You can leave the additional details at as default or specify your values. • Adapt Period (Minutes): Choose Global from the drop-down list, click On, and enter the period in minutes. • Shaping Rate Upstream: Choose Global from the drop-down list, click On, and enter the minimum, maximum, and default upstream bandwidth in Kbps. • Shaping Rate Downstream: Choose Global from the drop-down list, click On, and enter the minimum, maximum, downstream, and upstream bandwidth in Kbps.
Shaping Rate (kbps)	Choose Global from the drop-down list and configure the aggregate traffic transmission rate on the interface to be less than line rate, in kilobits per second (kbps). Range: 8 through 100000000

f. Configure ACL.

Table 112: ACL

Parameter Name	Description
IPv4 Ingress Access List	Enter the name of an IPv4 access list to packets being received on the interface.
IPv4 Egress Access List	Enter the name of an IPv4 access list to packets being transmitted on the interface.
IPv6 Ingress Access List	Enter the name of an IPv6 access list to packets being received on the interface.
IPv6 Egress Access List	Enter the name of an IPv6 access list to packets being transmitted on the interface.

g. Configure advanced parameters.

Table 113: Advanced

Parameter Name	Description
Shutdown	Click No to enable the interface.

Parameter Name	Description
Tracker / Tracker Group	Enter the name of a tracker or tracker group to track the status of transport interfaces that connect to the internet.
Service Provider	Specify the details of the service provider.
Bandwidth Upstream (Kbps)	Specify the bandwidth value to generate notifications when the bandwidth of traffic transmitted on a physical interface exceeds the value.
Bandwidth Downstream (Kbps)	Specify the bandwidth value to generate notifications when the bandwidth of traffic transmitted on a physical interface exceeds the value.
IP MTU	Enter the maximum MTU size of packets on the interface. Range: 576 through 1804 Default: 1500
TCP MSS	Enter the maximum segment size (MSS) of TCP SYN packets passing through the router. By default, the MSS is dynamically adjusted based on the interface or tunnel MTU such that TCP SYN packets are never fragmented. Range: 552 through 1460 bytes Default: 1500
TLOC Extension	Enter the name of a physical interface on the same router that connects to the WAN transport. This configuration binds the service-side interface to the WAN transport by enabling a device to access the opposite WAN transport connected to the neighbouring device using a TLOC-extension interface.
IP Directed Broadcast	From the drop-down list, select Global to enable IP Directed Broadcast. An IP directed broadcast is an IP packet whose destination address is a valid broadcast address for some IP subnet but which originates from a node that is not itself part of that destination subnet.

What to do next

Also see [Deploy a configuration group](#).

Configure DSL PPPoA Using a Configuration Group

Before you begin

On the **Configuration > Configuration Groups** page, choose **SD-WAN** as the solution type.

Procedure

Step 1 From the Cisco SD-WAN Manager menu, choose **Configuration > Configuration Groups**.

Step 2 Create and configure the DSL PPPoA parameters in a Transport and Management Profile.

- a. Configure basic configuration.

Table 114: Basic Configuration

Parameter Name	Description
Controller Slot*	Enter the slot number of the DSL controller, in the following format: <i>slot/subslot/port</i> (for example, 0/2/0)
Controller Mode	Select the operating mode of the DSL controller from the drop-down list: • ADSL1: Use ITU G.992.1 Annex A full-rate mode, which provides a downstream rate of 1.3 Mbps and an upstream rate of 1.8 Mbps. • ADSL2: Use ITU G.992.3 Annex A, Annex L, and Annex M, which provides a downstream rate of 12 Mbps and an upstream rate of 1.3 Mbps. • ADSL2+: Use ITU G.992.5 Annex A and Annex M, which provides a downstream rate of 24 Mbps and an upstream rate of 3.3 Mbps. • ANSI: Operating in ADSL2/2+ mode, as defined in ITU G.991.1, G.992.3, and G.992.5, Annex A and Annex M, and in VDSL2 mode, as defined in ITU-T G.993.2. • VDSL2: Operate in VDSL2 mode, as defined in ITU-T G.993.2, which uses frequencies of up to 30 MHz to provide a downstream rate of 200 Mbps and an upstream rate of 100 Mbps.
SRA	Disabled by default. Enable SRA to disable seamless rate adaptation on the interface. SRA adjusts the line rate based on current line conditions.
Dialer Pool Member*	Enter the number of the dialer pool to which the interface belongs. Range: 1 through 255

- b. Configure ATM.

Table 115: ATM

Parameter Name	Description
ATM Sub Interface Name*	The ATM Sub interface name is auto populated based on the controller slot. Enter a value for the ATM sub interface, in the format <i>subslot/port</i> (for example ATM0/2/0.100). In this example, ".100" is the sub interface value.
Sub Interface Description	Enter a description for the interface.

Parameter Name	Description
VPI/VCI*	Create an ATM permanent virtual circuit (PVC), in the following format: <i>vpi/vci</i> Enter values for the virtual path identifier (VPI) and the virtual channel identifier (VCI).
Encapsulation	Select the encapsulation type to use on the ATM PVC from the drop-down list: • AAL5 NLPID: Use NLPID multiplexing. • AAL5 SNAP: Multiplex two or more protocols on the same PVC. • AAL5 MUX: Dedicate the PVC to a single protocol.
PVC Mode	
VBR-NRT	Configure variable bit rate non-real-time parameters: • Peak Cell Rate: Enter a value from 48 through 1015 Kbps. • Sustainable Cell Rate: Enter the sustainable cell rate, in Kbps. • Maximum Burst Size: This size can be 1 through 65535.
VBR-RT	Configure variable bit rate real-time parameters: • Peak Cell Rate: Enter a value from 48 through 25000 Kbps. • Average Cell Rate: Enter the average cell rate, in Kbps. • Maximum Burst Size: This size can be 1 through 65535.
None	Don't configure variable bit rate parameters

c. Configure PPP.

Table 116: PPP

Parameter Name	Description
PPP Authentication Protocol	Select the authentication protocol used by the MLP: • PAP: Enter the username and password that are provided by your ISP. <i>username</i> can be up to 254 characters. • CHAP: Enter the hostname and password provided by your Internet Service Provider (ISP). <i>hostname</i> can be up to 254 characters. • PAP and CHAP: Configure both authentication protocols. Enter the login credentials for each protocol.

Parameter Name	Description
Authentication Type	Select the type authentication from one of the following options.: • Unidirectional: Only the side receiving the call (NAS) authenticates the remote side (client). The remote client does not authenticate the server. • Bidirectional: Each side independently sends an Authenticate-Request (AUTH-REQ) and receives either an Authenticate-Acknowledge (AUTH-ACK) or Authenticate-Not Acknowledged (AUTH-NAK).
CHAP Hostname*	Enter the CHAP hostname.
CHAP Password*	Enter the CHAP password.
PAP Hostname*	Enter the PAP hostname.
PAP Password*	Enter the PAP password.

d. Configure Tunnel.

Table 117: Tunnel

Parameter Name	Description
Tunnel Interface	
Per Tunnel QoS	Enable per tunnel QoS and choose from the following values to configure hub-to-spoke network topologies: • Spoke • Hub If you select hub topology, the following option appears: • Bandwidth Percentage: Enter a value for the bandwidth percentage. Default: 50
Color	Choose a color for the TLOC.
Color Description	Minimum supported release: Cisco Catalyst SD-WAN Manager Release 20.18.1 Enter a description associated to the TLOC color.
Groups	Enter the list of groups in the field.
Exclude Controller Group List	Set the Cisco SD-WAN Controllers that the tunnel interface is not allowed to connect to. Range: 0 through 100

Parameter Name	Description
Maximum Control Connections	Specify the maximum number of Cisco SD-WAN Controllers that the WAN tunnel interface can connect to. To have the tunnel establish no control connections, set the number to 0. Range: 0 through 8 Default: 2
Cisco SD-WAN Manager Connection Preference	Set the preference for using a tunnel interface to exchange control traffic with Cisco SD-WAN Manager. Range: 0 through 8 Default: 5
Tunnel TCP MSS	TCP MSS affects any packet containing an initial TCP header that flows through the router. When configured, TCP MSS is examined against the MSS exchanged in the three-way handshake. The MSS in the header is lowered if the configured TCP MSS setting is lower than the MSS in the header. If the MSS header value is already lower than the TCP MSS, the packets flow through unmodified. The host at the end of the tunnel uses the lower setting of the two hosts. To configure TCP MSS, provide a value that is 40 bytes lower than the minimum path MTU. Specify the MSS of TPC SYN packets passing through the Cisco vEdge device. By default, the MSS is dynamically adjusted based on the interface or tunnel MTU such that TCP SYN packets are never fragmented. Range: 552 to 1460 bytes
Border	From the drop-down list, select Global . Click On to set TLOC as border TLOC.
Validator As Stun Server	Click On to enable Session Traversal Utilities for NAT (STUN) to allow the tunnel interface to discover its public IP address and port number when the router is located behind a NAT.
Full Port Hop	Minimum release: Cisco IOS XE Catalyst SD-WAN Release 17.18.1a Enable full port hopping at the TLOC level to allow devices to establish connections with controllers by switching to the next port if the current port is blocked or non-functional. Default: Disabled
Port Hop	From the drop-down list, select Global . Click Off to allow port hopping on tunnel interface. Default: On , which disallows port hopping on a tunnel interface Starting from Cisco IOS XE Catalyst SD-WAN Release 17.18.1a, this field is deprecated. Instead use the Full Port Hop option. See the Full Port Hop field.
Low-Bandwidth Link	Click On to set the tunnel interface as a low-bandwidth link. Default: Off

Parameter Name	Description
Clear-Dont-Fragment	Configure Clear-Dont-Fragment for packets that arrive at an interface that has Don't Fragment configured. If these packets are larger than what MTU allows, they are dropped. If you clear the Don't Fragment bit, the packets are fragmented and sent. Click On to clear the Dont Fragment bit in the IPv4 packet header for packets being transmitted out of the interface. When the Dont Fragment bit is cleared, the router fragments packets larger than the MTU of the interface before sending the packets. the router fragments packets larger than the MTU of the interface before sending the packets. Note Clear-Dont-Fragment clears the Dont Fragment bit and the Dont Fragment bit is set. For packets not requiring fragmentation, the Dont Fragment bit is not affected.
Network Broadcast	From the drop-down list, select Global. Click On to accept and respond to network-prefix-directed broadcasts. Enable this parameter only if the Directed Broadcast is enabled on the LAN interface feature template. Default: Off
Carrier	From the drop-down list, select Global and select the carrier name or private network identifier to associate with the tunnel. Values: carrier1, carrier2, carrier3, carrier4, carrier5, carrier6, carrier7, carrier8, default. Default: default
Bind Loopback Tunnel	Enter the name of a physical interface to bind to a loopback interface. The interface name has the following format: ge slot/port
NAT Refresh Interval	Set the interval between NAT refresh packets sent on a DTLS or TLS WAN transport connection. Range: 1 through 60 seconds Default: 5 seconds
Hello Interval	Enter the interval between Hello packets sent on a DTLS or TLS WAN transport connection. Range: 100 through 10000 milliseconds Default: 1000 milliseconds (1 second)
Hello Tolerance	Enter the time to wait for a Hello packet on a DTLS or TLS WAN transport connection before declaring that transport tunnel to be down. Range: 12 through 60 seconds Default: 12 seconds

Parameter Name	Description
Last Resort Circuit	Select to use the tunnel interface as the circuit of last resort. Note It is assumed that an interface configured as a circuit of last resort is unavailable and is skipped while calculating the number of control connections. As a result, the cellular modem becomes dormant, and no traffic is sent over the circuit. When the configurations are activated on the edge device with cellular interfaces, all the interfaces begin the process of establishing control and BFD connections. When one or more of the primary interfaces establishes a BFD connection, the circuit of last resort shuts itself down. If the primary interfaces lose their connections to remote edges, the circuit of last resort activates itself, triggering a BFD TLOC Down alarm and a Control TLOC Down alarm on the edge device. The last resort interfaces are a backup circuit on edge device and are activated when all other transport links BFD sessions fail. In this mode, the radio interface is turned off, and no control or data connections exist over the cellular interface.
Allow Services	Click On or Off for each service to enable or disable the service on the cellular interface.
Encapsulation	

Parameter Name	Description
Encapsulation	Enable at least one of the following encapsulation methods: • IPsec: Enter a value to set the preference for directing traffic to the tunnel. A higher value is preferred over a lower value. Range: 0 through 4294967295 Default: 0 • IPsec Preference: From the drop-down list, select Global and enter a value to set the preference for directing traffic to the tunnel. A higher value is preferred over a lower value. Range: 0 through 4294967295 Default: 0 • IPsec Weight: From the drop-down list, select Global and enter a value to set weight for balancing traffic across multiple TLOCs. A higher value sends more traffic to the tunnel. Range: 1 through 255 Default: 1 • GRE: Enter a value to set GRE preference for TLOC. Range: 0 through 4294967295 • GRE Preference: From the drop-down list, select Global and enter a value to set the preference for directing traffic to the tunnel. A higher value is preferred over a lower value. Range: 0 through 4294967295 Default: 0 • GRE Weight: From the drop-down list, select Global and enter a value to set weight for balancing traffic across multiple TLOCs. A higher value sends more traffic to the tunnel. Range: 1 through 255 Default: 1

e. Configure NAT.

Table 118: NAT

Parameter Name	Description
UDP Timeout (Minutes)	Specify when NAT translations over UDP sessions time out. Range: 1 through 8947 minutes Default: 1 minute

Parameter Name	Description
TCP Timeout (Minutes)	Specify when NAT translations over TCP sessions time out. Range: 1 through 8947 minutes Default: 60 minutes (1 hour)

f. Configure QoS.

Table 119: QoS

Parameter Name	Description
Adaptive QoS	Enter adaptive QoS parameters. You can leave the additional details at as default or specify your values. • Adapt Period (Minutes): Choose Global from the drop-down list, click On, and enter the period in minutes. • Shaping Rate Upstream: Choose Global from the drop-down list, click On, and enter the minimum, maximum, and default upstream bandwidth in Kbps. • Shaping Rate Downstream: Choose Global from the drop-down list, click On, and enter the minimum, maximum, downstream, and upstream bandwidth in Kbps.
Shaping Rate (kbps)	Choose Global from the drop-down list and configure the aggregate traffic transmission rate on the interface to be less than line rate, in kilobits per second (kbps). Range: 8 through 100000000

g. Configure ACL.

Table 120: ACL

Parameter Name	Description
IPv4 Ingress Access List	Enter the name of an IPv4 access list to packets being received on the interface.
IPv4 Egress Access List	Enter the name of an IPv4 access list to packets being transmitted on the interface.
IPv6 Ingress Access List	Enter the name of an IPv6 access list to packets being received on the interface.
IPv6 Egress Access List	Enter the name of an IPv6 access list to packets being transmitted on the interface.

h. Configure advanced parameters.

Table 121: Advanced

Parameter Name	Description
Shutdown	Click No to enable the interface.

Parameter Name	Description
Tracker / Tracker Group	Enter the name of a tracker or tracker group to track the status of transport interfaces that connect to the internet.
Service Provider	Specify the details of the service provider.
Bandwidth Upstream (Kbps)	Specify the bandwidth value to generate notifications when the bandwidth of traffic transmitted on a physical interface exceeds the value.
Bandwidth Downstream (Kbps)	Specify the bandwidth value to generate notifications when the bandwidth of traffic transmitted on a physical interface exceeds the value.
IP MTU	Enter the maximum MTU size of packets on the interface. Range: 576 through 1804 Default: 1500
TCP MSS	Enter the maximum segment size (MSS) of TCP SYN packets passing through the router. By default, the MSS is dynamically adjusted based on the interface or tunnel MTU such that TCP SYN packets are never fragmented. Range: 552 through 1460 bytes Default: 1500
TLOC Extension	Enter the name of a physical interface on the same router that connects to the WAN transport. This configuration binds the service-side interface to the WAN transport by enabling a device to access the opposite WAN transport connected to the neighbouring device using a TLOC-extension interface.
IP Directed Broadcast	From the drop-down list, select Global to enable IP Directed Broadcast. An IP directed broadcast is an IP packet whose destination address is a valid broadcast address for some IP subnet but which originates from a node that is not itself part of that destination subnet.

What to do next

Also see [Deploy a configuration group](#).

Configure DSL PPPoE Using a Configuration Group

Before you begin

On the **Configuration > Configuration Groups** page, choose **SD-WAN** as the solution type.

Procedure

Step 1 From the Cisco SD-WAN Manager menu, choose **Configuration > Configuration Groups**.

Step 2 Create and configure the DSL PPPoE parameters in a Transport and Management Profile.

- a. Configure basic configuration.

Table 122: Basic Configuration

Parameter Name	Description
Controller Slot*	Enter the slot number of the controller, in the following format: <i>slot/subslot/port</i> (for example, 0/2/0)
Controller Mode	Select the operating mode of the DSL controller from the drop-down list: • ADSL1: Use ITU G.992.1 Annex A full-rate mode, which provides a downstream rate of 1.3 Mbps and an upstream rate of 1.8 Mbps. • ADSL2: Use ITU G.992.3 Annex A, Annex L, and Annex M, which provides a downstream rate of 12 Mbps and an upstream rate of 1.3 Mbps. • ADSL2+: Use ITU G.992.5 Annex A and Annex M, which provides a downstream rate of 24 Mbps and an upstream rate of 3.3 Mbps. • ANSI: Operating in ADSL2/2+ mode, as defined in ITU G.991.1, G.992.3, and G.992.5, Annex A and Annex M, and in VDSL2 mode, as defined in ITU-T G.993.2. • VDSL2: Operate in VDSL2 mode, as defined in ITU-T G.993.2, which uses frequencies of up to 30 MHz to provide a downstream rate of 200 Mbps and an upstream rate of 100 Mbps.
SRA	Disabled by default. Enable SRA to disable seamless rate adaptation on the interface. SRA adjusts the line rate based on current line conditions.
Dialer Pool Member*	Enter the number of the dialer pool to which the interface belongs. Range: 1 through 255

- b. Configure Ethernet.

Table 123: Ethernet

Parameter Name	Description
Ethernet Interface Name*	Enter the name of an ethernet interface. For IOS XE routers, you must spell out the interface names completely (for example, GigabitEthernet0/0/0).
Description	Enter a description for the interface.
VLAN ID	Enter the VLAN identifier of the Ethernet interface.

c. Configure PPP.

Table 124: PPP

Parameter Name	Description
PPP Authentication Protocol	Select the authentication protocol used by the MLP: • PAP: Enter the username and password that are provided by your ISP. <i>username</i> can be up to 254 characters. • CHAP: Enter the hostname and password provided by your Internet Service Provider (ISP). <i>hostname</i> can be up to 254 characters. • PAP and CHAP: Configure both authentication protocols. Enter the login credentials for each protocol.
Authentication Type	Select the type authentication from one of the following options: • Unidirectional: Only the side receiving the call (NAS) authenticates the remote side (client). The remote client does not authenticate the server. • Bidirectional: Each side independently sends an Authenticate-Request (AUTH-REQ) and receives either an Authenticate-Acknowledge (AUTH-ACK) or Authenticate-Not Acknowledged (AUTH-NAK).
CHAP Hostname*	Enter the CHAP hostname.
CHAP Password*	Enter the CHAP password.
PAP Hostname*	Enter the PAP hostname.
PAP Password*	Enter the PAP password.

d. Configure Tunnel.

Table 125: Tunnel

Parameter Name	Description
Tunnel Interface	
Per Tunnel QoS	Enable per tunnel QoS and choose from the following values to configure hub-to-spoke network topologies: • Spoke • Hub
Color	Select a color for the TLOC.
Color Description	Minimum supported release: Cisco Catalyst SD-WAN Manager Release 20.18.1 Enter a description associated to the TLOC color.
Groups	Enter the list of groups in the field.

Parameter Name	Description
Exclude Controller Group List	Set the Cisco SD-WAN Controllers that the tunnel interface is not allowed to connect to. Range: 0 through 100
Maximum Control Connections	Specify the maximum number of Cisco SD-WAN Controllers that the WAN tunnel interface can connect to. To have the tunnel establish no control connections, set the number to 0. Range: 0 through 8 Default: 2
Cisco SD-WAN Manager Connection Preference	Set the preference for using a tunnel interface to exchange control traffic with Cisco SD-WAN Manager. Range: 0 through 8 Default: 5
Tunnel TCP MSS	TCP MSS affects any packet that contains an initial TCP header that flows through the router. When configured, TCP MSS is examined against the MSS exchanged in the three-way handshake. The MSS in the header is lowered if the configured TCP MSS setting is lower than the MSS in the header. If the MSS header value is already lower than the TCP MSS, the packets flow through unmodified. The host at the end of the tunnel uses the lower setting of the two hosts. To configure TCP MSS, provide a value that is 40 bytes lower than the minimum path MTU. Specify the MSS of TPC SYN packets passing through the Cisco vEdge device. By default, the MSS is dynamically adjusted based on the interface or tunnel MTU such that TCP SYN packets are never fragmented. Range: 552 through 1460 bytes Default: None
Border	From the drop-down list, select Global . Click On to set TLOC as border TLOC.
Validator As Stun Server	Click On to enable Session Traversal Utilities for NAT (STUN) to allow the tunnel interface to discover its public IP address and port number when the router is located behind a NAT.
Full Port Hop	Minimum release: Cisco IOS XE Catalyst SD-WAN Release 17.18.1a Enable full port hopping at the TLOC level to allow devices to establish connections with controllers by switching to the next port if the current port is blocked or non-functional. Default: Disabled
Port Hop	From the drop-down list, select Global . Click Off to allow port hopping on tunnel interface. Default: On , which disallows port hopping on tunnel interface. Starting from Cisco IOS XE Catalyst SD-WAN Release 17.18.1a, this field is deprecated. Instead use the Full Port Hop option. See the Full Port Hop field.
Low-Bandwidth Link	Click On to set the tunnel interface as a low-bandwidth link. Default: Off

Parameter Name	Description
Clear-Dont-Fragment	Configure Clear-Dont-Fragment for packets that arrive at an interface that has Don't Fragment configured. If these packets are larger than what MTU allows, they are dropped. If you clear the Don't Fragment bit, the packets are fragmented and sent. Click On to clear the Dont Fragment bit in the IPv4 packet header for packets being transmitted out of the interface. When the Dont Fragment bit is cleared, the router fragments packets larger than the MTU of the interface before sending the packets. Note Clear-Dont-Fragment clears the Dont Fragment bit and the Dont Fragment bit is set. For packets not requiring fragmentation, the Dont Fragment bit is not affected.
Network Broadcast	From the drop-down list, select Global. Click On to accept and respond to network-prefix-directed broadcasts. Enable this parameter only if the Directed Broadcast is enabled on the LAN interface feature template. Default: Off
Carrier	From the drop-down list, select Global and select the carrier name or private network identifier to associate with the tunnel. Values: carrier1, carrier2, carrier3, carrier4, carrier5, carrier6, carrier7, carrier8, default. Default: default
Bind Loopback Tunnel	Enter the name of a physical interface to bind to a loopback interface. The interface name has the following format: geslot/port
NAT Refresh Interval	Set the interval between NAT refresh packets sent on a DTLS or TLS WAN transport connection. Range: 1 through 60 seconds Default: 5 seconds
Hello Interval	Enter the interval between Hello packets sent on a DTLS or TLS WAN transport connection. Range: 100 through 10000 milliseconds Default: 1000 milliseconds (1 second)
Hello Tolerance	Enter the time to wait for a Hello packet on a DTLS or TLS WAN transport connection before declaring that transport tunnel to be down. Range: 12 through 60 seconds Default: 12 seconds

Parameter Name	Description
Last Resort Circuit	Select to use the tunnel interface as the circuit of last resort. Note It is assumed that an interface configured as a circuit of last resort is unavailable and is skipped while calculating the number of control connections. As a result, the cellular modem becomes dormant, and no traffic is sent over the circuit. When the configurations are activated on the edge device with cellular interfaces, all the interfaces begin the process of establishing control and BFD connections. When one or more of the primary interfaces establishes a BFD connection, the circuit of last resort shuts itself down. If the primary interfaces lose their connections to remote edges, the circuit of last resort activates itself, triggering a BFD TLOC Down alarm and a Control TLOC Down alarm on the edge device. The last resort interfaces are a backup circuit on edge device and are activated when all other transport links BFD sessions fail. In this mode, the radio interface is turned off, and no control or data connections exist over the cellular interface.
Allow Services	Click On or Off for each service to allow or disallow the service on the cellular interface.
Encapsulation	

Parameter Name	Description
Encapsulation	Enable atleast one of the following encapsulation methods: • IPsec: Enter a value to set the preference for directing traffic to the tunnel. A higher value is preferred over a lower value. Range: 0 through 4294967295 Default: 0 • IPsec Preference: From the drop-down list, select Global and enter a value to set the preference for directing traffic to the tunnel. A higher value is preferred over a lower value. Range: 0 through 4294967295 Default: 0 • IPsec Weight: From the drop-down list, select Global and enter a value to set weight for balancing traffic across multiple TLOCs. A higher value sends more traffic to the tunnel. Range: 1 through 255 Default: 1 • GRE: Enter a value to set GRE preference for TLOC. Range: 0 through 4294967295 • GRE Preference: From the drop-down list, select Global and enter a value to set the preference for directing traffic to the tunnel. A higher value is preferred over a lower value. Range: 0 through 4294967295 Default: 0 • GRE Weight: From the drop-down list, select Global and enter a value to set weight for balancing traffic across multiple TLOCs. A higher value sends more traffic to the tunnel. Range: 1 through 255 Default: 1

e. Configure NAT.

Table 126: NAT

Parameter Name	Description
UDP Timeout (Minutes)	Specify when NAT translations over UDP sessions time out. Range: 1 through 65536 minutes Default: 1 minute

Parameter Name	Description
TCP Timeout (Minutes)	Specify when NAT translations over TCP sessions time out. Range: 1 through 65536 minutes Default: 60 minutes (1 hour)

f. Configure QoS.

Table 127: QoS

Parameter Name	Description
Adaptive QoS	Enter adaptive QoS parameters. You can leave the additional details at as default or specify your values. • Adapt Period (Minutes): Choose Global from the drop-down list, click On, and enter the period in minutes. • Shaping Rate Upstream: Choose Global from the drop-down list, click On, and enter the minimum, maximum, and default upstream bandwidth in Kbps. • Shaping Rate Downstream: Choose Global from the drop-down list, click On, and enter the minimum, maximum, downstream, and upstream bandwidth in Kbps.
Shaping Rate (kbps)	Choose Global from the drop-down list and configure the aggregate traffic transmission rate on the interface to be less than line rate, in kilobits per second (kbps). Range: 8 through 100000000

g. Configure ACL.

Table 128: ACL

Parameter Name	Description
IPv4 Ingress Access List	Enter the name of an IPv4 access list to packets being received on the interface.
IPv4 Egress Access List	Enter the name of an IPv4 access list to packets being transmitted on the interface.
IPv6 Ingress Access List	Enter the name of an IPv6 access list to packets being received on the interface.
IPv6 Egress Access List	Enter the name of an IPv6 access list to packets being transmitted on the interface.

h. Configure advanced parameters.

Table 129: Advanced

Parameter Name	Description
Shutdown	Click No to enable the interface.

Parameter Name	Description
Tracker / Tracker Group	Enter the name of a tracker or tracker group to track the status of transport interfaces that connect to the internet.
PPP Maximum Payload	Enter the maximum receive unit (MRU) value to be negotiated during PPP-over-Ethernet negotiation. Range: 64 through 1792 bytes
Service Provider	Specify the details of the service provider.
Bandwidth Upstream (Kbps)	Specify the bandwidth value to generate notifications when the bandwidth of traffic transmitted on a physical interface exceeds the value.
Bandwidth Downstream (Kbps)	Specify the bandwidth value to generate notifications when the bandwidth of traffic transmitted on a physical interface exceeds the value.
IP MTU	Enter the maximum MTU size of packets on the interface. Range: 576 through 1804 Default: 1500.
TCP MSS	Enter the maximum segment size (MSS) of TCP SYN packets passing through the router. By default, the MSS is dynamically adjusted based on the interface or tunnel MTU such that TCP SYN packets are never fragmented. Range: 552 through 1460 bytes Default: 1500
TLOC Extension	Enter the name of a physical interface on the same router that connects to the WAN transport. This configuration binds the service-side interface to the WAN transport by enabling a device to access the opposite WAN transport connected to the neighbouring device using a TLOC-extension interface.
IP Directed Broadcast	From the drop-down list, select Global to enable IP Directed Broadcast. An IP directed broadcast is an IP packet whose destination address is a valid broadcast address for some IP subnet but which originates from a node that is not itself part of that destination subnet.
Tracker / Tracker Group	Enter the name of a tracker or tracker group to track the status of transport interfaces that connect to the internet.

What to do next

Also see [Deploy a configuration group](#).

Configure VPN Interface Ethernet PPPoE Using a Configuration Group

Before you begin

On the **Configuration** > **Configuration Groups** page, choose **SD-WAN** as the solution type.

Procedure

Step 1 From the Cisco SD-WAN Manager menu, choose **Configuration** > **Configuration Groups**.

Step 2 Create and configure a Ethernet PPPoE feature under Transport VPN in a Transport and Management profile.

a) Configure basic PPPoE functionality.

Parameter Name	Description
Ethernet Interface Name *	Enter the name of an ethernet interface. For IOS XE routers, you must spell out the interface names completely (for example, GigabitEthernet0/0/0).
Description	Enter a description for the ethernet interface.
VLAN ID	Enter the VLAN identifier of the Ethernet interface.
Dialer Pool Member *	Enter the number of the dialer pool to which the interface belongs. Range: 1 through 255

b) Configure the PPP Authentication Protocol.

Parameter Name	Description
PPP Authentication Protocol*	Select the authentication protocol used by the MLP: • PAP: Enter the username and password that are provided by your ISP. <i>username</i> can be up to 254 characters. • CHAP: Enter the hostname and password provided by your Internet Service Provider (ISP). <i>hostname</i> can be up to 254 characters. • PAP and CHAP: Configure both authentication protocols. Enter the login credentials for each protocol.

Parameter Name	Description
Authentication Type	Select the type authentication from one of the following options.: • Unidirectional: Only the side receiving the call (NAS) authenticates the remote side (client). The remote client does not authenticate the server. • Bidirectional: Each side independently sends an Authenticate-Request (AUTH-REQ) and receives either an Authenticate-Acknowledge (AUTH-ACK) or Authenticate-Not Acknowledged (AUTH-NAK).
CHAP Hostname*	Enter the CHAP hostname.
CHAP Password*	Enter the CHAP password.
PAP Hostname*	Enter the PAP hostname.
PAP Password*	Enter the PAP password.

c) Configure a tunnel interface for the multilink interface.

Parameter Name	Description
Tunnel Interface	
Per Tunnel QoS	Enable per tunnel QoS and choose Spoke to configure the spoke network topology
Color	Select a color for the TLOC.
Color Description	Minimum supported release: Cisco Catalyst SD-WAN Manager Release 20.18.1 Enter a description associated to the TLOC color.
Groups	Enter the list of groups in the field.
Exclude Controller Group List	Set the Cisco SD-WAN Controllers that the tunnel interface is not allowed to connect to. Range: 0 through 100
Maximum Control Connections	Specify the maximum number of Cisco SD-WAN Controllers that the WAN tunnel interface can connect to. To have the tunnel establish no control connections, set the number to 0. Range: 0 through 8
Cisco SD-WAN Manager Connection Preference	Set the preference for using a tunnel interface to exchange control traffic with Cisco SD-WAN Manager. Range: 0 through 8 Default: 5

Parameter Name	Description
Tunnel TCP MSS	TCP MSS affects any packet that contains an initial TCP header that flows through the router. When configured, TCP MSS is examined against the MSS exchanged in the three-way handshake. The MSS in the header is lowered if the configured TCP MSS setting is lower than the MSS in the header. If the MSS header value is already lower than the TCP MSS, the packets flow through unmodified. The host at the end of the tunnel uses the lower setting of the two hosts. To configure TCP MSS, provide a value that is 40 bytes lower than the minimum path MTU. Specify the MSS of TPC SYN packets passing through the Cisco vEdge device. By default, the MSS is dynamically adjusted based on the interface or tunnel MTU such that TCP SYN packets are never fragmented. Range: 552 through 1460 bytes Default: None
Border	From the drop-down list, select Global . Click On to set TLOC as border TLOC.
Validator As Stun Server	Click On to enable Session Traversal Utilities for NAT (STUN) to allow the tunnel interface to discover its public IP address and port number when the router is located behind a NAT.
Full Port Hop	Minimum release: Cisco IOS XE Catalyst SD-WAN Release 17.18.1a Enable full port hopping at the TLOC level to allow devices to establish connections with controllers by switching to the next port if the current port is blocked or non-functional. Default: Disabled
Port Hop	From the drop-down list, select Global. Click Off to allow port hopping on tunnel interface. Default: On, which disallows port hopping on tunnel interface. Starting from Cisco IOS XE Catalyst SD-WAN Release 17.18.1a, this field is deprecated. Instead use the Full Port Hop option. See the Full Port Hop field.
Low-Bandwidth Link	Click On to set the tunnel interface as a low-bandwidth link. Default: Off
Clear-Dont-Fragment	Configure Clear-Dont-Fragment for packets that arrive at an interface that has Don't Fragment configured. If these packets are larger than what MTU allows, they are dropped. If you clear the Don't Fragment bit, the packets are fragmented and sent. Click On to clear the Dont Fragment bit in the IPv4 packet header for packets being transmitted out of the interface. When the Dont Fragment bit is cleared, the router fragments packets larger than the MTU of the interface before sending the packets. Note Clear-Dont-Fragment clears the Dont Fragment bit and the Dont Fragment bit is set. For packets not requiring fragmentation, the Dont Fragment bit is not affected.
Network Broadcast	From the drop-down list, select Global. Click On to accept and respond to network-prefix-directed broadcasts. Enable this parameter only if the Directed Broadcast is enabled on the LAN interface feature template. Default: Off

Parameter Name	Description
Carrier	From the drop-down list, select Global and select the carrier name or private network identifier to associate with the tunnel. Values: carrier1, carrier2, carrier3, carrier4, carrier5, carrier6, carrier7, carrier8, default. Default: default
Bind Loopback Tunnel	Enter the name of a physical interface to bind to a loopback interface. The interface name has the following format: ge slot/port
NAT Refresh Interval	Set the interval between NAT refresh packets sent on a DTLS or TLS WAN transport connection. Range: 1 through 60 seconds Default: 5 seconds
Hello Interval	Enter the interval between Hello packets sent on a DTLS or TLS WAN transport connection. Range: 100 through 10000 milliseconds Default: 1000 milliseconds (1 second)
Hello Tolerance	Enter the time to wait for a Hello packet on a DTLS or TLS WAN transport connection before declaring that transport tunnel to be down. Range: 12 through 60 seconds Default: 12 seconds
Last Resort Circuit	Select to use the tunnel interface as the circuit of last resort. Note It is assumed that an interface configured as a circuit of last resort is unavailable and is skipped while calculating the number of control connections. As a result, the cellular modem becomes dormant, and no traffic is sent over the circuit. When the configurations are activated on the edge device with cellular interfaces, all the interfaces begin the process of establishing control and BFD connections. When one or more of the primary interfaces establishes a BFD connection, the circuit of last resort shuts itself down. If the primary interfaces lose their connections to remote edges, the circuit of last resort activates itself, triggering a BFD TLOC Down alarm and a Control TLOC Down alarm on the edge device. The last resort interfaces are a backup circuit on edge device and are activated when all other transport links BFD sessions fail. In this mode, the radio interface is turned off, and no control or data connections exist over the cellular interface.
Allow Services	Click On or Off for each service to allow or disallow the service on the cellular interface.
Encapsulation	

Parameter Name	Description
Encapsulation	Enable at least one of the following encapsulation methods: IPsec: Enter a value to set the preference for directing traffic to the tunnel. A higher value is preferred over a lower value. Range: 0 through 4294967295 Default: 0 IPsec Preference: From the drop-down list, select Global and enter a value to set the preference for directing traffic to the tunnel. A higher value is preferred over a lower value. Range: 0 through 4294967295 Default: 0 IPsec Weight: From the drop-down list, select Global and enter a value to set weight for balancing traffic across multiple TLOCs. A higher value sends more traffic to the tunnel. Range: 1 through 255 Default: 1 GRE: Enter a value to set GRE preference for TLOC. Range: 0 through 4294967295 GRE Preference: From the drop-down list, select Global and enter a value to set the preference for directing traffic to the tunnel. A higher value is preferred over a lower value. Range: 0 through 4294967295 Default: 0 GRE Weight: From the drop-down list, select Global and enter a value to set weight for balancing traffic across multiple TLOCs. A higher value sends more traffic to the tunnel. Range: 1 through 255 Default: 1

d) Configure an interface to act as a NAT device for applications such as port forwarding.

Parameter Name	Description
UDP Timeout (Minutes)	Specify when NAT translations over UDP sessions time out. Range: 1 through 8947 minutes Default: 1 minute

Parameter Name	Description
TCP Timeout (Minutes)	Specify when NAT translations over TCP sessions time out. Range: 1 through 8947 minutes Default: 60 minutes (1 hour)

e) Configure QoS.

Parameter Name	Description
Adaptive QoS	Enter adaptive QoS parameters. You can leave the additional details at as default or specify your values. • Adapt Period (Minutes): Choose Global from the drop-down list, click On, and enter the period in minutes. • Shaping Rate Upstream: Choose Global from the drop-down list, click On, and enter the minimum, maximum, and default upstream bandwidth in Kbps. • Shaping Rate Downstream: Choose Global from the drop-down list, click On, and enter the minimum, maximum, downstream, and upstream bandwidth in Kbps.
Shaping Rate (kbps)	Choose Global from the drop-down list and configure the aggregate traffic transmission rate on the interface to be less than line rate, in kilobits per second (kbps). Range: 8 through 100000000

f) Configure ACL.

Parameter Name	Description
IPv4 Ingress Access List	Enter the name of an IPv4 access list to packets being received on the interface.
IPv4 Egress Access List	Enter the name of an IPv4 access list to packets being transmitted on the interface.
IPv6 Ingress Access List	Enter the name of an IPv6 access list to packets being received on the interface.
IPv6 Egress Access List	Enter the name of an IPv6 access list to packets being transmitted on the interface.

g) Configure additional tunnel interface parameters.

Parameter Name	Description
Shutdown	Choose No to enable the interface.
Tracker / Tracker Group	Enter the name of a tracker or tracker group to track the status of transport interfaces that connect to the internet.
Maximum Payload	Enter the maximum receive unit (MRU) value to be negotiated during PPP-over-Ethernet negotiation. Range: 64 through 1792 bytes

Parameter Name	Description
IP MTU	Enter the maximum MTU size of packets on the interface. Range: 576 through 1804 Default: 1500
TCP MSS	Enter the maximum segment size (MSS) of TCP SYN packets passing through the router. By default, the MSS is dynamically adjusted based on the interface or tunnel MTU such that TCP SYN packets are never fragmented. Range: 552 through 1460 bytes Default: 1500
TLOC Extension	Enter the name of a physical interface on the same router that connects to the WAN transport. This configuration binds the service-side interface to the WAN transport by enabling a device to access the opposite WAN transport connected to the neighbouring device using a TLOC-extension interface.
IP Directed Broadcast	From the drop-down list, select Global to enable IP Directed Broadcast. An IP directed broadcast is an IP packet whose destination address is a valid broadcast address for some IP subnet but which originates from a node that is not itself part of that destination subnet.
Tracker / Tracker Group	Enter the name of a tracker or tracker group to track the status of transport interfaces that connect to the internet.

What to do next

Also see [Deploy a configuration group](#).

VPN Interface Ethernet PPPoE

Use the PPPoE template for Cisco IOS XE Catalyst SD-WAN devices.

You configure PPPoE over GigabitEthernet interfaces on Cisco IOS XE routers, to provide PPPoE client support.

To configure interfaces on Cisco routers using Cisco SD-WAN Manager templates:

1. Create a VPN Interface Ethernet PPPoE feature template to configure Ethernet PPPoE interface parameters, as described in this section.
2. Create a VPN feature template to configure VPN parameters. See VPN help topic.

Navigate to the Template Screen and Name the Template

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Device Templates**, and click **Create Template**.



Note In Cisco vManage Release 20.7.x and earlier releases, **Device Templates** is titled **Device**.

3. From the **Create Template** drop-down list, choose **From Feature Template**.
4. From the **Device Model** drop-down list, select the type of device for which you are creating the template.
5. Click **Transport & Management VPN** or scroll to the **Transport & Management VPN** section.
6. Under **Additional VPN 0 Templates**, click **VPN Interface Ethernet PPPoE**.
7. From the **VPN Interface Ethernet PPPoE** drop-down list, click **Create Template**. The VPN Interface Ethernet PPPoE template form is displayed.

This form contains fields for naming the template, and fields for defining the Ethernet PPPoE parameters.



8. In **Template Name**, enter a name for the template.

The name can be up to 128 characters and can contain only alphanumeric characters.

9. In **Template Description**, enter a description of the template.

The description can be up to 2048 characters and can contain only alphanumeric characters.

When you first open a feature template, for each parameter that has a default value, the scope is set to Default (indicated by a check mark), and the default setting or value is shown. To change the default or to enter a value, click the **Scope** drop-down list and select one of the following:

Table 130:

Parameter Scope	Scope Description
Device Specific (indicated by a host icon)	Use a device-specific value for the parameter. For device-specific parameters, you cannot enter a value in the feature template. You enter the value when you attach a Cisco Catalyst SD-WAN device to a device template . When you click Device Specific, the Enter Key box opens. This box displays a key, which is a unique string that identifies the parameter in a CSV file that you create. This file is an Excel spreadsheet that contains one column for each key. The header row contains the key names (one key per column), and each row after that corresponds to a device and defines the values of the keys for that device. You upload the CSV file when you attach a Cisco Catalyst SD-WAN device to a device template. For more information, see Create a Template Variables Spreadsheet . To change the default key, type a new string and move the cursor out of the Enter Key box. Examples of device-specific parameters are system IP address, hostname, GPS location, and site ID.
Global (indicated by a globe icon)	Enter a value for the parameter, and apply that value to all devices. Examples of parameters that you might apply globally to a group of devices are DNS server, syslog server, and interface MTUs.

Configure PPPoE Functionality

To configure basic PPPoE functionality, click **Basic Configuration** and configure the following parameters. Required parameters are indicated with an asterisk.

Table 131:

Parameter Name	Description
Shutdown*	Click No to enable the GigabitEthernet interface.
Ethernet Interface Name	Enter the name of a GigabitEthernet interface. For IOS XE routers, you must spell out the interface names completely (for example, GigabitEthernet0/0/0).
VLAN ID	VLAN tag of the sub-interface.
Description	Enter a description of the Ethernet-PPPoE-enabled interface.
Dialer Pool Member	Enter the number of the dialer pool to which the interface belongs. <i>Range:</i> 100 to 255.
PPP Maximum Payload	Enter the maximum receive unit (MRU) value to be negotiated during PPP Link Control Protocol (LCP) negotiation. <i>Range:</i> 64 through 1792 bytes

To save the feature template, click **Save**.

Configure the PPP Authentication Protocol

To configure the PPP Authentication Protocol, click **PPP** and configure the following parameters. Required parameters are indicated with an asterisk.

Table 132:

Parameter Name	Description
PPP Authentication Protocol	Select the authentication protocol used by the MLP: • CHAP—Enter the hostname and password provided by your Internet Service Provider (ISP). <i>hostname</i> can be up to 254 characters. • PAP—Enter the username and password provided by your ISP. <i>username</i> can be up to 254 characters. • PAP and CHAP—Configure both authentication protocols. Enter the login credentials for each protocol. To use the same username and password for both, click Same Credentials for PAP and CHAP.

To save the feature template, click **Save**.

Create a Tunnel Interface

On IOS XE routers, you can configure up to eight tunnel interfaces. This means that each router can have up to eight TLOCs.

For the control plane to establish itself so that the overlay network can function, you must configure WAN transport interfaces in VPN 0.

To configure a tunnel interface for the multilink interface, select **Tunnel Interface** and configure the following parameters:

Table 133:

Parameter Name	Description
Tunnel Interface	Click On to create a tunnel interface.
Color	Select a color for the TLOC.
Color Description	Minimum supported release: Cisco Catalyst SD-WAN Manager Release 20.18.1 Enter a description associated to the TLOC color.
Control Connection	By default, Control Connection is set to On, which establishes a control connection for the TLOC. If the router has multiple TLOCs, click No to have the tunnel not establish control connection for the TLOC. Note We recommend a minimum of 650-700 Kbps bandwidth with default 1 sec hello-interval and 12 sec hello-tolerance parameters configured to avoid any data/packet loss in connection traffic. For each BFD session, an additional average sized BFD packet of 175 Bytes consumes 1.4 Kbps of bandwidth. A sample calculation of the required bandwidth for bidirectional BFD packet flow is given below: • 650 – 700 Kbps per device for control connections. • 175 Bytes (or 1.4 Kbps) per BFD session on the device (request) • 175 Bytes (or 1.4 Kbps) per BFD session on the device (response) If the path MTU discovery (PMTUD) is enabled, bandwidth for send/receive BFD packets per tunnel for every 30 secs: A 1500 Bytes BFD request packet is sent per tunnel every 30 secs: $1500 \text{ Bytes} * 8 \text{ bits/1 byte} * 1 \text{ packet} / 30 \text{ secs} = 400 \text{ bps (request)}$ A 147 Bytes BFD packet is sent in response: $147 \text{ Bytes} * 8 \text{ bits/1 byte} * 1 \text{ packet} / 30 \text{ secs} = 40 \text{ bps (response)}$ Therefore, a device with 775 BFD sessions (for example) requires a bandwidth of: $700k + (1.4k*775) + (400 *775) + (1.4k*775) + (40 *775) = \sim 3,5 \text{ MBps}$

Parameter Name	Description
Maximum Control Connections	Specify the maximum number of Cisco Catalyst SD-WAN Controllers that the WAN tunnel interface can connect to. To have the tunnel establish no control connections, set the number to 0. <i>Range: 0 through 8 Default: 2</i>
Cisco Catalyst SD-WAN Validator As STUN Server	Click On to enable Session Traversal Utilities for NAT (STUN) to allow the tunnel interface to discover its public IP address and port number when the router is located behind a NAT.
Exclude Controller Group List	Set the Cisco Catalyst SD-WAN Controllers that the tunnel interface is not allowed to connect to. <i>Range: 0 through 100</i>
Cisco SD-WAN Manager Connection Preference	Set the preference for using a tunnel interface to exchange control traffic with the Cisco SD-WAN Manager NMS. <i>Range: 0 through 8 Default: 5</i>
Full Port Hop	Minimum release: Cisco IOS XE Catalyst SD-WAN Release 17.18.1a Enable full port hopping at the TLOC level to allow devices to establish connections with controllers by switching to the next port if the current port is blocked or non-functional. Default: Disabled
Port Hop	Click On to enable port hopping, or click Off to disable it. When a router is behind a NAT, port hopping rotates through a pool of preselected OMP port numbers (called base ports) to establish DTLS connections with other routers when a connection attempt is unsuccessful. The default base ports are 12346, 12366, 12386, 12406, and 12426. To modify the base ports, set a port offset value. <i>Default: Enabled</i> Starting from Cisco IOS XE Catalyst SD-WAN Release 17.18.1a, this field is deprecated. Instead use the Full Port Hop option. See the Full Port Hop field.
Low-Bandwidth Link	Select to characterize the tunnel interface as a low-bandwidth link.
Allow Service	Select On or Off for each service to allow or disallow the service on the interface.

To configure additional tunnel interface parameters, click **Advanced Options** and configure the following parameters:

Table 134:

Parameter Name	Description
GRE	Use GRE encapsulation on the tunnel interface. By default, GRE is disabled. If you select both IPsec and GRE encapsulations, two TLOCs are created for the tunnel interface that have the same IP addresses and colors, but that differ by their encapsulation.

Parameter Name	Description
IPsec	Use IPsec encapsulation on the tunnel interface. By default, IPsec is enabled. If you select both IPsec and GRE encapsulations, two TLOCs are created for the tunnel interface that have the same IP addresses and colors, but that differ by their encapsulation.
IPsec Preference	Specify a preference value for directing traffic to the tunnel. A higher value is preferred over a lower value. <i>Range:</i> 0 through 4294967295. <i>Default:</i> 0
IPsec Weight	Enter a weight to use to balance traffic across multiple TLOCs. A higher value sends more traffic to the tunnel. <i>Range:</i> 1 through 255. <i>Default:</i> 1
Carrier	Select the carrier name or private network identifier to associate with the tunnel. <i>Values:</i> carrier1, carrier2, carrier3, carrier4, carrier5, carrier6, carrier7, carrier8, default. <i>Default:</i> default
Bind Loopback Tunnel	Enter the name of a physical interface to bind to a loopback interface.
Last-Resort Circuit	Select to use the tunnel interface as the circuit of last resort. Note An interface configured as a circuit of last resort is expected to be down and is skipped while calculating the number of control connections, the cellular modem becomes dormant, and no traffic is sent over the circuit. When the configurations are activated on the edge device with cellular interfaces, then all the interfaces begin the process of establishing control and BFD connections. When one or more of the primary interfaces establishes a BFD connection, the circuit of last resort shuts itself down. Only when all the primary interfaces lose their connections to remote edges, then the circuit of last resort activates itself triggering a BFD TLOC Down alarm and a Control TLOC Down alarm on the edge device. The last resort interfaces are used as backup circuit on edge device and are activated when all other transport links BFD sessions fail. In this mode the radio interface is turned off, and no control or data connections exist over the cellular interface. Note Configuring administrative distance values on primary interface routes is not supported.
NAT Refresh Interval	Enter the interval between NAT refresh packets sent on a DTLS or TLS WAN transport connection. <i>Range:</i> 1 through 60 seconds. <i>Default:</i> 5 seconds
Hello Interval	Enter the interval between Hello packets sent on a DTLS or TLS WAN transport connection. <i>Range:</i> 100 through 10000 milliseconds. <i>Default:</i> 1000 milliseconds (1 second)

Parameter Name	Description
Hello Tolerance	Enter the time to wait for a Hello packet on a DTLS or TLS WAN transport connection before declaring that transport tunnel to be down. <i>Range:</i> 12 through 60 seconds. <i>Default:</i> 12 seconds

Configure the Interface as a NAT Device

To configure an interface to act as a NAT device for applications such as port forwarding, select **NAT**, click **On** and configure the following parameters:

Table 135:

Parameter Name	Description
NAT	Click On to have the interface act as a NAT device.
Refresh Mode	Select how NAT mappings are refreshed, either outbound or bidirectional (outbound and inbound). <i>Default:</i> Outbound
UDP Timeout	Specify when NAT translations over UDP sessions time out. <i>Range:</i> 1 through 65536 minutes. <i>Default:</i> 1 minutes
TCP Timeout	Specify when NAT translations over TCP sessions time out. <i>Range:</i> 1 through 65536 minutes. <i>Default:</i> 60 minutes (1 hour)
Block ICMP	Select On to block inbound ICMP error messages. By default, a router acting as a NAT device receives these error messages. <i>Default:</i> Off
Respond to Ping	Select On to have the router respond to ping requests to the NAT interface's IP address that are received from the public side of the connection.

To create a port forwarding rule, click **Add New Port Forwarding Rule** and configure the following parameters. You can define up to 128 port-forwarding rules to allow requests from an external network to reach devices on the internal network.

Table 136:

Parameter Name	Description
Port Start Range	Enter a port number to define the port or first port in the range of interest. <i>Range:</i> 0 through 65535
Port End Range	Enter the same port number to apply port forwarding to a single port, or enter a larger number to apply it to a range of ports. <i>Range:</i> 0 through 65535
Protocol	Select the protocol to which to apply the port-forwarding rule, either TCP or UDP. To match the same ports for both TCP and UDP traffic, configure two rules.
VPN	Specify the private VPN in which the internal server resides. This VPN is one of the VPN identifiers in the overlay network. <i>Range:</i> 0 through 65527

Parameter Name	Description
Private IP	Specify the IP address of the internal server to which to direct traffic that matches the port-forwarding rule.

To save a port forwarding rule, click **Add**.

To save the feature template, click **Save**.

Apply Access Lists

To apply a rewrite rule, access lists, and policers to a router interface, click **ACL** and configure the following parameters:

Table 137:

Parameter Name	Description
Shaping rate	Configure the aggregate traffic transmission rate on the interface to be less than line rate, in kilobits per second (kbps).
QoS map	Specify the name of the QoS map to apply to packets being transmitted out the interface.
Rewrite Rule	Click On , and specify the name of the rewrite rule to apply on the interface.
Ingress ACL – IPv4	Click On , and specify the name of the access list to apply to IPv4 packets being received on the interface.
Egress ACL – IPv4	Click On , and specify the name of the access list to apply to IPv4 packets being transmitted on the interface.
Ingress ACL – IPv6	Click On , and specify the name of the access list to apply to IPv6 packets being received on the interface.
Egress ACL – IPv6	Click On , and specify the name of the access list to apply to IPv6 packets being transmitted on the interface.
Ingress Policer	Click On , and specify the name of the policer to apply to packets being received on the interface.
Egress Policer	Click On , and specify the name of the policer to apply to packets being transmitted on the interface.

To save the feature template, click **Save**.

Configure Other Interface Properties

To configure other interface properties, click **Advanced** and configure the following properties:

Table 138:

Parameter Name	Description
Bandwidth Upstream	For transmitted traffic, set the bandwidth above which to generate notifications. <i>Range:</i> 1 through $(2^{32} / 2) - 1$ kbps
Bandwidth Downstream	For received traffic, set the bandwidth above which to generate notifications. <i>Range:</i> 1 through $(2^{32} / 2) - 1$ kbps
IP MTU	Specify the maximum MTU size of packets on the interface. <i>Range:</i> 576 through 1804. <i>Default:</i> 1500 bytes
TCP MSS	Specify the maximum segment size (MSS) of TCP SYN packets passing through the router. By default, the MSS is dynamically adjusted based on the interface or tunnel MTU such that TCP SYN packets are never fragmented. <i>Range:</i> 552 to 1460 bytes. <i>Default:</i> None
TLOC Extension	Enter the name of the physical interface on the same router that connects to the WAN transport circuit. This configuration then binds this service-side interface to the WAN transport. A second router at the same site that itself has no direct connection to the WAN (generally because the site has only a single WAN connection) and that connects to this service-side interface is then provided with a connection to the WAN.
Tracker	Enter the name of a tracker to track the status of transport interfaces that connect to the internet.
IP Directed-Broadcast	Enables translation of a directed broadcast to physical broadcasts. An IP directed broadcast is an IP packet whose destination address is a valid broadcast address for some IP subnet but which originates from a node that is not itself part of that destination subnet.

To save the feature template, click **Save**.

Release Information

Introduced in Cisco SD-WAN Manager NMS in Release 18.4.1.

VPN Interface GRE

When a service, such as a firewall, is available on a device that supports only GRE tunnels, you can configure a GRE tunnel on the device to connect to the remote device by configuring a logical GRE interface. You then advertise that the service is available via a GRE tunnel, and you can create data policies to direct the appropriate traffic to the tunnel. GRE interfaces come up as soon as they are configured, and they stay up as long as the physical tunnel interface is up.

To configure GRE interfaces using Cisco SD-WAN Manager templates:

1. Create a VPN Interface GRE feature template to configure a GRE interface.
2. Create a VPN feature template to advertise a service that is reachable via a GRE tunnel, to configure GRE-specific static routes, and to configure other VPN parameters.

3. Create a data policy on the Cisco Catalyst SD-WAN Controller that applies to the service VPN, including a **set-service service-name local** command.

Navigate to the Template Screen and Name the Template

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Device Templates**, and click **Create Template**.



Note In Cisco vManage Release 20.7.x and earlier releases, **Device Templates** is titled **Device**.

3. From the **Create Template** drop-down list, select **From Feature Template**.
4. From the **Device Model** drop-down list, select the type of device for which you are creating the template.
5. To create a template for VPN 0 or VPN 512:
 - a. Click **Transport & Management VPN** or scroll to the **Transport & Management VPN** section.
 - b. Under **Additional VPN 0 Templates**, click **VPN Interface GRE**.
 - c. From the **VPN Interface GRE** drop-down list, click **Create Template**. The VPN Interface GRE template form is displayed.

This form contains fields for naming the template, and fields for defining the VPN Interface GRE parameters.
6. To create a template for VPNs 1 through 511, and 513 through 65530:
 - a. Click **Service VPN** or scroll to the **Service VPN** section.
 - b. Click the **Service VPN** drop-down list.
 - c. Under **Additional VPN templates**, click **VPN Interface GRE**.
 - d. From the **VPN Interface GRE** drop-down list, click **Create Template**. The VPN Interface GRE template form is displayed.

This form contains fields for naming the template, and fields for defining the VPN Interface GRE parameters.
7. In **Template Name**, enter a name for the template. The name can be up to 128 characters and can contain only alphanumeric characters.
8. In **Template Description**, enter a description of the template. The description can be up to 2048 characters and can contain only alphanumeric characters.

When you first open a feature template, for each parameter that has a default value, the scope is set to Default (indicated by a check mark), and the default setting or value is shown. To change the default or to enter a value, click the scope drop-down to the left of the parameter field and select the parameter scope.

Configuring a Basic GRE Interface

To configure a basic GRE interface, click **Basic Configuration** and then configure the following parameters. Parameters marked with an asterisk are required to configure a GRE interface.

Table 139:

Parameter Name	Description
Shutdown*	Click Off to enable the interface.
Interface Name*	Enter the name of the GRE interface, in the format gre number . <i>number</i> can be from 1 through 255.
Description	Enter a description of the GRE interface.
Source*	Enter the source of the GRE interface: - GRE Source IP Address—Enter the source IP address of the GRE tunnel interface. This address is on the local router. This address is on the local router. GRE keepalives can not be configured when source configured as IP address. - Tunnel Source Interface—Enter the physical interface that is the source of the GRE tunnel. GRE keepalives can not be configured when source configured as loopback interface. - If you selected the Source as Interface, enter the name of the source interface. If you enter a loopback interface, an additional field Tunnel Route-via Interface displays where you enter the egress interface name.
Destination*	Enter the destination IP address of the GRE tunnel interface. This address is on a remote device. If this tunnel connects to a Secure Internet Gateway (SIG), specify the URL for the SIG.
GRE Destination IP Address*	Enter the destination IP address of the GRE tunnel interface. This address is on a remote device
IPv4 Address	Enter an IPv4 address for the GRE tunnel.
IP MTU	Specify the maximum MTU size of packets on the interface. <i>Range: 576 through 1804 Default: 1500 bytes</i>
Clear-Dont-Fragment	Click On to clear the Don't Fragment bit in the IPv4 packet header for packets being transmitted out the interface.
TCP MSS	Specify the maximum segment size (MSS) of TCP SYN packets passing through the Cisco vEdge device. By default, the MSS is dynamically adjusted based on the interface or tunnel MTU such that TCP SYN packets are never fragmented. <i>Range: 552 to 1460 bytes Default: None</i>

To save the feature template, click **Save**.

CLI equivalent:

```

vpn vpn-id interface grenumber clear-dont-fragment description text
ip address ipv4-prefix/length keepalive seconds retries mtu bytes
policer policer-name (in | out)
    qos-map name rewrite-rule name shaping-rate name
    [no] shutdown tcp-mss-adjust bytes tunnel-destination ip-address
    ( tunnel-source ip-address | tunnel-source-interface interface-name)

```

Configure Interface Access Lists

To configure access lists on a GRE interface, click **ACL** and configure the following parameters:

Table 140:

Parameter Name	Description
Rewrite Rule	Click On , and specify the name of the rewrite rule to apply on the interface.
Ingress ACL – IPv4	Click On , and specify the name of the access list to apply to IPv4 packets being received on the interface.
Egress ACL – IPv4	Click On , and specify the name of the access list to apply to IPv4 packets being transmitted on the interface.
Ingress Policer	Click On , and specify the name of the policer to apply to packets being received on the interface.
Egress Policer	Click On , and specify the name of the policer to apply to packets being transmitted on the interface.

CLI equivalent:

```

vpn vpn-id interface grenumber access-list acl-list (in | out)
policer policer-name (in | out)
qos-map name rewrite-rule name shaping-rate name

```

Configure Tracker Interface

To configure a tracker interface to track the status of a GRE interface, select **Advanced** and configure the following parameter:

Table 141:

Parameter Name	Description
Tracker	Enter the name of a tracker to track the status of GRE interfaces that connect to the Internet.

Release Information

Introduced in Cisco SD-WAN Manager NMS Release 15.4.1.

Configure VPN Interface GRE on Transport Side Using a Configuration Group

Before you begin

On the **Configuration** > **Configuration Groups** page, choose **SD-WAN** as the solution type.

Procedure

Step 1 From the Cisco SD-WAN Manager menu, choose **Configuration** > **Configuration Groups**.

Step 2 Create and configure GRE.

- a. Configure GRE parameters.

Table 142: Basic Configuration

Field	Description
Interface Name (1..255)*	Enter the name of the GRE interface. Range: 1 through 255.
Interface Description	Enter a description of the GRE interface.
Tunnel Mode	Choose from one of the following GRE tunnel modes: • ipv4 underlay: GRE tunnel with IPv4 underlay. IPv4 underlay is the default value. • ipv6 underlay: GRE tunnel with IPv6 underlay.
Multiplexing	Choose Yes to enable multiplexing, in case of a tunnel in the transport VPN. Default: No
Preshared Key for IKE	Enter the preshared key (PSK) for authentication.

- b. Configure Tunnel fields.

Table 143: Tunnel

Field	Description
Source	Enter the source of the GRE interface: • IP Address: Enter the source IP address of the GRE tunnel interface. Based on the option you selected in the Tunnel Mode drop-down list, enter an IPv4 or an IPv6 address. This address is on the local router. • Interface: Enter the egress interface name for the GRE tunnel. • Tunnel Route Via*: Specify the tunnel route details to steer the GRE tunnel traffic through. Note If the Tunnel Source Interface type is a loopback interface, enter the interface for traffic to be routed to. You cannot use the tunnel route via option to configure IPsec tunnels on a cellular interface because cellular interfaces do not include a next hop IP address for the default route.
Destination	Enter the source of the GRE interface: • GRE Destination IP Address*: Enter the destination IP address of the GRE tunnel interface. This address is on a remote device. • IP Address: Based on the option you selected in the Tunnel Mode drop-down list, enter an IPv4 or an IPv6 address for the GRE tunnel. • Mask*: Enter the subnet mask. • IPv6 Address: Enter the destination IPv6 or address for the GRE tunnel.

c. Configure IKE fields.

Table 144: IKE

Field	Description
IKE Version	Enter 1 to choose IKEv1. Enter 2 to choose IKEv2. Default: IKEv1
IKE Integrity Protocol	Choose one of the following modes for the exchange of keying information and setting up IKE security associations: • Main: Establishes an IKE SA session before starting IPsec negotiations. • Aggressive: Negotiation is quicker, and the initiator and responder ID pass in the clear. Aggressive mode does not provide identity protection for communicating parties. Default: Main mode

Field	Description
IKE Rekey Interval	Specify the interval for refreshing IKE keys. Range: 3600 through 1209600 seconds (1 hour through 14 days) Default: 14400 seconds (4 hours)
IKE Cipher Suite	Specify the type of authentication and encryption to use during IKE key exchange. Values: aes128-cbc-sha1, aes128-cbc-sha2, aes256-cbc-sha1, aes256-cbc-sha2 Default: aes256-cbc-sha1
IKE Diffie-Hellman Group	Specify the Diffie-Hellman group to use in IKE key exchanges. Values: 2, 14, 15, 16, 19, 20, 21, 24 Default: 16
IKE ID for Local End Point	If the remote IKE peer requires a local endpoint identifier, specify it. Range: 1 through 64 characters Default: Source IP address of the tunnel
IKE ID for Remote End Point	If the remote IKE peer requires a remote end point identifier, specify it. Range: 1 through 64 characters Default: Destination IP address of the tunnel There is no default option if you have chosen IKEv2.

d. Configure IPSEC fields.

Table 145: IPSEC

Field	Description
IPsec Rekey Interval	Specify the interval for refreshing IKE keys. Range: 3600 through 1209600 seconds (1 hour through 14 days) Default: 3600 seconds
IPsec Replay Window	Specify the replay window size for the IPsec tunnel. Values: 64, 128, 256, 512, 1024, 2048, 4096, 8192 bytes Default: 512 bytes
IPsec Cipher Suite	Specify the authentication and encryption to use on the IPsec tunnel. Values: aes256-cbc-sha1, aes256-gcm, null-sha1 Default: aes256-gcm

Field	Description
Perfect Forward Secrecy	Specify the PFS settings to use on the IPsec tunnel by choosing one of the following values: • group-2: Use the 1024-bit Diffie-Hellman prime modulus group • group-14: Use the 2048-bit Diffie-Hellman prime modulus group • group-15: Use the 3072-bit Diffie-Hellman prime modulus group • group-16: Use the 4096-bit Diffie-Hellman prime modulus group • none: Disable PFS Default: group-16
DPD Interval	Specify the interval for IKE to send Hello packets on the connection. Range: 10 through 3600 seconds (1 hour) Default: 10 seconds
DPD Retries	Specify how many unacknowledged packets to accept before declaring an IKE peer to be dead and then removing the tunnel to the peer. Range: 2 through 60 Default: 3
Application	Choose an application from the drop-down list: • None • Sig

- e. Configure advanced fields.

Table 146: Advanced

Field	Description
Shutdown	Click Off to enable the interface.
IP MTU	Based on your choice in the Tunnel Mode option, specify the maximum MTU size of the IPv6 packets on the interface. Range: 576 through 9216 Default: 1500 bytes
TCP MSS	Based on your choice in the Tunnel Mode option, specify the maximum segment size (MSS) of TCP SYN packets passing through the Cisco IOS XE Catalyst SD-WAN device. By default, the MSS is dynamically adjusted based on the interface or tunnel MTU such that TCP SYN packets are never fragmented. Range: 552 through 1460 bytes Default: None

Field	Description
Clear-Dont-Fragment	Click On to clear the Don't Fragment bit in the IPv4 packet header for packets being transmitted out the interface.
Tunnel Protection	Choose Yes to enable tunnel protection. Default: No

What to do next

Also see [Deploy a configuration group](#).

Configure GRE on Service Side Using a Configuration Group

Before you begin

On the **Configuration > Configuration Groups** page, choose **SD-WAN** as the solution type.

Procedure

Step 1 From the Cisco SD-WAN Manager menu, choose **Configuration > Configuration Groups**.

Step 2 Create and configure GRE in Service Profile.

- a. Configure Basic Configuration fields.

Table 147: Basic Configuration

Field	Description
Interface Name (1..255)*	Enter the name of the GRE interface, in the format gre number . The value for number can be from 1 through 255.
Interface Description	Enter a description of the GRE interface.
Tunnel Mode	Choose from one of the following GRE tunnel modes: • ipv4 underlay: GRE tunnel with IPv4 underlay. IPv4 underlay is the default value. • ipv6 underlay: GRE tunnel with IPv6 underlay.
Preshared Key for IKE	Enter the preshared key (PSK) for authentication.

- b. Configure Tunnel Fields.

Table 148: Tunnel

Field	Description
Source	Enter the source of the GRE interface: • IP Address: Enter the source IP address of the GRE tunnel interface. Based on the option you selected in the Tunnel Mode drop-down list, enter an IPv4 or an IPv6 address. This address is on the local router. • Interface: Enter the egress interface name for the GRE tunnel. • Tunnel Route Via*: Specify the tunnel route details to steer the GRE tunnel traffic through. Note If the Tunnel Source Interface type is a loopback interface, enter the interface for traffic to be routed to. You cannot use the tunnel route via option to configure IPsec tunnels on a cellular interface because cellular interfaces do not include a next hop IP address for the default route.
Destination	Enter the source of the GRE interface: • GRE Destination IP Address*: Enter the destination IP address of the GRE tunnel interface. This address is on a remote device. • IP Address: Based on the option you selected in the Tunnel Mode drop-down list, enter an IPv4 or an IPv6 address for the GRE tunnel. • Mask*: Enter the subnet mask. • IPv6 Address: Enter the destination IPv6 or address for the GRE tunnel.

c. Configure IKE fields.

Table 149: IKE

Field	Description
IKE Version	Enter 1 to choose IKEv1. Enter 2 to choose IKEv2. Default: IKEv1
IKE Integrity Protocol	Choose one of the following modes for the exchange of keying information and setting up IKE security associations: • Main: Establishes an IKE SA session before starting IPsec negotiations. • Aggressive: Negotiation is quicker, and the initiator and responder ID pass in the clear. Aggressive mode does not provide identity protection for communicating parties. Default: Main mode

Field	Description
IKE Rekey Interval	Specify the interval for refreshing IKE keys. Range: 3600 through 1209600 seconds (1 hour through 14 days) Default: 14400 seconds (4 hours)
IKE Cipher Suite	Specify the type of authentication and encryption to use during IKE key exchange. Values: aes128-cbc-sha1, aes128-cbc-sha2, aes256-cbc-sha1, aes256-cbc-sha2 Default: aes256-cbc-sha1
IKE Diffie-Hellman Group	Specify the Diffie-Hellman group to use in IKE key exchanges. Values: 2, 14, 15, 16, 19, 20, 21, 24 Default: 16
IKE ID for Local End Point	If the remote IKE peer requires a local endpoint identifier, specify it. Range: 1 through 64 characters Default: Source IP address of the tunnel
IKE ID for Remote End Point	If the remote IKE peer requires a remote end point identifier, specify it. Range: 1 through 64 characters Default: Destination IP address of the tunnel There is no default option if you have chosen IKEv2.

d. Configure IPSEC fields.

Table 150: IPSEC

Field	Description
IPsec Rekey Interval (Seconds)	Specify the interval for refreshing IKE keys. Range: 3600 through 1209600 seconds (1 hour through 14 days) Default: 3600 seconds
IPsec Replay Window	Specify the replay window size for the IPsec tunnel. Values: 64, 128, 256, 512, 1024, 2048, 4096, 8192 bytes Default: 512 bytes
IPsec Cipher Suite	Specify the authentication and encryption to use on the IPsec tunnel. Values: aes256-cbc-sha1, aes256-gcm, null-sha1 Default: aes256-gcm

Field	Description
Perfect Forward Secrecy	Specify the PFS settings to use on the IPsec tunnel by choosing one of the following values: • group-2: Use the 1024-bit Diffie-Hellman prime modulus group • group-14: Use the 2048-bit Diffie-Hellman prime modulus group • group-15: Use the 3072-bit Diffie-Hellman prime modulus group • group-16: Use the 4096-bit Diffie-Hellman prime modulus group • none: Disable PFS Default: group-16
DPD Interval	Specify the interval for IKE to send Hello packets on the connection. Range: 10 through 3600 seconds (1 hour) Default: 10 seconds
DPD Retries	Specify how many unacknowledged packets to accept before declaring an IKE peer to be dead and then removing the tunnel to the peer. Range: 2 through 60 Default: 3
Application	Choose an application from the drop-down list: • None • Sig

e. Configure advanced fields.

Table 151: Advanced

Field	Description
Shutdown	Click Off to enable the interface.
IP MTU	Based on your choice in the Tunnel Mode option, specify the maximum MTU size of the IPv4 or IPv6 packets on the interface. Range: 576 through 9216 Default: 1500 bytes
TCP MSS	Specify the maximum segment size (MSS) of the IPv4 TCP SYN packets passing through the Cisco vEdge device. By default, the MSS is dynamically adjusted based on the interface or tunnel MTU such that TCP SYN packets are never fragmented. Range: 552 through 1460 bytes Default: None

Field	Description
IPv6 TCP MSS	Specify the maximum segment size (MSS) of the IPv6 TCP SYN packets passing through the Cisco vEdge device. By default, the MSS is dynamically adjusted based on the interface or tunnel MTU such that TCP SYN packets are never fragmented. Range: 552 through 1460 bytes Default: None
Clear-Dont-Fragment	Click On to clear the Don't Fragment bit in the IPv4 packet header for packets being transmitted out the interface.
Tunnel Protection	Choose Yes to enable tunnel protection. Default: No

What to do next

Also see [Deploy a configuration group](#).

VPN Interface IPsec (for Cisco vEdge Devices)

Use the VPN Interface IPsec feature template to configure IPsec tunnels on Cisco vEdge devices that are being used for Internet Key Exchange (IKE) sessions. You can configure IPsec on tunnels in the transport VPN (VPN 0) and in service VPNs (VPN 1 through 65530, except for 512).

Navigate to the Template Screen and Name the Template

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Device Templates**, and from the **Create Template** drop-down list, select **From Feature Template**.



Note In Cisco vManage Release 20.7.x and earlier releases, **Device Templates** is titled **Device**.

3. From the **Device Model** drop-down list, select the vEdge device for which you are creating the template.
4. Click **Transport and Management VPN** and the page scrolls to the Transport and Management VPN section.
5. Under **Additional VPN 0 Templates**, click **VPN Interface IPsec**.
6. From the **VPN Interface IPsec** drop-down list, click **Create Template**. The VPN Interface IPsec template form is displayed.

This form contains fields for naming the template, and fields for defining the VPN Interface IPsec parameters.

7. In **Template Name**, enter a name for the template. The name can be up to 128 characters and can contain only alphanumeric characters.
8. In **Template Description**, enter a description of the template. The description can be up to 2048 characters and can contain only alphanumeric characters.

When you first open a feature template, for each parameter that has a default value, the scope is set to Default (indicated by a check mark), and the default setting or value is shown. To change the default or to enter a value, click the **Scope** drop-down list and select one of the following:

Table 152:

Parameter Scope	Scope Description
Device Specific (indicated by a host icon)	Use a device-specific value for the parameter. For device-specific parameters, you cannot enter a value in the feature template. You enter the value when you attach a Viptela device to a device template . When you click Device Specific, the Enter Key box opens. This box displays a key, which is a unique string that identifies the parameter in a CSV file that you create. This file is an Excel spreadsheet that contains one column for each key. The header row contains the key names (one key per column), and each row after that corresponds to a device and defines the values of the keys for that device. You upload the CSV file when you attach a Viptela device to a device template. For more information, see Create a Template Variables Spreadsheet . To change the default key, type a new string and move the cursor out of the Enter Key box. Examples of device-specific parameters are system IP address, hostname, GPS location, and site ID.
Global (indicated by a globe icon)	Enter a value for the parameter, and apply that value to all devices. Examples of parameters that you might apply globally to a group of devices are DNS server, syslog server, and interface MTUs.

Configure a Basic IPsec Tunnel Interface

To configure an IPsec tunnel to use for IKE sessions, select the Basic Configuration tab and configure the following parameters. Parameters marked with an asterisk are required to configure an IPsec tunnel.

Table 153:

Parameter Name	Description
Shutdown*	Click No to enable the interface.
Interface Name*	Enter the name of the IPsec interface, in the format ipsec number . <i>number</i> can be from 1 through 256.
Description	Enter a description of the IPsec interface.
IPv4 Address*	Enter the IPv4 address of the IPsec interface, in the format <i>ipv4-prefix/length</i> . The address must be a /30.

Parameter Name	Description
Source*	Set the source of the IPsec tunnel that is being used for IKE key exchange: • Click IP Address—Enter the IPv4 address that is the source tunnel interface. This address must be configured in VPN 0. • Click Interface—Enter the name of the physical interface that is the source of the IPsec tunnel. This interface must be configured in VPN 0.
Destination: IPsec Destination IP Address/FQDN*	Set the destination of the IPsec tunnel that is being used for IKE key exchange. Enter either an IPv4 address or the fully qualified DNS name that points to the destination.
TCP MSS	Specify the maximum segment size (MSS) of TCP SYN packets passing through the Cisco vEdge device. By default, the MSS is dynamically adjusted based on the interface or tunnel MTU such that TCP SYN packets are never fragmented. <i>Range: 552 to 1460 bytes</i> <i>Default: None</i>
IP MTU	Specify the maximum MTU size of packets on the interface. <i>Range: 576 through 1804</i> <i>Default: 1500 bytes</i>

To save the feature template, click **Save**.

CLI equivalent:

```
vpn vpn-id
 interface ipsec number ip address ipv4-prefix/length mtu bytes
 no shutdown
 tcp-mss-adjust bytes tunnel-destination ipv4-address
 ( tunnel-source ip-address | tunnel-source-interface interface-name)
```

Configure Dead-Peer Detection

To configure IKE dead-peer detection to determine whether the connection to an IKE peer is functional and reachable, click **DPD** and the page scrolls to the section. Configure the following parameters:

Table 154:

Parameter Name	Description
DPD Interval	Specify the interval for IKE to send Hello packets on the connection. <i>Range: 0 to 30 seconds. Default: 10 seconds</i>
DPD Retries	Specify how many unacknowledged packets to accept before declaring an IKE peer to be dead and then tearing down the tunnel to the peer. <i>Range: 0 to 255. Default: 3</i>

To save the feature template, click **Save**.

CLI equivalent:

```
vpn vpn-id interface ipsec number dead-peer-detection seconds retries number
```

Configure IKE

To configure IKE, click **IKE** and configure the parameters discussed below.

When you create an IPsec tunnel on a Cisco vEdge device, IKE Version 1 is enabled by default on the tunnel interface. The following properties are also enabled by default for IKEv1:

- Authentication and encryption: AES-256 advanced encryption standard CBC encryption with the HMAC-SHA1 keyed-hash message authentication code algorithm for integrity
- Diffie-Hellman group number: 16
- Rekeying time interval: 4 hours
- SA establishment mode: Main

To modify IKEv1 parameters, configure the following:

Table 155:

Parameter Name	Description
IKE Version	Enter 1 to select IKEv1.
IKE Mode	Specify the IKE SA establishment mode. <i>Values:</i> Aggressive mode, Main mode <i>Default:</i> Main mode
IPsec Rekey Interval	Specify the interval for refreshing IKE keys. <i>Range:</i> 3600 through 1209600 seconds (1 hour through 14 days). <i>Default:</i> 14400 seconds (4 hours)
IKE Cipher Suite	Specify the type of authentication and encryption to use during IKE key exchange. <i>Values:</i> aes128-cbc-sha1, aes256-cbc-sha1. <i>Default:</i> aes256-cbc-sha1
IKE Diffie-Hellman Group	Specify the Diffie-Hellman group to use in IKE key exchange. <i>Values:</i> 1024-bit modulus, 2048-bit modulus, 3072-bit modulus, 4096-bit modulus. <i>Default:</i> 4096-bit modulus
IKE Authentication: Preshared Key	To use preshared key (PSK) authentication, enter the password to use with the preshared key.
IKE ID for Local End Point	If the remote IKE peer requires a local end point identifier, specify it. <i>Range:</i> <i>Default:</i> Tunnel's source IP address
IKE ID for Remote End Point	If the remote IKE peer requires a remote end point identifier, specify it. <i>Range:</i> 1 through 64 characters. <i>Default:</i> Tunnel's destination IP address

To save the feature template, click **Save**.

CLI equivalent:

```
vpn vpn-id interface ipsec number ike authentication-type type
    local-id id
    pre-shared-secret password
    remote-id id cipher-suite suite group number mode mode rekey-interval seconds
    version 1
```

To configure IKEv2, configure the following parameters:

Table 156:

Parameter Name	Description
IKE Version	Enter 2 to select IKEv2.
IPsec Rekey Interval	Specify the interval for refreshing IKE keys. <i>Range:</i> 3600 through 1209600 seconds (1 hour through 14 days). <i>Default:</i> 14400 seconds (4 hours)
IKE Cipher Suite	Specify the type of authentication and encryption to use during IKE key exchange. <i>Values:</i> aes128-cbc-sha1, aes256-cbc-sha1. <i>Default:</i> aes256-cbc-sha1
IKE Diffie-Hellman Group	Specify the Diffie-Hellman group to use in IKE key exchange. <i>Values:</i> 1024-bit modulus, 2048-bit modulus, 3072-bit modulus, 4096-bit modulus. <i>Default:</i> 4096-bit modulus
IKE Authentication: Preshared Key	To use preshared key (PSK) authentication, enter the password to use with the preshared key.
IKE ID for Local End Point	If the remote IKE peer requires a local end point identifier, specify it. <i>Range:</i> <i>Default:</i> Tunnel's source IP address
IKE ID for Remote End Point	If the remote IKE peer requires a remote end point identifier, specify it. <i>Range:</i> 1 through 64 characters. <i>Default:</i> Tunnel's destination IP address

To save the feature template, click **Save**.

CLI equivalent:

```
vpn vpn-id interface ipsec number ike authentication-type type
    local-id id
    pre-shared-secret password
    remote-id id cipher-suite suite group number rekey-interval seconds
    version 2
```

Configure IPsec Tunnel Parameters

To configure the IPsec tunnel that carries IKE traffic, click **IPsec** and configure the following parameters:

Table 157:

Parameter Name	Description
IPsec Rekey Interval	Specify the interval for refreshing IKE keys. <i>Range:</i> 3600 through 1209600 seconds (1 hour through 14 days). <i>Default:</i> 14400 seconds (4 hours)
IKE Replay Window	Specify the replay window size for the IPsec tunnel. <i>Values:</i> 64, 128, 256, 512, 1024, 2048, 4096, 8192 bytes. <i>Default:</i> 32 bytes
IPsec Cipher Suite	Specify the authentication and encryption to use on the IPsec tunnel. <i>Values:</i> aes256-cbc-sha1, aes256-gcm, null-sha1 . <i>Default:</i> aes256-gcm

Perfect Forward Secrecy	Specify the PFS settings to use on the IPsec tunnel. <i>Values:</i> • group-2: Use the 1024-bit Diffie-Hellman prime modulus group. • group-14: Use the 2048-bit Diffie-Hellman prime modulus group. • group-15: Use the 3072-bit Diffie-Hellman prime modulus group. • group-16: Use the 4096-bit Diffie-Hellman prime modulus group. • none: Disable PFS. <i>Default:</i> group-16
-------------------------	--

To save the feature template, click **Save**.

CLI equivalent:

```
vpn vpn-id interface ipsec number ipsec cipher-suite suite perfect-forward-secrecy
pfs-setting rekey-interval seconds replay-window number
```

Release Information

Introduced in Cisco SD-WAN Manager Release 17.2. In Release 17.2.3, add support for PFS. In Release 18.2, support for IPsec tunnels in VPN 0. In Release 18.4, standard IPsec support for IOS XE routers.

VPN Interface PPP

Point-to-Point Protocol (PPP) is a data link protocol used to establish a direct connection between two nodes. PPP properties are associated with a PPPoE-enabled interface on Cisco SD-WAN devices to connect multiple users over an Ethernet link.

To configure PPPoE on Cisco vEdge devices using Cisco SD-WAN Manager templates:

1. Create a VPN Interface PPP feature template to configure PPP parameters for the PPP virtual interface, as described in this section.
2. Create a VPN Interface PPP Ethernet feature template to configure a PPPoE-enabled interface. See VPN Interface PPP Ethernet.
3. Optionally, create a VPN feature template to modify the default configuration of VPN 0. See the VPN help topic.

Navigate to the Template Screen and Name the Template

1. From the Cisco SD-WAN Manager menu, choose **Configuration** > **Templates**.
2. Click **Device Templates**, and click **Create Template**.



Note In Cisco vManage Release 20.7.x and earlier releases, **Device Templates** is titled **Device**.

3. From the **Create Template** drop-down list, choose **From Feature Template**.
4. From the **Device Model** drop-down list, select the type of device for which you are creating the template.
5. Click **Transport & Management VPN** or scroll to the **Transport & Management VPN** section.
6. Under **Additional VPN 0 Templates**, click **VPN Interface PPP**.
7. From the **VPN Interface PPP** drop-down list, click **Create Template**. The VPN Interface PPP template form is displayed.

This form contains fields for naming the template, and fields for defining the VPN Interface PPP parameters.

8. In the **Template Name** field, enter a name for the template. The name can be up to 128 characters and can contain only alphanumeric characters.
9. In the **Template Description** field, enter a description of the template. The description can be up to 2048 characters and can contain only alphanumeric characters.

When you first open a feature template, for each parameter that has a default value, the scope is set to Default (indicated by a check mark), and the default setting or value is shown. To change the default or to enter a value, click the **Scope** drop-down list and select one of the following:

Table 158:

Parameter Scope	Scope Description
Device Specific (indicated by a host icon)	Use a device-specific value for the parameter. For device-specific parameters, you cannot enter a value in the feature template. You enter the value when you attach a Cisco vEdge device to a device template . When you click Device Specific, the Enter Key box opens. This box displays a key, which is a unique string that identifies the parameter in a CSV file that you create. This file is an Excel spreadsheet that contains one column for each key. The header row contains the key names (one key per column), and each row after that corresponds to a device and defines the values of the keys for that device. You upload the CSV file when you attach a Viptela device to a device template. For more information, see Create a Template Variables Spreadsheet . To change the default key, type a new string and move the cursor out of the Enter Key box. Examples of device-specific parameters are system IP address, hostname, GPS location, and site ID.
Global (indicated by a globe icon)	Enter a value for the parameter, and apply that value to all devices. Examples of parameters that you might apply globally to a group of devices are DNS server, syslog server, and interface MTUs.

Configure a PPP Virtual Interface

To configure a PPP virtual interface, click **Basic Configuration** and configure the following parameters. Parameters marked with an asterisk are required to configure the interface. You must also configure an authentication protocol and a tunnel interface for the PPP interface, and you must ensure that the maximum MTU for the PPP interface is 1492 bytes.

Table 159:

Parameter Name	Description
Shutdown*	Click No to enable the PPP virtual interface.
PPP Interface Name*	Enter the number of the PPP interface. It can be a number from 1 through 31.
Description	Enter a description for the PPP virtual interface.

Parameter Name	Description
Bandwidth Upstream	For transmitted traffic, set the bandwidth above which to generate notifications. <i>Range:</i> 1 through $(2^{32} / 2) - 1$ kbps
Bandwidth Downstream	For received traffic, set the bandwidth above which to generate notifications. <i>Range:</i> 1 through $(2^{32} / 2) - 1$ kbps
Block Non-Source IP	Click Yes to have the interface forward traffic only if the source IP address of the traffic matches the interface's IP prefix range.

To save the feature template, click **Save**.

CLI equivalent:

```
vpn 0
  interface pppnumber bandwidth-downstream kbps bandwidth-upstream kbps block-non-source-ip
  ppp
    no shutdown
```

Configure the Access Concentrator Name and Authentication Protocol

To configure the access concentrator name, click **PPP** and configure the following parameters:

Table 160:

Parameter Name	Description
AC Name	Name of the access concentrator used by PPPoE to route connections to the Internet.
Authentication Protocol	Select the authentication protocol used by PPPoE: • CHAP—Enter the hostname and password provided by your Internet Service Provider (ISP). <i>hostname</i> can be up to 255 characters. • PAP—Enter the username and password provided by your ISP. <i>username</i> can be up to 255 characters. • PAP and CHAP—Configure both authentication protocols. Enter the login credentials for each protocol. To use the same username and password for both, click Same Credentials for PAP and CHAP.

To save the feature template, click **Save**.

CLI equivalent:

```
vpn 0
  interface pppnumber ppp
    ac-name name
    authentication
      chap hostname name password password
      pap password password sent-username name
```

Create a Tunnel Interface

On Cisco vEdge devices, you can configure up to four tunnel interfaces. This means that each Cisco vEdge device can have up to four TLOCs.

For the control plane to establish itself so that the overlay network can function, you must configure WAN transport interfaces in VPN 0.

To configure a tunnel interface for the PPP interface, select the **Tunnel Interface** tab and configure the following parameters:

Table 161:

Parameter Name	Description
Tunnel Interface	Click On to create a tunnel interface.
Color	Select a color for the TLOC.
Control Connection	By default, Control Connection is set to On, which establishes a control connection for the TLOC. If the router has multiple TLOCs, click No to have the tunnel not establish control connection for the TLOC. Note We recommend a minimum of 650-700 Kbps bandwidth with default 1 sec hello-interval and 12 sec hello-tolerance parameters configured to avoid any data/packet loss in connection traffic. For each BFD session, an additional average sized BFD packet of 175 Bytes consumes 1.4 Kbps of bandwidth. A sample calculation of the required bandwidth for bidirectional BFD packet flow is given below: • 650 – 700 Kbps per device for control connections. • 175 Bytes (or 1.4 Kbps) per BFD session on the device (request) • 175 Bytes (or 1.4 Kbps) per BFD session on the device (response) If the path MTU discovery (PMTUD) is enabled, bandwidth for send/receive BFD packets per tunnel for every 30 secs: A 1500 Bytes BFD request packet is sent per tunnel every 30 secs: $1500 \text{ Bytes} * 8 \text{ bits/1 byte} * 1 \text{ packet} / 30 \text{ secs} = 400 \text{ bps (request)}$ A 147 Bytes BFD packet is sent in response: $147 \text{ Bytes} * 8 \text{ bits/1 byte} * 1 \text{ packet} / 30 \text{ secs} = 40 \text{ bps (response)}$ Therefore, a device with 775 BFD sessions (for example) requires a bandwidth of: $700k + (1.4k * 775) + (400 * 775) + (1.4k * 775) + (40 * 775) = \sim 3.5 \text{ MBps}$
Maximum Control Connections	Specify the maximum number of Cisco Catalyst SD-WAN Controller that the WAN tunnel interface can connect to. To have the tunnel establish no control connections, set the number to 0. <i>Range: 0 through 8 Default: 2</i>

Parameter Name	Description
vBond As STUN Server	Click On to enable Session Traversal Utilities for NAT (STUN) to allow the tunnel interface to discover its public IP address and port number when the Cisco vEdge device is located behind a NAT.
Exclude Controller Group List	Set the Cisco Catalyst SD-WAN Controller that the tunnel interface is not allowed to connect to. <i>Range: 0 through 100</i>
vManage Connection Preference	Set the preference for using a tunnel interface to exchange control traffic with the vManage NMS. <i>Range: 0 through 8 Default: 5</i>
Low-Bandwidth Link	Select to characterize the tunnel interface as a low-bandwidth link.
Allow Service	Select On or Off for each service to allow or disallow the service on the interface.

To configure additional tunnel interface parameters, click **Advanced Options** and configure the following parameters:

Table 162:

Parameter Name	Description
GRE	Use GRE encapsulation on the tunnel interface. By default, GRE is disabled. If you select both IPsec and GRE encapsulations, two TLOCs are created for the tunnel interface that have the same IP addresses and colors, but that differ by their encapsulation.
IPsec	Use IPsec encapsulation on the tunnel interface. By default, IPsec is enabled. If you select both IPsec and GRE encapsulations, two TLOCs are created for the tunnel interface that have the same IP addresses and colors, but that differ by their encapsulation.
IPsec Preference	Specify a preference value for directing traffic to the tunnel. A higher value is preferred over a lower value. <i>Range: 0 through 4294967295 Default: 0</i>
IPsec Weight	Enter a weight to use to balance traffic across multiple TLOCs. A higher value sends more traffic to the tunnel. <i>Range: 1 through 255 Default: 1</i>
Carrier	Select the carrier name or private network identifier to associate with the tunnel. <i>Values: carrier1, carrier2, carrier3, carrier4, carrier5, carrier6, carrier7, carrier8, default</i> <i>Default: default</i>
Bind Loopback Tunnel	Enter the name of a physical interface to bind to a loopback interface.
Last-Resort Circuit	Select to use the tunnel interface as the circuit of last resort.

Parameter Name	Description
NAT Refresh Interval	Enter the interval between NAT refresh packets sent on a DTLS or TLS WAN transport connection. <i>Range:</i> 1 through 60 seconds <i>Default:</i> 5 seconds
Hello Interval	Enter the interval between Hello packets sent on a DTLS or TLS WAN transport connection. <i>Range:</i> 100 through 10000 milliseconds <i>Default:</i> 1000 milliseconds (1 second)
Hello Tolerance	Enter the time to wait for a Hello packet on a DTLS or TLS WAN transport connection before declaring that transport tunnel to be down. <i>Range:</i> 12 through 60 seconds <i>Default:</i> 12 seconds

CLI equivalent:

```
vpn 0
  interface interface-name tunnel-interface allow-service service-name
bind interface-name
  carrier carrier-name
  color color encapsulation (gre | ipsec)
  preference number
  weight number hello-interval milliseconds hello-tolerance seconds
last-resort-circuit max-control-connections number nat-refresh-interval seconds
vbond-as-stun-server
```

Configure the Interface as a NAT Device

To configure an interface to act as a NAT device, click **NAT** and configure the following parameters:

Table 163:

Parameter Name	Description
NAT	Click On to have the interface act as a NAT device.
Refresh Mode	Select how NAT mappings are refreshed, either outbound or bidirectional (outbound and inbound). <i>Default:</i> Outbound
UDP Timeout	Specify when NAT translations over UDP sessions time out. <i>Range:</i> 1 through 65536 minutes <i>Default:</i> 1 minutes
TCP Timeout	Specify when NAT translations over TCP sessions time out. <i>Range:</i> 1 through 65536 minutes <i>Default:</i> 60 minutes (1 hour)
Block ICMP	Select On to block inbound ICMP error messages. By default, a Cisco vEdge device acting as a NAT device receives these error messages. <i>Default:</i> Off

Respond to Ping	Select On to have the Cisco vEdge device respond to ping requests to the NAT interface's IP address that are received from the public side of the connection.
-----------------	--

To create a port forwarding rule, click **Add New Port Forwarding Rule** and configure the following parameters. You can define up to 128 port-forwarding rules to allow requests from an external network to reach devices on the internal network.

Table 164:

Parameter Name	Description
Port Start Range	Enter a port number to define the port or first port in the range of interest. <i>Range:</i> 0 through 65535
Port End Range	Enter the same port number to apply port forwarding to a single port, or enter the larger number to apply it to a range or ports. <i>Range:</i> 0 through 65535
Protocol	Select the protocol to which to apply the port-forwarding rule, either TCP or UDP. To match the same ports for both TCP and UDP traffic, configure two rules.
VPN	Specify the private VPN in which the internal server resides. This VPN is one of the VPN identifiers in the overlay network. <i>Range:</i> 0 through 65535
Private IP	Specify the IP address of the internal server to which to direct traffic that matches the port-forwarding rule.

To save a port forwarding rule, click **Add**.

To save the feature template, click **Save**.

CLI equivalent:

```
vpn vpn-id
interface interface-name nat block-icmp-error port-forward port-start port-number1
port-end port-number2 proto (tcp | udp)
    private-ip-address ip-address private-vpn vpn-id refresh (bi-directional | outbound)

    respond-to-ping tcp-timeout minutes
    udp-timeout minutes
```

Apply Access Lists

To apply a rewrite rule, access lists, and policers to a router interface, select the **ACL** tab and configure the following parameters:

Table 165:

Parameter Name	Description
Rewrite Rule	Click On , and specify the name of the rewrite rule to apply on the interface.
Ingress ACL – IPv4	Click On , and specify the name of the access list to apply to IPv4 packets being received on the interface.

Parameter Name	Description
Egress ACL – IPv4	Click On , and specify the name of the access list to apply to IPv4 packets being transmitted on the interface.
Ingress ACL – IPv6	Click On , and specify the name of the access list to apply to IPv6 packets being received on the interface.
Egress ACL – IPv6	Click On , and specify the name of the access list to apply to IPv6 packets being transmitted on the interface.
Ingress Policer	Click On , and specify the name of the policer to apply to packets being received on the interface.
Egress Policer	Click On , and specify the name of the policer to apply to packets being transmitted on the interface.

To save the feature template, click **Save**.

CLI equivalent:

```
vpn 0
 interface pppnumber access-list acl-name (in | out)
   ipv6 access-list acl-name (in | out)
   policer policer-name (in |out)
   rewrite-rule name
```

Configure Other Interface Properties

To configure other interface properties, click **Advanced** and configure the following properties:

Table 166:

Parameter Name	Description
MAC Address	Specify a MAC address to associate with the interface, in colon-separated hexadecimal notation.
IP MTU	Specify the maximum MTU size of packets on the interface. <i>Range:</i> 576 through 1804 <i>Default:</i> 1500 bytes
TCP MSS	Specify the maximum segment size (MSS) of TCP SYN packets passing through the Cisco vEdge device. By default, the MSS is dynamically adjusted based on the interface or tunnel MTU such that TCP SYN packets are never fragmented. <i>Range:</i> 552 to 1460 bytes <i>Default:</i> None
Clear Dont Fragment	Click On to clear the Don't Fragment bit in the IPv4 packet header for packets being transmitted out the interface. When the DF bit is cleared, packets larger than that interface's MTU are fragmented before being sent.

Parameter Name	Description
TLOC Extension	Enter the name of the physical interface on the same router that connects to the WAN transport circuit. This configuration then binds this service-side interface to the WAN transport. A second Cisco vEdge device at the same site that itself has no direct connection to the WAN (generally because the site has only a single WAN connection) and that connects to this service-side interface is then provided with a connection to the WAN.
Tracker	Enter the name of a tracker to track the status of transport interfaces that connect to the internet.
ICMP Redirect	Click Disable to disable ICMP redirect messages on the interface. By default, an interface allows ICMP redirect messages.

To save the feature template, click **Save**.

CLI equivalent:

```
vpn vpn-id interface interface-name clear-dont-fragment icmp-redirect-disable
mac-address mac-address mtu bytes tcp-mss-adjust bytes tloc-extension
interface-name tracker tracker-name
```

Release Information

Introduced in vManage NMS in Release 15.3. In Release 16.3, add support for IPv6. In Release 17.1, support ability to configure both CHAP and PAP authentication on a PPP interface. In Release 17.2.2, add support for interface status tracking. In Release 18.2, add support for disabling ICMP redirect messages.

Configure VPN Interface Multilink Using Configuration Groups

Use the VPN Interface Multilink feature to configure multilink interface properties for Cisco SD-WAN Manager devices.

Before you begin

On the **Configuration > Configuration Groups** page, choose **SD-WAN** as the solution type.

Procedure

- Step 1** From the Cisco SD-WAN Manager menu, choose **Configuration > Configuration Groups**.
- Step 2** Create and configure VPN Interface Multilink in a service profile.
- a) Enter the basic configuration information.

Table 167: Basic Configuration

Parameter Name	Description
Interface Name	Enter the name of the multilink interface.

Parameter Name	Description
Multilink Group Number *	Enter the number of the multilink group. It must be the same as the number you enter in the multilink interface name parameter. Range: 1 through 65535
PPP Authentication Protocol	Select the authentication protocol used by the multilink interface: • CHAP: Enter the hostname and password provided by your Internet Service Provider (ISP). <i>hostname</i> can be up to 254 characters. • PAP: Enter the username and password provided by your ISP. <i>username</i> can be up to 254 characters. • PAP and CHAP: Configure both authentication protocols. Enter the login credentials for each protocol. To use the same username and password for both, click Same Credentials for PAP and CHAP.
Hostname *	Enter hostname for PPP CHAP Authentication.
CHAP Password *	Enter password for PPP CHAP Authentication.
IPv4 Address *	To configure a static address, click Static and enter an IPv4 address. To set the interface as a DHCP client so that the interface to receive its IP address from a DHCP server, click Dynamic. You can optionally set the DHCP distance to specify the administrative distance of routes learned from a DHCP server. Default: 1
Mask	Choose a value for the subnet mask.
IPv6 Address *	To configure a static address for an interface in VPN 0, click Static and enter an IPv6 address. To set the interface as a DHCP client so that the interface to receive its IP address from a DHCP server, click Dynamic. You can optionally set the DHCP distance to specify the administrative distance of routes learned from a DHCP server. The default DHCP distance is 1. You can optionally enable DHCP rapid commit, to speed up the assignment of IP addresses.

b) Enter multilink information

Table 168: Multilink

Parameter Name	Description
Add T1/E1 Interface	
T1	
Description	Enter a description for the T1 controller.

Parameter Name	Description
Slot*	Enter the number of the slot in slot/subslot/port format, where the T1 NIM is installed. For example, 0/1/0.
Framing	Enter the T1 frame type: • esf: Send T1 frames as extended superframes. This is the default. • sf: Send T1 frames as superframes. Superframing is sometimes called D4 framing.
Clock Source	Select the clock source: • line: Use phase-locked loop (PLL) on the interface. This is the default. When both T1 ports use line clocking and neither port is configured as the primary, by default, port 0 is the primary clock source and port 1 is the secondary clock source. • internal: Use the controller framer as the primary clock.
Line Code	Select the line encoding to use to send T1 frames: • ami: Use alternate mark inversion (AMI) as the linecode. AMI signaling uses frames grouped into superframes. • b8zs: Use bipolar 8-zero substitution as the linecode. This is the default. B8ZS uses frames that are grouped into extended superframes.
Cable Length	Select the cable length to configure the attenuation • short: Set the transmission attenuation for cables that are 660 feet or shorter. • long: Attenuate the pulse from the transmitter using pulse equalization and line buildout. You can configure a long cable length for cables longer than 660 feet. There is no default length.
E1	
Description	Enter a description for the E1 controller.
Slot*	Enter the number of the slot in slot/subslot/port format, where the E1 NIM is installed. For example, 0/1/0.
Framing	Enter the E1 frame type: • crc4: Use cyclic redundancy check 4 (CRC4). This is the default. • no-crc4: Do not use CRC4.

Parameter Name	Description
Clock Source	Select the clock source: • line: Use phase-locked loop (PLL) on the interface. This is the default. When both E1 ports use line clocking and neither port is configured as the primary, by default, port 0 is the primary clock source and port 1 is the secondary clock source. • internal: Use the controller framer as the primary clock.
Line Code	Select the line encoding to use to send E1 frames: • ami: Use alternate mark inversion (AMI) as the linecode. • hdb3: Use high-density bipolar 3 as the linecode. This is the default.
Add Channel Group	
Channel Group	To configure the serial WAN on the interface, enter a channel group number. Range: 0 through 30
Time Slot	To configure the serial WAN on the interface, enter a value for the timeslot. Range: 0 through 31
Add New A/S Serial Interface	
Interface Name	Enter the name of the serial interface.
Description	Enter a description for the serial interface.
Bandwidth	For transmitted traffic, set the bandwidth above which to generate notifications.
Clock Rate	Specify a value for the clock rate. Range: 1200 through 800000

c) Enter tunnel information

Table 169: Tunnel

Parameter Name	Description
Color	Choose a color for the TLOC.
Color Description	Minimum supported release: Cisco Catalyst SD-WAN Manager Release 20.18.1 Enter a description associated to the TLOC color.
Restrict	Enable this option to drop packets when a tunnel to the service is unreachable.
Groups	Enter the list of groups in the field.
Border	From the drop-down list, select Global . Click On to set TLOC as border TLOC.

Parameter Name	Description
Maximum Control Connections	Specify the maximum number of Cisco SD-WAN Controllers that the WAN tunnel interface can connect to. To have the tunnel establish no control connections, set the number to 0. Range: 0 through 8 Default: 2
Validator As Stun Server	Click On to enable Session Traversal Utilities for NAT (STUN) to allow the tunnel interface to discover its public IP address and port number when the router is located behind a NAT.
Exclude Controller Group List	Set the Cisco SD-WAN Controllers that the tunnel interface is not allowed to connect to. Range: 0 through 100
Cisco SD-WAN Manager Connection Preference	Set the preference for using a tunnel interface to exchange control traffic with Cisco SD-WAN Manager. Range: 0 through 8 Default: 5
Full Port Hop	Minimum release: Cisco IOS XE Catalyst SD-WAN Release 17.18.1a Enable full port hopping at the TLOC level to allow devices to establish connections with controllers by switching to the next port if the current port is blocked or non-functional. Default: Disabled
Port Hop	From the drop-down list, select Global . Click Off to allow port hopping on tunnel interface. Default: On , which disallows port hopping on tunnel interface Starting from Cisco IOS XE Catalyst SD-WAN Release 17.18.1a, this field is deprecated. Instead use the Full Port Hop option. See the Full Port Hop field.
Low-Bandwidth Link	Click On to set the tunnel interface as a low-bandwidth link. Default: Off
Network Broadcast	From the drop-down list, select Global . Click On to accept and respond to network-prefix-directed broadcasts. Enable this parameter only if the Directed Broadcast is enabled on the LAN interface feature template. Default: Off

Parameter Name	Description
Tunnel TCP MSS	TCP MSS affects any packet that contains an initial TCP header that flows through the router. When configured, TCP MSS is examined against the MSS exchanged in the three-way handshake. The MSS in the header is lowered if the configured TCP MSS setting is lower than the MSS in the header. If the MSS header value is already lower than the TCP MSS, the packets flow through unmodified. The host at the end of the tunnel uses the lower setting of the two hosts. To configure TCP MSS, provide a value that is 40 bytes lower than the minimum path MTU. Specify the MSS of TPC SYN packets passing through the Cisco vEdge device. By default, the MSS is dynamically adjusted based on the interface or tunnel MTU such that TCP SYN packets are never fragmented. Range: 552 through 1460 bytes

d) Enter ACL information.

Table 170: ACL

Parameter Name	Description
Ingress ACL - IPv4	Enter the name of an IPv4 access list to packets being received on the interface.
Egress ACL - IPv4	Enter the name of an IPv4 access list to packets being transmitted on the interface.
Igress ACL - IPv6	Enter the name of an IPv6 access list to packets being received on the interface.
Egress ACL - IPv6	Enter the name of an IPv6 access list to packets being transmitted on the interface.

e) Enter advanced information.

Table 171: Advanced

Parameter Name	Description
Shutdown	Click No to enable the multilink interface.
Description	Enter a description for the multilink interface.
PPP Authentication Type	Select the type authentication from one of the following options.: • Unidirectional: The server initiates the authentication. • Bidirectional: Both the client and the server can initiate the authentication.

Parameter Name	Description
TCP MSS	Specify the maximum segment size (MSS) of TCP SYN packets passing through the Cisco Catalyst SD-WAN device. By default, the MSS is dynamically adjusted based on the interface or tunnel MTU such that TCP SYN packets are never fragmented. Range: 500 through 1460 bytes Default: 536
Disable Fragmentation	Click On to disable fragmentation for PPP Multilink Protocol data units (PDUs).
Fragment Max Delay	Configure the delay between the transmission of fragments in a PPP Multilink Protocol link. Range: 0 through 1000 Default: No CLI Command
Interleaving Fragments	Enable interleave fragmentation for PPP Multilink Protocol data units (PDUs).
TLOC Extension	Enter the name of a physical interface on the same router that connects to the WAN transport. This configuration binds the service-side interface to the WAN transport by enabling a device to access the opposite WAN transport connected to the neighbouring device using a TLOC-extension interface.
IP MTU	Specify the maximum MTU size of packets on the interface. MLP encapsulation adds 6 extra bytes (4 header, 2 checksum) to each outbound packet. These overhead bytes reduce the effective bandwidth on the connection; therefore, the throughput for an MLP bundle is slightly less than an equivalent bandwidth connection that is not using MLP. Range: 576 through 1804 Default: 1500 bytes
IP Directed-Broadcast	Enable the translation of a directed broadcast to physical broadcasts.
Shaping Rate (Kbps)	Configure the aggregate traffic transmission rate on the interface to be less than line rate, in kilobits per second (kbps).

What to do next

Also see [Deploy a Configuration Group](#).

VPN Interface PPP Ethernet

Use the VPN Interface PPP Ethernet template for Cisco vEdge devices.

Point-to-Point Protocol (PPP) is a data link protocol used to establish a direct connection between two nodes. PPP properties are associated with a PPPoE-enabled interface on Cisco vEdge devices to connect multiple users over an Ethernet link.

To configure PPPoE on Cisco vEdge device using Cisco SD-WAN Manager templates:

1. Create a VPN Interface PPP Ethernet feature template to configure a PPPoE-enabled interface as described in this article.
2. Create a VPN Interface PPP feature template to configure PPP parameters for the PPP virtual interface. See the VPN Interface PPP help topic
3. Optionally, create a VPN feature template to modify the default configuration of VPN 0. See the VPN help topic.

Navigate to the Template Screen and Name the Template

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Device Templates**, and click **Create Template**.



Note In Cisco vManage Release 20.7.x and earlier releases, **Device Templates** is titled **Device**.

3. From the **Create Template** drop-down list, select **From Feature Template**.
4. From the **Device Model** drop-down list, select the type of device for which you are creating the template.
5. Click **Transport & Management VPN** or scroll to the **Transport & Management VPN** section.
6. Under **Additional VPN 0 Templates**, click **VPN Interface PPP**.
7. From the **VPN Interface PPP Ethernet** drop-down list, click **Create Template**. The **VPN Interface PPP Ethernet** template form is displayed.
This form contains fields for naming the template, and fields for defining the VPN Interface PPP parameters.
8. In the **Template Name** field, enter a name for the template. The name can be up to 128 characters and can contain only alphanumeric characters.
9. In the **Template Description** field, enter a description of the template. The description can be up to 2048 characters and can contain only alphanumeric characters.

When you first open a feature template, for each parameter that has a default value, the scope is set to Default (indicated by a check mark), and the default setting or value is shown. To change the default or to enter a value, click the scope drop-down list and select one of the following:

Table 172:

Parameter Scope	Scope Description
Device Specific (indicated by a host icon)	Use a device-specific value for the parameter. For device-specific parameters, you cannot enter a value in the feature template. You enter the value when you attach a Viptela device to a device template . When you click Device Specific, the Enter Key box opens. This box displays a key, which is a unique string that identifies the parameter in a CSV file that you create. This file is an Excel spreadsheet that contains one column for each key. The header row contains the key names (one key per column), and each row after that corresponds to a device and defines the values of the keys for that device. You upload the CSV file when you attach a Viptela device to a device template. For more information, see Create a Template Variables Spreadsheet. To change the default key, type a new string and move the cursor out of the Enter Key box. Examples of device-specific parameters are system IP address, hostname, GPS location, and site ID.
Global (indicated by a globe icon)	Enter a value for the parameter, and apply that value to all devices. Examples of parameters that you might apply globally to a group of devices are DNS server, syslog server, and interface MTUs.

Configure a Basic PPPoE-Enabled Interface

To create a PPPoE-enabled interface on a Cisco vEdge device, select the **Basic Configuration** tab and configure the following parameters. Parameters marked with an asterisk are required to configure the interface.

Table 173:

Parameter Name	Description
Shutdown*	Click No to enable the PPPoE-enabled interface.
Interface Name*	Enter the name of the physical interface in VPN 0 to associate with the PPP interface. For Cisco IOS XE Catalyst SD-WAN devices, you must spell out the interface names completely (for example, GigabitEthernet0/0/0), and you must configure all the router's interfaces even if you are not using them so that they are configured in the shutdown state and so that all default values for them are configured.
Description	Enter a description of the PPPoE-enabled interface.
IPv4 Configuration*	To configure a static address, click Static and enter an IPv4 address. To set the interface as a DHCP client so that the interface to receive its IP address from a DHCP server, click Dynamic. You can optionally set the DHCP distance to specify the administrative distance of routes learned from a DHCP server. The default DHCP distance is 1.

Parameter Name	Description
IPv6 Configuration*	To configure a static address for an interface in VPN 0, click Static and enter an IPv6 address. To set the interface as a DHCP client so that the interface to receive its IP address from a DHCP server, click Dynamic . You can optionally set the DHCP distance to specify the administrative distance of routes learned from a DHCP server. The default DHCP distance is 1. You can optionally enable DHCP rapid commit, to speed up the assignment of IP addresses.
DHCP Helper	Enter up to eight IP addresses for DHCP servers in the network, separated by commas, to have the interface be a DHCP helper. A DHCP helper interface forwards BOOTP (Broadcast) DHCP requests that it receives from the specified DHCP servers.
Bandwidth Upstream	For transmitted traffic, set the bandwidth above which to generate notifications. <i>Range:</i> 1 through $(2^{32} / 2) - 1$ kbps
Bandwidth Downstream	For received traffic, set the bandwidth above which to generate notifications. <i>Range:</i> 1 through $(2^{32} / 2) - 1$ kbps

To save the feature template, click **Save**.

CLI equivalent:

```
vpn 0
 interface pppnumber bandwidth-downstream kbps bandwidth-upstream kbps description text
 dhcp-helper ip-address
   ( ip address ipv4-prefix/length | ip-dhcp-client [dhcp-distance number])
   ( ipv6 address ipv6-prefix/length | ipv6 dhcp-client [dhcp-distance number] [
dhcp-rapid-commit]
   pppoe-client ppp-interface pppnumber
   [no] shutdown
```

Apply Access Lists

To configure a shaping rate to a PPPoE-enabled interface and to apply a QoS map, a rewrite rule, access lists, and policers to the interface, click **ACL/QOS** and configure the following parameters:

Table 174:

Parameter Name	Description
Shaping Rate	Configure the aggregate traffic transmission rate on the interface to be less than line rate, in kilobits per second (kbps).
QoS Map	Specify the name of the QoS map to apply to packets being transmitted out the interface.
Rewrite Rule	Click On , and specify the name of the rewrite rule to apply on the interface.
Ingress ACL – IPv4	Click On , and specify the name of the access list to apply to IPv4 packets being received on the interface.
Egress ACL – IPv4	Click On , and specify the name of the access list to apply to IPv4 packets being transmitted on the interface.

Parameter Name	Description
Ingress ACL – IPv6	Egress ACL – IPv6
Egress ACL – IPv6	Egress ACL – IPv6
Ingress Policer	Click On and specify the name of the policer to apply to packets being received on the interface.
Egress Policer	Click On , and specify the name of the policer to apply to packets being transmitted on the interface.

To save the feature temp

CLI equivalent:

```
vpn 0
  interface pppnumber access-list acl-list (in | out)
    policer policer-name (in | out)
    qos-map name rewrite-rule name shaping-rate name
```

Configure Other Interface Properties

To configure other interface properties, click **Advanced** and configure the following properties:

Table 175:

Parameter Name	Description
Duplex	Choose full or half to specify whether the interface runs in full-duplex or half-duplex mode. <i>Default:</i> Full
MAC Address	Specify a MAC address to associate with the interface, in colon-separated hexadecimal notation.
IP MTU	Specify the maximum MTU size of packets on the interface. <i>Range:</i> 576 through 1804 <i>Default:</i> 1500 bytes
PMTU Discovery	Click On to enable path MTU discovery on the interface. PMTU determines the largest MTU size that the interface supports so that packet fragmentation does not occur.
Flow Control	Select a setting for bidirectional flow control, which is a mechanism for temporarily stopping the transmission of data on the interface. <i>Values:</i> autonet, both, egress, ingress, none <i>Default:</i> autoneg
TCP MSS	Specify the maximum segment size (MSS) of TCP SYN packets passing through the Cisco vEdge device. By default, the MSS is dynamically adjusted based on the interface or tunnel MTU such that TCP SYN packets are never fragmented. <i>Range:</i> 552 to 1460 bytes <i>Default:</i> None

Parameter Name	Description
Speed	Specify the speed of the interface, for use when the remote end of the connection does not support autonegotiation. <i>Values:</i> 10, 100, or 1000 Mbps <i>Default:</i> Autonegotiate (10/100/1000 Mbps)
Static Ingress QoS	Specify a queue number to use for incoming traffic. <i>Range:</i> 0 through 7
ARP Timeout	Specify how long it takes for a dynamically learned ARP entry to time out. <i>Range:</i> 0 through 2678400 seconds (744 hours) <i>Default:</i> 1200 seconds (20 minutes)
Autonegotiation	Click Off to turn off autonegotiation. By default, an interface runs in autonegotiation mode.
TLOC Extension	Enter the name of a physical interface on the same router that connects to the WAN transport. This configuration then binds this service-side interface to the WAN transport. A second Cisco vEdge device at the same site that itself has no direct connection to the WAN (generally because the site has only a single WAN connection) and that connects to this service-side interface is then provided with a connection to the WAN.
Power over Ethernet (on Cisco vEdge 100m and Cisco vEdge 100wm routers)	Click On to enable PoE on the interface.
ICMP Redirect	Click Disable to disable ICMP redirect messages on the interface. By default, an interface allows ICMP redirect messages.

To save the feature template, click **Save**.

CLI equivalent:

```

vpn 0
  interface pppnumber arp-timeout seconds
    [no] autonegotiate duplex (full | half)
    flow-control control icmp-redirect-disable mac-address mac-address mtu bytes pmtu
pppoe-client
  ppp-interface pppnumber speed speed
  static-ingress-qos number tcp-mss-adjust bytes tloc-extension interface-name

```

Release Information

Introduced in vManage NMS Release 15.3. In Release 16.3, add support for IPv6. In Release 18.2, add support for disabling ICMP redirect messages.

Configure T1 or E1 Controller Using a Configuration Group

Before you begin

On the **Configuration > Configuration Groups** page, choose **SD-WAN** as the solution type.

Procedure

- Step 1** From the Cisco SD-WAN Manager menu, choose **Configuration > Configuration Groups**.
- Step 2** Create and configure the T1 or E1 network interface module (NIM) parameters in a Transport and Management Profile.
- a. Configure a T1 Controller.
- Parameters marked with an asterisk are mandatory.

Table 176: Configure a T1 Controller

Parameter Name	Description
Slot*	Enter the number of the slot in slot/subslot/port format, where the T1 NIM is installed. For example, 0/1/0.
Description	Enter a description for the controller.
Framing	It is an optional field. Enter the T1 frame type: • esf: Send T1 frames as extended superframes. This is the default. • sf: Send T1 frames as superframes. Superframing is sometimes called D4 framing.
Line Code	It is an optional field. Select the line encoding to use to send T1 frames: • ami: Use alternate mark inversion (AMI) as the linecode. AMI signaling uses frames grouped into superframes. • b8zs: Use bipolar 8-zero substitution as the linecode. This is the default. B8ZS uses frames that are grouping into extended superframes
Cable Length	Select the cable length to configure the attenuation • short: Set the transmission attenuation for cables that are 660 feet or shorter. • long: Attenuate the pulse from the transmitter using pulse equalization and line buildout. You can configure a long cable length for cables longer than 660 feet. There is no default length.
Clock Source	Select the clock source: • line: Use phase-locked loop (PLL) on the interface. This is the default. When both T1 ports use line clocking and neither port is configured as the primary, by default, port 0 is the primary clock source and port 1 is the secondary clock source. • internal: Use the controller framer as the primary clock. • loop-timed: • network:

b. Configure an E1 Controller.

Parameters marked with an asterisk are mandatory.

Table 177: Configure an E1 Controller

Parameter Name	Description
Slot*	Enter the number of the slot in slot/subslot/port format, where the E1 NIM is installed. For example, 0/1/0.
Description	Enter a description for the controller.
Framing	Enter the E1 frame type: • crc4: Use cyclic redundancy check 4 (CRC4). This is the default. • no-crc4: Do not use CRC4.
Line Code	Choose the line encoding to use to send E1 frames: • ami: Use alternate mark inversion (AMI) as the linecode. • hdb3: Use high-density bipolar 3 as the linecode. This is the default.
Clock Source	Choose the clock source: • internal: Use the controller framer as the primary clock. • line: Use phase-locked loop (PLL) on the interface. This is the default.

c. Configure Channel Group.

Table 178: Channel Group

Parameter Name	Description
Add Channel Group	To configure the serial WAN on the E1 interface, enter a channel group number and a value for the timeslot. • Channel Group: Enter a value for the channel group. Range: 0 through 30 • Time Slot: Type a value for the timeslot. Range: 0 through 31

What to do next

Also see [Deploy a configuration group](#).

Cellular Interfaces

To enable LTE connectivity, configure cellular interfaces on a router that has a cellular module. The cellular module provides wireless connectivity over a service provider's cellular network. One use case is to provide wireless connectivity for branch offices.

A cellular network is commonly used as a backup WAN link, to provide network connectivity if all the wired WAN tunnel interfaces on the router become unavailable. You can also use a cellular network as the primary WAN link for a branch office, depending on usage patterns within the branch office and the data rates supported by the core of the service provider's cellular network.

When you configure a cellular interface on a device, you can connect the device to the Internet or another WAN by plugging in the power cable of the device. The device then automatically begins the process of joining the overlay network, by contacting and authenticating with Cisco Catalyst SD-WAN Validators, Cisco Catalyst SD-WAN Controllers, and Cisco SD-WAN Manager systems.

vEdge routers support LTE and CDMA radio access technology (RAT) types.

Configure Cellular Interfaces Using a Configuration Groups

Before you begin

On the **Configuration > Configuration Groups** page, choose **SD-WAN** as the solution type.

Procedure

Step 1 From the Cisco SD-WAN Manager menu, choose **Configuration > Configuration Groups**.

Step 2 Create and configure a Transport VPN feature in Transport and Management profile.

Step 3 Create and configure Cellular Interface feature in Transport VPN.

a) Configure basic cellular interface functionality.

Field	Description
Shutdown*	Enable or disable the interface.
Interface Name*	Enter the name of the interface.
Description*	Enter a description of the cellular interface.
DHCP Helper	Enter up to four IP addresses for DHCP servers in the network, separated by commas, to have the interface be a DHCP helper. A DHCP helper interface forwards BOOTP (Broadcast) DHCP requests that it receives from the specified DHCP servers.

b) Configure a tunnel interface.

Field	Description
Tunnel Interface	Enable this option to create a tunnel interface.

Field	Description
Carrier	Choose the carrier name or private network identifier to associate with the tunnel. Values: carrier1, carrier2, carrier3, carrier4, carrier5, carrier6, carrier7, carrier8, default Default: default
Color	Choose a color for the TLOC.
Color Description	Minimum supported release: Cisco Catalyst SD-WAN Manager Release 20.18.1 Enter a description associated to the TLOC color.
Hello Interval	Enter the interval between Hello packets sent on a DTLS or TLS WAN transport connection. Range: 100 through 600000 milliseconds Default: 1000 milliseconds (1 second)
Hello Tolerance	Enter the time to wait for a Hello packet on a DTLS or TLS WAN transport connection before declaring that transport tunnel to be down. Range: 12 through 6000 seconds Default: 12 seconds
Last-Resort Circuit	Enable this option to use the tunnel interface as the circuit of last resort.
Restrict	Enable this option to limit the remote TLOCs that the local TLOC can establish BFD sessions with. When a TLOC is marked as restricted, a TLOC on the local router establishes tunnel connections with a remote TLOC only if the remote TLOC has the same color.
Group	Enter a group number. Range: 1 through 4294967295
Border	Enable this option to set the TLOC as a border TLOC.
Maximum Control Connections	Specify the maximum number of Cisco SD-WAN Controllers that the WAN tunnel interface can connect to. To have the tunnel establish no control connections, set the number to 0. Range: 0 through 100 Default: 2
NAT Refresh Interval	Enter the interval between NAT refresh packets sent on a DTLS or TLS WAN transport connection. Range: 1 through 60 seconds Default: 5 seconds

Field	Description
Validator As Stun Server	Enable Session Traversal Utilities for NAT (STUN) to allow the tunnel interface to discover its public IP address and port number when the Cisco IOS XE Catalyst SD-WAN device is located behind a NAT.
Exclude Controller Group List	Set the identifiers of one or more Cisco SD-WAN Controller groups that this tunnel is not allowed to connect to. Range: 1 through 100
Manager Connection Preference	Set the preference for using a tunnel interface to exchange control traffic with Cisco SD-WAN Manager. Range: 0 through 8 Default: 5
Full Port Hop	Minimum release: Cisco IOS XE Catalyst SD-WAN Release 17.18.1a Enable full port hopping at the TLOC level to allow devices to establish connections with controllers by switching to the next port if the current port is blocked or non-functional. Default: Disabled
Port Hop	Enable port hopping. When a router is behind a NAT, port hopping rotates through a pool of preselected OMP port numbers (called base ports) to establish DTLS connections with other routers when a connection attempt is unsuccessful. The default base ports are 12346, 12366, 12386, 12406, and 12426. To modify the base ports, set a port offset value. Default: Enabled Starting from Cisco IOS XE Catalyst SD-WAN Release 17.18.1a, this field is deprecated. Instead use the Full Port Hop option. See the Full Port Hop field.
Low-Bandwidth Link	Enable the low-bandwidth feature.
Tunnel TCP MSS	Specify the maximum segment size (MSS) of TCP SYN packets passing through the router. By default, the MSS is dynamically adjusted based on the interface or tunnel MTU such that TCP SYN packets are never fragmented. Range: 500 to 1460 bytes Default: None
Clear-Dont-Fragment	Enable this option to clear the Don't Fragment (DF) bit in the IPv4 packet header for packets being transmitted out the interface. When the DF bit is cleared, packets larger than the MTU of the interface are fragmented before being sent.
Network Broadcast	Enable this option to accept and respond to network-prefix-directed broadcasts.

Field	Description
Allow Service	Allow or disallow the following services on the interface: • All • BGP • DHCP • NTP • SSH • DNS • ICMP • HTTPS • OSPF • STUN • SNMP • NETCONF • BFD
Encapsulation	
GRE	Use GRE encapsulation on the tunnel interface. By default, GRE is disabled. If you select both IPsec and GRE encapsulations, two TLOCs are created for the tunnel interface that have the same IP addresses and colors, but that differ by their encapsulation.
GRE Preference	Specify a preference value for directing traffic to the tunnel. A higher value is preferred over a lower value. Range: 0 through 4294967295 Default: 0
GRE Weight	Enter a weight to use to balance traffic across multiple TLOCs. A higher value sends more traffic to the tunnel. Range: 1 through 255 Default: 1
IPsec	Use IPsec encapsulation on the tunnel interface. By default, IPsec is enabled. If you select both IPsec and GRE encapsulations, two TLOCs are created for the tunnel interface that have the same IP addresses and colors, but that differ by their encapsulation.

Field	Description
IPsec Preference	Specify a preference value for directing traffic to the tunnel. A higher value is preferred over a lower value. Range: 0 through 4294967295 Default: 0
IPsec Weight	Enter a weight to use to balance traffic across multiple TLOCs. A higher value sends more traffic to the tunnel. Range: 1 through 255 Default: 1

- c) Configure a cellular interface to act as a NAT device for applications such as port forwarding.

Field	Description
NAT	Enable this option to have the interface act as a NAT device.
UDP Timeout*	Specify when NAT translations over UDP sessions time out. Range: 1 through 8947 minutes Default: 1 minutes
TCP Timeout*	Specify when NAT translations over TCP sessions time out. Range: 1 through 8947 minutes Default: 60 minutes (1 hour)

- d) Configure static Address Resolution Protocol (ARP) table entries on the interface.

Field	Description
IP Address*	Enter the IP address for the ARP entry in dotted decimal notation or as a fully qualified host name.
MAC Address*	Enter the MAC address in colon-separated hexadecimal notation.

- e) Configure other interface properties.

Field	Description
MAC Address	Specify a MAC address to associate with the interface, in colon-separated hexadecimal notation.
IP MTU	Specify the maximum MTU size of packets on the interface. Range: 576 through 9216 Default: 1500 bytes

Field	Description
Interface MTU	Enter the maximum transmission unit size for frames received and transmitted on the interface. Range: 1500 through 9216 Default: 1500 bytes
TCP MSS	Specify the maximum segment size (MSS) of TCP SYN packets passing through the router. By default, the MSS is dynamically adjusted based on the interface or tunnel MTU such that TCP SYN packets are never fragmented. Range: 500 to 1460 bytes Default: None
TLOC Extension	Enter the name of a physical interface on the same router that connects to the WAN transport. This configuration then binds this service-side interface to the WAN transport. A second router at the same site that itself has no direct connection to the WAN (generally because the site has only a single WAN connection) and that connects to this service-side interface is then provided with a connection to the WAN. Note TLOC extension over L3 is supported only for Cisco IOS XE Catalyst SD-WAN devices. If configuring TLOC extension over L3 for a Cisco IOS XE Catalyst SD-WAN device, enter the IP address of the L3 interface.
Tracker	Tracking the interface status is useful when you enable NAT on a transport interface in VPN 0 to allow data traffic from the router to exit directly to the internet rather than having to first go to a router in a data center. In this situation, enabling NAT on the transport interface splits the TLOC between the local router and the data center into two, with one going to the remote router and the other going to the internet. When you enable transport tunnel tracking, Cisco Catalyst SD-WAN periodically probes the path to the internet to determine whether it is up. If Cisco Catalyst SD-WAN detects that this path is down, it withdraws the route to the internet destination, and traffic destined to the internet is then routed through the data center router. When Cisco Catalyst SD-WAN detects that the path to the internet is again functioning, the route to the internet is reinstalled. Enter the name of a tracker to track the status of transport interfaces that connect to the internet.

Field	Description
IP Directed-Broadcast	An IP directed broadcast is an IP packet whose destination address is a valid broadcast address for some IP subnet but which originates from a node that is not itself part of that destination subnet. A device that is not directly connected to its destination subnet forwards an IP directed broadcast in the same way it would forward unicast IP packets destined to a host on that subnet. When a directed broadcast packet reaches a device that is directly connected to its destination subnet, that packet is broadcast on the destination subnet. The destination address in the IP header of the packet is rewritten to the configured IP broadcast address for the subnet, and the packet is sent as a link-layer broadcast. If directed broadcast is enabled for an interface, incoming IP packets whose addresses identify them as directed broadcasts intended for the subnet to which that interface is attached are broadcast on that subnet.

What to do next

Also see [Deploy a Configuration Group](#).

Configure Cellular Interfaces Using Templates

To configure cellular interfaces using Cisco SD-WAN Manager templates:

1. Create a VPN Interface Cellular feature template to configure cellular module parameters, as described in this section.
2. Create a Cellular Profile template to configure the profiles used by the cellular modem.
3. Create a VPN feature template to configure VPN parameters.

Create VPN Interface Cellular

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Device Templates**.



Note In Cisco vManage Release 20.7.x and earlier releases, **Device Templates** is titled **Device**.

3. From the **Create Template** drop-down list, choose **From Feature Template**.
4. From the **Device Model** drop-down list, select the type of device for which you are creating the template.
5. Click **Transport & Management VPN** or scroll to the **Transport & Management VPN** section.
6. Under **Additional Cisco VPN 0 Templates**, click **VPN Interface Cellular**.

7. From the **VPN Interface Cellular** drop-down list, click **Create Template**. The VPN Interface Cellular template form is displayed.

This form contains fields for naming the template, and fields for defining the VPN Interface Cellular parameters.
8. In **Template Name**, enter a name for the template. The name can be up to 128 characters and can contain only alphanumeric characters.
9. In **Template Description**, enter a description of the template. The description can be up to 2048 characters and can contain only alphanumeric characters.

When you first open a feature template, for each parameter that has a default value, the scope is set to Default (indicated by a check mark), and the default setting or value is shown. To change the default or to enter a value, click the scope drop-down list.

Configure Basic Cellular Interface Functionality

To configure basic cellular interface functionality, click **Basic Configuration** and configure the following parameters. Parameters marked with an asterisk are required to configure an interface. You must also configure a tunnel interface for the cellular interface.

Table 179:

Parameter Name	Description
Shutdown*	Click No to enable the interface.
Technology	Cellular technology. The default is lte . Other values are auto and cdma . For ZTP to work, the technology must be auto . For Cisco ISR 1100 and ISR 1100X Series Routers operating with an LTE cellular module (LTE dongle), configure the value as lte .
Interface Name*	Enter the name of the interface. It must be cellular0 .
Profile ID*	Enter the identification number of the cellular profile. This is the profile identifier that you configure in the Cellular-Profile template. Range: 1 through 15.
Description	Enter a description of the cellular interface.
IPv4 Configuration	To configure a static address, click Static and enter an IPv4 address. To set the interface as a DHCP client so that the interface to receive its IP address from a DHCP server, click Dynamic . You can optionally set the DHCP distance to specify the administrative distance of routes learned from a DHCP server. The default DHCP distance is 1.
IPv6 Configuration	To configure a static address for an interface in VPN 0, click Static and enter an IPv6 address. To set the interface as a DHCP client so that the interface to receive its IP address from a DHCP server, click Dynamic . You can optionally set the DHCP distance to specify the administrative distance of routes learned from a DHCP server. The default DHCP distance is 1. You can optionally enable DHCP rapid commit, to speed up the assignment of IP addresses.

Parameter Name	Description
DHCP Helper	Enter up to four IP addresses for DHCP servers in the network, separated by commas, to have the interface be a DHCP helper. A DHCP helper interface forwards BOOTP (Broadcast) DHCP requests that it receives from the specified DHCP servers.
Block Non-Source IP	Click Yes to have the interface forward traffic only if the source IP address of the traffic matches the interface's IP prefix range.
Bandwidth Upstream	For transmitted traffic, set the bandwidth above which to generate notifications. Range: 1 through $(2^{32} / 2) - 1$ kbps
Bandwidth Downstream	For received traffic, set the bandwidth above which to generate notifications. Range: 1 through $(2^{32} / 2) - 1$ kbps
IP MTU*	Enter 1428 to set the MTU size, in bytes. This value must be 1428. You cannot use a different value.

To save the feature template, click **Save**.

CLI equivalent:

```
vpn 0
  interface cellular0
    bandwidth-downstream kbps bandwidth-upstream kbps block-non-source-ip ( ip address
ip-address/length | ip dhcp-client [dhcp-distance number])
    ( ipv6 address ipv6-prefix/length | ipv6 dhcp-client [dhcp-distance number]
[dhcp-rapid-comit])
    mtu 1428
    profile number
    no shutdown
```

Create a Tunnel Interface

To configure an interface in VPN 0 to be a WAN transport connection, you must configure a tunnel interface on the cellular interface. The tunnel, which provides security from attacks, is used to send the phone number. At a minimum, select **On** and select a color for the interface, as described in the previous section. You can generally accept the system defaults for the remainder of the tunnel interface settings.

To configure a tunnel interface, click **Tunnel**, and configure the following parameters. Parameters marked with an asterisk (*) are required to configure a cellular interface.

Parameter Name	Description
Tunnel Interface*	From the drop-down, select Global . Click On to create a tunnel interface.
Per-tunnel QoS	From the drop-down, select Global . Click On to create per-tunnel QoS. You can apply a Quality of Service (QoS) policy on individual tunnels, and is only supported for hub-to-spoke network topologies.
Per-tunnel QoS Aggregator	From the drop-down, select Global . Click On to create per-tunnel QoS. Note 'bandwidth downstream' is required for per-Tunnel QoS feature to take effect as spoke role.

Parameter Name	Description
Color*	From the drop-down, select Global . Select a color for the TLOC. The color typically used for cellular interface tunnels is lte .
Color Description	Minimum supported release: Cisco Catalyst SD-WAN Manager Release 20.18.1 Enter a description associated to the TLOC color.
Groups	From the drop-down, select Global . Enter the list of groups in the field.
Border	From the drop-down, select Global . Click On to set TLOC as border TLOC.
Maximum Control Connections	Set the maximum number of Cisco SD-WAN Controller that the WAN tunnel interface can connect to. To have the tunnel establish no control connections, set the number to 0. Range: 0 through 8 Default: 2
vBond As STUN Server	Click On to enable Session Traversal Utilities for NAT (STUN) to allow the tunnel interface to discover its public IP address and port number when the router is located behind a NAT.
Exclude Control Group List	Set the identifiers of one or more Cisco SD-WAN Controller groups that this tunnel is not allows to establish control connections with. Range: 0 through 100
vManage Connection Preference	Set the preference for using the tunnel to exchange control traffic with the Cisco SD-WAN Manager. Range: 0 through 9 Default: 5
Full Port Hop	Minimum release: Cisco IOS XE Catalyst SD-WAN Release 17.18.1a Enable full port hopping at the TLOC level to allow devices to establish connections with controllers by switching to the next port if the current port is blocked or non-functional. Default: Disabled
Port Hop	From the drop-down, select Global . Click Off to allow port hopping on tunnel interface. Default: On , which disallows port hopping on tunnel interface. Starting from Cisco IOS XE Catalyst SD-WAN Release 17.18.1a, this field is deprecated. Instead use the Full Port Hop option. See the Full Port Hop field.
Low-Bandwidth Link	Click On to set the tunnel interface as a low-bandwidth link. Default: Off

Parameter Name	Description
Allow Service	Click On or Off for each service to allow or disallow the service on the cellular interface.

To configure additional tunnel interface parameters, click **Advanced Options** and configure the following parameters:

Table 180:

Parameter Name	Description
GRE	From the drop-down, select Global . Click On to use GRE encapsulation on the tunnel interface. By default, GRE is disabled. If you select both IPsec and GRE encapsulations, two TLOCs are created for the tunnel interface that have the same IP addresses and colors, but that differ by their encapsulation.
GRE Preference	From the drop-down, select Global . Enter a value to set GRE preference for TLOC. Range: 0 to 4294967295
GRE Weight	From the drop-down, select Global . Enter a value to set GRE weight for TLOC. Default: 1
IPsec	From the drop-down, select Global . Click On to use IPsec encapsulation on the tunnel interface. By default, IPsec is enabled. If you select both IPsec and GRE encapsulations, two TLOCs are created for the tunnel interface that have the same IP addresses and colors, but that differ by their encapsulation.
IPsec Preference	From the drop-down, select Global . Enter a value to set the preference for directing traffic to the tunnel. A higher value is preferred over a lower value. Range: 0 through 4294967295. Default: 0
IPsec Weight	From the drop-down, select Global . Enter a value to set weight for balancing traffic across multiple TLOCs. A higher value sends more traffic to the tunnel. Range: 1 through 255. Default: 1
Carrier	From the drop-down, select Global . From the Carrier drop-down, select the carrier name or private network identifier to associate with the tunnel. Values: carrier1, carrier2, carrier3, carrier4, carrier5, carrier6, carrier7, carrier8, default. Default: default
Bind Loopback Tunnel	Enter the name of a physical interface to bind to a loopback interface. The interface name has the format ge slot/port .

Parameter Name	Description
Last-Resort Circuit	From the drop-down, select Global. Click On to use the tunnel interface as the circuit of last resort. By default, it is disabled. Note An interface configured as a circuit of last resort is expected to be down and is skipped while calculating the number of control connections, the cellular modem becomes dormant, and no traffic is sent over the circuit. When the configurations are activated on the edge device with cellular interfaces, then all the interfaces begin the process of establishing control and BFD connections. When one or more of the primary interfaces establishes a BFD connection, the circuit of last resort shuts itself down. Only when all the primary interfaces lose their connections to remote edges, then the circuit of last resort activates itself triggering a BFD TLOC Down alarm and a Control TLOC Down alarm on the edge device. The last resort interfaces are used as backup circuit on edge device and are activated when all other transport links BFD sessions fail. In this mode the radio interface is turned off, and no control or data connections exist over the cellular interface.
Hold Time	From the drop-down, select Global. Enter a value to set last resort hold down time for TLOC. Range: 100 to 10000 msec. Default: 7000 ms.
NAT Refresh Interval	Set the interval between NAT refresh packets sent on a DTLS or TLS WAN transport connection. Range: 1 through 60 seconds. Default: 5 seconds.
Hello Interval	Enter the interval between Hello packets sent on a DTLS or TLS WAN transport connection. Range: 100 through 10000 milliseconds. Default: 1000 milliseconds (1 second).
Hello Tolerance	Enter the time to wait for a Hello packet on a DTLS or TLS WAN transport connection before declaring that transport tunnel to be down. Range: 12 through 60 seconds. Default: 12 seconds.

To save the feature template, click **Save**.

CLI equivalent:

```

vpn 0
  interface cellular0
    tunnel-interface allow-service service-name
  bind interface-name carrier carrier-name
    color color encapsulation (gre | ipsec)
    preference number
    weight number exclude-controller-group-list number hello-interval milliseconds
    hello-tolerance seconds hold-time milliseconds low-bandwidth-link
max-control-connections number last-resort-circuit nat-refresh-interval seconds
vbond-as-stun-server vmanage-connection-preference number

```

Configure the Cellular Interface as a NAT Device

To configure a cellular interface to act as a NAT device for applications such as port forwarding, click **NAT**, and configure the following parameters:

Table 181:

Parameter Name	Description
NAT	Click On to have the interface act as a NAT device.
Refresh Mode	Select how NAT mappings are refreshed, either outbound or bidirectional (outbound and inbound). Default: Outbound
UDP Timeout	Specify when NAT translations over UDP sessions time out. Range: 1 through 65536 minutes. Default: 1 minute
TCP Timeout	Specify when NAT translations over TCP sessions time out. Range: 1 through 65536 minutes. Default: 60 minutes (1 hour)
Block ICMP	Select On to block inbound ICMP error messages. By default, a router acting as a NAT device receives these error messages. Default: Off
Respond to Ping	Select On to have the router respond to ping requests to the NAT interface's IP address that are received from the public side of the connection.

To create a port forwarding rule, click **Add New Port Forwarding Rule** and configure the following parameters. You can define up to 128 port-forwarding rules to allow requests from an external network to reach devices on the internal network.

Table 182:

Parameter Name	Description
Port Start Range	Enter a port number to define the port or first port in the range of interest. Range: 0 through 65535
Port End Range	Enter the same port number to apply port forwarding to a single port, or enter a larger number to apply it to a range of ports. Range: 0 through 65535
Protocol	Select the protocol to which to apply the port-forwarding rule, either TCP or UDP. To match the same ports for both TCP and UDP traffic, configure two rules.
VPN	Specify the private VPN in which the internal server resides. This VPN is one of the VPN identifiers in the overlay network. Range: 1 to 65525, excluding 512. For details see the VRF range behavior change described here .
Private IP	Specify the IP address of the internal server to which to direct traffic that matches the port-forwarding rule.

To save a port forwarding rule, click **Add**.

To save the feature template, click **Save**.

CLI equivalent:

```
vpn 0
interface cellular0
    nat block-icmp-error port-forward port-start port-number1 port-end port-number2
        proto (tcp | udp) private-ip-address ip address private-vpn vpn-id refresh
    (bi-directional | outbound)
        respond-to-ping tcp-timeout minutes
    udp-timeout minutes
```

Apply Access Lists

To configure a shaping rate to a cellular interface and to apply a QoS map, a rewrite rule, access lists, and policers to a router interface, click **ACL/QoS** and configure the following parameters:

Table 183: Access Lists Parameters

Parameter Name	Description
Shaping rate	Configure the aggregate traffic transmission rate on the interface to be less than line rate, in kilobits per second (kbps).
QoS map	Specify the name of the QoS map to apply to packets being transmitted out the interface.
Rewrite rule	Click On , and specify the name of the rewrite rule to apply on the interface.
Ingress ACL – IPv4	Click On , and specify the name of an IPv4 access list to packets being received on the interface.
Egress ACL– IPv4	Click On , and specify the name of an IPv4 access list to packets being transmitted on the interface.
Ingress ACL – IPv6	Click On , and specify the name of an IPv6 access list to packets being received on the interface.
Egress ACL– IPv6	Click On , and specify the name of an IPv6 access list to packets being transmitted on the interface.
Ingress policer	Click On , and specify the name of the policer to apply to packets being received on the interface.
Egress policer	Click On , and specify the name of the policer to apply to packets being transmitted on the interface.

To save the feature template, click **Save**.

CLI equivalent:

```
vpn 0
interface cellular0
    access-list acl-name (in | out)
    ipv6 access-list acl-name (in | out)
    policer policer-name (in |out)
    qos-map name rewrite-rule name shaping-rate name
```

Add ARP Table Entries

To configure static Address Resolution Protocol (ARP) table entries on the interface, click **ARP**. Then click **Add New ARP** and configure the following parameters:

Table 184:

Parameter Name	Description
IP Address	Enter the IP address for the ARP entry in dotted decimal notation or as a fully qualified host name.
MAC Address	Enter the MAC address in colon-separated hexadecimal notation.

To save the ARP configuration, click **Add**.

To save the feature template, click **Save**.

CLI equivalent:

```
vpn vpn-id interface irbnumber arp
ip address ip-address mac mac-address
```

Configure Other Interface Properties

To configure other interface properties, click **Advanced** and configure the following parameters.

Table 185: Cellular Interfaces Advanced Parameters

Parameter Name	Description
PMTU Discovery	Click On to enable path MTU discovery on the interface, to allow the router to determine the largest MTU size supported without requiring packet fragmentation.
TCP MSS	Specify the maximum segment size (MSS) of TCP SYN packets passing through the router. By default, the MSS is dynamically adjusted based on the interface or tunnel MTU such that TCP SYN packets are never fragmented. Range: 552 to 1460 bytes. Default: None.
Clear-Dont-Fragment	Click On to clear the Don't Fragment (DF) bit in the IPv4 packet header for packets being transmitted out the interface. When the DF bit is cleared, packets larger than that interface's MTU are fragmented before being sent.
Static Ingress QoS	Select a queue number to use for incoming traffic. Range: 0 through 7
ARP Timeout	Specify how long it takes for a dynamically learned ARP entry to time out. Range: 0 through 2678400 seconds (744 hours). Default: 1200 seconds (20 minutes)
Autonegotiate	Click Off to turn off autonegotiation. By default, an interface runs in autonegotiation mode.

Parameter Name	Description
TLOC Extension	Enter the name of a physical interface on the same router that connects to the WAN transport. This configuration then binds this service-side interface to the WAN transport. A second router at the same site that itself has no direct connection to the WAN (generally because the site has only a single WAN connection) and that connects to this service-side interface is then provided with a connection to the WAN.
Tracker	Enter the name of a tracker to track the status of transport interfaces that connect to the internet.
ICMP Redirect	Click Disable to disable ICMP redirect messages on the interface. By default, an interface allows ICMP redirect messages.

To save the feature template, click **Save**.

CLI equivalent:

```
vpn 0
 interface cellular0
 arp-timeout seconds
 [no] autonegotiate clear-dont-fragment icmp-redirect-disable mtu 1428
 pmtu static-ingress-qos number tcp-mss-adjust bytes
 tloc-extension interface-name tracker tracker-name
```

Release Information

Introduced in Cisco SD-WAN Manager in Release 16.1. In Release 16.2, add circuit of last resort and its associated hold time. In Release 16.3, add support for IPv6. In Release 17.2.2, add support for tracker interface status. In Release 18.2, add support for disabling ICMP redirect messages.

Configuring Cellular Interfaces Using the CLI

To configure a cellular interface on a Cisco vEdge device that has a cellular module:

1. Create a cellular profile:

```
vEdge(config)# cellular cellular number
vEdge(config-cellular)# profile profile-id
```

Each Cisco vEdge device has only one LTE module, so *number* must be 0. The profile identifier can be a value from 1 through 15.

2. If your ISP requires that you configure profile properties, configure one or more of the following:

```
vEdge(config-profile)# apn
name
vEdge(config-profile)# auth auth-method
vEdge(config-profile)# ip-addr ip-address
vEdge(config-profile)# name name
vEdge(config-profile)# pdn-type type
vEdge(config-profile)# primary-dns ip-address
vEdge(config-profile)# secondary-dns ip-address
vEdge(config-profile)# user-name username
vEdge(config-profile)# user-pass password
```

1. Create the cellular interface:

```
vEdge(config)# vpn 0 interface cellular0
```

2. Enable the cellular interface:

```
vEdge(config-interface)# no shutdown
```

3. For cellular interfaces, you must use a DHCP client to dynamically configure the IP address. This is the default option. To explicitly configure this:

```
vEdge(config-interface)# ip dhcp-client [dhcp-distance number]
```

number is the administrative distance of routes learned from a DHCP server. You can configure it to a value from 1 through 255.

4. Associate the cellular profile with the cellular interface:

```
vEdge(config-interface)# profile profile-id
```

The profile identifier is the number you configured in Step 1.

5. Set the interface MTU:

```
vEdge(config-interface)# mtu bytes
```

The MTU can be 1428 bytes or smaller.

6. By default, the radio access technology (RAT) type is LTE. For 2G/3G networks, change it to CDMA:

```
vEdge(config-interface)# technology cdma
```

If you are using the interface for ZTP, change the technology to **auto**:

```
vEdge(config-interface)# technology auto
```

7. Configure any other desired interface properties.

8. Create a tunnel interface on the cellular interface:

```
vEdge(config-interface)# tunnel-interface
vEdge(config-tunnel-interface)# color color
vEdge(config-tunnel-interface)# encapsulation (gre | ipsec)
```

9. By default, the tunnel interface associated with a cellular interface is not considered to be the circuit of last resort. To allow the tunnel to be the circuit of last resort:

```
vEdge(config-tunnel-interface)# last-resort-circuit
```

An interface configured as a circuit of last resort is expected to be down and is skipped while calculating the number of control connections, the cellular modem becomes dormant, and no traffic is sent over the circuit.

When the configurations are activated on the edge device with cellular interfaces, then all the interfaces begin the process of establishing control and BFD connections. When one or more of the primary interfaces establishes a BFD connection, the circuit of last resort shuts itself down.

Only when all the primary interfaces lose their connections to remote edges, then the circuit of last resort activates itself triggering a BFD TLOC Down alarm and a Control TLOC Down alarm on the edge device. The last resort interfaces are used as backup circuit on edge device and are activated when all other transport links BFD sessions fail. In this mode the radio interface is turned off, and no control or data connections exist over the cellular interface.

10. To minimize the amount of control plane keepalive traffic on the cellular interface, increase the Hello packet interval and tolerance on the tunnel interface:

```
vEdge(config-tunnel-interface) # hello-interval milliseconds
vEdge(config-tunnel-interface) # hello-tolerance seconds
```

The default hello interval is 1000 milliseconds, and it can be a time in the range 100 through 600000 milliseconds (10 minutes). The default hello tolerance is 12 seconds, and it can be a time in the range 12 through 600 seconds (10 minutes). To reduce outgoing control packets on a TLOC, it is recommended that on the tunnel interface you set the hello interval to 60000 milliseconds (10 minutes) and the hello tolerance to 600 seconds (10 minutes) and include the **no track-transport** to disable regular checking of the DTLS connection between the Cisco vEdge device and the vBond orchestrator. For a tunnel connection between a Cisco vEdge device and any controller device, the tunnel uses the hello interval and tolerance times configured on the Cisco vEdge device. This choice is made to minimize the amount of traffic sent over the tunnel, to allow for situations where the cost of a link is a function of the amount of traffic traversing the link. The hello interval and tolerance times are chosen separately for each tunnel between a Cisco vEdge device and a controller device. Another step taken to minimize the amount of control plane traffic is to not send or receive OMP control traffic over a cellular interface when other interfaces are available. This behavior is inherent in the software and is not configurable.

11. If the Cisco vEdge device has two or more cellular interfaces, you can minimize the amount of traffic between the vManage NMS and the cellular interfaces by setting one of the interfaces to be the preferred one to use when sending updates to the vManage NMS and receiving configurations from the vManage NMS:

```
vEdge(config-tunnel-interface) # vmanage-connection-preference number
```

The preference can be a value from 0 through 8. The default preference is 5. To have a tunnel interface never connect to the vManage NMS, set the number to 0. At least one tunnel interface on the Cisco vEdge device must have a nonzero vManage connection preference.

12. Configure any other desired tunnel interface properties.
13. To minimize the amount of data plane keepalive traffic on the cellular interface, increase the BFD Hello packet interval:

```
vEdge(bfd-color-lte) # hello-interval milliseconds
```

The default hello interval is 1000 milliseconds (1 second), and it can be a time in the range 100 through 300000 milliseconds (5 minutes).

To determine the status of the cellular hardware, use the **show cellular status** command.

To determine whether a Cisco vEdge device has a cellular module, use the **show hardware inventory** command.

To determine whether a cellular interface is configured as a last-resort circuit, use the **show control affinity config** and **show control local-properties** commands.



Note

If you want to remove a property from the cellular profile, delete the profile entirely from the configuration, and create it again with only the required parameters.



Note An interface configured as a circuit of last resort is expected to be down and is skipped while calculating the number of control connections, the cellular modem becomes dormant, and no traffic is sent over the circuit.

When the configurations are activated on the edge device with cellular interfaces, then all the interfaces begin the process of establishing control and BFD connections. When one or more of the primary interfaces establishes a BFD connection, the circuit of last resort shuts itself down.

Only when all the primary interfaces lose their connections to remote edges, then the circuit of last resort activates itself triggering a BFD TLOC Down alarm and a Control TLOC Down alarm on the edge device. The last resort interfaces are used as backup circuit on edge device and are activated when all other transport links BFD sessions fail. In this mode the radio interface is turned off, and no control or data connections exist over the cellular interface.

Best Practices for Configuring Cellular Interfaces

Cellular technology on edge devices can be used in a number of ways:

- **Circuit of last resort:** An interface configured as a circuit of last resort is expected to be down and is skipped while calculating the number of control connections, the cellular modem becomes dormant, and no traffic is sent over the circuit.

When the configurations are activated on the edge device with cellular interfaces, then all the interfaces begin the process of establishing control and BFD connections. When one or more of the primary interfaces establishes a BFD connection, the circuit of last resort shuts itself down.

Only when all the primary interfaces lose their connections to remote edges, then the circuit of last resort activates itself triggering a BFD TLOC Down alarm and a Control TLOC Down alarm on the edge device. The last resort interfaces are used as backup circuit on edge device and are activated when all other transport links BFD sessions fail. In this mode the radio interface is turned off, and no control or data connections exist over the cellular interface.

Use the **last-resort-circuit** command to configure a cellular interface to be a circuit of last resort.



Note **last-resort-circuit** is not limited to cellular interfaces.

The operating principle for cellular interfaces also applies to GigabitEthernet interfaces.

- **Active circuit:** You can choose to use a cellular interface as an active circuit, perhaps because it is the only last-mile circuit or to always keep the cellular interface active so that you can measure the performance of the circuit. In this scenario the amount of bandwidth utilized to maintain control and data connections over the cellular interface can become a concern. Here are some best practices to minimize bandwidth usage over a cellular interface:
 - **Increase control packet timers**—To minimize control traffic on a cellular interface, you can decrease how often protocol update messages are sent on the interface. OMP sends Update packets every second, by default. You can increase this interval to a maximum of 65535 seconds (about 18 hours) by including the **omp timers advertisement-interval** configuration command. BFD sends Hello packets every second, by default. You can increase this interval to a maximum of 5 minutes (300000 milliseconds) by including the **bfd color hello-interval** configuration command. (Note that you

specify the OMP Update packet interval in seconds and the BFD Hello packet interval in milliseconds.)

- **Prioritize Cisco SD-WAN Manager control traffic over a non-cellular interface:** When a edge device has both cellular and non-cellular transport interfaces, by default, the edge device chooses one of the interfaces to use to exchange control traffic with the Cisco SD-WAN Manager. You can configure the edge device to never use the cellular interface to exchange traffic with the Cisco SD-WAN Manager, or you can configure a lower preference for using the cellular interface for this traffic. You configure the preference by including the **vmanage-connection-preference** command when configuring the tunnel interface. By default, all tunnel interface have a Cisco SD-WAN Manager connection preference value of 5. The value can range from 0 through 8, where a higher value is more preferred. A tunnel with a preference value of 0 can never exchange control traffic with the Cisco SD-WAN Manager.

**Note**

At least one tunnel interface on the edge device must have a non-0 Cisco SD-WAN Manager connection preference value. Otherwise, the device has no control connections.

WiFi Radio

Use the WiFi Radio template for all devices that support wireless LANs (WLANs).

To configure WLAN radio parameters using Cisco SD-WAN Manager templates:

1. Create a WiFi Radio template to configure WLAN radio parameters, as described in this article.
2. Create a Wifi SSID template to configure an SSID and related parameters.

Create WLAN Feature Template

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Device Templates**, and click **Create Template**.

**Note**

In Cisco vManage Release 20.7.x and earlier releases, **Device Templates** is titled **Device**.

3. From the **Create Template** drop-down list, choose **From Feature Template**.
4. From the **Device Model** drop-down list, select the device model that supports wireless LANs (WLANs).
5. Click **WLAN**, or scroll to the **WLAN** section.
6. From the **WiFi Radio** drop-down list, click **Create Template**. The **WiFi Radio** template form is displayed. This form contains fields for naming the template, and fields for defining the WiFi Radio parameters.
7. In **Template Name**, enter a name for the template. The name can be up to 128 characters and can contain only alphanumeric characters.

8. In **Template Description**, enter a description of the template. The description can be up to 2048 characters and can contain only alphanumeric characters.

When you first open a feature template, for each parameter that has a default value, the scope is set to Default (indicated by a check mark), and the default setting or value is shown. To change the default or to enter a value, click the scope drop-down to the left of the parameter field.

Configure the WLAN Radio Frequency

To configure the WLAN radio frequency, click **Basic Configuration**, and configure the following parameters. Parameters marked with an asterisk are required to configure the radio.

Table 186:

Parameter Name	Description
Select Radio*	Select the radio band. It can be 2.4 GHz or 5 GHz.
Country*	Select the country where the router is installed.
Channel Bandwidth	Select the IEEE 802.11n and 802.11ac channel bandwidth. For a 5-GHz radio band, the default value is 80 MHz, and for 2.4 GHz, the default is 20 MHz.
Channel	Select the radio channel. The default is "auto", which automatically selects the best channel. For 5-GHz radio bands, you can configure dynamic frequency selection (DFS) channels.
Guard Interval	Select the guard interval. For a 5-GHz radio band, the default value is the short guard interval (SGI) of 400 ns, and for 2.4 GHz, the default is 800 ns.

To save the feature template, click **Save**.

CLI equivalent:

```
wlan frequency channel channel channel-bandwidth megahertz country country guard-interval nanoseconds
```

Release Information

Introduced in vManage NMS Release 16.3.

WiFi SSID

You can use the WiFi SSID template for all devices that support wireless LANs (WLANs)

To configure SSIDs on the WLAN radio using vManage templates:

1. Create a WiFi SSID template to configure the VAP interfaces to use as SSIDs, as described in this article.
2. Create a WiFi Radio template to configure WLAN radio parameters.
3. Create a Bridge template to assign the VAP interface to a bridging domain.
4. Create a device template that incorporates the WiFi Radio feature template and the Wifi SSID feature template.

Navigate to the Template Screen and Name the Template

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Device Templates**.



Note In Cisco vManage Release 20.7.x and earlier releases, **Device Templates** is titled **Device**.

3. From the **Create Template** drop-down list, choose **From Feature Template**.
4. From the **Device Model** drop-down list, select a device that supports wireless LANs (WLANs).
5. Click **WLAN**, or scroll to the **WLAN** section.
6. Under **Additional WiFi Radio Templates**, click **WiFi SSID**.
7. From the **WiFi SSID** drop-down list, click **Create Template**. The **WiFi SSID** template form is displayed. This form contains fields for naming the template, and fields for defining the WiFi SSID parameters.
8. In **Template Name**, enter a name for the template. The name can be up to 128 characters and can contain only alphanumeric characters.
9. In **Template Description**, enter a description of the template. The description can be up to 2048 characters and can contain only alphanumeric characters.

When you first open a feature template, for each parameter that has a default value, the scope is set to Default (indicated by a check mark), and the default setting or value is shown. To change the default or to enter a value, click the **Scope** drop-down list.

WLAN SSID Configuration

To configure SSIDs on a device, configure the following parameters in **Basic Configuration**. Parameters marked with an asterisk are required to configure the SSIDs.

Table 187:

Parameter Name	Description
Interface Name*	Select the VAP interface name.
Shutdown*	Click No to enable the interface.
Description (optional)	Enter a description for the interface.
SSID*	Enter the name of the SSID. It can be a string from 4 through 32 characters. The SSID must be unique. You can configure up to four SSIDs. Each SSID is called a virtual access point (VAP) interface. To a client, each VAP interfaces appears as a different access point (AP) with its own SSID. To provide access to different networks, assign each VAP to a different VLAN.

Parameter Name	Description
Maximum Clients	Enter the maximum number of clients allowed to connect to the WLAN. <i>Range:</i> 1 through 50 <i>Default:</i> 25
Data Security	Select the security type to enable user authentication or enterprise WPA security. For user authentication, select from WPA Personal, WPA/WPA2 Personal, or WPA2 Personal, and then enter a clear text or an AES-encrypted key. For enterprise security, select from WPA Enterprise, WPA/WPA2 Enterprise, or WPA2 Enterprise, and then enter a RADIUS server tag.
RADIUS Server	If you select one of the enterprise security methods based on using a RADIUS authentication server, enter the RADIUS server tag.
WPA Personal Key	If you select one of the personal security methods based on preshared keys, enter either a clear text or an AES-encrypted password.
Management Security	If you select one of the WPA2 security methods, select the encryption of management frames to be none, optional, or required.

To save the feature template, click **Save**.

CLI equivalent:

```
wlan frequency interface vapnumber data-security security
  description text mgmt-security security radius-servers tag
  no shutdown
  ssid ssid wpa-personal-key password
```

Release Information

Introduced in Cisco SD-WAN Manager Release 16.3.

Interface CLI Reference

CLI commands for configuring and monitoring system-wide parameters, interfaces, and SNMP on vEdge routers and vSmart controllers.

Interface Configuration Commands

Use the following commands to configure interfaces and interface properties in the Cisco SD-WAN overlay network. Interfaces must be configured on a per-VPN basis.

```
vpn vpn-id
  interface interface-name
    access-list acl-list (on vEdge routers only)
    arp
      ip ip-address mac mac-address
    arp-timeout seconds (on vEdge routers only)
    autonegotiate (on vEdge routers only)
    block-non-source-ip (on vEdge routers only)
    clear-dont-fragment
    dead-peer-detection interval seconds retries number (on vEdge routers only)
    description text
    dhcp-helper ip-address (on vEdge routers only)
```

```

dhcp-server (on vEdge routers only)
  address-pool prefix/length
  exclude ip-address
  lease-time seconds
  max-leases number
  offer-time minutes
  options
    default-gateway ip-address
    dns-servers ip-address
    domain-name domain-name
    interface-mtu mtu
    tftp-servers ip-address
  static-lease mac-address ip ip-address host-name hostname
dot1x
  accounting-interval seconds
  acct-req-attr attribute-number (integer integer | octet octet | string string)
  auth-fail-vlan vlan-id
  auth-order (mab | radius)
  auth-reject-vlan vlan-id
  auth-req-attr attribute-number (integer integer | octet octet | string string)
  control-direction direction
  das
    client ip-address
    port port-number
    require-timestamp
    secret-key password
    time-window seconds
    vpn vpn-id
  default-vlan vlan-id
  guest-vlan vlan-id
  host-mode (multi-auth | multi-host | single-host)
  mac-authentication-bypass
    allow mac-addresses
    server
  nas-identifier string
  nas-ip-address ip-address
  radius-servers tag
  reauthentication minutes
  timeout
    inactivity minutes
  wake-on-lan
duplex (full | half)
flow-control (bidirectional | egress | ingress)
ike (on vEdge routers only)
  authentication-type type
    local-id id
    pre-shared-secret password
    remote-id id
  cipher-suite suite
  group number
  mode mode
  rekey seconds
  version number
(ip address prefix/length | ip dhcp-client [dhcp-distance number])
(ipv6 address prefix/length | ipv6 dhcp-client [dhcp-distance number] [dhcp-rapid-commit])

ip address-list prefix/length (on vSmart controller containers only)
ip secondary-address ipv4-address (on vEdge routers only)
ipsec (on vEdge routers only)
  cipher-suite suite
  perfect-forward-secrecy pfs-setting
  rekey seconds
  replay-window number
keepalive seconds retries (on vEdge routers only)

```

```

mac-address mac-address
mtu bytes
nat (on vEdge routers only)
    block-icmp-error
    block-icmp-error
    direction (inside | outside)
    log-translations
    [no] overload
    port-forward port-start port-number1 port-end port-number2
    proto (tcp | udp) private-ip-address ip address private-vpn vpn-id
    refresh (bi-directional | outbound)
    respond-to-ping
    static source-ip ip-address1 translate-ip ip-address2 (inside | outside)
    static source-ip ip-address1 translate-ip ip-address2 source-vpn vpn-id protocol (tcp
| udp) source-port number translate-port number
    tcp-timeout minutes
    udp-timeout minutes
pmtu (on vEdge routers only)
policer policer-name (on vEdge routers only)
ppp (on vEdge routers only)
    ac-name name
    authentication (chap | pap) hostname name password password
pppoe-client (on vEdge routers only)
    ppp-interface name
profile profile-id (on vEdge routers only)
qos-map name (on vEdge routers only)
rewrite-rule name (on vEdge routers only)
shaping-rate name (on vEdge routers only)
shutdown
speed speed
static-ingress-qos number (on vEdge routers only)
tcp-mss-adjust bytes
technology technology (on vEdge routers only)
tloc-extension interface-name (on vEdge routers only)
tracker tracker-name (on vEdge routers only)
tunnel-interface
    allow-service service-name
    bind geslot/port (on vEdge routers only)
    carrier carrier-name
    color color [restrict]
    connections-limit number
    encapsulation (gre | ipsec) (on vEdge routers only)
        preference number
        weight number
    hello-interval milliseconds
    hello-tolerance seconds
    low-bandwidth-link (on vEdge routers only)
    max-control-connections number (on vEdge routers only)
    nat-refresh-interval seconds
    port-hop
    vbond-as-stun-server (on vEdge routers only)
    vmanage-connection-preference number (on vEdge routers only)
    tunnel-destination ip-address (GRE interfaces; on vEdge routers only)
    tunnel-destination (dns-name | ipv4-address) (IPsec interfaces; on vEdge routers only)
    (tunnel-source ip-address | tunnel-source-interface interface-name) (GRE interfaces;
on vEdge routers only)
    (tunnel-source ip-address | tunnel-source-interface interface-name) (IPsec interfaces;
on vEdge routers only)
    upgrade-confirm minutes
    vrrp group-name (on vEdge routers only)
        priority number
        timer seconds
    track-omp

```

Interface Monitoring Commands

Use the following commands to monitor interfaces:

show dhcp interface

show dhcp server

show interface

show interface arp-stats

show interface errors

show interface packet-sizes

show interface port-stats

show interface queue

show interface statistics

show vrrp

System Configuration Commands

Use the following commands to configure system-wide parameters:

```
banner
  login "text"
  motd "text"
system
  aaa
    admin-auth-order (local | radius | tacacs)
    auth-fallback
    auth-order (local | radius | tacacs)
    logs
      audit-disable
      netconf-disable
    radius-servers tag
    user user-name
      group group-name
      password password
    usergroup group-name
      task (interface | policy | routing | security | system) (read | write)
  admin-tech-on-failure
  archive
    interval minutes
    path file-path/filename
    ssh-id-file file-path/filename
    vpn vpn-id
  clock
    timezone timezone
  console-baud-rate rate
  control-session-pps rate
  description text
  device-groups group-name
  domain-id domain-id
  eco-friendly-mode (on vEdge Cloud routers only)
  gps-location (latitude decimal-degrees | longitude decimal-degrees)
  host-name string
  host-policer-pps rate (on vEdge routers only)
  icmp-error-pps rate
  idle-timeout minutes
  iptables-enable
```

```

location string
logging
  disk
    enable
    file
      name filename
      rotate number
      size megabytes
    priority priority
  host
    name (name | ip-address)
    port udp-port-number
    priority priority
    rate-limit number interval seconds
multicast-buffer-percent percentage (on vEdge routers only)
ntp
  keys
    authentication key-id md5 md5-key
    trusted key-id
  server (dns-server-address | ipv4-address)
    key key-id
    prefer
    source-interface interface-name
    version number
    vpn vpn-id
organization-name string
port-hop
port-offset number
radius
  retransmit number
  server ip-address
    auth-port port-number
    priority number
    secret-key key
    source-interface interface-name
    tag tag
    vpn vpn-id
  timeout seconds
route-consistency-check (on vEdge routers only)
site-id site-id
sp-organization-name name (on vBond orchestrators and vSmart controllers only)
system-ip ip-address
system-tunnel-mtu bytes
tacacs
  authentication authentication-type
  server ip-address
    auth-port port-number
    priority number
    secret-key key
    source-interface interface-name
    vpn vpn-id
  timeout seconds
tcp-optimization-enabled
timer
  dns-cache-timeout minutes
track-default-gateway
track-interface-tag number (on vEdge routers only)
track-transport
tracker tracker-name
  endpoint-dns-name dns-name
  endpoint-ip ip-address
  interval seconds
  multiplier number
  threshold milliseconds

```



```
upgrade-confirm minutes  
[no] usb-controller (on vEdge 1000 and vEdge 2000 routers only)  
vbond (dns-name | ip-address) [local] [port number] [ztp-server]
```

System Monitoring Commands on a Cisco vEdge device

Use the following commands to monitor system-wide parameters:

show aaa usergroup

show control local-properties

show logging

show ntp associations

show ntp peer

show orchestrator local-properties

show running-config system

show system status

show uptime

show users



CHAPTER 9

IPv6 Functionality

This chapter describes the options for enabling IPv6 functionality for Cisco Catalyst SD-WAN templates and policies. Use the information in this chapter if your deployment uses IPv6.

Configure IPv6 Functionality for an Interface or Subinterface Template

To configure IPv6 functionality for an interface or subinterface template, perform the following steps.

Cisco Catalyst SD-WAN supports dual stack: you can configure IPv4 and IPv6 in the same deployment. You can configure up to three global IPv6 addresses per interface.

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Feature Templates**, and click **Add Template** to select an appropriate device model.



Note In Cisco vManage Release 20.7.x and earlier releases, **Feature Templates** is titled as **Feature**.

3. Select **Cisco VPN Interface Ethernet** from the list of templates.
4. From **Basic Configuration**, click **IPv6** and configure the parameters that the following table describes:

Parameter Name	Description
Static	This radio button is selected by default because IPv6 addresses are static.
IPv6 Address	Enter the IPv6 address of the interface or subinterface.

Configure IPv6 Functionality for an OMP Template

To configure IPv6 functionality for an Overlay Management Protocol (OMP) template, follow these steps:

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Feature Templates**, and click **Add Template** to select an appropriate device model.



Note In Cisco vManage Release 20.7.x and earlier releases, **Feature Templates** is titled as **Feature**.

3. Select **Cisco OMP** from the list of templates.
4. Click **Advertise** and choose **IPv6** to configure the parameters that the following table describes:

Parameter Name	Description
Connected	Click Off to disable advertising connected routes to OMP. By default, connected routes are advertised to OMP.
Static	Click Off to disable advertising static routes to OMP. By default static routes are advertised to OMP.
BGP	Click On to advertise BGP routes to OMP. By default, BGP routes are not advertised to OMP.

Configure IPv6 Functionality for a BGP Template

To configure IPv6 functionality for a Border Gateway Protocol (BGP) template, follow these steps:

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Feature Templates**, and click **Add Template** to select an appropriate device model.



Note In Cisco vManage Release 20.7.x and earlier releases, **Feature Templates** is titled as **Feature**.

3. Select **Cisco BGP** from the list of templates.
4. Click **Unicast Address Family** and choose **IPv6** to configure the parameters that the following table describes:

Tab	Parameter Name	Description
	Maximum Paths	Specify the maximum number of parallel IBGP paths that can be installed into a route table to enable IBGP multipath load sharing. <i>Range: 0 to 32</i>
	Address Family	Enter the BGP IPv6 unicast address family.
RE-DISTRIBUTE		Click the Redistribute tab, and then click Add New Redistribute .
	Protocol	Select the protocols from which to redistribute routes into BGP, for all BGP sessions. Options are Connected, NAT, OMP, OSPF, and Static. At a minimum, select the following: • For service-side BGP routing, select OMP. By default, OMP routes are not redistributed into BGP. • For transport-side BGP routing, select Connected, and then under Route Policy, specify a route policy that has BGP advertise the loopback interface address to its neighbors.

Tab	Parameter Name	Description
	Route Policy	Enter the name of the route policy to apply to redistributed routes.
		Click Add to save the redistribution information.
NETWORK		Click the Network tab, and then click Add New Network .
	Network Prefix	Enter a network prefix, in the format of <i>prefix/length</i> , to be advertised by BGP.
		Click Add to save the network prefix.
AGGREGATE ADDRESS		Click the Aggregate Address tab, and then click Add New Aggregate Address .
	Aggregate Prefix	Enter the prefix of the addresses to aggregate for all BGP sessions, in the format <i>prefix/length</i> .
	AS Set Path	Click On to generate set path information for the aggregated prefixes.
	Summary Only	Click On to filter out more specific routes from BGP updates.
		Click Add to save the aggregate address.

1. In the Neighbor area, click **IPv6**, create a new neighbor or edit an existing one, and then configure the parameters that the following table describes.

Parameters marked with an asterisk are required.

Parameter Name	Description
IPv6 Address*	Specify the IPv6 address of the BGP neighbor.
Description	Enter a description of the BGP neighbor.
Remote AS*	Enter the AS number of the remote BGP peer.
Address Family	Select Global from the drop-down list, click On and select the address family. Enter the address family information.
Shutdown	To shut down a BGP neighbor when you push the template, select Global from the drop-down list and then click Yes . <i>Default: Off</i>

Configure IPv6 Functionality for a VRRP Template

To configure IPv6 functionality for a Virtual Router Redundancy Protocol (VRRP) template, follow these steps:

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Feature Templates**, and click **Add Template** to select an appropriate device model.



Note In Cisco vManage Release 20.7.x and earlier releases, **Feature Templates** is titled as **Feature**.

3. Select **Cisco VPN Interface Ethernet** from the list of templates.
4. Click **VRRP** and choose **IPv6**.
5. Click **New VRRP**.
6. Configure the parameters that the following table describes:

Parameter Name	Description
Group ID	Enter a virtual router ID, which represents a group of routers. Range: 1 through 255
Priority	Enter the priority level of the router within a VRRP group. • <i>Range:</i> 1 through 254 • <i>Default:</i> 100
Timer	Not used.
Track OMP	Select On to track the Overlay Management Protocol (OMP) session running on the WAN connection when determining the primary VRRP virtual router. <i>Default:</i> Off
Track Prefix List	Enter a value to track a list of IPv6 remote prefixes. This value is an alphanumeric string that is configured under Policy.
Link Local IPv6 Address	Enter a virtual link local IPv6 address, which represents the link local address of the group. The address should be in standard link local address format. For example, FE80::AB8.
Global IPv6 Address	Enter a virtual global unicast IPv6 address, which represents the global address of the group. The address should be an IPv6 global prefix address that has the same mask as the interface forwarding address on which the VRRP group is configured. For example, 2001::2/124. You can configure up to 3 global IPv6 addresses.

Configure IPv6 Functionality for an ACL Template or a QoS Template

To configure IPv6 functionality for an ACL and QoS template, follow these steps:

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Feature Templates**, and click **Add Template** to select an appropriate device model.



Note In Cisco vManage Release 20.7.x and earlier releases, **Feature Templates** is titled as **Feature**.

3. Select **Cisco VPN Interface Ethernet** from the list of templates.
4. From **ACL/QoS**, configure the parameters that the following table describes:

Parameter Name	Description
Ingress ACL – IPv6	Click On to enable the IPv6 ingress access list.
IPv6 Ingress Access List	Enter the name of the IPv6 ingress access list.
Egress ACL – IPv6	Click On to enable the IPv6 egress access list.
IPv6 Egress Access List	Enter the name of the IPv6 egress access list.

Configure IPv6 Functionality for a Logging Template

To configure IPv6 functionality for a Logging template, follow these steps:

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Feature Templates**, and click **Add Template** and then select an appropriate device model.



Note In Cisco vManage Release 20.7.x and earlier releases, **Feature Templates** is titled as **Feature**.

3. Select **Cisco Logging** from the list of templates.
4. From **Server**, click **IPv6**.
5. Configure the parameters that the following table describes.

Parameter Name	Description
IPv6 Hostname/IPv6 Address	Host name or IP address of the server to direct the logging information.
VPN ID	VPN ID of the VPN source interface.
Source Interface	Name of the source interface.
Priority	Choose the maximum severity of messages that are logged.

Configure IPv6 Functionality for a New Prefix List

To configure an IPv6 address for a new prefix list, follow these steps:

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Policies**.

2. From the **Custom Options** drop-down list, select **Lists**. You can make this selection for a Centralized Policy or a Localized Policy
3. Select **Prefix** from the list on the left and then select **New Prefix List**.
4. Click **IPv6** and enter the IPv6 address in **Add Prefix**.

Configure IPv6 Functionality for a Data Prefix

To configure an IPv6 address for a new prefix list, follow these steps:

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Policies**.
2. From the Custom Options drop-down list, select **Lists**. You can make this selection for a Centralized Policy or a Localized Policy
3. Select **Data Prefix** from the list on the left and then select **New Data Prefix List**.
4. From **Internet Protocol**, click **IPv6** and enter the IPv6 address in **Add Prefix**.

Configure IPv6 Functionality for a Centralized Policy

To configure a centralized policy to apply to IPv6 address families, follow these steps:

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Policies**.
2. From the Custom Options drop-down menu, select **Traffic Policy** under Centralized Policy.
3. Select **Traffic Data**.
4. Select **Add Policy** and click **Create New**.
5. Click **Sequence Type** and then select **Traffic Engineering**.
6. Click **Sequence Rule**.
7. From the **Protocol** drop-down list, select **IPv6** to apply the policy only to IPv6 address families, or select **Both** to apply the policy IPv4 and IPv6 address families.
8. Click **Sequence Type** and then select **QoS**.
9. Click **Sequence Rule**.
10. From the Protocol drop-down list, click **IPv6** to apply the policy only to IPv6 address families, or select **Both** to apply the policy IPv4 and IPv6 address families.

Configure IPv6 Functionality for a Localized Policy

To configure a localized policy to apply to IPv6 address families, follow these steps:

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Policies**.
2. From the **Custom Options** drop-down list, select **Access Control Lists** under Localized Policy.
3. Click **Add Access Control List Policy** and choose **Add IPv6 ACL Policy**. The policy you create will apply only to IPv6 address families.



CHAPTER 10

Configure a Cellular Gateway

- [Configure a Cellular Gateway, on page 441](#)
- [Information About Configuring a Cellular Gateway, on page 441](#)
- [Supported Cellular Gateway Devices, on page 442](#)
- [Configure a Cellular Gateway Using a Feature Template in Cisco SD-WAN Manager, on page 442](#)
- [Configure a Cellular Gateway Using a Configuration Group in Cisco SD-WAN Manager, on page 445](#)

Configure a Cellular Gateway

Table 188: Feature History

Feature Name	Release Information	Feature Description
Cellular Gateway Configuration	Cisco vManage Release 20.4.1 Cisco IOS XE Catalyst SD-WAN Release 17.4.1a (on devices)	This feature provides templates for configuring a supported cellular gateway as an IP pass-through device. This release supports the Cisco Cellular Gateway CG418-E and CG522-E.
Cellular Gateway Configuration Using a Configuration Group	Cisco Catalyst SD-WAN Manager Release 20.13.1 Cisco IOS CG Release 17.13.1	Added support for configuring cellular gateways using configuration groups. A new Create Cellular Gateway Group workflow creates a configuration group specifically for cellular gateways.

Information About Configuring a Cellular Gateway

You can configure a supported cellular gateway as an IP pass-through device. By positioning the configured device in an area in your facility that has a strong LTE signal, the signal can be extended over an Ethernet connection to a routing infrastructure in a location with a weaker LTE signal.

Secure Communication with Devices through a vmanage-admin Account

Cisco SD-WAN Manager communicates with devices, such as Cisco Catalyst Cellular Gateways, using a secure channel—either a datagram transport layer security (DTLS) tunnel or transport layer security (TLS)

tunnel. Within this secure channel, it communicates with the devices or controllers using the NETCONF protocol, within an SSH session. It uses an internal-use-only passwordless "vmanage-admin" user account on the device or controller. The vmanage-admin account is created during the initial device setup. Cisco SD-WAN Manager uses this secure channel for monitoring, configuring, and managing devices.

As noted, the vmanage-admin user accounts do not have any password associated with them, so Cisco SD-WAN Manager uses a passwordless procedure to log in to the account. To accomplish this, Cisco SD-WAN Manager generates an asymmetric encryption public-private key pair. During deployment of a device, Cisco SD-WAN Manager copies the public key that it has generated to the device. It sends the public key using a proprietary protocol, within a secure channel—a DTLS or TLS tunnel.

The activity that Cisco SD-WAN Manager performs using the vmanage-admin account appears in syslog messages and in the output of certain show commands. The syslog messages are logged with the same level of detail as activities performed through any other user account. The level of syslog detail depends on the syslog configuration of the device.

Cisco SD-WAN Manager requires the vmanage-admin account on devices in order to monitor, configure, and manage the devices. Removing, disabling, or altering this account on a device would prevent Cisco SD-WAN Manager from performing these activities, and is not supported.

Supported Cellular Gateway Devices

Cisco Catalyst Cellular Gateway models:

- CG418-E
- CG522-E

Configure a Cellular Gateway Using a Feature Template in Cisco SD-WAN Manager

Before You Begin

This procedure configures a cellular gateway using a feature template. For information about using a configuration group, see [Configure a Cellular Gateway Using a Configuration Group in Cisco SD-WAN Manager, on page 445](#).

Configure a Cellular Gateway Using a Feature Template

1. Create a device template for Cisco Cellular Gateway CG418-E devices.

See [Configure Devices](#) in the *Cisco Catalyst SD-WAN Systems and Interfaces Configuration Guide, Cisco IOS XE Catalyst SD-WAN Release 17.x*.

After you enter a description for the feature template:

- a. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
- b. Click **Device Templates**.



Note In Cisco vManage Release 20.7.x and earlier releases, **Device Templates** is titled **Device**.

- c. From the **Create Template** drop-down list choose **From Feature Template**.
- d. From the **Device Model** drop-down list select the type of device for which you are creating the template.
- e. Choose **Cellular Gateway > Cellular Gateway Platform > Create Template**. Then configure the Cellular Gateway Platform feature template as shown in the following table.

Table 189: Cellular Gateway Platform Template Parameters

Parameter Name	Description
Basic Configuration Tab	
Time Zone	Choose the time zone to use for the device. The device uses this time zone for clock synchronization when NTP is configured.
Management Interface	Enter the IPv4 address of the management interface for accessing the device.
Admin-Password	Enter the admin user password for logging in to the device by using an SSH client or a console port.
NTP-Servers	Configure one or more NTP servers to which the device synchronizes its clock.
Cellular Configuration Tab	
IP-Src-Violation	Choose v4 only , v6 only , or v4 and v6 to enable the IP source violation feature for the corresponding IP address types. Choose None if you do not want to enable this feature.
Auto-SIM	Choose On to enable the auto-SIM feature. When this feature is enabled, the device automatically detects the service provider to which SIMs in the device belong and automatically loads the appropriate firmware for that provider.
Primary SIM Slot	Choose the slot that contains the primary SIM card for the device. If the device loses service to this slot, it fails over to the secondary slot.
Failover-Timer (minutes)	Enter the number of minutes that the device waits before trying to communicate with the primary SIM slot after the device detects loss of service to this slot.

Parameter Name	Description
Max-Retry	Enter the number of consecutive unsuccessful attempts by the device to communicate with the primary SIM before failing over to the secondary slot

- f. Choose **Cellular Gateway** > **Cellular Gateway Profile** and choose **Create Template** from the Cellular Gateway Profile drop-down list. Then configure the Cellular Gateway Profile feature template as shown in the following table.

Table 190: Cellular Gateway Profile Template Parameters

Parameter Name	Description
Basic Configuration Tab	
SIM	Choose a SIM slot and configure the following options to create a profile for the SIM in that slot. This profile indicates to the service provider which of its cellular networks the SIM should attach to. • Profile ID: Enter a unique ID for the profile • Access Point Name: Enter the name of the access point for this profile • Packet Data Network Type: Choose the type of network for data services for this profile (IPv4, IPv6, or IPv4v6) • Authentication: Choose the authentication method that this profile uses for data, and enter the user name and password for this method in the Profile Username and Profile Password fields that display You can configure one profile for each SIM slot in the device.
Add Profile	Click to add an access point name (APN) profile that the cellular device uses to attach to a cellular network. You can add up to 16 profiles.
Profile ID	Enter a unique identifier for the profile. Valid values: Integers 1 through 16.
Access Point Name	Enter a name to identify the cellular access point.
Packet Data Network Type	Choose the packet data network (PDN) type of the cellular network (IPv4 , IPv6 , or IPv4v6).

Parameter Name	Description
Authentication	Choose the authentication method that is used to attach to the cellular access point (none , pap , chap , pap_chap).
Profile Username	If you choose an authentication method other than none , enter the user name to use for authentication when attaching to the cellular access point.
Password	If you choose an authentication method other than none , enter the password to use for authentication when attaching to the cellular access point.
Add	Click to add the profile your are configuring.
Advanced Configuration Tab	
Attach Profile	Choose the profile that the device uses to connect to the cellular network.
Cellular 1/1 Profile	Choose the profile that the device uses for data connectivity over the cellular network.

2. Attach the device template to the device.

For information, see [Attach and Detach a Device Template](#) in the *Cisco Catalyst SD-WAN Systems and Interfaces Configuration Guide, Cisco IOS XE Catalyst SD-WAN Release 17.x*.

Configure a Cellular Gateway Using a Configuration Group in Cisco SD-WAN Manager

Before You Begin

Create a configuration group for Cisco Catalyst Cellular Gateways using **Workflows > Create Cellular Gateway Group**. On the **Configuration Groups** page, the resulting configuration group is labelled **cellulargateway** in the **Device Solution** column.

For information about creating configuration groups and applying them to devices, see the [Using Configuration Groups](#) section of *Cisco Catalyst SD-WAN Configuration Groups, Cisco IOS XE Catalyst SD-WAN Release 17x*.

Configure a Cellular Gateway Using a Configuration Group

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Configuration Groups**.
2. Click ... adjacent to a configuration group for a Cisco Catalyst Cellular Gateway and choose **Edit**.
3. Open the **Global Profile** section and add (click **Add Global Profile Feature**) or edit (click ... and **Edit**) any of the following features.

- AAA feature:

Table 191: Local

Parameter Name	Description
Name	The account name is preset to admin and cannot be changed.
Password	Enter a password for login.

Table 192: TACACS

Parameter Name	Description
TACACS Configuration	Enable TACACS configuration. Click Add TACACS to add one or more TACACS servers.
Authentication	TACACS authentication option: • tacacs_ascii: Send authentication information in ASCII format. • tacacs_pap: Send authentication information using the password authentication protocol (PAP).
Timeout	Timeout for TACACS authentication. Range: 1 through 1000 seconds
TACACS	
IP Address	IP address of the TACACS server.
Auth Port	TCP port number to connect to the TACACS server. Default: 49
Secret Key	Encryption key for encrypting and decrypting traffic between the cellular gateway and the TACACS server. Configure the same key on the TACACS server.
Source Interface	Preconfigured as Cellular1/0, and cannot be changed. This is the only interface that the cellular gateway can use for communication with the TACACS server.
Priority	Priority level of the TACACS server. Zero is a default priority value and indicates the highest priority. If a cellular gateway is unable to establish a connection with the highest priority server, it attempts to connect to the server of the next highest priority. Range: 0 through 7

- Cellular feature:

Table 193: Cellular Settings

Parameter Name	Description
Primary Slot	Choose a SIM slot to designate it as primary. Range: 0, 1 Default: 0
SIM SLOT 0 Cellular Profile	
Profile Id	Profile ID. You can click Add to add multiple profiles.
Access Point Name	Access point name, from your service provider.
Authentication Method	Authentication method (none , pap , chap , pap_or_chap) indicated by your service provider.
Username	Username for authentication, as indicated by your service provider.
Password	Password for authentication, as indicated by your service provider.
Packet Data Network Type	Packet data network type (IPv4 , IPv6 , IPv4v6), as indicated by your service provider.
Attach Profile	Choose the attach profile from the defined profiles.
Data Profile	Choose the data profile from the defined profiles. You can use the same profile for the attach profile and data profile.
SIM SLOT 1 Cellular Profile	
See the fields described for SIM slot 0.	

- **Logging** feature:

Table 194: Disk

Parameter Name	Description
Disk File Rotate	Maximum number of log files to store locally. The device collects diagnostic monitor log files, which have a maximum size of 20 MB each, until the number of files reaches the rotate value. Then the device deletes the oldest file to make room for a new file. Range: 1 through 10
Disk File Size	Maximum file size for each log file that the device stores locally. After reaching the maximum size, the device creates a new log file, with a numerically sequenced filename. Range: 1 through 20 megabytes

Table 195: Servers

Parameter Name	Description
Server Name Type	Choose ipv4 or ipv6 , according to the server address type, or choose dns if you enter a server domain name in the Server Name Value field.
Server Name Value	IP address or domain name of the server.
Source IP	By default, this is the system IP address. You can choose the Device Specific option to specify per device.
Priority	Filter the type of log messages saved using one of the following priority options, listed from lowest to highest priority. Each priority option configures the device to save log messages of that priority and all higher priorities. For example, information is the lowest priority of message, so choosing information includes information log messages and all other log messages too. Choosing error excludes information, notice, and warn log messages, but includes error messages and all other log messages of higher priority (critical, alert, and emergency). From lowest to highest priority, the options are the following: • information • notice • warn • error • critical • alert • emergency

- **Network Protocol** feature:

Table 196: Basic Configuration

Parameter Name	Description
Passthrough	The cellular gateway operates in one of two modes: IP passthrough and NAT. In IP passthrough mode, the cellular gateway passes the public IP address assigned by the internet service provider (ISP) to a downstream device attached to the cellular gateway. Disabling the Passthrough option enables NAT, which gives the devices that are connected to the cellular gateway access to a DHCP server and to the local gateway. Note Enabling passthrough mode disables and hides the other fields in the Basic Configuration section.
DHCP Pool	
DHCP Pool	Enable a DHCP pool for NAT.
DHCP Network Pool	IP address pool, in classless interdomain routing (CIDR) format.
Lease Days	Days for DHCP lease time Range: 0 to 365
Lease Hours	Hours for DHCP lease time. Range: 0 to 23
Lease Minutes	Minutes for DHCP lease time. Range: 0 to 59
PAT Configuration	
PAT Configuration	Enable port address translation (PAT).
Add PAT Config	Click this to add one or more PAT configurations.
Description	Description of the PAT configuration.
Protocol	Choose TCP or UDP .
LocalAddress	IPv4 format address.
LocalPort	Port number. Range: 0 to 65535
InterfaceName	Preconfigured as Cellular1/0, which is the WAN interface for the cellular gateway.

Parameter Name	Description
GlobalPort	Global port number. Range: 1 to 65535

Table 197: NTP Servers

Parameter Name	Description
NTP	To configure a network time protocol (NTP) server, enter an IPv4 address or a DNS name. Maximum number of NTP servers: 4

4. (Optional) To add CLI configuration commands, do the following:
 - a. Open the **CLI Add-on Profile**.
 - b. Click **Add Feature**.
 - c. In the **Type** drop-down list, choose **Config**.
 - d. Enter a name for the feature.
 - e. Enter a CLI configuration.
 - f. Click **Save**.



Note CLI configuration commands in the CLI Add-on Profile override any configuration done using the Global Profile.



CHAPTER 11

Track Static Routes for Service VPNs

Table 198: Feature History

Feature Name	Release Information	Description
Static Route Tracker for Service VPNs for Cisco vEdge Devices	Cisco SD-WAN Release 20.4.1 Cisco vManage Release 20.4.1	This feature enables you to configure IPv4 static route endpoint tracking for service VPNs. For static routes, endpoint tracking determines whether the configured endpoint is reachable before adding that route to the route table of the device.
TCP/UDP Endpoint Tracker and Dual Endpoint Static Route Tracker for Cisco vEdge devices	Cisco SD-WAN Release 20.7.1 Cisco vManage Release 20.7.1	This feature enables you to configure the TCP/UDP static route endpoint trackers. Using this feature you can also configure IPv4, TCP/UDP dual endpoint static-route tracker groups for service VPNs to enhance the reliability of probes.

- [Information About Static Route Tracking, on page 451](#)
- [Restrictions for IPv4 Static Route Tracking, on page 452](#)
- [Configure Tracker Group Using a Configuration Group, on page 452](#)
- [Workflow to Configure IPv4 Static Route Tracking, on page 453](#)
- [Configure Static Routes Using CLI, on page 457](#)
- [Configuration Examples for Static Route Tracking Using the CLI, on page 459](#)
- [Verify Static Route Tracking Configuration Using CLI, on page 461](#)

Information About Static Route Tracking

Static-route tracking for service VPNs enables you to track the availability of the configured endpoint address to determine if the static route can be included in the routing table of a device. This is applicable when a site uses a static route in a service VPN to advertise its route over Overlay Management Protocol (OMP). The

static route tracker periodically sends ICMP ping probes to the configured endpoint. If the tracker does not receive a response, the static route is not included in the routing table and is not advertised to OMP. You can configure an alternative next-hop address or a static route with a higher administrative distance to provide a backup path. This path is advertised over OMP.

**Note**

- From Cisco SD-WAN Release 20.7.1, you can configure TCP/UDP individual endpoint trackers and configure a tracker group with dual endpoints (using two trackers), and associate the trackers and tracker group to a static route. Dual endpoints help in avoiding false negatives that might be introduced because of the unavailability of the routes.
- From Cisco IOS XE Catalyst SD-WAN Release 17.15.1a only UP/DOWN status changes are reported to Cisco SD-WAN Manager, while RTT value changes are not. This optimization improves efficiency in extensive networks.

Restrictions for IPv4 Static Route Tracking

- Only one endpoint tracker is supported per static route per next-hop address.
- IPv6 static routes are not supported.
- You cannot link the same endpoint-tracker to static routes in different VPNs. Endpoint-tracker is identified by a name and can be used for multiple static routes in a single VPN.

Configure Tracker Group Using a Configuration Group

Before you begin

On the **Configuration > Configuration Groups** page, choose **SD-WAN** as the solution type.

Procedure

- Step 1** From the Cisco SD-WAN Manager menu, choose **Configuration > Configuration Groups**.
- Step 2** Create and configure a tracker group in a Service Profile.

Table 199: Tracker Group

Field	Description
Tracker Elements*	This field is displayed only if you chose Tracker Type as the Tracker Group . Add the existing interface tracker names, separated with a space. When you add this tracker to the template, the tracker group is associated with these individual trackers, and you can then associate the tracker group to an interface.

Field	Description
Tracker Boolean	This field is displayed only if you chose Tracker Type as the Tracker Group. Select AND or OR. OR is the default boolean operation. An OR ensures that the transport interface status is reported as active if either one of the associated trackers of the tracker group reports that the interface is active. If you select the AND operation, the transport-interface status is reported as active if both the associated trackers of the tracker group report that the interface is active.

What to do next

Also see [Deploy a configuration group](#).

Workflow to Configure IPv4 Static Route Tracking

1. Configure an endpoint tracker using the System template.
2. Configure a static route using the VPN template.
3. Apply the tracker to the next-hop address.

Create a Static Route Tracker

Use the **System Template** to create a tracker for static routes.

1. From Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Feature Templates**.



Note In Cisco vManage Release 20.7.x and earlier releases, **Feature Templates** is titled **Feature**.

3. Navigate to the **Cisco System** template for the device.



Note For information about creating a System template, see [Create System Template](#).

4. Click **Tracker**. Click **New Endpoint Tracker** to configure the tracker parameters.

Table 200: Tracker Parameters

Field	Description
Name	Name of the tracker. The name can be up to 128 alphanumeric characters.
Threshold	Wait time for the probe to return a response before declaring that the configured endpoint is down. Range is from 100 to 1000 milliseconds. Default is 300 milliseconds.
Interval	Time interval between probes to determine the status of the configured endpoint. Default is 60 seconds (1 minute). Range is from 10 to 600 seconds.
Multiplier	Number of times probes are sent before declaring that the endpoint is down. Range is from 1 to 10. Default is 3.
Tracker Type	From the drop-down, choose Global. From the Tracker Type field drop-down, choose Static Route. From Cisco SD-WAN Release 20.7.1, you can configure a tracker group with dual endpoints on Cisco vEdge devices and associate this tracker group to a static route.
Endpoint Type	Choose endpoint type IP Address. Note Configuring the tracker type Static Route using endpoint URL or endpoint DNS name is not supported.
End-Point Type: IP Address	IP address of the static route end point. This is the destination on the internet to which the router sends probes to determine the status of the route.

5. Click **Add**.
6. Click **Save**.
7. To create a tracker group, click **New Endpoint Tracker**.

From the **Tracker Type** drop-down list, choose **tracker-group** and configure the tracker group parameters.



Note Ensure that you have created two trackers to form a tracker group.

Table 201: Tracker Group Parameters

Fields	Description
Name	Name of the tracker group.

Fields	Description
Tracker Type	From the drop-down, choose Global. From the Tracker Type field drop-down, choose Static Route. From Cisco SD-WAN Release 20.7.1, you can configure a tracker group with dual endpoints on Cisco vEdge devices and associate this tracker group to a static route.
Tracker Elements	This field is displayed only if you chose Tracker-group as the tracker type. Add the existing interface tracker names (separated by a space). When you add this tracker to the template, the tracker group is associated with these individual trackers, and you can then associate the tracker group to a static route.
Tracker Boolean	From the drop-down list, choose Global. This field is displayed only if you chose tracker-group as the Tracker Type. By default, the OR option is selected. Choose AND or OR. OR ensures that the static route status is reported as active if either one of the associated trackers of the tracker group report that the route is active. If you select AND, the static route status is reported as active if both the associated trackers of the tracker group report that the route is active.

8. Click **Add**.

9. Click **Save**.



Note Complete all the mandatory actions before you save the template.

Configure a Next Hop Static Route with Tracker

Use the **VPN** template to associate a tracker to a static route next hop.



Note You can apply only one tracker per static route next hop.

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Feature Templates**.



Note In Cisco vManage Release 20.7.x and earlier releases, **Feature Templates** is titled **Feature**.

3. Navigate to the **Cisco VPN Template** for the device.



Note For information about creating a VPN template, see [Create VPN Template](#).

4. Enter **Template Name** and **Description** as required.
5. In Basic Configuration, by default, VPN is set to 0. Set a VPN value within (Range: 1 to 65525, excluding 512. For details see the VRF range behavior change described [here](#).) for service VPNs, for service-side data traffic on Cisco IOS XE Catalyst SD-WAN devices.



Note You can configure static route tracker only on service VPNs.

6. Click **IPv4 Route**.
7. Click **New IPv4 Route**.
8. In the **IPv4 Prefix** field, enter a value.
9. Click **Next Hop**.
10. Click **Add Next Hop** and enter values for the fields listed in the table.

Parameter Name	Description
Address	Specify the next-hop IPv4 address.
Distance	Specify the administrative distance for the route.
Tracker	Enter the name of the gateway tracker to determine whether the next hop is reachable before adding that route to the route table of the device.
Add Next Hop	Enter the name of the gateway tracker with the next hop address to determine whether the next hop is reachable before adding that route to the route table of the device.

11. Click **Add** to create the static route with the next-hop tracker.



Note Configuring a static route with a next-hop 'X.X.X.255' is not supported.
Cisco vEdge device does not implement RFC 3021.

12. Click **Save**.



Note You need to fill all the mandatory fields in the form to save the VPN template.

Monitor Static Route Tracker Configuration

View Static Route Tracker

To view information about a static tracker on a transport interface:

1. From the Cisco SD-WAN Manager menu, choose **Monitor** > **Devices**.
Cisco vManage Release 20.6.x and earlier: From the Cisco SD-WAN Manager menu, choose **Monitor** > **Network**.
2. Choose a device from the list of devices.
3. Click **Real Time**.
4. From the **Device Options** drop-down list, choose **Static Route Tracker Info**.

Configure Static Routes Using CLI

The following sections provide information about how to configure static routes using the CLI.

Configure a Static Route Tracker



Note You can configure static route tracking using the Cisco SD-WAN Manager CLI Add-on feature templates and CLI device templates. For more information on configuring using CLI templates, see [CLI Templates](#).

```
Device# config terminal
Device(config)# system tracker <tracker-name>
Device(config-tracker-trackername)# tracker-type <tracker-type>
Device(config-tracker-trackername)# endpoint-ip <ip-address>
Device(config-tracker-trackername)# threshold <value>
Device(config-tracker-trackername)# multiplier <value>
Device(config-tracker-trackername)# interval <value>
Device(config-tracker-trackername)# exit
```

Configure a Static Route Tracker with TCP Port as the Endpoint

```
Device# config terminal
Device(config)# system tracker <tracker-name>
Device(config-tracker-trackername)# tracker-type <tracker-type>
Device(config-tracker-trackername)# endpoint-ip <ip-address> tcp <port-number>
Device(config-tracker-trackername)# threshold <value>
Device(config-tracker-trackername)# multiplier <value>
Device(config-tracker-trackername)# interval <value>
Device(config-tracker-trackername)# exit
```

Configure a Static Route Tracker with UDP Port as the Endpoint

```
Device# config terminal
Device(config)# system tracker <tracker-name>
```

```

Device(config-tracker-trackername) # tracker-type <tracker-type>
Device(config-tracker-trackername) # endpoint-ip <ip-address> udp <port-number>
Device(config-tracker-trackername) # threshold <value>
Device(config-tracker-trackername) # multiplier <value>
Device(config-tracker-trackername) # interval <value>
Device(config-tracker-trackername) # exit

```

Configure Tracker Groups



Note You can create tracker groups to probe static routes from Cisco SD-WAN Release 20.7.1 and Cisco vManage Release 20.7.1.

```

Device# config terminal
Device(config) # system tracker <tracker-name1>
Device(config-tracker-trackername1) # tracker-type <tracker-type>
Device(config-tracker-trackername1) # endpoint-ip <ip-address> tcp <port-number>
Device(config-tracker-trackername1) # threshold <value>
Device(config-tracker-trackername1) # multiplier <value>
Device(config-tracker-trackername1) # interval <value>
Device(config-tracker-trackername1) # exit

Device(config) # system tracker <tracker-name2>
Device(config-tracker-trackername2) # tracker-type <tracker-type>
Device(config-tracker-trackername2) # endpoint-ip <ip-address> udp <port-number>
Device(config-tracker-trackername2) # threshold <value>
Device(config-tracker-trackername2) # multiplier <value>
Device(config-tracker-trackername2) # interval <value>
Device(config-tracker-trackername2) # exit

Device(config) # system tracker <tracker-group-name>
Device(config-tracker-tracker-group-name) # tracker-type <tracker-group>
Device(config-tracker-tracker-group-name) # tracker-elements <tracker-name1> <tracker-name2>
Device(config-tracker-tracker-group-name) # boolean {and | or}
Device(config-tracker-tracker-group-name) # exit

```

Configure a Next Hop Static Route with Tracker

```

Device(config) # system
Device(config) # vpn <vpn-number>
Device(config-vpn-vpn-number) # ip route <ipv4address/prefix> <ip-address>
<administrative-distance> tracker <tracker-name>

```



Note Configuring a static route with a next-hop 'X.X.X.255' is not supported.
Cisco vEdge device does not implement RFC 3021.

**Note**

- Use the **ip route** command to bind a tracker or tracker group with a static route and to configure a backup route for administrative distance that is higher than the default value of 1.
- You can apply only one tracker to an endpoint.
- A tracker group can have a mix of endpoint trackers. For example, you can create a tracker group with an IP address tracker and UDP tracker.

Configuration Examples for Static Route Tracking Using the CLI

Configure Tracker

This example shows how to configure a single static route tracker:

```
config terminal
!
system tracker tracker1
!
tracker-type static-route
endpoint-ip 10.1.1.1
threshold 100
multiplier 5
interval 60
exit
!
vpn 1
ip route 192.0.2.0/24 10.20.24.17 tracker tracker1
ip route 172.16.0.0/12 10.20.24.16 100
```

This example shows how to configure a tracker with TCP port as endpoint:

```
config terminal
!
system tracker tcp-10001
!
tracker-type static-route
endpoint-ip 10.0.0.1 tcp 10001
threshold 100
interval 10
multiplier 1
exit
!
vpn 1
ip route 192.0.0.4/24 10.20.25.18 tracker tcp-10001
```

This example shows how to configure a tracker with UDP port as endpoint:

```
config terminal
!
system tracker udp-10001
!
tracker-type static-route
endpoint-ip 10.0.0.1 udp 10001
threshold 100
```

```

        interval      10
        multiplier    1
        exit
!
vpn 1
ip route 192.0.0.4/24 10.20.30.19 tracker udp-10001

```

Configure Tracker Groups

This example shows how to configure a tracker group with two trackers (two endpoints). You can create tracker groups to probes static routes from Cisco SD-WAN Release 20.7.1.

```

config terminal
!
system tracker tcp-10001
!
    tracker-type static-route
    endpoint-ip 10.1.1.1 tcp 10001
    threshold 100
    multiplier 5
    interval 20
!
system tracker udp-10002
!
    tracker-type static-route
    endpoint-ip 10.2.2.2 udp 10002
    threshold 100
    multiplier 5
    interval 20
!
system tracker group-tcp-10001-udp-10002
!
    tracker-type tracker-group
    boolean and
    tracker-elements tcp-10001 udp-10002
    exit
!
vpn 1
ip route 192.168.2.0/16 10.20.24.17 tracker group-tcp-10001-udp-10002
ip route 192.168.2.0/16 10.20.24.16 100

```



Note

- You must configure an administrative distance when you are configuring through CLI templates.
- Use the **ip route** command to bind the tracker or tracker group with a static route and to configure a backup route for administrative distance when it is higher than the default value of 1.
- You can apply only one tracker to an endpoint.
- Configuring a static route with a next-hop 'X.X.X.255' is not supported.
Cisco vEdge device does not implement RFC 3021.

Verify Static Route Tracking Configuration Using CLI

Command Verification

Use the following command to verify if the configuration is committed. The following sample configuration shows tracker definition for a static route tracker and its application to an IPv4 static route:

```
Device# show running-config system tracker
system
  tracker tracker1
  endpoint-ip 10.1.1.1
  interval 60
  multiplier 5
  tracker-type static-route

  tracker tracker2
  endpoint-ip 10.1.1.12
  interval 40
  multiplier 2
  tracker-type static-route
```

Use the following command to verify the IPv4 route:

```
Device# show running-config vpn 1 ip route

vpn 1
  ip route 10.20.30.0/24 10.20.30.1
  ip route 192.168.2.0/16 10.20.24.16 100
  ip route 192.168.2.0/16 10.20.24.17 tracker tracker1
!
```

The following is a sample output from the **show tracker static-route** command displaying individual static route tracker status:

```
Device# show tracker static-route
TRACKER          RTT IN
NAME      VPN  STATUS  MSEC
-----
tcp-10001  1      UP       0
udp-10002  1      UP       0
```

The following is a sample output from the **show tracker static-route-group** command displaying tracker group status:

```
Device# show tracker static-route-group

TRACKER NAME          VPN  BOOLEAN  STATUS  TRACKER
ELEMENT              NAME      STATUS   RTT
-----
group-tcp-10001-udp-10002  1    and      UP      tcp-10001  UP      0
                        udp-10002  UP      0
```

The following is a sample output from the **show ip route static** command:

```
Device# show ip route static
Codes Proto-sub-type:
  IA -> ospf-intra-area, IE -> ospf-inter-area,
  E1 -> ospf-external1, E2 -> ospf-external2,
  N1 -> ospf-nssa-external1, N2 -> ospf-nssa-external2,
  e -> bgp-external, i -> bgp-internal
Codes Status flags:
```

Verify Static Route Tracking Configuration Using CLI

F -> fib, S -> selected, I -> inactive,
 B -> blackhole, R -> recursive, L -> import

		PROTOCOL	NEXTHOP			NEXTHOP			NEXTHOP
VPN	PREFIX	PROTOCOL	SUB TYPE	IF NAME	ADDR	VPN	TLOC IP	COLOR	
ENCAP	STATUS								
1	192.168.2.0/16	STATIC	-	ge0/4	10.20.24.17	-	-	-	
-	F,S								
1	192.168.2.0/16	STATIC	-	ge0/4	10.20.24.16	-	-	-	
-	F,S								



CHAPTER 12

VRRP Interface Tracking

Table 202: Feature History

Feature Name	Release Information	Description
VRRP Interface Tracking for Cisco vEdge Devices	Cisco SD-WAN Release 20.4.1 Cisco vManage Release 20.4.1	This feature enables VRRP to set the edge as active or standby based on the WAN Interface or SIG tracker events and increase the TLOC preference value on a new VRRP active to ensure traffic symmetry, for Cisco vEdge Devices. In this release, you can configure VRRP interface tracking using only the CLI template.
VRRP Interface Tracking for Cisco vEdge Devices.	Cisco SD-WAN Release 20.7.1 Cisco vManage Release 20.7.1	Starting this release, you can configure VRRP interface tracking through Cisco SD-WAN Manager feature template on Cisco vEdge Devices.

- [Information About VRRP Interface Tracking, on page 464](#)
- [Restrictions and Limitations, on page 464](#)
- [VRRP Tracking Use Cases, on page 465](#)
- [Configure Object Tracker Using a Configuration Group, on page 465](#)
- [Workflow to Configure VRRP Tracking, on page 466](#)
- [Configure an Object Tracker, on page 467](#)
- [Configure VRRP for a VPN Interface Template and Associate Interface Object Tracker, on page 467](#)
- [Configure VRRP Tracking Using CLI Templates, on page 468](#)
- [Configuration Example for VRRP Object Tracking Using CLI, on page 469](#)
- [Configuration Examples for SIG Object Tracking, on page 470](#)
- [Verify VRRP Tracking, on page 470](#)

Information About VRRP Interface Tracking

The Virtual Router Redundancy Protocol (VRRP) is a LAN-side protocol that provides redundant gateway service for switches and other IP end stations. In Cisco Catalyst SD-WAN, you can configure VRRP on interfaces and subinterfaces, within a VPN.

For more information, see [Configuring VRRP](#).

The VRRP Tracking feature enables switching to a backup or a secondary VRRP router in the following scenarios:

- If a single tunnel (or two tunnels - when you configure redundancy using Transport Locators (TLOC)) on a vEdge device goes down. In this case, the VRRP priority decrements and the secondary router becomes the primary router. VRRP notifies this change to the overlay through Overlay Management Protocol (OMP).
- VRRP can track up to one interface object or Secure Internet Gateway (SIG) object for a group. The interface object can have up to four interfaces. Hence, a group can track up to four tunnel interfaces. The VRRP priority decrements only if all the interfaces of an interface object go down.

Restrictions and Limitations

- VRRP is only supported with service-side VPNs. If you are using subinterfaces, configure VRRP physical interfaces in VPN 0.
- VRRP tracking is enabled on either a physical uplink interface or a logical tunnel interface (IPSEC or GRE or both).
- The VRRP Tracking feature does not support IP prefix as an object.
- You can track a maximum of four interfaces simultaneously using a single tracker. VRRP state transition gets triggered only if all four interfaces go down.
- You can use the same tracker under multiple VRRP groups or VPNs.
- You cannot configure **tloc-change** and **increase-preference** on more than one VRRP group.
- In Cisco SD-WAN release 20.6.1 and earlier releases, you can configure VRRP tracking only through Cisco vManage CLI template.

**Note**

Starting from Cisco SD-WAN release 20.7.1, you can configure VRRP tracking using Cisco vManage feature template as well.

**Note**

In Cisco SD-WAN release 20.6.1 and earlier releases, to update any existing VRRP configuration and add VRRP tracking, convert the configuration and the VRRP tracking commands to the CLI template.

VRRP Tracking Use Cases

The VRRP state is determined based on the tunnel link status. If the tunnel or interface is down on the primary VRRP, then the traffic is directed to the secondary VRRP. The secondary VRRP router in the LAN segment becomes primary VRRP to provide gateway for the service-side traffic.

Zscaler Tunnel Use Case 1—Primary VRRP, Single Internet Provider

The primary and secondary Zscaler tunnels are connected through a single internet provider to the primary VRRP. The primary and secondary VRRP routers are connected through using TLOC extension. In this scenario, the VRRP state transition occurs if the primary and secondary tunnels go down on primary VRRP. The predetermined priority value decrements when the tracking object is down, which triggers the VRRP state transition. To avoid asymmetric routing, VRRP notifies this change to the Overlay through OMP.

Zscaler Tunnel Use Case 2—VRRP Routers in TLOC Extension, Dual Internet Providers

The primary and secondary VRRP routers are configured in TLOC extension high availability mode. The primary and secondary Zscaler tunnels are directly connected with primary and secondary VRRP routers, respectively, using dual internet providers. In this scenario too, the VRRP state transition occurs if the primary and secondary tunnels go down on primary VRRP. The predetermined priority value decrements when the tracking object is down, which triggers the VRRP state transition. VRRP notifies this change to the Overlay through OMP.

TLOC Preference

Transport Locators (TLOCs) connect an OMP route to a physical location. A TLOC is directly reachable using an entry in the routing table of the physical network, or represented by a prefix beyond a NAT device.

The TLOC change preference is an optional configuration under VRRP group. If you configure TLOC change preference value using the `tloc-change-pref` command, the value increases by 1 when a node becomes the primary node. The configured or default TLOC preference is applied back on standby state.



Note

We recommend that you use the same TLOC preference value for all TLOCs in a site. For a Cisco vEdge device, the default TLOC preference for the tunnel interface can be modified irrespective of whether VRRP is configured or not. However, if you want to use the VRRP tracking feature and utilize the advantage of TLOC preference values for VRRP tracking, ensure that the default tunnel preference is same on both the VRRP routers.

Configure Object Tracker Using a Configuration Group

Before you begin

On the **Configuration > Configuration Groups** page, choose **SD-WAN** as the solution type.

Procedure

Step 1 From the Cisco SD-WAN Manager menu, choose **Configuration > Configuration Groups**.

Step 2 Create and configure a object tracker in a Service Profile.

Configure basic settings.

Table 203: Basic Settings

Parameter Name	Description
Tracker Type*	
Interface	Configure the following interface values: • Object tracker ID*: Enter the object tracker ID number. Range: 1-1000 • Interface name*: Enter the global or device-specific tracker interface name. For example, Gigabitethernet1 or Gigabitethernet2.
SIG	Object tracker ID* : Enter the object tracker ID number.
Route	Configure the route details: • Object tracker ID*: Enter the object tracker ID number. Range: 1-1000 • Route IP*: Enter the IPv4 address of the route. • Route IP Mask*: Select a value for the subnet mask. • VPN: Enter a value for the VPN.

What to do next

Also see [Deploy a configuration group](#).

Workflow to Configure VRRP Tracking

1. Configure an object tracker. For more information, see [Configure an Object Tracker, on page 467](#).
2. Configure VRRP for a VPN Interface template and associate the object tracker with the template. For more information, see [Configure VRRP for a VPN Interface Template and Associate Interface Object Tracker, on page 467](#).

Configure an Object Tracker

Use the **System** template to configure an object tracker.

1. From the Cisco SD-WAN Manager menu, choose **Configuration** > **Templates**.
2. Click **Feature**.
3. Navigate to the **System** template for the device.



Note To create a **System** template, see [Create System Template](#)

4. Click **Tracker**, and click **New Object Tracker** to configure the tracker parameters.

Table 204: Tracker Parameters

Field	Description
Tracker Type	Choose Interface or SIG to configure the Object tracker.
Tracker List	Enter the name of the tracker list.
Interface	Choose global or device-specific tracker interface name.

5. Click **Add**.
6. Click **Save**.

Configure VRRP for a VPN Interface Template and Associate Interface Object Tracker

To configure VRRP for a **VPN** template, do the following:

1. From the Cisco SD-WAN Manager menu, choose **Configuration** > **Templates**.
2. Click **Feature Templates**.



Note In Cisco vManage Release 20.7.x and earlier releases, **Feature Templates** is titled **Feature**.

3. Navigate to the **VPN Interface Ethernet** template for the device.



Note For information about creating a new **VPN Interface Ethernet** template, see [Configure VPN Ethernet Interface](#).

4. Click **VRRP** and choose **IPv4**.
5. Click **New VRRP** to create a new VRRP or edit the existing VRRP and configure the following parameters:

Parameter Name	Description
TLOC Preference Change	(Optional) Choose On or Off to set whether the TLOC preference can be changed or not.

6. Click the **Add Tracking Object** link, and in the **Tracking Object** dialog box that is displayed, click **Add Tracking Object**.
7. In the **Tracker Name** field, enter the name of the tracker.
8. From the **Action** drop-down list, choose **Decrement** and enter the **Decrement Value**.
9. Click **Add**.
10. Click **Add** to save the VRRP details.
11. Click **Save**.

Configure VRRP Tracking Using CLI Templates

You can configure VRRP tracking using the CLI add-on feature templates and CLI device templates. For more information, see [CLI Templates](#).

VRRP Object Tracking Using CLI

Configure Track List Interface

Use the following configuration to add an interface to a track list using Cisco SD-WAN Manager device CLI template:

```
Device# config terminal
Device(config)# system
Device(config-system)# track-list zsl interface ge0/1 gre1 ipsec1
Device(config-track-list-zsl)# commit
Device(config-system-tracker-list-zsl)# exit
Device(config-system)# exit
```

Configure Interface Tracking and Priority Decrement

```
Device(config)# vpn 1
Device(config-vpn-1)# name vpn-name
Device(config-vpn-1)# interface ge0/2
Device(config-interface-ge0/2)# ip address 172.16.10.1/24
Device(config-interface-ge0/2)# no shutdown
Device(config-interface-ge0/2)# vrrp 100
Device(config-vrrp-100)# track zsl decrement 10
Device(config-vrrp-track-zsl)# exit
Device(config-vrrp-100)# ipv4 172.16.10.100
Device(config-vrrp-100)# tloc-change-pref
```

SIG Container Tracking

The following example shows how to configure a track list and tracking for SIG containers using the Cisco SD-WAN Manager device CLI template.



Note In SIG Object Tracking, you can only set *global* as the variable for Service Name.

Configure Track List for SIG Container

```
Device# config terminal
Device(config)# system
Device(config-system)# track-list SIG sig-container global
Device(config-system-tracker-list-SIG)# exit
Device(config-system)# exit
```

Configure SIG Container Tracking and Priority Decrement

```
Device(config)# vpn 1
Device(config-vpn-1)# name vpn-name
Device(config-vpn-1)# interface ge0/2
Device(config-interface-ge0/2)# ip address 172.16.10.1/24
Device(config-interface-ge0/2)# no shutdown
Device(config-interface-ge0/2)# vrrp 100
Device(config-vrrp-100)# track SIG decrement 10
Device(config-vrrp-track-zs1)# exit
Device(config-vrrp-100)# ipv4 172.16.10.100
Device(config-vrrp-100)# tloc-change-pref
```

Configure SIG Container Tracking for VRRP Group

```
Device(config-vpn-1)# int ge0/4
Device(config-interface-ge0/4)# vrrp 10
Device(config-vrrp-10)# track SIG decrement 10
Device(config-track-SIG)# commit
Commit complete.
Device(config-track-SIG)#
```

Configuration Example for VRRP Object Tracking Using CLI

Interface Object Tracking Using CLI

This example shows how to add an interface to a track list using Cisco vManage device CLI template:

```
Configure terminal
system
track-list zs1 interface ge0/1 gre1 ipsec1
commit
exit
```

Configure Interface Tracking and Priority Decrement

```
vpn 1
name vpn-name
interface ge0/2
```

```
ip address 172.16.10.1/24
no shutdown
vrrp 100
track zsl decrement 10
exit
ipv4 172.16.10.100
tloc-change-pref
```

Configuration Examples for SIG Object Tracking

Configure Track List for SIG Container

```
config terminal
system
track-list SIG sig-container global
exit
exit
```

Configure SIG Container Tracking and Priority Decrement

```
vpn 1
name vpn-name
interface ge0/2
ip address 172.16.10.1/24
no shutdown
vrrp 100
track SIG decrement 10
exit
ipv4 172.16.10.100
tloc-change-pref
```

Verify VRRP Tracking

Device# show vrrp

The following is a sample output for the **show vrrp** command:

```
vrrp vpn 1
  interfaces ge0/4
    groups 10
      virtual-ip          10.1.1.2
      virtual-mac         00:00:5e:00:01:0a
      priority            100
      real-priority        100
      vrrp-state           init
      omp-state            up
      advertisement-timer  1
      primary-down-timer   3
      last-state-change-time 0000-00-00T00:00:00+00:00
```

Device# show vrrp detail

The following is a sample output for the **show vrrp detail** command:

```
OMP status: up

group-id: 10, track-omp: no, initialized: yes
address: 10.20.24.1
track-prefix-list: -, resolved: -
```

```

state: Primary, down-reason: none, cfg-priority: 100, priority: 100
adv-timer: 1, primary-down-timer: 3, sock-fd: 23, addr-count: 1
adv-timer: Enabled (e: 4 v: 10 c: 1)
primary-down-timer: Disabled (e: -1 v: 30 c: 3)
virtual-mac: 0x0 0x0 0x5e 0x0 0x1 0xa
TLOC Change Preference: Configured
TLOC Change Preference value: 1
TLOC Real Preference value: 1
Group current adaptive priority: 0
Total Tracking object : 1 (head: 0x7f0f6d6771c0)
Group Address: 0x7f0f6d624100
  Name: zs1
  Decrement: 18
  Adaptive direction: 0
  List Entry :0x7f0f6d687230

```

Track List:

```

Name: zs1
Total Tracking Objects: 0
VRRP Daemon: 0x7f0f6d68e140
Tracking Object: 0x7f0f6d677270
  Type: 1
  VRRP Daemon: 0x7f0f6d68e140
  Total Interface: 1
    Interface: ge0_1(0x7f0f6d66a700)
    Interface Created: Yes
    Operational State: UP

```

Device# **show run system**

The following is a sample output for the **show run system** command:

```

system
host-name                vm6
system-ip                172.16.255.16
site-id                  600
no admin-tech-on-failure
route-consistency-check
organization-name        "vIPtela Inc Regression"
track-list SIG
  container global
!
track-list zs1
  track-interface ge0/1 ge0/7
!

```




CHAPTER 13

Configure a Cisco vEdge Device as an NTP Parent

Table 205: Feature History

Feature Name	Release Information	Feature Description
Configure a Cisco vEdge Device as an NTP Parent and Optionally to Support NTP in Symmetric Active Mode.	Cisco SD-WAN Release 20.4.1 Cisco vManage Release 20.4.1	This feature enables configuring a Cisco vEdge device as an NTP parent and configuring the device to support NTP in symmetric active mode.

You can configure a Cisco vEdge device as an NTP parent. You also can configure an NTP parent device to support NTP in symmetric active mode.

- [Configure an NTP Parent, on page 473](#)
- [Configure Support for NTP in Symmetric Active Mode, on page 474](#)

Configure an NTP Parent

Starting with Cisco SD-WAN Release 20.4.1, you can configure a supported Cisco vEdge device as an NTP parent device by using the device CLI template. A device that is configured in this way acts as the NTP server to which other nodes in the deployment synchronize their clocks. You can configure multiple devices as NTP parents. The NTP server functionality is supported for IPv4, but not for IPv6.

You also can configure a device that is configured as an NTP parent device to support NTP in symmetric active mode. See "Configure Support for NTP in Symmetric Active Mode."

Use the following commands to configure device as an NTP parent device using a Cisco vEdge device device CLI template. For more information about configuring device CLI template, see "Create a Device CLI Template" in *Systems and Interfaces Configuration Guide*.

```
Device# config terminal
Device(config)# system
Device(config-system) ntp
Device(config-ntp) # parent
Device(config-parent) # enable
Device(config-parent) # source-interface loopback511
Device(config-parent) # stratum 6
Device(config-parent) # vpn 511
Device(config-parent) # exit
```

Restrictions and Limitations

- You can configure a device as an NTP parent only through a Cisco SD-WAN Manager CLI template. Cisco SD-WAN Manager feature templates do not support this configuration.
- The source interface must be in the same VPN that the vpn keyword defines.

Verify Configuration

Use the following show command to verify NTP parent configuration. The sample output shows that the server also is configured to support NTP in symmetric active mode.

```
Device# show running-config system ntp

system
 ntp
  keys
    authentication 101 md5 $8$vV6PtHeLdiEcLqDNLqV/mCWN5X92yT8PUPowDCQgS4c=
    authentication 108 md5 $8$NTzFC6sRZiFUyeHw/pOY2dEoiO6dpxphedDs7YnRKeuY=
    trusted 101 108
  !
parent
 enable
 stratum 6
 source-interface loopback511
 vpn 511
exit
server 10.20.25.1
 source-interface ge0/1
 vpn 511
 version 4
exit
peer 172.16.10.100
 key 101
 vpn 511
 version 4
 source-interface ge0/1
exit
```

Configure Support for NTP in Symmetric Active Mode

Starting with Cisco SD-WAN Release 20.4.1, you can configure a Cisco vEdge device that is configured as an NTP parent to support NTP in symmetric active mode by using the device CLI template. When a device is configured in this way, it synchronizes its time with another device that is defined with this mode if it cannot reach its original NTP parent.

Use the following commands to configure a device to support NTP in symmetric active mode by using a Cisco vManage device CLI template. For more information about configuring device CLI template, see "Create a Device CLI Template" in *Systems and Interfaces Configuration Guide*.

```
Device# config terminal
Device# system
Device(config-system) ntp
Device(config-ntp)# peer 172.16.10.1
Device(config-peer)# key 101
Device(config-peer)# vpn 511
Device(config-peer)# version 4
Device(config-parent)# source-interface ge0/1
Device(config-parent)# exit
```

Restrictions and Limitations

- You can configure a device support NTP in symmetric active mode only through a Cisco SD-WAN Manager CLI template. Cisco SD-WAN Manager feature templates do not support this configuration.
- You can configure up to two devices to support NTP in symmetric active mode.
- A device that is configured as an NTP peer should also be configured as an NTP parent.
- The source interface must be in the same VPN that the vpn keyword defines.
- Each peer must use the same source interface.

Use the following show command to verify NTP parent functional configuration. In the **show ntp peer** command output, the server with the .LOCL. REFID is the NTP parent.

```
Device# show ntp peer
```

INDEX	REMOTE	REFID	ST	TYPE	WHEN	POLL	REACH	DELAY	OFFSET	JITTER
1	*10.20.25.1	.GNSS.	1	u	186	1024	377	226.712	0.793	2.381
2	172.16.10.1	(loop)	3	s	760	1024	376	0.126	-1.307	1.397
3	172.16.10.10	.LOCL.	6	1	52h	64	0	0.000	0.000	0.000



CHAPTER 14

Dynamic On-Demand Tunnels

Table 206: Feature History

Feature Name	Release Information	Description
Dynamic On-Demand Tunnels	Cisco SD-WAN Release 20.3.1 Cisco vManage Release 20.3.1	This feature enables you to configure an Inactive state for tunnels between edge devices, reducing performance demands on devices and reducing network traffic.
Dynamic On-Demand Tunnels with Transport Gateways	Cisco Catalyst SD-WAN Control Components Release 20.12.1 Cisco IOS XE Catalyst SD-WAN Release 17.12.1a	A transport gateway can serve as the hub between two spoke devices, providing the backup route that is necessary for spoke-to-spoke on-demand tunnels to operate. Using a transport gateway as the hub simplifies the process of enabling on-demand tunnels. This method does not require any change to control policy on Cisco SD-WAN Controllers.

- [Information About On-Demand Tunnels, on page 477](#)
- [Prerequisites for On-Demand Tunnels, on page 480](#)
- [Restrictions for On-Demand Tunnels, on page 484](#)
- [Configure On-Demand Tunnels, on page 485](#)
- [Monitor the Status of On-Demand Tunnels, on page 489](#)

Information About On-Demand Tunnels

Cisco Catalyst SD-WAN supports dynamic on-demand tunnels between any two Cisco Catalyst SD-WAN spoke devices. These tunnels are triggered to be set up only when there is traffic between the two devices. After the flow of traffic between the devices stops, a user-configurable inactivity timer starts, and after the configured time, the tunnel between the devices is removed. The on-demand link between the two devices is

then considered to be inactive. In this inactive state, it does not use network bandwidth and does not affect device performance.

Backup Route and Reactivating the Tunnel

To enable two spoke device peers to use on-demand tunnels, they must have an alternate route, a backup route, through a hub. Using the backup route, either spoke device can resume traffic flow between the two spokes, which reactivates the tunnel to handle the traffic directly from peer to peer.

Advantages

On-demand tunnels offer the following advantages:

- Improved performance, especially for less-powerful platforms operating in a full-mesh network.
- Improved latency in hub-and-spoke deployments when on-demand tunnels are used between spokes.
- Reduced bandwidth use in the network because tunnels in Inactive state do not require Bidirectional Forwarding Detection (BFD) probes, so there is less BFD traffic produced in the network.
- Direct tunnels between spokes, while also optimizing CPU and memory usage.

Mechanism

When you configure a site to use dynamic tunnels, the on-demand functionality is enabled. In this mode of operation, Cisco Catalyst SD-WAN edge routers do not bring up direct tunnels to other sites that are also enabled with on-demand functionality.

Cisco Catalyst SD-WAN selects one or more edge routers (typically centrally located routers) to act as backup forwarding node(s), providing a secondary path for traffic between two nodes. The backup node(s) are not enabled for on-demand. All on-demand sites form static tunnels with the backup node(s). The backup node(s) provide a static backup route for traffic between two nodes that have on-demand enabled.

The first packet of traffic between two nodes is routed through the static backup path, and triggers the on-demand tunnel to become active between the sites. The backup path continues to forward traffic until the direct path becomes active.

All on-demand sites learn the TLOCs and prefixes of all other on-demand remote sites. The prefixes also have a backup path set up through Cisco Catalyst SD-WAN Controller control policy. So in the control plane, the on-demand tunnel network has the same state as a full-mesh tunnel network, including a backup path.

The control plane downloads to the data plane, routes, with the backup path and remote TLOCs that represent a potential direct path between any two sites, but it does not set up a direct path tunnel to remote TLOCs.

All prefixes learned from remote sites must have a backup path. A less-specific aggregate route from a hub site is not a valid backup path. The specific prefix advertised from the remote branch must also be advertised as a backup from the hub.

If this specific prefix is missing as a backup, the on-demand tunnel setup will fail.

Therefore, the backup path must be a static tunnel that is always UP. This static tunnel should include both the specific prefixes from the remote branch and any necessary aggregate routes to ensure proper tunnel establishment and maintenance.

Traffic from either end of the on-demand tunnel triggers setting up the tunnel. This enables on-demand tunnels to accommodate network address translation (NAT) traversal.

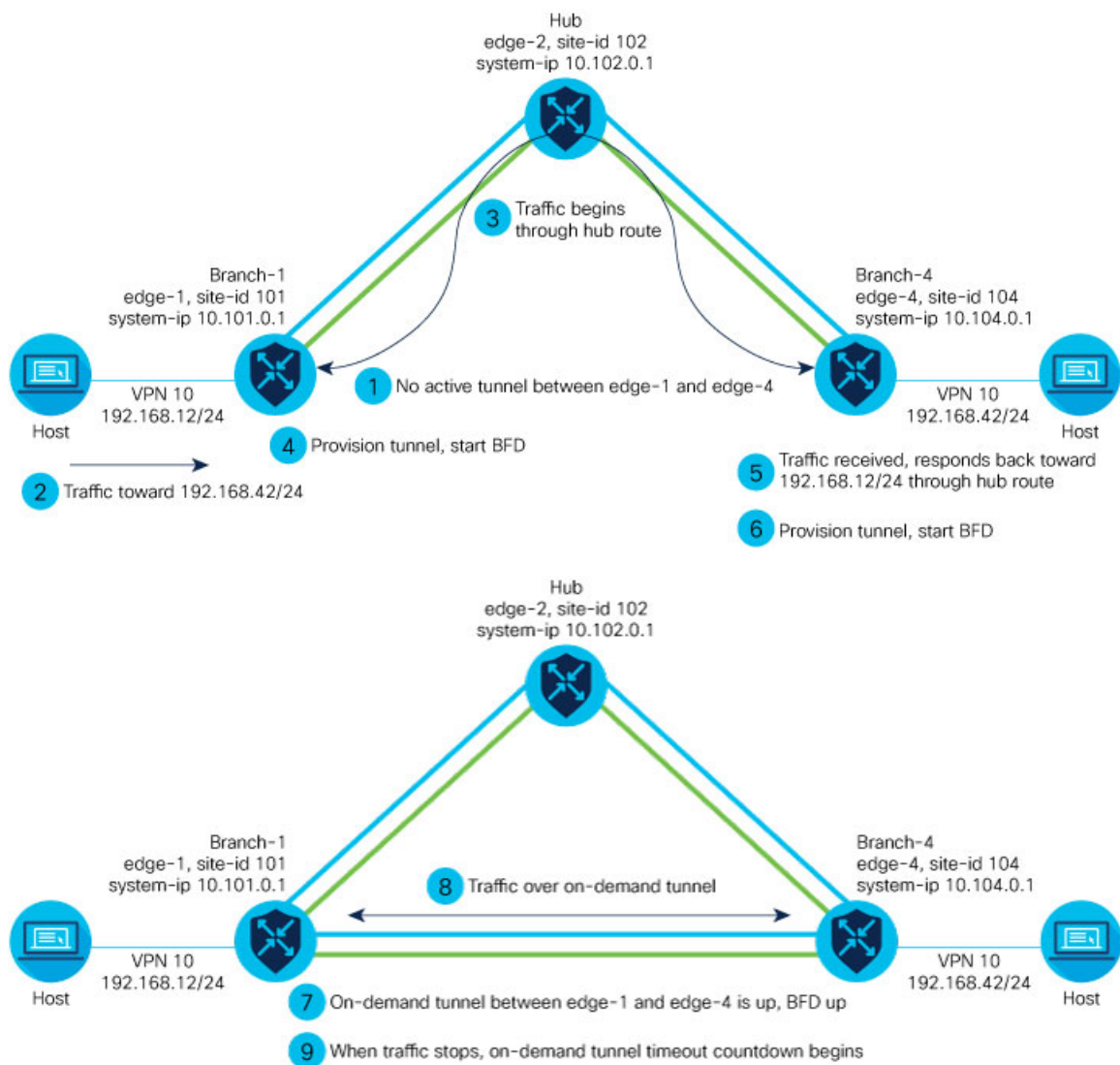
The on-demand tunnel feature introduces two states for the on-demand branch site:

- **Inactive:** The on-demand tunnel is not set up with the remote site. There is no active traffic to or from the remote site. Remote site TLOCs are inactive - no bidirectional forwarding detection (BFD) is set up, the prefix is installed with the inactive paths, and the backup path is set as the path to forward any traffic. The inactive path detects flows and triggers a direct site-to-site tunnel to be set up.
- **Active:** The on-demand direct site-to-site tunnel is set up to the remote site. There is active traffic to or from the remote site. This state is identical to the case of a typical tunnel, where the remote TLOCs have BFD set up, and the prefix is installed with the direct path tunnel. In this state, tunnel activity is tracked. If there is no traffic for the “idle time” duration (default 10 minutes), the direct site-to-site tunnel is removed and the state changes to Inactive.

Steps in Illustrations

The figures below show the following steps that occur between two edge routers with an on-demand tunnel configured.

1. There is no active tunnel between the two edge routers. edge-1 and edge-4 are in Inactive state.
 2. The host behind edge-1 initiates traffic toward the host behind edge-4.
 3. edge-1 forwards the traffic through the backup path using the hub or backup node to edge-4.
 4. edge-1 provisions the on-demand tunnel and begins bidirectional forwarding detection (BFD). edge-4 is now in Active state on edge-1.
 5. When edge-4 receives the return traffic for the host behind edge-1, it forwards the traffic through the backup path using the hub or backup node to edge-1.
 6. edge-4 provisions the on-demand tunnel and begins BFD. edge-1 is now in Active state on edge-4.
 7. At this point, the on-demand tunnel between edge-1 and edge-4 is up, and BFD is up.
 8. Traffic between the two edge devices takes the direct route through the on-demand tunnel.
 9. Both edge-1 and edge-4 track the traffic activity on the on-demand tunnel in both directions.
- If there is no traffic for the idle timeout duration, the on-demand tunnel is deleted, and the edge-1 and edge-4 devices go back to the Inactive state.



On-Demand Tunnels with a Transport Gateway

A transport gateway can serve as the hub between two spoke devices, providing the backup route that is necessary for spoke-to-spoke on-demand tunnels to operate. Using a transport gateway as the hub simplifies the process of enabling on-demand tunnels. This method does not require configuring control policy on Cisco SD-WAN Controllers.

For information about configuration, see [Configure On-Demand Tunnels Using a Transport Gateway, on page 487](#).

Prerequisites for On-Demand Tunnels

There are several prerequisites for using on-demand tunnels:

- [Configure a Centralized Control Policy for On-Demand Tunnels, on page 485](#)

- [Prerequisites: OMP Settings, on page 481](#)
- [Prerequisites: Hub Device Traffic Engineering Service, on page 482](#)
- [Prerequisites: Spoke Device ECMP Limit, on page 483](#)

Prerequisites: OMP Settings

The Cisco Catalyst SD-WAN Controller send-path-limit must be more than the default 4.

When on-demand tunnels are enabled, spokes use backup paths through the hub, so a higher path limit is necessary. The direct paths as well as the backup paths need to be advertised. To accommodate this, increase the Cisco Catalyst SD-WAN Controller send-path-limit to advertise all available paths. We recommend to use the maximum possible value.



Note If there are too many Hub TLOCs configured in the on-demand tunnel control policy, the recommended value for send-path-limit is not enough always. In such cases, the on-demand tunnel feature will not work at all.

Starting from Cisco vManage Release 20.8.1 and Cisco IOS XE Catalyst SD-WAN Release 17.8.1a, the maximum send-path-limit is 32. In Cisco vManage Release 20.7.x and earlier releases, the maximum send-path-limit is 16.

For information about configuring Cisco SD-WAN Controller send-path-limit, see the routing configuration guides on the [Cisco Catalyst SD-WAN Configuration Guides](#) page.

Configure the OMP Send Path Limit Using a Feature Template

For a Cisco Catalyst SD-WAN Controller in Managed mode, use this procedure to configure the OMP send path limit.

To confirm that it is in Managed mode, from the Cisco SD-WAN Manager menu, choose **Configuration > Control Components**. For a **Controller** row, the **Managed By** column shows a template name.

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Feature Templates**.



Note In Cisco vManage Release 20.7.x and earlier releases, **Feature Templates** is titled **Feature**.

3. Edit or create an OMP template for device type Controller. In some earlier releases, the device type is called vSmart.



Note Any functioning Cisco Catalyst SD-WAN deployment has at least one vSmart OMP template configured.

4. In **Basic Configuration**, set the **Number of Paths Advertised per Prefix** to 16 (recommended).
From Cisco SD-WAN Release 20.8.1, the maximum is 32.

Configure the OMP Send Path Limit for an Edge Device, Using CLI Commands

Use a CLI add-on profile, CLI add-on feature template, or CLI template to execute these CLI commands. By default, CLI templates and the CLI add-on profile execute commands in global configuration mode.

For an edge device, use these commands to configure an OMP send path limit.

1. Enter OMP configuration mode.

```
omp
```

2. Set the send path limit. We recommend 16.

From Cisco SD-WAN Release 20.8.1, the maximum is 32.

```
send-path-limit 16
```

Example configuration:

```
sdwan
omp
  send-path-limit 16
```

Prerequisites: Hub Device Traffic Engineering Service

On the hub device, the Traffic Engineering service (service TE) must be enabled.

This ensures that the Cisco Catalyst SD-WAN Overlay Management Protocol (OMP) on the spoke devices accepts the backup path through the hub, which is being added as an intermediate path between the two spoke devices. Without this, the backup path through the hub would be considered invalid and unresolved by the spoke devices.

Enable the Traffic Engineering Service Using Cisco SD-WAN Manager

1. In Cisco SD-WAN Manager, open **Configuration > Templates**.
2. Click **Feature Templates**.



Note In Cisco vManage Release 20.7.x and earlier releases, **Feature Templates** is titled **Feature**.

3. Click **Add Template**.
4. Select a platform.
5. From **VPN**, select **VPN**.
6. Ensure that in **Basic Configuration**, the **VPN** field is set to 0.
7. From **Service**, click **New Service** and select **TE**.
8. Click **Add**, and then click **Update**. The TE service appears in the table of services.
9. Apply the VPN-0 template to the hub.

Enable the Traffic Engineering Service, Using CLI Commands (Cisco IOS XE Catalyst SD-WAN Devices)

Use a CLI add-on profile, CLI add-on feature template, or CLI template to execute these CLI commands. By default, CLI templates and the CLI add-on profile execute commands in global configuration mode.

```
sdwan
 service TE vrf global
exit
```

Enable the Traffic Engineering Service, Using CLI Commands (Cisco vEdge Devices)

```
vpn 0
 service TE
exit
```

Prerequisites: Spoke Device ECMP Limit

On spoke devices, the ECMP limit must be more than the default 4. Recommended: 16

When on-demand tunnels are enabled, spoke devices create both direct and backup paths. To accommodate the need for more paths, increase the ECMP limit .

Configure the ECMP Limit Using a Feature Template

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Feature Templates**.



Note In Cisco vManage Release 20.7.x and earlier releases, **Feature Templates** is titled **Feature**.

3. Click **Add Template**.
4. Select a device and click **Cisco OMP**.
5. In **Basic Configuration**, set the **ECMP Limit** field to 16 (recommended).

Configure the ECMP Limit Using a Configuration Group

On the **Configuration > Configuration Groups** page, choose the SD-WAN solution type.

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Configuration Groups**.
2. Do one of these:
 - Edit a profile directly:
In the System Profile tab, create (**Add New**) or edit a System profile.
 - Edit a profile in a configuration group:
Open a configuration group and edit the System profile.
3. In the system profile, create or edit an **OMP** feature.
4. In the **Basic Configuration** section, use the **ECMP Limit** field to configure the equal-cost multi-path (ECMP) limit. We recommend 16.

Configure the ECMP Limit, Using CLI Commands

Use a CLI add-on profile, CLI add-on feature template, or CLI template to execute these CLI commands. By default, CLI templates and the CLI add-on profile execute commands in global configuration mode.

```
sdwan
omp
no shutdown
ecmp-limit      16
```



Note You can view the current `ecmp-limit` using the **show running-config omp** command.

Restrictions for On-Demand Tunnels

- On-demand tunnel Performance Routing (PfR) statistics collection starts fresh every time an on-demand tunnel is setup. PfR statistics are not cached for deleted on-demand tunnels after idle timeout.
- Out of order (OOO) packets may occur when traffic moves from the backup path to the direct on-demand tunnel. Packets are forwarded by the Cisco Catalyst SD-WAN router as they are received.
- Unidirectional flows do not trigger on-demand tunnel setup. They continue to use the backup path.
- Multicast traffic does not trigger on-demand tunnel setup. It continues to use the backup path.
- Do not configure a data policy that applies a **set tloc-list** action to an on-demand site TLOC. If configured, traffic will be dropped.
- On-demand tunnels are not supported when the Pair Wise Key (PWK) IPSEC feature is enabled.
- All TLOCs in the system are reset (disabled and then enabled) when you execute **on-demand enable** or **no on-demand enable**.
- When an edge device provisions on-demand tunnels, it provisions to all the TLOCs on the remote edge device.
- For a multi-home site to be in on-demand mode, you must configure on-demand enable on all of the systems at the site.
- All edge devices using on-demand tunnels are kept active if there is a service or user traffic on any on-demand tunnel in either direction.
- On-demand tunnels can be enabled between two sites only if both sites are enabled with on-demand mode.
- The first packet to any host behind a remote site triggers on-demand tunnel setup to that remote site. Return traffic from that host triggers tunnel setup in the opposite direction.
- All prefixes from on-demand remote sites must also have a backup path configured. If not, sites will not be able set up on-demand tunnels. The backup path is a static tunnel and must be always UP.
- The setup or removal of on-demand tunnels does not affect overlay route (OMP) updates by Cisco Catalyst SD-WAN Controller, or service/LAN-side route updates (examples: OSPF or BGP).
- If either the local site or the remote site is not in on-demand mode, static tunnels are set up between the sites.

Configure On-Demand Tunnels

The following procedures describe how to configure on-demand tunnels using different methods, including using control policy, or a simpler method using a transport gateway as a hub.

Configure On-Demand Tunnels Using Control Policy

To configure on-demand tunnels using the control policy method, do the following:

1. Configure a control policy, as described in [Configure a Centralized Control Policy for On-Demand Tunnels, on page 485](#).
2. On spoke devices, enable on-demand tunnels, as described in [Enable On-Demand Tunnels on a Spoke Device Using a Template, on page 488](#) and [Enable On-Demand Tunnels Using a CLI Template, on page 489](#).

Configure a Centralized Control Policy for On-Demand Tunnels

Before You Begin

This procedure configures a centralized control policy on a Cisco Catalyst SD-WAN Controller to enable on-demand tunnels.

- The Cisco Catalyst SD-WAN Controller centralized control policy must include the **tloc-action backup** action.

Explanation: This ensures that the backup path through the hub for communication between all of the spoke devices.

- The Cisco Catalyst SD-WAN Controller centralized control policy must accept all spoke prefix routes.
- The Cisco Catalyst SD-WAN Controller centralized control policy must accept TLOCs of all spokes.

For information about configuring a Cisco SD-WAN Controller **centralized control policy**, see the policies configuration guides on the [Cisco Catalyst SD-WAN Configuration Guides page](#).

- When configuring on-demand tunnels using a transport gateway, do not use the control policy procedure described here. For information, see [Configure On-Demand Tunnels Using a Transport Gateway, on page 487](#).

Configure a Centralized Control Policy for On-Demand Tunnels Using Cisco SD-WAN Manager

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Policies**.
2. Select **Centralized Policy**.
3. Click **Add Policy**.
4. In the left pane, click **Site**.
5. Click **Next**.
6. Click **Add Topology** and select **Custom Control (Route & TLOC)**.

7. Enter a name and description for the topology.
8. Click **Sequence Type**.
9. In the **Add Control Policy** pop-up window, choose **Route**.
10. Click **Sequence Rule** to create a sequence.
11. Click **Match**.
12. Among the match conditions, click **Site**.
13. In the **Match Conditions** area, click the **Site List** menu and choose a site list.
14. Click **Actions**.
15. Click **Accept**.
16. Among the actions, click **TLOC Action**.
17. In the **Actions** area, click the **TLOC Action** menu and choose **Backup**.
18. Among the actions, click **TLOC**.
19. In the **Actions** area, click the **TLOC List** menu and choose or create a TLOC list.
20. Click **Save Match and Actions**.
21. Click **Default Action**.
22. In the **Default Action** area, click the pencil icon to edit.
23. Near the **Actions** label, click **Accept**.
24. Click **Save Control Policy**.
25. Click **Next** twice.
26. In the Topology tab, click **New Site/WAN Region List**.
27. Click **Outbound Site List** and choose a site list that defines the sites at which you are enabling on-demand tunnels.
28. Adjacent to the site list, click **Add**.
29. Enter a name and description for the policy.
30. Click **Save Policy**.

Configure Centralized Control Policy for On-Demand Tunnels Using a CLI Policy

Prerequisite: The Cisco SD-WAN Controller must be managed by Cisco SD-WAN Manager.

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Classic > Policies**.
2. Open **Centralized Policy**.
3. From **Custom Options**, choose **Centralized Policy > CLI Policy**.
4. Click **Add Policy**.
5. Enter the CLI commands for the policy.

Example:

```

control-policy Dynamic-Tunnel-Control-Policy
  sequence 100
    match route
      site-list Branches
    !
    action accept
    set
      tloc-action backup
      tloc-list Hub-TLOCs
    !
  !
  sequence 200
    match tloc
  !
  action accept
  !
  default-action accept
  !
lists
  site-list Branches
    site-id 200
    site-id 300
  !
  tloc-list Hub-TLOCs
    tloc 10.0.0.1 color mpls encaps ipsec
    tloc 10.0.0.1 color public-internet encaps ipsec
  !
!
!
apply-policy
  site-list Branches
  control-policy Dynamic-Tunnel-Control-Policy out
  !
!

```

Configure On-Demand Tunnels Using a Transport Gateway

Before You Begin

- On Cisco SD-WAN Controllers, configure the send path limit, as described in [Prerequisites: OMP Settings, on page 481](#).
- On spoke devices, configure the ECMP limit, as described in [Prerequisites: Spoke Device ECMP Limit, on page 483](#).
- When using a transport gateway as a hub to support on-demand tunnels, there is no need to create or modify a control policy. Do not use the procedure described in [Configure a Centralized Control Policy for On-Demand Tunnels, on page 485](#).

Configure On-Demand Tunnels Using Transport Gateways

1. On a router serving as the hub, providing a backup route between spokes, enable transport gateway functionality, as described in , in the [Transport Gateway](#) section of the *Cisco Catalyst SD-WAN Routing Configuration Guide*.
2. On spoke devices, enable on-demand tunnels and configure the idle timeout, as described in [Enable On-Demand Tunnels on a Spoke Device Using a Template, on page 488](#).

Enable On-Demand Tunnels on a Spoke Device Using a Configuration Group

Before you begin

On the **Configuration > Configuration Groups** page, choose the SD-WAN solution type.

Procedure

Step 1 From the Cisco SD-WAN Manager menu, choose **Configuration > Configuration Groups**.

Step 2 Do one of these:

- Edit a profile directly:

In the System Profile tab, create (**Add New**) or edit a System profile.

- Edit a profile in a configuration group:

Open a configuration group and edit the System profile.

Step 3 In the System profile, create (**Add New**) or edit a **Basic** feature.

Step 4 In the **Advanced** section, use the **On Demand Tunnel** control to enable on-demand tunnels.

Enable On-Demand Tunnels on a Spoke Device Using a Template

Before You Begin

- See the [Prerequisites for On-Demand Tunnels](#).
- Do not enable on-demand on the hub device.
- On the spoke devices, enable on-demand at the system level. In the case of multi-homed sites, enable on-demand on all systems at the site.

Enable On-Demand Tunnels on a Spoke Device

1. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
2. Click **Feature Templates**.



Note In Cisco vManage Release 20.7.x and earlier releases, **Feature Templates** is titled **Feature**.

3. Click **Add Template**.
4. Select a device.
5. From **Basic Information**, select **Cisco System**.
6. Click **Advanced**.

7. Enable **On-demand Tunnel**.
8. (optional) Configure the **On-demand Tunnel Idle Timeout** time. The default idle timeout value is 10 minutes. Range: 1 to 65535 minutes
9. Attach the System feature template to the device template for the spoke device.

Enable On-Demand Tunnels Using a CLI Template

For more information about using CLI templates, see [CLI Add-On Feature Templates](#) and [CLI Templates](#). By default, CLI templates execute commands in global configuration mode.

Before You Begin

- See [Prerequisites for On-Demand Tunnels](#).
- Do not enable on-demand on the hub device

Enable On-Demand Tunnels

On the spoke devices, enable on-demand tunnels at the system level. In the case of multi-homed sites, enable on-demand on all systems in the site.

The default idle timeout value is 10 minutes. Range: 1 to 65535 minutes

Example

```
system
  on-demand enable
  on-demand idle-timeout 10
```

Monitor the Status of On-Demand Tunnels

The following sections describe procedures for monitoring the status of on-demand tunnels.

View the Current Status of On-Demand Tunnels Using Cisco SD-WAN Manager

1. From the Cisco SD-WAN Manager menu, choose **Monitor** > **Devices**.
Cisco vManage Release 20.6.x and earlier: From the Cisco SD-WAN Manager menu, choose **Monitor** > **Network**.
2. Select a device.
3. Select **Real Time**.
4. For **Device Options**, select one of the following:
 - **On Demand Local**: Displays the status of on-demand tunnels on the specified device.
 - **On Demand Remote**: Displays the status of on-demand tunnels on the specified device, and on all connected devices.

The output is equivalent to executing the `show [sdwan] system on-demand [remote-system] [system-ip ip-address]` CLI command. It displays the status of on-demand tunnels.

View a Chart of the On-Demand Tunnel Status Over Time in Cisco SD-WAN Manager

1. From the Cisco SD-WAN Manager menu, choose **Monitor** > **Devices**.
Cisco vManage Release 20.6.x and earlier: From the Cisco SD-WAN Manager menu, choose **Monitor** > **Network**.
2. Select a device.
3. From **WAN**, choose **Tunnel**.
4. From the **Chart Options** drop-down list, select **On-Demand Tunnel Status**. The chart shows the status of tunnels as ACTIVE or INACTIVE. INACTIVE indicates that an on-demand tunnel is in Inactive mode.

View the Route to a Destination Device

Viewing the route between routers A and B can show whether the route is using an on-demand tunnel. On router A, use the `traceroute` command and enter router B as the destination. The command output shows whether the current route includes a hop at a hub device or whether the route is directly to the destination.

In the following examples, the router IP addresses are as follows:

- Router A: 10.1.1.1
- Router B: 10.1.1.2
- Hub device: 10.100.1.100

No Active On-Demand Tunnel

In the following example, there is no active on-demand tunnel between routers A and B, so the route includes the hub device. Note that it takes two hops to reach router B.

```
RouterA#traceroute vrf 1 10.1.1.2 numeric
Type escape sequence to abort.
Tracing the route to 10.1.1.2
VRF info: (vrf in name/id, vrf out name/id)
 1 10.100.1.100 10 msec 8 msec 0 msec
 2 10.1.1.2 2 msec * 1 msec
```

Active On-Demand Tunnel

In the following example, there is an active on-demand tunnel between routers A and B, so the route from router A and to router B is direct.

```
RouterA#traceroute vrf 1 10.1.1.2 numeric
Type escape sequence to abort.
Tracing the route to 10.1.1.2
VRF info: (vrf in name/id, vrf out name/id)
 1 10.1.1.2 1 msec
```

View OMP Routes

Viewing OMP routes can show the status of on-demand tunnels between two routers. Use the **show sdwan omp routes** command and view the **STATUS** column. The following table shows the possible values for this column, depending on whether an on-demand tunnel is active or not between two routers:

Table 207: Status of Routes, with or without an Active On-Demand Tunnel Between Two Routers

On-Demand Tunnel Between Routers A and B	STATUS for OMP Routes Between Routers A and B	STATUS for Backup Routes (through the Hub)
Not active	I, U, IA (installed, unresolved, and inactive)	C, I, R (chosen, installed, and resolved)
Active	C, I, R (chosen, installed, and resolved)	R (resolved)



CHAPTER 15

Cisco Catalyst SD-WAN Multitenancy

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Cisco Catalyst SD-WAN Multitenancy

Table 208: Feature History

Feature Name	Release Information	Description
Tenant Device Forecasting	Cisco SD-WAN Release 20.6.1 Cisco vManage Release 20.6.1	With this feature, a service provider can control the number of WAN edge devices a tenant can add to their overlay network. By doing so, the provider can utilize Cisco Catalyst SD-WAN controller resources efficiently. For more information, see Tenant Device Forecasting .
Migrate Multitenant Cisco Catalyst SD-WAN Overlay	Cisco SD-WAN Release 20.6.1 Cisco vManage Release 20.6.1	This feature enables you to migrate a multitenant Cisco Catalyst SD-WAN overlay comprising shared Cisco SD-WAN Manager instances and Cisco SD-WAN Validator, and tenant-specific Cisco SD-WAN Controllers to a multitenant overlay comprising shared Cisco SD-WAN Manager instances, Cisco SD-WAN Validator, and Cisco SD-WAN Controllers. For more information, see Migrate Multitenant Cisco Catalyst SD-WAN Overlay .
Enhanced Cisco Catalyst SD-WAN Manager Dashboard for Multitenancy	Cisco IOS XE Catalyst SD-WAN Release 17.12.1a Cisco Catalyst SD-WAN Manager Release 20.12.1	This feature ensures a consistent user experience across tenant and service provider dashboards. The Cisco Catalyst SD-WAN Manager dashboard provides visibility into the available resources on shared devices.
RADIUS and TACACS Support for Multitenancy	Cisco IOS XE Catalyst SD-WAN Release 17.12.1a Cisco Catalyst SD-WAN Control Components Release 20.12.1	This feature enables support for Remote Authentication Dial-In User Service (RADIUS) and Terminal Access Controller Access Control System (TACACS) authentication in a multitenant deployment on WAN edge devices. For more information, see Information about RADIUS and TACACS Support for Multitenancy .

Feature Name	Release Information	Description
Migration of a Tenant from a Multitenant Overlay to a Single-Tenant Deployment	Cisco IOS XE Catalyst SD-WAN Release 17.13.1a Cisco Catalyst SD-WAN Manager Release 20.13.1	This feature supports the migration of a tenant from a multitenant overlay to a single-tenant deployment. To migrate a tenant between two Cisco Catalyst SD-WAN deployments, move the tenant configurations, statistical data, and WAN edge devices from one deployment to another.
Multitenancy Support for Cisco Catalyst Cellular Gateways	Cisco IOS CG Release 17.14.1 Cisco Catalyst SD-WAN Control Components Release 20.14.1	Added multitenancy support for Cisco Catalyst Cellular Gateways.
Support for Remote AAA and Webhook Notifications Over Management VPN 512	Cisco IOS XE Catalyst SD-WAN Release 17.16.1a Cisco Catalyst SD-WAN Manager Release 20.16.1	This feature provides increased security by allowing you to configure tenants to receive AAA traffic and webhook notifications over management VPN 512.
Configure RADIUS and TACACS Servers to Receive Authentication Requests Over Management VPN 512	Cisco IOS XE Catalyst SD-WAN Release 17.16.1a Cisco Catalyst SD-WAN Manager Release 20.16.1	This feature provides increased security by allowing you to configure tenants to send and receive AAA traffic over WAN transport VPN 0 or management VPN 512. For more information, see Configure Remote AAA .
Support for Multitenancy Providers to View Tenant's Alarms and Events	Cisco IOS XE Catalyst SD-WAN Release 17.16.1a Cisco Catalyst SD-WAN Manager Release 20.16.1	This feature allows providers to view tenant alarms and events. Multitenancy providers can enable or disable this feature for each tenant.
Support for Tenant to Access to Controller-Specific Information	Cisco IOS XE Catalyst SD-WAN Release 17.16.1a	This feature enables tenants to access their controller-specific restricted information, which enhances visibility and simplifies management and troubleshooting within the Cisco Catalyst SD-WAN network. This improved visibility aids in addressing issues related to OMP, routing, control connections, and more.

Information about Cisco Catalyst SD-WAN multitenancy

With Cisco Catalyst SD-WAN multitenancy, a service provider can manage multiple customers, called tenants, from Cisco SD-WAN Manager. The tenants share the same set of underlying Cisco SD-WAN controllers: Cisco SD-WAN Manager, Cisco Catalyst SD-WAN Validator, and Cisco Catalyst SD-WAN Controller. The tenant data is logically isolated on these shared controllers.

The service provider accesses Cisco SD-WAN Manager using a domain name mapped to the IP address of a Cisco SD-WAN Manager cluster and manages the multitenant deployment. Each tenant is provided a subdomain to access a tenant-specific Cisco SD-WAN Manager view and manage the tenant deployment. For example, a service provider using the domain name `managed-sp.com`, can assign tenants Customer1 and Customer2 the subdomains `customer1.managed-sp.com` and `customer2.managed-sp.com` and manage them on the same set of Cisco SD-WAN controllers, instead of providing each customer a single tenant setup with a dedicated set of Cisco SD-WAN controllers.

Key features of Cisco Catalyst SD-WAN multitenancy include:

- Full enterprise multitenancy:
 - Cisco Catalyst SD-WAN supports multitenancy and offers enterprises the flexibility of segregated roles such as service provider and tenants. Service providers can use multitenancy to provide Cisco Catalyst SD-WAN service offerings to their customers.
- Multitenancy:
 - Multitenant Cisco SD-WAN Manager
 - Multitenant Cisco Catalyst SD-WAN Validators
 - Multitenant Cisco Catalyst SD-WAN Controllers
- Tenant-specific WAN edge devices.
- Security:
 - Send and receive AAA traffic over management VPN 512 from Cisco IOS XE Catalyst SD-WAN Release 17.16.1a.
- Overlapping VPN numbers:
 - A particular VPN or a set of common VPNs is assigned to a specific tenant, with their own configurations and monitoring dashboard environment. These VPN numbers can overlap where they are used by other tenants.
- On-prem and cloud deployment models:
 - Cisco Catalyst SD-WAN controllers can be deployed in an organization data center on servers running the VMware ESXi 6.7 or later, or the Kernel-based Virtual Machine (KVM) hypervisor. Cisco Catalyst SD-WAN controllers can also be hosted on Amazon Web Services (AWS) servers by Cisco CloudOps.
- Tenant-specific Cisco SD-WAN Analytics:
 - Cisco SD-WAN Analytics is a cloud-based service that offers insights into the performance of applications and the underlying SD-WAN network infrastructure. Each tenant can obtain Cisco

SD-WAN Analytics insights for their overlay network by requesting a tenant-specific Cisco SD-WAN Analytics instance and enabling data collection on Cisco SD-WAN Manager. The service provider must enable cloud services on Cisco SD-WAN Manager in the provider view to facilitate the onboarding of the Cisco SD-WAN Analytics instance for the tenant overlay network.



Note Starting from Cisco SD-WAN Manager Release 20.13.1, a single node multitenancy is supported. Expand a single node cluster into a multitenant deployment.

Multitenant Cisco SD-WAN Manager

Cisco SD-WAN Manager is deployed and configured by the service provider. The provider enables multitenancy and creates a Cisco SD-WAN Manager cluster to serve tenants. Only the provider can access a Cisco SD-WAN Manager instance through the SSH terminal.

In the Provider view, Cisco SD-WAN Manager offers service providers an overall view of the SD-WAN multitenant deployment and allows a provider to manage all the Cisco Catalyst SD-WAN Validator and Cisco Catalyst SD-WAN Controller devices. Cisco SD-WAN Manager also allows service providers to monitor and manage the deployments of each tenant through Provider-as-Tenant view.

In the Tenant view, Cisco SD-WAN Manager allows individual tenants to:

- Monitor and manage their own deployment through a dashboard.
- Deploy and configure WAN edge devices.
- Configure custom policies on Cisco Catalyst SD-WAN Controllers.

Cisco Catalyst SD-WAN Control Component infrastructure settings are not displayed in Tenant view.

Multitenant Cisco Catalyst SD-WAN Validators

Cisco Catalyst SD-WAN Validators are deployed and configured by the service provider. Only the provider can access a Cisco Catalyst SD-WAN Validator through the SSH terminal.

Cisco Catalyst SD-WAN Validators serve WAN edge devices of multiple tenants as the devices are added to the overlay network.

Multitenant Cisco Catalyst SD-WAN Controllers

Cisco Catalyst SD-WAN Controllers are deployed by the service provider. Only the provider can create and attach device and feature templates to Cisco Catalyst SD-WAN Controllers, and can access a Cisco Catalyst SD-WAN Controller through the SSH terminal.

- When a tenant is created, Cisco SD-WAN Manager assigns two Cisco Catalyst SD-WAN Controllers for the tenant. The Cisco Catalyst SD-WAN Controllers form an active-active cluster.

Each tenant is assigned only two Cisco Catalyst SD-WAN Controllers. Before a tenant is created, two Cisco Catalyst SD-WAN Controllers must be available to serve the tenant.

- When more than one pair of Cisco Catalyst SD-WAN Controllers are available to serve a tenant, Cisco SD-WAN Manager assigns to the tenant the pair of Cisco Catalyst SD-WAN Controllers connected to the lowest number of forecast devices. If two pairs of Cisco SD-WAN Controllers are connected to the

same number of devices, Cisco SD-WAN Manager assigns to the tenant the pair of Cisco SD-WAN Controllers serving the lowest number of tenants.

- From Cisco vManage Release 20.9.1, while onboarding a tenant to a multitenant deployment, you can choose the pair of multitenant Cisco SD-WAN Controllers that serve the tenant. After onboarding a tenant, you can migrate the tenant to a different pair of multitenant Cisco SD-WAN Controllers, if necessary. For more information, see [Flexible Tenant Placement on Multitenant Cisco SD-WAN Controllers](#).
- Each pair of Cisco Catalyst SD-WAN Controllers can serve up to 24 tenants.
- Tenants can configure custom policies on the Cisco Catalyst SD-WAN Controllers assigned to them. Cisco SD-WAN Manager notifies the Cisco Catalyst SD-WAN Controllers to pull the policy templates. Cisco Catalyst SD-WAN Controllers pull the templates and deploy the policy configuration for the specific tenant.
- Only the provider can view events, audit logs, and OMP alarms for a Cisco Catalyst SD-WAN Controller on Cisco SD-WAN Manager.



Note Starting from Cisco SD-WAN Manager Release 20.16.1, a provider can view alarms and events for the sites and devices in its tenancy.

Tenant-specific WAN edge devices

A tenant or the provider acting on behalf of a tenant can:

- Add WAN edge devices to the tenant network.
- Configure the devices.
- Remove the devices from the tenant network.
- Access the device through the SSH terminal.

A provider can manage the WAN edge devices only from [provider-as-tenant](#) view. In the [provider](#) view, Cisco SD-WAN Manager does not show any WAN edge device information.

Cisco SD-WAN Manager reports WAN edge device events, logs, and alarms only in the [tenant](#) and the provider-as-tenant views.

Single tenant environments

A single tenant environment exclusively manages, and is responsible for, its own Cisco Catalyst SD-WAN Control Components and devices. All configured resources are visible to the single tenant administrator in the Cisco SD-WAN Manager interface.

Cloud-delivered Catalyst SD-WAN

Cloud-delivered Catalyst SD-WAN operates as a tenant within a multitenant environment rather than as a single tenant. Cloud-delivered Catalyst SD-WAN users do not see controller infrastructure settings in Cisco SD-WAN Manager. Their available information is limited to their own components and WAN edge devices.

For more information on Cloud-delivered Catalyst SD-WAN, consult the [Cloud-delivered Cisco SD-WAN Getting Started Guide](#).

Feature availability

Some Cisco Catalyst SD-WAN features are available only in specific tenancy configurations.

Table 209: SD-WAN feature availability

Feature	Single Tenant	Multitenant	Cloud-delivered Catalyst SD-WAN
Identity Services Engine (ISE) integration	Yes	Yes	No
Intent-based hub and spoke topology	Yes	No	No
Controller Group Affinity	Yes	No	No
Cisco Duo multifactor authentication (MFA) support	Yes	No	No
SD-Routing support without an SD-WAN Controller	Yes	No	No
Data stream and packet capture setting changes at tenant level	—	No, supported at provider level	No, supported at provider level

Information About Tenant Migration in a Multitenant Deployment

(Minimum supported releases: Cisco IOS XE Catalyst SD-WAN Release 17.6.1a and Cisco vManage Release 20.6.1) Migration of a single-tenant overlay to a multitenant deployment is only supported with the Cisco Catalyst SD-WAN controllers deployed on-premises.

(Minimum supported releases: Cisco IOS XE Catalyst SD-WAN Release 17.13.1a and Cisco Catalyst SD-WAN Manager Release 20.13.1) Migration of a tenant from a multitenant overlay to a single-tenant deployment is supported.

The tenant migration involves export of tenant data from the source Cisco SD-WAN Manager instance and import of data to the destination Cisco SD-WAN Manager instance. After the data migration is complete, the tenant WAN edge devices with active control connections with the source Cisco SD-WAN Manager migrate and form connections with the destination Cisco SD-WAN Manager.

User Roles in Multitenant Environment

A multi-tenant environment includes the service provider and tenant roles. Each role has distinct privileges, views, and functions.

Provider Role

The provider role entitles system-wide administrative privileges. A user with the provider role has the default username **admin**. The provider user can access Cisco SD-WAN Manager using the domain name of the service provider or by using the Cisco SD-WAN Manager IP address. When using a domain name, the domain name has the format `https://managed-sp.com`.

The **admin** user is part of the user group **netadmin**. Users in this group are permitted to perform all operations on the controllers and the WAN edge devices of the tenants. You can add additional users to the **netadmin** group.

You cannot modify the privileges of the **netadmin** group. On Cisco SD-WAN Manager, you can view the privileges of the user group from the **Administration > Manage Users > User Groups** page.



Note When you create a new provider user in Cisco SD-WAN Manager, including a **netadmin** user, by default, the user is not allowed SSH access to the Cisco SD-WAN Manager VM. To enable SSH access, configure SSH authentication using a AAA template and push the template to Cisco SD-WAN Manager. For more information on enabling SSH authentication, see [SSH Authentication using Cisco SD-WAN Manager on Cisco vEdge Devices](#).

For more information about configuring users and user groups, see [Configure User Access and Authentication](#).

Cisco SD-WAN Manager offers two views to a provider:

• Provider View

When a provider user logs in to multi-tenant Cisco SD-WAN Manager as **admin** or another **netadmin** user, Cisco SD-WAN Manager presents the provider view and displays the provider dashboard.

You can perform the following functions from the provider view:

- Provision and manage Cisco SD-WAN Manager, Cisco SD-WAN Validators and Cisco SD-WAN Controllers.
- Add, modify, or delete tenants.
- Monitor the overlay network.
- Starting from Cisco Catalyst SD-WAN Manager Release 20.16.1, view alarms and events for the sites and devices of its tenants.

• Provider-as-Tenant View

When a provider user selects a specific tenant from the **Select Tenant** drop-down list at the top of the provider dashboard, Cisco SD-WAN Manager presents the provider-as-tenant view and displays the tenant dashboard for the selected tenant. The provider user has the same view of Cisco SD-WAN Manager as a tenant user would when logged in as **tenantadmin**. From this view, the provider can manage the tenant deployment on behalf of the tenant.

In the provider dashboard, a table of tenants presents a status summary for each tenant. A provider user can also launch the provider-as-tenant view by clicking on a tenant name in this table.

Tenant Role

The tenant role entitles tenant administrative privileges. A user with the tenant role has the default username **tenantadmin**. The default password is **Cisco#123@Viptela**. We recommend that you change the default password on first login. For information on changing the default password, see [Hardware and Software Installation](#).

The **tenantadmin** user is part of the user group **tenantadmin**. Users in this group are permitted to perform all operations on the WAN edge devices of the tenants. You can add additional users to the **tenantadmin** group.

You cannot modify the privileges of the **tenantadmin** group. On Cisco SD-WAN Manager, you can view the privileges of the user group from the **Administration > Manage Users > User Groups** page.

For more information about configuring users and user groups, see [Configure User Access and Authentication](#).

A tenant user can log in to Cisco SD-WAN Manager using a dedicated URL and the default username **tenantadmin**. For example, the dedicated URL of a tenant could be `https://customer1.managed-sp.com` for a provider using the domain name `https://managed-sp.com`. When the user logs in, Cisco SD-WAN Manager presents the tenant view and displays the tenant dashboard.



Tip If you cannot access the dedicated tenant URL, update the subdomain details in the `/etc/hosts` file on the local machine. Alternatively, if you use an external DNS server, add a DNS entry for the tenant subdomain.

A tenant user with administrative privileges can perform the following functions:

- Provision and manage tenant routers
- Monitor overlay network of the tenant
- Create custom policies on the assigned Cisco SD-WAN Controller
- Upgrade the software on the tenant routers.
- Starting from Cisco Catalyst SD-WAN Manager Release 20.16.1, view tenant-specific information of controller connections and OMP statistics in a Cisco Catalyst SD-WAN network.

Provider and Tenant Remote Servers and Images

Cisco Catalyst SD-WAN Manager Release 20.14.1 and Earlier Releases

In these releases, remote servers and images operate as follows:

- Only the provider can add remote servers and images.
- The remote servers and images are visible to all tenants. Tenants can use the remote servers and images but can't edit them.

Cisco Catalyst SD-WAN Manager Release 20.15.1

In these releases, remote servers and images operate as follows:

- A tenant can add a remote server and remote image for both software images and virtual images. The remote server and image are visible only to the corresponding tenant and not to the provider or other tenants.
- The provider can add a remote server, a remote image, and a local image for both software images and virtual images in Cisco SD-WAN Manager.

Supported Devices and Controller Specifications

The following Cisco Catalyst SD-WAN edge devices support multitenancy.

Table 210: Supported Devices

Platform	Device Models
Cisco vEdge device	• vEdge 100, vEdge 100b, vEdge 100m, vEdge 100wm, vEdge 1000, vEdge 2000, vEdge 5000, vEdge Cloud • ISR1100-6G/ISR1100-4G, ISR1100-4GLTENA, ISR1100-4GLTEGB
Cisco Catalyst Cellular Gateways	(From Cisco IOS CG Release 17.14.1 and Cisco Catalyst SD-WAN Control Components Release 20.14.1) • CG418-E • CG522-E

The following hypervisors are supported for multitenancy:

- VMware ESXi 6.7 or later
- KVM
- AWS (cloud-hosted and managed by Cisco CloudOps)
- Microsoft Azure (cloud-hosted and managed by Cisco CloudOps)

From Cisco vManage Release 20.6.1, a multitenant Cisco SD-WAN Manager instance can have one of the following three personas. The personas enable a predefined set of services on the Cisco SD-WAN Manager instance.

Table 211: Cisco SD-WAN Manager Personas

Persona	Services
Compute+Data	Cluster Oracle, Service Proxy, Messaging Service, Coordination Service, Configuration Database, Data Collection Agent, Statistics Database, and Application Server

Persona	Services
Data	Cluster Oracle, Service Proxy, Application Server, Data Collection Agent, and Statistics Database
Compute	Cluster Oracle, Service Proxy, Messaging Service, Coordination Service, Configuration Database, and Application Server

The supported hardware specifications for the Cisco SD-WAN Validator, Cisco SD-WAN Manager, and the Cisco SD-WAN Controllers are as follows:

Hardware Specifications to Support 50 Tenants and 1000 Devices

For more information on supported hardware specifications for the Cisco SD-WAN Validator, Cisco SD-WAN Manager, and the Cisco SD-WAN Controllers see, [Cisco Catalyst SD-WAN Controller Compatibility Matrix and Recommended Computing Resources](#).

Hardware Specifications to Support 75 Tenants and 2500 Devices

For more information on supported hardware specifications for the Cisco SD-WAN Validator, Cisco SD-WAN Manager, and the Cisco SD-WAN Controllers see, [Cisco Catalyst SD-WAN Controller Compatibility Matrix and Recommended Computing Resources](#).

Hardware Specifications to Support 100 Tenants and 5000 Devices

For more information on supported hardware specifications for the Cisco SD-WAN Validator, Cisco SD-WAN Manager, and the Cisco SD-WAN Controllers see, [Cisco Catalyst SD-WAN Controller Compatibility Matrix and Recommended Computing Resources](#).

Hardware Specifications to Support 150 Tenants and 7500 Devices

For more information on supported hardware specifications for the Cisco SD-WAN Validator, Cisco SD-WAN Manager, and the Cisco SD-WAN Controllers see, [Cisco Catalyst SD-WAN Controller Compatibility Matrix and Recommended Computing Resources](#).

Restrictions

- Feature availability

See the feature availability described in [Information about Cisco Catalyst SD-WAN multitenancy, on page 496](#).

- Connecting to a device by SSH

Do not use a user-configured system IP address to connect to a device through SSH. Instead, use the IP address of the `vmanage_system` interface; this IP address is assigned by SD-WAN Manager.

To find the IP address of the `vmanage_system` interface, use one of the following methods:

- Launch the device SSH terminal from SD-WAN Manager and find the `vmanage_system` IP address from the first line of the log-in prompt.

- Run the **show interface description** command and find the `vmanage_system` IP address from the command output.

- Adding tenants

If you add a second tenant immediately after adding a tenant, SD-WAN Manager adds them sequentially, and not in parallel.

- Re-adding a previously invalidated device

If you are adding a WAN edge device that you had previously invalidated and deleted from an overlay network, you must reset the device software after adding the device. To reset the software on a Cisco vEdge device, use the command **request platform software reset**.

- Minimum SD-WAN Manager cluster for multitenant scenario

For Cisco IOS XE Catalyst SD-WAN Release 17.12.1a and earlier releases, single node SD-WAN Manager is not supported on a multitenant deployment. A minimum of 3-Node SD-WAN Manager cluster is required for a multitenant deployment.

- Cannot upgrade devices while upgrading SD-WAN Controller or SD-WAN Validator

When a SD-WAN Controller or SD-WAN Validator upgrade is in progress, upgrade of tenant edge devices is not supported.

- Controller group

The Controller group feature is not supported in multitenant mode.

- Site ID of edge devices

The WAN edge device's site ID must be different from the SD-WAN Control Component site ID, when the SD-WAN Manager has different public and private IP addresses.

Restrictions for Migration of a Tenant from a Multitenant Overlay to a Single-Tenant Deployment

Minimum supported releases: Cisco IOS XE Catalyst SD-WAN Release 17.13.1a and Cisco Catalyst SD-WAN Manager Release 20.13.1

- Change in the tenant organization name is not supported when the tenant moves from the Cisco Catalyst SD-WAN source to destination deployment.
- Tenant migration with multitenant WAN edge devices is not supported.
- Data traffic loss is expected during migration as devices are migrating from one set of SD-WAN Controllers to another.
- All user passwords are set to the default Cisco password on the destination overlay. The default password is **Cisco#123@Viptela**.
- Statistical data of the tenant that can be relearned by destination SD-WAN Manager is not migrated.
- The migration procedure does not support multiple imports on the same destination SD-WAN Manager. Reinitialize the destination SD-WAN Manager to allow import again.

Initial Setup for Multitenancy

Prerequisites

- Download and install software versions as recommended in the following table:

Table 212: Minimum Software Prerequisites for Cisco Catalyst SD-WAN Multitenancy

Device	Software Version
Cisco SD-WAN Manager	Cisco vManage Release 20.6.1
Cisco SD-WAN Validator	Cisco SD-WAN Release 20.6.1
Cisco SD-WAN Controller	Cisco SD-WAN Release 20.6.1
Cisco vEdge Device	Cisco SD-WAN Release 20.6.1

A configuration in which one or more controllers, or WAN edge devices, are running software versions earlier than those mentioned in the table above is not supported.

- Do not migrate an existing single-tenant Cisco SD-WAN Manager instance into multitenant mode, even if you invalidate or delete all devices from the existing Cisco SD-WAN Manager instance. Instead, download and install a new Cisco SD-WAN Manager software image.



Note After you enable Cisco SD-WAN Manager for multitenancy, you cannot migrate it back to single tenant mode.

- Follow the recommended hardware specifications in the *Supported Devices and Controller Specifications* section of this document.
- Log in to Cisco SD-WAN Manager as the provider **admin** user.

1. Create Cisco SD-WAN Manager cluster.

- To support 50 tenants and 1000 devices across all tenants, [create a 3-node Cisco SD-WAN Manager Multitenant cluster](#).
- To support 100 tenants and 5000 devices across all tenants, [create a 6-node Cisco SD-WAN Manager Multitenant cluster](#).
- From Cisco SD-WAN Release 20.6.3, Cisco vManage Release 20.6.3, to support 150 tenants and 7500 devices across all tenants, [create a 6-node Cisco SD-WAN Manager Multitenant cluster](#).

2. Create and configure Cisco SD-WAN Validator instances. See [Deploy Cisco SD-WAN Validator](#).

While configuring Cisco SD-WAN Validator instances, configure the service provider organization name (`sp-organization-name`) and the organization name (`organization-name`). See [Configure Organization Name in Cisco SD-WAN Validator](#).

```
sp-organization-name multitenancy
organization-name multitenancy
```

3. Create Cisco SD-WAN Controller instances. See [Deploy the Cisco SD-WAN Controller](#).
 - To support 50 tenants and 1000 devices across all tenants, deploy 6 Cisco SD-WAN Controller instances.
 - To support 100 tenants and 5000 devices across all tenants, deploy 10 Cisco SD-WAN Controller.
 - From Cisco SD-WAN Release 20.6.3, Cisco vManage Release 20.6.3, to support 150 tenants and 7500 devices across all tenants, deploy 16 Cisco SD-WAN Controllers.
- a. [Add Cisco SD-WAN Controllers](#) to the overlay network.
4. Onboard new tenants. See [Add a New Tenant, on page 517](#).

Create a 3-Node Cisco SD-WAN Manager Multitenant Cluster

1. Download the Cisco vManage Release 20.6.1 or later software image from [Cisco Software Download](#).
2. Create three Cisco SD-WAN Manager instances (say vManage1, vManage2, and vManage3) by installing the downloaded software image file. See [Deploy Cisco SD-WAN Manager](#).



Important

- Deploy Cisco SD-WAN Manager servers having the hardware specifications in the table *Hardware Specifications to Support 50 Tenants and 1000 Devices* from the *Supported Devices and Controller Specifications* section of this document.
- Choose the **Compute+Data** persona for each Cisco SD-WAN Manager instance.

3. Complete the following operations on vManage1:
 - a. Configure the following using the CLI:
 - System IP address
 - Site ID
 - Service Provider organization name (`sp-organization-name`)
 - Organization-name
 - Cisco SD-WAN Validator IP address
 - VPN 0 Transport/Tunnel interface
 - VPN 0 Out-of-band (OOB) interface: Ensure that you assign a static IP address to this interface. Do not enable DHCP.
 - VPN 512 Management interface



Note

Configure only one default route in VPN 0.

- b. [Enable Multitenancy on Cisco SD-WAN Manager, on page 510](#).

- c. (Optional) Using the CLI, install the Root CA certificate for vManage1.



Note Skip this step if you are using a Symantec or Cisco PKI certificate.

- d. Complete the following through the Cisco SD-WAN Manager:
1. [Generate a Certificate Signing Request](#)
 2. After Symantec or your enterprise root CA has signed the certificate, [install the signed certificate](#).
- e. [Configure the Cluster IP Address of the Cisco SD-WAN Manager Server](#).
- Before proceeding to the next step, ensure that the **Manager IP Address** field on the **Administration > Cluster Management** page shows the OOB interface address.
4. Complete the following operations on vManage2 and vManage 3:



Important Do not enable multitenancy on vManage2 and vManage3.

- a. Configure the following using the CLI:
- System IP address
 - Site ID
 - Service Provider organization name (sp-organization-name)
 - Organization-name
 - Cisco SD-WAN Validator IP address
 - VPN 0 Transport/Tunnel interface
 - VPN 0 Out-of-band (OOB) interface: Ensure that you assign a static IP address to this interface. Do not enable DHCP.
 - VPN 512 Management interface
- b. (Optional) Using the CLI, install the Root CA certificate for vManage1.



Note Skip this step if you are using a Symantec or Cisco PKI certificate.

- c. Complete the following through the Cisco SD-WAN Manager:
1. [Generate a Certificate Signing Request](#)
 2. After Symantec or your enterprise root CA has signed the certificates, [install signed certificate](#).
- d. [Log in to the Cisco SD-WAN Manager Web Application Server](#).

- e. Ping the OOB interfaces on the other two Cisco SD-WAN Manager instances and ensure they are reachable.
- f. [Configure the Cluster IP Address of the Cisco SD-WAN Manager Server.](#)
Before proceeding to the next step, ensure that the **Manager IP Address** field on the **Administration > Cluster Management** page shows the OOB interface address.

5. Log in to the vManage1 GUI and [add vManage2 to the cluster.](#)

vManage2 reboots before being added to the cluster.

While vManage2 is being added to the cluster, on the **Administration > Cluster Management** page, the **Configure Status** for vManage2 shows **Pending**. You can monitor the System Generated Cluster Sync transaction to check the progress of the adding vManage2 to the cluster.

When the operation is completed, on the **Administration > Cluster Management** page, you can view both vManage1 and vManage2, and their node personas.

6. Repeat **Step 5** and add vManage3 to the cluster.



Note After rebooting, you have to select persona (non-cloud setup) from CLI and services starts running on the node according to the selected persona.

Create a 6-Node Cisco SD-WAN Manager Multitenant Cluster

1. Download the Cisco vManage Release 20.6.1 or later software image from [Cisco Software Download](#).
2. Create six Cisco SD-WAN Manager instances by installing the downloaded software image file. See [Deploy Cisco SD-WAN Manager](#).



Important

- To support 100 tenants and 5000 devices across all tenants, deploy Cisco SD-WAN Manager servers having the hardware specifications in the table *Hardware Specifications to Support 100 Tenants and 5000 Devices* from the *Supported Devices and Controller Specifications* section of this document.

From Cisco SD-WAN Release 20.6.3, Cisco vManage Release 20.6.3, to support 150 tenants and 7500 devices across all tenants, deploy Cisco SD-WAN Manager servers having the hardware specifications in the table *Hardware Specifications to Support 150 Tenants and 7500 Devices* from the *Supported Devices and Controller Specifications* section of this document.

- Choose the **Compute+Data** persona for three Cisco SD-WAN Manager instances (say vManage1, vManage2, and vManage 3). Choose the **Data** persona for the other three Cisco SD-WAN Manager instances (say vManage4, vManage5, and vManage6).

3. Complete the following operations on vManage1:
 - a. Configure the following using the CLI:
 - System IP address
 - Site ID

- Service Provider organization name (`sp-organization-name`)
- Organization-name
- Cisco SD-WAN Validator IP address
- VPN 0 Transport/Tunnel interface
- VPN 0 Out-of-band (OOB) interface: Ensure that you assign a static IP address to this interface. Do not enable DHCP.
- VPN 512 Management interface



Note Configure only one default route in VPN 0.

- [Enable Multitenancy on Cisco SD-WAN Manager, on page 510.](#)
- (Optional) Using the CLI, install the Root CA certificate for vManage1.



Note Skip this step if you are using a Symantec or Cisco PKI certificate.

- Complete the following through the Cisco SD-WAN Manager:
 - [Generate a Certificate Signing Request](#)
 - After Symantec or your enterprise root CA has signed the certificate, [install the signed certificate](#).
 - [Configure the Cluster IP Address of the Cisco SD-WAN Manager Server.](#)

Before proceeding to the next step, ensure that the **Manager IP Address** field on the **Administration > Cluster Management** page shows the OOB interface address.
4. Complete the following operations on vManage2 through vManage6:



Important Do not enable multitenancy on vManage2 through vManage6.

- Configure the following using the CLI:
 - System IP address
 - Site ID
 - Service Provider organization name (`sp-organization-name`)
 - Organization-name
 - Cisco SD-WAN Validator IP address
 - VPN 0 Transport/Tunnel interface

- VPN 0 Out-of-band (OOB) interface: Ensure that you assign a static IP address to this interface. Do not enable DHCP.
 - VPN 512 Management interface
- b. (Optional) Using the CLI, install the Root CA certificate for vManage1.



Note Skip this step if you are using a Symantec or Cisco PKI certificate.

- c. Complete the following through the Cisco SD-WAN Manager:
1. [Generate a Certificate Signing Request](#)
 2. After Symantec or your enterprise root CA has signed the certificate, [install the signed certificate](#).
- d. [Log in to the Cisco SD-WAN Manager Web Application Server](#).
- e. Ping the OOB interfaces on the other Cisco SD-WAN Manager instances and ensure they are reachable.
- f. [Configure the Cluster IP Address of the Cisco SD-WAN Manager Server](#).
- Before proceeding to the next step, ensure that the **Manager IP Address** field on the **Administration > Cluster Management** page shows the OOB interface address.
5. Log in to the vManage1 GUI and [add vManage2 to the cluster](#).
vManage2 reboots before being added to the cluster.
- While vManage2 is being added to the cluster, on the **Administration > Cluster Management** page, the **Configure Status** for vManage2 shows **Pending**. You can monitor the System Generated Cluster Sync transaction to check the progress of the adding vManage2 to the cluster.
- When the operation is completed, on the **Administration > Cluster Management** page, you can view both vManage1 and vManage2, and their node personas.
6. Repeat **Step 5** and add vManage3 through vManage6 to the cluster.

Enable Multitenancy on Cisco SD-WAN Manager



Note Administrator triggered disaster recovery is supported for multitenant clusters from Cisco vManage Release 20.6.1 or later releases.

Prerequisites

Do not migrate an existing single-tenant Cisco SD-WAN Manager into multitenant mode, even if you invalidate or delete all devices from the existing Cisco SD-WAN Manager. Instead, download and install a new software image of Cisco vManage Release 20.6.1 or a later release.



Note After you enable multitenancy on Cisco SD-WAN Manager, you cannot migrate it back to single tenant mode.

1. Launch Cisco SD-WAN Manager using the URL `https://vmanage-ip-address:port`. Log in as the provider **admin** user.
2. From the Cisco SD-WAN Manager menu, choose **Administration > Settings > Tenancy Mode**. If you are using Cisco Catalyst SD-WAN Manager Release 20.12.x or earlier, click **Edit**.
3. In the **Tenancy** field, click **Multitenant**.
4. In the **Domain** field, enter the domain name of the service provider (for example, managed-sp.com).
5. Enter a **Cluster Id** (for example, cluster-1 or 123456).
6. Click **Save**. If you are using Cisco Catalyst SD-WAN Manager Release 20.12.x or earlier, click **Proceed** to confirm that you want to change the tenancy mode.

Cisco SD-WAN Manager reboots in multitenant mode and when a provider user logs in to Cisco SD-WAN Manager, the provider dashboard appears.



Note The **Domain** and **Cluster Id** values created in steps 5 and 6 serve as the Provider FQDN. Ensure these values conform to current DNS naming conventions. You can not modify these values after the configuration is saved. To change these values, a new Cisco SD-WAN Manager cluster need to be deployed. For more details on Provider and Tenant DNS requirements, see step 3.d in [Add a New Tenant](#).

Add Cisco SD-WAN Controller

1. Log in to Cisco SD-WAN Manager as the provider **admin** user.
2. From the Cisco SD-WAN Manager menu, choose **Configuration > Devices**.
3. Click **Controllers**.



Note Starting from Cisco IOS XE Catalyst SD-WAN Release 17.13.1a, the **Controllers** tab is renamed as the **Control Components** tab to stay consistent with Cisco Catalyst SD-WAN rebranding.

4. Click **Add Controller**.
5. In the **Add Controller** dialog box, do the following:
 - a. In the **Controller Management IP Address** field, enter the system IP address of the Cisco SD-WAN Controller.
 - b. Enter the **Username** and **Password** required to access the Cisco SD-WAN Controller.
 - c. Select the protocol to use for control-plane connections. The default is **DTLS**.
If you select **TLS**, enter the port number to use for TLS connections. The default is 23456.

- d. Check the **Generate CSR** check box for Cisco SD-WAN Manager to create a Certificate Signing Request.
 - e. Click **Add**.
6. From the Cisco SD-WAN Manager menu, choose **Configuration > Certificates**.
For the newly added Cisco SD-WAN Controller, the **Operation Status** reads **CSR Generated**.
 - a. For the newly added Cisco SD-WAN Controller, click **More Options** icon and click **View CSR**.
 - b. Submit the CSR to the Certificate Authority (CA) and obtain a signed certificate.
 7. From the Cisco SD-WAN Manager menu, choose **Configuration > Certificates**.
 8. Click **Install Certificate**.
 9. In the **Install Certificate** dialog box, paste the **Certificate Text** or click **Select a file** upload the certificate file. Click **Install**.

Cisco SD-WAN Manager installs the certificate on the Cisco SD-WAN Controller. Cisco SD-WAN Manager also sends the serial number of the certificate to other controllers.

On the **Configuration > Certificates** page, the **Operation Status** for the newly added Cisco SD-WAN Controller reads as **Validator Updated**.

On the **Configuration > Devices** page, the new controller is listed in the Controller table with the controller type, hostname of the controller, IP address, site ID, and other details. The **Mode** is set to **CLI**.
 10. Change the mode of the newly added Cisco SD-WAN Controller to **Manager Mode** by attaching a template to the device.
 - a. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
For more information on configuration using CLI template, see [Device Configuration-Based CLI Templates](#).
 - b. Click **Device Templates**.



Note In Cisco vManage Release 20.7.x and earlier releases, **Device Templates** is titled as **Device**

- c. Find the template to be attached to the Cisco SD-WAN Controller.
- d. Click ..., and click **Attach Devices**.
- e. In the **Attach Devices** dialog box, move the new controller to the **Selected Device** list and click **Attach**.
- f. Verify the **Config Preview** and click **Configure Devices**.

Cisco SD-WAN Manager pushes the configuration from the template to the new controller.

In the **Configuration > Devices** page, the **Mode** for the Cisco SD-WAN Controller shows **Manager Mode**. The new Cisco SD-WAN Controller is ready to be used in your multitenant deployment.

Expand a Multitenant Deployment to Support More Tenants and Tenant Devices

As a service provider, suppose you have deployed a C to the overlay to support up to 100 tenants and 5000 devices. From Cisco SD-WAN Release 20.6.3, Cisco vManage Release 20.6.3, you can expand the Cisco SD-WAN Manager cluster and add additional Cisco SD-WAN Controllers to the overlay to support up to 150 tenants and 7500 devices.

Prerequisites

A multitenant Cisco Catalyst SD-WAN overlay that supports up to 50 tenants and 1000 devices, deployed according to the steps outlined in the *Initial Setup for Multitenancy* section of this document.

1. [Expand the existing 3-node Cisco SD-WAN Manager cluster to a 6-node cluster.](#)
2. To support up to 100 tenants and 5000 devices, you must have 10 Cisco SD-WAN Controllers in the overlay. So, deploy 4 Cisco SD-WAN Controllers in addition to the 6 existing Cisco SD-WAN Controllers in the overlay.

To support up to 150 tenants and 7500 devices, you must have 16 Cisco SD-WAN Controllers in the overlay. So, deploy 10 Cisco SD-WAN Controllers in addition to the 6 existing Cisco SD-WAN Controllers in the overlay.

- a. Create Cisco SD-WAN Controller instances. See [Deploy the Cisco SD-WAN Controller](#).
- b. [Add Cisco SD-WAN Controllers](#) to the overlay network.

You can now add more tenants or allow your existing tenants to add more devices subject to the relevant limits.



Note Starting from Cisco SD-WAN Manager Release 20.13.1, you can expand a single node cluster into 3 or 6 node clusters.

Expand a 3-Node Cluster to a 6-node Cluster



Note You can only expand a 3-node Cisco SD-WAN Manager cluster to a 6-node Cisco SD-WAN Manager cluster. Expansion of the 3-node cluster to other cluster sizes is not supported.

1. To support 100 tenants and 5000 devices: Upgrade the three Cisco SD-WAN Manager servers in the existing 3-node cluster to the hardware specifications in the table *Hardware Specifications to Support 100 Tenants and 5000 Devices* from the *Supported Devices and Controller Specifications* section of this document.

From Cisco SD-WAN Release 20.6.3, Cisco vManage Release 20.6.3, to support 150 tenants and 7500 devices: Upgrade the three Cisco SD-WAN Manager servers in the existing 3-node cluster to the hardware

specifications in the table *Hardware Specifications to Support 150 Tenants and 7500 Devices* from the *Supported Devices and Controller Specifications* section of this document.

2. Download the Cisco vManage Release 20.6.1 or a later release software image from [Cisco Software Download](#).
3. Create three Cisco SD-WAN Manager instances (say vManage1, vManage2, and vManage3) by installing the downloaded software image file. See [Deploy Cisco SD-WAN Manager](#).



Important

- Deploy Cisco SD-WAN Manager servers having the hardware specifications in the table *Hardware Specifications to Support 100 Tenants and 5000 Devices* from the *Supported Devices and Controller Specifications* section of this document.

From Cisco SD-WAN Release 20.6.3, Cisco vManage Release 20.6.3, to support 150 tenants and 7500 devices, deploy Cisco SD-WAN Manager servers having the hardware specifications in the table *Hardware Specifications to Support 150 Tenants and 7500 Devices* from the *Supported Devices and Controller Specifications* section of this document.

- Choose the **Data** persona for each Cisco SD-WAN Manager instance.

4. Complete the following operations on vManage1 through vManage3:



Important

Do not enable multitenancy on vManage1 through vManage3.

- a. Configure the following using the CLI:

- System IP address
- Site ID
- Service Provider organization name (`sp-organization-name`)
- Organization-name
- Cisco SD-WAN Validator IP address
- VPN 0 Transport/Tunnel interface
- VPN 0 Out-of-band (OOB) interface: Ensure that you assign a static IP address to this interface. Do not enable DHCP.
- VPN 512 Management interface



Note

Configure only one default route in VPN 0.

- b. (Optional) Using the CLI, install the Root CA certificate for vManage1.



Note Skip this step if you are using a Symantec or Cisco PKI certificate.

- c. Complete the following through the Cisco SD-WAN Manager:
 1. [Generate a Certificate Signing Request](#)
 2. After Symantec or your enterprise root CA has signed the certificate, [install the signed certificate](#).
- d. [Log in to the Cisco SD-WAN Manager Web Application Server](#).
- e. Ping the OOB interfaces on the other Cisco SD-WAN Manager instances and ensure they are reachable.
- f. [Configure the Cluster IP Address of the Cisco SD-WAN Manager Server](#).

Before proceeding to the next step, ensure that the **Manager IP Address** field on the **Administration > Cluster Management** page shows the OOB interface address.

5. Log in to the GUI of the existing 3-node Cisco SD-WAN Manager cluster and [add vManage1 to the cluster](#).

vManage1 reboots before being added to the cluster.

While vManage1 is being added to the cluster, on the **Administration > Cluster Management** page, the **Configure Status** for vManage1 shows **Pending**. You can monitor the System Generated Cluster Sync transaction to check the progress of the adding vManage1 to the cluster.

When the operation is completed, on the **Administration > Cluster Management** page, you can view vManage1 and its node persona listed along with the three Cisco SD-WAN Manager instances that were part of the original 3-node cluster.

6. Repeat **Step 4** and add vManage2 and vManage3 to the cluster.

Manage Tenants

Tenant Device Forecasting

While adding a new tenant to the multitenant Cisco Catalyst SD-WAN deployment, a service provider can forecast the number of WAN edge devices that the tenant may deploy in their overlay network. Cisco SD-WAN Manager enforces this forecast limit. If the tenant tries to add devices beyond this limit, Cisco SD-WAN Manager responds with an appropriate error message and the device addition fails.

In a multitenant deployment, a tenant can add a maximum of 1000 devices to their overlay network.



Note From Cisco SD-WAN Release 20.6.2, Cisco vManage Release 20.6.2, you can modify the device forecast for a tenant after the tenant is added. This modification is not supported in Cisco SD-WAN Release 20.6.1, Cisco vManage Release 20.6.1.

Benefits:

- The service provider can ensure that the Cisco Catalyst SD-WAN controller resources are used more efficiently.
- Depending on the configuration, a multitenant deployment can support a fixed number of WAN edge devices across all tenants. By forecasting the number of devices a tenant may add, the service provider can assign a quota for each tenant from the overall pool of edge devices that the deployment can support.

View Specific Cisco Catalyst SD-WAN Controller Information in Tenant Dashboard

Minimum supported releases: Cisco IOS XE Catalyst SD-WAN Release 17.16.1a and Cisco Catalyst SD-WAN Manager Release 20.16.1

When adding a new tenant or modifying an existing tenant's information, providers can allow tenants to view their specific Cisco SD-WAN Controller information. The enhanced visibility into controller information allows tenants more efficient monitoring, management, and troubleshooting using Cisco Catalyst SD-WAN Manager. For more information about configuring tenant's access to Cisco Catalyst SD-WAN Controller information, see the following sections:

- [Add a New Tenant, on page 517](#)
- [Modify Tenant Information, on page 521](#)

View Tenant Alarms in Provider Dashboard

Minimum supported releases: Cisco IOS XE Catalyst SD-WAN Release 17.16.1a and Cisco Catalyst SD-WAN Manager Release 20.16.1

1. Log in to Cisco SD-WAN Manager as the provider **admin** user.
2. From the Cisco SD-WAN Manager menu, choose **Operated as Provider > Provider > Logs**.
3. In the **Alarms** tab, toggle **View tenant alarms**.
4. Optional: Filter alarms using **Advanced Filter** based on **Tenant**, **Object Type**, **Object List**, **Severity**, and **Type**.
5. Optional: Click the bell icon on top to view all the notifications for tenant's alarms.

What to do next:

Subscribe to tenant's alarm notifications.

View Tenant Events in Provider Dashboard

Minimum supported releases: Cisco IOS XE Catalyst SD-WAN Release 17.16.1a and Cisco Catalyst SD-WAN Manager Release 20.16.1

1. Log in to Cisco SD-WAN Manager as the provider **admin** user.
2. From the Cisco SD-WAN Manager menu, choose **Operated as Provider > Provider > Logs**.

3. In the **Events** tab, toggle **View tenant events**.
4. Optional: Filter events using **Advanced Filter** based on **Tenant**, **Object Type**, **Object List**, **Severity**, and **Type**.

Add a New Tenant

Prerequisites

- At least two Cisco SD-WAN Controllers must be operational and in the `Manager` mode before you can add new tenants.

A Cisco SD-WAN Controller enters the `Manager` mode when you push a template onto the controller from Cisco SD-WAN Manager. A Cisco SD-WAN Controller in the `CLI` mode cannot serve multiple tenants.

- Each pair of Cisco SD-WAN Controllers can serve a maximum of 24 tenants and a maximum of 1000 tenant devices. Ensure that there at least two Cisco SD-WAN Controllers that can serve a new tenant. If no pair of Cisco SD-WAN Controllers in the deployment can serve a new tenant, add two Cisco SD-WAN Controllers and change their mode to `Manager`.
- If you add a second tenant immediately after adding a tenant, Cisco SD-WAN Manager adds them sequentially, and not in parallel.
- Each tenant must have a unique Virtual Account (VA) on **Plug and Play Connect** on [Cisco Software Central](#). The tenant VA should belong to the same Smart Account (SA) as the provider VA.
- For an on-premises deployment, create a **Validator** controller profile for the tenant on **Plug and Play Connect**. The fields in the following table are mandatory.

Table 213: Controller Profile Fields

Field	Description/Value
Profile Name	Enter a name for the controller profile.
Multi-Tenancy	From the drop-down list, select Yes .
SP Organization Name	Enter the provider organization name.
Organization Name	Enter the tenant organization name in the format <SP Org Name>-<Tenant Org Name>. Note The organization name can be up to 50 characters. Note A mismatch of organization name format of the controller profile and the tenant creation leads to a failure in device sync up.
Primary Controller	Enter the host details for the primary Cisco SD-WAN Validator.

If you are using a Cisco-hosted multitenant environment, see [Add a new tenant in Cisco-hosted multitenant environment](#)

1. Log in to Cisco SD-WAN Manager as the provider **admin** user.
2. From the Cisco SD-WAN Manager menu, choose **Administration > Tenant Management**.
3. Click **Add Tenant**. In the **Add Tenant** dialog box:

- a. Enter a name for the tenant.

- b. Enter a description of the tenant.

The description can be up to 256 characters and can contain only alphanumeric characters.

- c. Enter the name of the organization.

The organization name is case-sensitive. Each tenant or customer must have a unique organization name.

Enter the organization name in the following format:

```
<SP Org Name>-<Tenant Org Name>
```

For example, if the provider organization name is 'multitenancy' and the tenant organization name is 'Customer1', while adding the tenant, enter the organization name as **multitenancy-Customer1**.



Note The organization name can be up to 50 characters.

A mismatch of organization name format of the controller profile and the tenant creation leads to a failure in device sync up.

- d. In the **URL Subdomain Name** field, enter the fully qualified sub-domain name of the tenant.

- The sub-domain name must include the domain name of the service provider. For example, for the managed-sp.com service provider, a valid domain name can be customer1.managed-sp.com.



Note The service provider name is shared amongst all tenants. Hence, ensure that the URL naming convention follows the same domain name convention that was provided while enabling multitenancy from **Administration > Settings > Tenancy Mode**.

- For an on-premises deployment, add the fully qualified sub-domain name of the tenant to the DNS. Map the fully qualified sub-domain name to the IP addresses of the three Cisco SD-WAN Manager instances in the Cisco SD-WAN Manager cluster.
- **Provider Level:** Create DNS A record and map it to the IP addresses of the Cisco SD-WAN Manager instances running in the Cisco SD-WAN Manager cluster. The A record is derived from the domain and Cluster ID that was created in steps 5 and 6 in [Enable Multitenancy on Cisco SD-WAN Manager](#). For example, if domain is **sdwan.cisco.com** and Cluster ID is **vmanage123**, then A record will need to be configured as **vmanage123.sdwan.cisco.com**.



Note If you fail to update DNS entries, it will result in authentication errors when logging in to Cisco SD-WAN Manager. Validate DNS is configured correctly by executing **nslookup vmanage123.sdwan.cisco.com**.

- **Tenant Level:** Create DNS CNAME records for each tenant created and map them to the FQDN created at the Provider Level. For example, if domain is **sdwan.cisco.com** and tenant name is **customer1** the CNAME record will need to be configured as **customer1.sdwan.cisco.com**.



Note Cluster ID is not required for CNAME record. Validate DNS is configured correctly by executing **nslookup customer1.sdwan.cisco.com**.

For a cloud deployment, the fully qualified sub-domain name of the tenant is automatically added to the DNS as part of the tenant creation process. After you add a tenant, it could take up to an hour before the fully qualified sub-domain name of the tenant can be resolved by the DNS.

- e. In the **Number of Devices** field, enter the number of WAN edge devices that the tenant can deploy. If the tenant tries to add WAN edge devices beyond this number, Cisco SD-WAN Manager reports an error and the device addition fails.
- f. Starting from Cisco Catalyst SD-WAN Manager Release 20.16.1, toggle **Controller Visibility** to enable the following in your tenant's dashboard:
 - In the **Monitor** dashboard, the **Controller Components** card with the **Controller** option that has an active hyperlink to view tenant-specific controller information
 - In the **Devices** page, controller details relevant to tenant
 - In the **Certificates** page, certificate details relevant to tenant
 - In **Devices > Real-time, Control Connections** and OMP statistics relevant to tenant.
- g. Click **Save**.

The **Create Tenant** screen appears, and the **Status** of the tenant creation reads **In progress**. To view status messages related to the creation of a tenant, click the > button to the left of the status.

Cisco SD-WAN Manager does the following:

- creates the tenant
- assigns two Cisco SD-WAN Controllers to serve the tenant and pushes a CLI template to these controllers to configure tenant information
- sends the tenant and Cisco SD-WAN Controller information to Cisco SD-WAN Validator.

What to do next:

After the **Status** column changes to **Success**, you can view the tenant information on the **Administration > Tenant Management** page.

Add a new tenant in Cisco-hosted multitenant environment

1. Log in to Cisco SD-WAN Manager as the provider **admin** user.
2. From the Cisco SD-WAN Manager menu, choose **Administration > Tenant Management**.
3. Click **Add Tenant**. In the **Add Tenant** dialog box:
 - a. Enter a name for the tenant. It must be the same as the Virtual Account used for the tenant.
 - b. Enter a description.
 - c. Enter the name of the organization in the following format:

<Provider Org Name>-<Tenant Org Name>

For example, if the provider organization name is 'EFT20.17-VA-Main – 841534' and the tenant organization name is 'T1', enter the organization name as **EFT20.17-VA-Main – 841534-T1**. The tenant organization name can be T1 for Tenant 1, T2 for Tenant 2, and so on.



Note The organization name can be up to 50 characters.

- d. Enter the sub-domain URL in FQDN format.
 - The sub-domain name must include `sdwan.cisco.com`. For example, a sub-domain name can be `Eftt1.sdwan.cisco.com`.
 - Ensure the sub-domain name is unique by performing a `nslookup` or `ping` on the expected domain. If the domain can be found, use a new URL.

The FQ sub-domain name of the tenant is automatically added to the DNS as part of the tenant creation process. After you add a tenant, it could take up to an hour before the fully qualified sub-domain name of the tenant is resolved by the DNS.

- e. In the **Number of Devices** field, enter the number of WAN edge devices that the tenant can deploy. If the tenant tries to add WAN edge devices beyond this number, Cisco SD-WAN Manager reports an error and the device addition fails.
- f. Choose Auto placement or manual option for controller assignment.
- g. Click **Save**.

What to do next:

After the process is complete, Cisco SD-WAN Manager automatically generates the controller profile in the tenant Virtual Account and the FQDN. You will receive an email after completion of the process.

View Tenant Information

Minimum supported releases: Cisco IOS XE Catalyst SD-WAN Release 17.12.1a and Cisco Catalyst SD-WAN Manager Release 20.12.1

You can view the following tenant information from the **Tenant Management > Tenants** page:

- **Tenant Name**
- **Description**
- **Controllers**
- **Forecasted Edge Count**
- **Total Edge Count:** Total number of both multitenant and single-tenant edge devices.
- **Multi Tenant WAN Edge Devices:** To view the number of multitenant edge device, click the non-zero number.
- **Tenant-Provider VPN Mapping:** To view the tenant and device VPN mappings for the tenant, click the non-zero number.
- **Service Connector:** View the multitenant edge device that provides a VXLAN connectivity to tenants.
- **Notifications:** View whether webhook notifications are managed by the tenant or a provider.
- **AAA:** View whether remote AAA is managed by the tenant or a provider.
- **ISE:** Starting from Cisco IOS XE Catalyst SD-WAN Release 17.18.2, you can view if ISE is configured for a tenant or a provider.
- **Controller Visibility:** View whether controller visibility is enabled or disabled.

Modify Tenant Information

1. Log in to Cisco SD-WAN Manager as the provider **admin** user.
2. From the Cisco SD-WAN Manager menu, choose **Administration > Tenant Management**.
3. In the left pane, click the name of the tenant.
The tenant information is displayed in a pane on the right.
4. To modify tenant data, do as follows:
 - a. In the right pane, click the pencil icon.
 - b. In the **Edit Tenant** dialog box, you can modify the following:
 - **Description:** The description can be up to 256 characters and can contain only alphanumeric characters.
 - **Forecasted Edge Count:** The number of WAN edge devices that the tenant can deploy.
A tenant can add a maximum of 1000 devices.



Note This option is available from Cisco SD-WAN Release 20.6.2, Cisco vManage Release 20.6.2.

If you increase the number of devices that a tenant can deploy, you must add the required number of device licenses to the tenant virtual account on **Plug and Play Connect** on [Cisco Software Central](#).

Before you increase the number of devices that a tenant can deploy, ensure that the Cisco SD-WAN Controller pair assigned to the tenant can support this increased number. A pair of Cisco SD-WAN Controllers can support a maximum of 24 tenants and 1000 devices across all these tenants.

- **Subdomain URL:** Modify the fully qualified sub-domain name of the tenant.
- c. (Optional) **Configure Services:** Set up service connectors to connect Cisco SD-WAN Manager to on-premises services such as AAA or custom webhook servers via the Cisco Catalyst SD-WAN overlay.
- **Service Connector IP:** Choose the service connector IP address from the drop-down list.
 - **Service Connector Interface Name:** Choose the VXLAN tunnel endpoint (interface name) from the drop-down list.
 - **Provider Managed - External Services:** Enable services to enforce provider-managed or tenant-managed configuration.
 - **Notifications:** Configure alarm notifications through custom webhook channels. Choose **Yes** to enable notifications and configure a provider-managed multitenant setup. The default value is **No**, making it a tenant-managed setup.
 - **AAA:** Configure remote AAA for managing tenants. Choose **Yes** to enable remote AAA and configure a provider-managed multitenant setup. The default value is **No**, making it a tenant-managed setup.
 - **Identity Services Engine:** Configure ISE for providers or tenants. Choose **Yes** for provider-managed services. Choose **No** for tenant-managed services.
- d. Starting from Cisco Catalyst SD-WAN Manager Release 20.16.1, toggle **Controller Visibility** to enable the following in your tenant's dashboard:
- In the **Monitor** dashboard, the **Controller Components** card with the **Controller** option that has an active hyperlink to view tenant-specific controller information
 - In the **Devices** page, controller details relevant to tenant
 - In the **Certificates** page, certificate details relevant to tenant
 - In **Devices > Real-time, Control Connections** and OMP statistics relevant to tenant.
- e. Click **Save**

Delete a Tenant

Before you delete a tenant, delete all tenant WAN edge devices. See [Delete a WAN Edge Device from a Tenant Network](#), on page 529.

1. Log in to Cisco SD-WAN Manager as the provider **admin** user.
2. From the Cisco SD-WAN Manager menu, choose **Administration > Tenant Management**.
3. In the left pane, click the name of the tenant.
The tenant information is displayed in a pane on the right.
4. To delete the tenant, do as follows:
 - a. In the right pane, click the trash icon.
 - b. In the **Delete Tenant** dialog box, enter the provider **admin** password and click **Save**.

Cisco SD-WAN Manager Dashboard for Multitenancy

After enabling Cisco SD-WAN Manager for multitenancy, you can view the multitenant dashboard when you log in to Cisco SD-WAN Manager. Cisco SD-WAN Manager multitenant dashboard is a portal where the provider or tenant can view and provision the underlying system.

The bar at the top of every Cisco SD-WAN Manager multitenant screen includes icons that allow smooth navigation.

View Cisco Catalyst SD-WAN Validator Health Dashlet

Minimum supported releases: Cisco Catalyst SD-WAN Manager Release 20.12.1

You can view the state of each Cisco Catalyst SD-WAN Validator, which is the cumulative state of the Cisco Catalyst SD-WAN Validator in a selected time window, and the number of Cisco Catalyst SD-WAN Validators in each state in the **Validator Health** dashlet on **Monitor Overview** dashboard.

You can filter the Cisco Catalyst SD-WAN Validator dashlet view based on the health status using the drop-down list for **Good Devices**, **Fair Devices**, and **Poor Devices** and also for **CPU Load**.

Click **View Details** to open the **Monitor > Devices** page to view the device health in table view.

View Cisco SD-WAN Manager Health Dashlet

Minimum supported releases: Cisco Catalyst SD-WAN Manager Release 20.12.1

You can view the state of Cisco SD-WAN Manager in the **Manager Health** dashlet on **Monitor Overview** dashboard.

You can filter the Cisco SD-WAN Manager dashlet view based on the health status using the drop-down list for **Good Devices**, **Fair Devices**, and **Poor Devices** and also for **CPU Load**.

Click **View Details** to open the **Monitor > Devices** page to view the device health in table view.

View Cisco Catalyst SD-WAN Controller Health Dashlet

Minimum supported releases: Cisco Catalyst SD-WAN Manager Release 20.12.1

You can view the list of tenants hosted on a particular device by clicking the **Controller** bar. You can view the state of each Cisco Catalyst SD-WAN Controller, which is the cumulative state of the Cisco Catalyst SD-WAN Controller in a selected time window, and the number of Cisco Catalyst SD-WAN Controllers in each state in the **Controller Health** dashlet on **Monitor Overview** dashboard.

You can filter the Cisco Catalyst SD-WAN Controller dashlet view based on the health status using the drop-down list for **Good Devices**, **Fair Devices**, and **Poor Devices** and also for **CPU Load**.

Click **View Details** to open the **Monitor > Devices** page to view the device health in table view.

View Multi Tenant WAN Edge Health Dashlet

Minimum supported releases: Cisco Catalyst SD-WAN Manager Release 20.12.1

You can view the state of each WAN edge device, which is the cumulative state of the devices in a selected time window, and the number of WAN edge devices in each state in the **Multi Tenant WAN Edge Health** dashlet on **Monitor Overview** dashboard.

You can view the list of tenants hosted on a particular device by clicking the multi-tenant WAN edge device bar. You can filter the Multi Tenant WAN Edge Health dashlet view based on the health status using the drop-down list for **Good Devices**, **Fair Devices**, and **Poor Devices** and also for **CPU Load**.

Click **View Details** to open the **Monitor > Devices** page to view the device health in table view.

View Tenant Activity, Device, and Network Information

When you log in to a multitenant Cisco SD-WAN Manager as an administrator, the provider dashboard displays the following components. To return to the provider dashboard from other Cisco SD-WAN Manager screens, click **Dashboard**.

- Device pane — runs across the top of the multitenant dashboard screen. The device pane displays the number of active Cisco Catalyst SD-WAN Controllers, Cisco SD-WAN Validator, and Cisco SD-WAN Manager instances, the connectivity status of devices, and information on certificates that have expired or about to expire.
- Tenants pane — displays the total number of tenants and a summary of the control status, site health, router health, and Cisco Catalyst SD-WAN Controller status of all tenants.
- Table of tenants in the overlay network — List of individual tenants, with separate information about the control status, site health, WAN edge device health, and Cisco Catalyst SD-WAN Controller status for each tenant.

To display tenant-specific status summary information,

1. Click a tenant name from the tenant list.

A dialog box opens on the right side of the screen that provides additional information about the status of the tenant.

2. To access the tenant dashboard for the selected tenant, click **<Tenant name> Dashboard**.

Cisco SD-WAN Manager presents the provider-as-tenant view and displays the tenant dashboard. To return to the provider view, click **Provider** at the top of page.

3. To close the dialog box, click the tenant name from the tenant list.

View Detailed Information of a Tenant Setup

Cisco SD-WAN Manager displays the tenant dashboard, which provides information about a tenant deployment when

- a provider **admin** user selects a specific tenant from the **Select Tenant** drop-down list in the provider dashboard. This view is called the provider-as-tenant view.
- a **tenantadmin** user logs in to Cisco SD-WAN Manager. This view is called the tenant view.

View All Network Connections in the Tenant Overlay Network

The **Device** pane runs across the top of the tenant dashboard and displays the number of control connections from Cisco SD-WAN Manager to the Cisco SD-WAN Controllers and routers in the overlay network of a tenant. For each WAN edge device, the Device pane shows

- Total number of control connections between Cisco SD-WAN Controllers and WAN edge devices
- Number of valid control connections between Cisco SD-WAN Controllers and WAN edge devices
- Number of invalid control connections between Cisco SD-WAN Controllers and WAN edge devices

Click a connection number, or the **Up** or **Down** arrow, to display a table with detailed information about each connection. Click the **More Actions** icon at the right of each table row to access the **Device Dashboard** or **Real Time** view from the **Monitor > Network** screen, or access the **Tools > SSH Terminal** Screen.

Click a connection number, or the **Up** or **Down** arrow, to display a table with detailed information about each connection. Click the **More Actions** icon at the right of each table row to access the **Device Dashboard** or **Real Time** view from the **Monitor > Devices** screen, or access the **Tools > SSH Terminal** Screen.



Note In Cisco vManage Release 20.6.1 and earlier releases, you can view information related to the **Monitor > Devices** page under the **Monitor > Network** page.

View Information About Device Reboots

The **Reboot** pane displays the total number of reboots in the last 24 hours for all devices in the network. It includes soft and cold reboots and reboots that occurred as a result of power-cycling a device. For each reboot, the following information is listed:

- System IP and hostname of the device that rebooted.
- Time when the device was rebooted.
- Reason for the device reboot

If the same device reboots more than once, each reboot option is reported separately.

Click the **Reboot** pane to open the **Reboot** dialog box. In the **Reboot** dialog box, click the **Crashes** tab. For all device crashes, the following information is listed:

- System IP and hostname of the device on which the crash occurred.
- Crash index of the device
- Core time when the device crashed.
- File name of the device crash log

View Network Connections

The **Control Status** pane displays whether Cisco SD-WAN Controller and WAN edge devices are connected. Each Cisco SD-WAN Controller must connect to all other Cisco SD-WAN Controllers in the network. Each WAN edge device must connect to the maximum number of configured Cisco SD-WAN Controllers. The **Control Status** pane displays three network connection counts:

- Control Up — total number of devices with the required number of operational control plane connections to a Cisco SD-WAN Controller
- Partial — total number of devices with some, but not all, operational control plane connection to Cisco SD-WAN Controllers.
- Control Down — total number of devices with no control plane connection to a Cisco SD-WAN Controller

To display a table with device details, click a row from the **Control Status** dialog box. Click the **More Actions** icon at the right of each table row to access the **Device Dashboard** or **Real Time** view from the **Monitor > Devices** screen.



Note InCisco vManage Release 20.6.1 and earlier releases, you can view information related to the **Monitor > Devices** page under the **Monitor > Network** page.

View State of Data Connections for a Site

The **Site Health** pane displays the state of data connections for a site. When a site has multiple WAN edge devices, this pane displays the state for the entire site and not for individual devices. The Site Health pane displays three connectivity states:

- Full WAN Connectivity — total number of sites where all BFD sessions on all routers are in the up state.
- Partial WAN Connectivity — total number of sites where tunnel and all BFD sessions on all routers are in the down state. These sites still have limited data plane connectivity.
- No WAN Connectivity — total number of sites where all BFD sessions on all routers are in the down state. These sites have no data plane connectivity.

To display a table with detailed information about each site, node, or tunnel, click a row from the **Site Health** dialog box. Click the **More Actions** icon at the right of each row in the table to access the **Device Dashboard** or **Real Time** view from the **Monitor > Devices** screen, or access the **Tools > SSH Terminal** screen.



Note InCisco vManage Release 20.6.1 and earlier releases, you can view information related to the **Monitor > Devices** page under the **Monitor > Network** page.

View Interface Usage for WAN Edge Interfaces

The **Transport Interface Distribution** pane displays interface usage in the last 24 hours for all WAN edge interfaces in VPN 0. It includes all TLOC interfaces. Click the pane to view details of interface usage in the **Transport Interface Distribution** dialog box.

View WAN Edge Device Counts

The **WAN Edge Inventory** pane provides four WAN edge device counts:

- **Total** — total number of authorized serial numbers for WAN edge devices that have been uploaded on Cisco SD-WAN Manager. The serial number is uploaded on the **Configuration > Devices** screen.
- **Authorized** — total number of authorized WAN edge devices in the overlay network. These WAN edge devices are marked as **Valid** in the **Configuration > Certificates > WAN Edge List** screen.
- **Deployed** — total number of deployed WAN edge devices. These are WAN edge devices that are marked as **Valid** and are now operational in the network.
- **Staging** — total number of WAN edge devices you configure at a staging site before they are made a part of the overlay network. These routers do not take part in any routing decisions and do not affect network monitoring through Cisco SD-WAN Manager.

Click the pane to view hostname, system IP, site ID, and other details of each router from the **WAN Edge Inventory** dialog box.

View Aggregated State of WAN Edge Devices

The **WAN Edge Health** pane offers an aggregated view of the state of WAN edge devices by providing a count of the number of devices in each state, therefore describing the health of the hardware nodes. The three WAN edge device states are:

- **Normal** — number of WAN edge devices with memory, hardware, and CPU in normal state. Using less than 70% of total memory or total CPU is classified as, normal.
- **Warning** — number of WAN edge devices with memory, hardware, or CPU in warning state. Using between 70% and 90% of total memory or total CPU is classified as, warning.
- **Error** — number of WAN edge devices with memory, hardware, or CPU in error state. Using more than 90% of total memory or total CPU is classified as, error.

Click a number or the WAN edge device state to display a table with the last 12 or 24 hours of memory usage, CPU utilization, and hardware-related alarms, including temperature, power supply, and PIM modules. Click the **More Actions** icon at the right of each row in the table to access the following:

- **Hardware Environment**
- **Real Time** view from the **Monitor > Network** screen



Note

In Cisco vManage Release 20.6.1 and earlier releases, you can view information related to the **Monitor > Devices** page under the **Monitor > Network** page.

- **Tools > SSH Terminal** screen.

View WAN Edge Device Loss, Latency, Jitter

The **Transport Health** pane displays the aggregated average loss, latency, and jitters for all links and all combinations of colors (for example, all LTE-to-LTE links, all LTE-to-3G links).

From the **Type** drop-down arrow, choose loss, latency, or jitter.

Click the  icon to select a time period for which to display the transport health.

Click the  icon to open the **Transport Health** dialog box. This dialog box displays a more detailed view. To display information in a tabular format, click the **Details** tab. You can choose to change the displayed health type and time period.

View SAIE Flow Information of WAN Edge Devices

The **Top Applications** pane displays SD-WAN Application Intelligence Engine (SAIE) flow information for traffic transiting routers in the overlay network.



Note

- In Cisco vManage Release 20.7.x and earlier releases, the SAIE flow is known as deep packet inspection (DPI).
- The SAIE flow information is shown only for the last 24 hours. To view SAIE flow information for a time before the last 24 hours, you must check the information for the specific device.

Click the  icon to select a time period for which to display data. From the **VPN** drop-down list, select a VPN to display SAIE information for all flows in that VPN.

Click the  icon to open the **Top Applications** dialog box. This dialog box displays a more detailed view of the same information. You can change the VPN and time period.

View Tunnels Data

The **Application-Aware Routing** pane allows you to choose the following tunnel criteria from the **Type** drop-down arrow:

- Loss
- Latency
- Jitter

Based on the tunnel criteria, the pane displays the 10 worst tunnels. For example, if you choose loss, the pane shows 10 tunnels with the greatest average loss over the last 24 hours.

Click the  icon against a row to display a graphical representation of the data. Select a time period for which to display data or click **Custom** to display a drop-down arrow for specifying a custom time period.

Click the  icon to open the **Application-Aware Routing** dialog box. This dialog box displays the 25 worst tunnels based on criteria you choose from the **Type** drop-down arrow, the criteria being loss, latency, and jitter.

Manage Tenant WAN Edge Devices

Add a WAN Edge Device to a Tenant Network



Note If you are adding a WAN edge device that you had previously invalidated and deleted from an overlay network, you must reset the device software after adding the device. To reset the software on a Cisco vEdge device, use the command **request platform software reset**.

1. Log in to Cisco SD-WAN Manager.

If you're a provider user, log in as the **admin**. In the provider dashboard, choose a tenant from the drop-down list to enter the provider-as-tenant view.

If you're a tenant user, log in as the **tenantadmin**.

2. Upload the device serial number file to Cisco SD-WAN Manager.

3. Validate the device and send details to controllers.

4. Create a configuration template for the device and attach the device to the template.

While configuring the device, configure the service provider organization name and the tenant organization name as in the following example:

```
sp-organization-name multitenancy
organization-name multitenancy-Customer1
```



Note Enter the `organization-name` in the format `<SP Org Name>-<Tenant Org Name>`.

5. Bootstrap the device using bootstrap configuration generated through Cisco SD-WAN Manager or manually create the initial configuration on the device.
6. If you are using Enterprise Certificates to authenticate the device, download the CSR from Cisco SD-WAN Manager and get the CSR signed by the Enterprise CA. Install the certificate on Cisco SD-WAN Manager.

Delete a WAN Edge Device from a Tenant Network

1. Log in to Cisco SD-WAN Manager.

If you're a provider user, log in as the **admin**. In the provider dashboard, choose a tenant from the drop-down list to enter the provider-as-tenant view.

If you're a tenant user, log in as the **tenantadmin**.

2. Detach the device from any configuration templates.
3. [Delete a WAN Edge Router](#).

Tenant-Specific Policies on Cisco SD-WAN Controller

A provider **admin** user (from the Cisco SD-WAN Manager provider-as-tenant view) or a **tenantadmin** user (from the Cisco SD-WAN Manager tenant view) can create and deploy tenant-specific policies on the Cisco SD-WAN Controller serving the tenant. The user can configure a CLI policy or create the policy using the UI policy configuration wizard.

When you activate or deactivate a policy,

1. Cisco SD-WAN Manager identifies the Cisco SD-WAN Controllers serving the tenant.
2. Cisco SD-WAN Manager notifies the Cisco SD-WAN Controllers to pull the policy configuration.
3. Cisco SD-WAN Controllers pull and deploy the policy configuration.
4. Cisco SD-WAN Manager reports the status of the policy pull by the Cisco SD-WAN Controllers.

Manage Tenant Data

Back Up Tenant Data

The tenant data backup solution of Cisco SD-WAN Manager multitenancy provides the following functionalities:

- [Create, list, and extract configuration backup files.](#)
- Back up configuration database of a specific tenant with an option to restore it later. See [Back up and restore configuration database.](#)
- Delete back up files of a tenant stored in Cisco SD-WAN Manager. For deleting tenant data backup files, see [Delete tenant data backup file.](#)

The following factors are applicable when using data backup solution:

- The tenant data backup solution operations can be performed by a tenant administrator in the tenant view and or by a provider administrator in the provider-as-tenant view. To know how to access tenant dashboard through different views, see [User Roles in Multitenant Environment, on page 499.](#)
- A tenant is allowed to perform the following backup operations at a particular time and must complete an operation before starting a new operation:
 - Back up a single configuration database
 - Download the backup file.
 - Restore or import backup files
 - Delete backup files.
 - List backup files
- A tenant backup file format is as follows:

```
Bkup_tenantId_MMDDYY-HHMMSS_taskIdWithoutDash.tar.gz
```

- The tenant data backup operation is a readonly operation on the configuration database. However, to ensure data consistency and prevent data loss, do not perform any major changes on the network while the operation is in progress.
- Multiple tenants can perform back-up and restore operations in parallel.
- A tenant is not allowed to perform other backup operations when the restore operation of the tenant database is in-progress. So, a tenant can perform a single backup operation and when this operation is in-progress, all new backup operation requests are rejected.

The remaining tenants can continue with their backup operations.

- A tenant must perform backup and restore operations on Cisco SD-WAN Manager instances running identical Cisco SD-WAN Manager software versions.
- A tenant can store a maximum of three backup files in Cisco SD-WAN Manager and can download to store them outside Cisco SD-WAN Manager repository. If the tenant already has three backup files, a subsequent backup operation results in the earliest backup file being deleted and a new backup file being generated.
- Ensure that the following parameter values match in both the backup file and the setup where tenant has requested for a restore operation:
 - Tenant Id
 - Organization Name
 - SP Organization Name
- The tenant data backup solution creates a task in the tenant view of Cisco SD-WAN Manager. Therefore, the tenant can monitor the progress of the operation from the task view of the tenant dashboard.
- A provider cannot back up provider data using this solution. Therefore, the provider can back up all tenants information at once by backing up all tenants configuration database using CLI.

Create, Extract, and List Configuration Data Backup File

1. Log in to Cisco SD-WAN Manager.

If you're a provider user, log in as the **admin**. In the provider dashboard, choose a tenant from the drop-down list to enter the provider-as-tenant view.

If you're a tenant user, log in as the **tenantadmin**.

2. In the address bar, modify the URL path with `dataservice` for the REST API connection.

Example: `https://<tenant_URL>/dataservice`

3. Create a configuration backup file by using the following API:

`https://<tenant_URL>/dataservice/tenantbackup/export.`

4. If the configuration backup file has been created successfully, Cisco SD-WAN Manager task view indicates that the backup file has been generated. You can view the process identifier of the created process or task.

Example:

```
{
  "processId": "72d69805-b987-436f-9b7a-afef2f3f9061",
  "status": "in-progress"
}
```

5. Verify the task status using the obtained process identifier.

Example:

```
https://<tenant_URL>/dataservice/device/action/status/72d69805-b987-436f-9b7a-afef2f3f9061
```

The verification generates the details of the task in the JSON file format.

6. After the task is completed, extract or download the backed-up file available under the **data** section of the JSON task file.

Example: To extract or download the backup file, use the following API:

```
https://<tenant_URL>/dataservice/tenantbackup/download/1570057020772/backup_1570057020772_100919-181838.tar.gz
```

7. List backup files stored in Cisco SD-WAN Manager using the following API.

Example: `https://<tenant_URL>/dataservice/tenantbackup/list`

Restore and Delete Tenant Data Backup File

Before you begin:

To run the restore and delete tenant data backup files API, you can download and install Postman tool or any other alternative tool for testing http applications and services. In this document, the procedure to restore and delete tenant data backup files has been explained using the Postman tool. Postman is a software tool used as an API development environment. You can download the tool from the Postman website.

1. Open Google Chrome, or another browser, and enable developer mode on it.
2. Log in to Cisco SD-WAN Manager.

If you're a provider user, log in as the **admin**. In the provider dashboard, choose a tenant from the drop-down list to enter the provider-as-tenant view.

If you're a tenant user, log in as the **tenantadmin**.

3. To get header information of the restore API, do as follows:
 - a. On the right side of the screen, click the **Network** tab to get the network capture view.
 - b. In the network capture view, click the **Name** column to sort the listed items.
 - c. Search and click **index.html**.
 - d. Click the **Headers** tab and expand **Request Headers**.
 - e. Choose all text under **Request Headers** and copy it to the clipboard.
4. Import backup files through the Postman UI:
 - a. Open the Postman UI.
 - b. To disable SSL certificate verification, click **Postman > Preferences > General > Request**. Turn off **SSL Certificate Verification**.
 - c. In the Postman UI, create a new tab.

- d. Click **Headers** and then click **Bulk Edit**.
- e. Paste the text that was copied in step 3 from the **Request Headers** block into an editable form.
- f. From the **GET** method drop-down list, choose **POST**.

- g. In the **Paste request URL** field, paste the dedicated URL of the tenant and include `dataservice/tenantbackup/import`.

Example: `https://customer1.managed-sp.com/dataservice/tenantbackup/import`

- h. Click the **Body** tab and select **form-data**.
- i. Under **KEY** column, enter `bakup.tar.gz`
- j. Under **VALUE** column, click **Select Files** and select a backup file to be imported.
- k. To run the API, click **Send**.

In the **Response** section of the Postman UI, you can view the JSON information that indicates the file that was restored.

5. Monitor the restoration of backup files in either of the following ways:

- a. Use Cisco SD-WAN Manager task view that indicates if backup file has been imported successfully. You can view the process identifier of the created process or task.

Example:

```
{
  "processId": "40adb6c0-eacc-4ad4-ba6c-2c2da2e96d1d",
  "status": "Import Successfully Submitted for tenant 1579026919487"
}
```

- b. Use the following URL to get the status,

`https://<tenant_URL>/dataservice/device/action/status/<processId>`

Example:

`https://customer1.managed-sp.com/dataservice/device/action/status/40adb6c0-eacc-4ad4-ba6c-2c2da2e96d1d`

6. Delete tenant data backup file through Postman UI.

- a. In the Postman UI, create a new tab.
- b. Click **Headers** and then click **Bulk Edit**.
- c. Paste the text that was copied in step 3 from the **Request Headers** block into an editable form.
- d. From the **GET** method drop-down list, choose **DELETE**.
- e. In the **Paste request URL** field, paste the dedicated URL of the tenant and include `dataservice/tenantbackup/delete?fileName=' filename '`. The filename can either be name of the backup file or all.

Example:

`https://customer1.managed-sp.com/dataservice/tenantbackup/delete?fileName=1579026919487_012820-180712_c09230904bf40adb6c0e5068b03002.tar.gz`

Example: `https://customer1.managed-sp.com/dataservice/tenantbackup/delete?fileName=all`

- f. To run the API, click **Send**.

In the **Response** section of the Postman UI, you can view the JSON information that indicates the files that were deleted.

Example:

```
{
  "Deleted": [
    "bkup_1579026919487_012820-180712_c09230904dfc40edb0d1e50b68b03002.tar.gz"
  ]
}
```

View OMP Statistics per Tenant on a Cisco SD-WAN Controller

1. Log in to Cisco SD-WAN Manager as the provider **admin** user.
2. From the Cisco SD-WAN Manager menu, choose **Monitor > Devices**.
Cisco vManage Release 20.6.x and earlier: From the Cisco SD-WAN Manager menu, choose **Monitor > Network**.
3. In the table of devices, click on the hostname of a Cisco SD-WAN Controller.
4. In the left pane, click **Real Time**.
5. In the **Device Options** field, enter **OMP** and select the OMP statistics you wish to view.
6. In the **Select Filters** dialog box, click **Show Filters**.
7. Enter the **Tenant Name** and click **Search**.

Cisco SD-WAN Manager displays the selected OMP statistics for the particular tenant.

View Tenants Associated with a Cisco SD-WAN Controller

1. Log in to Cisco SD-WAN Manager as the provider **admin** user.
2. Click a **Controller** connection number to display a table with detailed information about each connection.
Cisco SD-WAN Manager displays a table that provides a summary of the Cisco SD-WAN Controllers and their connections.
3. For a Cisco SD-WAN Controller, click ... and click **Tenant List**.
Cisco SD-WAN Manager displays a summary of tenants associated with the Cisco SD-WAN Controller.

Migrate Single-Tenant Cisco Catalyst SD-WAN Overlay to Multitenant Cisco Catalyst SD-WAN Deployment

Before You Begin

- Before you begin the migration,

- Migration of a single-tenant overlay to a multitenant deployment is only supported with the Cisco Catalyst SD-WAN controllers deployed on-premises. Migration is yet to be supported with cloud-hosted Cisco Catalyst SD-WAN controllers.
- Ensure that the edge devices in the single-tenant deployment can reach the Cisco SD-WAN Validator in the multitenant deployment
- Ensure that the template, routing, and policy configuration on the edge devices is synchronized with the current configuration on Cisco SD-WAN Manager
- Configure a maintenance window for the single-tenant overlay before performing this procedure. See [Configure or Cancel SD-WAN Manager Server Maintenance Window](#).
- Minimum software requirements for the single-tenant overlay to be migrated:

Device	Software Version
Cisco SD-WAN Manager	Cisco vManage Release 20.6.1
Cisco SD-WAN Validator	Cisco SD-WAN Release 20.6.1
Cisco Catalyst SD-WAN Controller	Cisco SD-WAN Release 20.6.1
Cisco vEdge Device	Cisco SD-WAN Release 20.6.1

- Minimum software requirements for the multitenant deployment to which the single-tenant overlay must be migrated:

Device	Software Version
Cisco SD-WAN Manager	Cisco vManage Release 20.6.1
Cisco SD-WAN Validator	Cisco SD-WAN Release 20.6.1
Cisco Catalyst SD-WAN Controller	Cisco SD-WAN Release 20.6.1
Cisco vEdge Device	Cisco SD-WAN Release 20.6.1

- The software versions of the Cisco Catalyst SD-WAN controllers and WAN edge devices must be identical in both the single-tenant and multitenant deployments.
- We recommend that you use a custom script or a third-party application like Postman to execute the API calls.

Migration Procedure

1. Export the single-tenant deployment and configuration data from a Cisco SD-WAN Manager instance controlling the overlay.

Method	POST
URL	<code>https://ST-vManage-IP-address</code>
Endpoint	<code>/dataservice/tenantmigration/export</code>
Authorization	Admin user credentials.

Body	Required Format: Raw JSON <pre>{ "desc": <tenant_description>, "name": <tenant_name>, "subdomain": <tenant_name>.<domain>, "orgName": <tenant_orgname > }</pre> Field Description: • desc: A description of the tenant. The description can be up to 256 characters and can contain only alphanumeric characters. • name: Unique name for the tenant in the multitenant deployment. • subdomain: Fully qualified sub-domain name of the tenant. The sub-domain name must include the domain name of the service provider. For example, if <code>managed-sp.com</code> is the domain name of service provider, and the tenant name is <code>Customer1</code>, the tenant sub-domain name would be <code>customer1.managed-sp.com</code>. • orgName: Name of the tenant organization. The organization name is case-sensitive.
Response	Format: JSON <pre>{ "processId": <vManage_process_ID>, }</pre>

While exporting the data, Cisco SD-WAN Manager attempts to detach any CLI templates from the edge devices in preparation for the migration to the multitenant deployment. If prompted by Cisco SD-WAN Manager, detach CLI templates from the edge devices and execute the export API call again.

2. Check the status of the data export task in Cisco SD-WAN Manager. When the task succeeds, download the data using the URL

`https://ST-vManage-IP-address/dataservice/tenantmigration/download/default.tar.gz`

3. On a multitenant Cisco SD-WAN Manager instance, import the data exported from the single-tenant overlay.

Method	POST
URL	<code>https://MT-vManage-IP-address</code>
Endpoint	<code>/dataservice/tenantmigration/import</code>
Authorization	Provider Admin user credentials.
Body	Required Format: form-data Key Type: File Value: <code>default.tar.gz</code>

Response	Format: JSON <pre>{ "processId": <vManage_process_ID>, "migrationTokenURL": <token_URL>, }</pre>
----------	---

When the task succeeds, on the multitenant Cisco SD-WAN Manager, you can view the devices, templates, and policies imported from the single-tenant overlay.

4. Obtain the migration token using the token URL obtained in response to the API call in **Step 3**.

Method	GET
URL	<code>https://MT-vManage-IP-address</code>
Endpoint	migrationTokenURL obtained in Step 3 .
Authorization	Provider Admin user credentials.
Response	The migration token as a large blob of encoded text.

5. On the single-tenant Cisco SD-WAN Manager instance, initiate the migration of the overlay to the multitenant deployment.

Method	POST
URL	<code>https://ST-vManage-IP-address</code>
Endpoint	<code>dataservice/tenantmigration/networkMigration</code>
Authorization	Admin user credentials.
Body	Required Format: Raw text Content: Migration token obtained in Step 4 .
Response	Format: JSON <pre>{ "processId": <vManage_process_ID>, }</pre>

In Cisco SD-WAN Manager, check the status of the migration task. As part of the migration task, the address of the multitenant Cisco SD-WAN Validator, and the service provider and tenant organization names are pushed to the WAN edge devices of the single-tenant overlay. If the task succeeds, WAN edge devices form control connections to controllers in the multitenant deployment; the WAN edge devices are no longer connected to the controllers of the single-tenant overlay.

Attach any CLI templates detached from the edge devices (in Step 1) after migration to the multitenant deployment. Before you attach the templates, update the Cisco SD-WAN Validator IP address and the Organization name to match the configuration of the multitenant deployment.



Note In the single-tenant deployment, if Cisco SD-WAN Manager-signed certificates are installed on cloud-based WAN edge devices, the certificates are cleared when the devices are migrated to the multitenant deployment. You must re-certify the devices on the multitenant Cisco SD-WAN Manager. If enterprise certificates are installed on the cloud-based WAN edge devices, the certificates are not affected by the migration. For more information, see [Enterprise Certificates](#).

Migrate a Tenant from a Multitenant Cisco Catalyst SD-WAN Overlay to Single-Tenant Cisco Catalyst SD-WAN Deployment

Before You Begin

- Manually migrate the serial number of the WAN edge device associated to a virtual account on the source Cisco SD-WAN Manager overlay in Cisco PNP to the destination virtual account.
- Ensure that you manually create the controller profile on the destination virtual account for on-prem to on-prem or cloud to on-prem deployments.
- Ensure that the source and destination Cisco SD-WAN Manager instances have the same Certificate Authority (CA). If not, recertify the devices after the migration is complete.
- Ensure that you check the CPU, memory, and disk size requirements of the destination overlay Cisco SD-WAN Controllers before the migration to meet the WAN edge forecast requirements.
- Ensure that there is no overlap between the configured system IP addresses of edge devices and the destination overlay controllers.
- Ensure that the destination single-tenant Cisco SD-WAN Manager does not have any configurations before migration. You can configure only mandatory admin settings and all other configurations can be done after data import.
- Ensure that the Cisco SD-WAN Control Components in the source and destination overlays are using the same software release. The migration process does not check for a software release mismatch and a mismatch blocks the import of tenant data, causing the migration to fail.
- Ensure that all devices in a tenant have connectivity to the Cisco SD-WAN Validator in the destination single-tenant overlay. The migration procedure supports a Cisco SD-WAN Validator on the single-tenant deployment configured either with IP or DNS.

Push any required static route configuration to the devices before initiating any of the migration steps.

- Ensure that the WAN edge devices that are configured using CLI, device template, or configuration groups, have an IP host mapping to the Cisco SD-WAN Validator in the destination single-tenant overlay.
- Ensure that there are valid control connections from Cisco SD-WAN Manager to the WAN edge devices in the source overlay.
- Configure a maintenance window for the multitenant overlay before performing this procedure. See [Configure or Cancel SD-WAN Manager Server Maintenance Window](#).

- We recommend that you use a custom script or a third-party application like Postman to execute the API calls.

Migration Procedure

1. Export the multitenant deployment configuration and statistical data from a Cisco SD-WAN Manager instance controlling the source overlay.

Method	POST
URL	<code>https://MT-vManage-IP-address</code>
Endpoint	<code>/dataservice/tenantmigration/export</code>
Authorization	Administrator user credentials.

Body	Required Format: Raw JSON Example: <pre>{ "name": "tenant1", "desc": "This is tenant1", "orgName": "vIPtela Inc MT to ST Migration Regression-Tenant1 Inc", "subDomain": "tenant1.mtreg.com", "wanEdgeForecast": 100, "migrationKey="tenant1TenantMigrationKey123", "isDestinationOverlayMT": false }</pre> Field descriptions: Note Ensure that the <code>name</code>, <code>desc</code>, <code>orgName</code>, <code>subdomain</code>, and <code>wanEdgeForecast</code> match the tenant you wish to migrate. • <code>name</code>: Unique name for the tenant in the multitenant deployment. The name should be between 8-32 characters and can contain only alphanumeric characters. • <code>desc</code>: Description of the tenant. The description can be up to 256 characters and can contain only alphanumeric characters. • <code>orgName</code>: Name of the tenant organization. The organization name is case-sensitive. • <code>subdomain</code>: Fully qualified sub-domain name of the tenant. The sub-domain name must include the domain name of the service provider. For example, if <code>managed-sp.com</code> is the domain name of service provider, and the tenant name is <code>Customer1</code>, the tenant sub-domain name would be <code>customer1.managed-sp.com</code>. • <code>wanEdgeForecast</code>: Number of WAN edge devices that the tenant can deploy. • <code>migrationKey</code>: Migration key which is used to encrypt sensitive data during migration. The migration key should be between 8-32 characters and can contain only alphanumeric characters. • <code>isDestinationOverlayMT</code>: Boolean variable which specifies if the migration is happening to a multitenant overlay or not.
Response	Format: JSON <pre>{ "processId": <vManage_process_ID>, }</pre>

2. Check the status of the data export task in Cisco SD-WAN Manager. When the task is successfully complete, download the data from the following URL:

<https://MT-vManage-IP-address/dataservice/tenantmigration/download/default.tar.gz>

3. Import the data to the single-tenant instance, as follows:

- a. Execute the following API:

Method	POST
URL	<code>https://ST-vManage-IP-address</code>
Endpoint	<code>/dataservice/tenantmigration/import/{migrationKey}</code> Use the same migration key specified earlier.
Authorization	Provider administrator user credentials.
Body	Required Format: form-data Key Type: File Value: <code>default.tar.gz</code>
Response	Format: JSON <pre>{ "processId": <vManage_process_ID>, "migrationTokenURL": <token_URL>, }</pre>

- b. When the task is complete, on the single-tenant Cisco SD-WAN Manager, you can view the devices, templates, and policies imported from the multitenant overlay.

- After the import, update information related to the device templates, policies, and other deployment-specific parameters. Check and update the administrator settings as some of the administrator settings specific to the source overlay are not exported. The import does not override the administrator settings that are already configured in destination Cisco SD-WAN Manager.
- If a centralized policy is present on the source tenant, the migration copies the policy to the destination overlay. We recommend creating Cisco SD-WAN Controller templates and attaching them to the devices. Apply the centralized policy to devices in the destination overlay before proceeding.
- Obtain the migration token using the token URL from the previous step.

Method	GET
URL	<code>https://ST-vManage-IP-address</code>
Endpoint	<code>migrationTokenURL</code> obtained in the previous step.
Authorization	Provider administrator user credentials.
Response	The migration token as a large encoded text.

- On the multitenant Cisco SD-WAN Manager instance, initiate the migration of the overlay to the single-tenant deployment.

Method	POST
URL	<code>https://MT-vManage-IP-address</code>
Endpoint	<code>dataservice/tenantmigration/networkMigration</code>
Authorization	Administrator user credentials.

Body	Required Format: Raw text Content: Migration token obtained in the previous step.
Response	Format: JSON <pre>{ "processId": <vManage_process_ID>, }</pre>

In Cisco SD-WAN Manager, check the status of the migration task. When the task succeeds, WAN edge devices form control connections to controllers in the single-tenant deployment; the WAN edge devices are no longer connected to the controllers of the multitenant overlay.

8. After the migration is successfully complete, perform the following tasks:

- If WAN edge devices have Cisco SD-WAN Manager signed certificates in the source setup, the certificates are cleared from the device during migration and control connections are lost. Recertify the devices in the destination.
- The passwords are updated to the default password in the destination overlay for users created on a tenant in the source overlay. Make any configuration changes specific to the destination overlay.
- Delete the tenant on the source overlay after migration and verification is complete.

Migrate Multitenant Cisco Catalyst SD-WAN Overlay

Prerequisites

Minimum software requirements for Cisco Catalyst SD-WAN controllers and WAN edge devices in the multitenant overlay to be migrated:

Device	Software Version
Cisco SD-WAN Manager	Cisco vManage Release 20.3.3
Cisco SD-WAN Validator	Cisco SD-WAN Release 20.3.3
Cisco SD-WAN Controller	Cisco SD-WAN Release 20.3.3
Cisco vEdge Device	Cisco SD-WAN Release 20.3.3

Restrictions

- This migration procedure applies only to Cisco Catalyst SD-WAN controllers deployed on premises.
- The multitenant overlay can only be migrated to a setup in which Cisco SD-WAN Manager instances run Cisco vManage Release 20.6.1 software and Cisco Catalyst SD-WAN controllers run Cisco SD-WAN Release 20.6.1 software.
- This migration procedure cannot be used to merge two or more multitenant overlays. Only one multitenant overlay can be migrated to the new setup at a time.

Migration Procedure

1. Upgrade the software on the three Cisco SD-WAN Manager instances in the cluster to Cisco vManage Release 20.6.1. For more information, see [Upgrade Cisco SD-WAN Manager Cluster](#).



Note Run the command **request nms configuration-db upgrade** on only one of the Cisco SD-WAN Manager instances.

2. After the Cisco SD-WAN Manager software is upgraded to Cisco vManage Release 20.6.1, log in to the Cisco SD-WAN Manager.
You're prompted to set a new password.
 - a. Enter a new password that adheres to the password guidelines.
3. Upload the Cisco SD-WAN Release 20.6.1 software to Cisco SD-WAN Manager. For more information, see [Add an Image to the Software Repository](#).
4. Upgrade the Cisco SD-WAN Validator software to Cisco SD-WAN Release 20.6.1. For more information, see [Upgrade the Software Image on a Device](#).
5. Create two Cisco SD-WAN Controller instances running Cisco SD-WAN Release 20.6.1 software. See [Deploy the Cisco SD-WAN Controller](#).



Note With two Cisco SD-WAN Controller instances, you can support up to 24 tenants. To support up to 50 tenants, create six Cisco SD-WAN Controller instances.

- a. [Add Cisco SD-WAN Controllers](#) to the overlay network.

The **Provider Dashboard** shows the new Cisco SD-WAN Controllers running Cisco SD-WAN Release 20.6.1 software. The **Tenant Dashboard** shows the older Cisco SD-WAN Controllers running Cisco SD-WAN Release 20.3.3 software.

6. Enable the maintenance window on Cisco SD-WAN Manager. For more information, see [Configure or Cancel SD-WAN Manager Server Maintenance Window](#).
A maintenance window of 3 to 4 hours is recommended.
7. Migrate the tenant configuration from the older tenant-specific Cisco SD-WAN Controllers running Cisco SD-WAN Release 20.3.3 software to the new shared Cisco SD-WAN Controllers running Cisco SD-WAN Release 20.6.1 software.

Method	POST
URL	<code>https://<vmanageip>:<port></code>
Endpoint	<code>dataservice/tenant/vsmart-mt/migrate</code>
Authorization	Provider admin user credentials.

Body	Required Format: Raw JSON { }
Response	Format: JSON { "processId": <vManage_process_ID>, }

In Cisco SD-WAN Manager, check the status of the migration task using the `processId` from the API response. During the migration task, the following changes are affected:

- a. The older Cisco SD-WAN Controllers are invalidated and deleted from the overlay network.
 - b. In the tenant view, the older Cisco SD-WAN Controllers are removed from the **Tenant Dashboard**, and the **Devices** and the **Certificates** page.
 - c. The tenant WAN edge devices are connected to the new Cisco SD-WAN Controllers.
8. (Optional) Upgrade the Cisco vEdge device software to Cisco SD-WAN Release 20.6.1. For more information, see [Upgrade the Software Image on a Device](#) and [Activate a New Software Image](#).



Tip It is not necessary to upgrade the tenant WAN edge device software in the same maintenance window in which you migrate the multitenant overlay. However, we recommend that you upgrade the tenant WAN edge device software within a few weeks of the migration.

Verify the Migration

1. In the provider view, perform the following checks:
 - a. From the **Main Dashboard** page, verify whether the tenant WAN edge devices are connected to the new multitenant Cisco SD-WAN Controllers.
 - b. [View Tenants Associated with a Cisco SD-WAN Controller, on page 534.](#)
 - c. On the Cisco SD-WAN Controller CLI, run the command **show control connections**. In the command output, verify that control connections are established between the Cisco SD-WAN Controller and the tenant WAN edge devices.
2. In the provider-as-tenant view, verify whether the multitenant Cisco SD-WAN Controllers appear on the **Tenant Dashboard**.

Upgrade Cisco Catalyst SD-WAN Controller and Edge Device Software

Prerequisites

Minimum software requirements for Cisco Catalyst SD-WAN controllers and WAN edge devices:

Device	Software Version
Cisco SD-WAN Manager	Cisco vManage Release 20.4.1 or later
Cisco SD-WAN Validator	Cisco SD-WAN Release 20.4.1 or later
Cisco Catalyst SD-WAN Controller	Cisco SD-WAN Release 20.4.1 or later
Cisco vEdge Device	Cisco SD-WAN Release 20.4.1 or later

Upgrade Procedure

1. Upgrade the software on the three Cisco SD-WAN Manager instances in the cluster to Cisco vManage Release 20.6.1 or a later release. For more information, see [Upgrade Cisco SD-WAN Manager Cluster](#).



Note Skip the step to upgrade the configuration-db service using the command **request nms configuration-db upgrade**.

2. After the Cisco SD-WAN Manager software is upgraded to Cisco vManage Release 20.6.1 or a later release, log in to the Cisco SD-WAN Manager.
3. Upload the Cisco SD-WAN Release 20.6.1 or a later release software to Cisco SD-WAN Manager. For more information, see [Add an Image to the Software Repository](#).
4. Upgrade the Cisco SD-WAN Validator software to Cisco SD-WAN Release 20.6.1 or a later release. For more information, see [Upgrade the Software Image on a Device](#) and [Activate a New Software Image](#).
5. Enable maintenance window on Cisco SD-WAN Manager. For more information, see [Configure or Cancel SD-WAN Manager Server Maintenance Window](#).
6. Upgrade the Cisco Catalyst SD-WAN Controller software to Cisco SD-WAN Release 20.6.1 or a later release. For more information, see [Upgrade the Software Image on a Device](#) and [Activate a New Software Image](#).
7. Upgrade the Cisco vEdge device software to Cisco SD-WAN Release 20.6.1 or a later release. For more information, see [Upgrade the Software Image on a Device](#) and [Activate a New Software Image](#).



Tip We recommend that you upgrade the WAN edge device software in the same maintenance window. If the WAN edge device software is not upgraded within the OMP graceful restart window, traffic may be lost.

Multitenant Cisco SD-WAN Manager: Disaster Recovery



Note Administrator triggered disaster recovery is supported for multitenant clusters from Cisco vManage Release 20.6.1 or later releases.

If a Multitenant Cisco SD-WAN Manager cluster or the data center hosting the Cisco SD-WAN Manager nodes in the cluster fail, you can recover from the failure by activating a standby Cisco SD-WAN Manager cluster. You can perform disaster recovery as follows:

1. Deploy and configure a standby Cisco SD-WAN Manager cluster.

The standby Cisco SD-WAN Manager cluster is not part of the overlay network and is not active.

2. Back up the configuration database of the active Cisco SD-WAN Manager cluster periodically.

Choose a Cisco SD-WAN Manager node in the cluster that hosts the configuration database service and back up the configuration database.

3. If the active Cisco SD-WAN Manager cluster fails, restore the most recent configuration database on the standby Cisco SD-WAN Manager cluster, activate the standby Cisco SD-WAN Manager cluster, and remove the previously active Cisco SD-WAN Manager cluster from the overlay network.

Choose a Cisco SD-WAN Manager node in the cluster that hosts the configuration database service and restore the configuration database backed up from the previously active Cisco SD-WAN Manager cluster.

To test disaster recovery, you can simulate a scenario in which the active Cisco SD-WAN Manager cluster fails. One way to simulate such a failure would be by disabling the tunnel interface as described in this document.

Prerequisites

- The number of Cisco SD-WAN Manager nodes in the active and standby clusters must be identical.
- Each Cisco SD-WAN Manager node in the active and standby clusters must run the same Cisco SD-WAN Manager software release.
- Each Cisco SD-WAN Manager node in the active and standby clusters must be able to connect to the WAN transport IP address of the Cisco SD-WAN Validator in the overlay network.
- Initially, the tunnel interfaces of the Cisco SD-WAN Manager nodes in the standby cluster must be disabled.
- The Cisco SD-WAN Manager nodes in the standby cluster must be certified.
- The clock of every Cisco SD-WAN Manager node in the standby cluster must be synchronized with the clocks of the Cisco Catalyst SD-WAN controllers and WAN edge devices in the overlay network. If NTP is configured on the overlay, configure the same on the standby Cisco SD-WAN Manager nodes.
- The Cisco SD-WAN Manager nodes in the active and standby clusters should use identical neo4j credentials.

Restrictions

- Do not interrupt any active processes while backing up the configuration database.
- If you wish to enable SD-AVC, you must do so before the restoring the configuration database on standby Cisco SD-WAN Manager node.

Configure a Standby Cisco SD-WAN Manager Cluster

1. Configure the standby Cisco SD-WAN Manager nodes with a similar running configuration as the active Cisco SD-WAN Manager nodes. Install local certificates on the standby Cisco SD-WAN Manager nodes.



Note The running configuration on a standby Cisco SD-WAN Manager is usually identical to that of an active Cisco SD-WAN Manager node. However, you must ensure that settings such as the system IP address and the tunnel interface IP address are unique.

2. On the standby Cisco SD-WAN Manager nodes, shut down the transport interface in VPN 0: On the CLI, include the **shutdown** command in the transport interface configuration.
3. Create a standby cluster using the standby Cisco SD-WAN Manager nodes.

With the standby Cisco SD-WAN Manager nodes configured in this manner, the overlay network is not aware of the standby Cisco SD-WAN Manager cluster.

Back Up the Active Cisco SD-WAN Manager Cluster Configuration

Back up the full configuration database of the active Cisco vManage cluster periodically. Additionally, take snapshots of the active Cisco SD-WAN Manager virtual machines.

1. Choose an active Cisco SD-WAN Manager node that hosts the configuration database service and export a backup the configuration database. On the CLI of the Cisco SD-WAN Manager node, run the following command: **request nms configuration-db backup path file-path**

The command backs up the configuration database in a `.tar.gz` file and saves the file in the specified *file-path*.

In the following example, the database is backed up to a file named `db_backup.tar.gz` in the `/home/admin/` directory.

```
Active-vManage# request nms configuration-db backup path /home/admin/db_backup
Successfully saved database to /home/admin/db_backup.tar.gz
```

2. Choose a standby Cisco SD-WAN Manager node that hosts the configuration database service and copy the configuration database backup to this node.

In the following example, `db_backup.tar.gz` is copied from the active Cisco SD-WAN Manager node to the `/home/admin/` directory of a standby Cisco SD-WAN Manager node.

```
Active-vManage# request execute vpn 512 scp /home/admin/db_backup.tar.gz
admin@10.126.93.92:/home/admin
The authenticity of host '10.126.93.92 (10.126.93.92)' can't be established.
ECDSA key fingerprint is SHA256:jTjJWQ0UNHv1rBUxWzNjd8mUz819gPf51MeopsgDlAc.
Are you sure you want to continue connecting (yes/no)? yes
Warning: Permanently added '10.126.93.92' (ECDSA) to the list of known hosts.
viptela 18.4.5
```

```
admin@10.126.93.92's password:
db_backup.tar.gz          100% 399KB 4.4MB/s 00:00
```

Restore Cisco SD-WAN Manager Cluster Using the Configuration Database Backup

Restore the most recent backup of the configuration database from the active Cisco SD-WAN Manager cluster on the standby Cisco SD-WAN Manager node to which you copied this backup.



Note

- The restore operation does not restore all the information included in the configuration database. Cisco SD-WAN Manager configurations such as users and repositories must be configured on the standby Cisco SD-WAN Manager node after the configuration database is restored using the backup.
- When you complete the steps that follow, the previously active Cisco SD-WAN Manager nodes cannot be reused. To reuse the nodes, you must perform additional steps that are beyond the scope of this document.

1. On the CLI of the standby Cisco SD-WAN Manager node, run the following command: **request nms configuration-db restore path file-path**

In the following example, the configuration database is restored using the backup file `db_backup.tar.gz`.

```
Standby-vManage# request nms configuration-db restore path /home/admin/db_backup.tar.gz
Configuration database is running in a standalone mode
Importing database...Successfully restored database
```

2. Verify that the appropriate services are running on the standby Cisco SD-WAN Manager nodes: On the CLI of each standby Cisco SD-WAN Manager node, run the **request nms all status** command. From the command output, verify the services running on the node.
3. Verify that every standby Cisco SD-WAN Manager node has a list of all the active and standby Cisco SD-WAN Manager nodes.
 - a. From the Cisco SD-WAN Manager menu, choose **Configuration > Devices > Controllers**.



Note

Starting from Cisco IOS XE Catalyst SD-WAN Release 17.13.1a, the **Controllers** tab is renamed as the **Control Components** tab to stay consistent with Cisco Catalyst SD-WAN rebranding.

- b. Verify that the page displays all active and standby Cisco SD-WAN Manager nodes.
4. On the standby Cisco SD-WAN Manager nodes, enable the transport interface on VPN 0.

Use one of the following two methods:

- a. Enable the transport interface in VPN 0: On the CLI of each standby Cisco SD-WAN Manager node, run the **no shutdown** command.

```
Active-vManage# config
Active-vManage(config)# vpn 0 interface interface-name
Active-vManage(config-interface)# no shutdown
Active-vManage(config-interface)# commit and-quit
```

- b. Activate the tunnel interface in VPN 0: On the CLI of each standby Cisco SD-WAN Manager node, run the **tunnel-interface** command.

```
Active-vManage# config
Active-vManage(config)# vpn 0 interface interface-name
Active-vManage(config-interface)# tunnel-interface
Active-vManage(config-interface)# commit and-quit
```

5. Add each standby Cisco SD-WAN Manager node to the overlay network.
 - a. From the Cisco SD-WAN Manager menu, choose **Configuration > Devices**.
 - b. Click **Controllers**.



Note Starting from Cisco IOS XE Catalyst SD-WAN Release 17.13.1a, the **Controllers** tab is renamed as the **Control Components** tab to stay consistent with Cisco Catalyst SD-WAN rebranding.

- c. For a Cisco SD-WAN Validator, click ... and click **Edit**.
 - d. In the **Edit** dialog box, enter the following details of the Cisco SD-WAN Validator: WAN transport IP address, username, and password.
 - e. Repeat **Step 5c** and **Step 5d** for every Cisco SD-WAN Validator.

6. Disconnect the active Cisco SD-WAN Manager nodes from the overlay network.



Note In a lab environment, where you are simulating a disaster scenario, you can perform this step. However, if you cannot reach Cisco SD-WAN Manager instances in an actual disaster scenario, you may not be able to perform this step and can omit the step.

Use one of the following two methods:

- a. Shut down the transport interface in VPN 0: On the CLI of each active Cisco SD-WAN Manager node, run the **shutdown** command.

```
Active-vManage# config
Active-vManage(config)# vpn 0 interface interface-name
Active-vManage(config-interface)# shutdown
Active-vManage(config-interface)# commit and-quit
```

- b. Deactivate the tunnel interface in VPN 0: On the CLI of each active Cisco SD-WAN Manager node, run the **no tunnel-interface** command.

```
Active-vManage# config
Active-vManage(config)# vpn 0 interface interface-name
Active-vManage(config-interface)# no tunnel-interface
Active-vManage(config-interface)# commit and-quit
```

7. From the standby Cisco SD-WAN Manager, send the updated controller and device list to the Cisco SD-WAN Validator.

Send the list of controllers:

- a. From the Cisco SD-WAN Manager menu, choose **Configuration > Certificates**.

- b. Click **Controllers**.



Note Starting from Cisco IOS XE Catalyst SD-WAN Release 17.13.1a, the **Controllers** tab is renamed as the **Control Components** tab to stay consistent with Cisco Catalyst SD-WAN rebranding.

- c. Click **Send to Validator**.

Wait for the configuration task to complete. When the task is complete,

- The standby Cisco SD-WAN Manager nodes become the active Cisco SD-WAN Manager nodes.
- The previously active Cisco SD-WAN Manager nodes are no longer part of the overlay network.
- The active Cisco SD-WAN Manager nodes have the configuration from the most recent configuration database backup.
- Every controller establishes connection with the other controllers in the network.

- d. Click **WAN Edge List**.

- e. Click **Send to Controllers**.

8. Verify that the following are intact:

- Policies
- Templates
- Controller and WAN edge device lists

9. Verify the valid Cisco SD-WAN Manager nodes.

- a. Log in to the CLI of each Cisco SD-WAN Validator and run the **show orchestrator valid-vmanage-id** command.

In the command output, verify that the chassis numbers of the active and previously active Cisco SD-WAN Manager nodes are listed.

- b. Log in to the CLI of a WAN edge device and run the **show control valid-vmanage-id** command.

In the command output, verify that the chassis numbers of the active and previously active Cisco SD-WAN Manager nodes are listed. Also, check whether the device is connected to the active Cisco SD-WAN Manager nodes and the Cisco Catalyst SD-WAN Controller.

10. Invalidate the previously active Cisco SD-WAN Manager nodes.



Note After you invalidate the Cisco SD-WAN Manager nodes, the nodes cannot be reused without performing additional steps that are beyond the scope of this document.

- a. From the Cisco SD-WAN Manager menu, choose **Configuration > Certificates**.
- b. Click **Controllers**.



Note Starting from Cisco IOS XE Catalyst SD-WAN Release 17.13.1a, the **Controllers** tab is renamed as the **Control Components** tab to stay consistent with Cisco Catalyst SD-WAN rebranding.

c. For each previously active Cisco SD-WAN Manager node, click ... and click **Invalidate**.

11. Verify the valid Cisco SD-WAN Manager nodes.

a. Log in to the CLI of each Cisco SD-WAN Validator and run the **show orchestrator valid-vmanage-id** command.

In the command output, verify that the chassis numbers of only the active Cisco SD-WAN Manager nodes are listed.

b. Log in to the CLI of WAN edge device and run the **show control valid-vmanage-id** command.

In the command output, verify that the chassis numbers of only the active Cisco SD-WAN Manager nodes are listed. Also, check whether the device is connected to the active Cisco SD-WAN Manager nodes and the Cisco Catalyst SD-WAN Controller.

The Cisco SD-WAN Manager cluster that was initially the standby cluster is now the active Cisco SD-WAN Manager cluster.

Multitenant Cisco SD-WAN Manager: Disaster Recovery in an Overlay Network with Virtual Routers

If a Multitenant Cisco SD-WAN Manager cluster or the data center hosting the Cisco SD-WAN Manager nodes in the cluster fail, you can recover from the failure by activating a standby Cisco SD-WAN Manager cluster. You can perform disaster recovery as follows:

1. Deploy and configure a standby Cisco SD-WAN Manager cluster.

The standby Cisco SD-WAN Manager cluster is not part of the overlay network and is not active.

2. Back up the configuration database of the active Cisco SD-WAN Manager cluster periodically.

Choose a Cisco SD-WAN Manager node in the cluster that hosts the configuration database service and back up the configuration database.

3. If the active Cisco SD-WAN Manager cluster fails, restore the most recent configuration database on the standby Cisco SD-WAN Manager cluster, activate the standby Cisco SD-WAN Manager cluster, and remove the previously active Cisco SD-WAN Manager cluster from the overlay network.

Choose a Cisco SD-WAN Manager node in the cluster that hosts the configuration database service and restore the configuration database backed up from the previously active Cisco SD-WAN Manager cluster.

To test disaster recovery, you can simulate a scenario in which the active Cisco SD-WAN Manager cluster fails. One way to simulate such a failure would be by disabling the tunnel interface as described in this document.

The following disaster recovery procedure applies to an overlay network in which Cisco vEdge Cloud routers are deployed at branch locations.

Prerequisites

- The number of Cisco SD-WAN Manager nodes in the active and standby clusters must be identical.
- Each Cisco SD-WAN Manager node in the active and standby clusters must run the same Cisco SD-WAN Manager software release.
- Each Cisco SD-WAN Manager node in the active and standby clusters must be able to connect to the WAN transport IP address of the Cisco SD-WAN Validator in the overlay network.
- Initially, the tunnel interfaces of the Cisco SD-WAN Manager nodes in the standby cluster must be disabled.
- The Cisco SD-WAN Manager nodes in the standby cluster must be certified.
- The clock of every Cisco SD-WAN Manager node in the standby cluster must be synchronized with the clocks of the Cisco Catalyst SD-WAN controllers and WAN edge devices in the overlay network. If NTP is configured on the overlay, configure the same on the standby Cisco SD-WAN Manager nodes.
- The Cisco SD-WAN Manager nodes in the active and standby clusters should use identical neo4j credentials.

Restrictions

- Do not interrupt any active processes while backing up the configuration database.
- If you wish to enable SD-AVC, you must do so before the restoring the configuration database on standby Cisco SD-WAN Manager node.

Configure a Standby Cisco SD-WAN Manager Cluster

1. Configure the standby Cisco SD-WAN Manager nodes with a similar running configuration as the active Cisco SD-WAN Manager nodes. Install local certificates on the standby Cisco SD-WAN Manager nodes.



Note The running configuration on a standby Cisco SD-WAN Manager is usually identical to that of an active Cisco SD-WAN Manager node. However, you must ensure that settings such as the system IP address and the tunnel interface IP address are unique.

2. On the standby Cisco SD-WAN Manager nodes, shut down the transport interface in VPN 0: On the CLI, include the **shutdown** command in the transport interface configuration.
3. Create a standby cluster using the standby Cisco SD-WAN Manager nodes.

With the standby Cisco SD-WAN Manager nodes configured in this manner, the overlay network is not aware of the standby Cisco SD-WAN Manager cluster.

Back Up the Active Cisco SD-WAN Manager Cluster Configuration

Back up the full configuration database of the active Cisco vManage cluster periodically. Additionally, take snapshots of the active Cisco SD-WAN Manager virtual machines.

1. Choose an active Cisco SD-WAN Manager node that hosts the configuration database service and export a backup the configuration database. On the CLI of the Cisco SD-WAN Manager node, run the following command: **request nms configuration-db backup path file-path**

The command backs up the configuration database in a `.tar.gz` file and saves the file in the specified *file-path*.

In the following example, the database is backed up to a file named `db_backup.tar.gz` in the `/home/admin/` directory.

```
Active-vManage# request nms configuration-db backup path /home/admin/db_backup
Successfully saved database to /home/admin/db_backup.tar.gz
```

2. Choose a standby Cisco SD-WAN Manager node that hosts the configuration database service and copy the configuration database backup to this node.

In the following example, `db_backup.tar.gz` is copied from the active Cisco SD-WAN Manager node to the `/home/admin/` directory of a standby Cisco SD-WAN Manager node.

```
Active-vManage# request execute vpn 512 scp /home/admin/db_backup.tar.gz
admin@10.126.93.92:/home/admin
The authenticity of host '10.126.93.92 (10.126.93.92)' can't be established.
ECDSA key fingerprint is SHA256:jTjJWQ0UNHv1rBUxWzNjd8mUz819gPf51MeopsgDlAc.
Are you sure you want to continue connecting (yes/no)? yes
Warning: Permanently added '10.126.93.92' (ECDSA) to the list of known hosts.
viptela 18.4.5

admin@10.126.93.92's password:
db_backup.tar.gz                                100% 399KB 4.4MB/s 00:00
```

Restore Cisco SD-WAN Manager Cluster Using the Configuration Database Backup

Restore the most recent backup of the configuration database from the active Cisco SD-WAN Manager cluster on the standby Cisco SD-WAN Manager node to which you copied this backup.



Note

- The restore operation does not restore all the information included in the configuration database. Cisco SD-WAN Manager configurations such as users and repositories must be configured on the standby Cisco SD-WAN Manager node after the configuration database is restored using the backup.
- When you complete the steps that follow, the previously active Cisco SD-WAN Manager nodes cannot be reused. To reuse the nodes, you must perform additional steps that are beyond the scope of this document.

1. On the CLI of the standby Cisco SD-WAN Manager node, run the following command: **request nms configuration-db restore path file-path**

In the following example, the configuration database is restored using the backup file `db_backup.tar.gz`.

```
Standby-vManage# request nms configuration-db restore path /home/admin/db_backup.tar.gz
Configuration database is running in a standalone mode
Importing database...Successfully restored database
```

2. Verify that the appropriate services are running on the standby Cisco SD-WAN Manager nodes: On the CLI of each standby Cisco SD-WAN Manager node, run the **request nms all status** command. From the command output, verify the services running on the node.

3. Verify that every standby Cisco SD-WAN Manager node has a list of all the active and standby Cisco SD-WAN Manager nodes.
 - a. From the Cisco SD-WAN Manager menu, choose **Configuration > Devices > Controllers**.



Note Starting from Cisco IOS XE Catalyst SD-WAN Release 17.13.1a, the **Controllers** tab is renamed as the **Control Components** tab to stay consistent with Cisco Catalyst SD-WAN rebranding.

- b. Verify that the page displays all active and standby Cisco SD-WAN Manager nodes.
4. Log in to the CLI of each Cisco SD-WAN Validator and run the **show orchestrator valid-vmanage-id** command.

In the command output, verify that the chassis numbers of the active and previously active Cisco SD-WAN Manager nodes are listed.

5. Log in to the CLI of Cisco vEdge Cloud router and run the **show control valid-vmanage-id** command.
In the command output, verify that the chassis numbers of the active and previously active Cisco SD-WAN Manager nodes are listed.

6. On the standby Cisco SD-WAN Manager nodes, enable the transport interface on VPN 0.

Use one of the following two methods:

- a. Enable the transport interface in VPN 0: On the CLI of each standby Cisco SD-WAN Manager node, run the **no shutdown** command.

```
Active-vManage# config
Active-vManage(config)# vpn 0 interface interface-name
Active-vManage(config-interface)# no shutdown
Active-vManage(config-interface)# commit and-quit
```

- b. Activate the tunnel interface in VPN 0: On the CLI of each standby Cisco SD-WAN Manager node, run the **tunnel-interface** command.

```
Active-vManage# config
Active-vManage(config)# vpn 0 interface interface-name
Active-vManage(config-interface)# tunnel-interface
Active-vManage(config-interface)# commit and-quit
```

7. Add each standby Cisco SD-WAN Manager node to the overlay network.
 - a. From the Cisco SD-WAN Manager menu, choose **Configuration > Devices**.
 - b. Click **Controllers**.



Note Starting from Cisco IOS XE Catalyst SD-WAN Release 17.13.1a, the **Controllers** tab is renamed as the **Control Components** tab to stay consistent with Cisco Catalyst SD-WAN rebranding.

- c. For a Cisco SD-WAN Validator, click ... and click **Edit**.
 - d. In the **Edit** dialog box, enter the following details of the Cisco SD-WAN Validator: WAN transport IP address, username, and password.

- e. Repeat **Step 7c** and **Step 7d** for every Cisco SD-WAN Validator.

8. Disconnect the active Cisco SD-WAN Manager nodes from the overlay network.



Note In a lab environment, where you are simulating a disaster scenario, you can perform this step. However, if you cannot reach Cisco SD-WAN Manager instances in an actual disaster scenario, you may not be able to perform this step and can omit this step.

Use one of the following two methods:

- a. Shut down the transport interface in VPN 0: On the CLI of each active Cisco SD-WAN Manager node, run the **shutdown** command.

```
Active-vManage# config
Active-vManage(config)# vpn 0 interface interface-name
Active-vManage(config-interface)# shutdown
Active-vManage(config-interface)# commit and-quit
```

- b. Deactivate the tunnel interface in VPN 0: On the CLI of each active Cisco SD-WAN Manager node, run the **no tunnel-interface** command.

```
Active-vManage# config
Active-vManage(config)# vpn 0 interface interface-name
Active-vManage(config-interface)# no tunnel-interface
Active-vManage(config-interface)# commit and-quit
```

9. From the standby Cisco SD-WAN Manager, send the updated controller and device list to the Cisco SD-WAN Validator.

Send the list of controllers:

- a. From the Cisco SD-WAN Manager menu, choose **Configuration > Certificates**.
- b. Click **Controllers**.



Note Starting from Cisco IOS XE Catalyst SD-WAN Release 17.13.1a, the **Controllers** tab is renamed as the **Control Components** tab to stay consistent with Cisco Catalyst SD-WAN rebranding.

- c. Click **Send to Validator**.

Wait for the configuration task to complete. When the task is complete,

- The standby Cisco SD-WAN Manager nodes become the active Cisco SD-WAN Manager nodes.
- The previously active Cisco SD-WAN Manager nodes are no longer part of the overlay network.
- The active Cisco SD-WAN Manager nodes have the configuration from the most recent configuration database backup.
- Every controller establishes connection with the other controllers in the network.

- d. Click **WAN Edge List**.

- e. Click **Send to Controllers**.
10. Verify that the following are intact:
 - Policies
 - Templates
 - Controller and WAN edge device lists
 11. Verify the valid Cisco SD-WAN Manager nodes.
 - a. Log in to the CLI of each Cisco SD-WAN Validator and run the **show orchestrator valid-vmanage-id** command.
 In the command output, verify that the chassis numbers of the active and previously active Cisco SD-WAN Manager nodes are listed.
 - b. Log in to the CLI of a Cisco vEdge Cloud router and run the **show control valid-vmanage-id** command.
 In the command output, verify that the chassis numbers of the active and previously active Cisco SD-WAN Manager nodes are listed. Also, check whether the device is connected to the active Cisco SD-WAN Manager nodes and the Cisco Catalyst SD-WAN Controller.
 12. Invalidate the previously active Cisco SD-WAN Manager nodes.
 The previously active Cisco SD-WAN Manager is the certificate issuer for the cloud WAN edge devices. The active Cisco SD-WAN Manager issues certificates to the cloud WAN edge devices only after the previously active Cisco SD-WAN Manager nodes are invalidated.

**Note**

- After you invalidate the Cisco SD-WAN Manager nodes, the nodes cannot be reused without performing additional steps that are beyond the scope of this document.
- When you invalidate the previously active Cisco SD-WAN Manager nodes, Cisco SD-WAN Manager marks the nodes as invalid and sends an update to all controllers. However, Cisco SD-WAN Manager does not send an updated list of valid Cisco SD-WAN Manager UUIDs to Cisco SD-WAN Validator immediately because the previously active Cisco SD-WAN Manager is the CA for the cloud WAN edge devices. So, the output of the **show orchestrator valid-vmanage-id** command on a Cisco SD-WAN Validator includes the UUIDs of the invalidated Cisco SD-WAN Manager nodes.

Cisco SD-WAN Manager has a scheduled task that runs every 24 hours and checks to see if all the cloud WAN edges have been moved to the active Cisco SD-WAN Manager. Cisco SD-WAN Manager sends the updated list of valid Cisco SD-WAN Manager UUIDs to Cisco SD-WAN Validator only after the cloud WAN edge devices have been moved to the active Cisco SD-WAN Manager. After this list is received, the output of the **show orchestrator valid-vmanage-id** command on a Cisco SD-WAN Validator does not include the UUIDs of the invalidated Cisco SD-WAN Manager nodes.

- a. From the Cisco SD-WAN Manager menu, choose **Configuration > Certificates**.
- b. Click **Controllers**.



Note Starting from Cisco IOS XE Catalyst SD-WAN Release 17.13.1a, the **Controllers** tab is renamed as the **Control Components** tab to stay consistent with Cisco Catalyst SD-WAN rebranding.

c. For each previously active Cisco SD-WAN Manager node, click ... and click **Invalidate**.

13. Verify the valid Cisco SD-WAN Manager nodes after 24 hours.

a. Log in to the CLI of each Cisco SD-WAN Validator and run the **show orchestrator valid-vmanage-id** command.

In the command output, verify that the chassis numbers of only the active Cisco SD-WAN Manager nodes are listed.

b. Log in to the CLI of WAN edge device and run the **show control valid-vmanage-id** command.

In the command output, verify that the chassis numbers of only the active Cisco SD-WAN Manager nodes are listed. Also, check whether the device is connected to the active Cisco SD-WAN Manager nodes and the Cisco Catalyst SD-WAN Controllers.

The Cisco SD-WAN Manager cluster that was initially the standby cluster is now the active Cisco SD-WAN Manager cluster.

Multitenant Cisco SD-WAN Manager: Disaster Recovery After a Failed Data Center Becomes Operational

If a Multitenant Cisco SD-WAN Manager cluster or the data center hosting the Cisco SD-WAN Manager nodes in the cluster fail, you can recover from the failure by activating a standby Cisco SD-WAN Manager cluster. You can perform disaster recovery as follows:

1. Deploy and configure a standby Cisco SD-WAN Manager cluster.

The standby Cisco SD-WAN Manager cluster is not part of the overlay network and is not active.

2. Back up the configuration database of the active Cisco SD-WAN Manager cluster periodically.

Choose a Cisco SD-WAN Manager node in the cluster that hosts the configuration database service and back up the configuration database.

3. If the active Cisco SD-WAN Manager cluster fails, restore the most recent configuration database on the standby Cisco SD-WAN Manager cluster, activate the standby Cisco SD-WAN Manager cluster, and remove the previously active Cisco SD-WAN Manager cluster from the overlay network.

Choose a Cisco SD-WAN Manager node in the cluster that hosts the configuration database service and restore the configuration database backed up from the previously active Cisco SD-WAN Manager cluster.

To test disaster recovery, you can simulate a scenario in which the active Cisco SD-WAN Manager cluster fails. One way to simulate such a failure would be by disabling the tunnel interface as described in this document.

The following procedure applies to a scenario in which an initially active Cisco SD-WAN Manager cluster or the data center hosting the cluster failed and the standby Cisco SD-WAN Manager cluster was configured

to be the active Cisco SD-WAN Manager cluster. If the cluster that was initially active becomes operational again, it serves as a standby cluster. By completing the following procedure, you can turn this standby cluster into the active cluster.

Check the Configuration of the Standby Cisco SD-WAN Manager

1. Check whether the running configuration of the standby Cisco SD-WAN Manager nodes is similar to the running configuration of the active Cisco SD-WAN Manager nodes. Local certificates must be installed on the standby Cisco SD-WAN Manager nodes.



Note The running configuration on a standby Cisco SD-WAN Manager is usually identical to that of an active Cisco SD-WAN Manager node. However, you must ensure that settings such as the system IP address and the tunnel interface IP address are unique.

2. On the standby Cisco SD-WAN Manager nodes, shut down the transport interface in VPN 0: On the CLI, include the **shutdown** command in the transport interface configuration.

With the standby Cisco SD-WAN Manager nodes configured in this manner, the overlay network is not aware of the standby Cisco SD-WAN Manager cluster.

Back Up the Active Cisco SD-WAN Manager Cluster Configuration

Back up the full configuration database of the active Cisco vManage cluster periodically. Additionally, take snapshots of the active Cisco SD-WAN Manager virtual machines.

1. Choose an active Cisco SD-WAN Manager node that hosts the configuration database service and export a backup the configuration database. On the CLI of the Cisco SD-WAN Manager node, run the following command: **request nms configuration-db backup path file-path**

The command backs up the configuration database in a `.tar.gz` file and saves the file in the specified *file-path*.

In the following example, the database is backed up to a file named `db_backup.tar.gz` in the `/home/admin/` directory.

```
Active-vManage# request nms configuration-db backup path /home/admin/db_backup
Successfully saved database to /home/admin/db_backup.tar.gz
```

2. Choose a standby Cisco SD-WAN Manager node that hosts the configuration database service and copy the configuration database backup to this node.

In the following example, `db_backup.tar.gz` is copied from the active Cisco SD-WAN Manager node to the `/home/admin/` directory of a standby Cisco SD-WAN Manager node.

```
Active-vManage# request execute vpn 512 scp /home/admin/db_backup.tar.gz
admin@10.126.93.92:/home/admin
The authenticity of host '10.126.93.92 (10.126.93.92)' can't be established.
ECDSA key fingerprint is SHA256:jTjJWQ0UNHv1rBUxWzNjd8mUz819gPf5lMeopsgDlAc.
Are you sure you want to continue connecting (yes/no)? yes
Warning: Permanently added '10.126.93.92' (ECDSA) to the list of known hosts.
viptela 18.4.5

admin@10.126.93.92's password:
db_backup.tar.gz                                100% 399KB 4.4MB/s 00:00
```

Restore Cisco SD-WAN Manager Cluster Using the Configuration Database Backup

Restore the most recent backup of the configuration database from the active Cisco SD-WAN Manager cluster on the standby Cisco SD-WAN Manager node to which you copied this backup.



Note

- The restore operation does not restore all the information included in the configuration database. Cisco SD-WAN Manager configurations such as users and repositories must be configured on the standby Cisco SD-WAN Manager node after the configuration database is restored using the backup.
- When you complete the steps that follow, the previously active Cisco SD-WAN Manager nodes cannot be reused. To reuse the nodes, you must perform additional steps that are beyond the scope of this document.

1. On the CLI of the standby Cisco SD-WAN Manager node, run the following command: **request nms configuration-db restore path *file-path***

In the following example, the configuration database is restored using the backup file `db_backup.tar.gz`.

```
Standby-vManage# request nms configuration-db restore path /home/admin/db_backup.tar.gz
Configuration database is running in a standalone mode
Importing database...Successfully restored database
```

2. Verify that the appropriate services are running on the standby Cisco SD-WAN Manager nodes: On the CLI of each standby Cisco SD-WAN Manager node, run the **request nms all status** command. From the command output, verify the services running on the node.
3. Verify that every standby Cisco SD-WAN Manager node has a list of all the active and standby Cisco SD-WAN Manager nodes.
 - a. From the Cisco SD-WAN Manager menu, choose **Configuration > Devices > Controllers**.



Note

Starting from Cisco IOS XE Catalyst SD-WAN Release 17.13.1a, the **Controllers** tab is renamed as the **Control Components** tab to stay consistent with Cisco Catalyst SD-WAN rebranding.

- b. Verify that the page displays all active and standby Cisco SD-WAN Manager nodes.
4. On the standby Cisco SD-WAN Manager nodes, enable the transport interface on VPN 0. Use one of the following two methods:
 - a. Enable the transport interface in VPN 0: On the CLI of each standby Cisco SD-WAN Manager node, run the **no shutdown** command.

```
Active-vManage# config
Active-vManage(config)# vpn 0 interface interface-name
Active-vManage(config-interface)# no shutdown
Active-vManage(config-interface)# commit and-quit
```

- b. Activate the tunnel interface in VPN 0: On the CLI of each standby Cisco SD-WAN Manager node, run the **tunnel-interface** command.

```
Active-vManage# config
Active-vManage(config)# vpn 0 interface interface-name
```

```
Active-vManage(config-interface) # tunnel-interface
Active-vManage(config-interface) # commit and-quit
```

5. Add each standby Cisco SD-WAN Manager node to the overlay network.
 - a. From the Cisco SD-WAN Manager menu, choose **Configuration > Devices**.
 - b. Click **Controllers**.



Note Starting from Cisco IOS XE Catalyst SD-WAN Release 17.13.1a, the **Controllers** tab is renamed as the **Control Components** tab to stay consistent with Cisco Catalyst SD-WAN rebranding.

- c. For a Cisco SD-WAN Validator, click ... and click **Edit**.
 - d. In the **Edit** dialog box, enter the following details of the Cisco SD-WAN Validator: WAN transport IP address, username, and password.
 - e. Repeat **Step 5c** and **Step 5d** for every Cisco SD-WAN Validator.
6. Disconnect the active Cisco SD-WAN Manager nodes from the overlay network.



Note In a lab environment, where you are simulating a disaster scenario, you can perform this step. However, if you cannot reach Cisco SD-WAN Manager instances in an actual disaster scenario, you may not be able to perform this step and can omit the step.

Use one of the following two methods:

- a. Shut down the transport interface in VPN 0: On the CLI of each active Cisco SD-WAN Manager node, run the **shutdown** command.

```
Active-vManage# config
Active-vManage(config) # vpn 0 interface interface-name
Active-vManage(config-interface) # shutdown
Active-vManage(config-interface) # commit and-quit
```

- b. Deactivate the tunnel interface in VPN 0: On the CLI of each active Cisco SD-WAN Manager node, run the **no tunnel-interface** command.

```
Active-vManage# config
Active-vManage(config) # vpn 0 interface interface-name
Active-vManage(config-interface) # no tunnel-interface
Active-vManage(config-interface) # commit and-quit
```

7. From the standby Cisco SD-WAN Manager, send the updated controller and device list to the Cisco SD-WAN Validator.

Send the list of controllers:

- a. From the Cisco SD-WAN Manager menu, choose **Configuration > Certificates**.
- b. Click **Controllers**.



Note Starting from Cisco IOS XE Catalyst SD-WAN Release 17.13.1a, the **Controllers** tab is renamed as the **Control Components** tab to stay consistent with Cisco Catalyst SD-WAN rebranding.

c. Click **Send to Validator**.

Wait for the configuration task to complete. When the task is complete,

- The standby Cisco SD-WAN Manager nodes become the active Cisco SD-WAN Manager nodes.
- The previously active Cisco SD-WAN Manager nodes are no longer part of the overlay network.
- The active Cisco SD-WAN Manager nodes have the configuration from the most recent configuration database backup.
- Every controller establishes connection with the other controllers in the network.

d. Click **WAN Edge List**.

e. Click **Send to Controllers**.

8. Verify that the following are intact:

- Policies
- Templates
- Controller and WAN edge device lists

9. Verify the valid Cisco SD-WAN Manager nodes.

a. Log in to the CLI of each Cisco SD-WAN Validator and run the **show orchestrator valid-vmanage-id** command.

In the command output, verify that the chassis numbers of the active and previously active Cisco SD-WAN Manager nodes are listed.

b. Log in to the CLI of a WAN edge device and run the **show control valid-vmanage-id** command.

In the command output, verify that the chassis numbers of the active and previously active Cisco SD-WAN Manager nodes are listed. Also, check whether the device is connected to the active Cisco SD-WAN Manager nodes and the Cisco Catalyst SD-WAN Controller.

10. Invalidate the previously active Cisco SD-WAN Manager nodes.



Note After you invalidate the Cisco SD-WAN Manager nodes, the nodes cannot be reused without performing additional steps that are beyond the scope of this document.

a. From the Cisco SD-WAN Manager menu, choose **Configuration > Certificates**.

b. Click **Controllers**.



Note Starting from Cisco IOS XE Catalyst SD-WAN Release 17.13.1a, the **Controllers** tab is renamed as the **Control Components** tab to stay consistent with Cisco Catalyst SD-WAN rebranding.

c. For each previously active Cisco SD-WAN Manager node, click ... and click **Invalidate**.

11. Verify the valid Cisco SD-WAN Manager nodes.

a. Log in to the CLI of each Cisco SD-WAN Validator and run the **show orchestrator valid-vmanage-id** command.

In the command output, verify that the chassis numbers of only the active Cisco SD-WAN Manager nodes are listed.

b. Log in to the CLI of WAN edge device and run the **show control valid-vmanage-id** command.

In the command output, verify that the chassis numbers of only the active Cisco SD-WAN Manager nodes are listed. Also, check whether the device is connected to the active Cisco SD-WAN Manager nodes and the Cisco Catalyst SD-WAN Controllers.

The Cisco SD-WAN Manager cluster that was initially the standby cluster is now the active Cisco SD-WAN Manager cluster.

Replace Faulty Cisco SD-WAN Controller

To replace a faulty Cisco SD-WAN Controller with a new instance, follow these steps:

1. Create a Cisco SD-WAN Controller instance. See [Deploy the Cisco SD-WAN Controller](#).
2. [Add the Cisco SD-WAN Controller](#) to the overlay network.
3. From the Cisco SD-WAN Manager menu, choose **Configuration > Devices**.
4. Click **Controllers**.



Note Starting from Cisco IOS XE Catalyst SD-WAN Release 17.13.1a, the **Controllers** tab is renamed as the **Control Components** tab to stay consistent with Cisco Catalyst SD-WAN rebranding.

5. For the faulty Cisco SD-WAN Controllers, click ... and click **Invalidate**.

The **Invalidate** dialog box appears.



Note If you have not added a new Cisco SD-WAN Controller that can replace the faulty Cisco SD-WAN Controller, Cisco SD-WAN Manager indicates this through an error message. Click **Cancel** in the **Invalidate** dialog box and add a new Cisco SD-WAN Controller before invalidating the faulty instance.

6. In the **Invalidate** dialog box, do the following:

- a. Check the **Replace Controller** check box.
- b. From the **Select Controller** drop-down list, choose the new Cisco SD-WAN Controller that should replace the faulty instance.
- c. Click **Invalidate**.

Cisco SD-WAN Manager launches the **Invalidate Device** and **Push CLI Template Configuration** task. When these tasks are completed, the faulty Cisco SD-WAN Controller is invalidated and removed from the overlay network. The tenants that were served by the faulty Cisco SD-WAN Controller are now served by the new Cisco SD-WAN Controller that you chose as the replacement.

RADIUS and TACACS Support for Multitenancy

Information about RADIUS and TACACS Support for Multitenancy

From Cisco IOS XE Catalyst SD-WAN Release 17.12.1a, Cisco SD-WAN Manager supports for RADIUS and TACACS servers in a multitenant deployment.

RADIUS

RADIUS is a distributed client and server system that secures networks that have authorized access. RADIUS clients run on supported Cisco devices and send authentication requests to a central RADIUS server, which contains all user authentication and network service access information.

TACACS

TACACS is a security application that provides centralized validation of users attempting to gain access to an access point. Unlike RADIUS, TACACS does not authenticate wireless client devices accessing the network through an access point.

TACACS provides for separate and modular authentication, authorization, and accounting. Each service can be tied into its own database to take advantage of other services available on that server or on the network.

TACACS administered through the AAA security services can provide these services:

- **Authentication:** Provides complete control of authentication through login and password dialog, challenge and response, and messaging support.
- **Authorization:** Provides fine-grained control of user privileges for the duration of the session, including access control, session duration, or protocol support. You can also enforce restrictions on the commands to execute a TACACS authorization feature.
- **Accounting:** Collects and sends information used for billing, auditing, and reporting to the TACACS daemon. Network managers can use the accounting feature to track administrator activity for security audit or to provide information for user billing. Accounting records include administrator identities, start and stop times, executed commands, number of packets, and number of bytes.

**Note**

NAT is not supported between Cisco SD-WAN Manager and the multitenant connector.

Prerequisites for Cloud Multitenant with On-Prem Per-tenant AAA and Provider AAA

Prerequisites for Cloud Multitenant with On-Prem Per-tenant AAA and Provider AAA

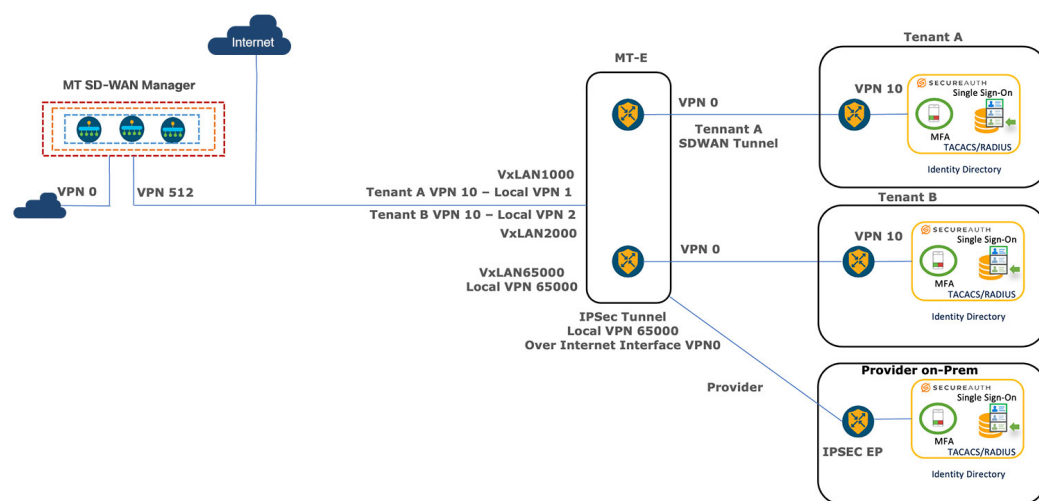
- A Multitenant edge connector is onboarded in Cisco SD-WAN Manager.
- Provider can have tenant configurations only through a device or feature template.
- The edge connector is on the same premises as the controllers.
- Cisco SD-WAN Manager is configured with VPN 512 interface.
- VxLAN tunnels must use the VPN 512 interface as the underlay.
- In a Cisco SD-WAN Manager cluster, there is a VxLAN tunnel created between each Cisco SD-WAN Manager node and the edge connector.
- A provider's RADIUS and TACACS server cannot be shared with the tenant.
- RADIUS and TACACS server authentication is within the tenant network.
- Multiple RADIUS and TACACS servers are used for the same tenant.
- A tenant's RADIUS and TACACS server is on-prem or cloud-hosted.



Note You must configure an external AAA server and provide mapping between the user and the Viptela groups to authentication. For example, Viptela-Group-Name as basic, tenantadmin, or operator.

The following illustration shows the architecture of the cloud multitenancy with on-prem per tenant and provider AAA.

Figure 1: Cloud Multitenant with On-Prem Per-tenant and Provider AAA



Workflows to Configure Remote AAA

Enable Multitenancy

1. From the Cisco SD-WAN Manager menu, choose **Administration** > **Settings**.
2. Click **Edit** adjacent to the **Tenancy Mode Tenancy Mode**. If you are using Cisco Catalyst SD-WAN Manager Release 20.12.x or earlier, click **Edit**.
3. Click **Multitenant**.
4. In the **Domain** field, enter the domain name of the service provider (for example, managed-sp.com).
5. Enter a **Cluster Id** (for example, cluster-1 or 123456).
6. Click **Save**. If you are using Cisco Catalyst SD-WAN Manager Release 20.12.x or earlier, click **Proceed** to confirm that you want to change the tenancy mode.

Cisco SD-WAN Manager reboots in multitenant mode and when a provider user logs in to Cisco SD-WAN Manager, the provider dashboard appears.

For information about configuring AAA using feature templates for a single tenant, see [Configuring AAA using Cisco SD-WAN Manager Template](#).

Configure the Tenant

To on board the Edge Connector in a Cisco SD-WAN Manager provider, perform the following steps:

1. From the Cisco SD-WAN Manager menu, choose **Administration** > **Tenant Management**.
2. Click **Edit** adjacent to the **Tenant**.
The **Edit Tenant** window is displayed.
3. Enter the description in the **Description** field.
4. Enter the edge number in the **Forecasted Edge** field.
5. Enter the sub-domain URL in the **URL Subdomain** field.
6. Enable the **Edge Connector** option.
7. Choose the **Edge Connector IP** from the drop-down list.
8. Choose the VXLAN tunnel endpoint from the **Edge Connector VTEP Interface Name** drop-down list.
For a cloud deployment, it is essential to provision the Cloud-hosted SD-WAN Control Components and gateway with the third interface **GigabitEthernet3** for AAA. When selecting the VXLAN tunnel endpoint from the **Edge Connector VTEP Interface Name** drop-down list, choose **GigabitEthernet3**. Selecting any other interface, such as **VPN 0** or **VPN 512**, might result in misconfiguration.
9. Click **Save**.

Configure Remote AAA

Cisco SD-WAN Manager reboots in multitenant mode and when a provider user logs in to Cisco SD-WAN Manager, the provider dashboard appears.

1. From the Cisco SD-WAN Manager menu, choose **Administration > Manage Users**.
2. Click **Remote AAA**.
3. Expand the **Remote AAA** tab to configure remote AAA.
4. Enter the order in which to attempt different authentication methods in the **Authentication Order** field.
5. Choose the option in **Authentication Fallback** to fallback if higher-priority authentication fails.
6. Choose the **Admin Authentication Order** to authenticate a tenantadmin user according to the authentication order.
7. Enable or disable audit logs in the **Disable Audit Logs** field.
8. Enable or disable user accounting in the **Enable/disable user accounting** field.
9. Click **Save** to save the changes.

Configure RADIUS

1. From the Cisco SD-WAN Manager menu, choose **Administration > Manage Users**.
2. Click **Remote AAA**.
3. Expand the **RADIUS** tab to configure a RADIUS server.
4. Enter the duration to wait for replies from the RADIUS server in the **Timeout** field.
5. Enter the number of times you want to contact a RADIUS server in the **Retransmit Count** field.
6. Click **New RADIUS Server** to add a new RADIUS server.

Field	Description
Radius Server Details	
Address	Enter the IP address of the RADIUS server.
Authentication Port	Enter the port to connect to the RADIUS server.
Accounting Port	Enter the port used to connect to the server.
Key	Enter the password to access the RADIUS server.
Secret Key	Enter the AES encrypted key to access the RADIUS server.
Priority	Enter the server priority.
VPN ID	Enter the VPN in which the RADIUS server. Note This option is available in Cisco Catalyst SD-WAN Manager Release 20.15.x and earlier.

Field	Description
VPN IP Subnet	Enter the VPN IP subnet (VXLAN tunnel VPN IP subnet) in which the RADIUS server is located. Note This option is available in Cisco Catalyst SD-WAN Manager Release 20.15.x and earlier.
Network Connectivity Note This Network Connectivity option is available from Cisco Catalyst SD-WAN Manager Release 20.16.1.	
Source VPN	Choose from these VPN options, the location of the RADIUS server: • 0: Transport VPN. • 512: Management VPN. Configure these fields only if you have specified the tenant to manage the AAA server. These fields do not appear if the multitenant setup is configured to be managed by the provider. • (Optional) Source Subnet: Enter the VPN IP subnet that is unique within the tenant network. • (Optional) Destination VPN: Enter the VPN in which the RADIUS server is located.

7. Click **Save** and **Add**.

Configure TACACS

1. From the Cisco SD-WAN Manager menu, choose **Administration > Manage Users**.
2. Click **Remote AAA**.
3. Expand the **TACACS** tab to configure TACACS.
4. Enter the duration to wait for replies from the TACACS server in the **Timeout** field.
5. Choose the TACACS authentication type from the **Authentication** drop-down list.
 - ASCII
 - PAP
6. Click **New TACACS Server** to add a new TACACS server.

Field	Description
TACACS Server Details	
Address	Enter the IP address of the TACACS server.

Field	Description
Authentication Port	Enter the port to connect to a TACACS server.
Key	Enter the password to access the TACACS server.
Secret Key	Enter the AES encrypted key to access the TACACS server.
Priority	Enter the TACACS server priority.
VPN ID	Enter the VPN in which the TACACS server. Note This option is available in Cisco Catalyst SD-WAN Manager Release 20.15.x and earlier.
VPN IP Subnet	Enter the VPN IP subnet in which the TACACS server is located. Note This option is available in Cisco Catalyst SD-WAN Manager Release 20.15.x and earlier.
Network Connectivity Note The Network Connectivity option is available from Cisco Catalyst SD-WAN Manager Release 20.16.1.	
Source VPN	Choose from these VPN options, the location of the TACACS server: • 0: Transport VPN. • 512: Management VPN. Configure these fields only if you have specified the tenant to manage the AAA server. These fields do not appear if the multitenant setup is configured to be managed by the provider. • (Optional) Source Subnet: Enter the VPN IP subnet that is unique within the tenant network. • (Optional) Destination VPN: Enter the VPN in which the TACACS server is located.

7. Click **Add**.

Verify RADIUS and TACACS Configuration for Multitenancy

The following is a RADIUS and TACACS configuration example on Cisco IOS XE Catalyst SD-WAN devices through CLI:

```
Device# interface GigabitEthernet4
description VTEP Interface
```



```
no shutdown

arp timeout 1200

ip address 172.1.1.101 255.255.255.0

no ip redirects

ip mtu      1500

load-interval 30

mtu          1500

negotiation auto

exit
```

Use the **show ip interface brief** command to show the AAA configuration:

```
Device# show ip interface brief | i <VPN IP SUBNET of VxTunnel>
```

Where <VPN IP SUBNET of VXTunnel> is one that is configured under Tenant - > Administration
-> Remote AAA

The output shows one tunnel per one node. If there are three nodes in a cluster, the output displays three tunnels in the subnet.

Configure ISE

Before you begin

Choose who manages ISE, provider or tenant.

1. From the SD-WAN Manager menu, choose **Administration > Tenant Management**.
2. Go to the tenant and from the **Identity Service Engine** dropdown, choose one of these:
 - For Provider-managed services, choose **Yes**.
 - For Tenant-managed services, choose **No**.

Procedure

-
- | | |
|---------------|--|
| Step 1 | Log in to Cisco SD-WAN Manager as the provider or tenant admin user. |
| Step 2 | Follow the steps in Configure Cisco ISE in Cisco SD-WAN Manager for Multitenancy . |
-



CHAPTER 16

Flexible Tenant Placement on Multitenant Cisco Catalyst SD-WAN Controllers

Table 214: Feature History

Feature Name	Release Information	Description
Flexible Tenant Placement on Multitenant Cisco Catalyst SD-WAN Controllers	Cisco vManage Release 20.9.1	With this feature, while onboarding a tenant to a multitenant deployment, you can choose the pair of multitenant Cisco SD-WAN Controllers that serve the tenant. After onboarding a tenant, you can migrate the tenant to a different pair of multitenant Cisco SD-WAN Controller, if necessary.

- [Information About Flexible Tenant Placement on Multitenant Cisco SD-WAN Controllers, on page 571](#)
- [Restrictions for Flexible Tenant Placement on Multitenant Cisco SD-WAN Controllers, on page 573](#)
- [Assign Cisco SD-WAN Controllers to Tenants During Onboarding, on page 573](#)
- [Update Cisco SD-WAN Controllers Placement For a Tenant, on page 578](#)

Information About Flexible Tenant Placement on Multitenant Cisco SD-WAN Controllers

Automatic Tenant Placement by Cisco SD-WAN Manager

In Cisco vManage Release 20.8.x and earlier releases, when you onboard a tenant, Cisco SD-WAN Manager assigns a pair of multitenant Cisco SD-WAN Controllers to the tenant based on an internal algorithm that considers factors such as the following:

- number of tenant WAN edge devices that you forecast for the tenant
- number of tenants served by a pair of multitenant Cisco SD-WAN Controllers
- number of WAN edge devices connected to a pair of multitenant Cisco SD-WAN Controllers

After the tenant is onboarded, if the tenant needs to add more devices than you originally forecast, you can modify the forecast if the pair of multitenant Cisco SD-WAN Controllers serving the tenant can accommodate these additional WAN edge devices. If the Cisco SD-WAN Controllers cannot accommodate the additional WAN edge devices, you must delete the tenant and onboard the tenant again with the revised device forecast so that Cisco SD-WAN Manager assigns a suitable pair of Cisco SD-WAN Controllers. If none of the pairs of multitenant Cisco SD-WAN Controllers can accommodate the revised device forecast, add a new pair of Cisco SD-WAN Controllers and then onboard the tenant.

Flexible Tenant Placement by Provide Admin User

From Cisco vManage Release 20.9.1, while onboarding a tenant, you have the flexibility to choose the pair of multitenant Cisco SD-WAN Controllers that are assigned to the tenant. Automatic tenant placement by Cisco SD-WAN Manager continues to be the default behavior with flexible tenant placement as an optional configuration.

To help you with flexible tenant placement, Cisco SD-WAN Manager lists available multitenant Cisco SD-WAN Controllers and provides the following details, as a percentage, for each controller:

- number of tenants assigned
- number of tenant WAN edge devices connected
- memory utilized
- CPU utilized

A multitenant Cisco SD-WAN Controller can serve a maximum of 24 tenants and 1000 tenant WAN edge devices across all the tenants. While onboarding a tenant, choose a pair of controllers that can be assigned one more tenant and can also connect to the number of WAN edge devices forecast for the tenant.

After the tenant is onboarded, if the tenant needs to add more devices than you originally forecast and the assigned pair of multitenant Cisco SD-WAN Controllers cannot connect to these additional WAN edge devices, you can migrate the tenant to another pair of Cisco SD-WAN Controllers that can serve one more tenant and accommodate the revised WAN edge device forecast for the tenant. If none of the multitenant Cisco SD-WAN Controllers pairs can accommodate the revised device forecast, you can migrate other tenants to alternative Cisco SD-WAN Controllers so that the controller utilization is efficient and the tenant assignment is optimal. If the optimization doesn't create the capacity required to accommodate the revised device forecast for the tenant, add a new pair of Cisco SD-WAN Controllers and then migrate the tenant.

Benefits of Flexible Tenant Placement on Multitenant Cisco SD-WAN Controllers

- Choose Cisco SD-WAN Controllers deployed in different failure zones to reduce the probability of both the controllers failing simultaneously. In a cloud environment, choose controllers deployed in different regions.
- Choose Cisco SD-WAN Controllers deployed in the same geographical region as the tenant WAN edge devices to reduce latency.
- Choose Cisco SD-WAN Controllers based on the CPU, DRAM, and hard disk resources allocated, and the utilization of these resources.
- Migrate a tenant to a different Cisco SD-WAN Controller to accommodate changes in the tenant device forecast.

Restrictions for Flexible Tenant Placement on Multitenant Cisco SD-WAN Controllers

If you wish to migrate a tenant to different pair of Cisco SD-WAN Controllers, you must change the Cisco SD-WAN Controllers assigned to the tenant one at a time. Doing so ensures that one of the Cisco SD-WAN Controllers is available to the tenant WAN edge devices during the migration and prevents disruptions in traffic.

Assign Cisco SD-WAN Controllers to Tenants During Onboarding

Prerequisites

- At least two Cisco SD-WAN Controllers must be operational and in Cisco SD-WAN Manager before you can add new tenants.

A Cisco SD-WAN Controller enters the **Manager** mode when you push a template to the controller from Cisco SD-WAN Manager. A Cisco SD-WAN Controller in the **CLI** mode cannot serve multiple tenants.

- Each pair of Cisco SD-WAN Controllers can serve a maximum of 24 tenants and a maximum of 1000 tenant devices. Ensure that there are at least two Cisco SD-WAN Controllers that can serve a new tenant. If no pair of Cisco SD-WAN Controllers in the deployment can serve a new tenant, add two Cisco SD-WAN Controllers and change their mode to **Manager**.
- Add up to 16 tenants in a single operation. If you add more than one tenant, during the **Add Tenant** task, Cisco SD-WAN Manager adds the tenants one after another and not in parallel.

While an **Add Tenant** task is in progress, do not perform a second tenant addition operation. If you do so, the second Add Tenant task fails.

- Each tenant must have a unique Virtual Account (VA) on Plug and Play Connect on [Cisco Software Central](#). The tenant VA should belong to the same Smart Account (SA) as the provider VA.
- For an on-premises deployment, create a Cisco SD-WAN Validator controller profile for the tenant on Plug and Play Connect. The fields in the following table are mandatory.

Field	Description
Profile Name	Enter a name for the controller profile.
Multi-Tenancy	From the drop-down list, select Yes .
SP Organization Name	Enter the provider organization name.
Organization Name	Enter the tenant organization name in the format <SP Org Name>-<Tenant Org Name>. The organization name can be up to 50 characters.
Primary Controller	Enter the host details for the primary Cisco SD-WAN Validator.

For a cloud deployment, the Cisco SD-WAN Validator controller profile is created automatically as part of the tenant creation process.

1. Log in to Cisco SD-WAN Manager as the provider **admin** user.
2. From the Cisco SD-WAN Manager menu, choose **Administration > Tenant Management**.
3. Click **Add Tenant**.
4. In the **Add Tenant** slide-in pane, click **New Tenant**.
5. Configure the following tenant details:

Field	Description
Name	Enter a name for the tenant. For a cloud deployment, the tenant name should be same as the tenant VA name on Plug and Play Connect.
Description	Enter a description for the tenant. The description can have up to 256 characters and can contain only alphanumeric characters.
Organization Name	Enter the name of the tenant organization. The organization name can have up to 50 characters. The organization name is case-sensitive. Each tenant or customer must have a unique organization name. Enter the organization name in the following format: <SP Org Name>-<Tenant Org Name> For example, if the provider organization name is 'managed-sp' and the tenant organization name is 'customer1', while adding the tenant, enter the organization name as 'managed-sp-customer1'.

Field	Description
URL Subdomain	

Field	Description
	Enter the fully qualified subdomain name of the tenant. - The subdomain name must include the domain name of the service provider. For example, for the managed-sp.com service provider, a valid domain name for customer1 is customer1.managed-sp.com. Note The service provider name is shared amongst all tenants. Ensure that the URL naming convention follows the same domain name convention that was followed while enabling multitenancy using Administration > Settings > Tenancy Mode. - For an on-premises deployment, add the fully qualified subdomain name of the tenant to the DNS. Map the fully qualified subdomain name to the IP addresses of the three Cisco SD-WAN Manager instances in the Cisco SD-WAN Manager cluster. - Provider DNS: Create a DNS A record and map it to the IP addresses of the Cisco SD-WAN Manager instances running in the Cisco SD-WAN Manager cluster. The A record is derived from the provider's domain name and the cluster ID that was created while enabling multitenancy on Cisco SD-WAN Manager. For example, if the provider's domain name is <code>sdwan.cisco.com</code> and the cluster ID is <code>vmanage123</code>, configure the A record as <code>vmanage123.sdwan.cisco.com</code>. Note If you fail to add the DNS A record, you will experience authentication errors when logging in to Cisco SD-WAN Manager. Validate that the DNS is configured correctly by using the nslookup command. Example: <code>nslookup vmanage123.sdwan.cisco.com</code>. - Tenant DNS: Create DNS CNAME records for each tenant that you created and map them to the provider FQDN. For example, if the provider's domain name is <code>sdwan.cisco.com</code> and tenant name is <code>customer1</code>, configure the CNAME record as <code>customer1.sdwan.cisco.com</code>. Cluster ID is not required in the CNAME record. Validate that the DNS is configured correctly by using the nslookup command. Example: <code>nslookup customer1.sdwan.cisco.com</code>. - For a cloud deployment, the fully qualified subdomain name

Field	Description								
	of the tenant is automatically added to the DNS as part of the tenant creation process. After you add a tenant, it could take up to an hour before the fully qualified subdomain name of the tenant can be resolved by the DNS.								
Forecasted Devices	Enter the number of WAN edge devices that the tenant can add to the overlay. If the tenant tries to add WAN edge devices beyond this number, Cisco SD-WAN Manager reports an error and the device addition fails.								
Select two Controllers	- Automatic tenant placement: Ensure that the Select two Controllers field has the value Autoplacement. This is the default configuration. - Flexible tenant placement: a. Click the Select two Controllers drop-down list. Cisco SD-WAN Manager lists the hostnames of the available Cisco SD-WAN Controllers. For each Cisco SD-WAN Controller, Cisco SD-WAN Manager shows whether the controller is reachable and reports the following utilization details: <table> <tr> <td>Tenant hosting capacity</td><td>Each Cisco SD-WAN Controller can serve a maximum of 24 tenants. Tenant hosting capacity represents the number of tenants to which the Cisco SD-WAN Controller is assigned in the form of a percentage. This value indicates whether you can assign another tenant to this controller.</td></tr> <tr> <td>Used device capacity</td><td>Each Cisco SD-WAN Controller can support a maximum of 1000 tenant WAN edge devices. Used device capacity represents the number of tenant WAN edge devices connected to the Cisco SD-WAN Controller in the form of a percentage of the maximum capacity (1000 WAN edge devices). This value indicates whether the Cisco SD-WAN Controller can support the number of devices forecast for the tenant that you are onboarding.</td></tr> <tr> <td>Memory utilized</td><td>This value represents memory consumption as a percentage.</td></tr> <tr> <td>CPU utilized</td><td>This value represents CPU usage as a percentage.</td></tr> </table> b. Select two Cisco SD-WAN Controllers to assign to the tenant based on the utilization details. To select a Cisco SD-WAN Controller, check the check box adjacent to its hostname.	Tenant hosting capacity	Each Cisco SD-WAN Controller can serve a maximum of 24 tenants. Tenant hosting capacity represents the number of tenants to which the Cisco SD-WAN Controller is assigned in the form of a percentage. This value indicates whether you can assign another tenant to this controller.	Used device capacity	Each Cisco SD-WAN Controller can support a maximum of 1000 tenant WAN edge devices. Used device capacity represents the number of tenant WAN edge devices connected to the Cisco SD-WAN Controller in the form of a percentage of the maximum capacity (1000 WAN edge devices). This value indicates whether the Cisco SD-WAN Controller can support the number of devices forecast for the tenant that you are onboarding.	Memory utilized	This value represents memory consumption as a percentage.	CPU utilized	This value represents CPU usage as a percentage.
Tenant hosting capacity	Each Cisco SD-WAN Controller can serve a maximum of 24 tenants. Tenant hosting capacity represents the number of tenants to which the Cisco SD-WAN Controller is assigned in the form of a percentage. This value indicates whether you can assign another tenant to this controller.								
Used device capacity	Each Cisco SD-WAN Controller can support a maximum of 1000 tenant WAN edge devices. Used device capacity represents the number of tenant WAN edge devices connected to the Cisco SD-WAN Controller in the form of a percentage of the maximum capacity (1000 WAN edge devices). This value indicates whether the Cisco SD-WAN Controller can support the number of devices forecast for the tenant that you are onboarding.								
Memory utilized	This value represents memory consumption as a percentage.								
CPU utilized	This value represents CPU usage as a percentage.								

6. To save the tenant configuration, click **Save**.
7. To add another tenant, repeat Step 4 to Step 6.
8. To onboard tenants to the deployment, click **Add**.

Cisco SD-WAN Manager initiates the Create Tenant Bulk task to onboard the tenants.

As part of this task, Cisco SD-WAN Manager performs the following activities:

- creates the tenant
- assigns two Cisco SD-WAN Controllers to serve the tenant and pushes a CLI template to these controllers to configure tenant information
- sends the tenant and Cisco SD-WAN Controller information to Cisco SD-WAN Validator

When the task is successfully completed, you can view the tenant information, including the Cisco SD-WAN Controller and Cisco SD-WAN Validators assigned to the tenant, on the **Administration > Tenant Management** page.

Update Cisco SD-WAN Controllers Placement For a Tenant

You can migrate a tenant to a different pair of Cisco SD-WAN Controllers from the controllers that are currently assigned to the tenant. For instance, if you need to increase the tenant WAN edge device forecast and the controllers assigned to the tenant cannot connect to these revised number of tenant WAN edge devices, you can migrate the tenant to a pair of controllers that can accommodate the revised forecast.

If you wish to migrate a tenant to different pair of Cisco SD-WAN Controllers, you must change the Cisco SD-WAN Controllers that are assigned to the tenant one at a time. Doing so ensures that one of the Cisco SD-WAN Controllers is available to the tenant WAN edge devices during the migration and prevents disruptions in traffic.

1. Log in to Cisco SD-WAN Manager as the provider **admin** user.
2. From the Cisco SD-WAN Manager menu, choose **Administration > Tenant Management**.
3. For the tenant you wish to migrate to a different controller, click ... adjacent to the tenant organization name.
4. Click **Update Controller Placement**.
5. In the **Update Controller Placement** slide-in pane, configure the following:

Field	Description	
Source Controller (currently applied)	a. Click the Source Controller (currently applied) drop-down list. Cisco SD-WAN Manager lists the hostnames of the Cisco SD-WAN Controllers assigned to the tenant. For each Cisco SD-WAN Controller, Cisco SD-WAN Manager shows whether the controller is reachable and reports the following utilization details:	
	Tenant hosting capacity	Each Cisco SD-WAN Controller can serve a maximum of 24 tenants. Tenant hosting capacity represents the number of tenants to which the Cisco SD-WAN Controller is assigned in the form of a percentage. This value indicates whether you can assign another tenant to this controller.
	Used device capacity	Each Cisco SD-WAN Controller can support a maximum of 1000 tenant WAN edge devices. Used device capacity represents the number of tenant WAN edge devices connected to the Cisco SD-WAN Controller in the form of a percentage of the maximum capacity (1000 devices). This value indicates whether the Cisco SD-WAN Controller can support the number of devices forecast for the tenant that you are onboarding.
	Memory utilized	This value represents memory consumption as a percentage.
	CPU utilized	This value represents CPU usage as a percentage.
	b. Check the check box adjacent to the hostname of one of the Cisco SD-WAN Controllers assigned to the tenant.	

Field	Description								
Destination Controller	a. Click the Destination Controller drop-down list. Cisco SD-WAN Manager lists the hostnames of the available Cisco SD-WAN Controllers that are not assigned to the tenant. For each Cisco SD-WAN Controller, Cisco SD-WAN Manager shows whether the controller is reachable and reports the following utilization details: <table border="1"> <tr> <td>Tenant hosting capacity</td><td>Each Cisco SD-WAN Controller can serve a maximum of 24 tenants. Tenant hosting capacity represents the number of tenants to which the Cisco SD-WAN Controller is assigned in the form of a percentage. This value indicates whether you can assign another tenant to this controller.</td></tr> <tr> <td>Used device capacity</td><td>Each Cisco SD-WAN Controller can support a maximum of 1000 tenant WAN edge devices. Used device capacity represents the number of tenant WAN edge devices connected to the Cisco SD-WAN Controller in the form of a percentage of the maximum capacity (1000 devices). This value indicates whether the Cisco SD-WAN Controller can support the number of devices forecast for the tenant that you are onboarding.</td></tr> <tr> <td>Memory utilized</td><td>This value represents memory consumption as a percentage.</td></tr> <tr> <td>CPU utilized</td><td>This value represents CPU usage as a percentage.</td></tr> </table> b. Check the check box adjacent to the hostname of the Cisco SD-WAN Controller you want to assign to the tenant. If you select a Cisco SD-WAN Controller that does not have the required capacity to serve the tenant devices, the update operation fails.	Tenant hosting capacity	Each Cisco SD-WAN Controller can serve a maximum of 24 tenants. Tenant hosting capacity represents the number of tenants to which the Cisco SD-WAN Controller is assigned in the form of a percentage. This value indicates whether you can assign another tenant to this controller.	Used device capacity	Each Cisco SD-WAN Controller can support a maximum of 1000 tenant WAN edge devices. Used device capacity represents the number of tenant WAN edge devices connected to the Cisco SD-WAN Controller in the form of a percentage of the maximum capacity (1000 devices). This value indicates whether the Cisco SD-WAN Controller can support the number of devices forecast for the tenant that you are onboarding.	Memory utilized	This value represents memory consumption as a percentage.	CPU utilized	This value represents CPU usage as a percentage.
Tenant hosting capacity	Each Cisco SD-WAN Controller can serve a maximum of 24 tenants. Tenant hosting capacity represents the number of tenants to which the Cisco SD-WAN Controller is assigned in the form of a percentage. This value indicates whether you can assign another tenant to this controller.								
Used device capacity	Each Cisco SD-WAN Controller can support a maximum of 1000 tenant WAN edge devices. Used device capacity represents the number of tenant WAN edge devices connected to the Cisco SD-WAN Controller in the form of a percentage of the maximum capacity (1000 devices). This value indicates whether the Cisco SD-WAN Controller can support the number of devices forecast for the tenant that you are onboarding.								
Memory utilized	This value represents memory consumption as a percentage.								
CPU utilized	This value represents CPU usage as a percentage.								

6. Click **Update**.

7. To change the other Cisco SD-WAN Controller that is assigned to the tenant, repeat Step 3 to Step 6.

Cisco SD-WAN Manager initiates the **Tenant Controller Update** task to assign the selected Cisco SD-WAN Controller to the tenant, migrating the tenant details from the Cisco SD-WAN Controller that was previously assigned. When the task is successfully completed, you can view the tenant information, including the Cisco SD-WAN Controllers assigned to the tenant, on the **Administration > Tenant Management** page.



CHAPTER 17

Cisco Catalyst SD-WAN Multitenancy (Cisco SD-WAN Releases 20.4.x and 20.5.x)

Table 215: Feature History

Feature Name	Release Information	Feature Description
Cisco Catalyst SD-WAN Multitenancy	Cisco SD-WAN Release 20.4.1 Cisco vManage Release 20.4.1	With Cisco Catalyst SD-WAN multitenancy, a service provider can manage multiple customers, called tenants, from Cisco SD-WAN Manager. In a multitenant Cisco Catalyst SD-WAN deployment, tenants share Cisco SD-WAN Manager instances, Cisco Catalyst SD-WAN Validators and Cisco Catalyst SD-WAN Controllers. Tenant data is logically isolated on these shared resources.

- [Overview of Cisco Catalyst SD-WAN Multitenancy, on page 581](#)
- [User Roles in Multitenant Environment, on page 584](#)
- [Hardware Supported and Specifications, on page 585](#)
- [Initial Setup for Multitenancy, on page 587](#)
- [Manage Tenants, on page 590](#)
- [Cisco SD-WAN Manager Dashboard for Multitenancy, on page 593](#)
- [Manage Tenant WAN Edge Devices, on page 598](#)
- [Tenant-Specific Policies on Cisco Catalyst SD-WAN Controllers, on page 599](#)
- [Manage Tenant Data, on page 599](#)
- [View OMP Statistics per Tenant on a Cisco SD-WAN Controller, on page 603](#)
- [View Tenants Associated with a Cisco SD-WAN Controller, on page 603](#)
- [Migrate Single-Tenant Cisco Catalyst SD-WAN Overlay to Multitenant Cisco Catalyst SD-WAN Deployment, on page 604](#)

Overview of Cisco Catalyst SD-WAN Multitenancy

With Cisco Catalyst SD-WAN multitenancy, a service provider can manage multiple customers, called tenants, from Cisco SD-WAN Manager. The tenants share Cisco SD-WAN Manager instances, Cisco Catalyst SD-WAN Validators, and Cisco Catalyst SD-WAN Controllers. The domain name of the service provider has subdomains for each tenant. For example, the `multitenancy.com` service provider can manage the tenants `Customer1` (`Customer1.multitenancy.com`) and `Customer2` (`Customer2.multitenancy.com`).

Following are the key features of Cisco Catalyst SD-WAN multitenancy:

- Full enterprise multitenancy: Cisco Catalyst SD-WAN supports multitenancy and offers enterprises the flexibility of segregated roles such as service provider and tenants. Service providers can use multitenancy to provide Cisco Catalyst SD-WAN service offerings to their customers.
- Multi-tenant Cisco SD-WAN Manager:
 - Cisco SD-WAN Manager is deployed and configured by the service provider. The provider enables multitenancy and creates a Cisco SD-WAN Manager cluster to serve tenants. Only the provider can access a Cisco SD-WAN Manager instance through the SSH terminal.



Note To connect to a device through SSH, use the IP address of the `vmanage_system` interface; this IP address is assigned by Cisco SD-WAN Manager. Do not use a user-configured system IP address to connect to a device through SSH.

You can find the IP address of the `vmanage_system` interface from the output of the **show interface description** command. Alternatively, you can launch the device SSH terminal from Cisco SD-WAN Manager and find the `vmanage_system` IP address from the first line of the log-in prompt.

- Cisco SD-WAN Manager offers service providers an overall view of the SD-WAN multi-tenant deployment and allows a provider to manage the shared Cisco Catalyst SD-WAN Validator and Cisco Catalyst SD-WAN Controller devices. Cisco SD-WAN Manager also allows service providers to monitor and manage the deployments of each tenant.
- Cisco SD-WAN Manager allows tenants to monitor and manage their deployment. Through Cisco SD-WAN Manager, tenants can deploy and configure WAN edge devices. Tenants can also configure custom policies on assigned Cisco Catalyst SD-WAN Controllers.
- Multi-tenant Cisco Catalyst SD-WAN Validators:
 - Cisco Catalyst SD-WAN Validators are deployed and configured by the service provider. Only the provider can access a Cisco Catalyst SD-WAN Validator through the SSH terminal.



Note To connect to a device through SSH, use the IP address of the `vmanage_system` interface; this IP address is assigned by Cisco SD-WAN Manager. Do not use a user-configured system IP address to connect to a device through SSH.

You can find the IP address of the `vmanage_system` interface from the output of the **show interface description** command. Alternatively, you can launch the device SSH terminal from Cisco SD-WAN Manager and find the `vmanage_system` IP address from the first line of the log-in prompt.

- Cisco Catalyst SD-WAN Validators serve WAN edge devices of multiple tenants as the devices are added to the overlay network.
- Multi-tenant Cisco Catalyst SD-WAN Controllers:

- Cisco Catalyst SD-WAN Controllers are deployed by the service provider. Only the provider can create and attach device and feature templates to Cisco Catalyst SD-WAN Controllers, and can access a Cisco Catalyst SD-WAN Controller through the SSH terminal.

**Note**

To connect to a device through SSH, use the IP address of the `vmanage_system` interface; this IP address is assigned by Cisco SD-WAN Manager. Do not use a user-configured system IP address to connect to a device through SSH.

You can find the IP address of the `vmanage_system` interface from the output of the **show interface description** command. Alternatively, you can launch the device SSH terminal from Cisco SD-WAN Manager and find the `vmanage_system` IP address from the first line of the log-in prompt.

- When a tenant is created, Cisco SD-WAN Manager assigns two Cisco Catalyst SD-WAN Controllers for the tenant. The Cisco Catalyst SD-WAN Controllers form an active-active cluster.

Each tenant is assigned only two Cisco Catalyst SD-WAN Controllers. Before a tenant is created, two Cisco Catalyst SD-WAN Controllers must be available to serve the tenant.

- Each pair of Cisco Catalyst SD-WAN Controllers can serve a maximum of 24 tenants.
- Tenants can configure custom policies on the Cisco Catalyst SD-WAN Controllers assigned to them. Cisco SD-WAN Manager notifies the Cisco Catalyst SD-WAN Controllers to pull the policy templates. Cisco Catalyst SD-WAN Controllers pull the templates and deploy the policy configuration for the specific tenant.
- Only the provider can view events, audit logs, and OMP alarms for a Cisco Catalyst SD-WAN Controller on Cisco SD-WAN Manager.

- WAN Edge Devices:

- A tenant or the provider acting on behalf of a tenant can add WAN edge devices to the tenant network, configure the devices, and remove the devices from the tenant network, or access the device through the SSH terminal.

**Note**

To connect to a device through SSH, use the IP address of the `vmanage_system` interface; this IP address is assigned by Cisco SD-WAN Manager. Do not use a user-configured system IP address to connect to a device through SSH.

You can find the IP address of the `vmanage_system` interface from the output of the **show interface description** command. Alternatively, you can launch the device SSH terminal from Cisco SD-WAN Manager and find the `vmanage_system` IP address from the first line of the log-in prompt.

- A provider can manage the WAN edge devices only from [provider-as-tenant](#) view. In the [provider](#) view, Cisco SD-WAN Manager does not present any WAN edge device information.
- Cisco SD-WAN Manager reports WAN edge device events, logs, and alarms only in the [tenant](#) and the provider-as-tenant views.

- Overlapping VPN numbers: A particular VPN or a set of common VPNs is assigned to a specific tenant, with their own configurations and monitoring dashboard environment. These VPN numbers can overlap where they are used by other tenants.
- On-prem and cloud deployment models: Cisco Catalyst SD-WAN controllers can be deployed in an organization data center on servers running the VMware vSphere ESXi or the Kernel-based Virtual Machine (KVM) hypervisor. Cisco Catalyst SD-WAN controllers can also be deployed in the cloud on Amazon Web Services (AWS) servers.

User Roles in Multitenant Environment

A multi-tenant environment includes the service provider and tenant roles. Each role has distinct privileges, views, and functions.

Provider Role

The provider role entitles system-wide administrative privileges. A user with the provider role has the default username **admin**. The provider user can access Cisco SD-WAN Manager using the domain name of the service provider or by using the Cisco SD-WAN Manager IP address. When using a domain name, the domain name has the format `https://multitenancy.com`.

The **admin** user is part of the user group **netadmin**. Users in this group are permitted to perform all operations on the controllers and the Cisco Catalyst SD-WAN devices of the tenants. You can add additional users to the **netadmin** group.

You cannot modify the privileges of the **netadmin** group. On Cisco SD-WAN Manager, you can view the privileges of the user group from the **Administration > Manage Users > User Groups** page.



Note When you create a new provider user in Cisco SD-WAN Manager, including a **netadmin** user, by default, the user is not allowed SSH access to the Cisco SD-WAN Manager VM. To enable SSH access, configure SSH authentication using a AAA template and push the template to Cisco SD-WAN Manager. For more information on enabling SSH authentication, see [SSH Authentication using vManage on Cisco vEdge Devices](#).

For more information about configuring users and user groups, see [Configure User Access and Authentication](#).

Cisco SD-WAN Manager offers two views to a provider:

• Provider View

When a provider user logs in to multi-tenant Cisco SD-WAN Manager as **admin** or another **netadmin** user, Cisco SD-WAN Manager presents the provider view and displays the provider dashboard.

You can perform the following functions from the provider view:

- Provision and manage Cisco SD-WAN Manager, Cisco Catalyst SD-WAN Validators and Cisco Catalyst SD-WAN Controllers.
- Add, modify, or delete tenants.
- Monitor the overlay network.

• Provider-as-Tenant View

When a provider user selects a specific tenant from the **Select Tenant** drop-down list at the top of the provider dashboard, Cisco SD-WAN Manager presents the provider-as-tenant view and displays the tenant dashboard for the selected tenant. The provider user has the same view of Cisco SD-WAN Manager as a tenant user would when logged in as **tenantadmin**. From this view, the provider can manage the tenant deployment on behalf of the tenant.

In the provider dashboard, a table of tenants presents a status summary for each tenant. A provider user can also launch the provider-as-tenant view by clicking on a tenant name in this table.

Tenant Role

The tenant role entitles tenant administrative privileges. A user with the tenant role has the default username **tenantadmin**. The default password is **Cisco#123@Viptela**. We recommend that you change the default password on first login. For information on changing the default password, see [Hardware and Software Installation](#).

The **tenantadmin** user is part of the user group **tenantadmin**. Users in this group are permitted to perform all operations on the WAN edge devices of the tenants. You can add additional users to the **tenantadmin** group.

You cannot modify the privileges of the **tenantadmin** group. On Cisco SD-WAN Manager, you can view the privileges of the user group from the **Administration > Manage Users > User Groups** page.

For more information about configuring users and user groups, see [Configure User Access and Authentication](#).

A tenant user can log in to Cisco SD-WAN Manager using a dedicated URL and the default username **tenantadmin**. For example, the dedicated URL of a tenant could be `https://Customer1.multitenancy.com` for a provider using the domain name `https://multitenancy.com`. When the user logs in, Cisco SD-WAN Manager presents the tenant view and displays the tenant dashboard.

A tenant user with administrative privileges can perform the following functions:

- Provision and manage tenant routers
- Monitor overlay network of the tenant
- Create custom policies on the assigned Cisco Catalyst SD-WAN Controllers
- Upgrade the software on the tenant routers.

Hardware Supported and Specifications

The following platforms support multitenancy.

Table 216: Router Models

Platform	Router Models
Cisco vEdge device	- vEdge 100, vEdge 100b, vEdge 100m, vEdge 100wm, vEdge 1000, vEdge 2000, vEdge 5000, vEdge Cloud - ISR1100-6G/ISR1100-4G, ISR1100-4GLTENA, ISR1100-4GLTEGB

The following hypervisors and deployment model are supported for multitenancy.

Table 217: Deployment Model

Specification	Description
Supported hypervisors	VMware, KVM, AWS (cloud-hosted by Cisco)
Cisco SD-WAN Manager Deployment Model	Cluster, 3 Cisco SD-WAN Manager instances with each instance running all Cisco SD-WAN Manager services.

The supported hardware specifications for the Cisco Catalyst SD-WAN Validator, Cisco SD-WAN Manager, and the Cisco Catalyst SD-WAN Controller are as follows:

Table 218: On-prem Deployment

Server	Cisco SD-WAN Manager	Cisco Catalyst SD-WAN Validator	Cisco Catalyst SD-WAN Controller
Deployment Model	Cluster	N/A	Non-containerized
Number of Instances	3	2	2 per 24 tenants
CPU	32 vCPU	4 vCPU	8 vCPU
DRAM	72 GB	4 GB	16 GB
Hard Disk	1 TB	10 GB	16 GB
NMS Service Distribution	Some services run on all three Cisco SD-WAN Manager instances in the cluster, while some services run on only one of the three instances in the cluster. Therefore, the CPU load may vary among the instances.	N/A	N/A



Note If DPI is enabled, we recommend that the aggregated DPI data across all Cisco SD-WAN Managere instances not exceed 350 GB per day.

Initial Setup for Multitenancy

Prerequisites

- Download and install software versions as recommended in the following table:

Table 219: Software Prerequisites for Cisco Catalyst SD-WAN Multitenancy

Device	Software Version
Cisco SD-WAN Manager	Cisco vManage Release 20.4.1
Cisco Catalyst SD-WAN Validator	Cisco SD-WAN Release 20.4.1
Cisco Catalyst SD-WAN Controller	Cisco SD-WAN Release 20.4.1
Cisco vEdge Device	Cisco SD-WAN Release 20.4.1

A configuration in which one or more controllers, or WAN edge devices, are running software versions earlier than those mentioned in the table above is not supported.

- Do not migrate an existing single-tenant Cisco SD-WAN Manager instance into multitenant mode, even if you invalidate or delete all devices from the existing Cisco SD-WAN Manager instance. Instead, download and install a new Cisco SD-WAN Manager software image.



Note After you enable Cisco SD-WAN Manager for multitenancy, you cannot migrate it back to single tenant mode.

- Log in to Cisco SD-WAN Manager as the provider **admin** user.
- Create three Cisco SD-WAN Manager instances and associated configuration templates. See [Deploy Cisco SD-WAN Manager](#).
 - While configuring Cisco SD-WAN Manager instances, configure the service provider organization name (`sp-organization-name`) and the organization name (`organization-name`).

Example:

```
sp-organization-name multitenancy
organization-name multitenancy
```
 - Configure one of the Cisco SD-WAN Manager instances to support multitenancy. See [Enable Multitenancy on Cisco SD-WAN Manager, on page 588](#)

3. Create a Cisco SD-WAN Manager cluster consisting of three Cisco SD-WAN Manager instances. See [Cluster Management](#).
 - The Cisco SD-WAN Manager cluster must have three Cisco SD-WAN Manager instances. A cluster with more than three instances or fewer than three instances is not a supported configuration for Cisco Catalyst SD-WAN multitenancy.
 - While creating the Cisco SD-WAN Manager cluster, add the Cisco SD-WAN Manager instance configured to support multitenancy before adding the other two Cisco SD-WAN Manager instances.
4. Certify all instances of Cisco SD-WAN Manager. See [Generate Cisco SD-WAN Manager Certificate](#).
5. Create and configure Cisco SD-WAN Validator instances. See [Deploy Cisco SD-WAN Validator](#).
While configuring Cisco SD-WAN Validator instances, configure the service provider organization name (`sp-organization-name`) and the organization name (`organization-name`). See [Configure Organization Name in Cisco SD-WAN Validator](#).

```
sp-organization-name multitenancy
organization-name multitenancy
```
6. Create Cisco SD-WAN Controller instances. See [Deploy the Cisco SD-WAN Controllers](#).
To support 50 tenants and 1000 devices across all tenants, deploy 6 Cisco SD-WAN Controller instances. To support 100 tenants and 5000 devices across all tenants, deploy 12 Cisco SD-WAN Controllers.
 - a. [Add Cisco SD-WAN Controllers](#) to the overlay network.
7. Onboard new tenants. See [Add a New Tenant](#).

Enable Multitenancy on Cisco SD-WAN Manager

1. Launch Cisco SD-WAN Manager using the URL `https://vmanage-ip-address:port`. Log in as the provider **admin** user.
2. From the Cisco SD-WAN Manager menu, choose **Administration > Settings > Tenancy Mode**. If you are using Cisco Catalyst SD-WAN Manager Release 20.12.x or earlier, click **Edit**.
3. In the **Tenancy** field, click **Multitenant**.
4. In the **Domain** field, enter the domain name of the service provider (for example, `multitenancy.com`).
5. Enter a **Cluster Id** (for example, `cluster-1` or `123456`).
6. Click **Save**. If you are using Cisco Catalyst SD-WAN Manager Release 20.12.x or earlier, click **Proceed** to confirm that you want to change the tenancy mode.

Cisco SD-WAN Manager reboots in multitenant mode and when a provider user logs in to Cisco SD-WAN Manager, the provider dashboard appears.

Add Cisco SD-WAN Controller

1. Log in to Cisco SD-WAN Manager as the provider **admin** user.
2. From the Cisco SD-WAN Manager menu, choose **Configuration > Devices**.

3. Click **Controllers**.



Note Starting from Cisco IOS XE Catalyst SD-WAN Release 17.13.1a, the **Controllers** tab is renamed as the **Control Components** tab to stay consistent with Cisco Catalyst SD-WAN rebranding.

4. Click **Add Controller**.
5. In the **Add Controller** dialog box, do the following:
 - a. In the **Controller Management IP Address** field, enter the system IP address of the Cisco SD-WAN Controller.
 - b. Enter the **Username** and **Password** required to access the Cisco SD-WAN Controller.
 - c. Select the protocol to use for control-plane connections. The default is **DTLS**.
If you select **TLS**, enter the port number to use for TLS connections. The default is 23456.
 - d. Check the **Generate CSR** check box for Cisco SD-WAN Manager to create a Certificate Signing Request.
 - e. Click **Add**.
6. From the Cisco SD-WAN Manager menu, choose **Configuration > Certificates**.
For the newly added Cisco SD-WAN Controller, the **Operation Status** reads **CSR Generated**.
 - a. For the newly added Cisco SD-WAN Controller, click **More Options** icon and click **View CSR**.
 - b. Submit the CSR to the Certificate Authority (CA) and obtain a signed certificate.
7. From the Cisco SD-WAN Manager menu, choose **Configuration > Certificates**.
8. Click **Install Certificate**.
9. In the **Install Certificate** dialog box, paste the **Certificate Text** or click **Select a file** upload the certificate file. Click **Install**.

Cisco SD-WAN Manager installs the certificate on the Cisco SD-WAN Controller. Cisco SD-WAN Manager also sends the serial number of the certificate to other controllers.

On the **Configuration > Certificates** page, the **Operation Status** for the newly added Cisco SD-WAN Controller reads as **Validator Updated**.

On the **Configuration > Devices** page, the new controller is listed in the Controller table with the controller type, hostname of the controller, IP address, site ID, and other details. The **Mode** is set to **CLI**.
10. Change the mode of the newly added Cisco SD-WAN Controller to **Manager** by attaching a template to the device.
 - a. From the Cisco SD-WAN Manager menu, choose **Configuration > Templates**.
 - b. Click **Device Templates**.



Note In Cisco vManage Release 20.7.x and earlier releases, **Device Templates** is titled as **Device**

- c. Find the template to be attached to the Cisco SD-WAN Controller.
- d. Click ..., and click **Attach Devices**.
- e. In the **Attach Devices** dialog box, move the new controller to the **Selected Device** list and click **Attach**.
- f. Verify the **Config Preview** and click **Configure Devices**.

Cisco SD-WAN Manager pushes the configuration from the template to the new controller.

In the **Configuration > Devices** page, the **Mode** for the Cisco SD-WAN Controller shows **Manager**. The new Cisco SD-WAN Controller is ready to be used in your multitenant deployment.

Manage Tenants

Add a New Tenant

Prerequisites

- At least two Cisco SD-WAN Controllers must be operational and in the `Manager` mode before you can add new tenants.

A Cisco SD-WAN Controller enters the `Manager` mode when you push a template onto the controller from Cisco SD-WAN Manager. A Cisco SD-WAN Controller in the `CLI` mode cannot serve multiple tenants.

- Each pair of Cisco SD-WAN Controllers can serve a maximum of 24 tenants. Ensure that there at least two Cisco SD-WAN Controllers that can serve a new tenant. If no pair of Cisco SD-WAN Controllers in the deployment can serve a new tenant, add two Cisco SD-WAN Controllers and change their mode to `Manager`.
- If you add a second tenant immediately after adding a tenant, Cisco SD-WAN Manager adds them sequentially, and not in parallel.
- Each tenant must have a unique Virtual Account (VA) on **Plug and Play Connect** on [Cisco Software Central](#). The tenant VA should belong to the same Smart Account (SA) as the provider VA.
- For an on-premises deployment, create a Cisco SD-WAN Validator controller profile for the tenant on **Plug and Play Connect**. The fields in the following table are mandatory.

Table 220: Controller Profile Fields

Field	Description/Value
Profile Name	Enter a name for the controller profile.
Multi-Tenancy	From the drop-down list, select Yes .

Field	Description/Value
SP Organization Name	Enter the provider organization name.
Organization Name	Enter the tenant organization name in the format <SP Org Name>-<Tenant Org Name>. Note The organization name can be up to 50 characters.
Primary Controller	Enter the host details for the primary Cisco SD-WAN Validator.

For a cloud deployment, the Cisco SD-WAN Validator controller profile is created automatically as part of the tenant creation process.

1. Log in to Cisco SD-WAN Manager as the provider **admin** user.
2. From the Cisco SD-WAN Manager menu, choose **Administration > Tenant Management**.
3. Click **Add Tenant**. In the **Add Tenant** dialog box:
 - a. Enter a name for the tenant.
For a cloud deployment, the tenant name should be same as the tenant VA name on **Plug and Play Connect**.
 - b. Enter a description of the tenant.
The description can be up to 256 characters and can contain only alphanumeric characters.
 - c. Enter the name of the organization.
The organization name is case-sensitive. Each tenant or customer must have a unique organization name.

Enter the organization name in the following format:
`<SP Org Name>-<Tenant Org Name>`

For example, if the provider organization name is 'multitenancy' and the tenant organization name is 'Customer1', while adding the tenant, enter the organization name as **multitenancy-Customer1**.



Note The organization name can be up to 50 characters.

- d. In the **URL Subdomain Name** field, enter the fully qualified sub-domain name of the tenant.
 - The sub-domain name must include the domain name of the service provider. For example, for the `multitenancy.com` service provider, a valid domain name can be `Customer1.multitenancy.com`.



Note The service provider name is shared amongst all tenants. Hence, ensure that the URL naming convention follows the same domain name convention that was provided while enabling multitenancy from the Cisco SD-WAN Manager **Administration > Settings > Tenancy Mode** GUI navigation path.

- For an on-premises deployment, add the fully qualified sub-domain name of the tenant to the DNS. Map the fully qualified sub-domain name to the IP addresses of the three Cisco SD-WAN Manager instances in the Cisco SD-WAN Manager cluster.

When creating fully qualified domain names (FQDN) the following DNS entries are required:

- **Provider Level:** Create DNS A record and map it to the IP addresses of the Cisco SD-WAN Manager instances running in the Cisco SD-WAN Manager cluster. The A record is derived from the domain and Cluster ID that was created in steps 5 and 6 in [Enable Multitenancy on Cisco SD-WAN Manager](#). For example, if domain is **sdwan.cisco.com** and Cluster ID is **vmmanage123**, then A record will need to be configured as **vmmanage123.sdwan.cisco.com**.



Note If you fail to update DNS entries, it will result in authentication errors when logging in to Cisco SD-WAN Manager. Validate DNS is configured correctly by executing **nslookup vmmanage123.sdwan.cisco.com**.

- **Tenant Level:** Create DNS CNAME records for each tenant created and map them to the FQDN created at the Provider Level. For example, if domain is **sdwan.cisco.com** and tenant name is **customer1** the CNAME record will need to be configured as **customer1.sdwan.cisco.com**.



Note Cluster ID is not required for CNAME record. Validate DNS is configured correctly by executing **nslookup customer1.sdwan.cisco.com**.

For a cloud deployment, the fully qualified sub-domain name of the tenant is automatically added to the DNS as part of the tenant creation process. After you add a tenant, it could take up to an hour before the fully qualified sub-domain name of the tenant can be resolved by the DNS.

e. Click **Save**.

The **Create Tenant** screen appears, and the **Status** of the tenant creation reads **In progress**. To view status messages related to the creation of a tenant, click the > button to the left of the status.

Cisco SD-WAN Manager does the following:

- creates the tenant
- assigns two Cisco SD-WAN Controllers to serve the tenant and pushes a CLI template to these controllers to configure tenant information
- sends the tenant and Cisco SD-WAN Controller information to Cisco SD-WAN Validators.

What to do next:

After the **Status** column changes to **Success**, you can view the tenant information on the **Administration > Tenant Management** page.

Modify Tenant Information

1. Log in to Cisco SD-WAN Manager as the provider **admin** user.
2. From the Cisco SD-WAN Manager menu, choose **Administration > Tenant Management**.
3. In the left pane, click the name of the tenant.
The tenant information is displayed in a pane on the right.
4. To modify tenant data, do as follows:
 - a. In the right pane, click the pencil icon.
 - b. In the **Edit Tenant** dialog box, modify the tenant name, description, or domain name.
 - c. Click **Save**

Delete a Tenant

Before you delete a tenant, delete all tenant WAN edge devices. See [Delete a WAN Edge Device from a Tenant Network, on page 599](#).

1. Log in to Cisco SD-WAN Manager as the provider **admin** user.
2. From the Cisco SD-WAN Manager menu, choose **Administration > Tenant Management**.
3. In the left pane, click the name of the tenant.
The tenant information is displayed in a pane on the right.
4. To delete the tenant, do as follows:
 - a. In the right pane, click the trash icon.
 - b. In the **Delete Tenant** dialog box, enter the provider **admin** password and click **Save**.

Cisco SD-WAN Manager Dashboard for Multitenancy

After enabling Cisco SD-WAN Manager for multitenancy, you can view the multitenant dashboard when you log in to Cisco SD-WAN Manager. Cisco SD-WAN Manager multitenant dashboard is a portal where the provider or tenant can view and provision the underlying system.

The bar at the top of every Cisco SD-WAN Manager multitenant screen includes icons that allow smooth navigation.

View Tenant Activity, Device, and Network Information

When you log in to a multitenant Cisco SD-WAN Manager as an administrator, the provider dashboard displays the following components. To return to the provider dashboard from other Cisco SD-WAN Manager screens, click **Dashboard** at the left bar.

- **Device pane** — runs across the top of the multitenant dashboard screen. The device pane displays the number of active Cisco Catalyst SD-WAN Controllers, Cisco Catalyst SD-WAN Validators, and Cisco SD-WAN Manager instances, the connectivity status of devices, and information on certificates that have expired or about to expire.
- **Tenants pane** — displays the total number of tenants and a summary of the control status, site health, router health, and Cisco Catalyst SD-WAN Controller status of all tenants.
- **Table of tenants in the overlay network** — List of individual tenants, with separate information about the control status, site health, WAN edge device health, and Cisco Catalyst SD-WAN Controller status for each tenant.

To display tenant-specific status summary information,

1. Click a tenant name from the tenant list.

A dialog box opens on the right side of the screen that provides additional information about the status of the tenant.

2. To access the tenant dashboard for the selected tenant, click <Tenant name> **Dashboard**.

Cisco SD-WAN Manager presents the provider-as-tenant view and displays the tenant dashboard. To return to the provider view, click **Provider** at the top of page.

3. To close the dialog box, click the tenant name from the tenant list.

View Detailed Information of a Tenant Setup

Cisco SD-WAN Manager displays the tenant dashboard, which provides information about a tenant deployment when

- a provider **admin** user selects a specific tenant from the **Select Tenant** drop-down list in the provider dashboard. This view is called the provider-as-tenant view.
- a **tenantadmin** user logs in to Cisco SD-WAN Manager. This view is called the tenant view.

View All Network Connections in the Tenant Overlay Network

The **Device** pane runs across the top of the tenant dashboard and displays the number of control connections from Cisco SD-WAN Manager to the Cisco SD-WAN Controllers and routers in the overlay network of a tenant. For each WAN edge device, the Device pane shows

- Total number of control connections between Cisco SD-WAN Controllers and WAN edge devices
- Number of valid control connections between Cisco SD-WAN Controllers and WAN edge devices
- Number of invalid control connections between Cisco SD-WAN Controllers and WAN edge devices

Click a connection number, or the **Up** or **Down** arrow, to display a table with detailed information about each connection. Click the **More Actions** icon at the right of each table row to access the **Device Dashboard** or **Real Time** view from the **Monitor > Devices** screen, or access the **Tools > SSH Terminal** Screen.



Note In Cisco vManage Release 20.6.x and earlier releases, **Real Time** view is part of the **Monitor > Network** screen.

View Information About Device Reboots

The **Reboot** pane displays the total number of reboots in the last 24 hours for all devices in the network. It includes soft and cold reboots and reboots that occurred as a result of power-cycling a device. For each reboot, the following information is listed:

- System IP and hostname of the device that rebooted.
- Time when the device was rebooted.
- Reason for the device reboot

If the same device reboots more than once, each reboot option is reported separately.

Click the **Reboot** pane to open the **Reboot** dialog box. In the **Reboot** dialog box, click **Crashes**. For all device crashes, the following information is listed:

- System IP and hostname of the device on which the crash occurred.
- Crash index of the device
- Core time when the device crashed.
- File name of the device crash log

View Network Connections

The **Control Status** pane displays whether Cisco SD-WAN Controller and WAN edge devices are connected. Each Cisco SD-WAN Controller must connect to all other Cisco SD-WAN Controllers in the network. Each WAN edge device must connect to the maximum number of configured Cisco SD-WAN Controllers. The **Control Status** pane displays three network connection counts:

- Control Up — total number of devices with the required number of operational control plane connections to a Cisco SD-WAN Controller
- Partial — total number of devices with some, but not all, operational control plane connection to Cisco SD-WAN Controllers.
- Control Down — total number of devices with no control plane connection to a Cisco SD-WAN Controller

To display a table with device details, click a row from the **Control Status** dialog box. Click the **More Actions** icon at the right of each table row to access the **Device Dashboard** or **Real Time** view from the **Monitor > Devices** screen.



Note In Cisco vManage Release 20.6.x and earlier releases, **Real Time** view is part of the **Monitor > Network** screen.

View State of Data Connections for a Site

The **Site Health** pane displays the state of data connections for a site. When a site has multiple WAN edge devices, this pane displays the state for the entire site and not for individual devices. The Site Health pane displays three connectivity states:

- Full WAN Connectivity — total number of sites where all BFD sessions on all routers are in the up state.
- Partial WAN Connectivity — total number of sites where tunnel and all BFD sessions on all routers are in the down state. These sites still have limited data plane connectivity.
- No WAN Connectivity — total number of sites where all BFD sessions on all routers are in the down state. These sites have no data plane connectivity.

To display a table with detailed information about each site, node, or tunnel, click a row from the **Site Health** dialog box. Click the **More Actions** icon at the right of each row in the table to access the **Device Dashboard** or **Real Time** view from the **Monitor > Devices** screen, or access the **Tools > SSH Terminal** screen.



Note In Cisco vManage Release 20.6.x and earlier releases, **Real Time** view is part of the **Monitor > Network** screen.

View Interface Usage for WAN Edge Interfaces

The **Transport Interface Distribution** pane displays interface usage in the last 24 hours for all WAN edge interfaces in VPN 0. It includes all TLOC interfaces. Click the pane to view details of interface usage in the **Transport Interface Distribution** dialog box.

View WAN Edge Device Counts

The **WAN Edge Inventory** pane provides four WAN edge device counts:

- Total — total number of authorized serial numbers for WAN edge devices that have been uploaded on Cisco SD-WAN Manager. The serial number is uploaded on the **Configuration > Devices** screen.
- Authorized — total number of authorized WAN edge devices in the overlay network. These WAN edge devices are marked as **Valid** in the **Configuration > Certificates > WAN Edge List** screen.
- Deployed — total number of deployed WAN edge devices. These are WAN edge devices that are marked as **Valid** and are now operational in the network.
- Staging — total number of WAN edge devices you configure at a staging site before they are made a part of the overlay network. These routers do not take part in any routing decisions and do not affect network monitoring through Cisco SD-WAN Manager.

Click the pane to view hostname, system IP, site ID, and other details of each router from the **WAN Edge Inventory** dialog box.

View Aggregated State of WAN Edge Devices

The **WAN Edge Health** pane offers an aggregated view of the state of WAN edge devices by providing a count of the number of devices in each state, therefore describing the health of the hardware nodes. The three WAN edge device states are:

- **Normal** — number of WAN edge devices with memory, hardware, and CPU in normal state. Using less than 70% of total memory or total CPU is classified as, normal.
- **Warning** — number of WAN edge devices with memory, hardware, or CPU in warning state. Using between 70% and 90% of total memory or total CPU is classified as, warning
- **Error** — number of WAN edge devices with memory, hardware, or CPU in error state. Using more than 90% of total memory or total CPU is classified as, error.

Click a number or the WAN edge device state to display a table with the last 12 or 24 hours of memory usage, CPU utilization, and hardware-related alarms, including temperature, power supply, and PIM modules. Click the **More Actions** icon at the right of each row in the table to access the following:

- **Hardware Environment**
- **Real Time** view from the **Monitor > Devices** screen
Cisco vManage Release 20.6.x and earlier: **Real Time** view from the **Monitor > Network** screen
- **Tools > SSH Terminal** screen.

View WAN Edge Device Loss, Latency, Jitter

The **Transport Health** pane displays the aggregated average loss, latency, and jitters for all links and all combinations of colors (for example, all LTE-to-LTE links, all LTE-to-3G links).

From the **Type** drop-down arrow, choose loss, latency, or jitter.

Click the  icon to select a time period for which to display the transport health.

Click the  icon to open the **Transport Health** dialog box. This dialog box displays a more detailed view. To display information in a tabular format, click **Details**. You can choose to change the displayed health type and time period.

View DPI Flow Information of WAN Edge Devices

The **Top Applications** pane displays DPI flow information for traffic transiting routers in the overlay network.



Note DPI flow information is shown only for the last 24 hours. To view DPI flow information for a time before the last 24 hours, you must check the information for the specific device.

Click the  icon to select a time period for which to display data. From the **VPN** drop-down list, select a VPN to display DPI information for all flows in that VPN.

Click the  icon to open the **Top Applications** dialog box. This dialog box displays a more detailed view of the same information. You can change the VPN and time period.

View Tunnels Data

The **Application-Aware Routing** pane allows you to choose the following tunnel criteria from the **Type** drop-down arrow:

- Loss
- Latency
- Jitter

Based on the tunnel criteria, the pane displays the 10 worst tunnels. For example, if you choose loss, the pane shows 10 tunnels with the greatest average loss over the last 24 hours.

Click the  icon against a row to display a graphical representation of the data. Select a time period for which to display data or click **Custom** to display a drop-down arrow for specifying a custom time period.

Click the  icon to open the **Application-Aware Routing** dialog box. This dialog box displays the 25 worst tunnels based on criteria you choose from the **Type** drop-down arrow, the criteria being loss, latency, and jitter.

Manage Tenant WAN Edge Devices

Add a WAN Edge Device to a Tenant Network

1. Log in to Cisco SD-WAN Manager.

If you're a provider user, log in as the **admin**. In the provider dashboard, choose a tenant from the drop-down list to enter the provider-as-tenant view.

If you're a tenant user, log in as the **tenantadmin**.

2. Upload the device serial number file to Cisco SD-WAN Manager.

3. Validate the device and send details to controllers.

4. Create a configuration template for the device and attach the device to the template.

While configuring the device, configure the service provider organization name and the tenant organization name as in the following example:

```
sp-organization-name multitenancy
organization-name multitenancy-Customer1
```



Note Enter the `organization-name` in the format `<SP Org Name>-<Tenant Org Name>`.

5. Bootstrap the device using bootstrap configuration generated through Cisco SD-WAN Manager or manually create the initial configuration on the device.
6. If you are using Enterprise Certificates to authenticate the device, download the CSR from Cisco SD-WAN Manager and get the CSR signed by the Enterprise CA. Install the certificate on Cisco SD-WAN Manager.

Delete a WAN Edge Device from a Tenant Network

1. Log in to Cisco SD-WAN Manager.
If you're a provider user, log in as the **admin**. In the provider dashboard, choose a tenant from the drop-down list to enter the provider-as-tenant view.
If you're a tenant user, log in as the **tenantadmin**.
2. Detach the device from any configuration templates.
3. [Delete a WAN Edge Router](#).

Tenant-Specific Policies on Cisco Catalyst SD-WAN Controllers

A provider **admin** user (from the Cisco SD-WAN Manager provider-as-tenant view) or a **tenantadmin** user (from the Cisco SD-WAN Manager tenant view) can create and deploy tenant-specific policies on the Cisco Catalyst SD-WAN Controllers serving the tenant. The user can configure a CLI policy or create the policy using the UI policy configuration wizard.

When you activate or deactivate a policy,

1. Cisco SD-WAN Manager identifies the Cisco Catalyst SD-WAN Controllers serving the tenant.
2. Cisco SD-WAN Manager notifies the Cisco Catalyst SD-WAN Controllers to pull the policy configuration.
3. Cisco Catalyst SD-WAN Controllers pull and deploy the policy configuration.
4. Cisco SD-WAN Manager reports the status of the policy pull by the Cisco Catalyst SD-WAN Controllers.

Manage Tenant Data

Back Up Tenant Data

The tenant data backup solution of Cisco SD-WAN Manager multitenancy provides the following functionalities:

- [Create, list, and extract configuration backup files](#).
- Back up configuration database of a specific tenant with an option to restore it later. See [Back up and restore configuration database](#).
- Delete back up files of a tenant stored in Cisco SD-WAN Manager. For deleting tenant data backup files, see [Delete tenant data backup file](#).

The following factors are applicable when using data backup solution:

- The tenant data backup solution operations can be performed by a tenant administrator over tenant view and as a provider. To know how to access tenant dashboard through different views, see [User Roles in Multitenant Environment](#), on page 584.

- A tenant is allowed to perform the following backup operations at a particular time and must complete an operation before starting a new operation:
 - Back up a single configuration database
 - Download the backup file.
 - Restore or import backup files
 - Delete backup files.
 - List backup files

- A tenant backup file format is as follows:

```
Bkup_tenantId_MMDDYY-HHMMSS_taskIdWithoutDash.tar.gz
```

- The tenant data backup operation is a readonly operation on the configuration database. However, to ensure data consistency and prevent data loss, do not perform any major changes on the network.
- When a backup or restore operation for a specific tenant is in-progress, other tenants are allowed to perform the backup and restore operations smoothly.
- A tenant is not allowed to perform other backup operations when the restore operation of the tenant database is in-progress. So, a tenant can perform a single backup operation and when this operation is in-progress, all new backup operation requests are rejected.

The remaining tenants can continue with their backup operations.

- A tenant must use the same Cisco SD-WAN Manager version for backup generation and restore operation.
- A tenant can store a maximum of three backup files in Cisco SD-WAN Manager and can download to store them outside Cisco SD-WAN Manager repository. If the tenant already has three backup files, a subsequent backup operation results in the earliest backup file being deleted and a new backup file being generated.
- Ensure that the following parameter values match in both the backup file and the setup where tenant has requested for a restore operation:
 - Tenant Id
 - Organization Name
 - SP Organization Name
- The tenant data backup solution creates a task in the tenant view of Cisco SD-WAN Manager. Therefore, the tenant can monitor the progress of the operation from the task view of the tenant dashboard.
- A provider cannot back up provider data using this solution. Therefore, the provider can back up all tenants information at once by backing up all tenants configuration database using CLI.

Create, Extract, and List Configuration Data Backup File

1. Log in to Cisco SD-WAN Manager.

If you're a provider user, log in as the **admin**. In the provider dashboard, choose a tenant from the drop-down list to enter the provider-as-tenant view.

If you're a tenant user, log in as the **tenantadmin**.

2. In the address bar, modify the URL path with `dataservice` for the REST API connection.

Example: `https://<tenant_URL>/dataservice`

3. Create a configuration backup file by using the following API:

`https://<tenant_URL>/dataservice/tenantbackup/export.`

4. If the configuration backup file has been created successfully, Cisco SD-WAN Manager task view indicates that the backup file has been generated. You can view the process identifier of the created process or task.

Example:

```
{
  "processId": "72d69805-b987-436f-9b7a-afef2f3f9061",
  "status": "in-progress"
}
```

5. Verify the task status using the obtained process identifier.

Example:

`https://<tenant_URL>/dataservice/device/action/status/72d69805-b987-436f-9b7a-afef2f3f9061`

The verification generates the details of the task in the JSON file format.

6. After the task is completed, extract or download the backed-up file available under the **data** section of the JSON task file.

Example: To extract or download the backup file, use the following API:

`https://<tenant_URL>/dataservice/tenantbackup/download/1570057020772/backup_1570057020772_100919-181838.tar.gz`

7. List backup files stored in Cisco SD-WAN Manager using the following API.

Example: `https://<tenant_URL>/dataservice/tenantbackup/list`

Restore and Delete Tenant Data Backup File

Before you begin:

To run the restore and delete tenant data backup files API, you can download and install Postman tool or any other alternative tool for testing http applications and services. In this document, the procedure to restore and delete tenant data backup files has been explained using the Postman tool. Postman is a software tool used as an API development environment. You can download the tool from the Postman website.

1. Open Google Chrome, or another browser, and enable developer mode on it.
2. Log in to Cisco SD-WAN Manager.

If you're a provider user, log in as the **admin**. In the provider dashboard, choose a tenant from the drop-down list to enter the provider-as-tenant view.

If you're a tenant user, log in as the **tenantadmin**.

3. To get header information of the restore API, do as follows:
 - a. On the right side of the screen, click the **Network** tab to get the network capture view.
 - b. In the network capture view, click the **Name** column to sort the listed items.
 - c. Search and click **index.html**.

- d. Click the **Headers** tab and expand **Request Headers**.
 - e. Choose all text under **Request Headers** and copy it to the clipboard.
4. Import backup files through the Postman UI:
 - a. Open the Postman UI.
 - b. To disable SSL certificate verification, click **Postman > Preferences > General > Request**. Turn off **SSL Certificate Verification**.
 - c. In the Postman UI, create a new tab.
 - d. Click **Headers** and then click **Bulk Edit**.
 - e. Paste the text that was copied in step 3 from the **Request Headers** block into an editable form.
 - f. From the **GET** method drop-down list, choose **POST**.
 - g. In the **Paste request URL** field, paste the dedicated URL of the tenant and include `dataservice/tenantbackup/import`.
Example: `https://Customer1.multitenancy.com/dataservice/tenantbackup/import`
 - h. Click the **Body** tab and select **form-data**.
 - i. Under **KEY** column, enter `bakup.tar.gz`
 - j. Under **VALUE** column, click **Select Files** and select a backup file to be imported.
 - k. To run the API, click **Send**.
 In the **Response** section of the Postman UI, you can view the JSON information that indicates the file that was restored.
5. Monitor the restoration of backup files in either of the following ways:
 - a. Use Cisco SD-WAN Manager task view that indicates if backup file has been imported successfully. You can view the process identifier of the created process or task.
Example:

```
{
  "processId": "40adb6c0-eacc-4ad4-ba6c-2c2da2e96d1d",
  "status": "Import Successfully Submitted for tenant 1579026919487"
}
```
 - b. Use the following URL to get the status,
`https://<tenant_URL>/dataservice/device/action/status/<processId>`
Example:
`https://Customer1.multitenancy.com/dataservice/device/action/status/40adb6c0-eacc-4ad4-ba6c-2c2da2e96d1d`
6. Delete tenant data backup file through Postman UI.
 - a. In the Postman UI, create a new tab.
 - b. Click **Headers** and then click **Bulk Edit**.
 - c. Paste the text that was copied in step 3 from the **Request Headers** block into an editable form.
 - d. From the **GET** method drop-down list, choose **DELETE**.

- e. In the **Paste request URL** field, paste the dedicated URL of the tenant and include `dataservice/tenantbackup/delete?fileName='filename'`. The filename can either be name of the backup file or all.

Example:

```
https://Customer1.multitenancy.com/dataservice/tenantbackup/delete?fileName=bkup_1579026919487_012820-180712_c09230904dfc40edb0d1e50b68b03002.tar.gz
```

Example:

```
https://Customer1.multitenancy.com/dataservice/tenantbackup/delete?fileName=all
```

- f. To run the API, click **Send**.

In the **Response** section of the Postman UI, you can view the JSON information that indicates the files that were deleted.

Example:

```
{
  "Deleted": [
    "bkup_1579026919487_012820-180712_c09230904dfc40edb0d1e50b68b03002.tar.gz"
  ]
}
```

View OMP Statistics per Tenant on a Cisco SD-WAN Controller

1. Log in to Cisco SD-WAN Manager as the provider **admin** user.
2. From the Cisco SD-WAN Manager menu, choose **Monitor > Devices**.
Cisco vManage Release 20.6.x and earlier: From the Cisco SD-WAN Manager menu, choose **Monitor > Network**.
3. In the table of devices, click on the hostname of a Cisco SD-WAN Controller.
4. In the left pane, click **Real Time**.
5. In the **Device Options** field, enter **OMP** and select the OMP statistics you wish to view.
6. In the **Select Filters** dialog box, click **Show Filters**.
7. Enter the **Tenant Name** and click **Search**.

Cisco SD-WAN Manager displays the selected OMP statistics for the particular tenant.

View Tenants Associated with a Cisco SD-WAN Controller

1. Log in to Cisco SD-WAN Manager as the provider **admin** user.
2. Click a **Controller** connection number to display a table with detailed information about each connection.
Cisco SD-WAN Manager displays a table that provides a summary of the Cisco SD-WAN Controllers and their connections.
3. For a Cisco SD-WAN Controller, click ... and click **Tenant List**.
Cisco SD-WAN Manager displays a summary of tenants associated with the Cisco SD-WAN Controller.

Migrate Single-Tenant Cisco Catalyst SD-WAN Overlay to Multitenant Cisco Catalyst SD-WAN Deployment

Table 221: Feature History

Feature Name	Release Information	Description
Migrate Single-Tenant Cisco Catalyst SD-WAN Overlay to Multitenant Cisco Catalyst SD-WAN Deployment	Cisco SD-WAN Release 20.5.1 Cisco vManage Release 20.5.1	This feature enables you to migrate a single-tenant Cisco Catalyst SD-WAN overlay to a multitenant deployment using a sequence of Cisco SD-WAN Manager API calls.

Before You Begin

- Before you begin the migration,
 - Ensure that the edge devices in the single-tenant deployment can reach the Cisco SD-WAN Validator in the multitenant deployment
 - Ensure that the template, routing, and policy configuration on the edge devices is synchronized with the current configuration on Cisco SD-WAN Manager
 - Ensure that the Certificate Authority (CA) on both single-tenant and multitenant Cisco SD-WAN Managers are same.
 - Configure a maintenance window for the single-tenant overlay before performing this procedure. See [Configure or Cancel vManage Server Maintenance Window](#).
- Minimum software requirements for the single-tenant overlay to be migrated:

Device	Software Version
Cisco SD-WAN Manager	Cisco vManage Release 20.5.1
Cisco SD-WAN Validator	Cisco SD-WAN Release 20.5.1
Cisco SD-WAN Controller	Cisco SD-WAN Release 20.5.1
Cisco vEdge Device	Cisco SD-WAN Release 20.4.1

- Minimum software requirements for the multitenant deployment to which the single-tenant overlay must be migrated:

Device	Software Version
Cisco SD-WAN Manager	Cisco vManage Release 20.5.1
Cisco SD-WAN Validator	Cisco SD-WAN Release 20.5.1
Cisco SD-WAN Controller	Cisco SD-WAN Release 20.5.1
Cisco vEdge Device	Cisco SD-WAN Release 20.5.1

- We recommend that you use a custom script or a third-party application like Postman to execute the API calls.

Migration Procedure

1. Export the single-tenant deployment and configuration data from a Cisco SD-WAN Manager instance controlling the overlay.

Method	POST
URL	<code>https://ST-vManage-IP-address</code>
Endpoint	<code>/dataservice/tenantmigration/export</code>
Authorization	Admin user credentials.
Body	Required Format: Raw JSON <pre>{ "desc": <tenant_description>, "name": <tenant_name>, "subdomain": <tenant_name>.<domain>, "orgName": <tenant_orgname > }</pre> Field Description: • desc: A description of the tenant. The description can be up to 256 characters and can contain only alphanumeric characters. • name: Unique name for the tenant in the multitenant deployment. • subdomain: Fully qualified sub-domain name of the tenant. The sub-domain name must include the domain name of the service provider. For example, if <code>multitenancy.com</code> is the domain name of service provider, and the tenant name is <code>Customer1</code>, the tenant sub-domain name would be <code>Customer1.multitenancy.com</code>. • orgName: Name of the tenant organization. The organization name is case-sensitive.
Response	Format: JSON <pre>{ "processId": <vManage_process_ID>, }</pre>

While exporting the data, Cisco SD-WAN Manager attempts to detach any CLI templates from the edge devices in preparation for the migration to the multitenant deployment. If prompted by Cisco SD-WAN Manager, detach CLI templates from the edge devices and execute the export API call again.

2. Check the status of the data export task in Cisco SD-WAN Manager. When the task succeeds, download the data using the URL
`https://ST-vManage-IP-address/dataservice/tenantmigration/download/default.tar.gz`
3. On a multitenant Cisco SD-WAN Manager instance, import the data exported from the single-tenant overlay.

Method	POST
URL	<code>https://MT-vManage-IP-address</code>
Endpoint	<code>/dataservice/tenantmigration/import</code>
Authorization	Provider Admin user credentials.
Body	Required Format: form-data Key Type: File Value: <code>default.tar.gz</code>
Response	Format: JSON <pre>{ "processId": <vManage_process_ID>, "migrationTokenURL": <token_URL>, }</pre>

When the task succeeds, on the multitenant Cisco SD-WAN Manager, you can view the devices, templates, and policies imported from the single-tenant overlay.

- Obtain the migration token using the token URL obtained in response to the API call in **Step 3**.

Method	GET
URL	<code>https://MT-vManage-IP-address</code>
Endpoint	migrationTokenURL obtained in Step 3 .
Authorization	Provider Admin user credentials.
Response	The migration token as a large blob of encoded text.

- On the single-tenant Cisco SD-WAN Manager instance, initiate the migration of the overlay to the multitenant deployment.

Method	POST
URL	<code>https://ST-vManage-IP-address</code>
Endpoint	<code>dataservice/tenantmigration/networkMigration</code>
Authorization	Admin user credentials.
Body	Required Format: Raw text Content: Migration token obtained in Step 4 .
Response	Format: JSON <pre>{ "processId": <vManage_process_ID>, }</pre>

In Cisco SD-WAN Manager, check the status of the migration task. As part of the migration task, the address of the multitenant Cisco SD-WAN Validator, and the service provider and tenant organization

names are pushed to the WAN edge devices of the single-tenant overlay. If the task succeeds, WAN edge devices form control connections to controllers in the multitenant deployment; the WAN edge devices are no longer connected to the controllers of the single-tenant overlay.

Attach any CLI templates detached from the edge devices (in Step 1) after migration to the multitenant deployment. Before you attach the templates, update the Cisco SD-WAN Validator IP address and the Organization name to match the configuration of the multitenant deployment.



Note In the single-tenant deployment, if Cisco SD-WAN Manager-signed certificates are installed on cloud-based WAN edge devices, the certificates are cleared when the devices are migrated to the multitenant deployment. You must re-certify the devices on the multitenant Cisco SD-WAN Manager. If enterprise certificates are installed on the cloud-based WAN edge devices, the certificates are not affected by the migration. For more information, see [Enterprise Certificates](#).



CHAPTER 18

Appendix: Cisco SD-WAN Manager How-Tos

- [How to Load a Custom Cisco SD-WAN Manager Application Server Logo, on page 609](#)

How to Load a Custom Cisco SD-WAN Manager Application Server Logo

To change the Cisco SD-WAN Manager web application server logo and load a new custom logo, use the **request nms application-server update-logo** command.

The logo image is located in the upper left corner of all Cisco SD-WAN Manager web application server screens. You can load two files, a larger version, which is displayed on wider browser screens, and a smaller version, which is displayed when the screen size narrows. Both files must be PNG files located on the local device, and both must be 1 MB or smaller in size. For best resolution, it is recommended that the image for the large logo be 180 x 33 pixels, and for the small logo 30 x 33 pixels.

